

Pauli semiwebs of the double-checking circuits

A basis for the $d = 3$ and $d = 5$ ZX diagrams

Contents

Each following page shows one basis element. Products of the 90 elements at $d = 3$, or the 264 elements at $d = 5$, generate every semiweb of the corresponding graph, up to overall Pauli phase. The basis is not unique. The counts depend on the chosen ZX graph.

	Count	Pages
$d = 3$: incoming X_q, Z_q	14	2–15
$d = 3$: additional output support	14	16–29
$d = 3$: no boundary support	62	30–91
$d = 5$: incoming X_q, Z_q	38	92–129
$d = 5$: additional output support	38	130–167
$d = 5$: no boundary support	188	168–355

“No boundary support” means that the labels on all input and output wires are I . These semiwebs may still have defects. The full Pauli-web subspaces have dimensions 12 and 37; their subspaces with no boundary support have dimensions 6 and 19.

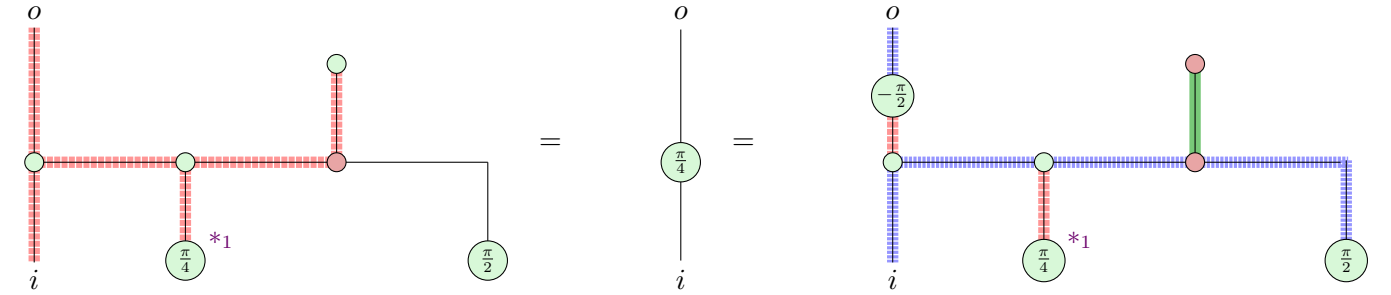
Circuit

The $d = 5$ headings F1–F7 and U1–U7 identify the original fold and unfold CNOT groups. Gates within a group use separate drawing columns to avoid overlapping connections.

Sources

The definitions and drawing styles follow Kissinger and van de Wetering, *ZX-Flow*, arXiv:2603.09580, Sec. 3. The double-checking circuits are from the construction of Gidney, Shutty and Jones, arXiv:2409.17595. The $d = 5$ graph retains the mixed $T/T^\dagger$ pattern of the SOFT source.

Reading a diagram (diagram from <https://github.com/tqec/tqec>)



Both circuits give T , up to a scalar, with an $S^\dagger$ correction on the right. The overlays show different semiwebs.

Legend I — X Y Z Z spider X spider
Up to normalisation: $-\langle \frac{\pi}{4} \rangle = T$ ($-\pi/4$: $T^\dagger$); $-\langle \frac{\pi}{2} \rangle = S$ ($-\pi/2$: $S^\dagger$); = CNOT; measurement outcome 0: $-\langle \circ \rangle = \langle + |$ (X basis), $-\langle \bullet \rangle = \langle 0 |$ (Z basis); preparations $\bullet = |0\rangle$ and $\circ = |+\rangle$; resources $\langle \frac{\pi}{4} \rangle = |T\rangle$, $\langle \frac{\pi}{2} \rangle = |Y\rangle$.

At every spider, the incident Pauli labels satisfy

$$w_v |v(\theta_v)\rangle = \lambda_v |v(\theta_v + \delta_v)\rangle.$$

A nonzero phase shift δ_v modulo 2π is a *defect*. No defects gives a Pauli web. Numbered violet stars mark the affected nodes; the phase shifts are listed below each diagram. The phases printed inside the spiders are the original phases. A π defect at a preparation or measurement changes $|+\rangle$ or $\langle +|$ to $|-\rangle$ or $\langle -|$.

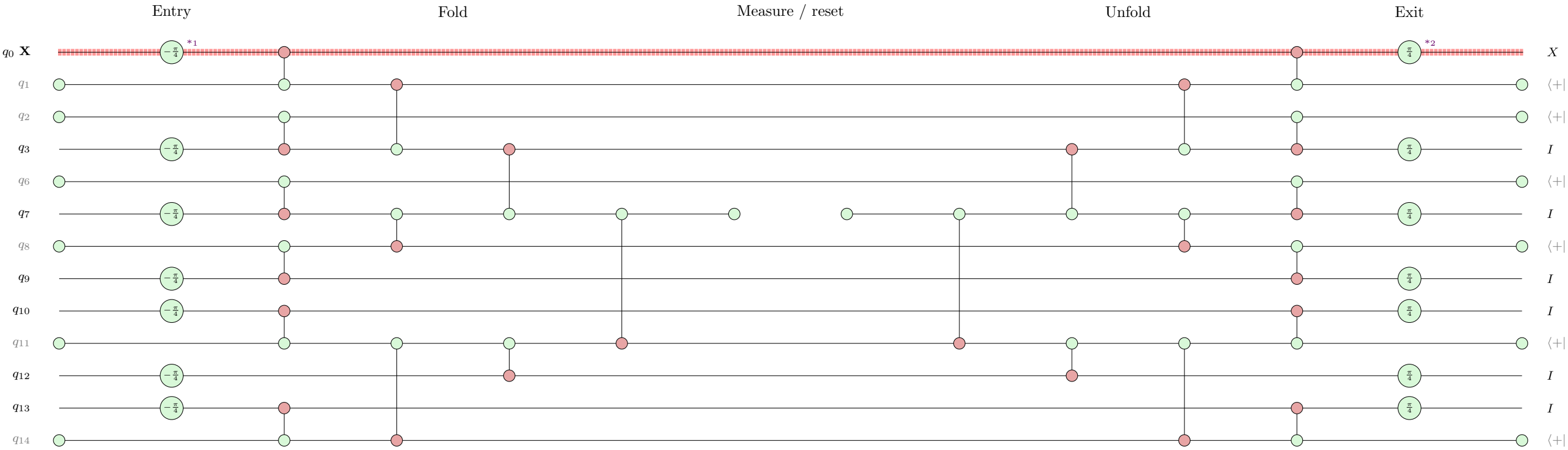
The boundary labels and scalar satisfy

$$K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

K represents the selected double-checking stage, and K_{δ} has the marked phases changed. Multiplication includes the Pauli scalar phases, and phase changes must be composed. Adding two defect shifts computed on the original spider is not generally correct.

Each TikZ file records the affected node, qubit and phase change in comments. The star is positioned relative to that named node.

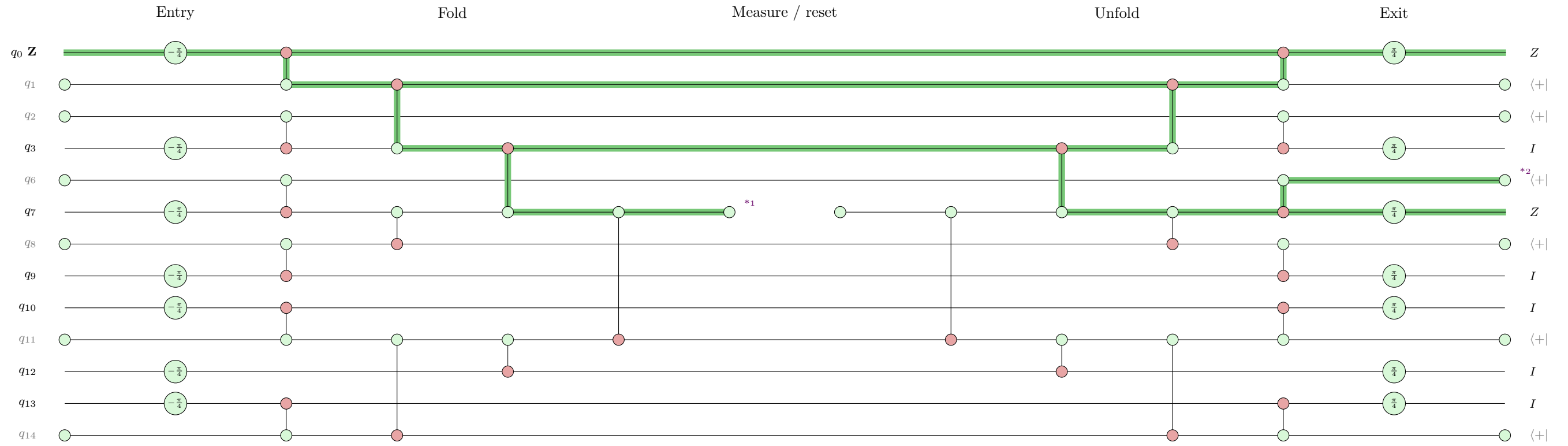
$d = 3$: incoming X_0
Incoming Paulis. 2 phase defects.



$$P_{\text{in}} = X_0, \quad P_{\text{out}} = X_0, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_0
Incoming Paulis. 2 phase defects.

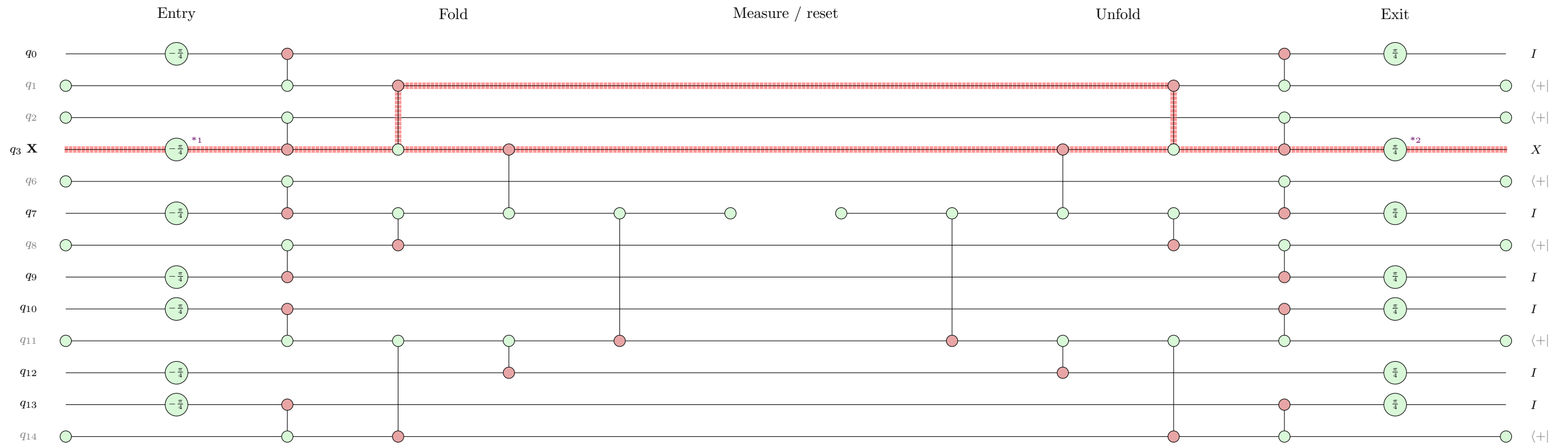


$$P_{\text{in}} = Z_0, \quad P_{\text{out}} = Z_0 Z_7, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: incoming X_3

Incoming Paulis. 2 phase defects.

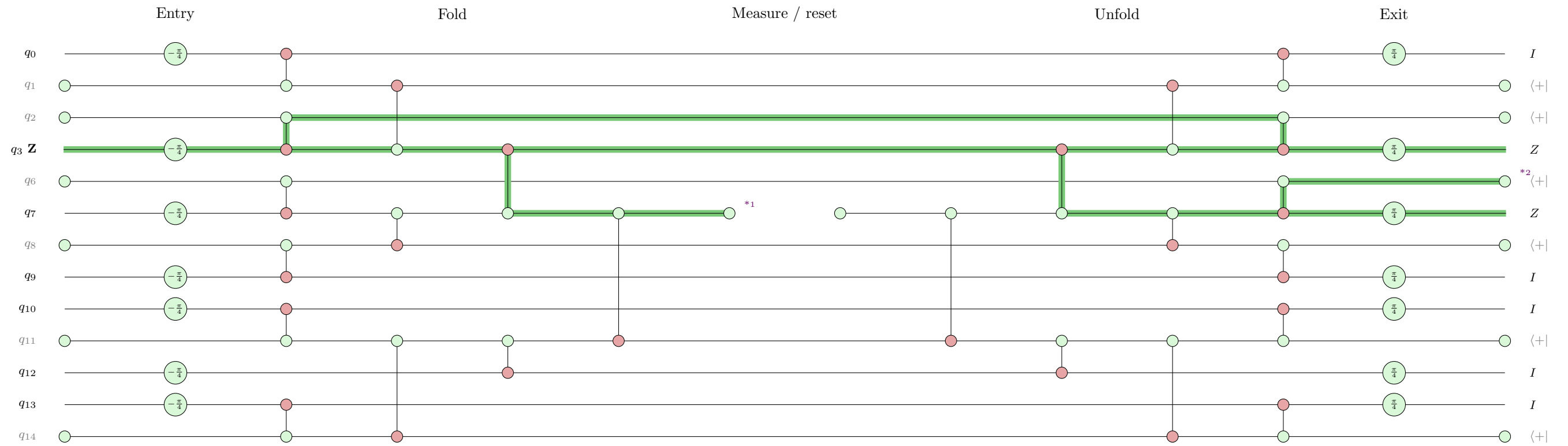


$$P_{\text{in}} = X_3, \quad P_{\text{out}} = X_3, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_3

Incoming Paulis. 2 phase defects.

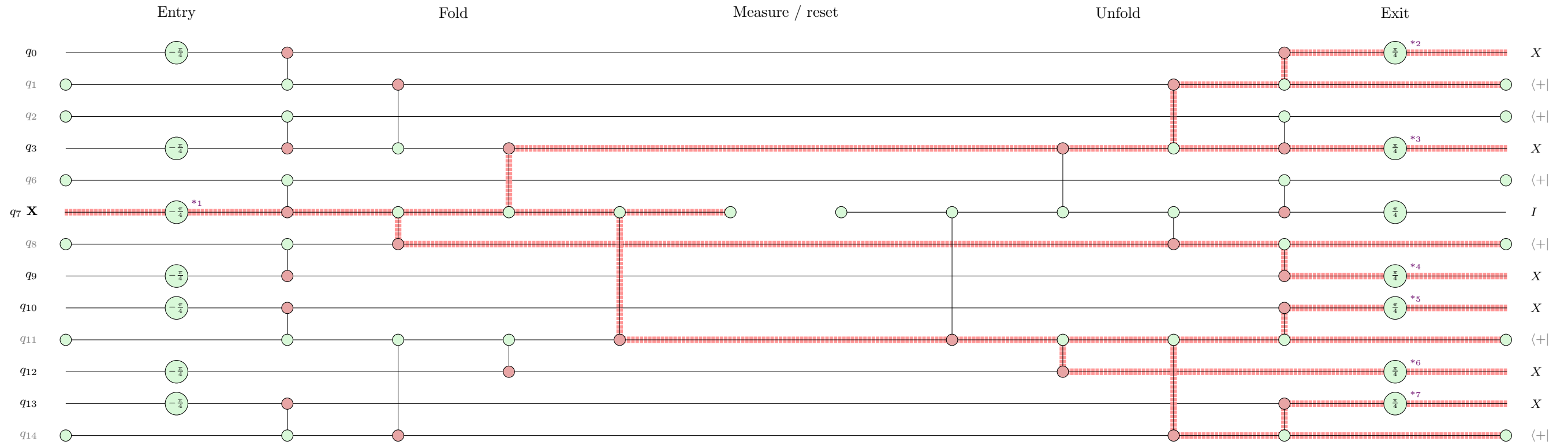


$$P_{\text{in}} = Z_3, \quad P_{\text{out}} = Z_3 Z_7, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: incoming X_7

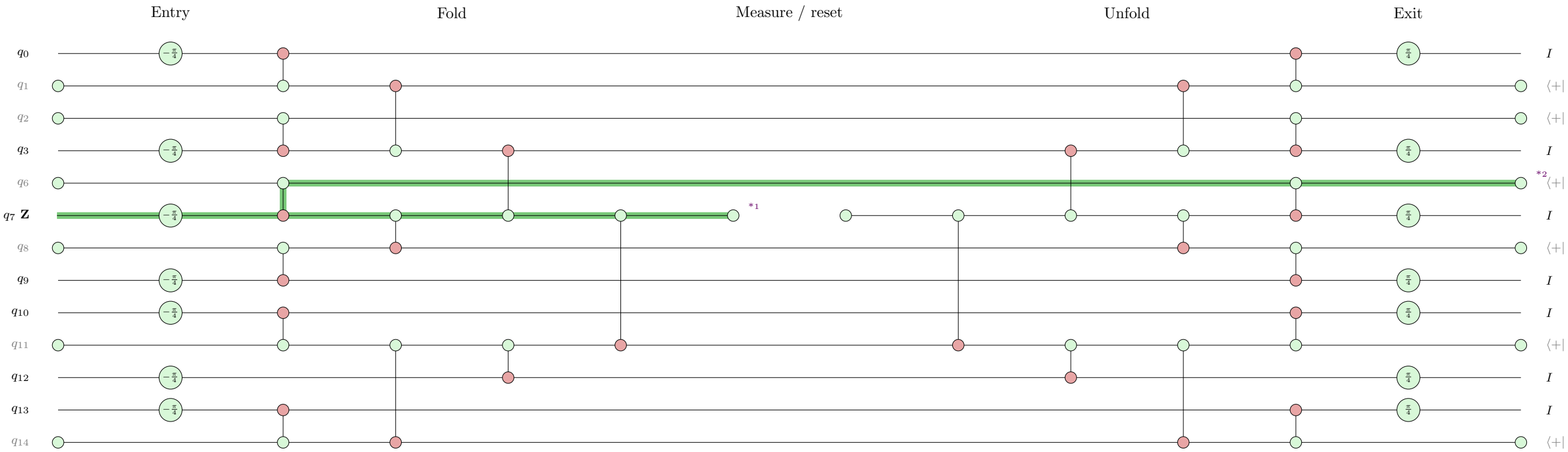
Incoming Paulis. 7 phase defects.



$$P_{\text{in}} = X_7, \quad P_{\text{out}} = X_0 X_3 X_9 X_{10} X_{12} X_{13}, \quad \lambda = e^{i\frac{5\pi}{4}}, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$; $\delta_3 = -\frac{\pi}{2}$; $\delta_4 = -\frac{\pi}{2}$; $\delta_5 = -\frac{\pi}{2}$; $\delta_6 = -\frac{\pi}{2}$; $\delta_7 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_7
 Incoming Paulis. 2 phase defects.

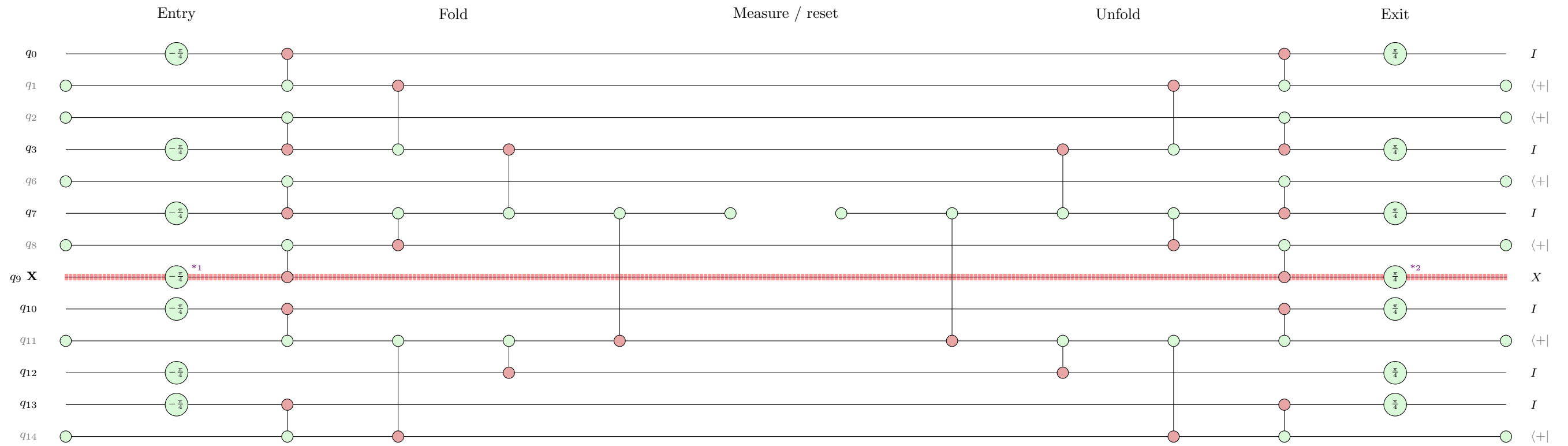


$$P_{\text{in}} = Z_7, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: incoming X_9

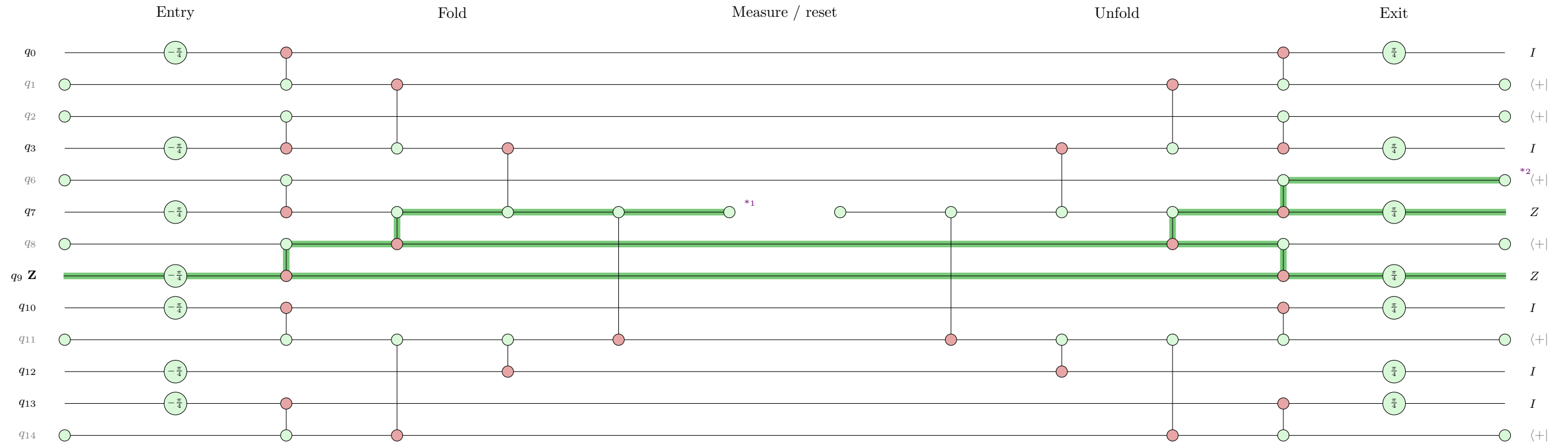
Incoming Paulis. 2 phase defects.



$$P_{\text{in}} = X_9, \quad P_{\text{out}} = X_9, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_9
Incoming Paulis. 2 phase defects.

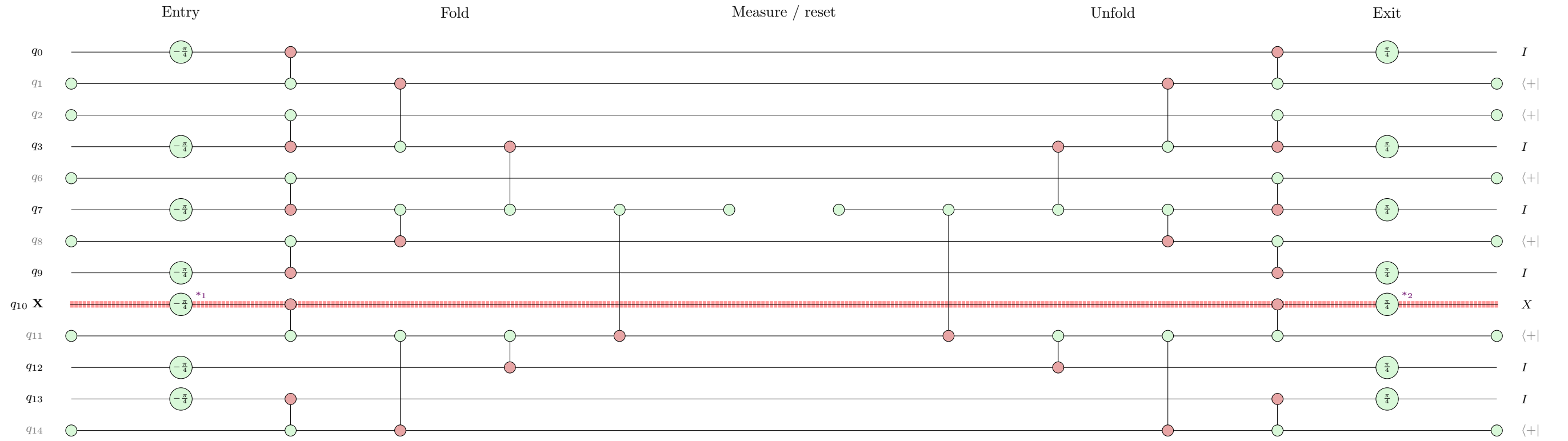


$$P_{\text{in}} = Z_9, \quad P_{\text{out}} = Z_7 Z_9, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: incoming X_{10}

Incoming Paulis. 2 phase defects.

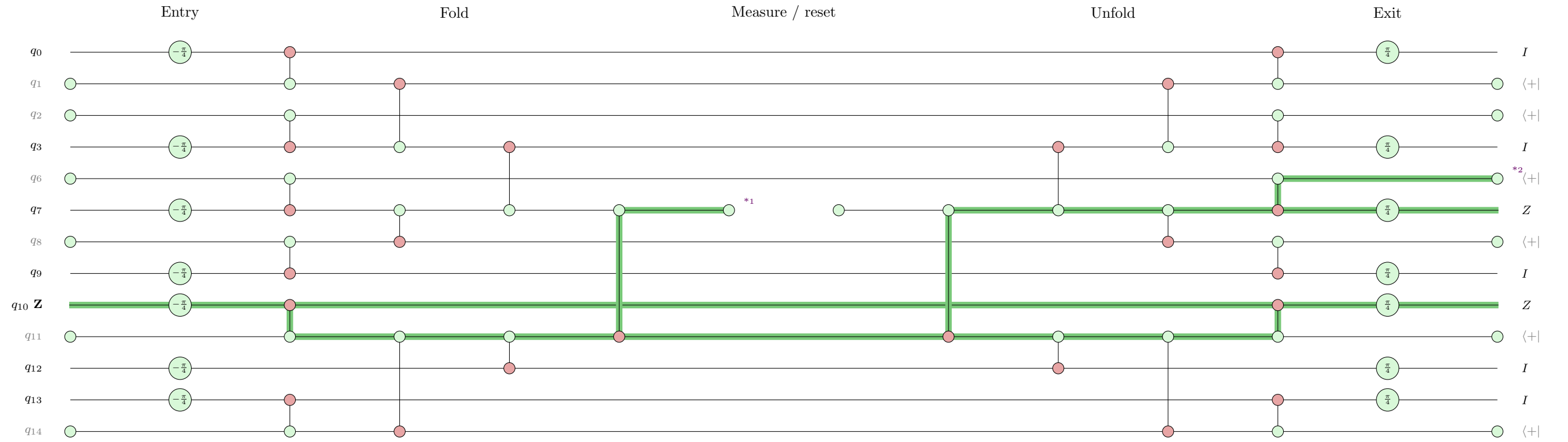


$$P_{\text{in}} = X_{10}, \quad P_{\text{out}} = X_{10}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_{10}

Incoming Paulis. 2 phase defects.

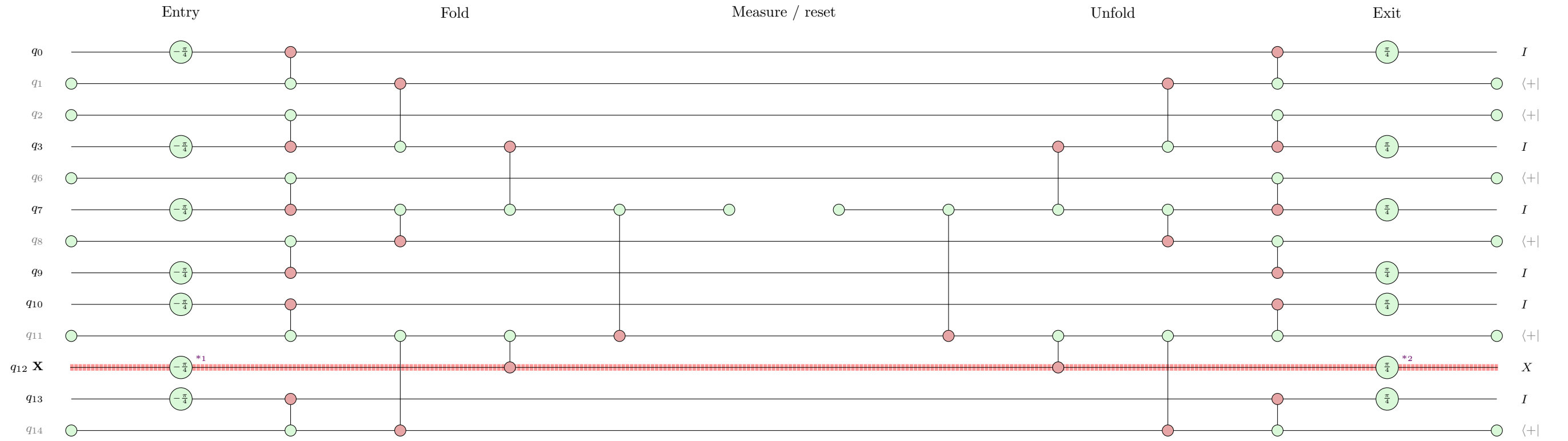


$$P_{\text{in}} = Z_{10}, \quad P_{\text{out}} = Z_7 Z_{10}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: incoming X_{12}

Incoming Paulis. 2 phase defects.

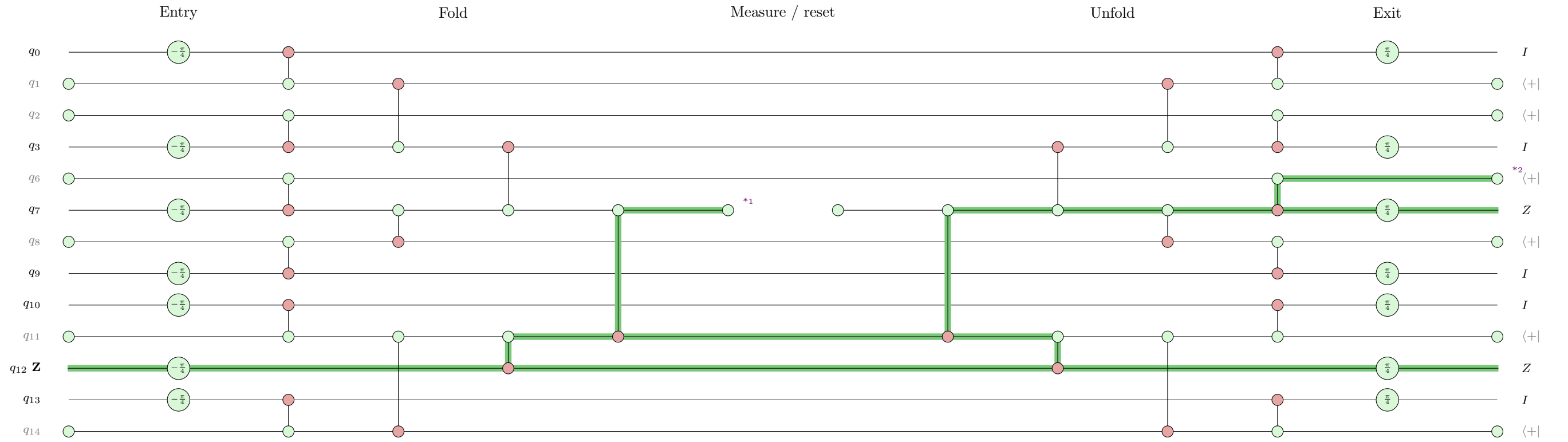


$$P_{\text{in}} = X_{12}, \quad P_{\text{out}} = X_{12}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_{12}

Incoming Paulis. 2 phase defects.

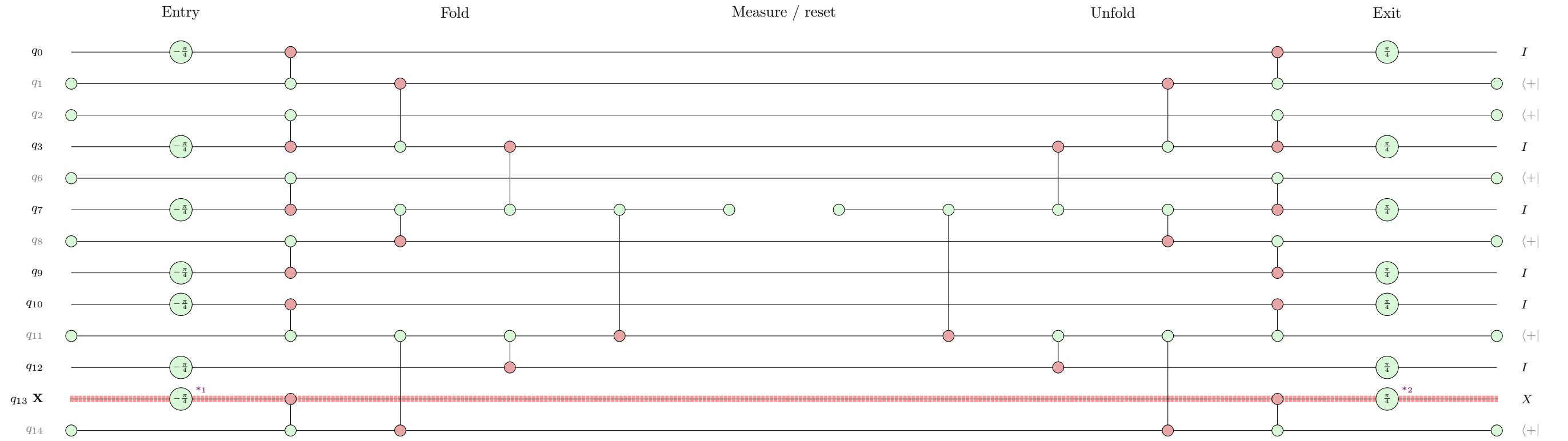


$$P_{\text{in}} = Z_{12}, \quad P_{\text{out}} = Z_7 Z_{12}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: incoming X_{13}

Incoming Paulis. 2 phase defects.

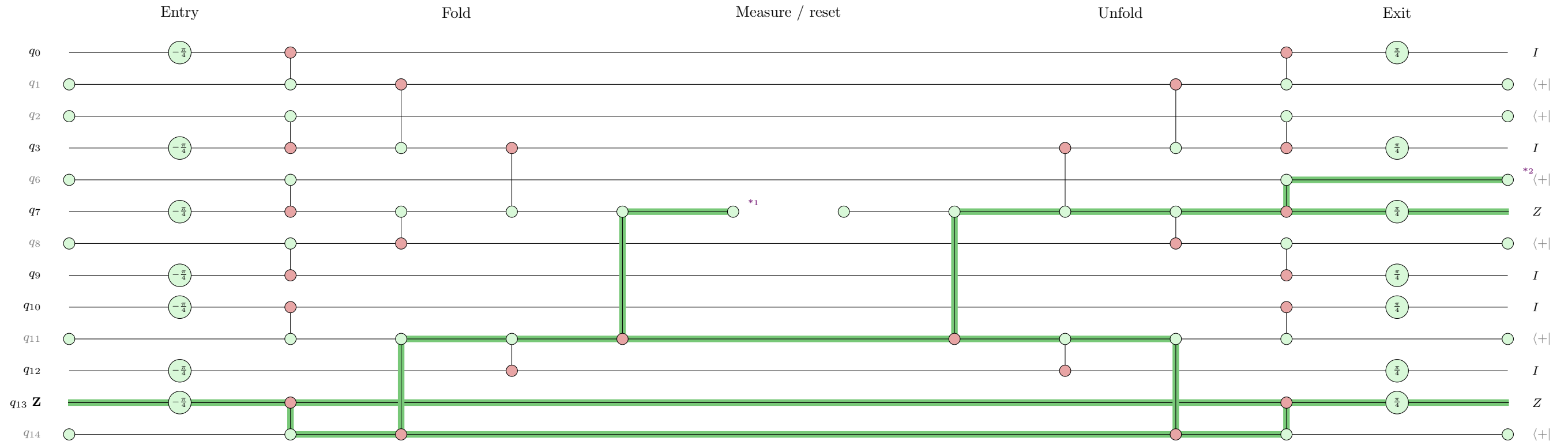


$$P_{\text{in}} = X_{13}, \quad P_{\text{out}} = X_{13}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: incoming Z_{13}

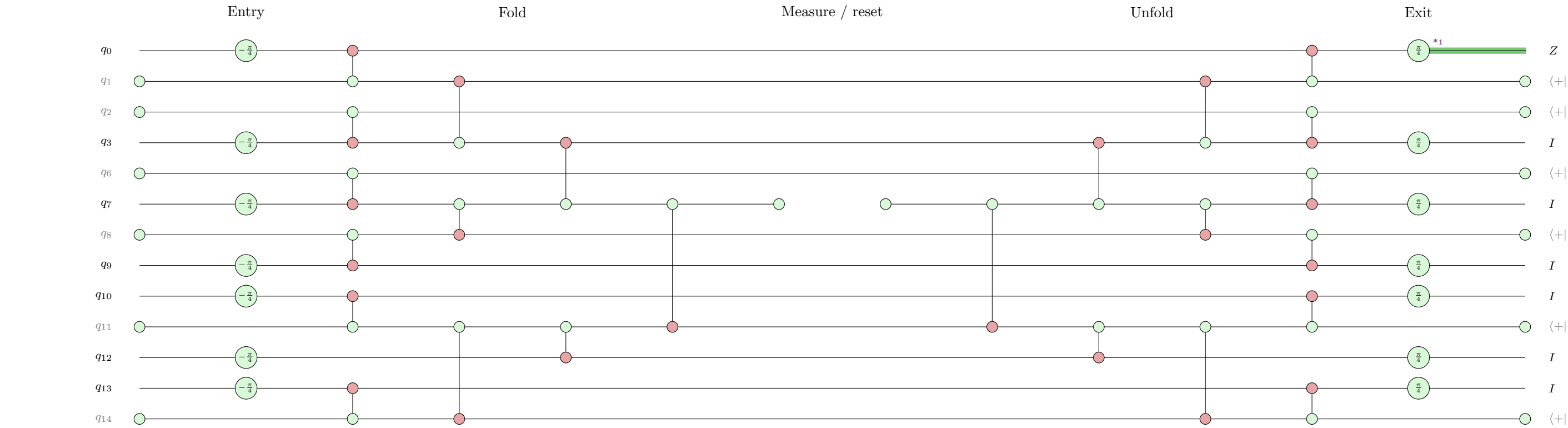
Incoming Paulis. 2 phase defects.



$$P_{\text{in}} = Z_{13}, \quad P_{\text{out}} = Z_7 Z_{13}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

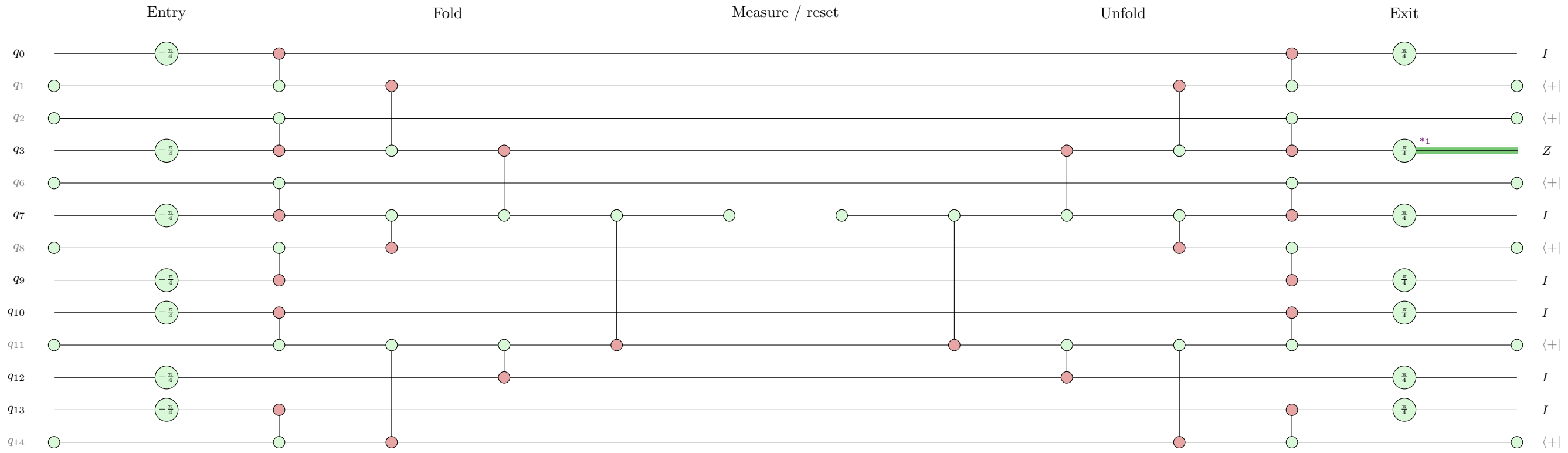
$d = 3$: basis element 15
 Additional output support. 1 phase defect.



$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_0, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

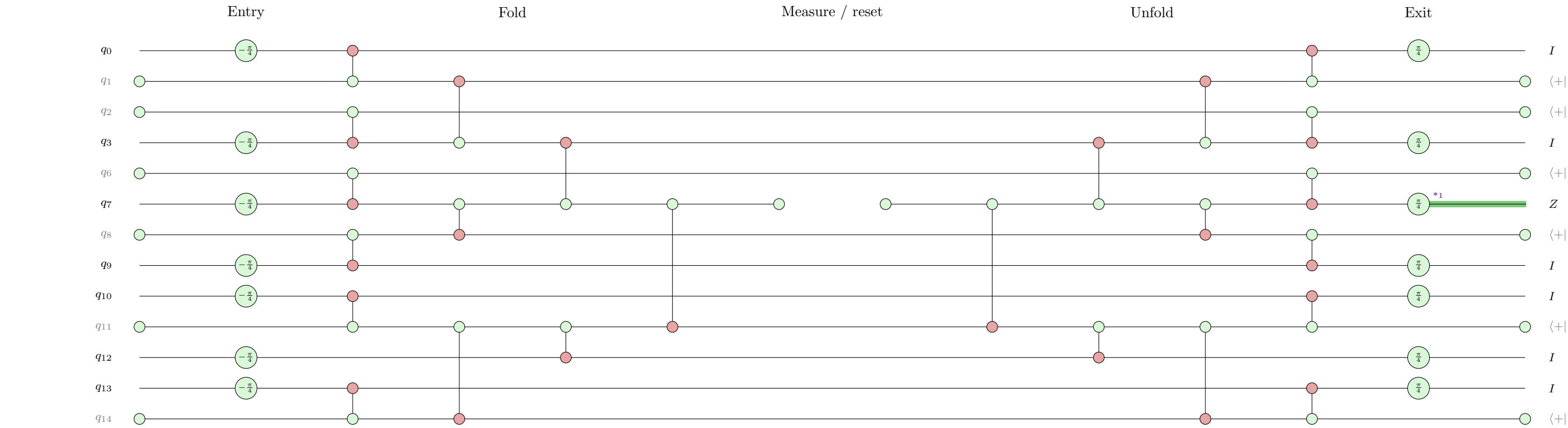
Phase changes: $\delta_1 = \pi$.

$d = 3$: basis element 16
 Additional output support. 1 phase defect.



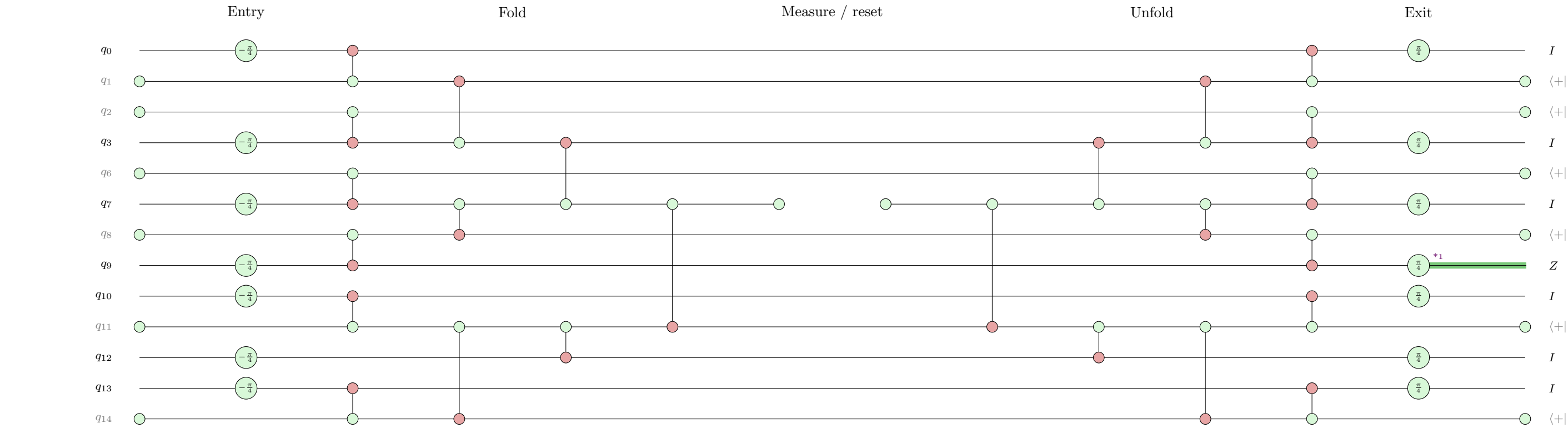
$P_{\text{in}} = I, \quad P_{\text{out}} = Z_3, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi$.

$d = 3$: basis element 17
 Additional output support. 1 phase defect.



$P_{\text{in}} = I, \quad P_{\text{out}} = Z_7, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi$.

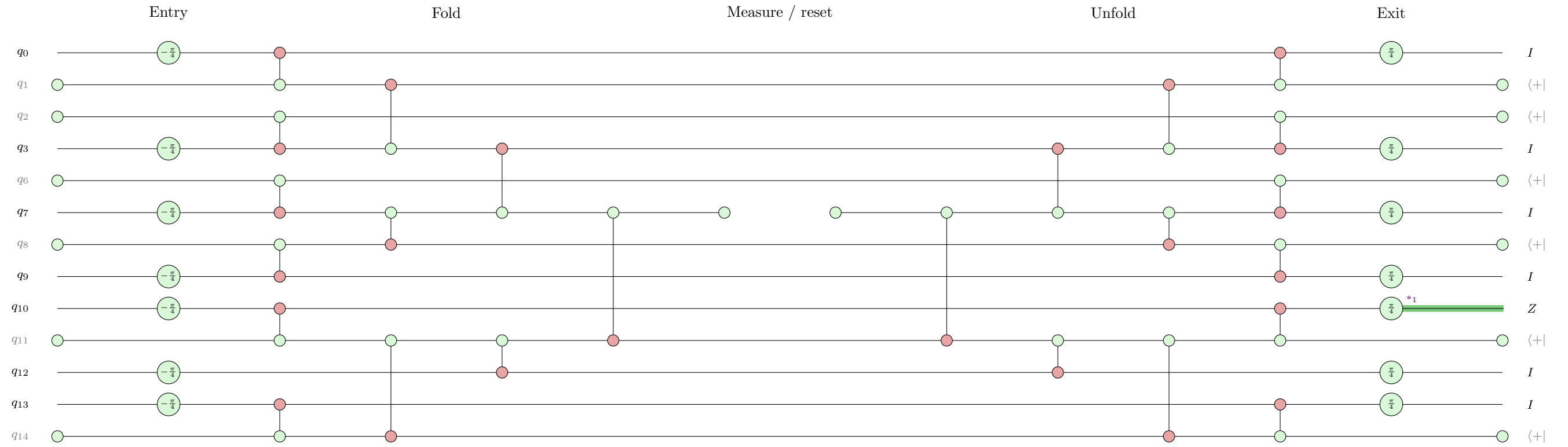
$d = 3$: basis element 18
 Additional output support. 1 phase defect.



$P_{\text{in}} = I, \quad P_{\text{out}} = Z_9, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi.$

$d = 3$: basis element 19

Additional output support. 1 phase defect.

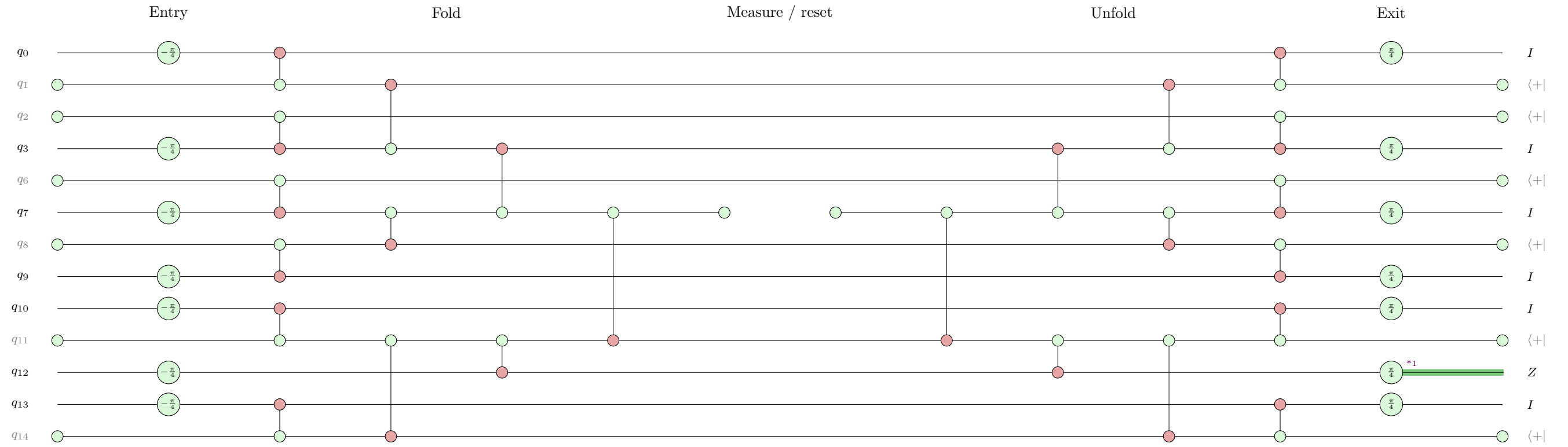


$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{10}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$.

$d = 3$: basis element 20

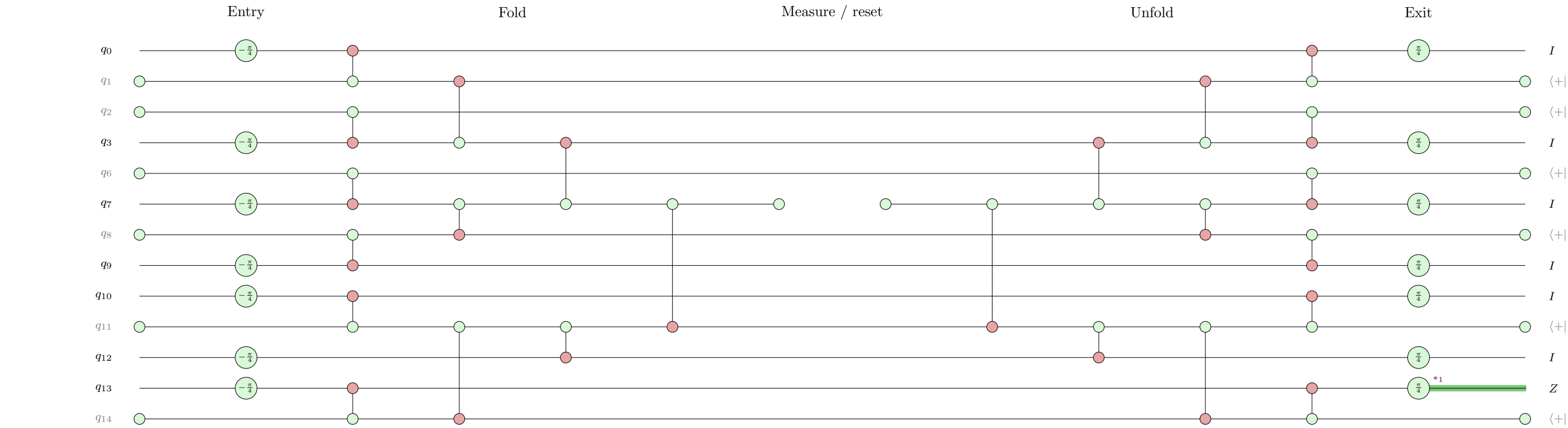
Additional output support. 1 phase defect.



$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{12}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

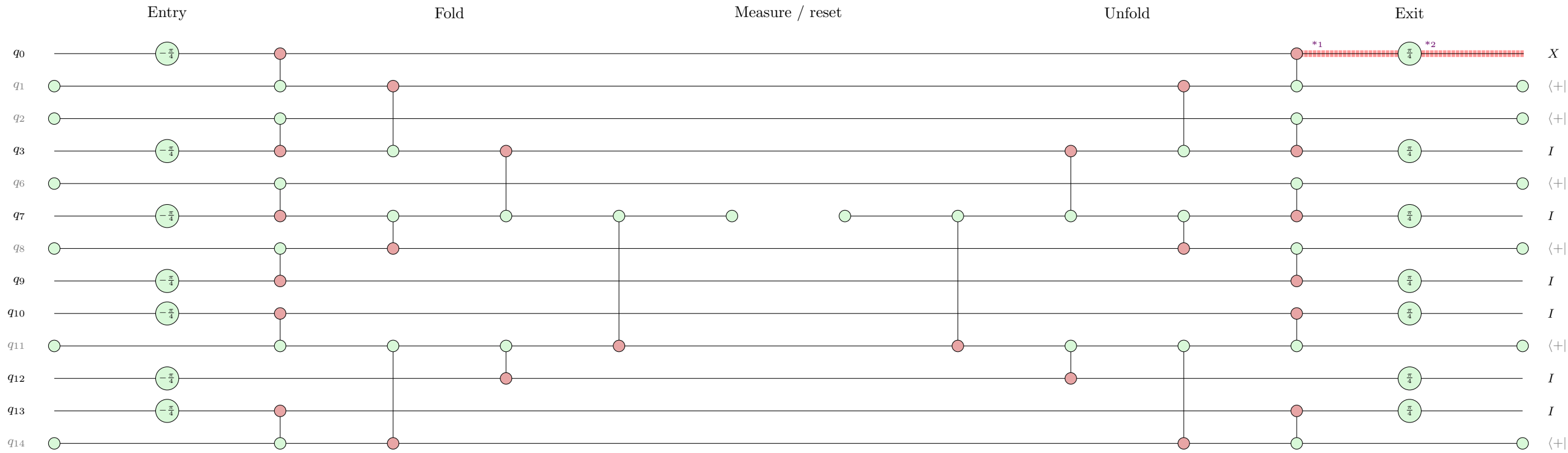
Phase changes: $\delta_1 = \pi$.

$d = 3$: basis element 21
 Additional output support. 1 phase defect.



$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{13}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi$.

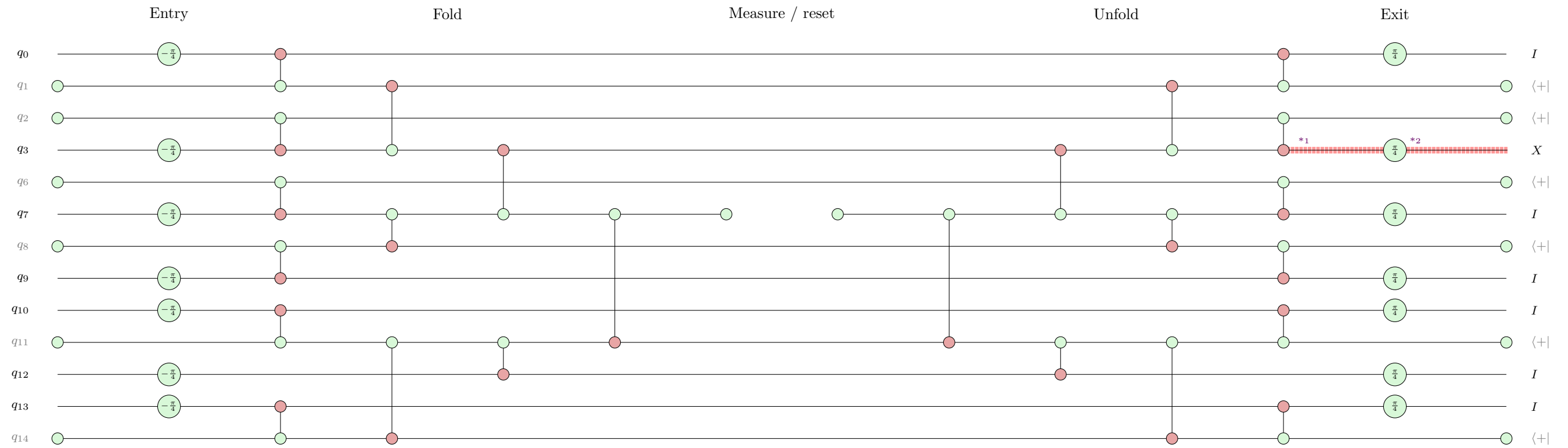
$d = 3$: basis element 22
 Additional output support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = X_0, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = -\frac{\pi}{2}.$

$d = 3$: basis element 23

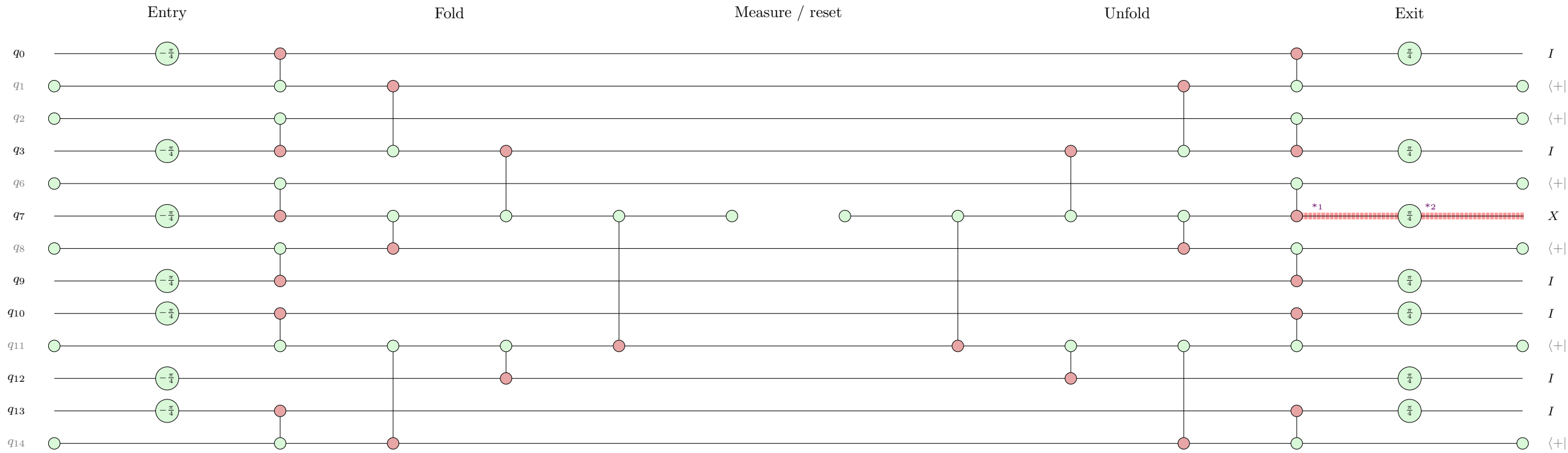
Additional output support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = X_3, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

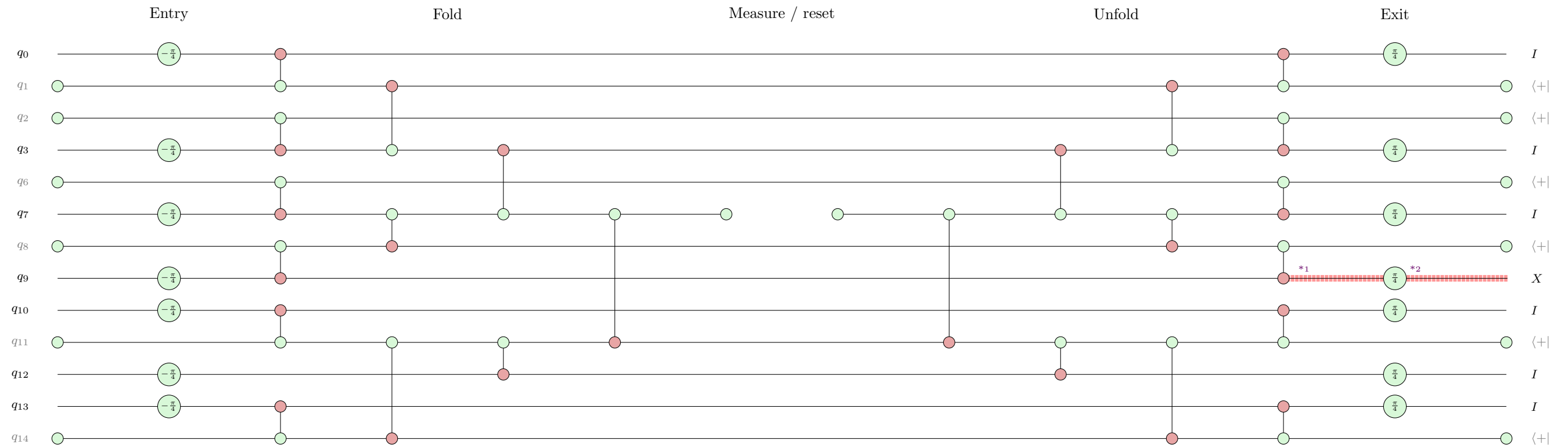
$d = 3$: basis element 24
 Additional output support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = X_7, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = -\frac{\pi}{2}.$

$d = 3$: basis element 25

Additional output support. 2 phase defects.

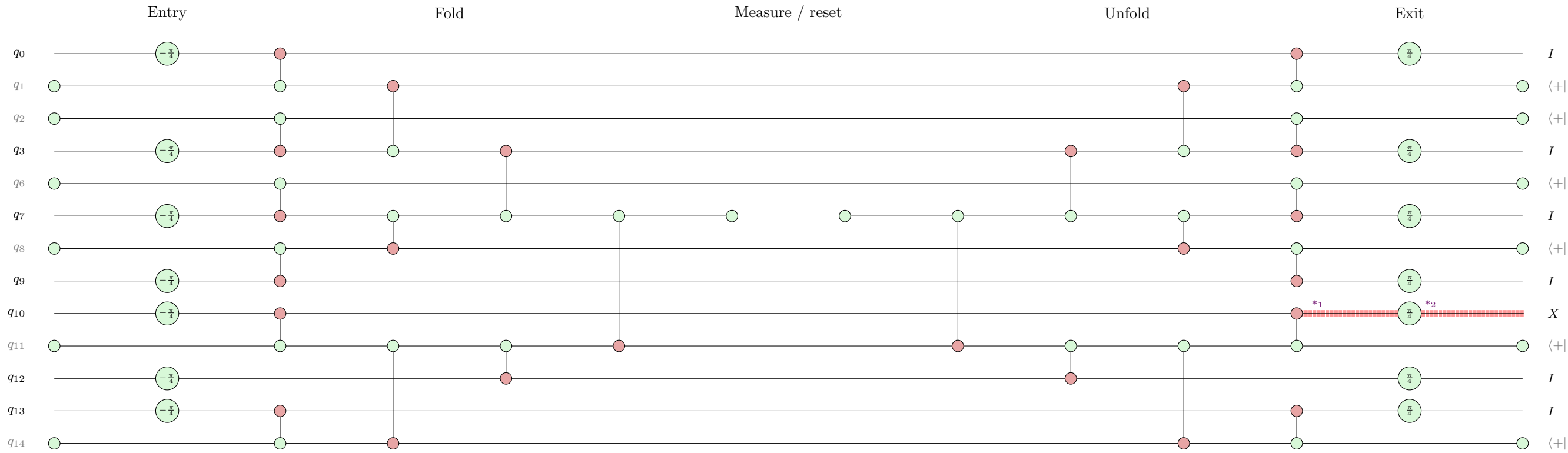


$$P_{\text{in}} = I, \quad P_{\text{out}} = X_9, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: basis element 26

Additional output support. 2 phase defects.

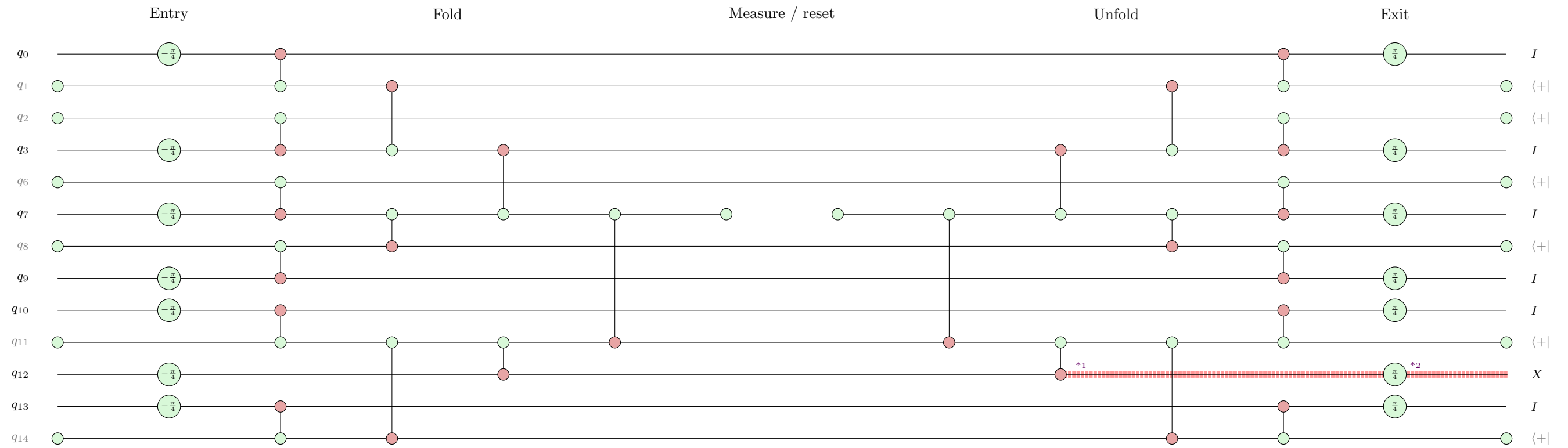


$P_{\text{in}} = I, \quad P_{\text{out}} = X_{10}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$

Phase changes: $\delta_1 = \pi; \quad \delta_2 = -\frac{\pi}{2}.$

$d = 3$: basis element 27

Additional output support. 2 phase defects.

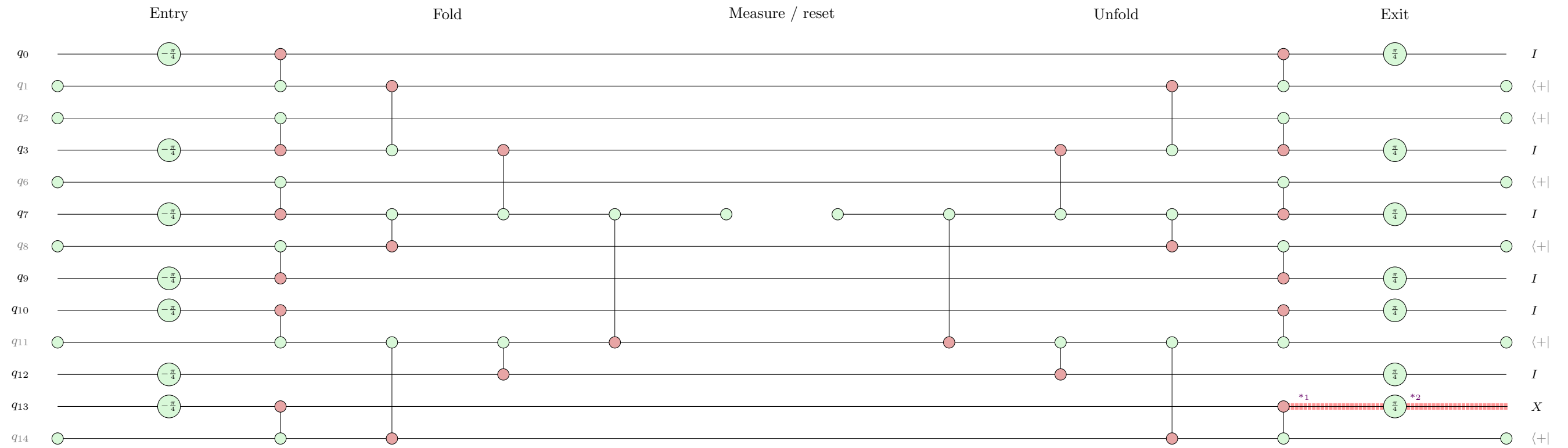


$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{12}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: basis element 28

Additional output support. 2 phase defects.

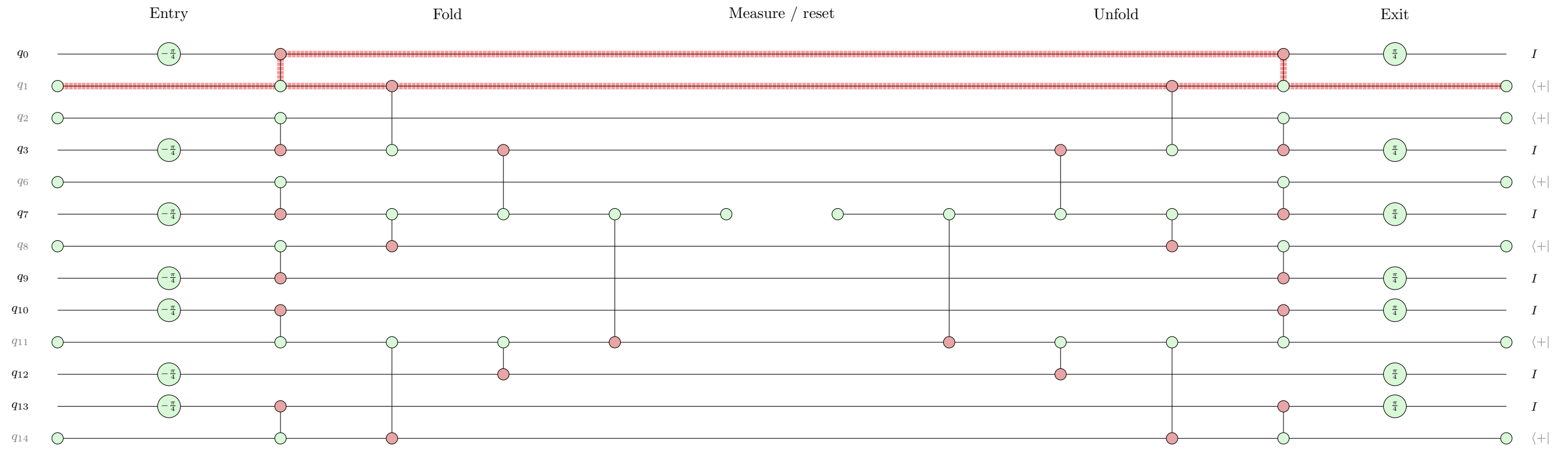


$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{13}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

$d = 3$: basis element 29

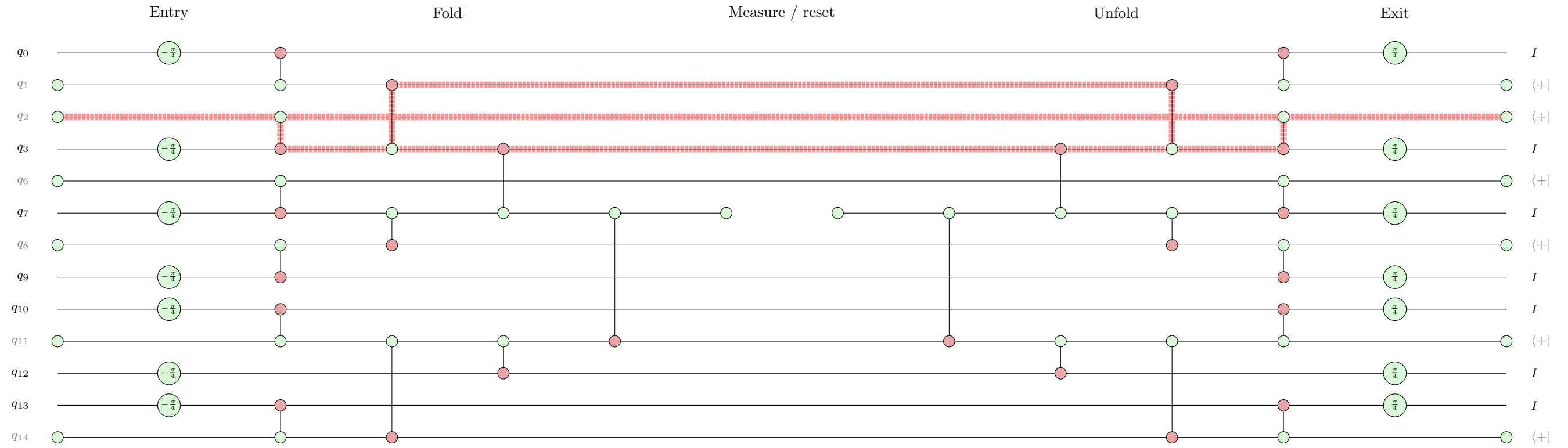
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 3$: basis element 30

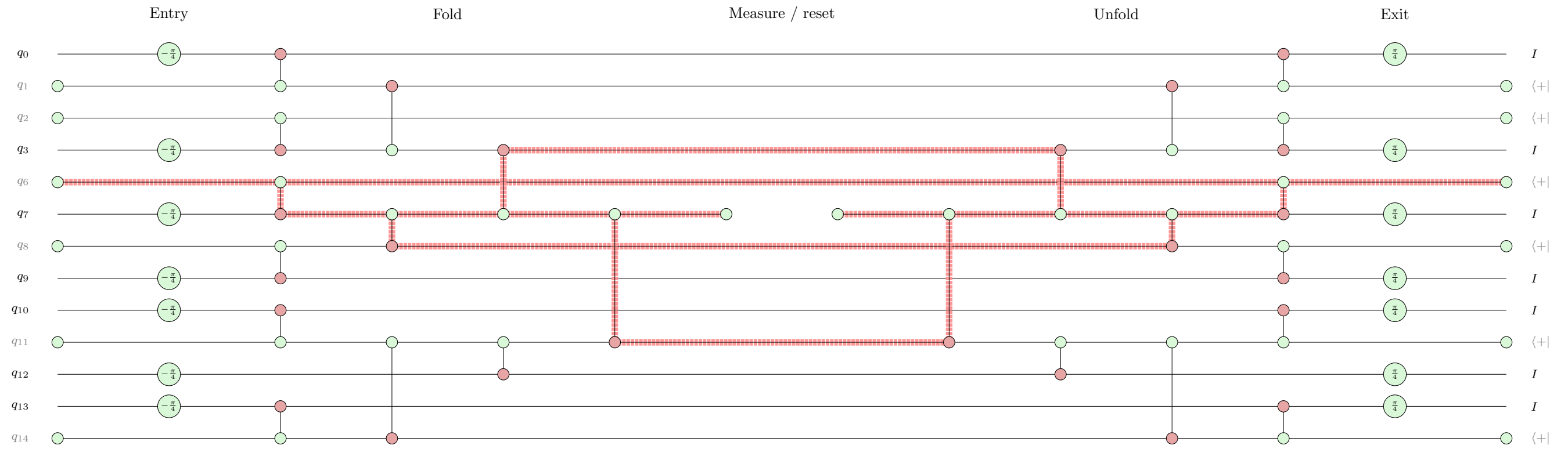
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

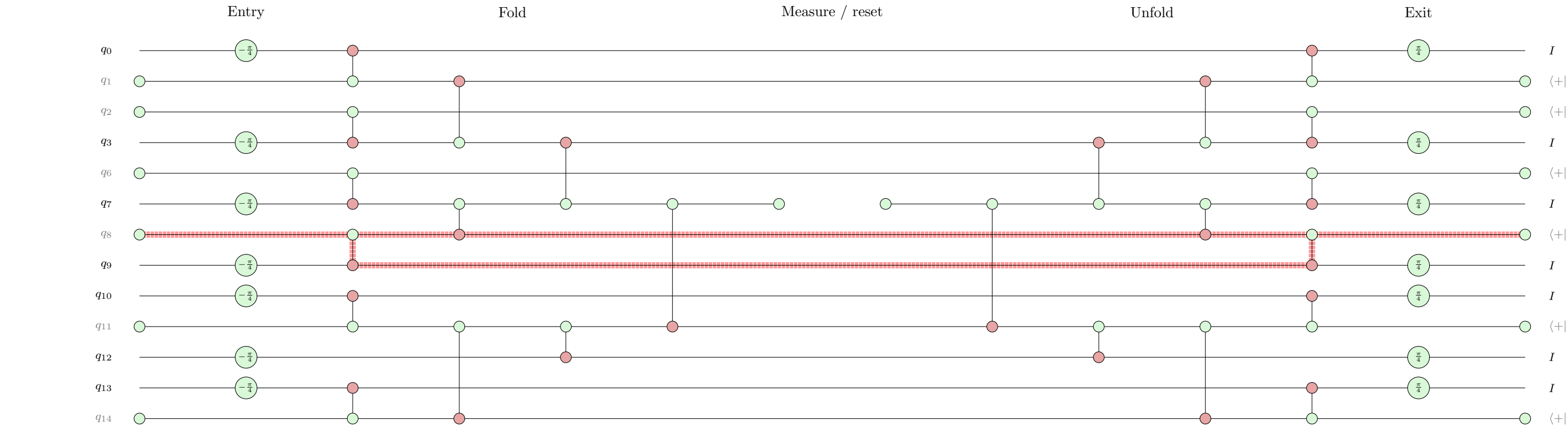
$d = 3$: basis element 31

No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

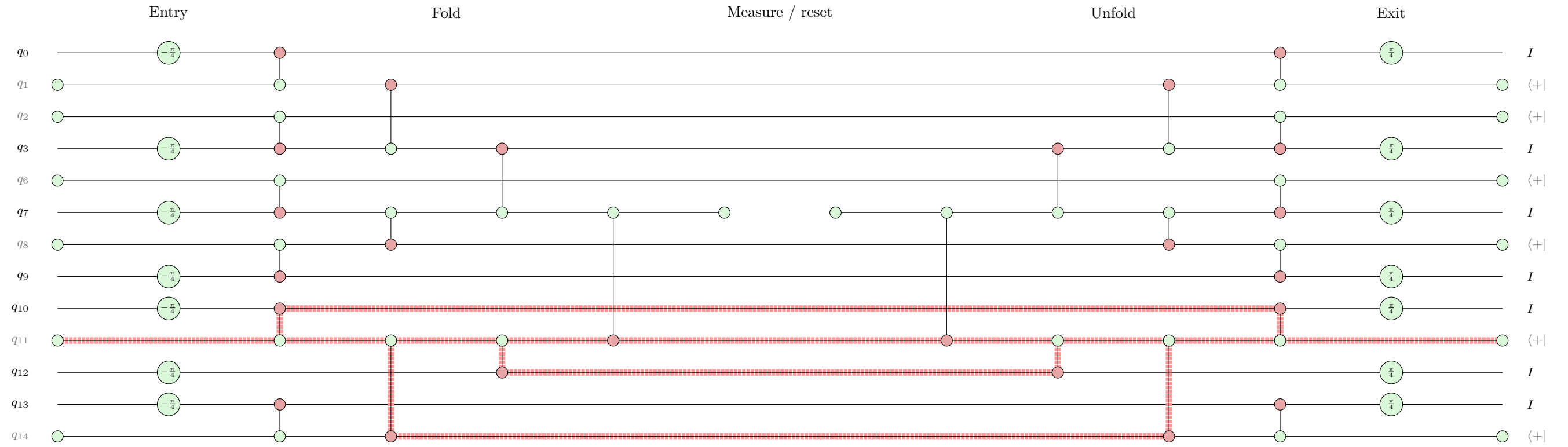
$d = 3$: basis element 32
 No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 3$: basis element 33

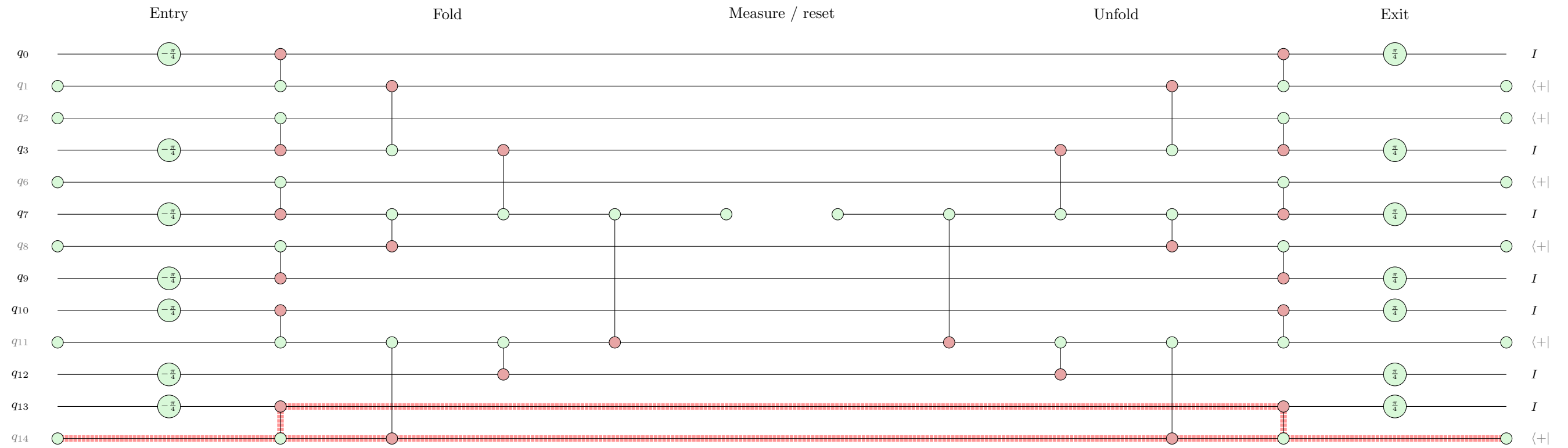
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

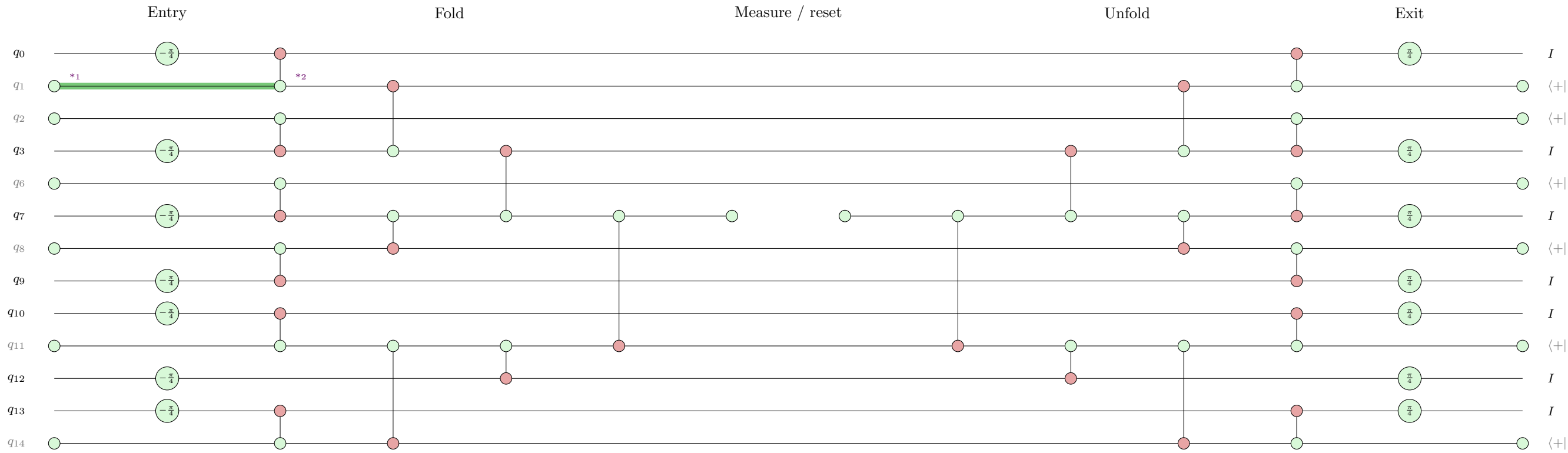
$d = 3$: basis element 34

No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 3$: basis element 35
 No boundary support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

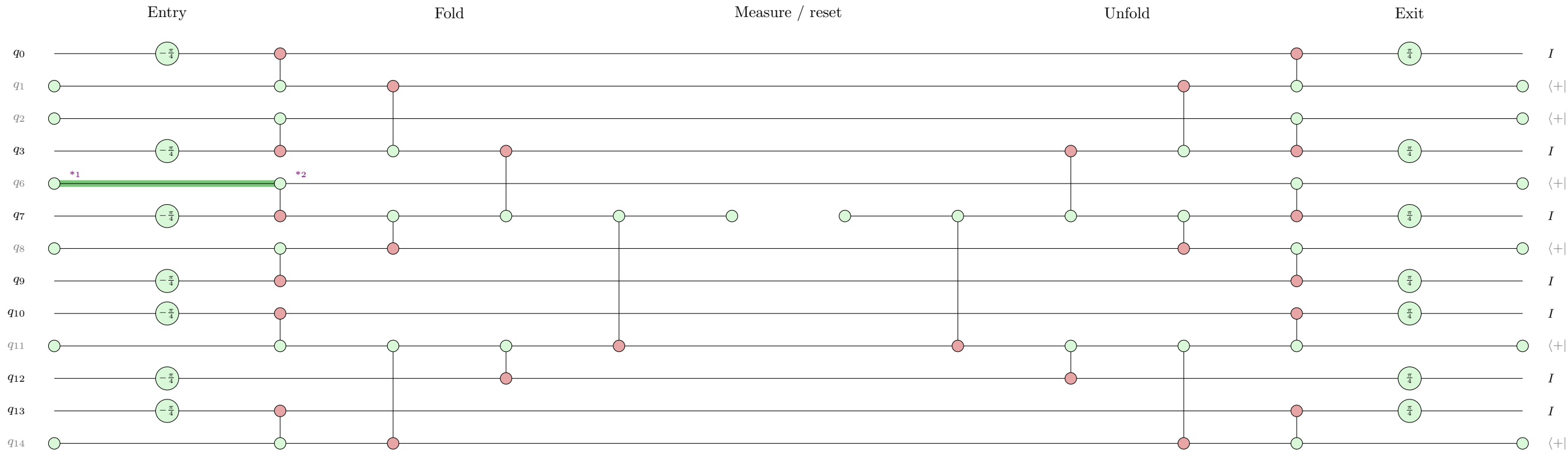
$d = 3$: basis element 36
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: basis element 37
 No boundary support. 2 phase defects.



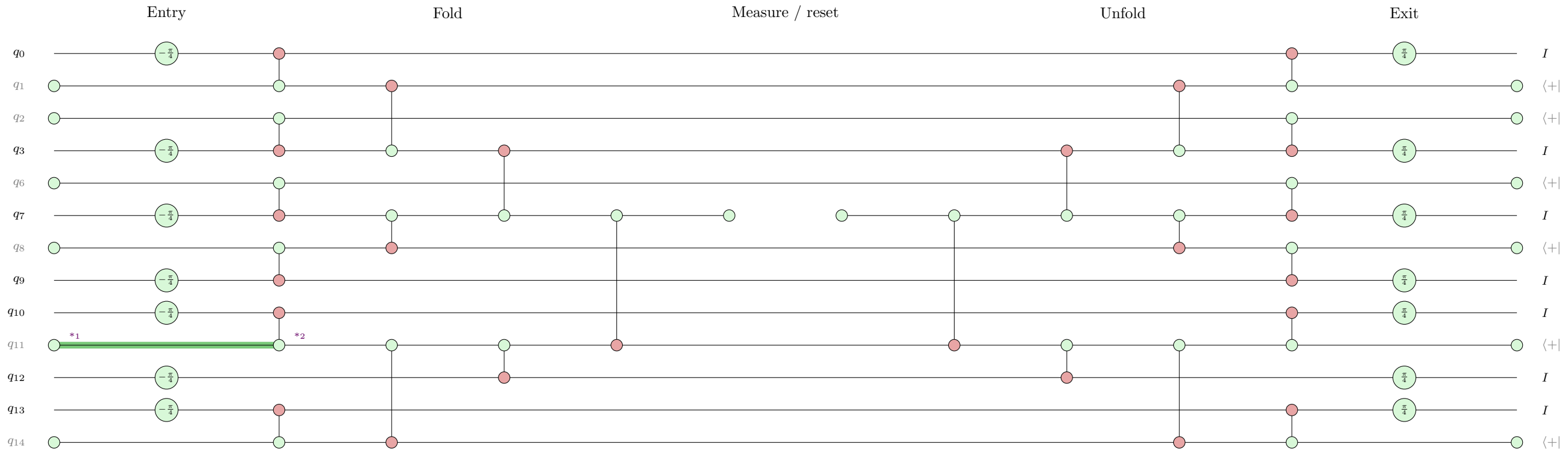
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 38
 No boundary support. 2 phase defects.



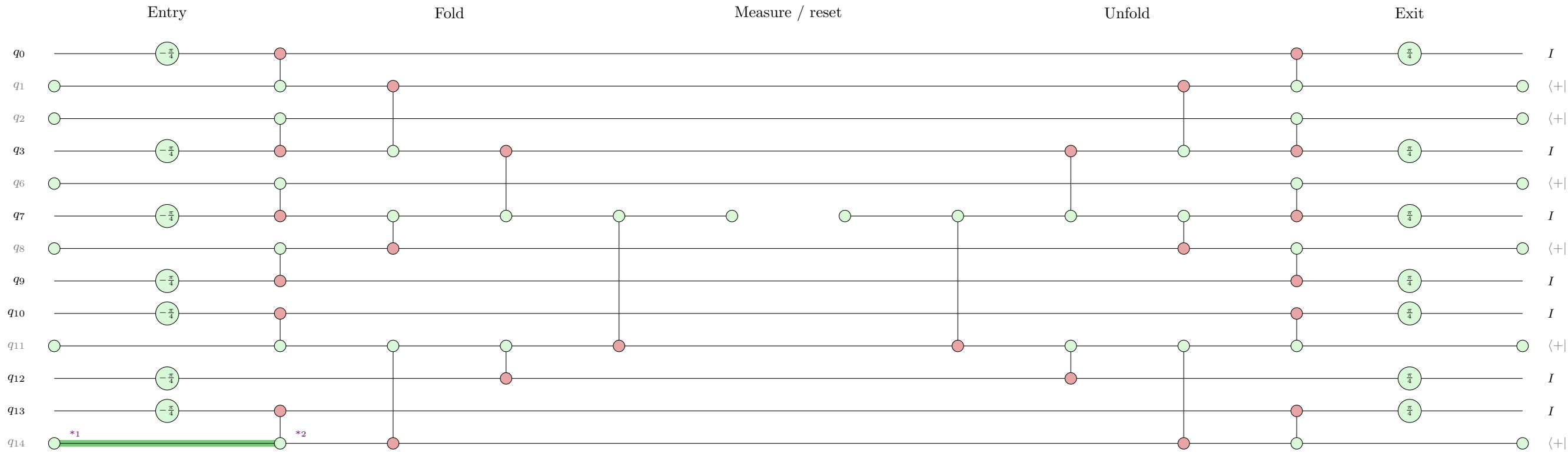
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 39
 No boundary support. 2 phase defects.



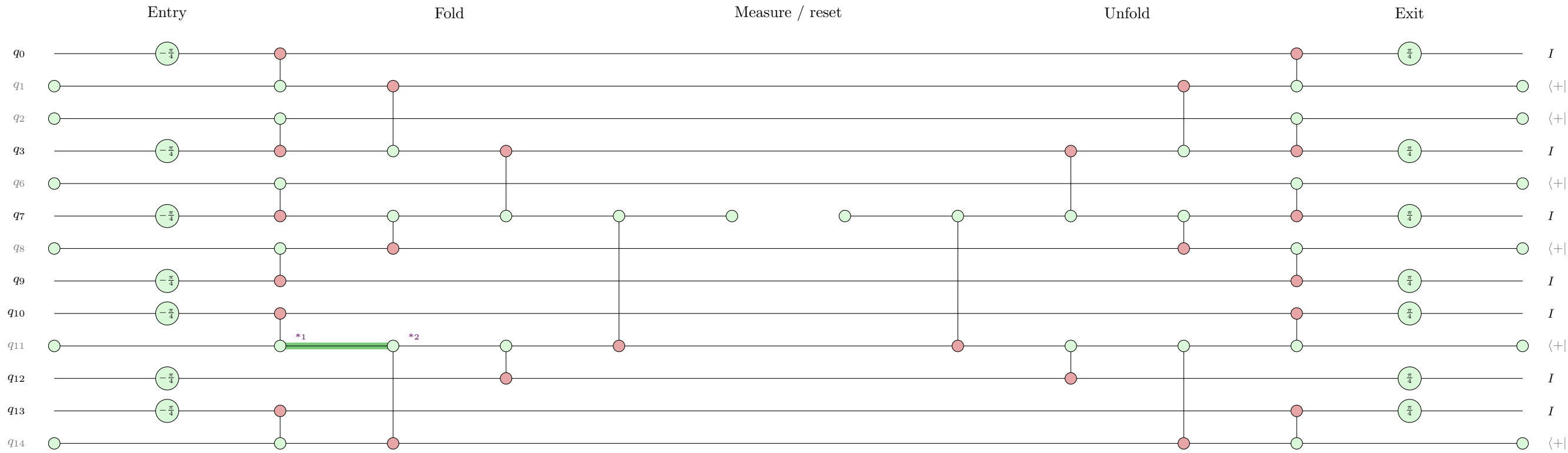
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 40
 No boundary support. 2 phase defects.



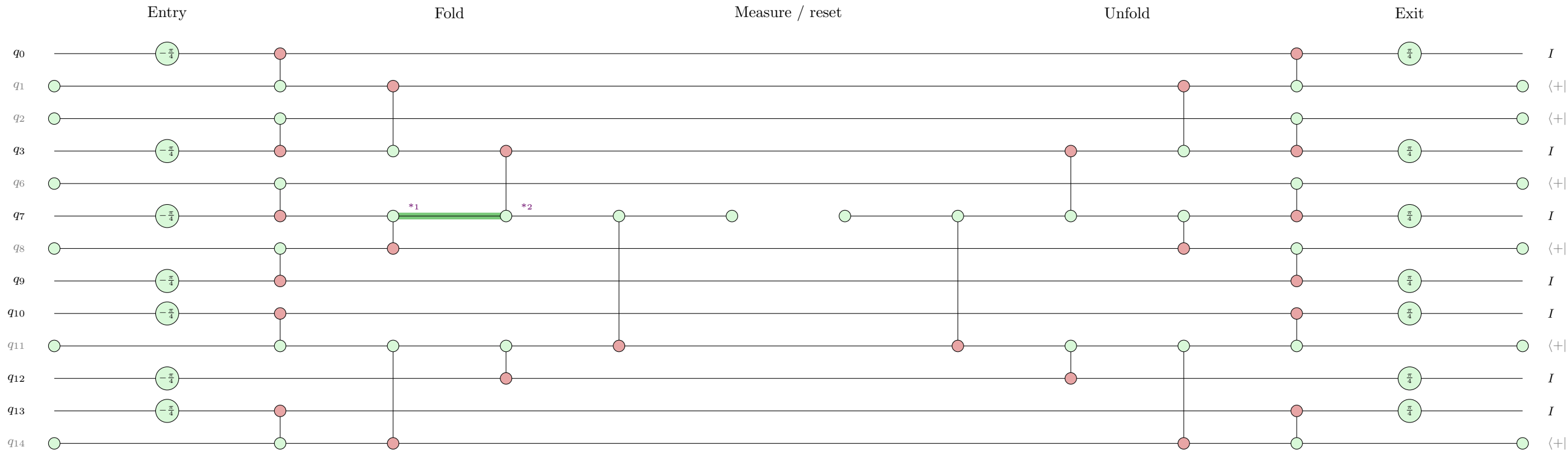
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 41
 No boundary support. 2 phase defects.



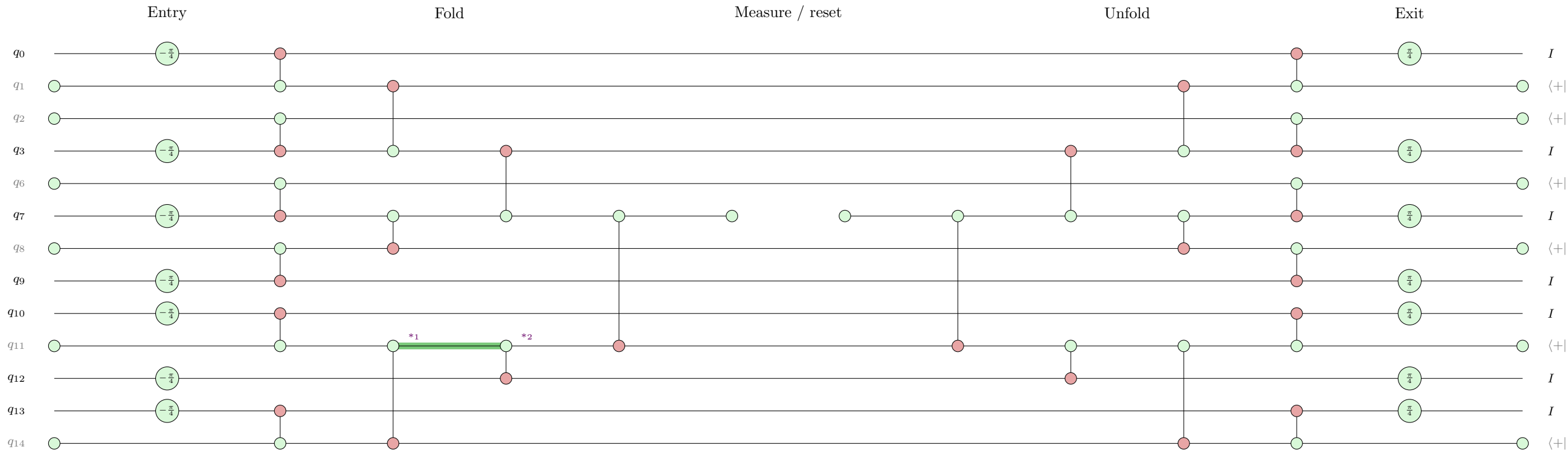
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 42
 No boundary support. 2 phase defects.



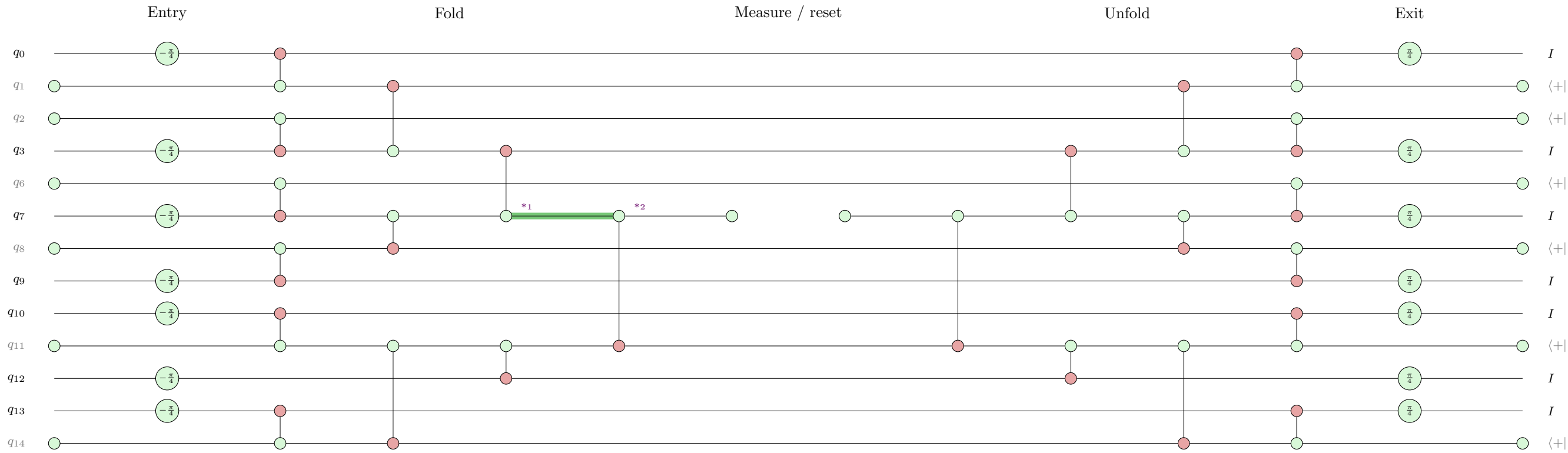
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 43
 No boundary support. 2 phase defects.



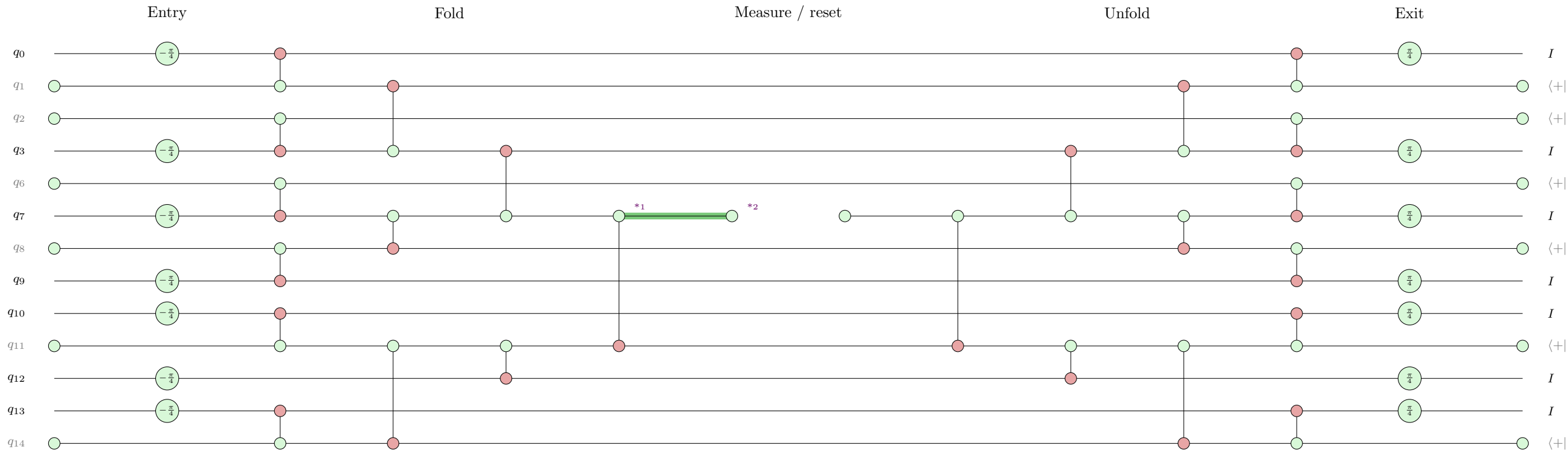
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 44
 No boundary support. 2 phase defects.



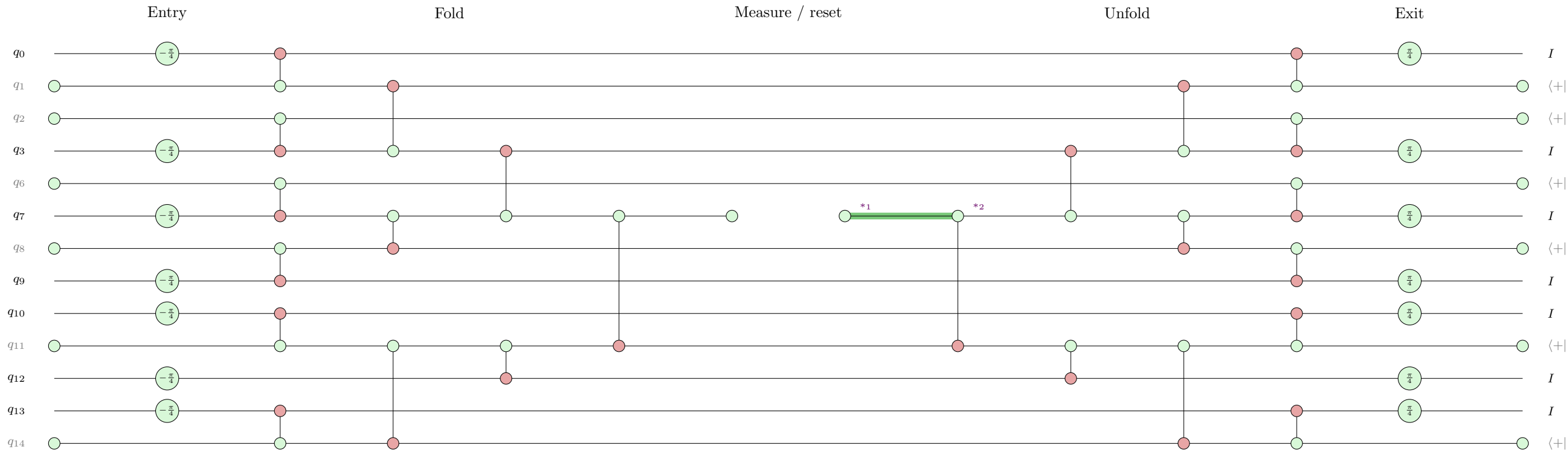
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 45
 No boundary support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

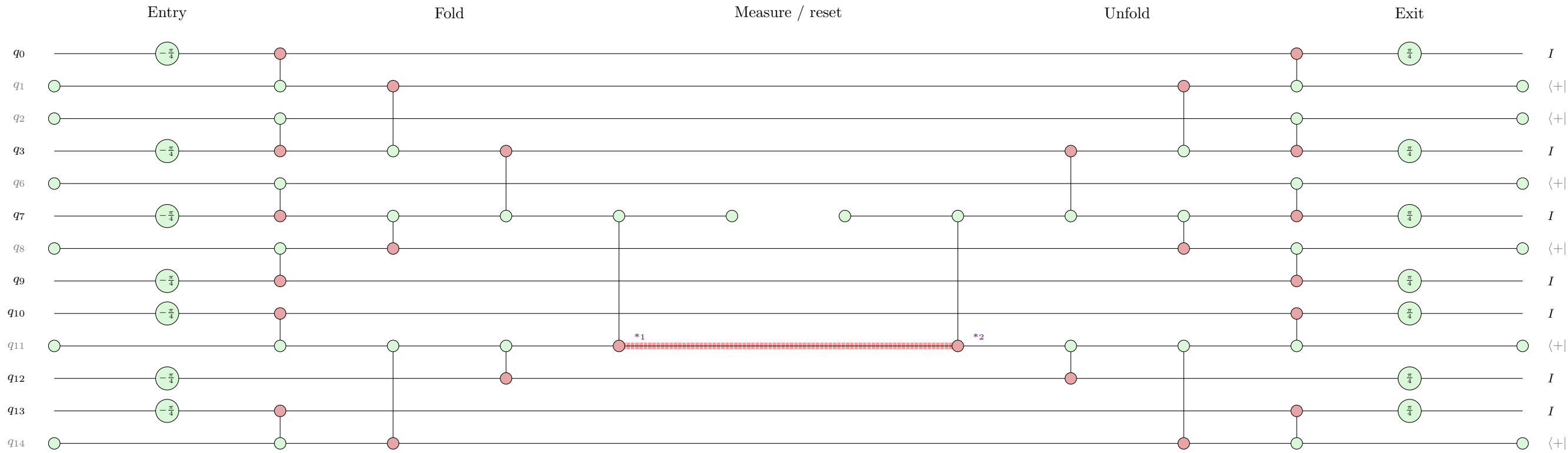
$d = 3$: basis element 46
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

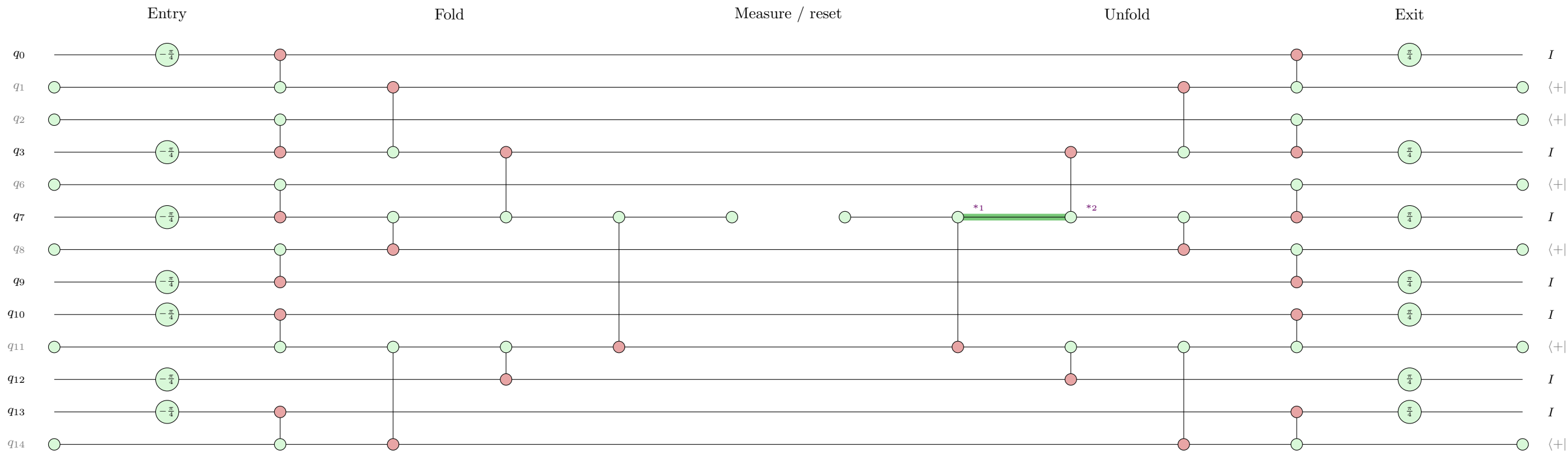
Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: basis element 47
 No boundary support. 2 phase defects.



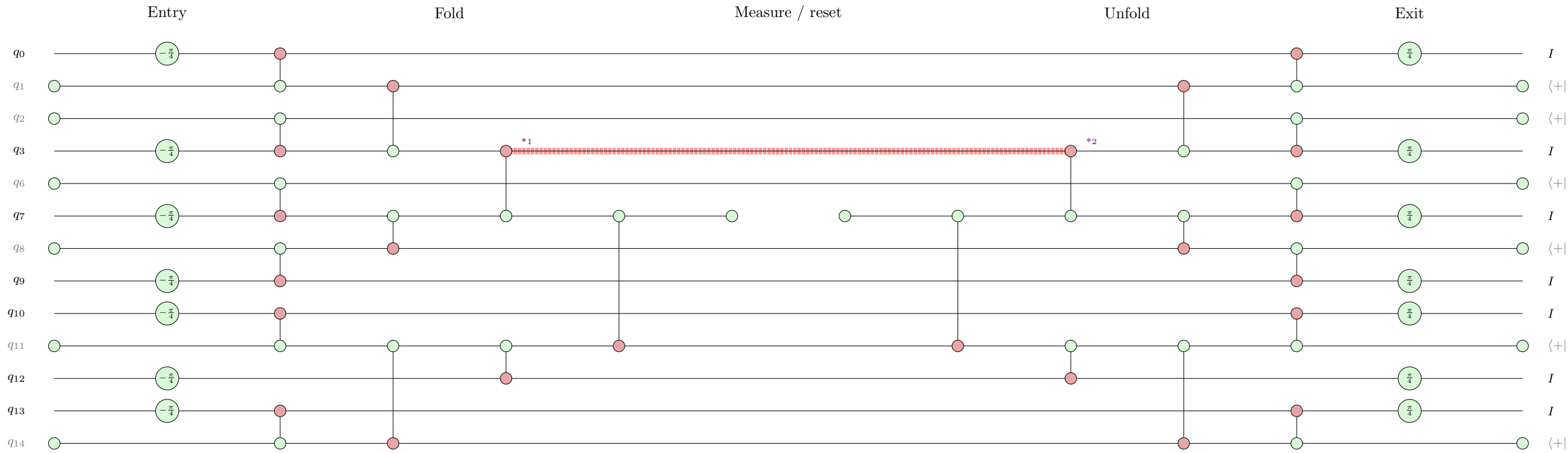
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 48
 No boundary support. 2 phase defects.



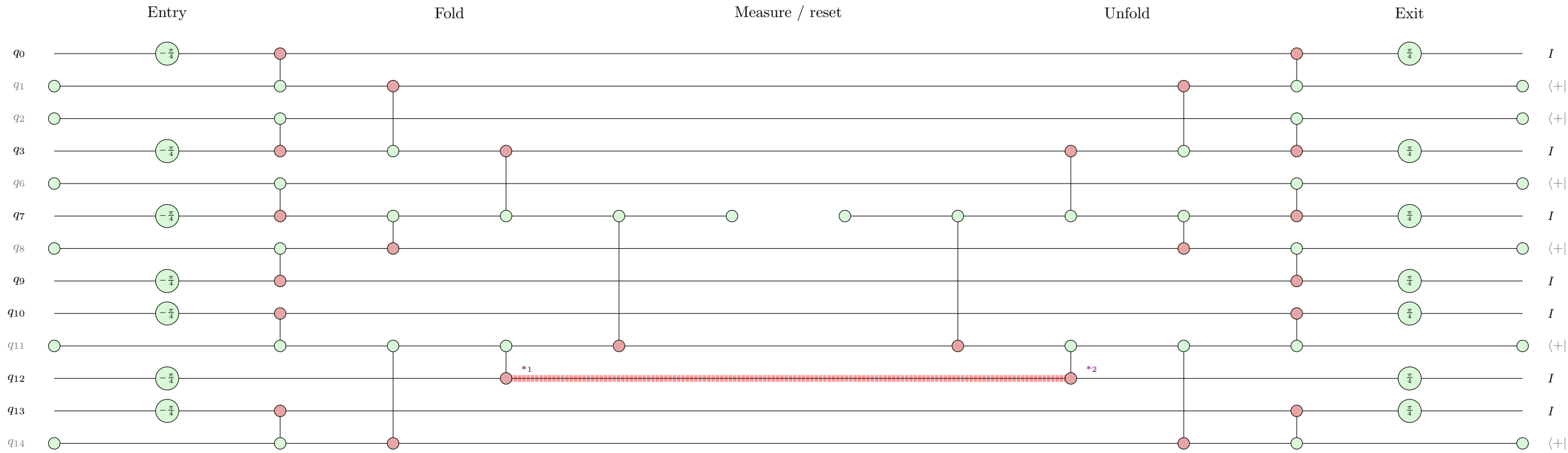
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 49
 No boundary support. 2 phase defects.



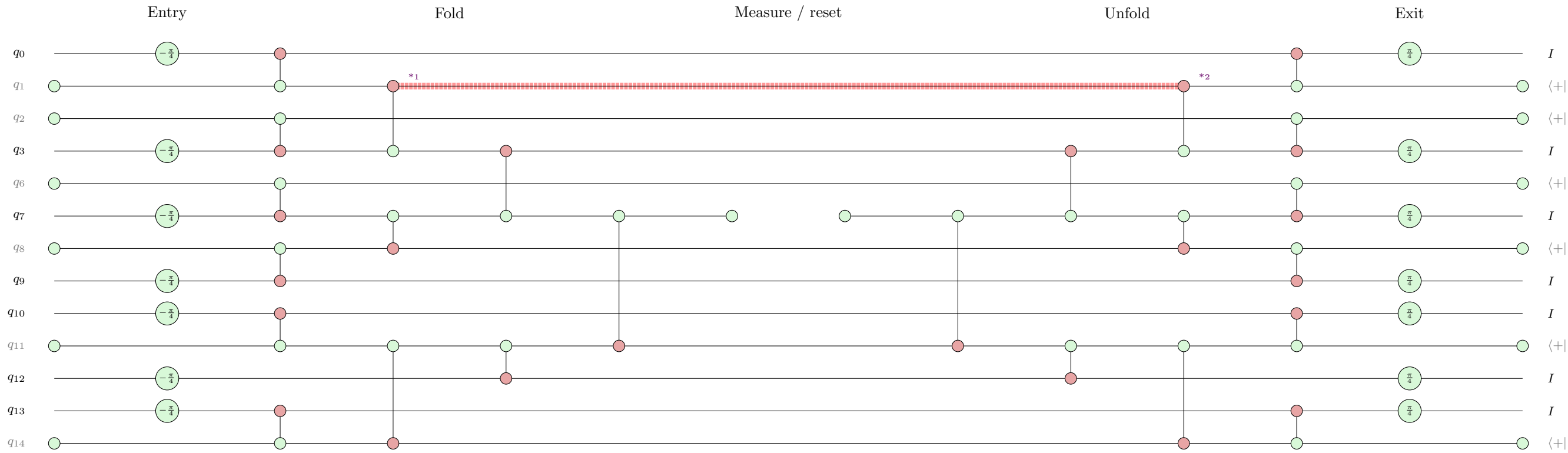
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 50
 No boundary support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

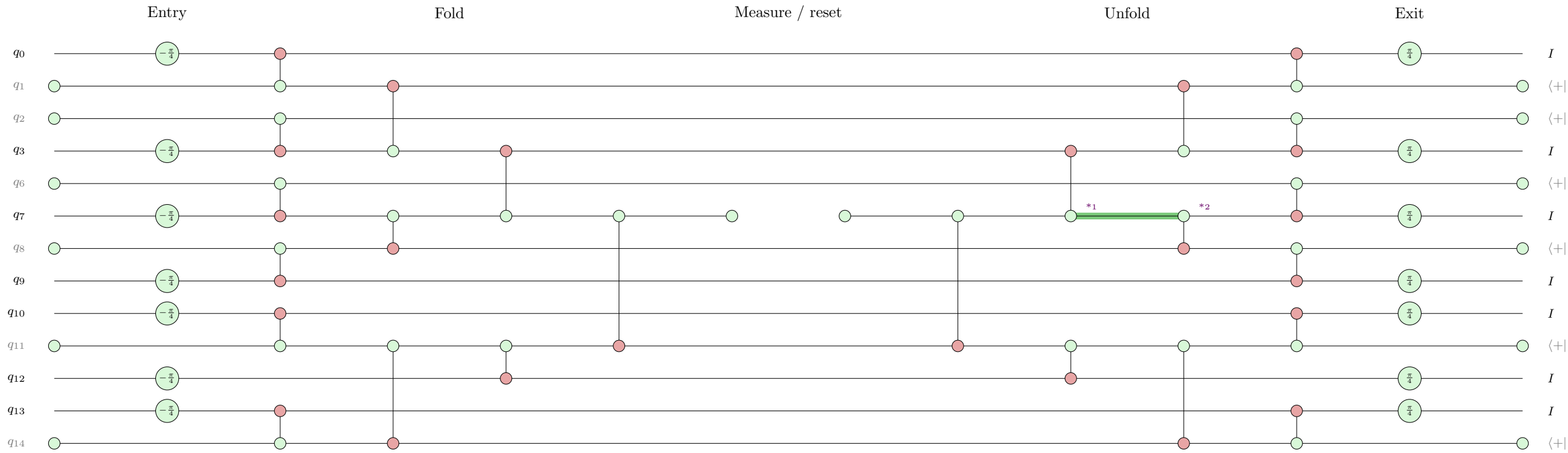
$d = 3$: basis element 51
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

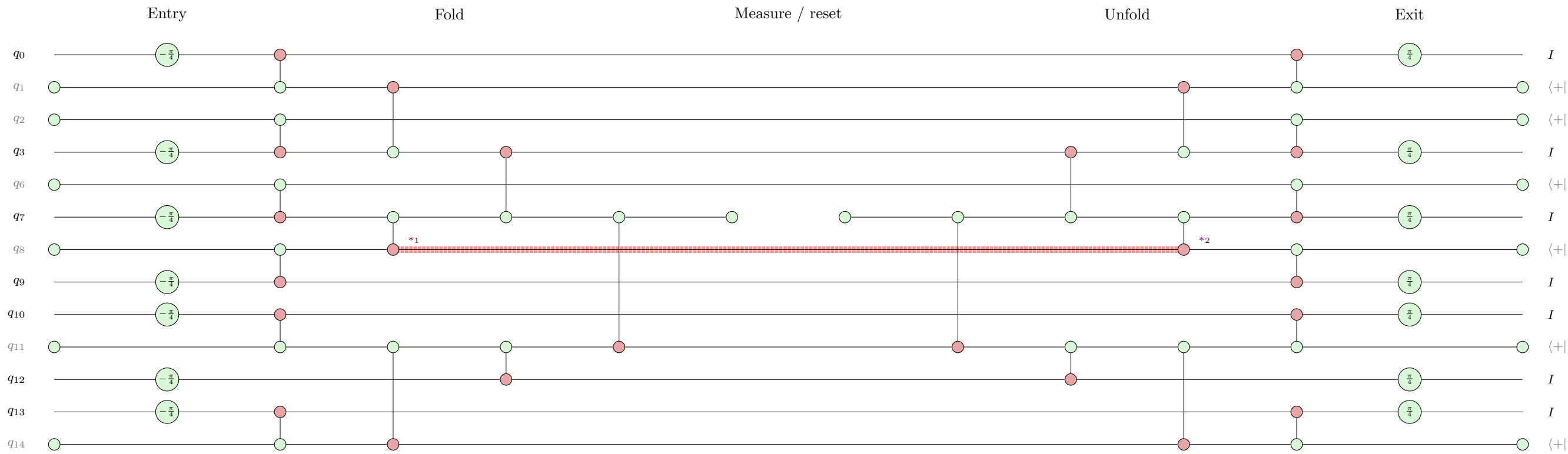
Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: basis element 52
 No boundary support. 2 phase defects.



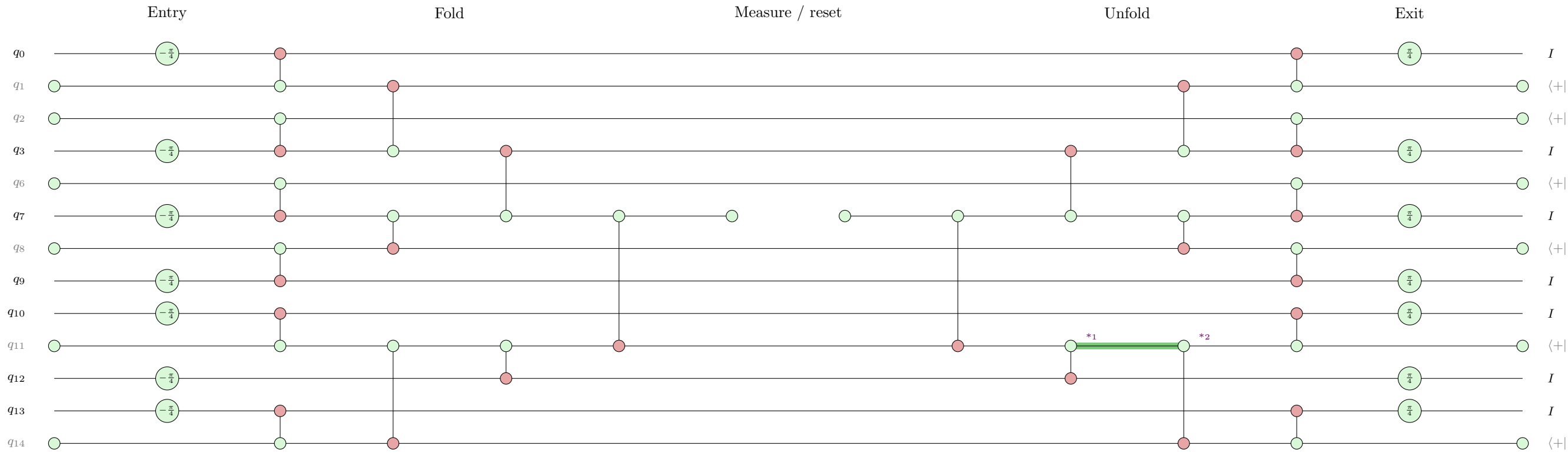
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 53
 No boundary support. 2 phase defects.



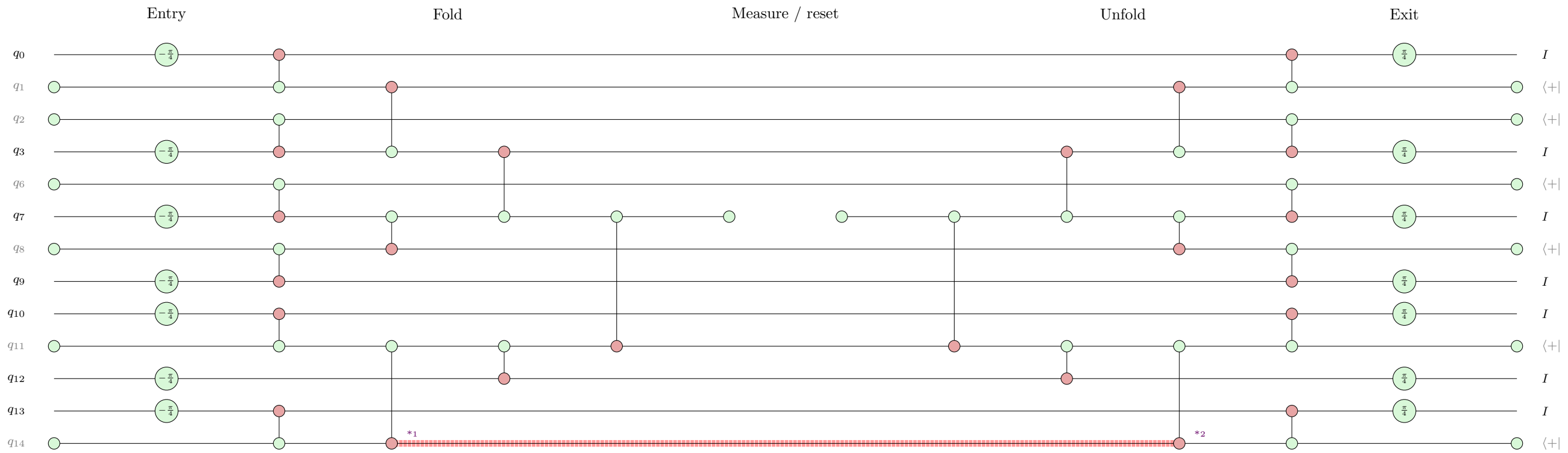
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 54
 No boundary support. 2 phase defects.



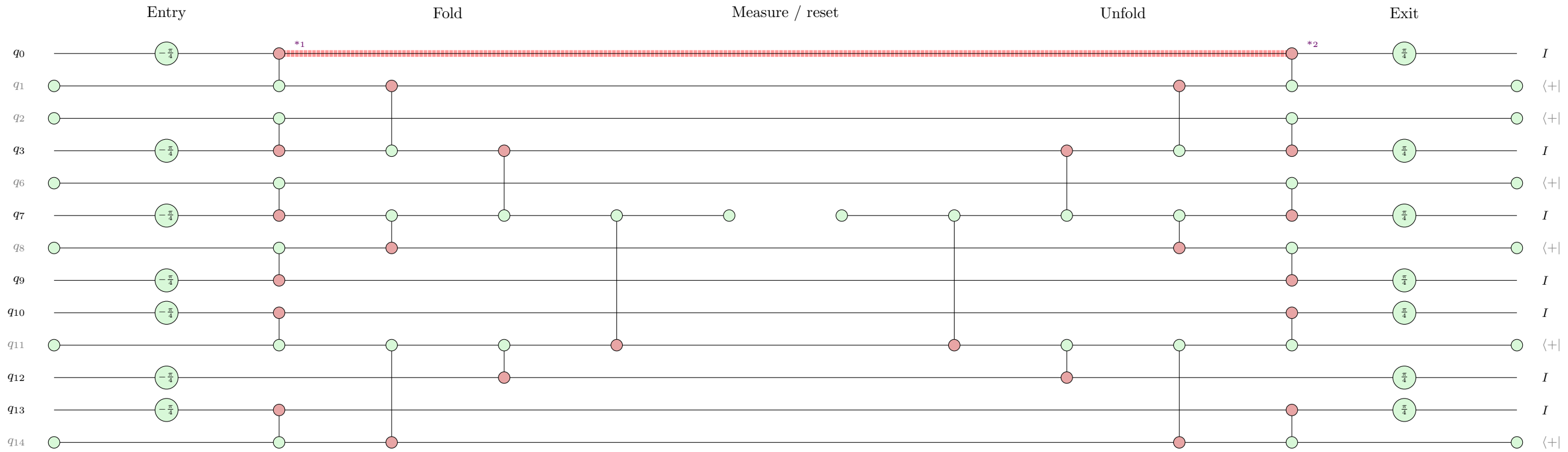
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 55
 No boundary support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

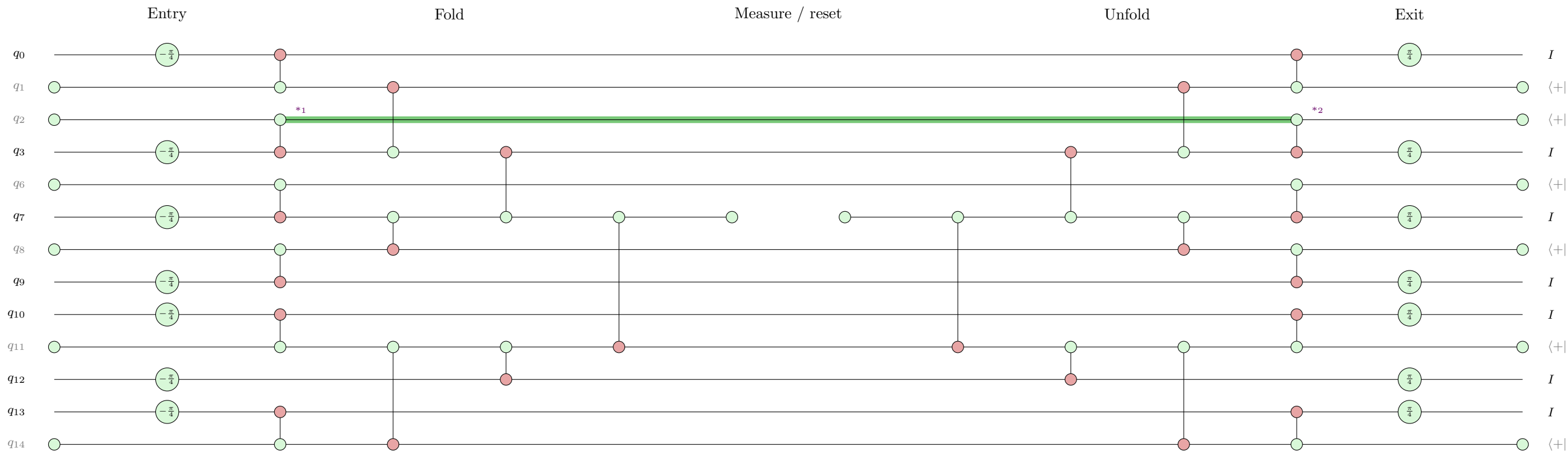
$d = 3$: basis element 56
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

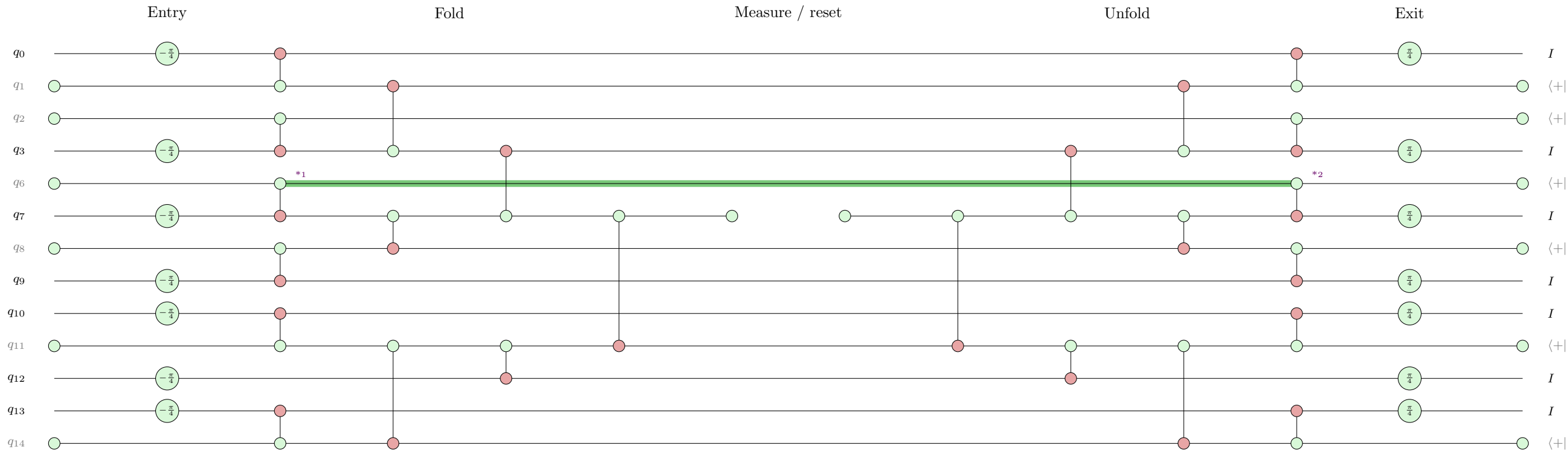
Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: basis element 57
 No boundary support. 2 phase defects.



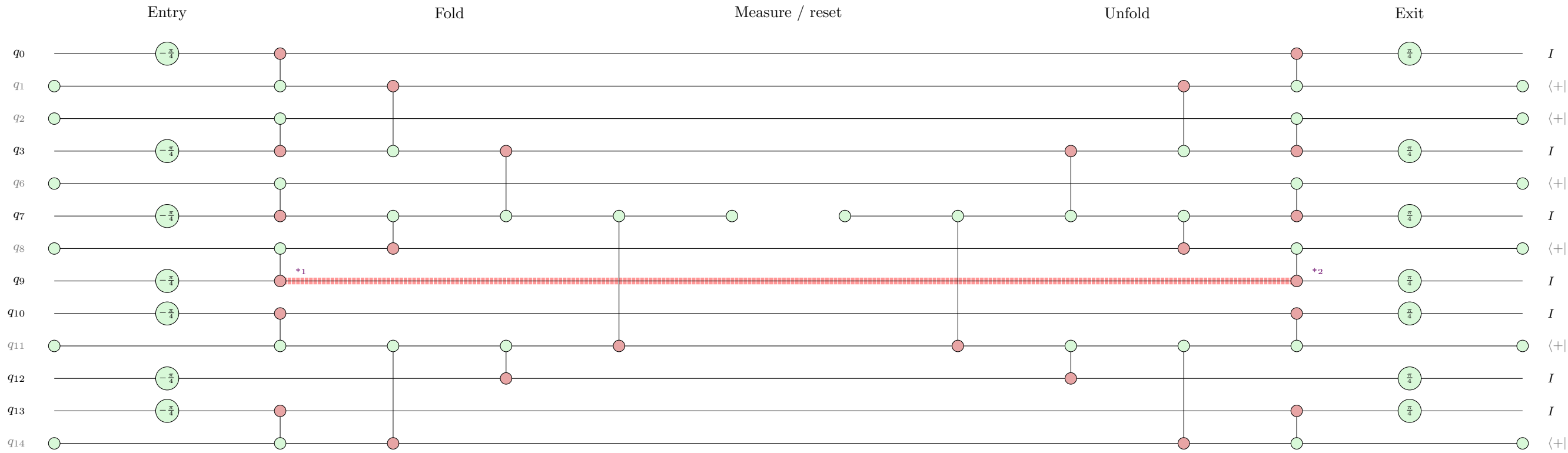
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 58
 No boundary support. 2 phase defects.



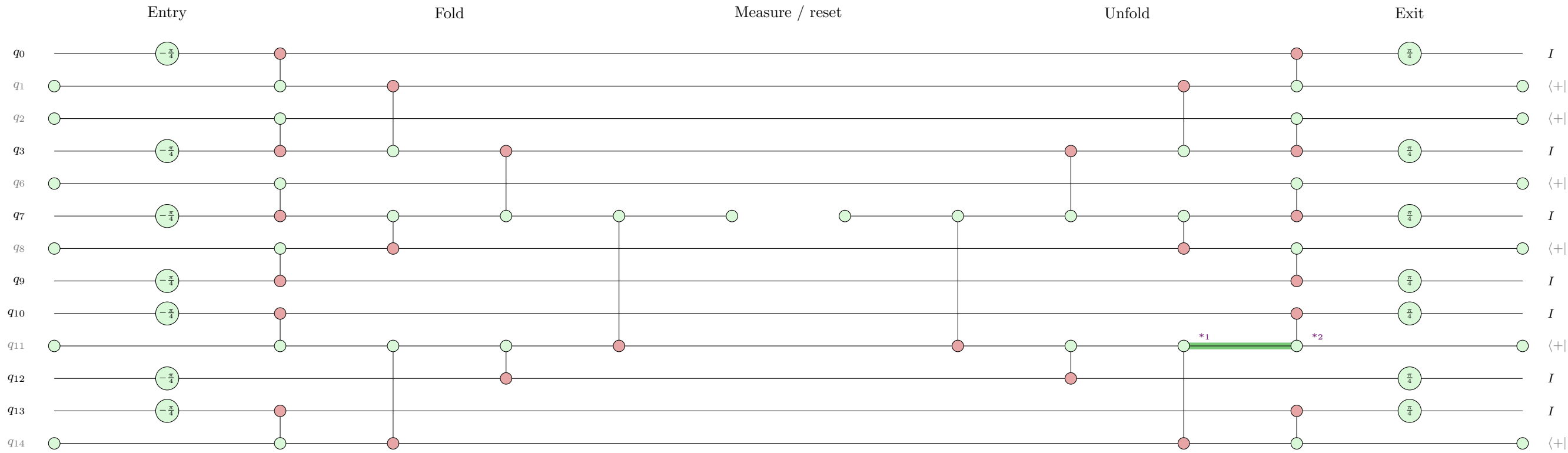
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 59
 No boundary support. 2 phase defects.



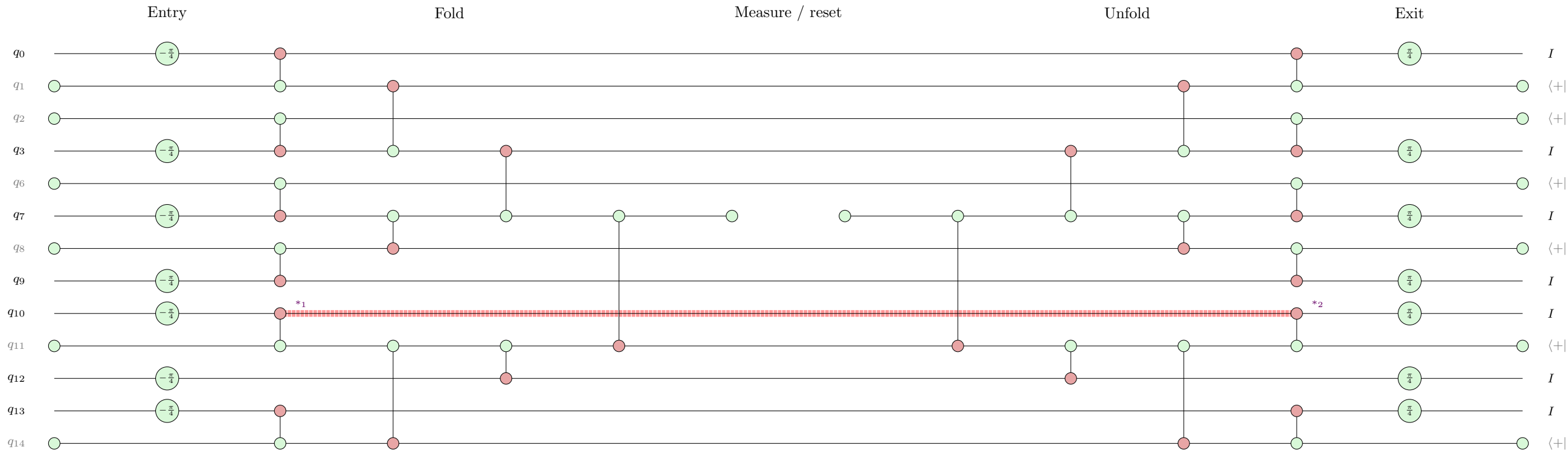
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 60
 No boundary support. 2 phase defects.



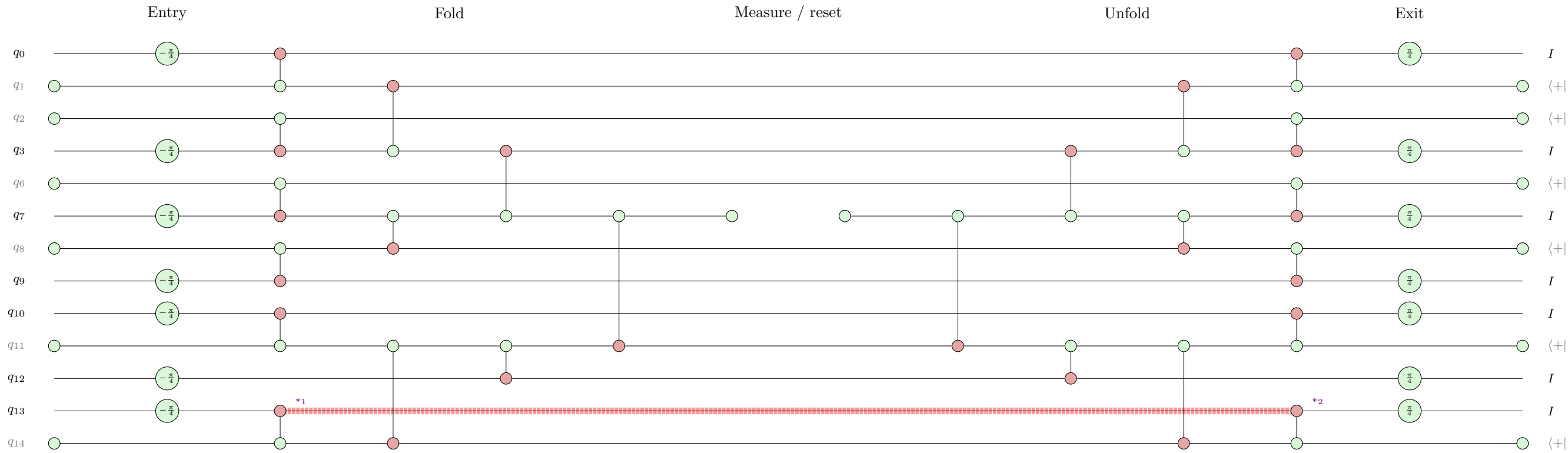
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 61
 No boundary support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

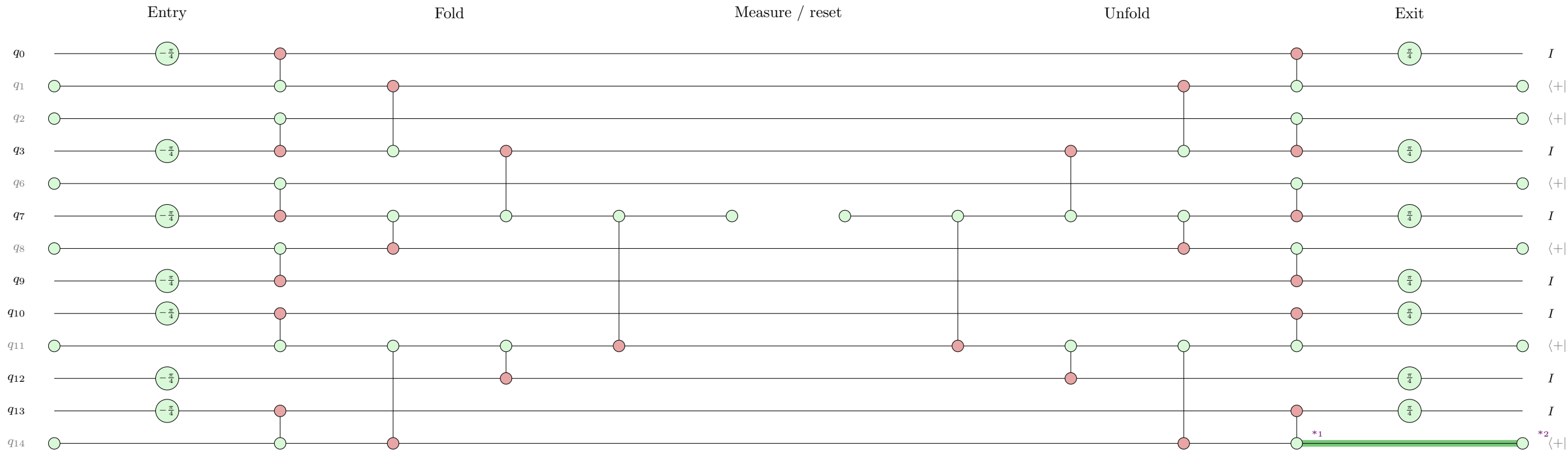
$d = 3$: basis element 62
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

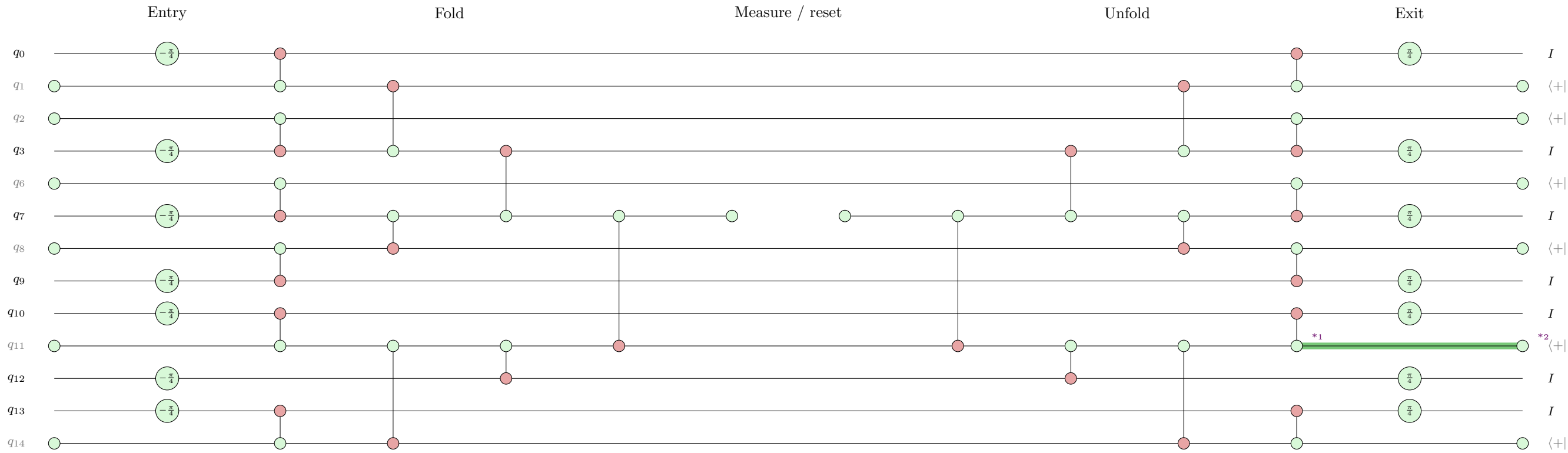
$d = 3$: basis element 63
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

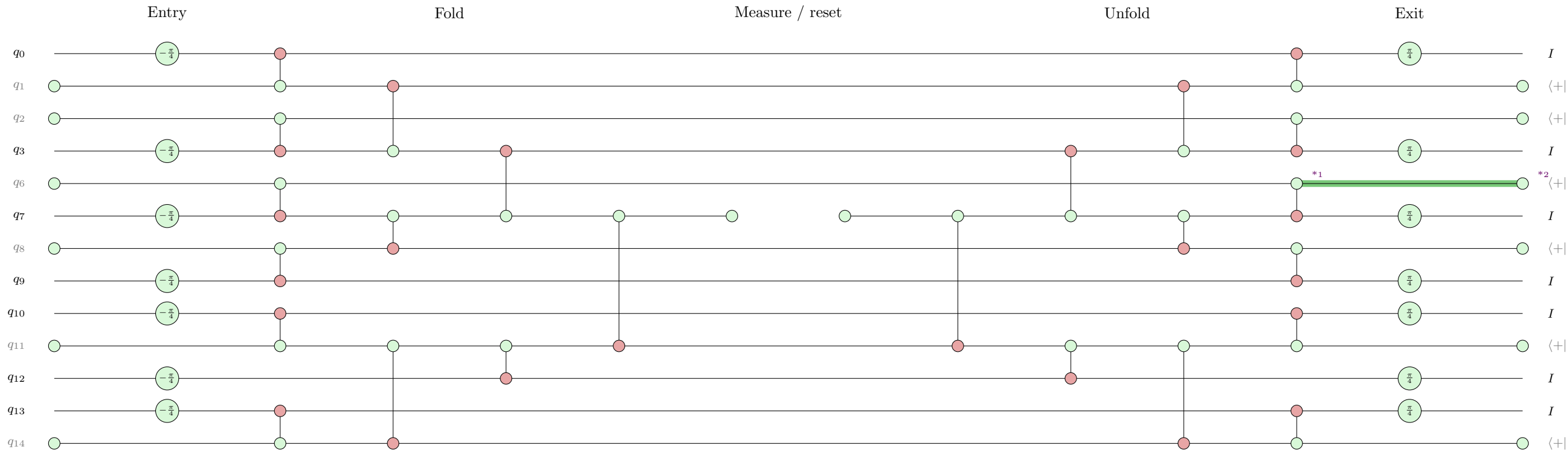
Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: basis element 64
 No boundary support. 2 phase defects.



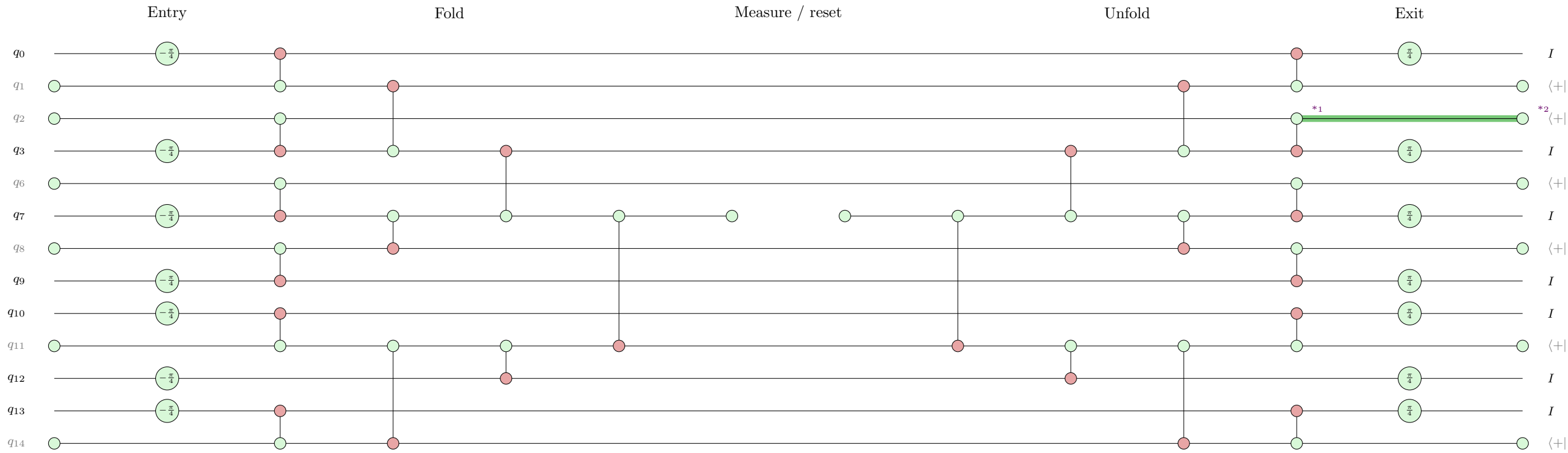
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 65
 No boundary support. 2 phase defects.



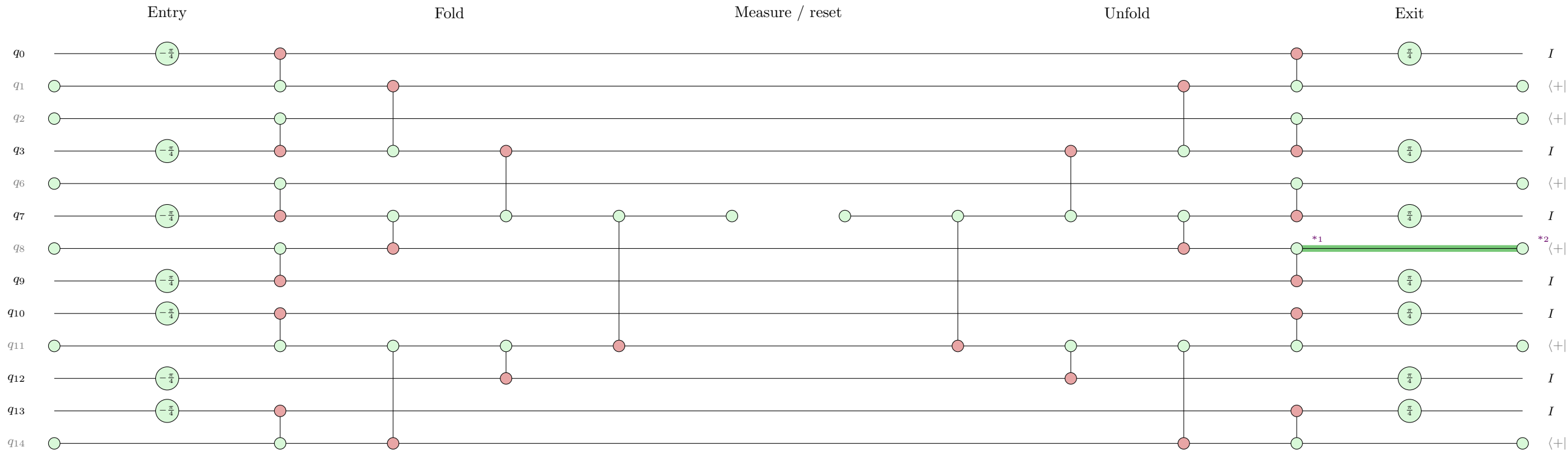
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 66
 No boundary support. 2 phase defects.



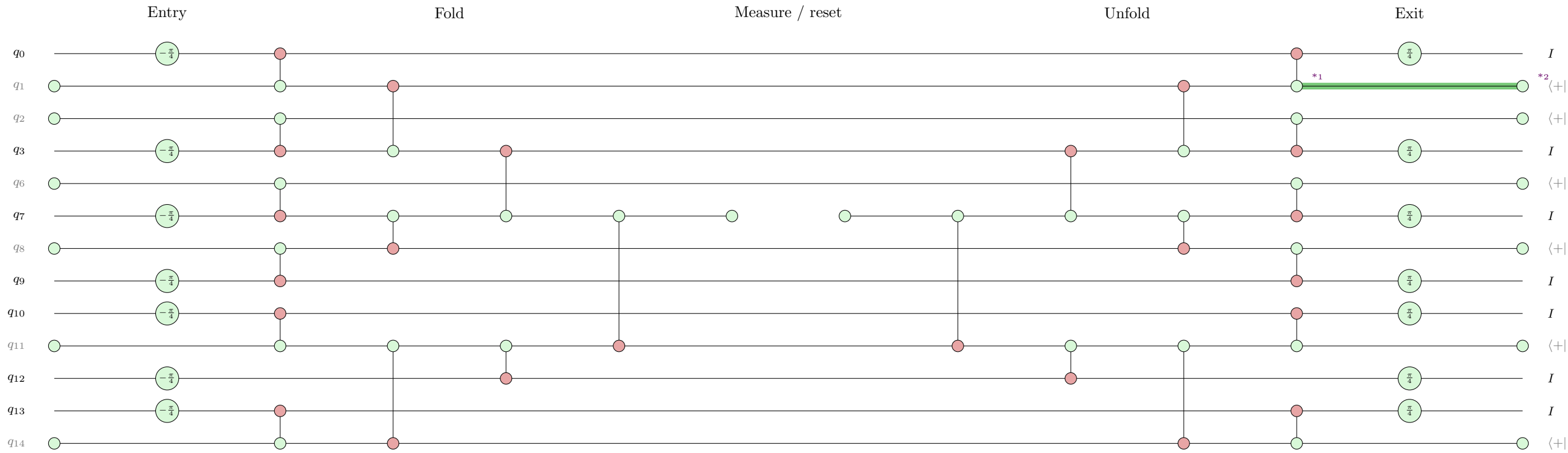
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 67
 No boundary support. 2 phase defects.



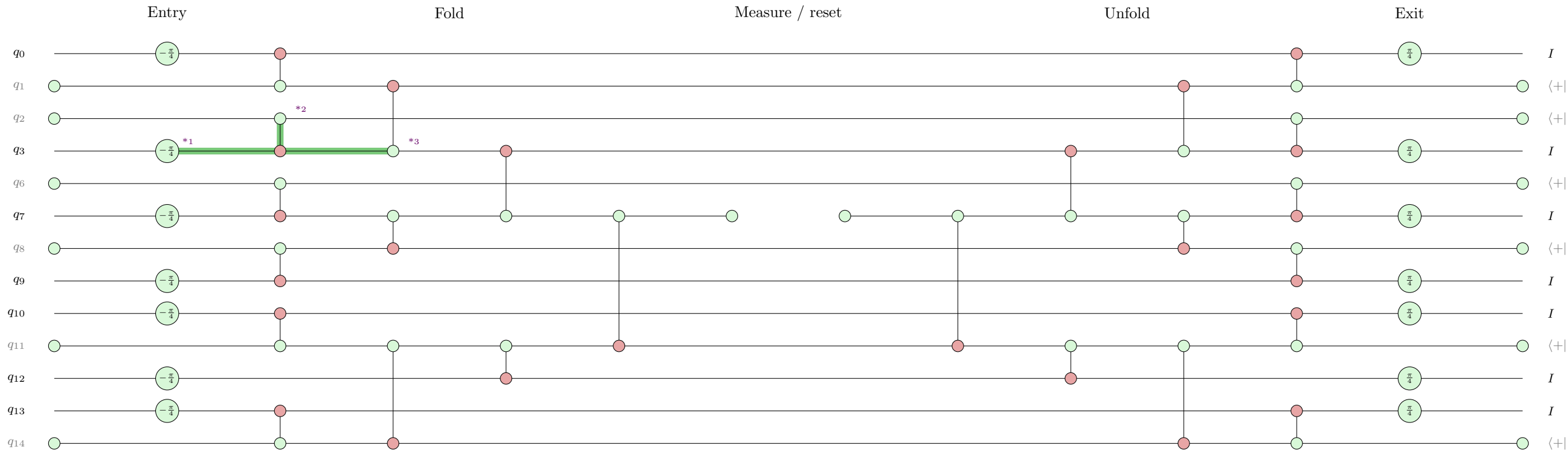
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 68
 No boundary support. 2 phase defects.



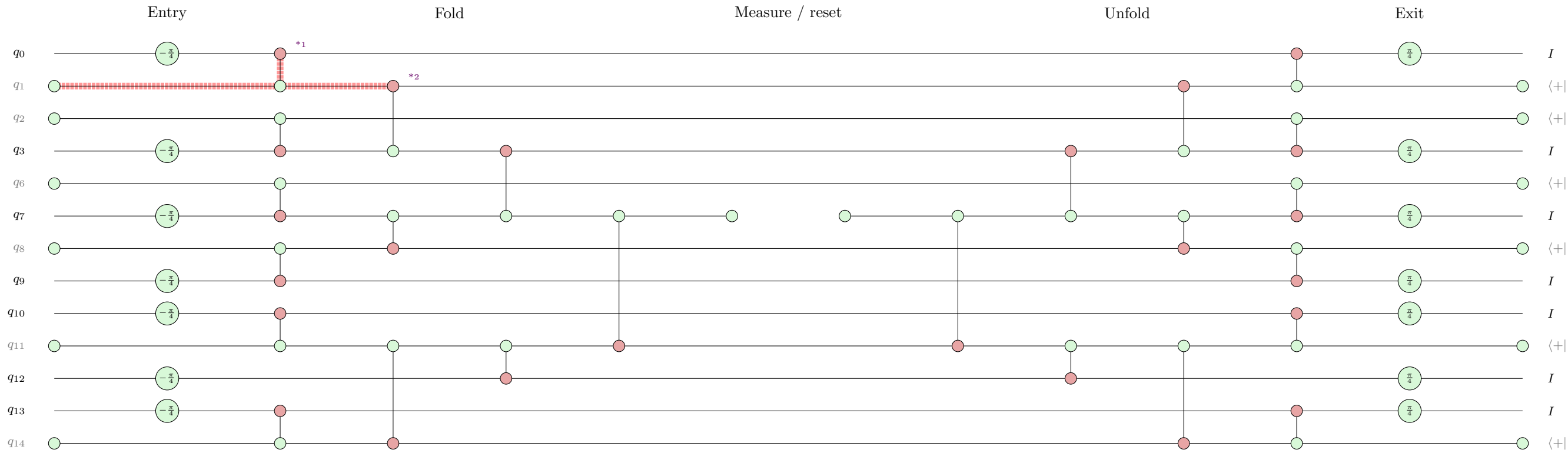
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 69
 No boundary support. 3 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi.$

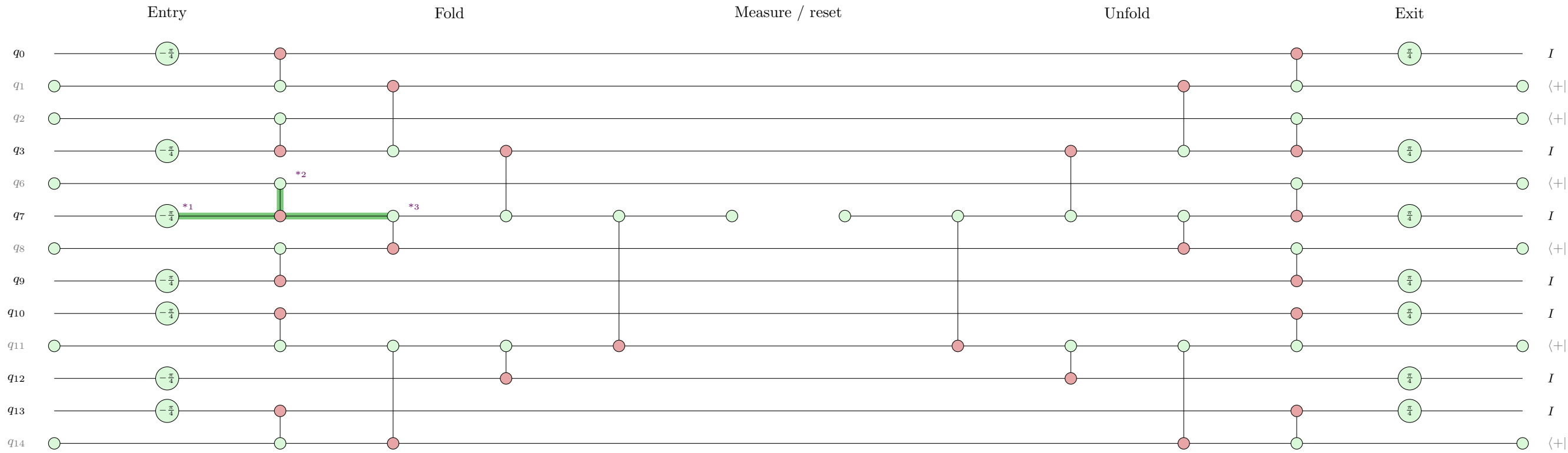
$d = 3$: basis element 70
 No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

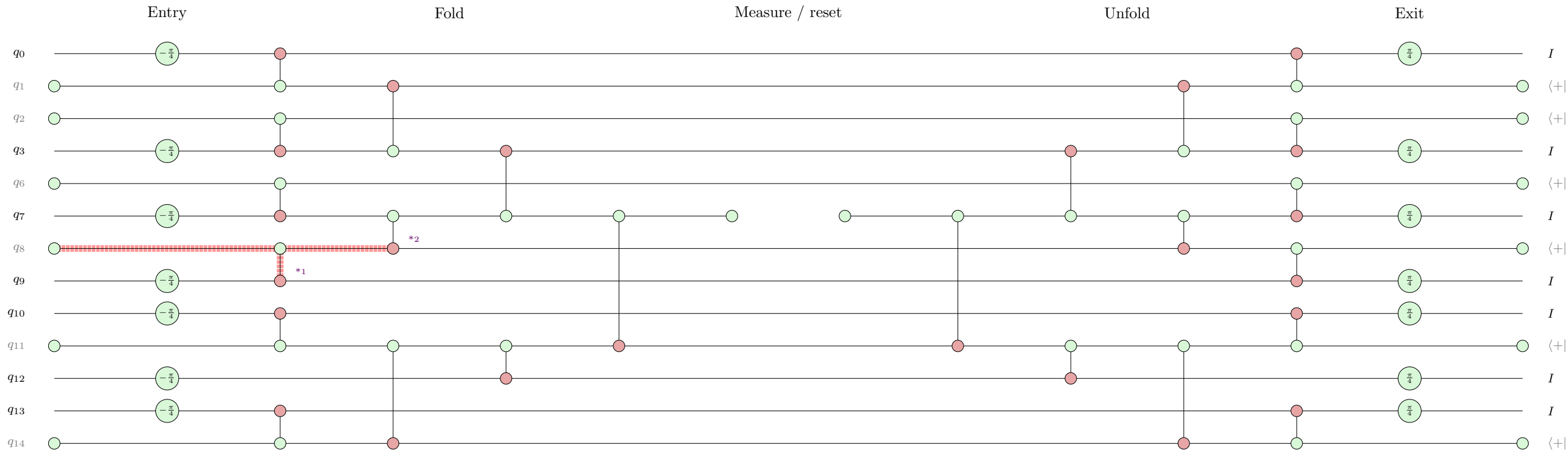
Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 3$: basis element 71
 No boundary support. 3 phase defects.



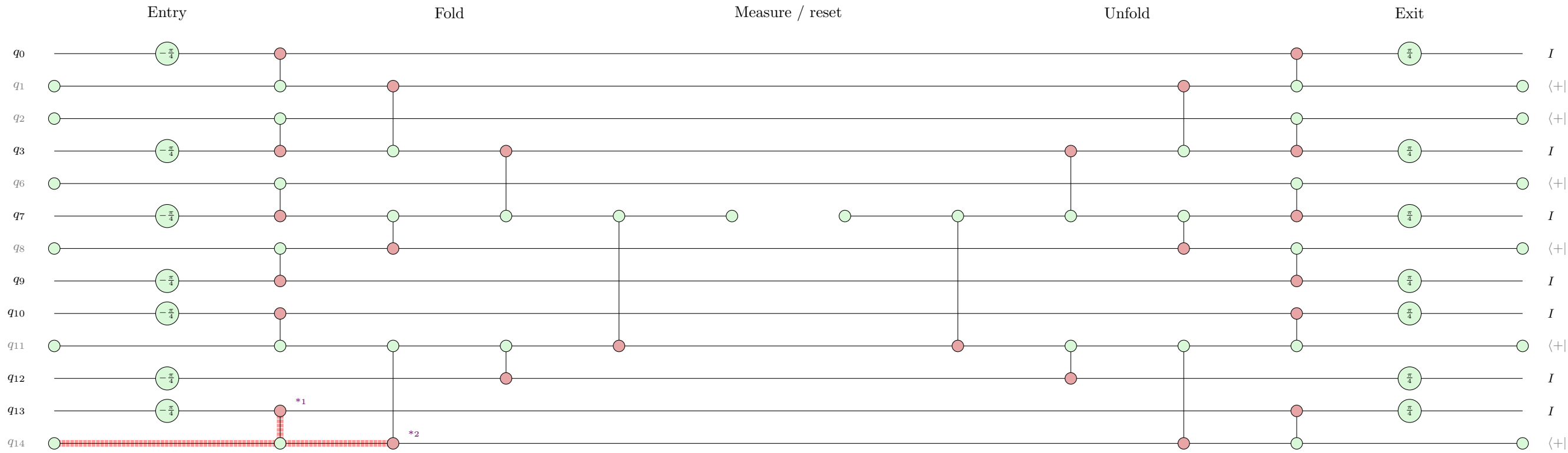
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi.$

$d = 3$: basis element 72
 No boundary support. 2 phase defects.



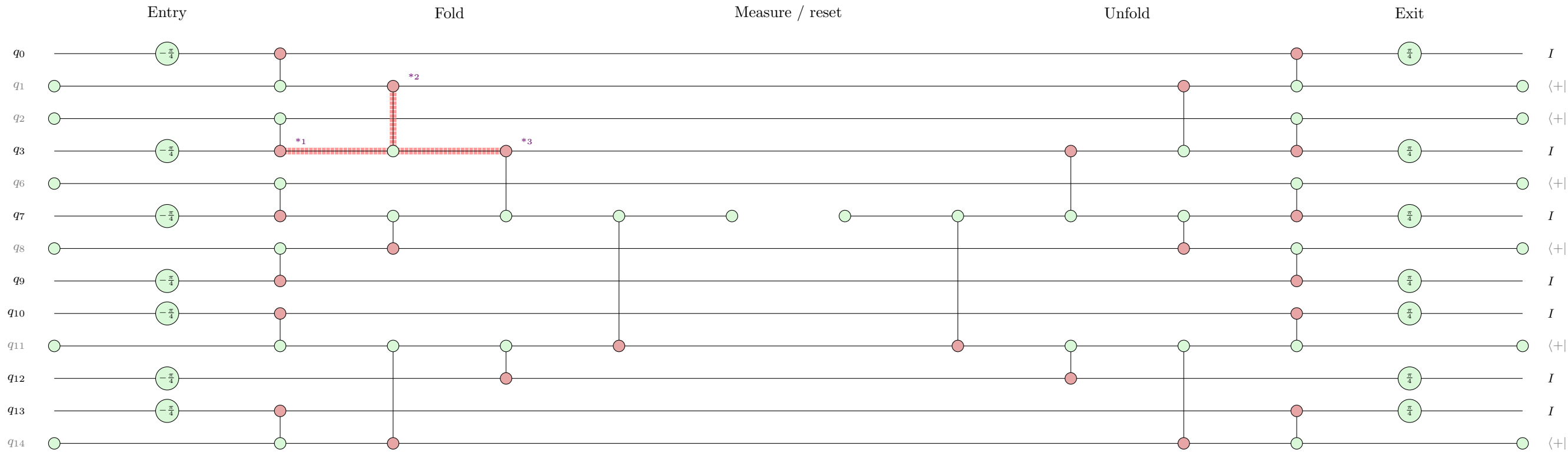
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 73
 No boundary support. 2 phase defects.



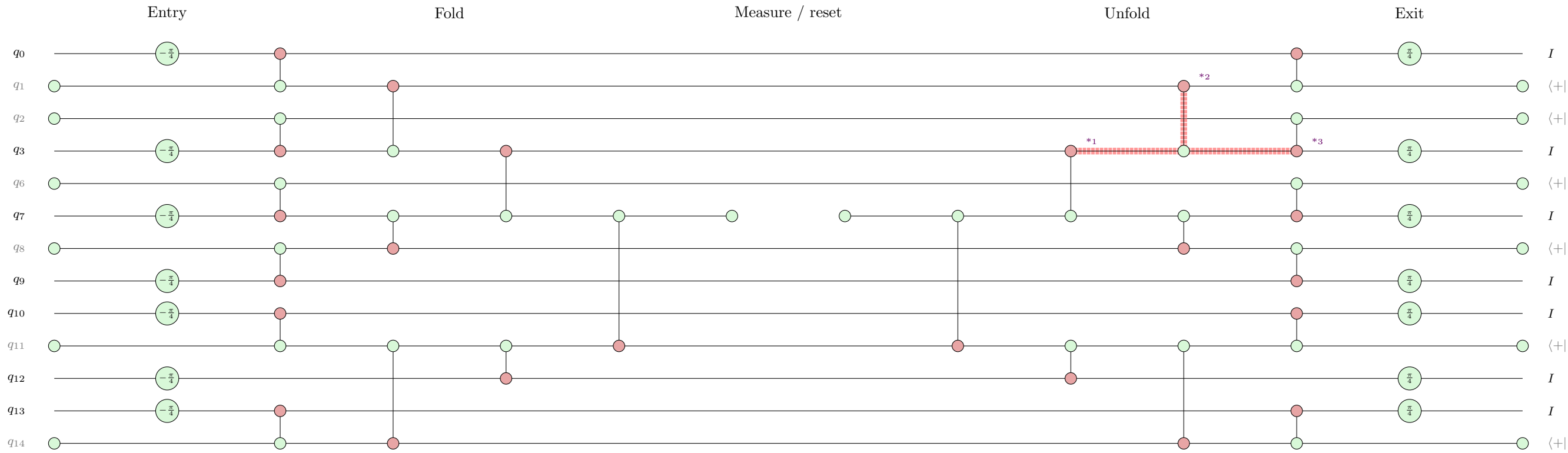
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 74
 No boundary support. 3 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi.$

$d = 3$: basis element 75
 No boundary support. 3 phase defects.



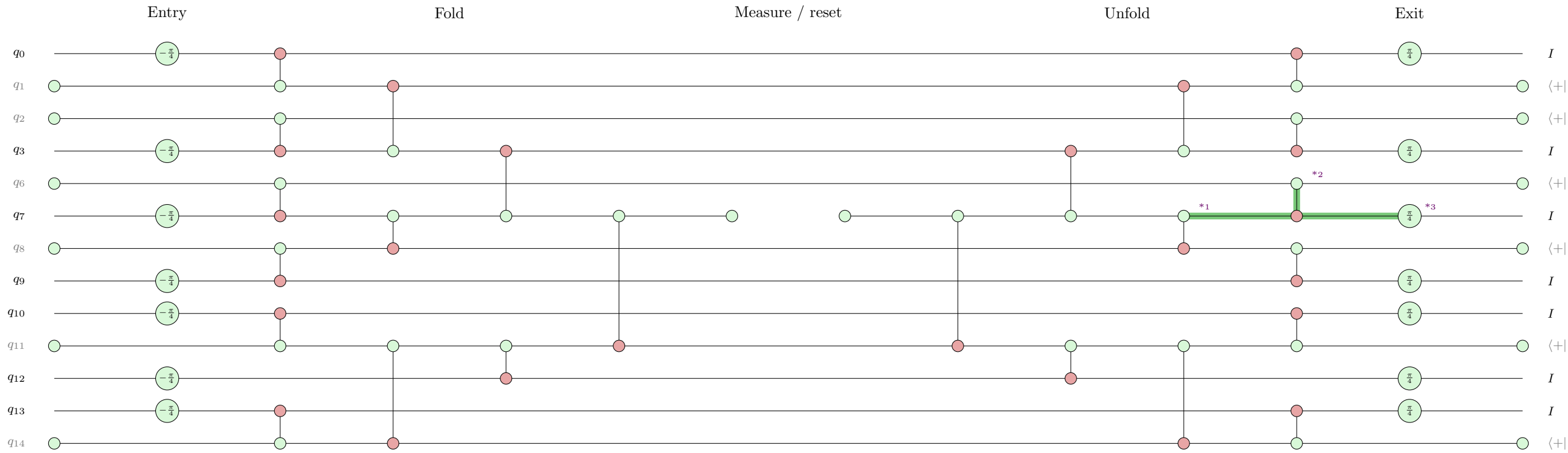
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi.$

$d = 3$: basis element 76
 No boundary support. 3 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi.$

$d = 3$: basis element 77
 No boundary support. 3 phase defects.



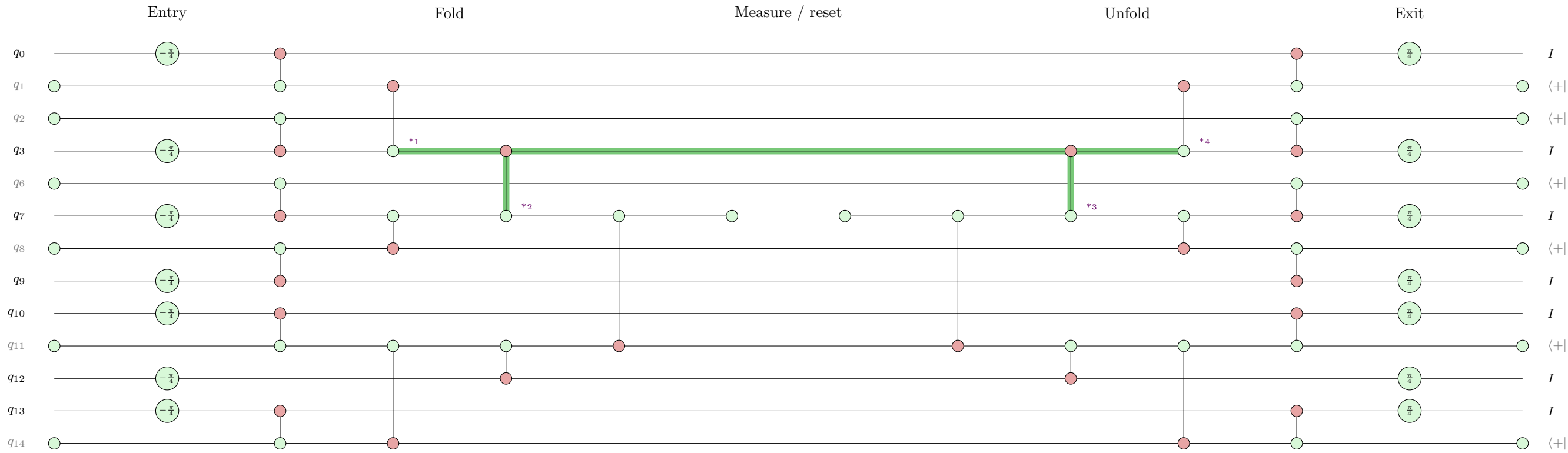
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi.$

$d = 3$: basis element 78
 No boundary support. 4 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

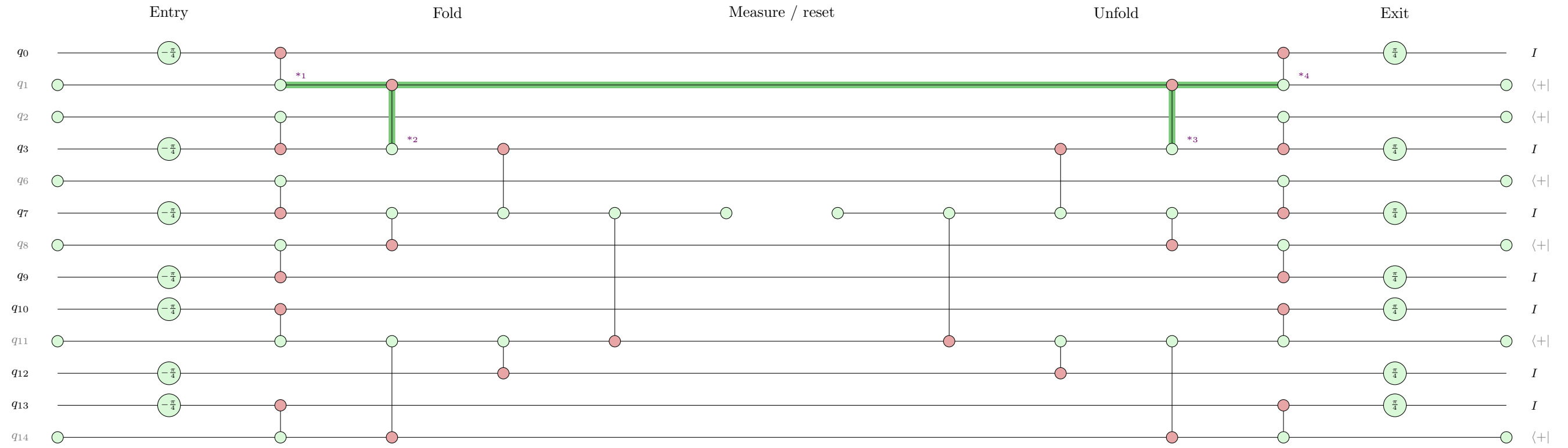
$d = 3$: basis element 79
 No boundary support. 4 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\boldsymbol{\delta}}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

$d = 3$: basis element 80

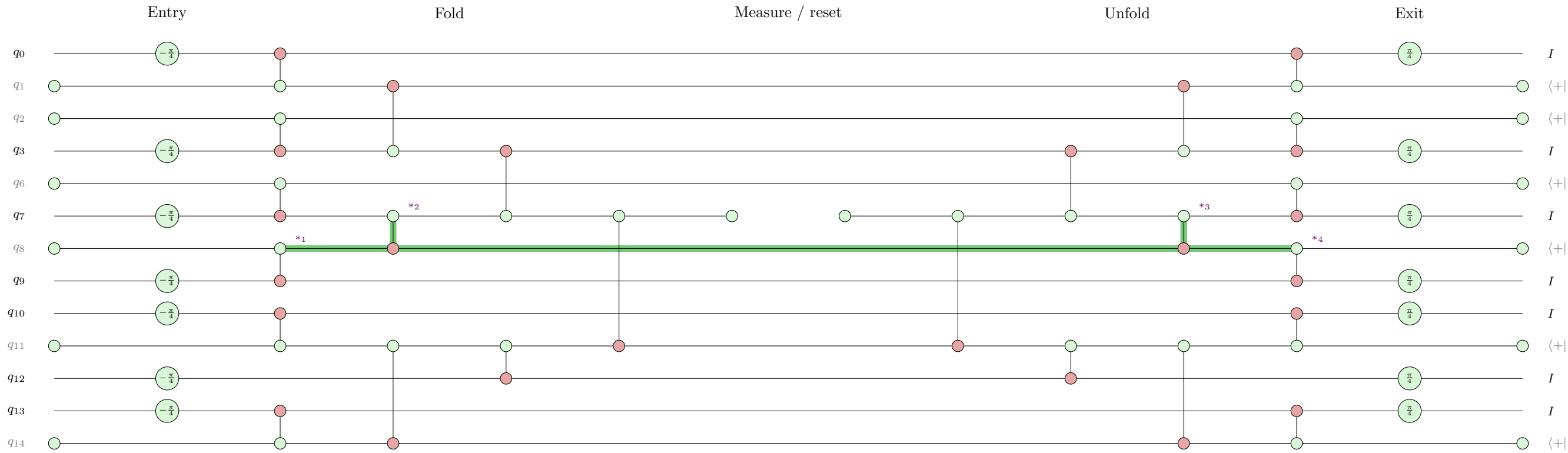
No boundary support. 4 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

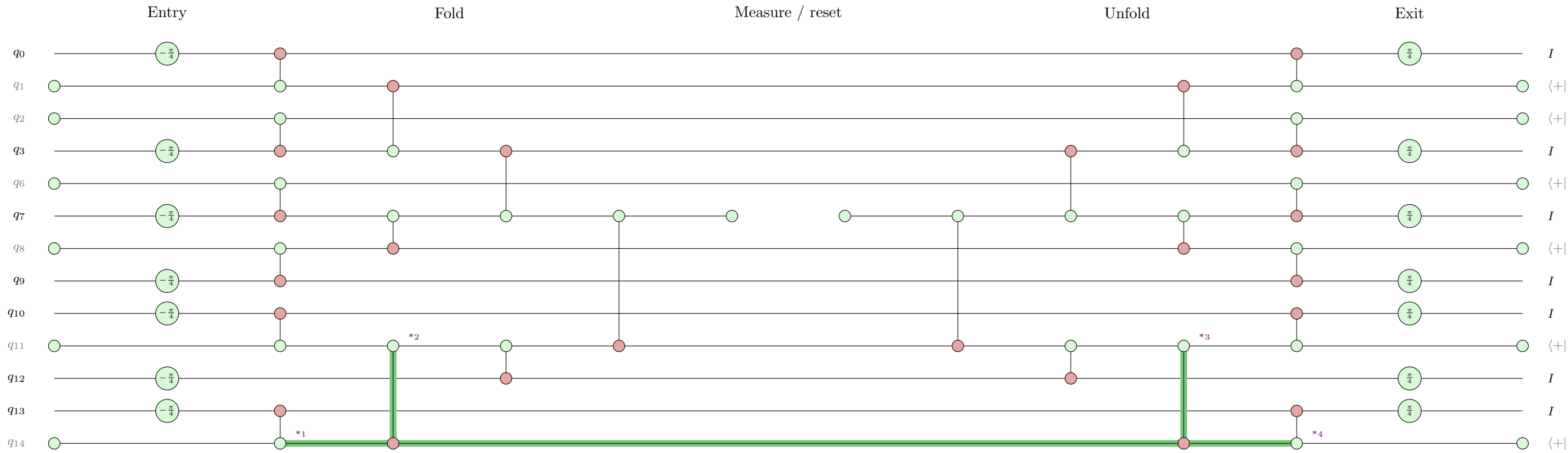
Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

$d = 3$: basis element 81
 No boundary support. 4 phase defects.



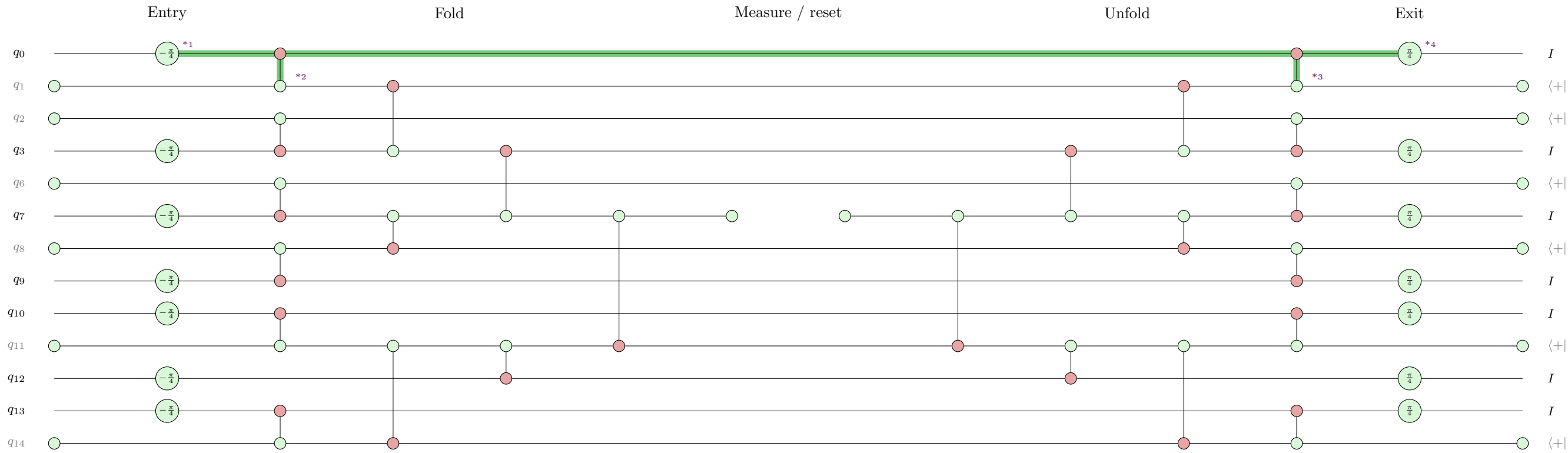
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

$d = 3$: basis element 82
 No boundary support. 4 phase defects.



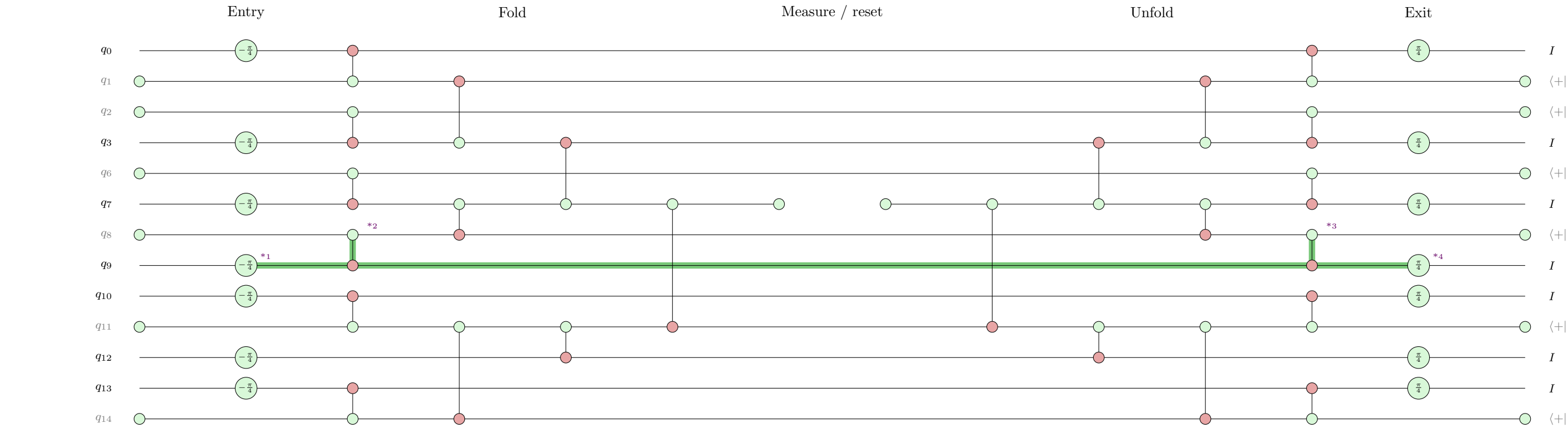
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

$d = 3$: basis element 83
 No boundary support. 4 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

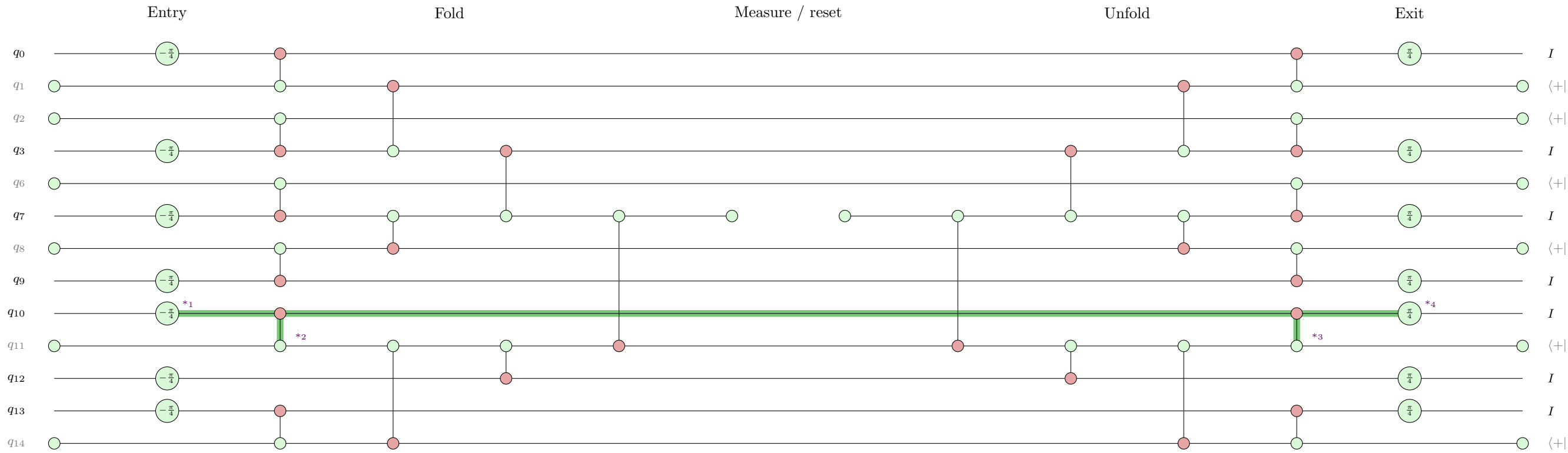
$d = 3$: basis element 84
 No boundary support. 4 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

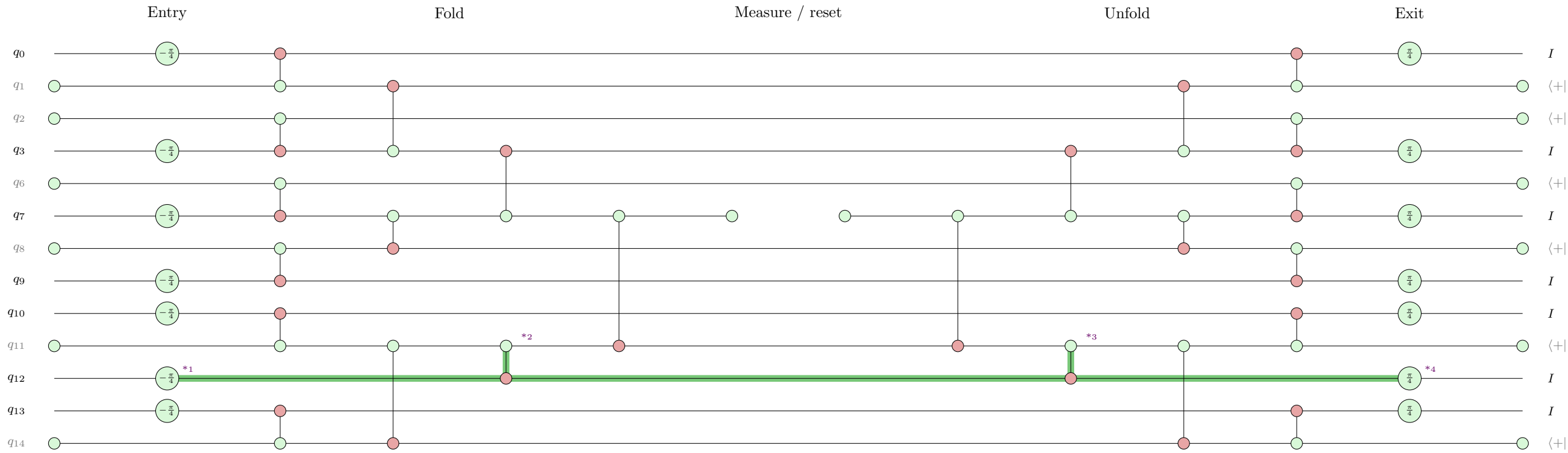
Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

$d = 3$: basis element 85
 No boundary support. 4 phase defects.



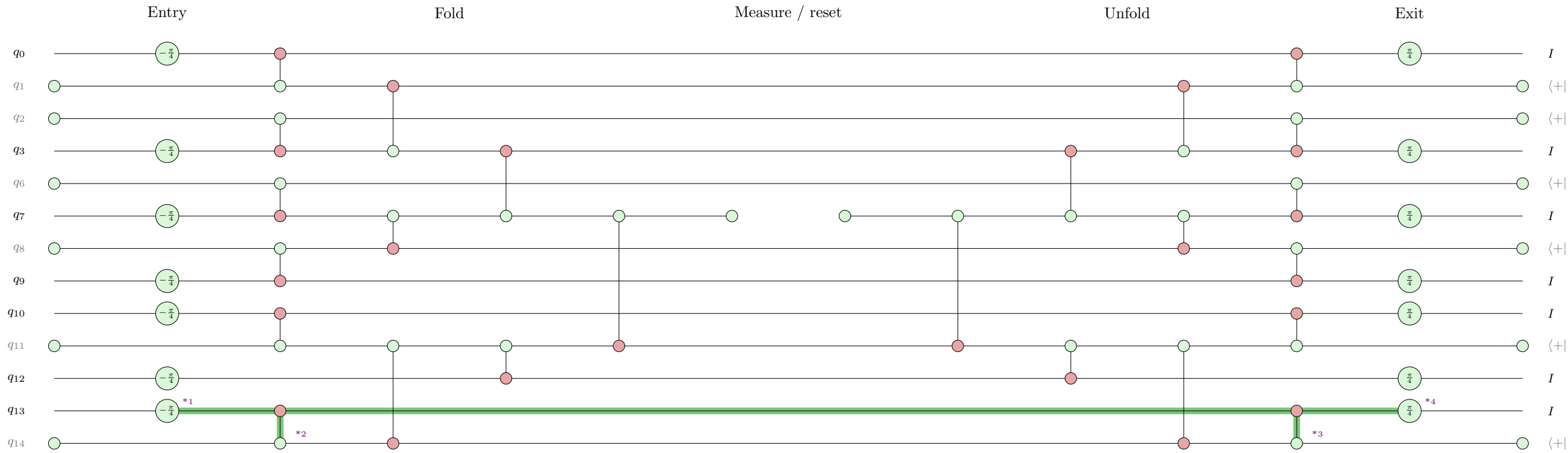
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

$d = 3$: basis element 86
 No boundary support. 4 phase defects.



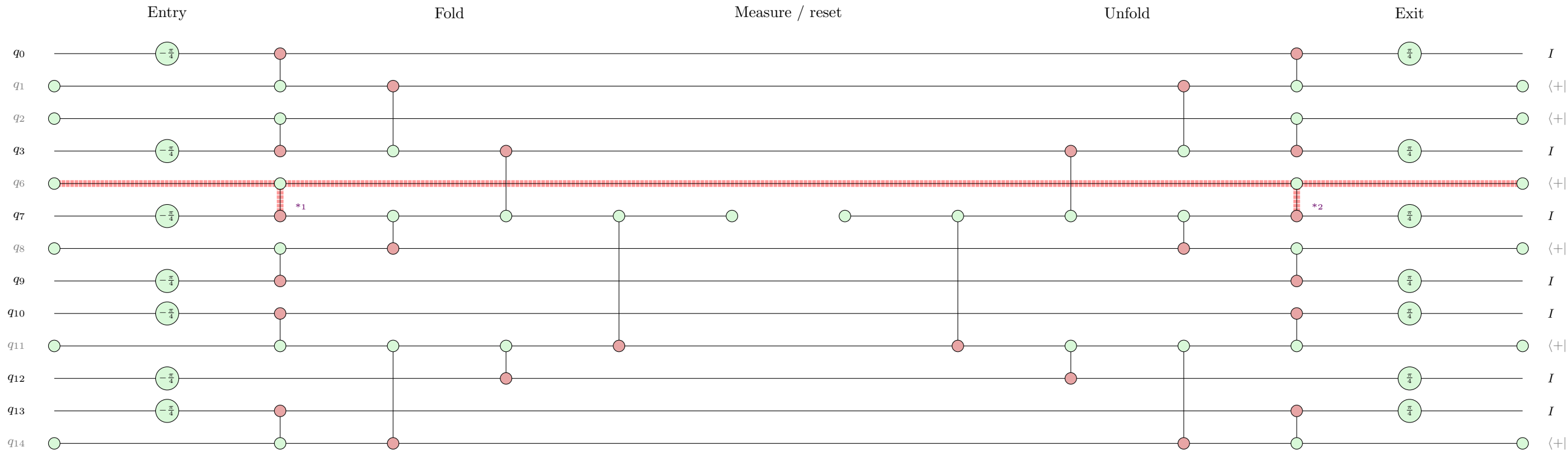
$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

$d = 3$: basis element 87
 No boundary support. 4 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi.$

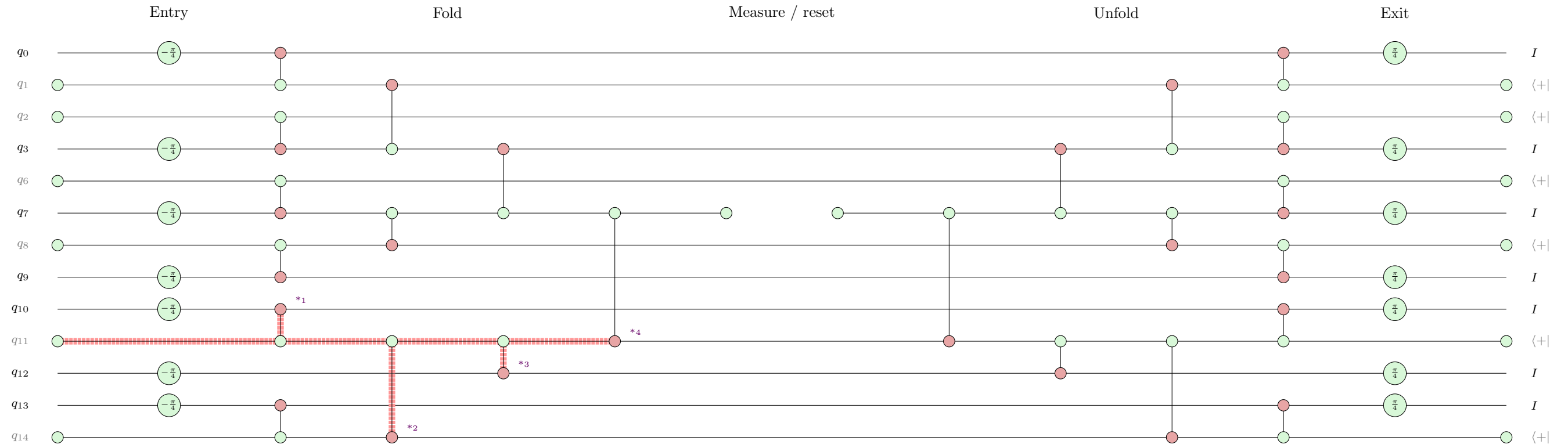
$d = 3$: basis element 88
 No boundary support. 2 phase defects.



$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$
 Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi.$

$d = 3$: basis element 89

No boundary support. 4 phase defects.

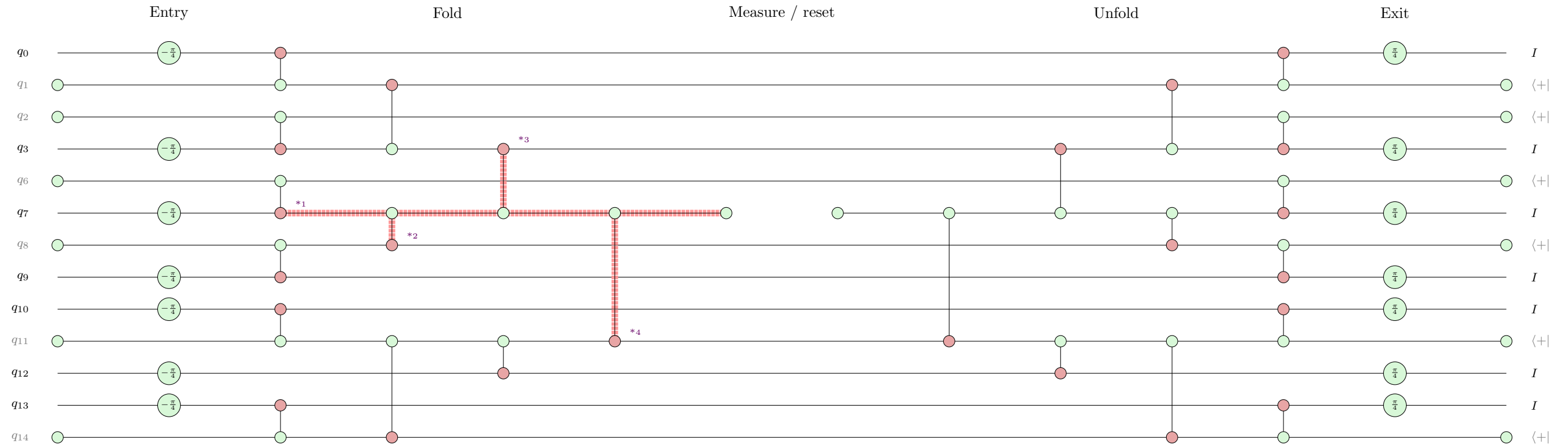


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

$d = 3$: basis element 90

No boundary support. 4 phase defects.

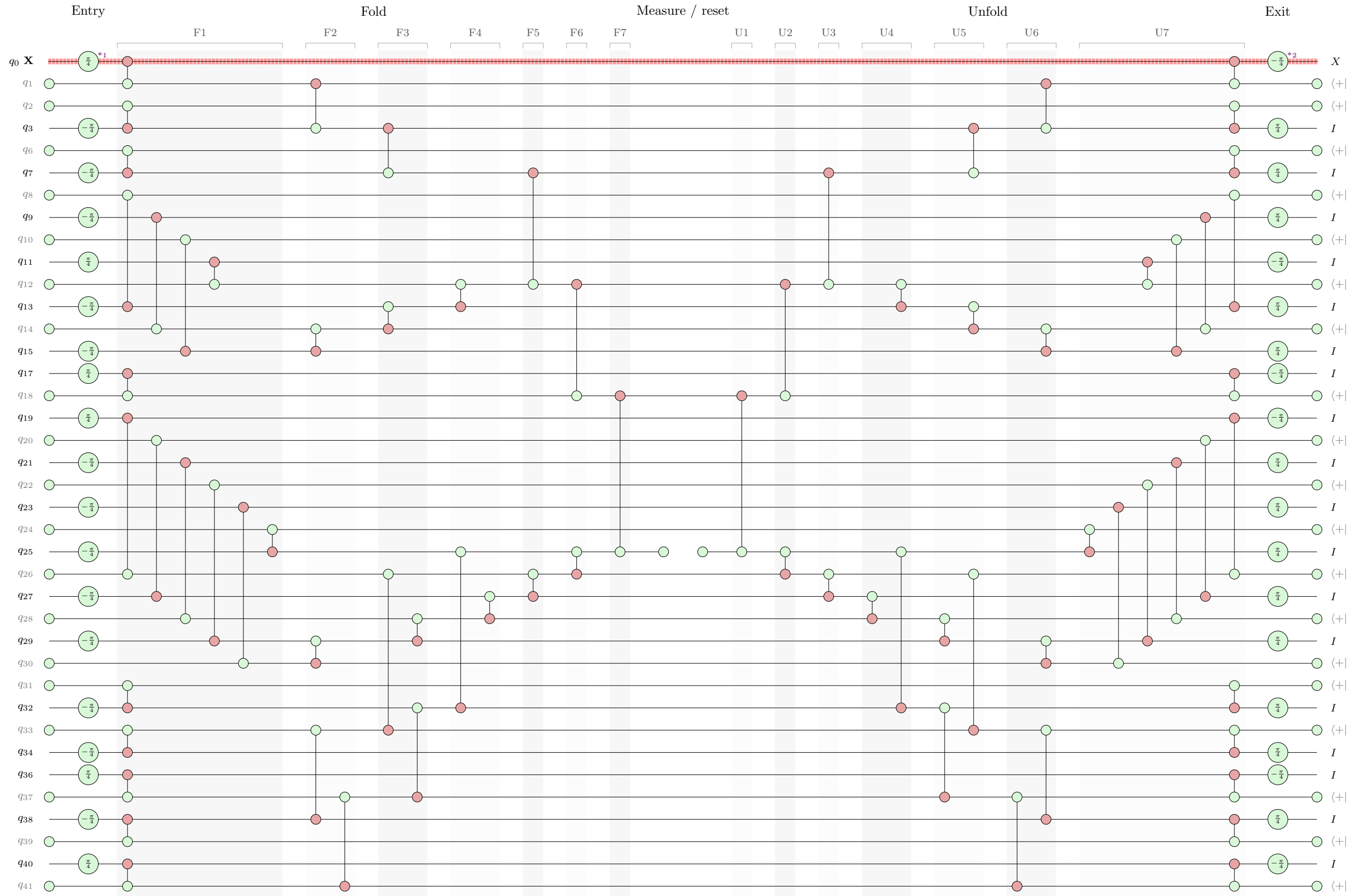


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

$d = 5$: incoming X_0

Incoming Paulis. 2 phase defects.

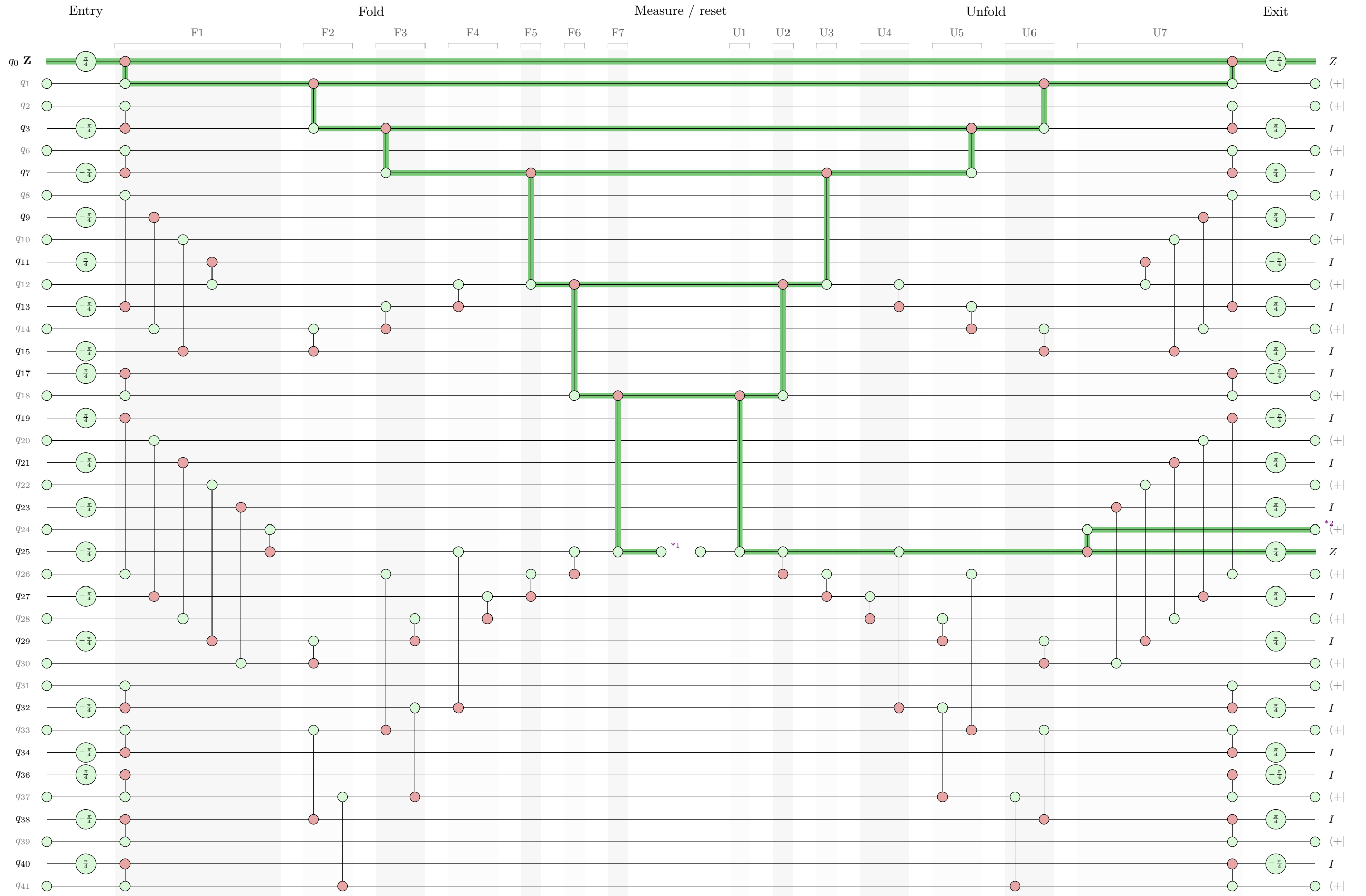


$$P_{\text{in}} = X_0, \quad P_{\text{out}} = X_0, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = -\frac{\pi}{2}$; $\delta_2 = \frac{\pi}{2}$.

d5_001.tikz

$d = 5$: incoming Z_0
Incoming Paulis. 2 phase defects.

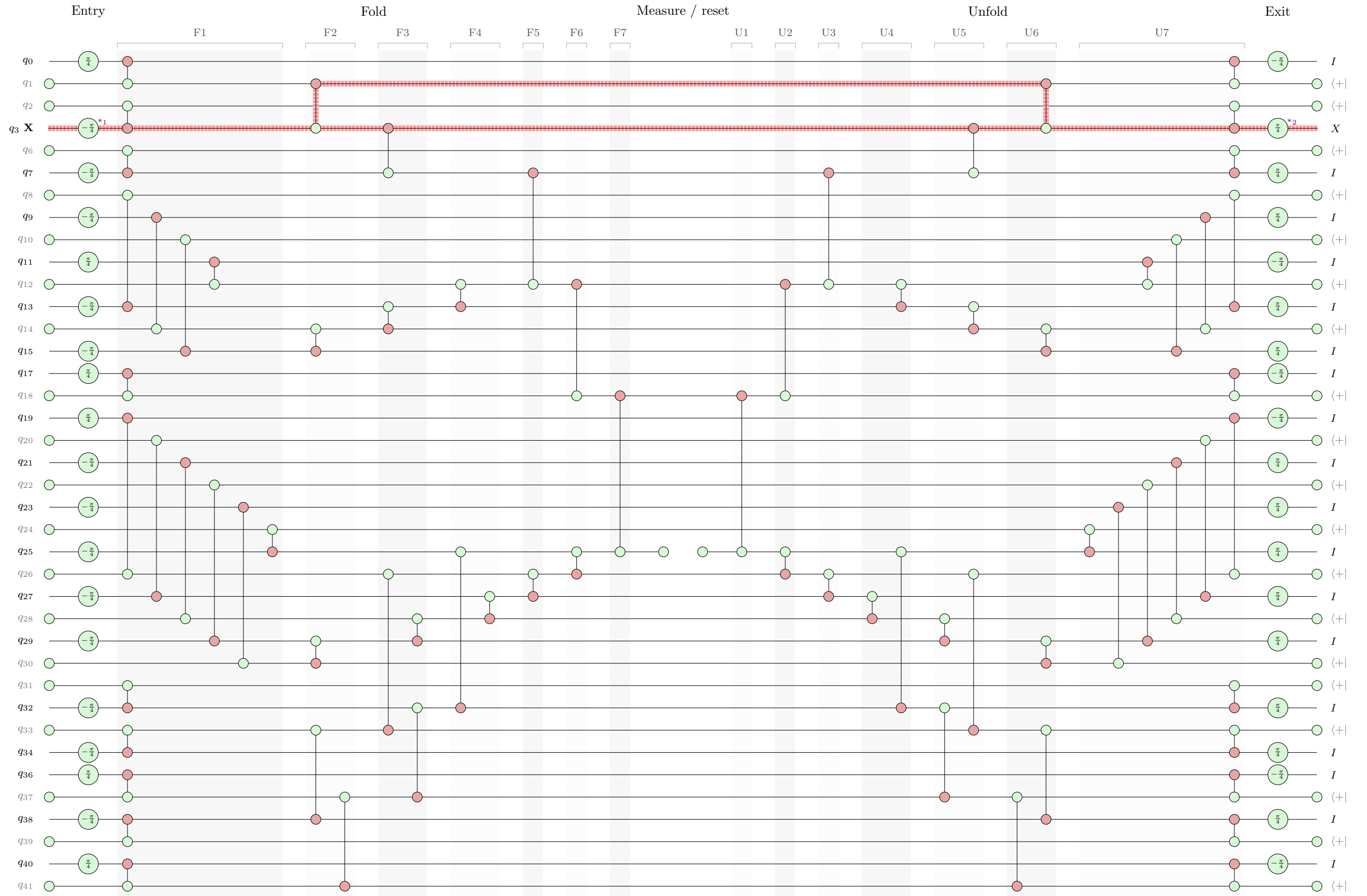


$$P_{\text{in}} = Z_0, \quad P_{\text{out}} = Z_0 Z_{25}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: incoming X_3

Incoming Paulis. 2 phase defects.

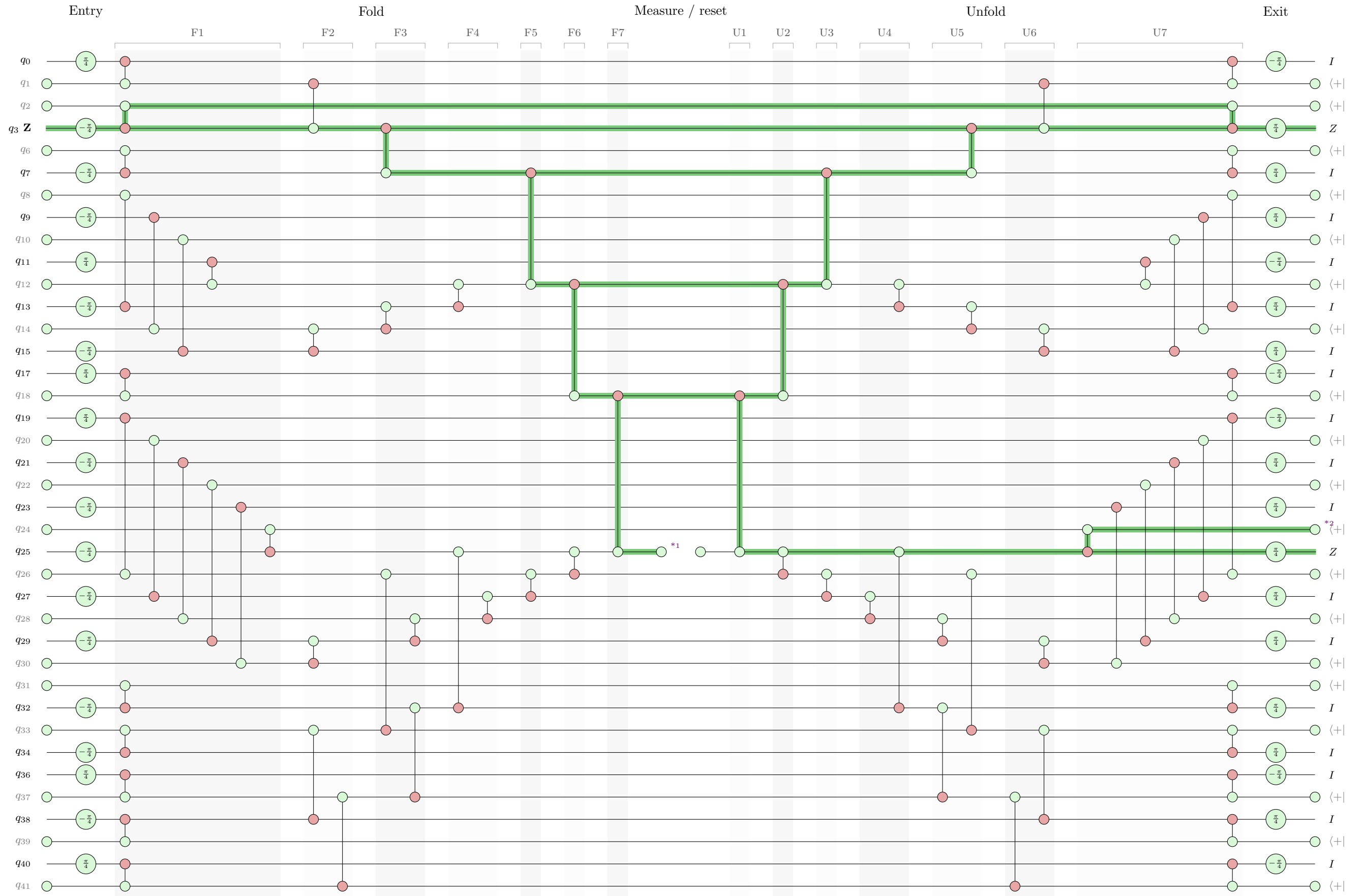


$$P_{\text{in}} = X_3, \quad P_{\text{out}} = X_3, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_003.tikz

$d = 5$: incoming Z_3
Incoming Paulis. 2 phase defects.



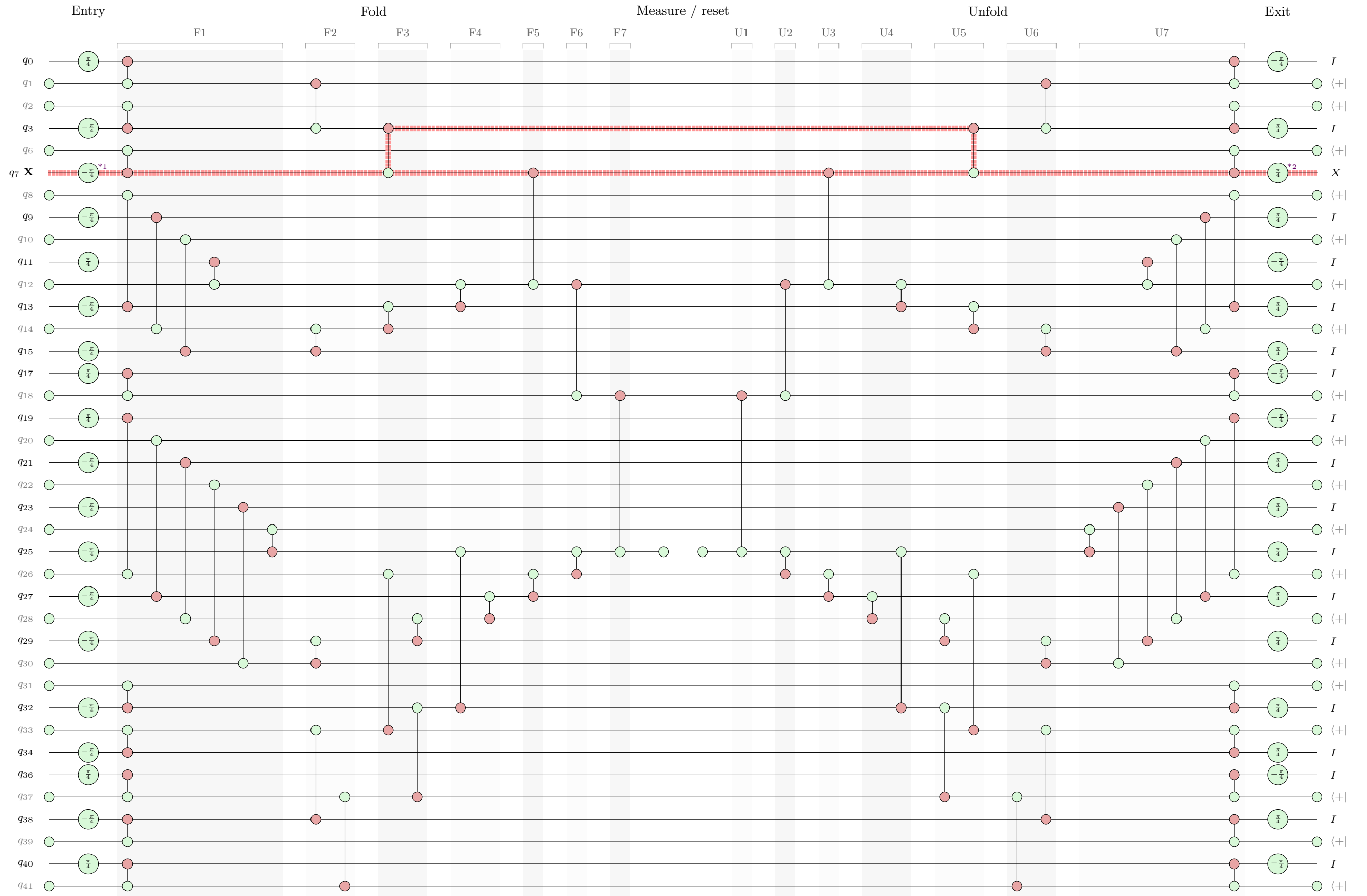
$$P_{\text{in}} = Z_3, \quad P_{\text{out}} = Z_3 Z_{25}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_004.tikz

$d = 5$: incoming X_7

Incoming Paulis. 2 phase defects.

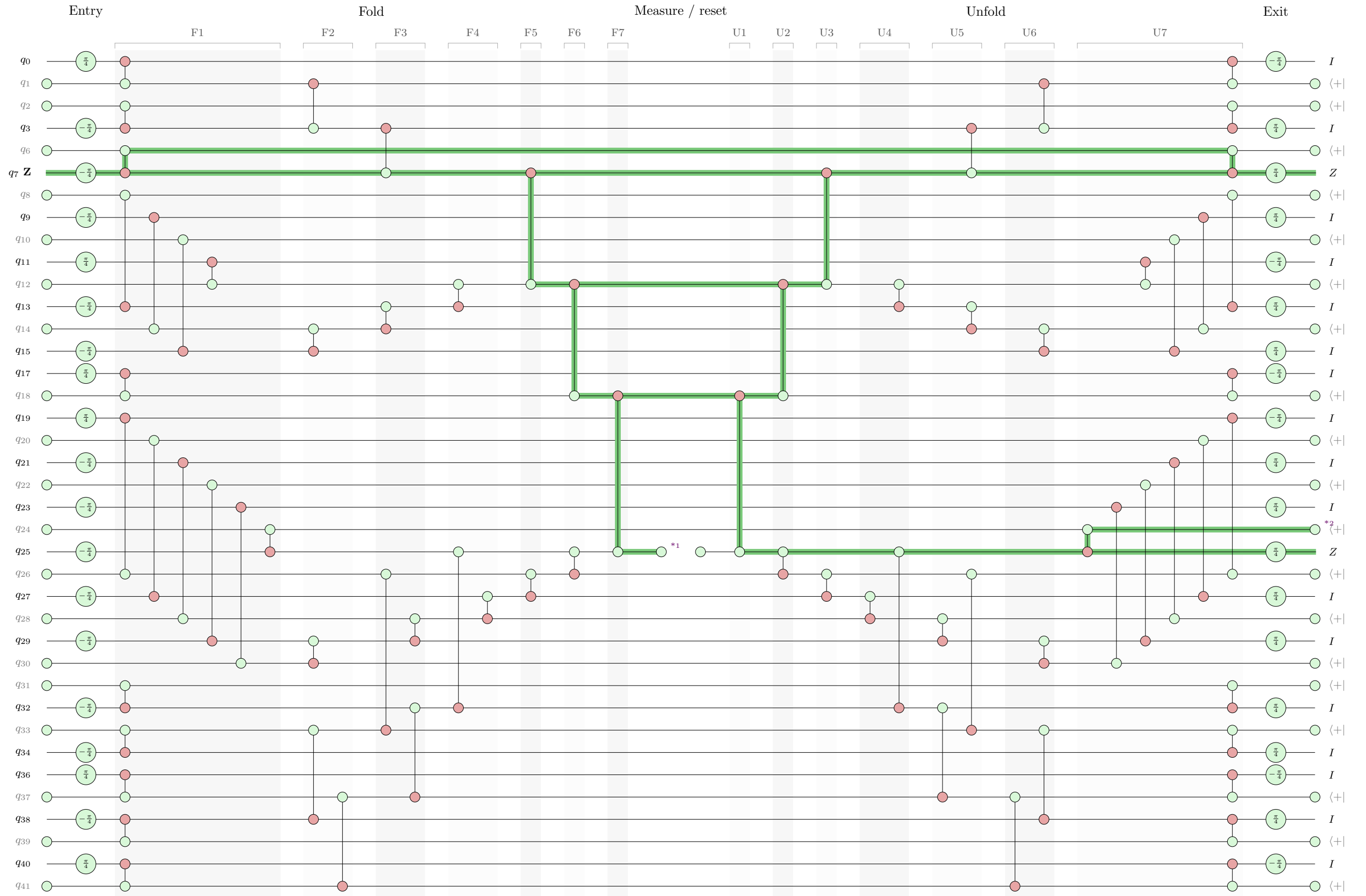


$$P_{\text{in}} = X_7, \quad P_{\text{out}} = X_7, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_005.tikz

$d = 5$: incoming Z_7
Incoming Paulis. 2 phase defects.

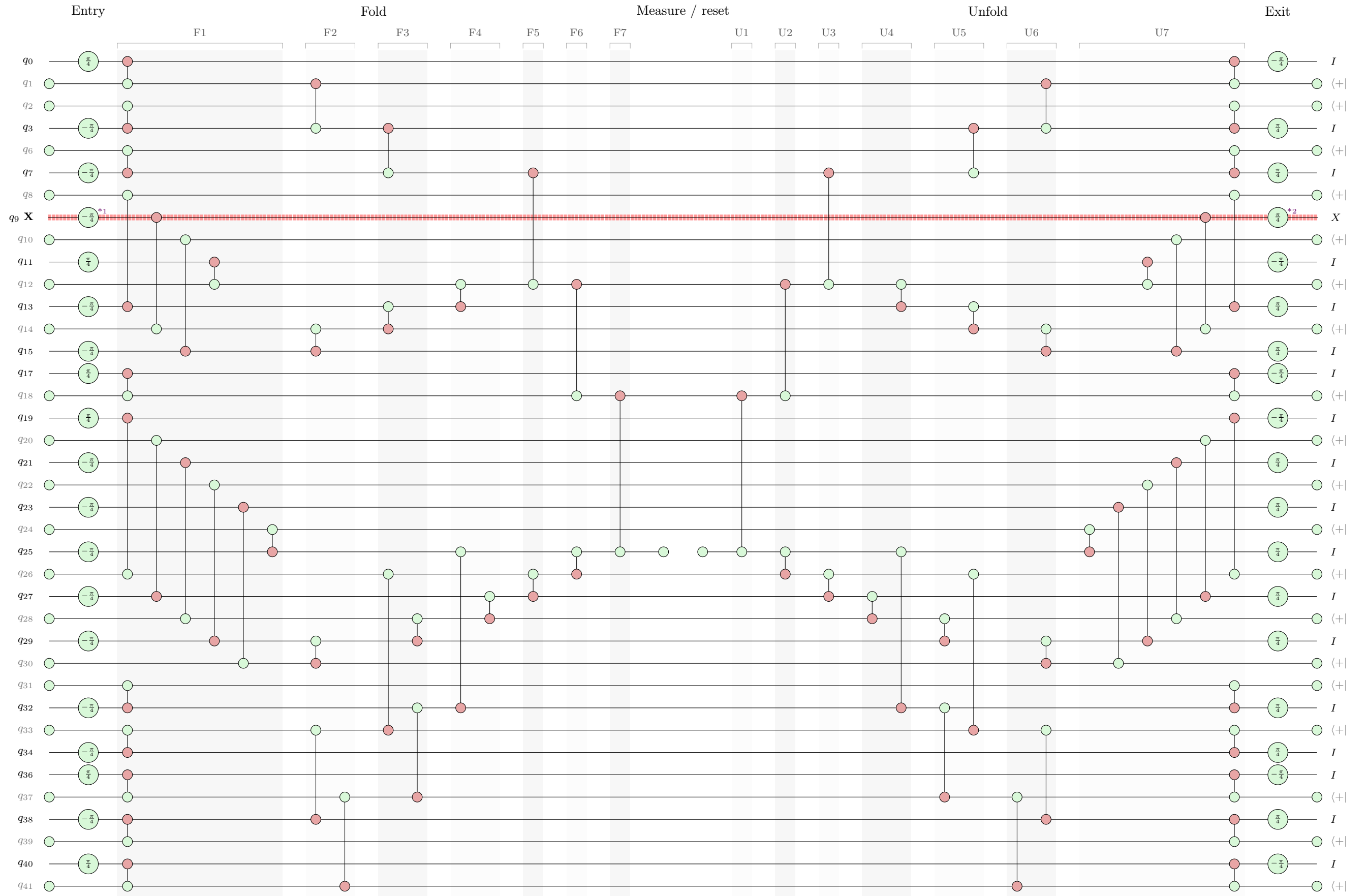


$$P_{\text{in}} = Z_7, \quad P_{\text{out}} = Z_7 Z_{25}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: incoming X_9

Incoming Paulis. 2 phase defects.

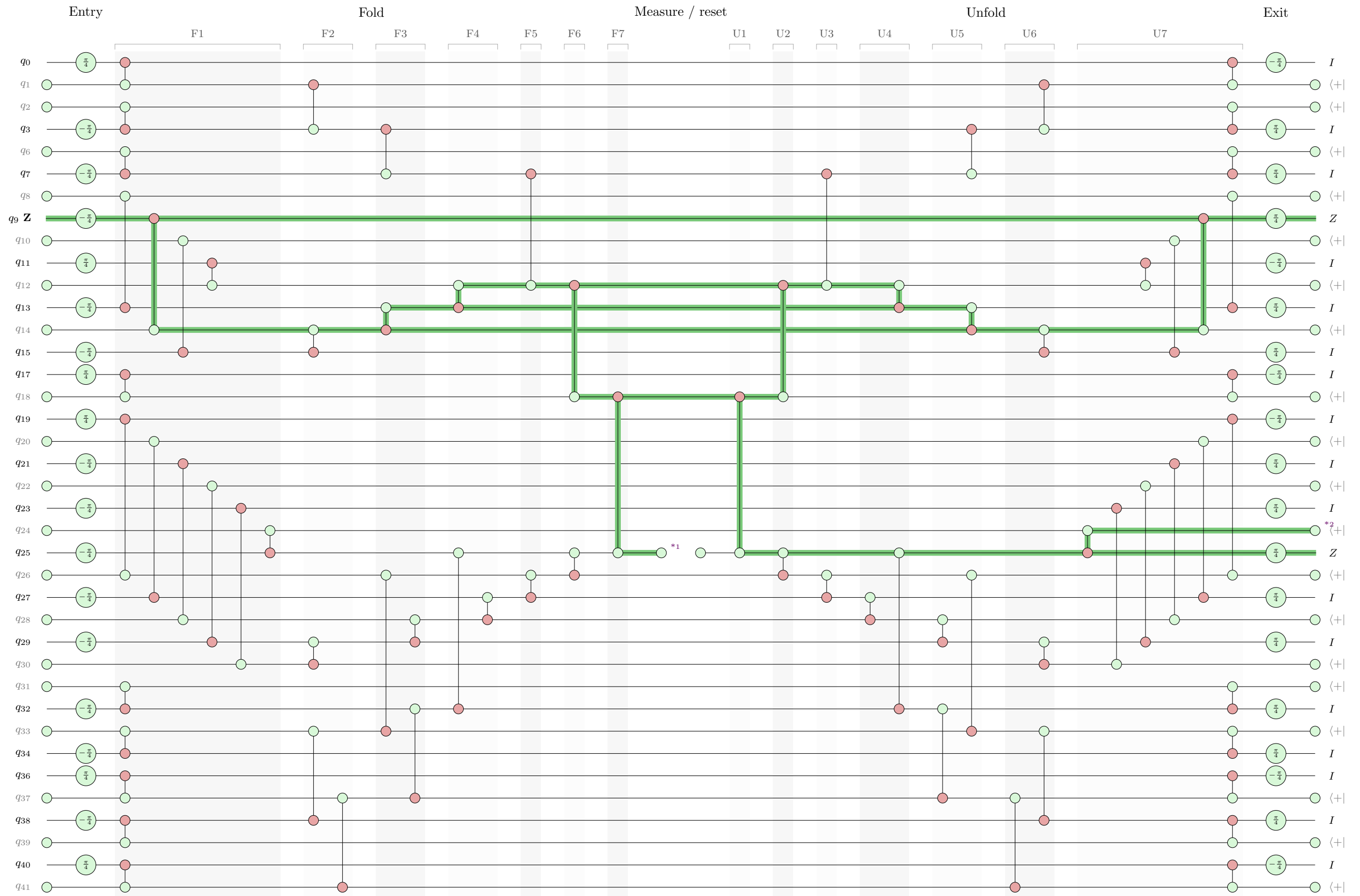


$$P_{\text{in}} = X_9, \quad P_{\text{out}} = X_9, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_007.tikz

$d = 5$: incoming Z_9
Incoming Paulis. 2 phase defects.

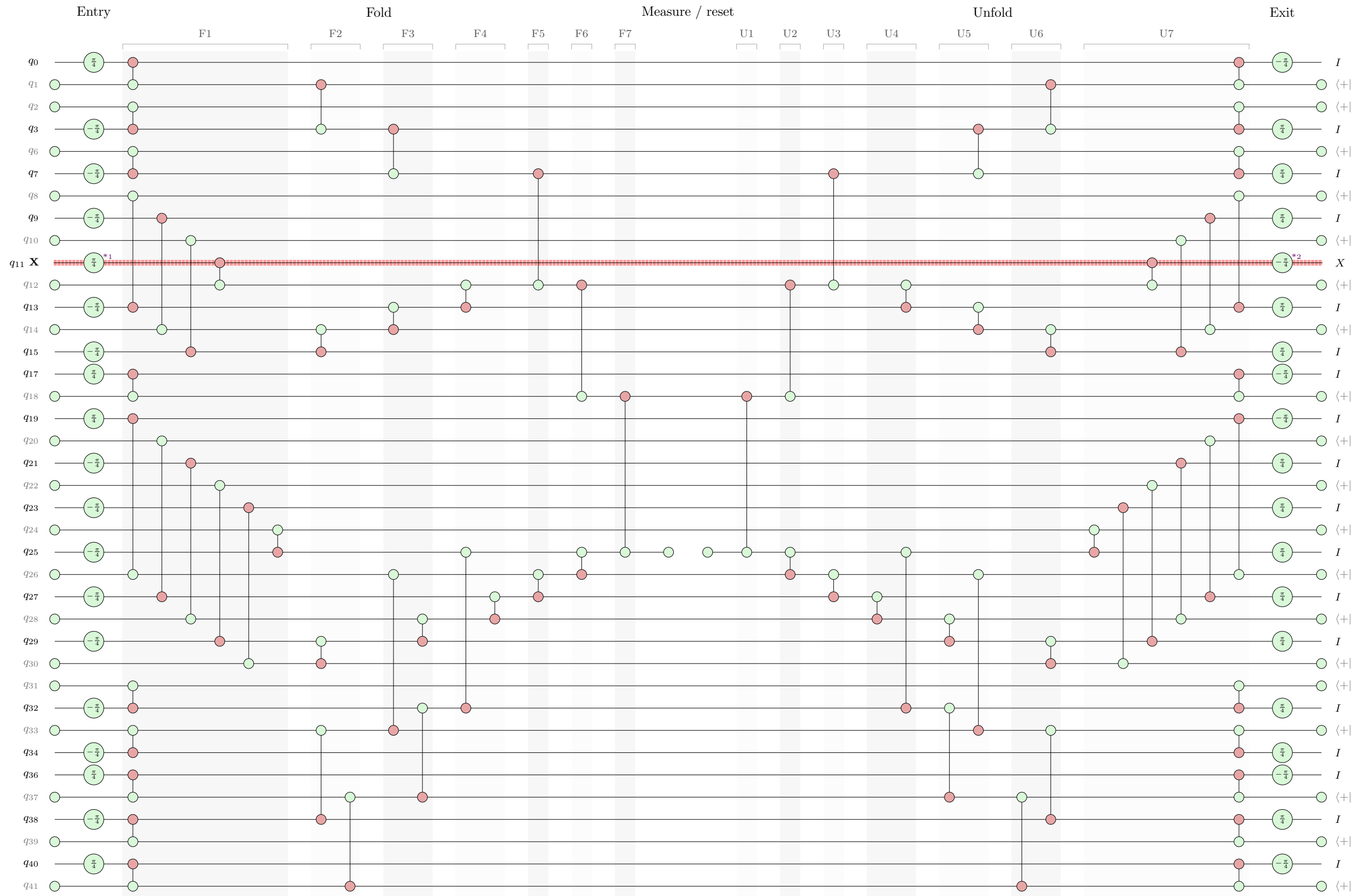


$$P_{\text{in}} = Z_9, \quad P_{\text{out}} = Z_9 Z_{25}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: incoming X_{11}

Incoming Paulis. 2 phase defects.

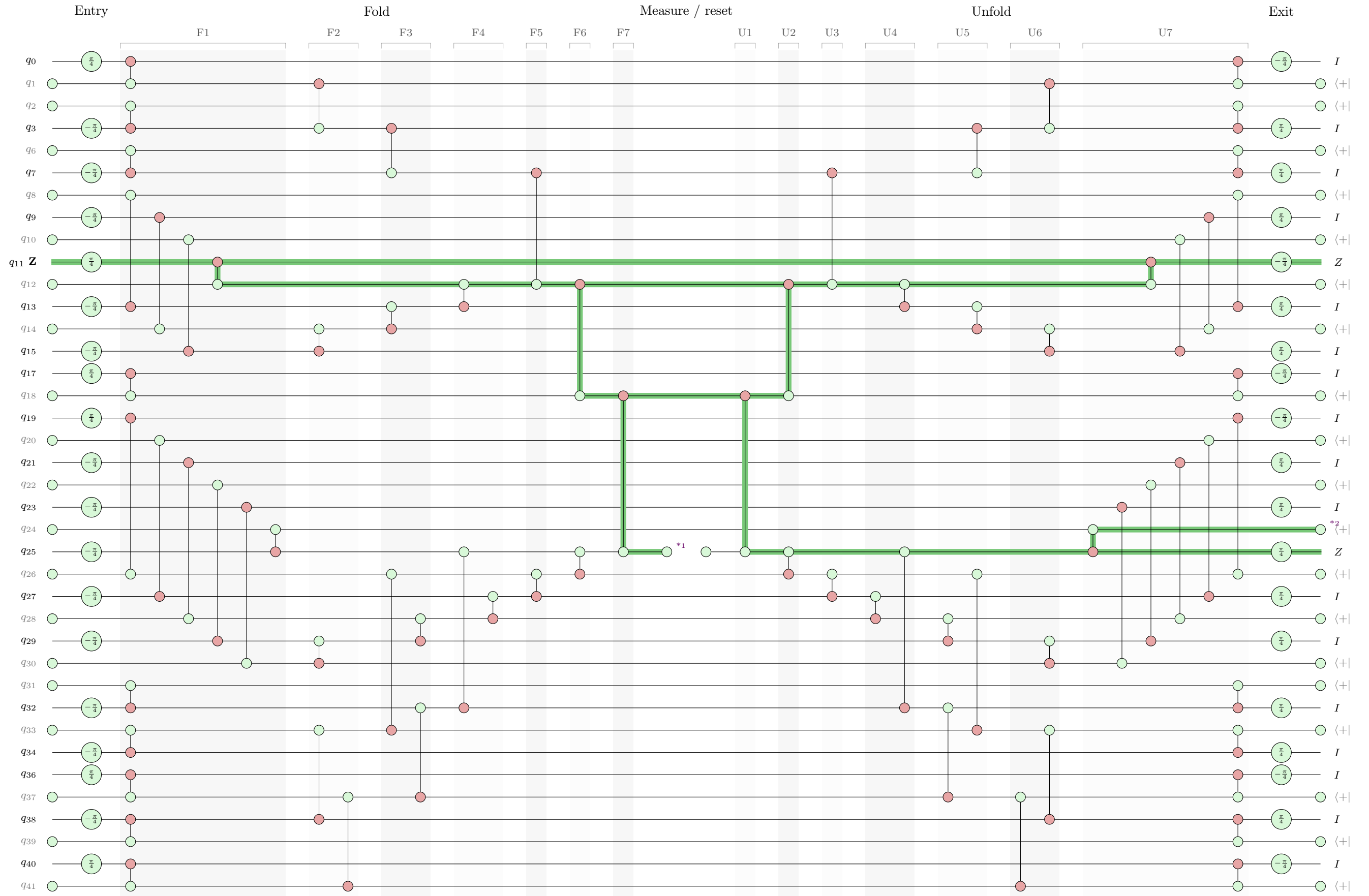


$$P_{\text{in}} = X_{11}, \quad P_{\text{out}} = X_{11}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = -\frac{\pi}{2}$; $\delta_2 = \frac{\pi}{2}$.

d5_009.tikz

$d = 5$: incoming Z_{11}
Incoming Paulis. 2 phase defects.



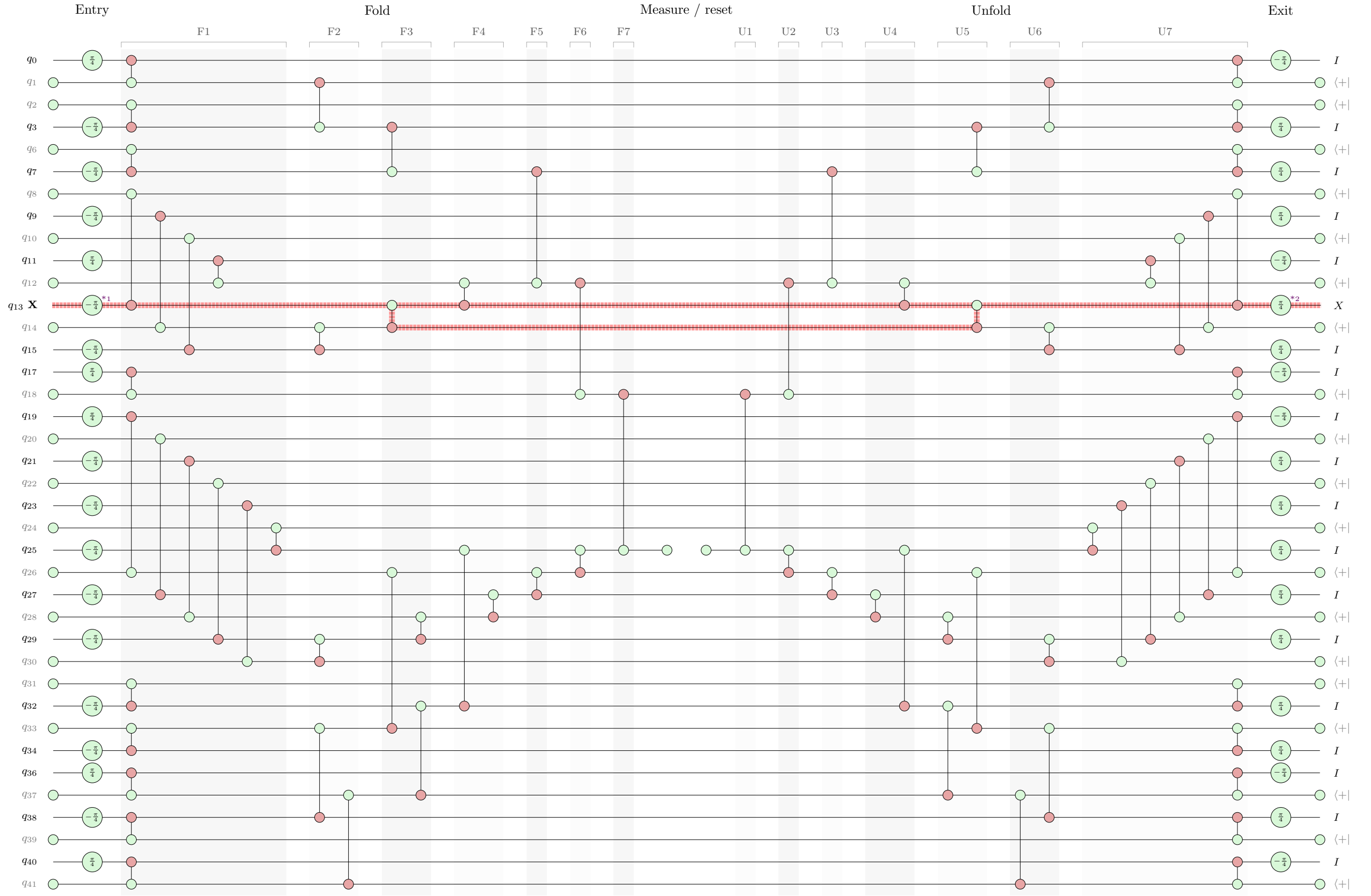
$$P_{\text{in}} = Z_{11}, \quad P_{\text{out}} = Z_{11}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_010.tikz

$d = 5$: incoming X_{13}

Incoming Paulis. 2 phase defects.



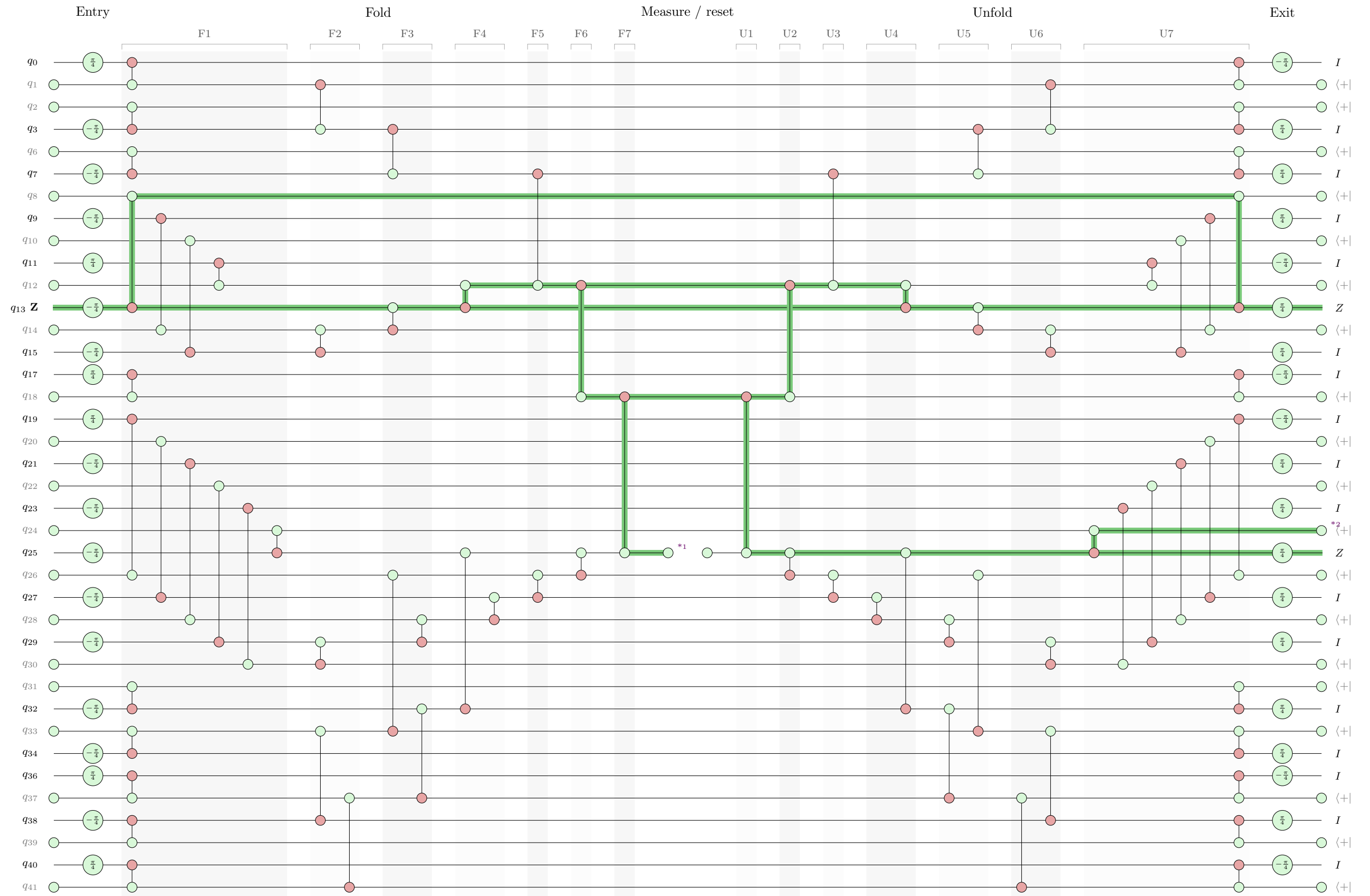
$$P_{\text{in}} = X_{13}, \quad P_{\text{out}} = X_{13}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_011.tikz

$d = 5$: incoming Z_{13}

Incoming Paulis. 2 phase defects.



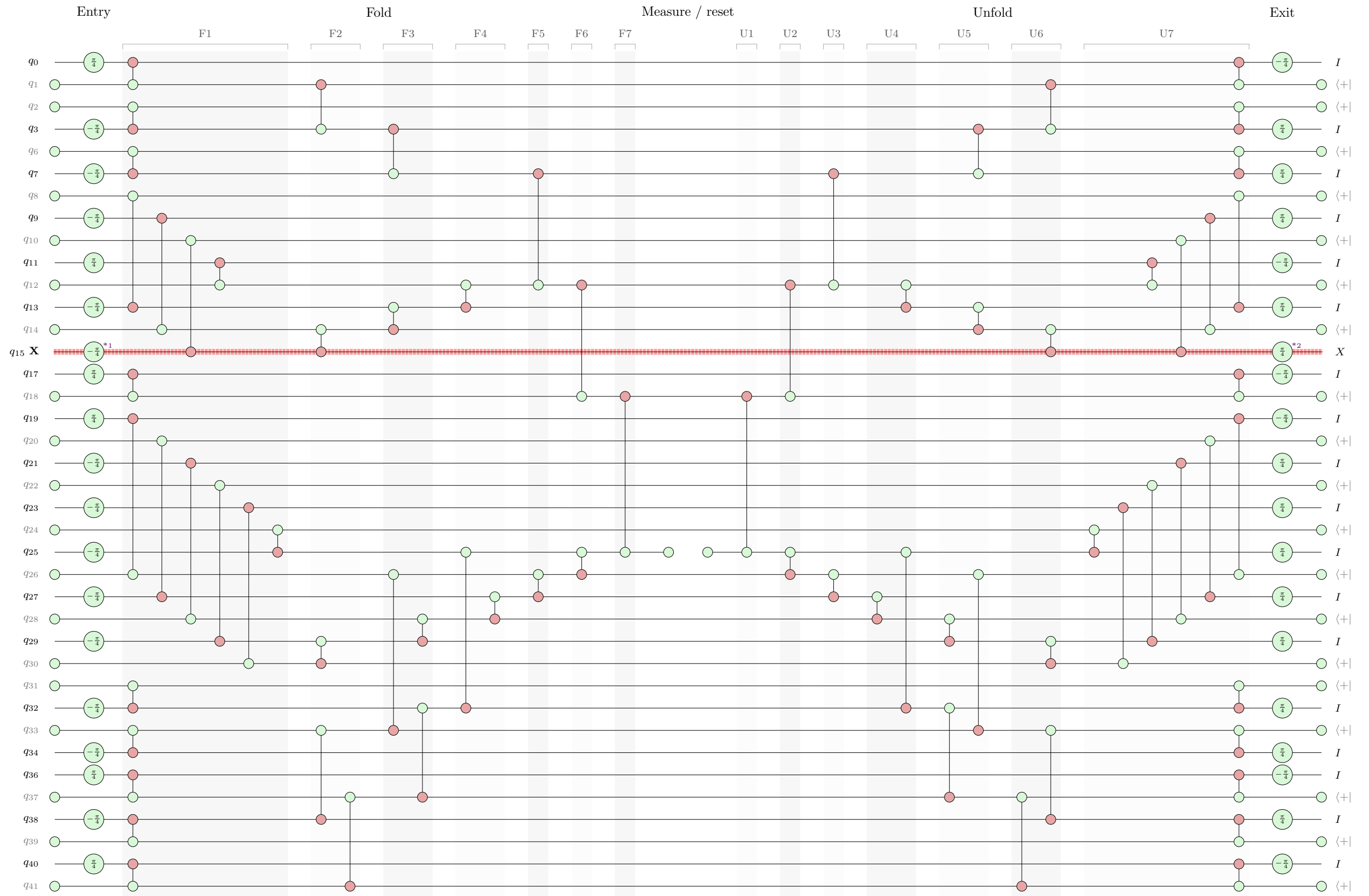
$$P_{\text{in}} = Z_{13}, \quad P_{\text{out}} = Z_{13}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_012.tikz

$d = 5$: incoming X_{15}

Incoming Paulis. 2 phase defects.

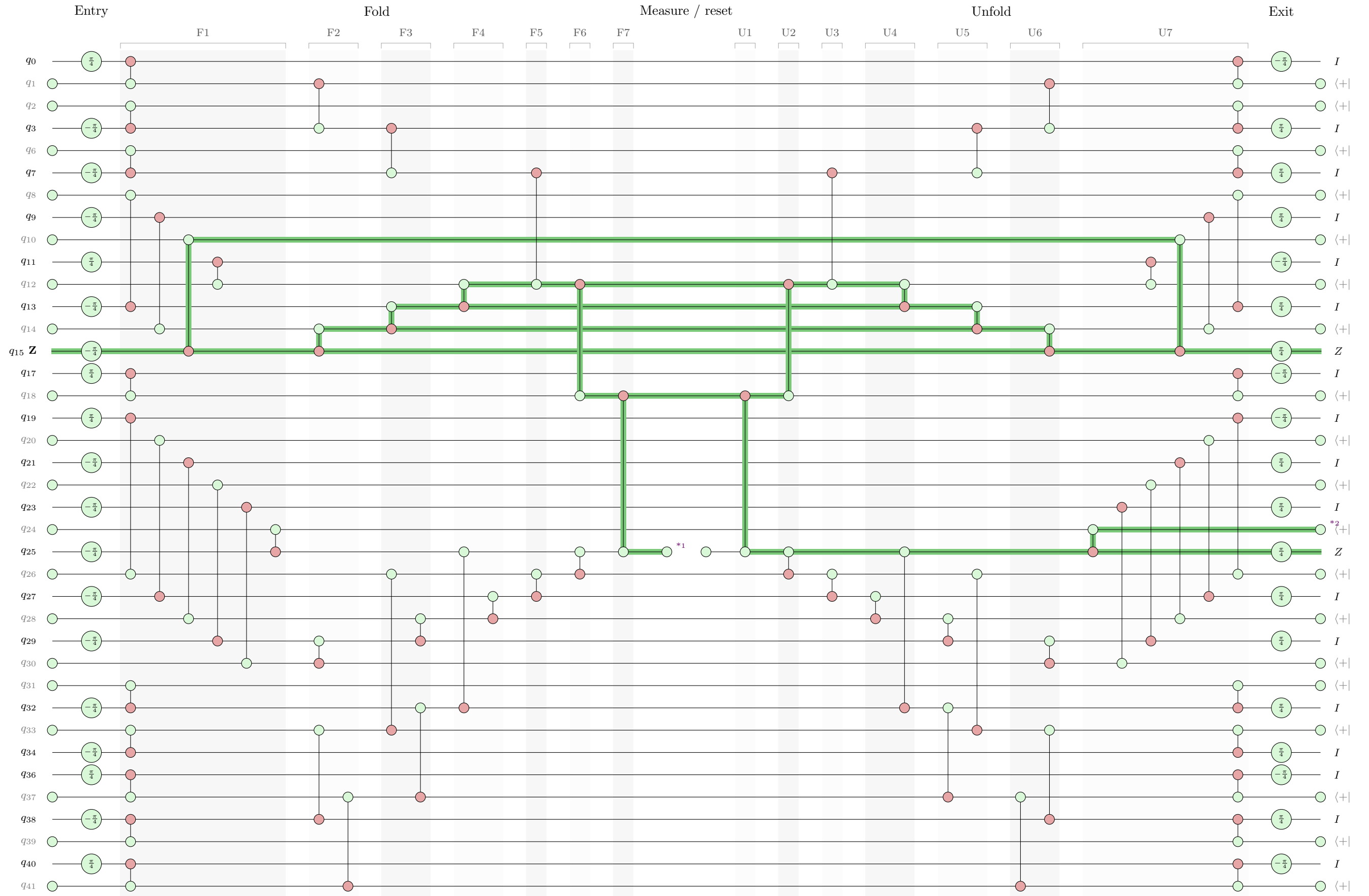


$$P_{\text{in}} = X_{15}, \quad P_{\text{out}} = X_{15}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_013.tikz

$d = 5$: incoming Z_{15}
Incoming Paulis. 2 phase defects.



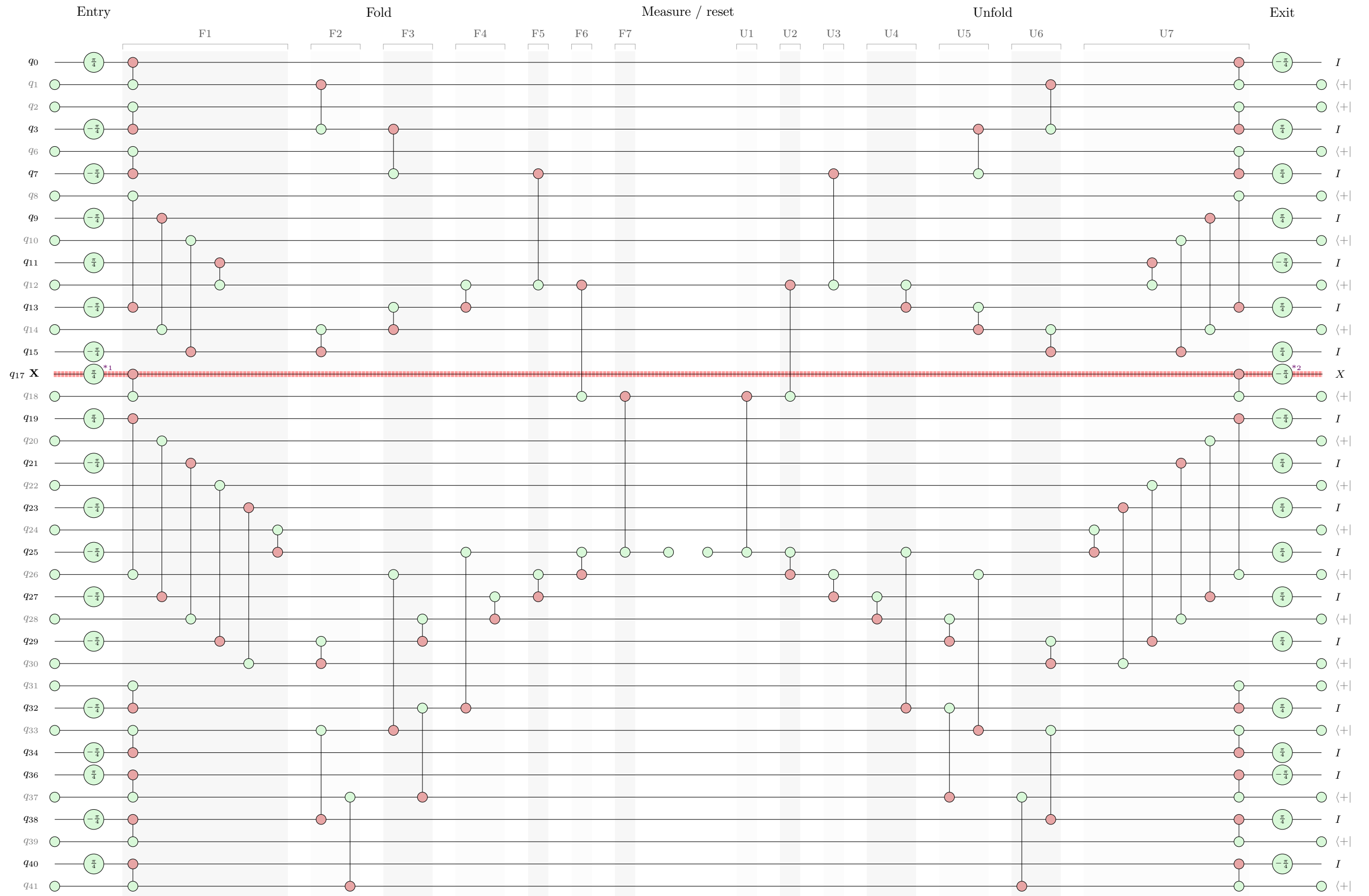
$$P_{\text{in}} = Z_{15}, \quad P_{\text{out}} = Z_{15}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_014.tikz

$d = 5$: incoming X_{17}

Incoming Paulis. 2 phase defects.



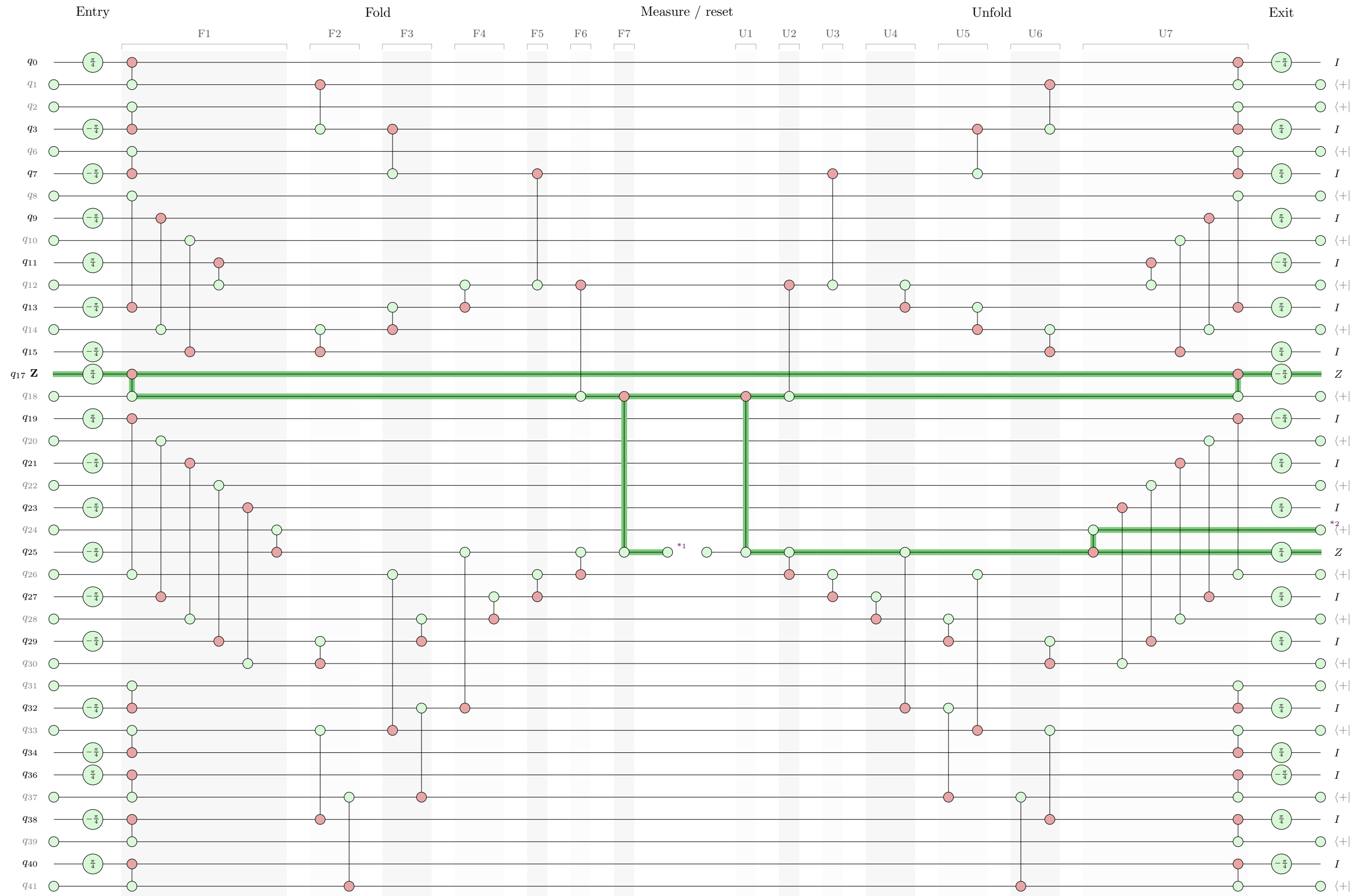
$$P_{\text{in}} = X_{17}, \quad P_{\text{out}} = X_{17}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = -\frac{\pi}{2}$; $\delta_2 = \frac{\pi}{2}$.

d5_015.tikz

$d = 5$: incoming Z_{17}

Incoming Paulis. 2 phase defects.



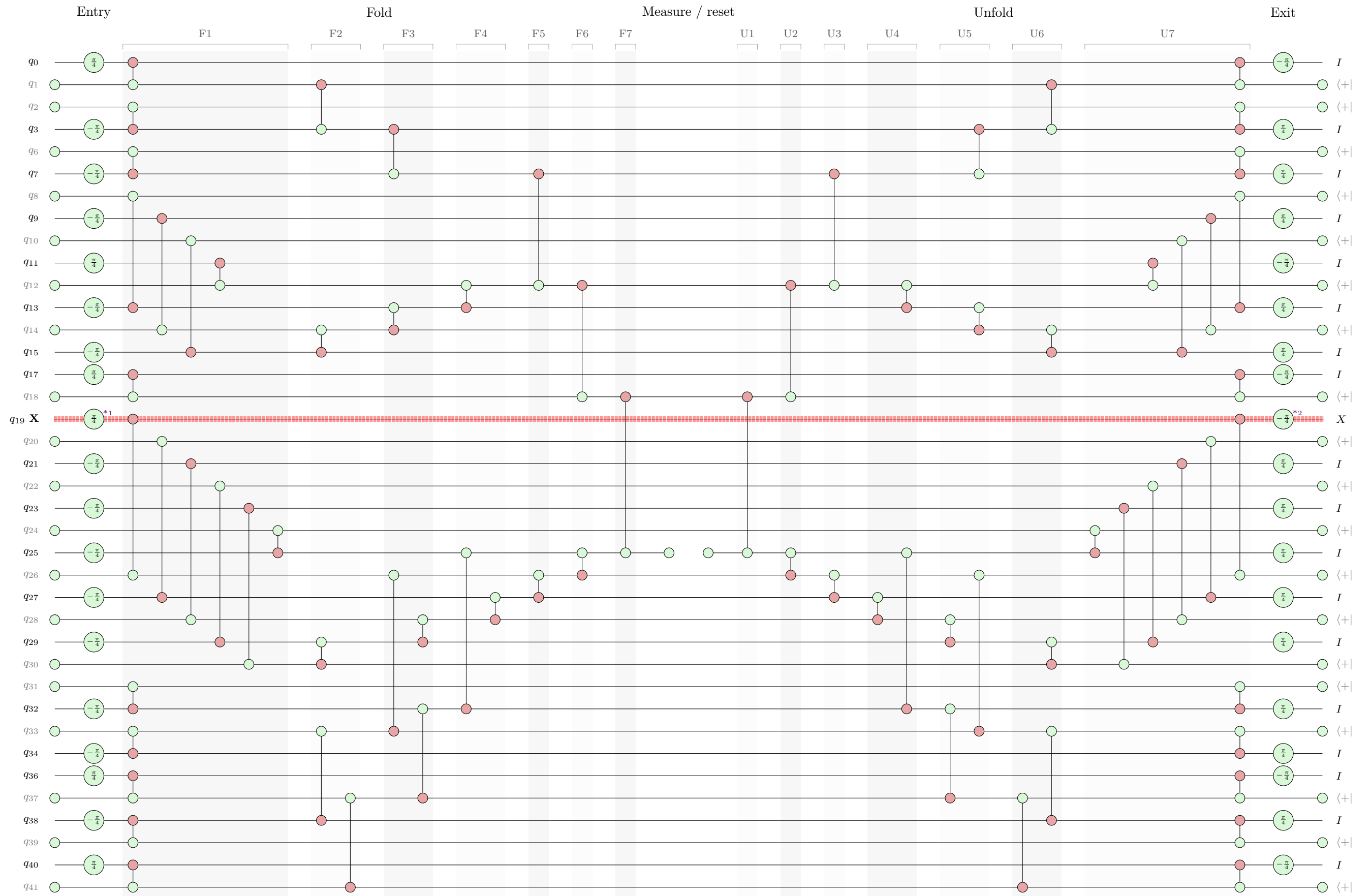
$$P_{\text{in}} = Z_{17}, \quad P_{\text{out}} = Z_{17}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_016.tikz

$d = 5$: incoming X_{19}

Incoming Paulis. 2 phase defects.



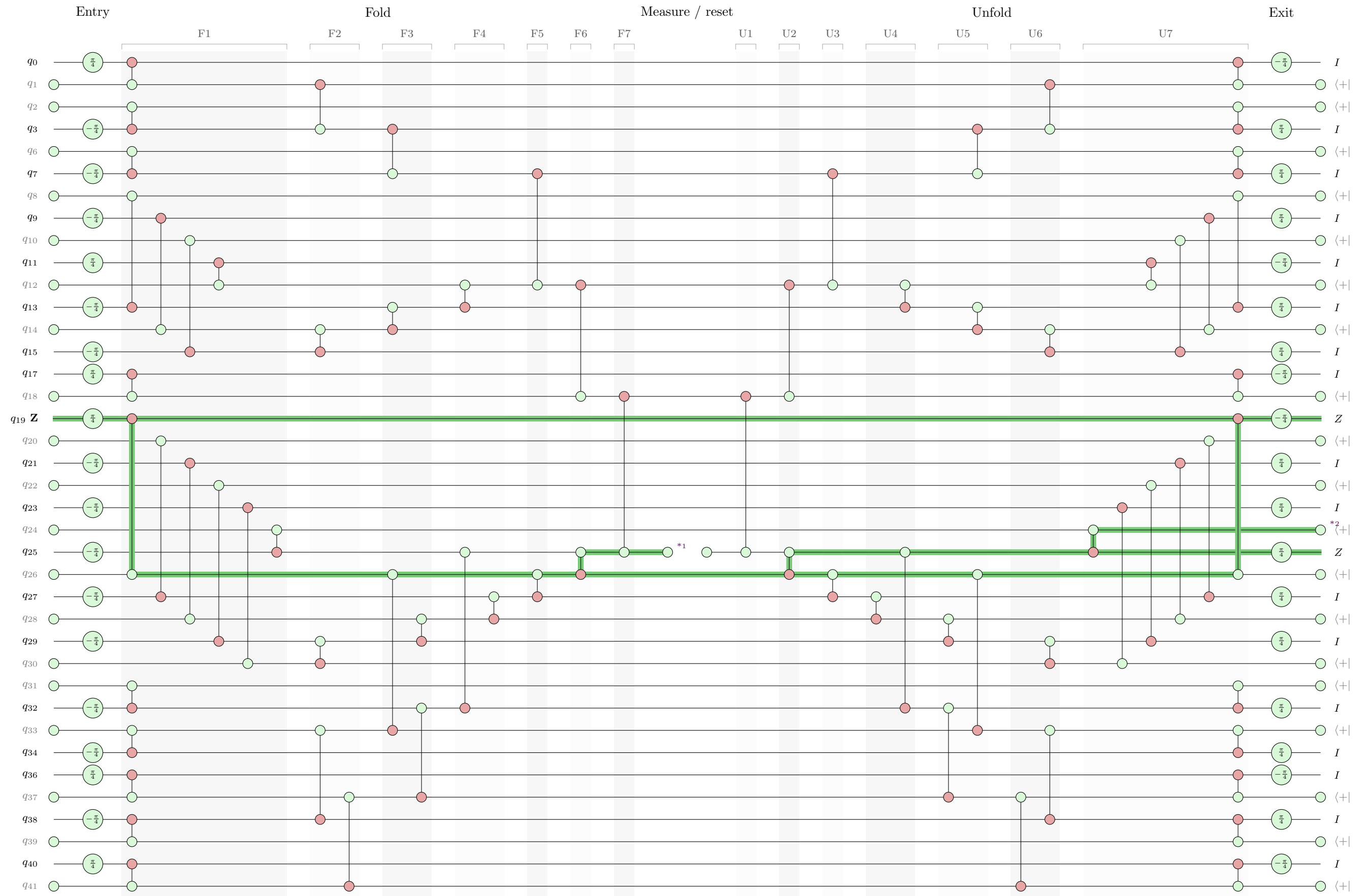
$$P_{\text{in}} = X_{19}, \quad P_{\text{out}} = X_{19}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = -\frac{\pi}{2}$; $\delta_2 = \frac{\pi}{2}$.

d5_017.tikz

$d = 5$: incoming Z_{19}

Incoming Paulis. 2 phase defects.

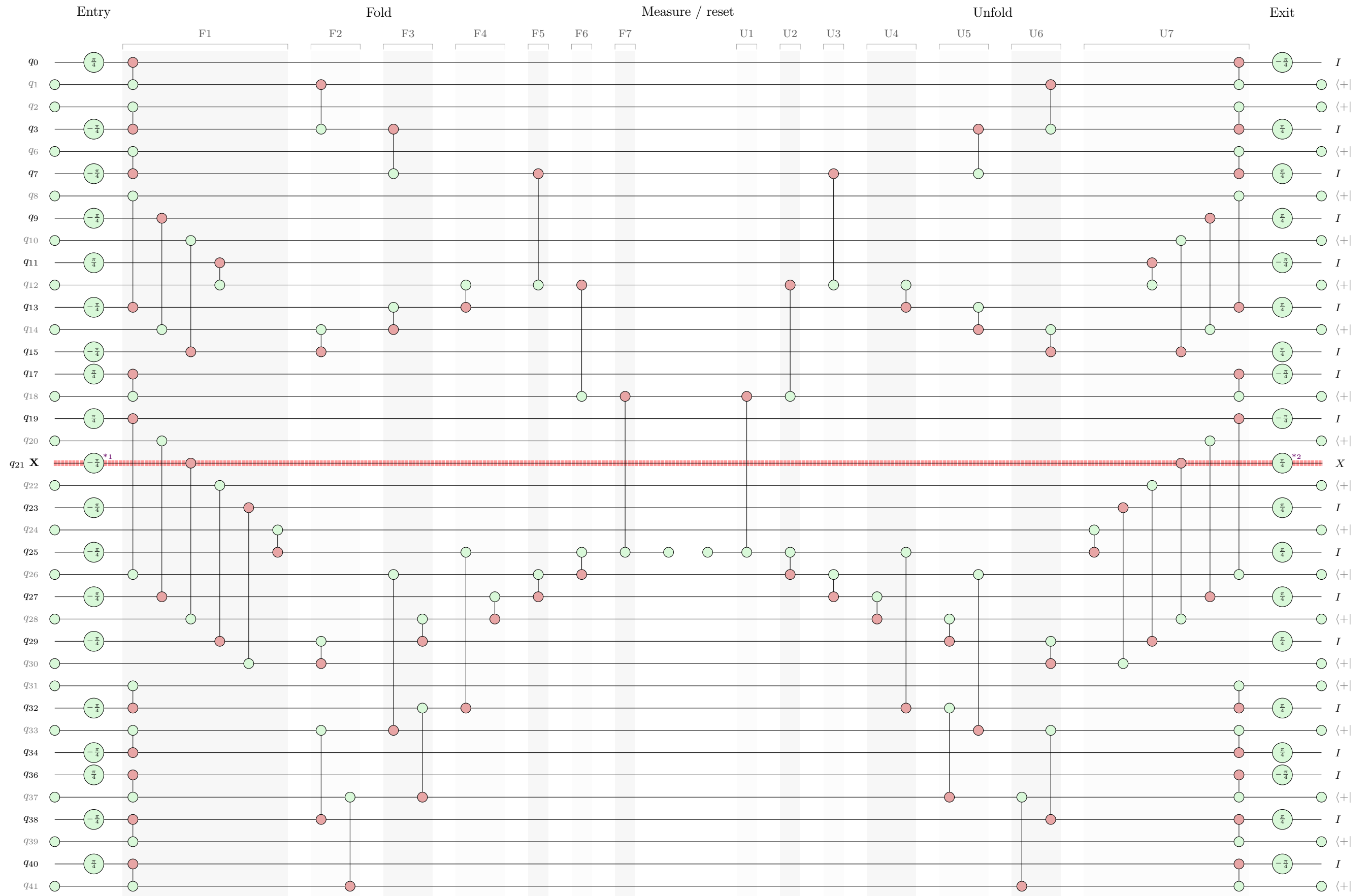


$$P_{\text{in}} = Z_{19}, \quad P_{\text{out}} = Z_{19}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_018.tikz

$d = 5$: incoming X_{21}
Incoming Paulis. 2 phase defects.



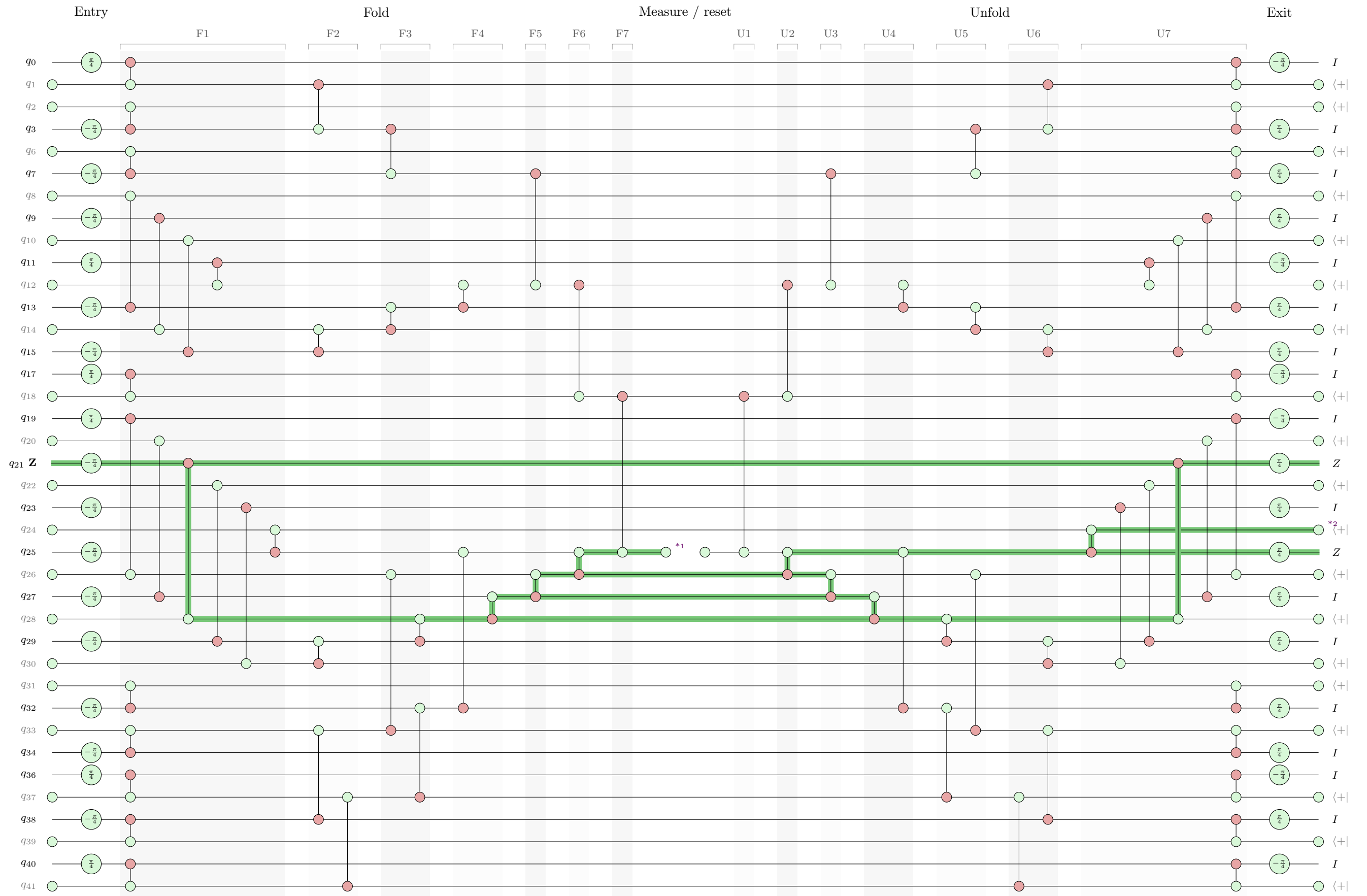
$$P_{\text{in}} = X_{21}, \quad P_{\text{out}} = X_{21}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_019.tikz

$d = 5$: incoming Z_{21}

Incoming Paulis. 2 phase defects.



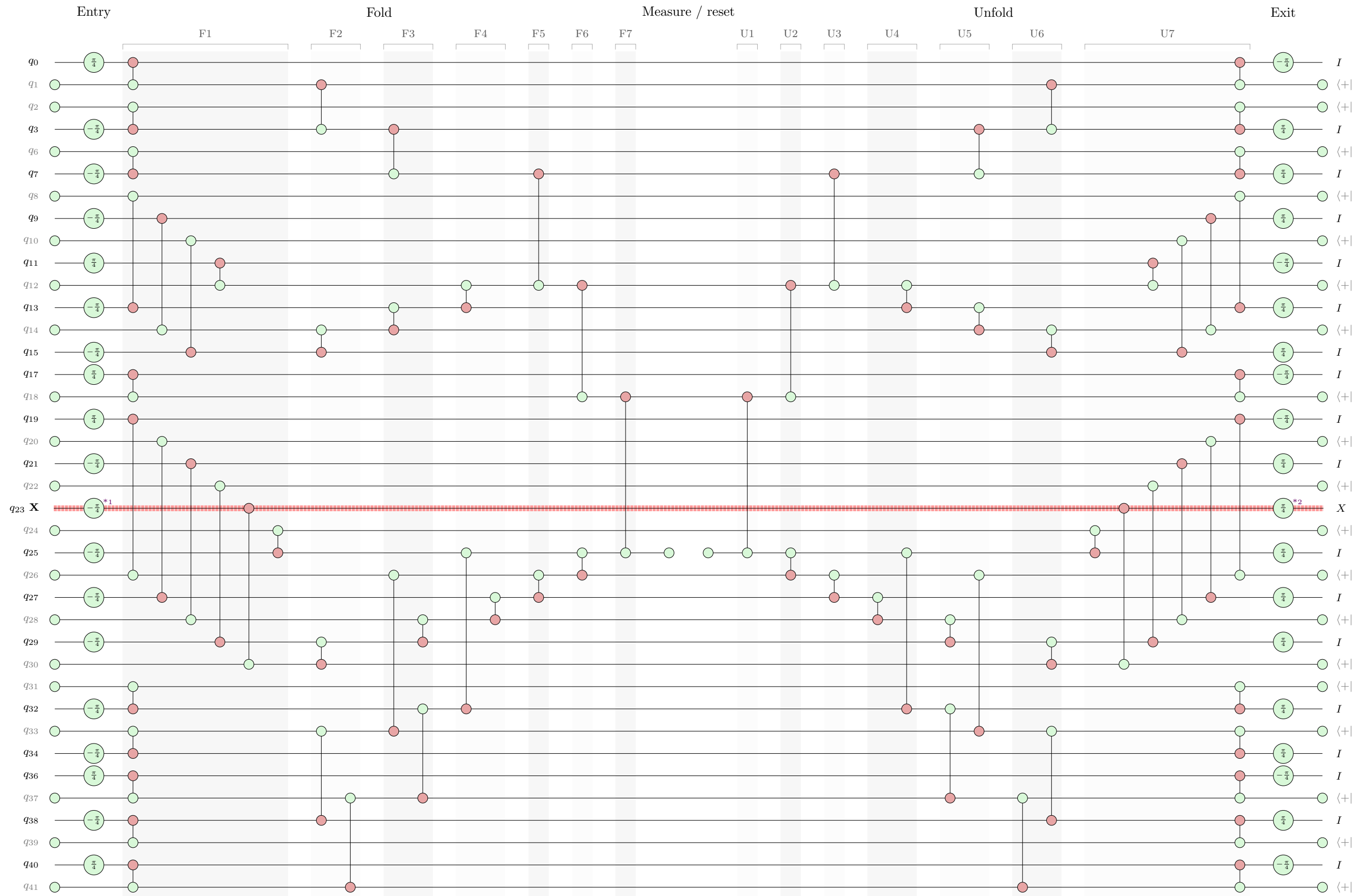
$$P_{\text{in}} = Z_{21}, \quad P_{\text{out}} = Z_{21}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_020.tikz

$d = 5$: incoming X_{23}

Incoming Paulis. 2 phase defects.



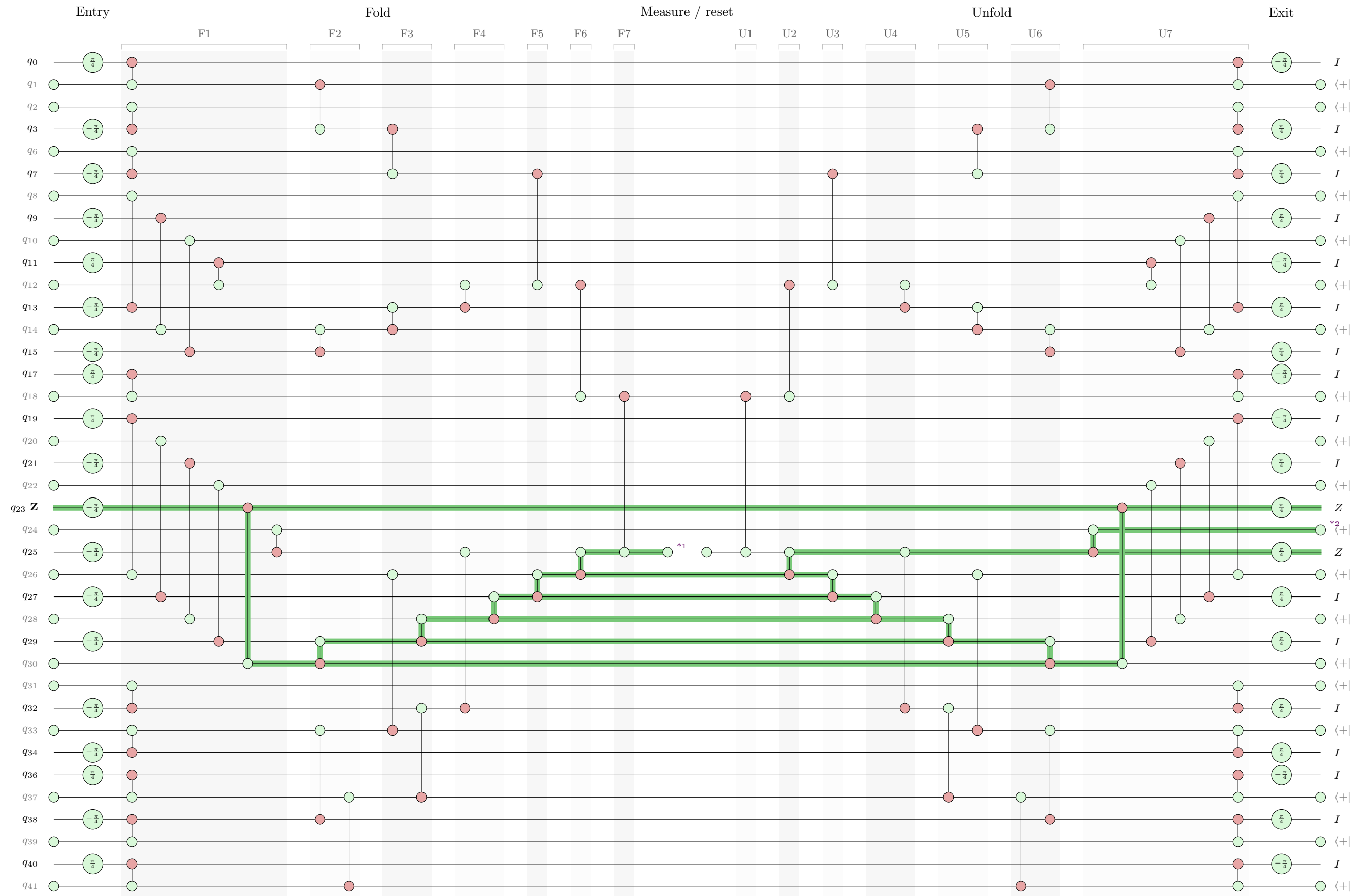
$$P_{\text{in}} = X_{23}, \quad P_{\text{out}} = X_{23}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_021.tikz

$d = 5$: incoming Z_{23}

Incoming Paulis. 2 phase defects.



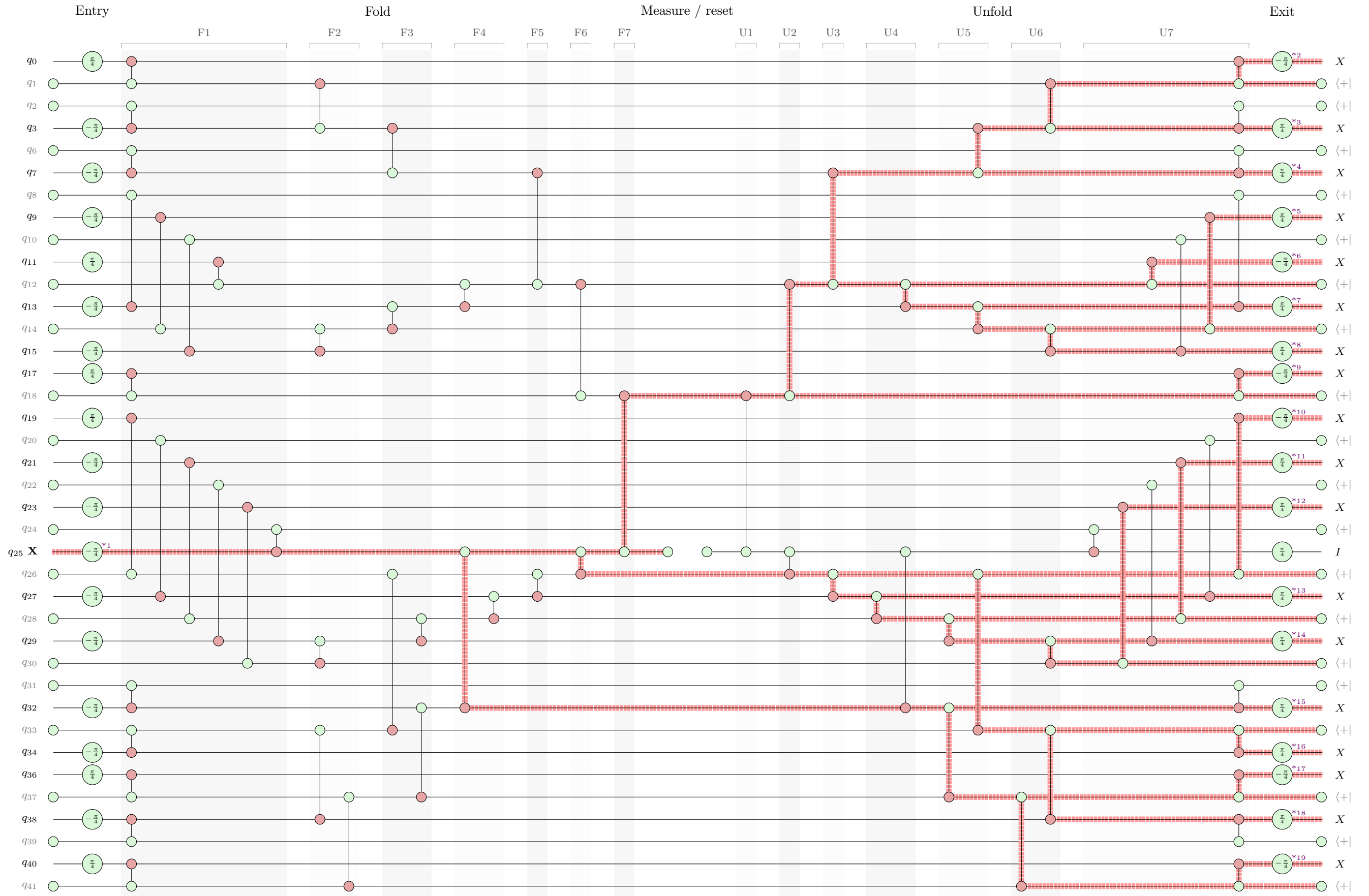
$$P_{\text{in}} = Z_{23}, \quad P_{\text{out}} = Z_{23}Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_022.tikz

$d = 5$: incoming X_{25}

Incoming Paulis. 19 phase defects.



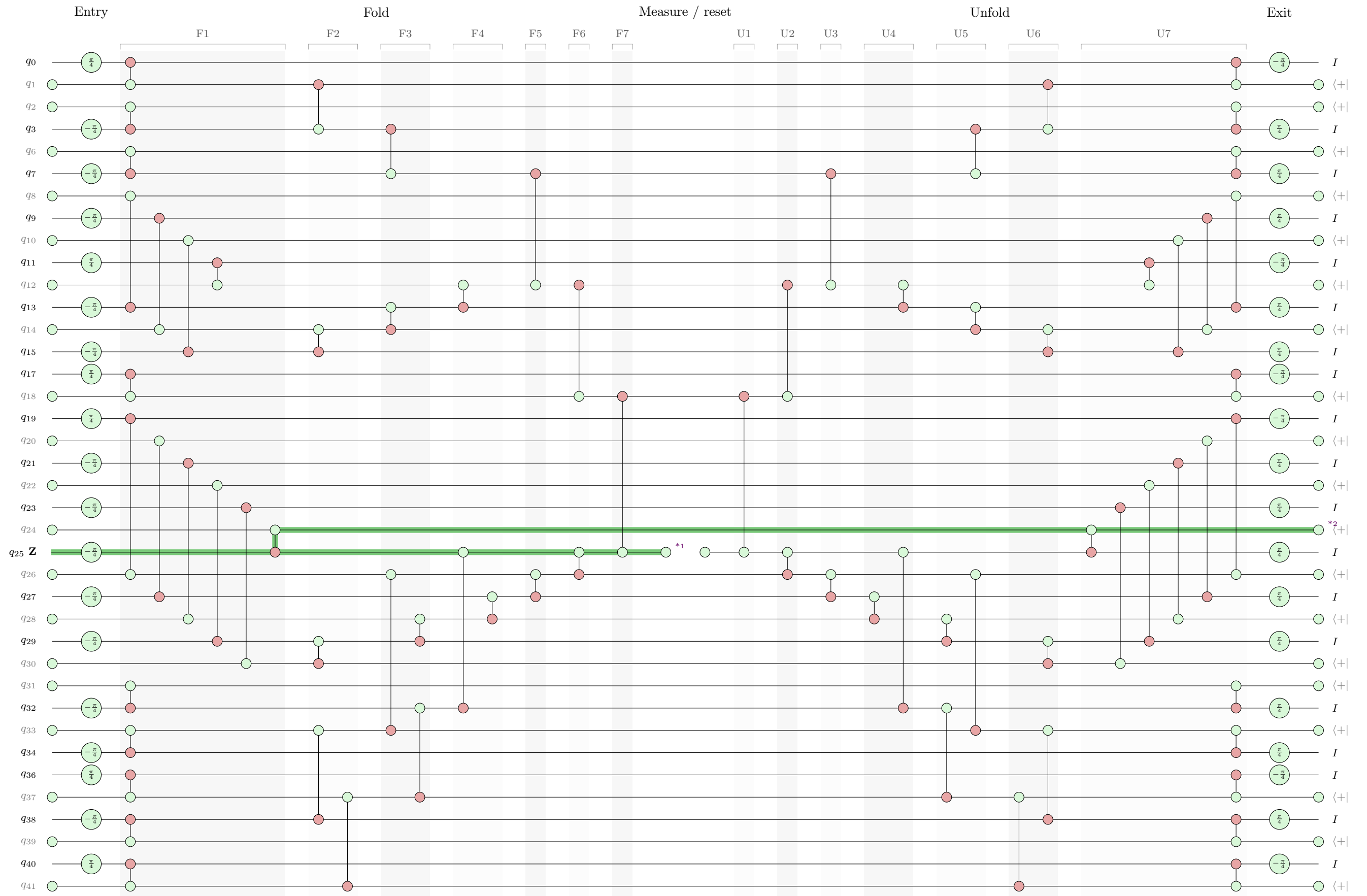
$$P_{\text{in}} = X_{25}, \quad P_{\text{out}} = X_0 X_3 X_7 X_9 X_{11} X_{13} X_{15} X_{17} X_{19} X_{21} X_{23} X_{27} X_{29} X_{32} X_{34} X_{36} X_{38} X_{40}, \quad \lambda = e^{i\frac{5\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K\delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}; \quad \delta_2 = \frac{\pi}{2}; \quad \delta_3 = -\frac{\pi}{2}; \quad \delta_4 = -\frac{\pi}{2}; \quad \delta_5 = -\frac{\pi}{2}; \quad \delta_6 = \frac{\pi}{2}; \quad \delta_7 = -\frac{\pi}{2}; \quad \delta_8 = -\frac{\pi}{2}; \quad \delta_9 = \frac{\pi}{2}; \quad \delta_{10} = \frac{\pi}{2}; \quad \delta_{11} = -\frac{\pi}{2}; \quad \delta_{12} = -\frac{\pi}{2}; \quad \delta_{13} = -\frac{\pi}{2}; \quad \delta_{14} = -\frac{\pi}{2}; \quad \delta_{15} = -\frac{\pi}{2}; \quad \delta_{16} = -\frac{\pi}{2}; \quad \delta_{17} = \frac{\pi}{2}; \quad \delta_{18} = -\frac{\pi}{2}; \quad \delta_{19} = \frac{\pi}{2}.$

d5_023.tikz

$d = 5$: incoming Z_{25}

Incoming Paulis. 2 phase defects.



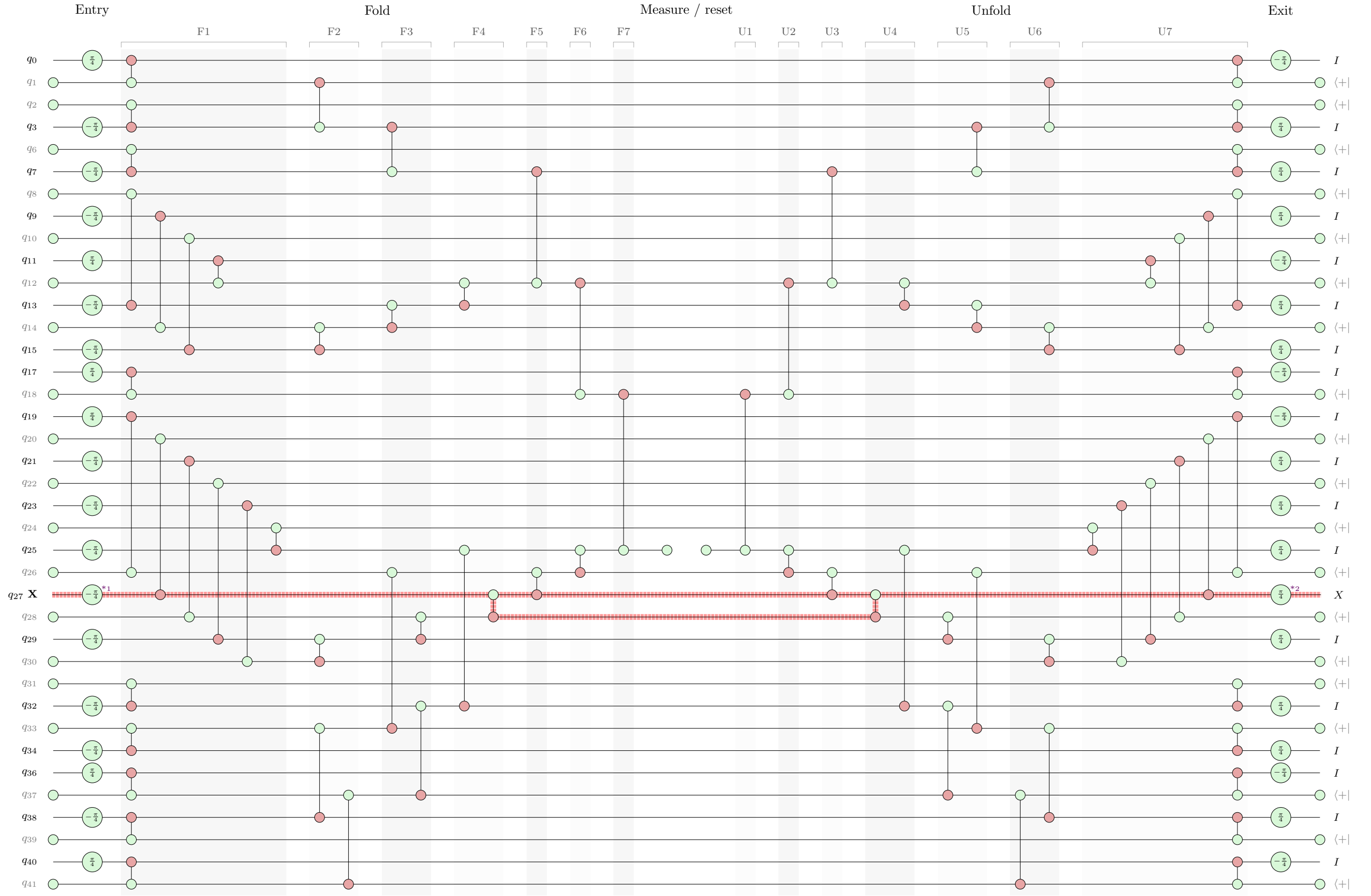
$$P_{\text{in}} = Z_{25}, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_024.tikz

$d = 5$: incoming X_{27}

Incoming Paulis. 2 phase defects.



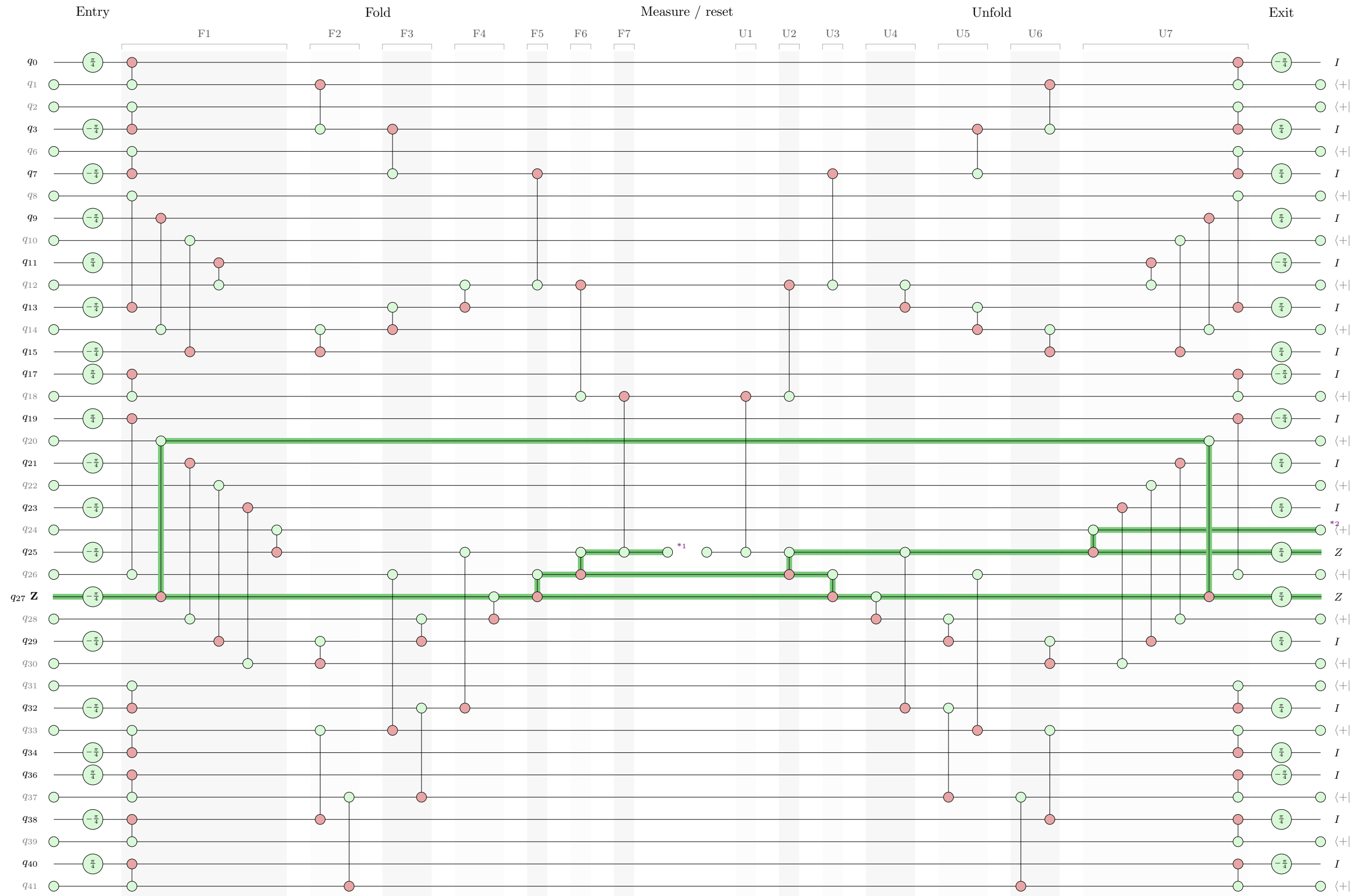
$$P_{\text{in}} = X_{27}, \quad P_{\text{out}} = X_{27}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_025.tikz

$d = 5$: incoming Z_{27}

Incoming Paulis. 2 phase defects.



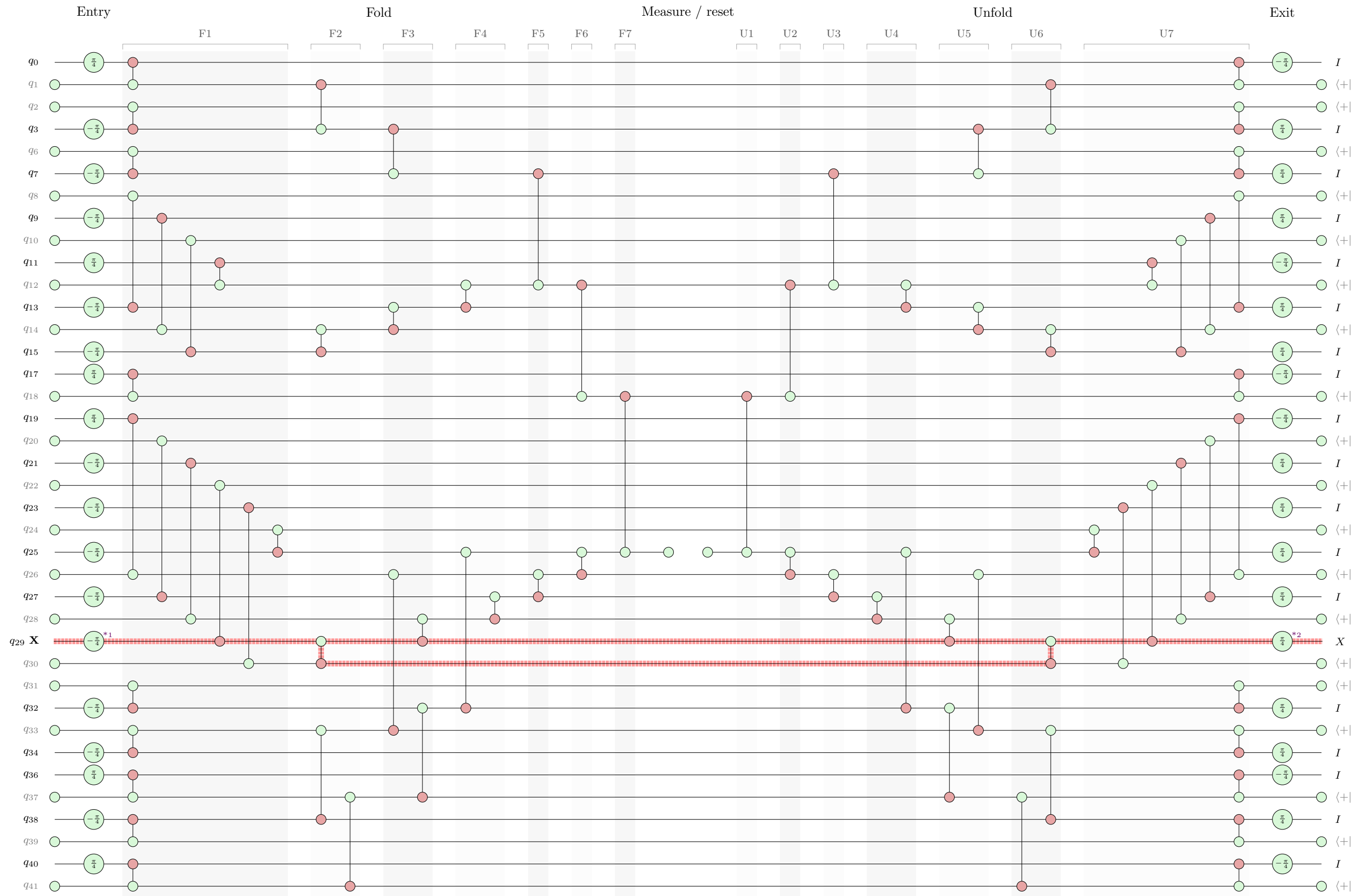
$$P_{\text{in}} = Z_{27}, \quad P_{\text{out}} = Z_{25}Z_{27}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_026.tikz

$d = 5$: incoming X_{29}

Incoming Paulis. 2 phase defects.



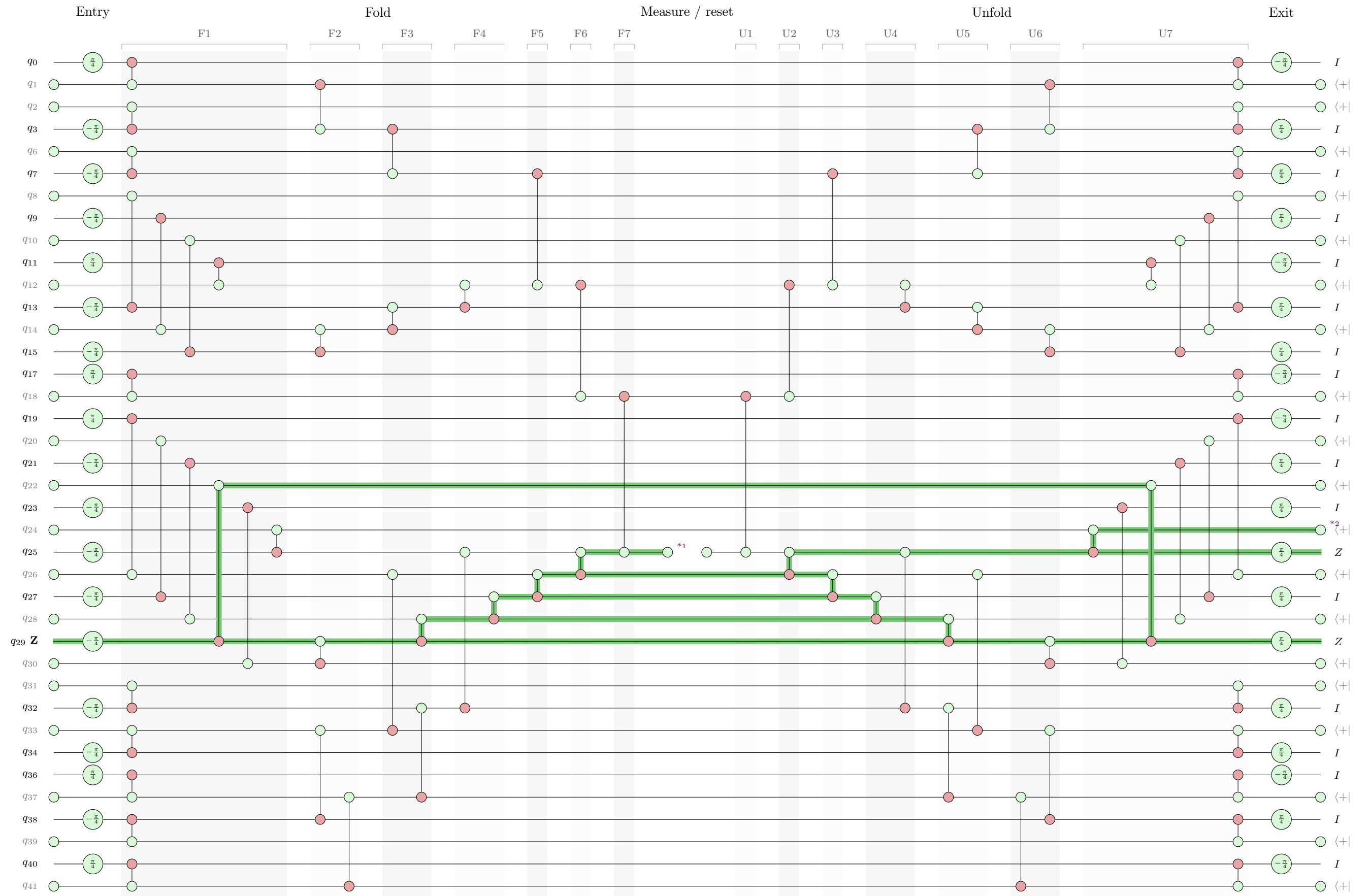
$$P_{\text{in}} = X_{29}, \quad P_{\text{out}} = X_{29}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_027.tikz

$d = 5$: incoming Z_{29}

Incoming Paulis. 2 phase defects.



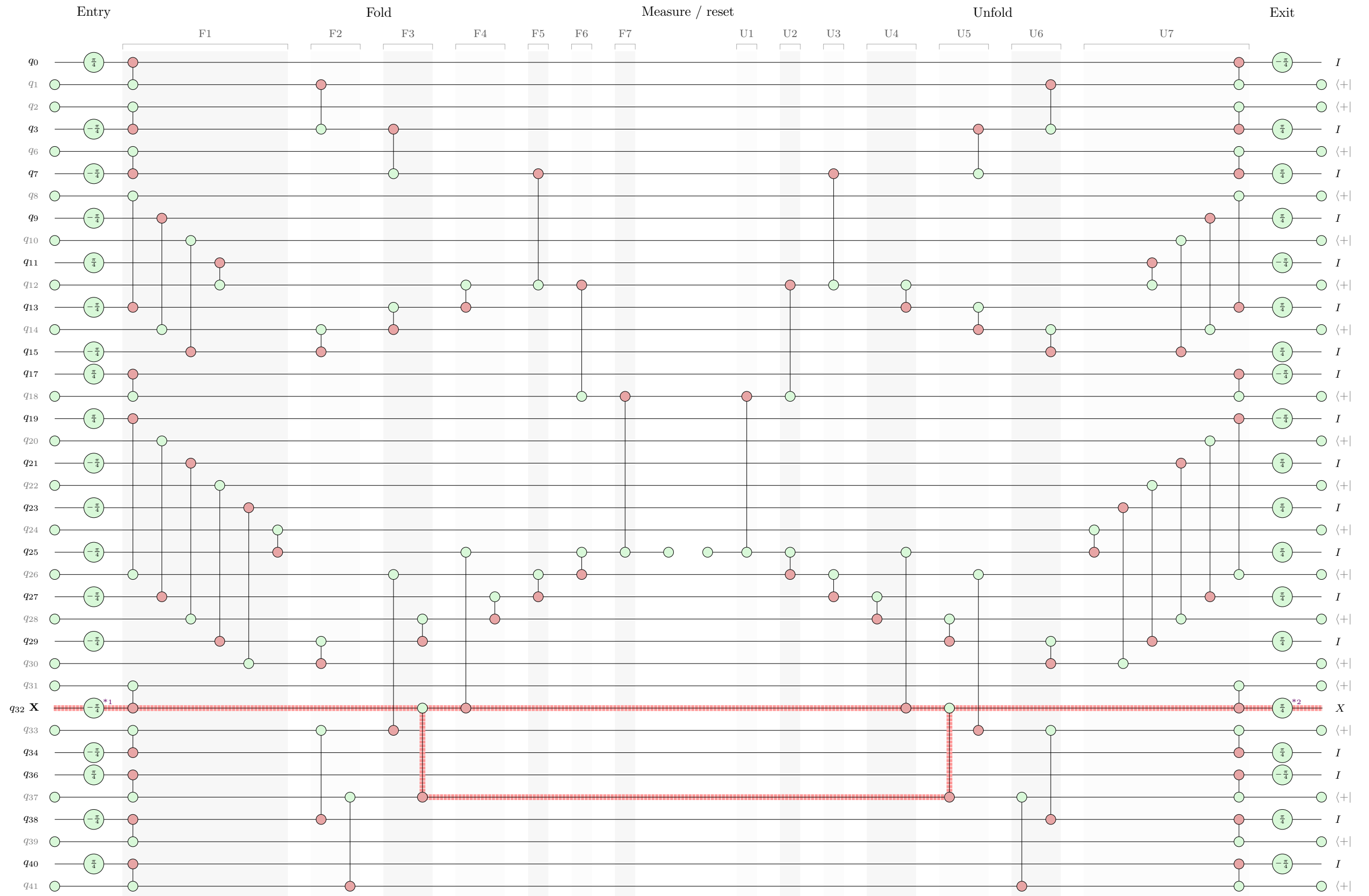
$$P_{\text{in}} = Z_{29}, \quad P_{\text{out}} = Z_{25}Z_{29}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_028.tikz

$d = 5$: incoming X_{32}

Incoming Paulis. 2 phase defects.



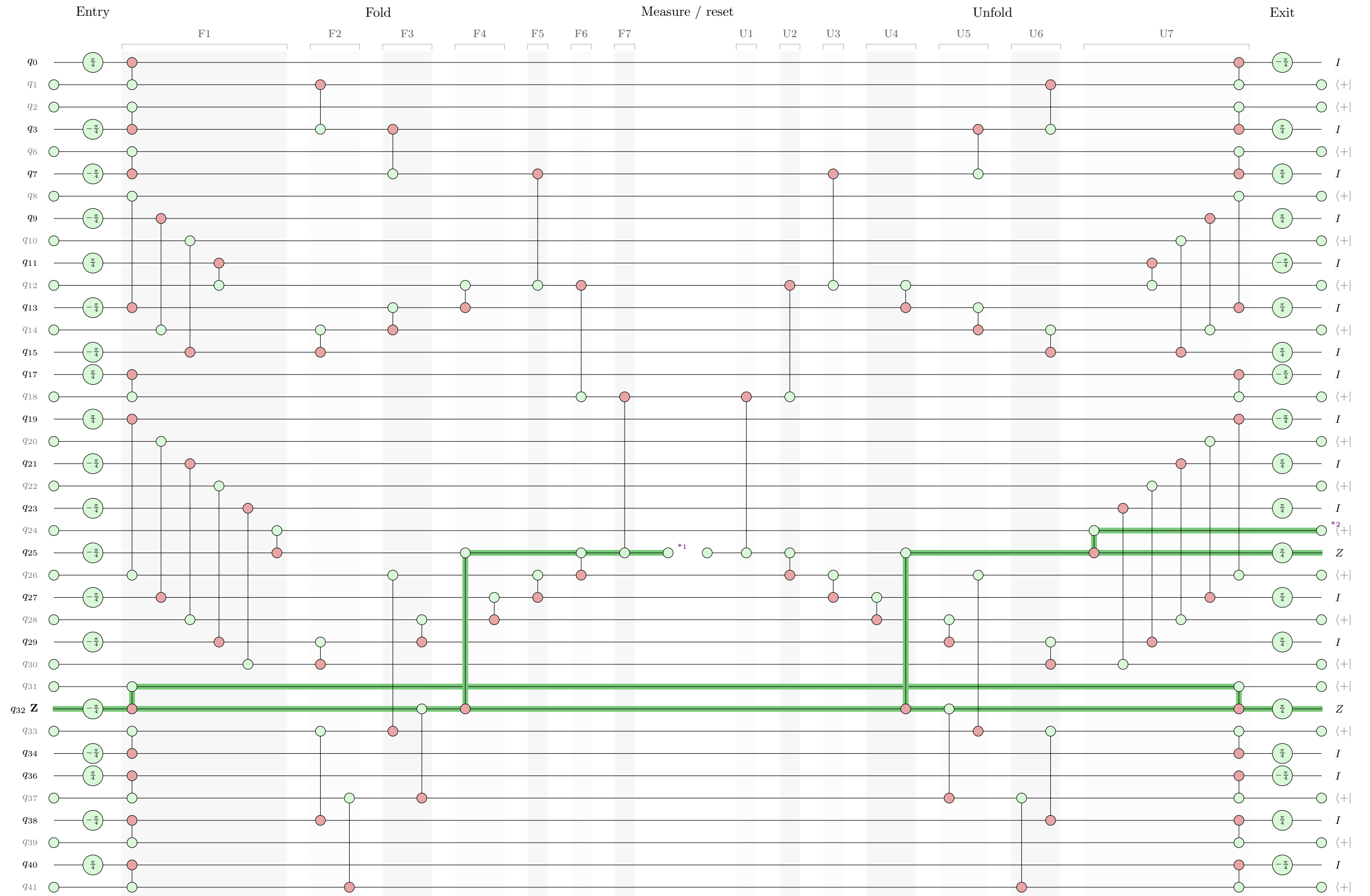
$$P_{\text{in}} = X_{32}, \quad P_{\text{out}} = X_{32}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_029.tikz

$d = 5$: incoming Z_{32}

Incoming Paulis. 2 phase defects.



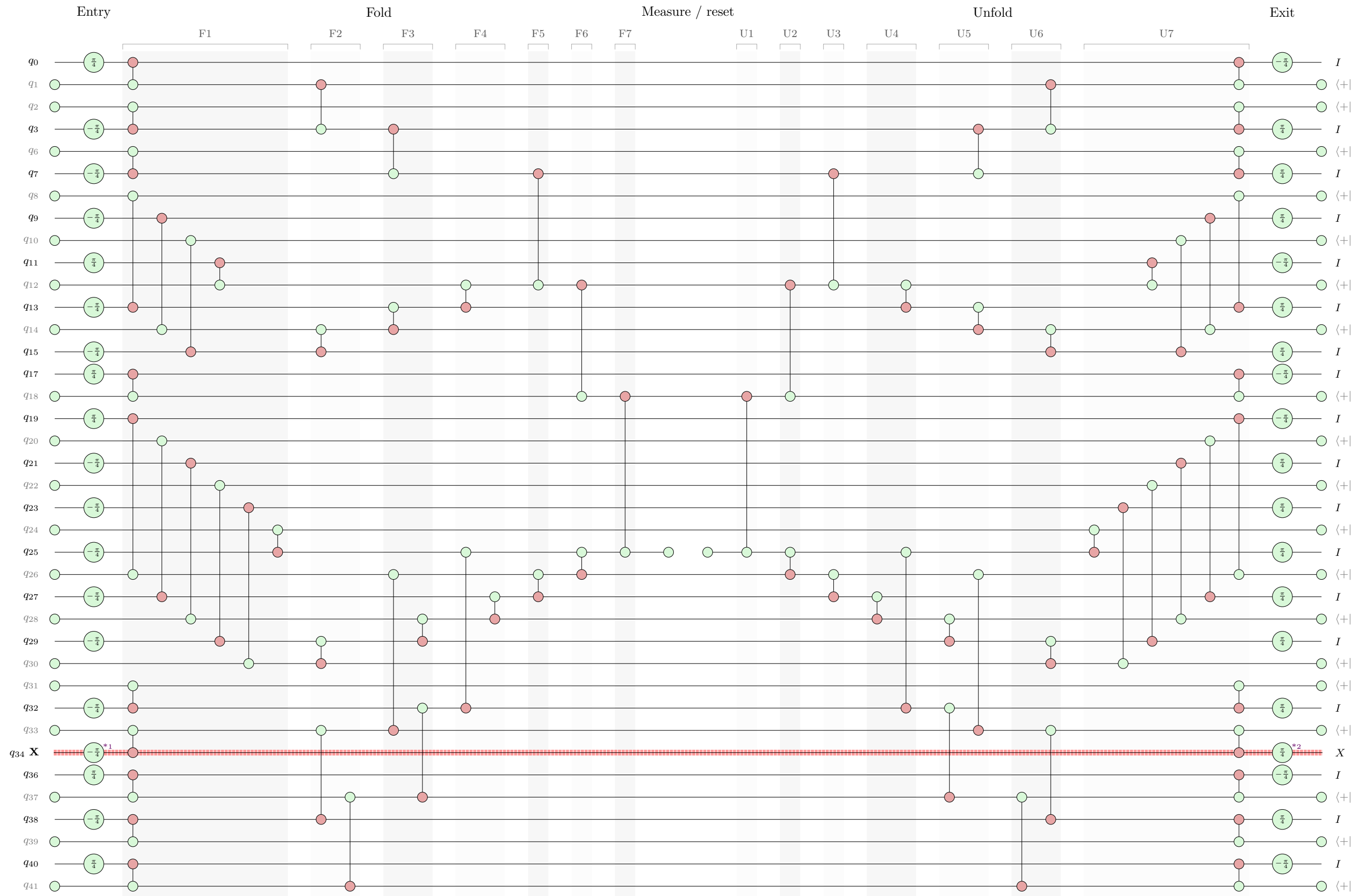
$$P_{\text{in}} = Z_{32}, \quad P_{\text{out}} = Z_{25}Z_{32}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_030.tikz

$d = 5$: incoming X_{34}

Incoming Paulis. 2 phase defects.

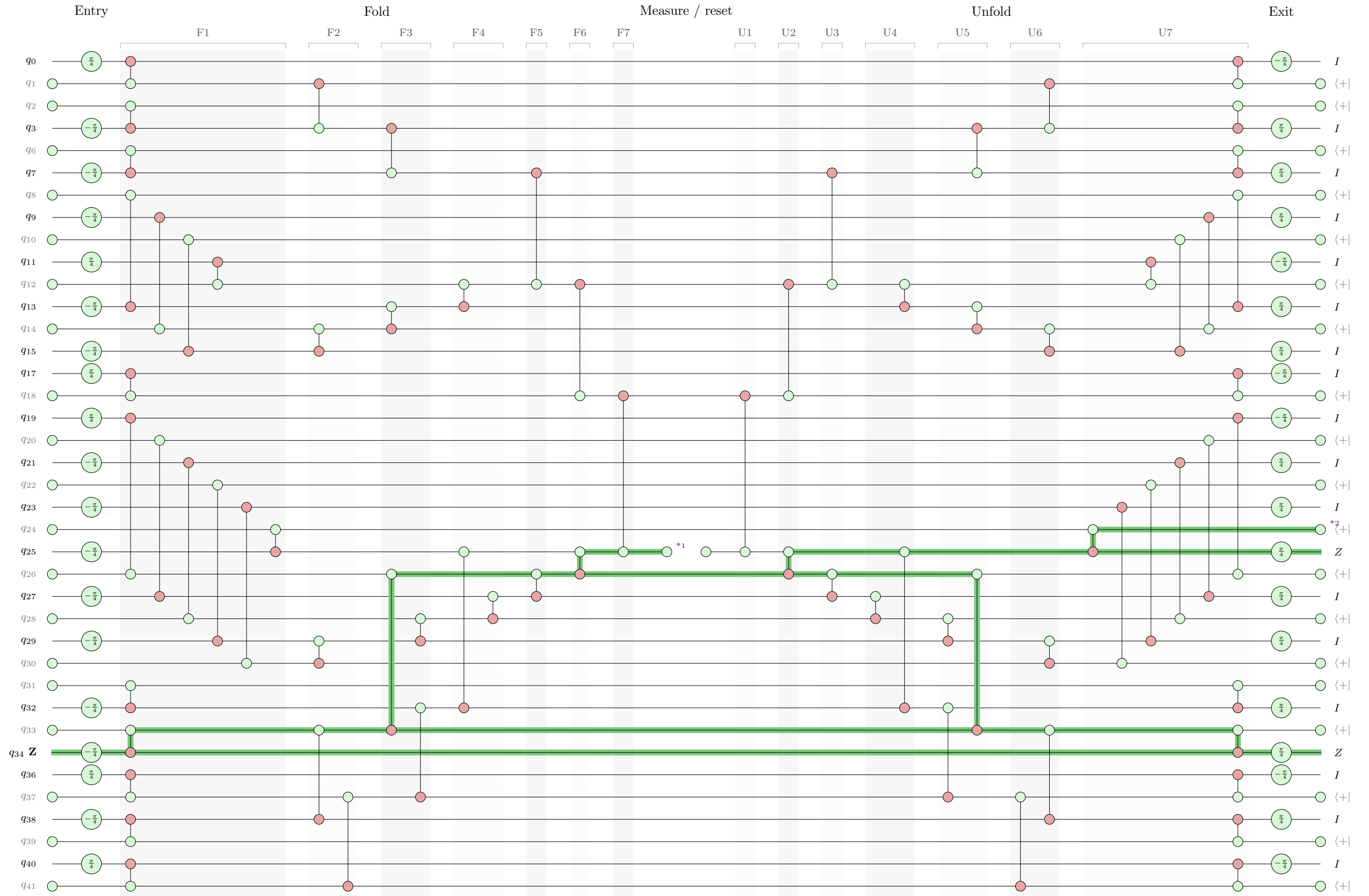


$$P_{\text{in}} = X_{34}, \quad P_{\text{out}} = X_{34}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_031.tikz

$d = 5$: incoming Z_{34}
Incoming Paulis. 2 phase defects.



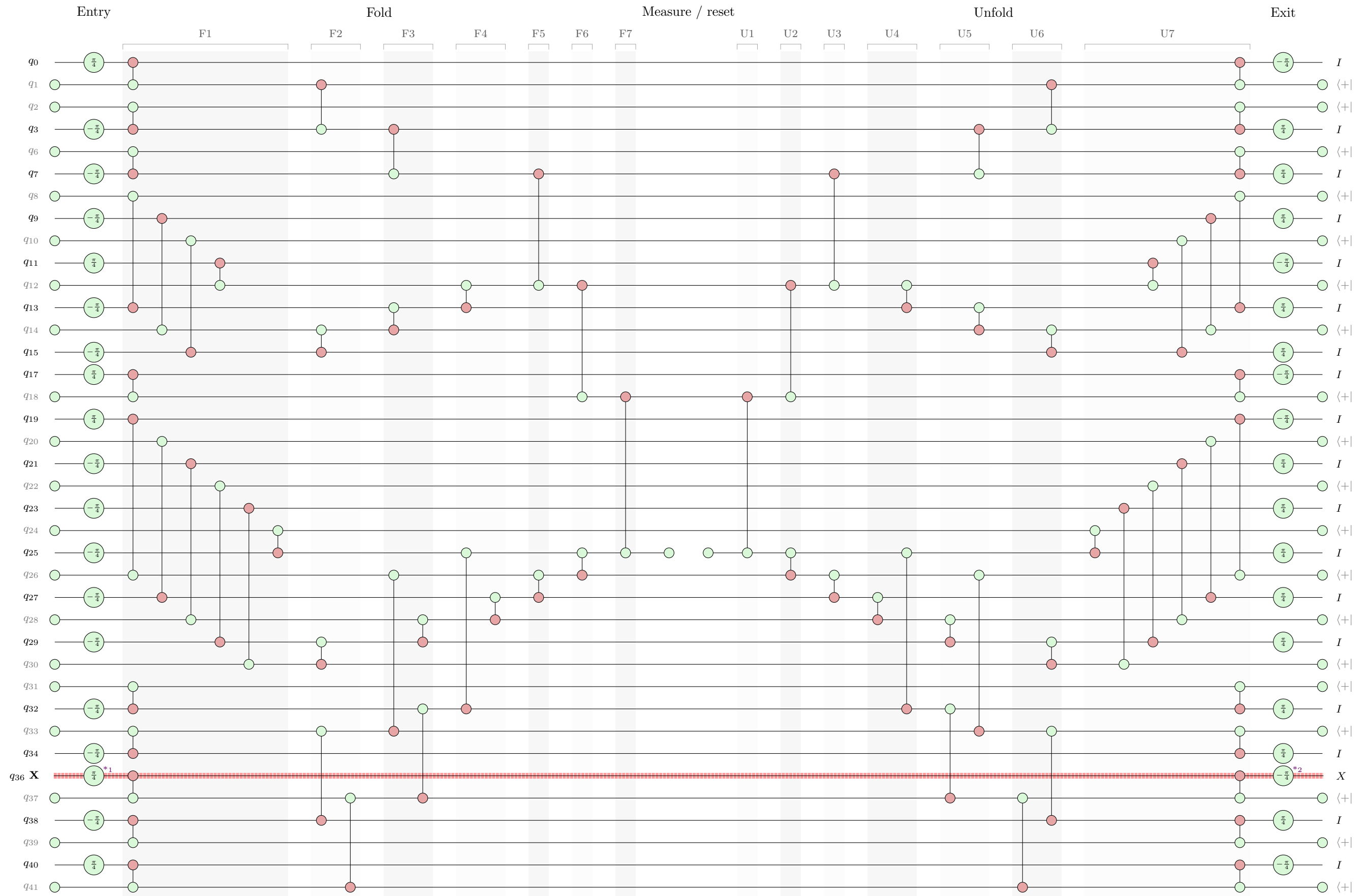
$$P_{\text{in}} = Z_{34}, \quad P_{\text{out}} = Z_{25}Z_{34}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_032.tikz

$d = 5$: incoming X_{36}

Incoming Paulis. 2 phase defects.



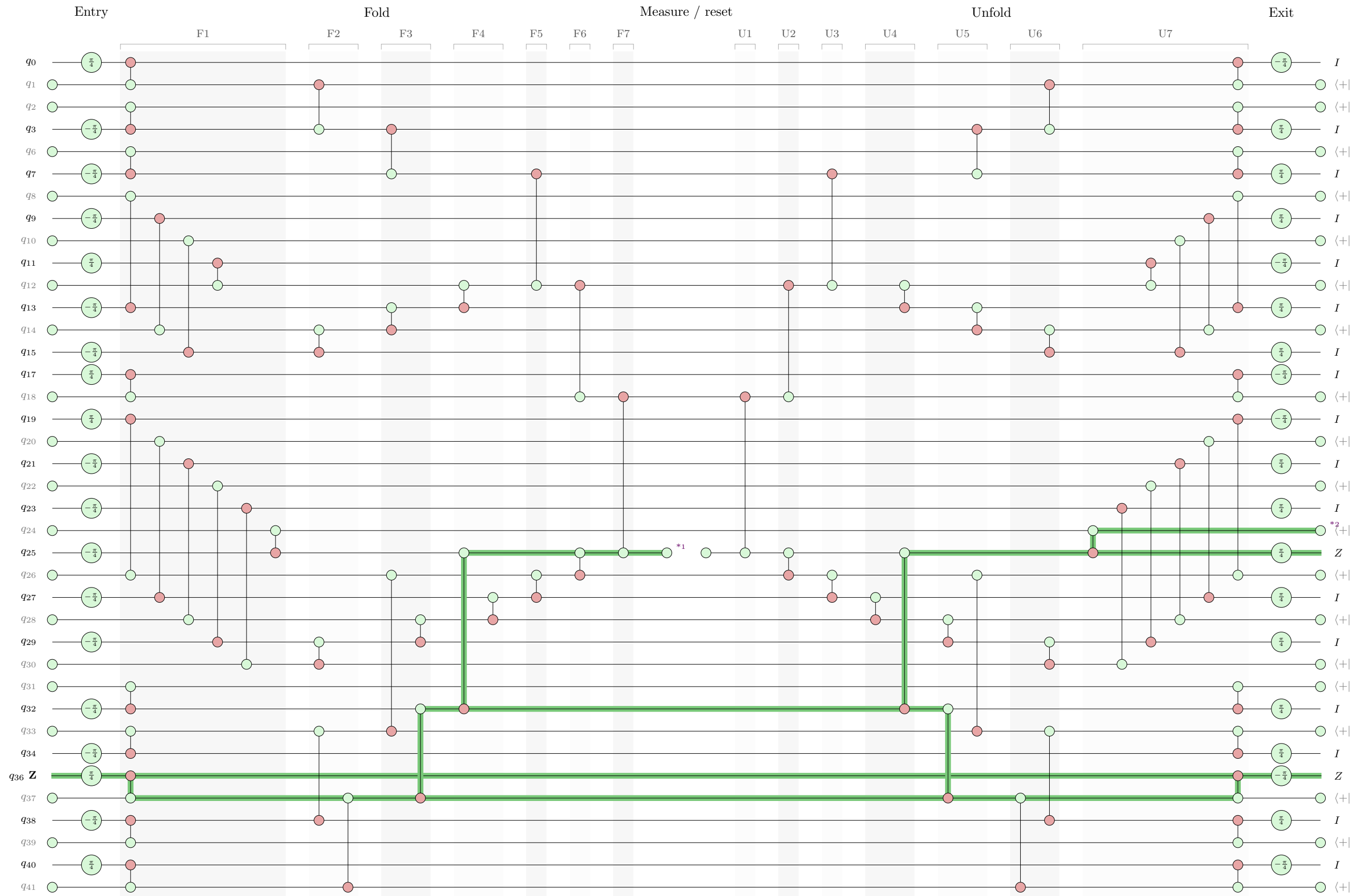
$$P_{\text{in}} = X_{36}, \quad P_{\text{out}} = X_{36}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = -\frac{\pi}{2}$; $\delta_2 = \frac{\pi}{2}$.

d5_033.tikz

$d = 5$: incoming Z_{36}

Incoming Paulis. 2 phase defects.



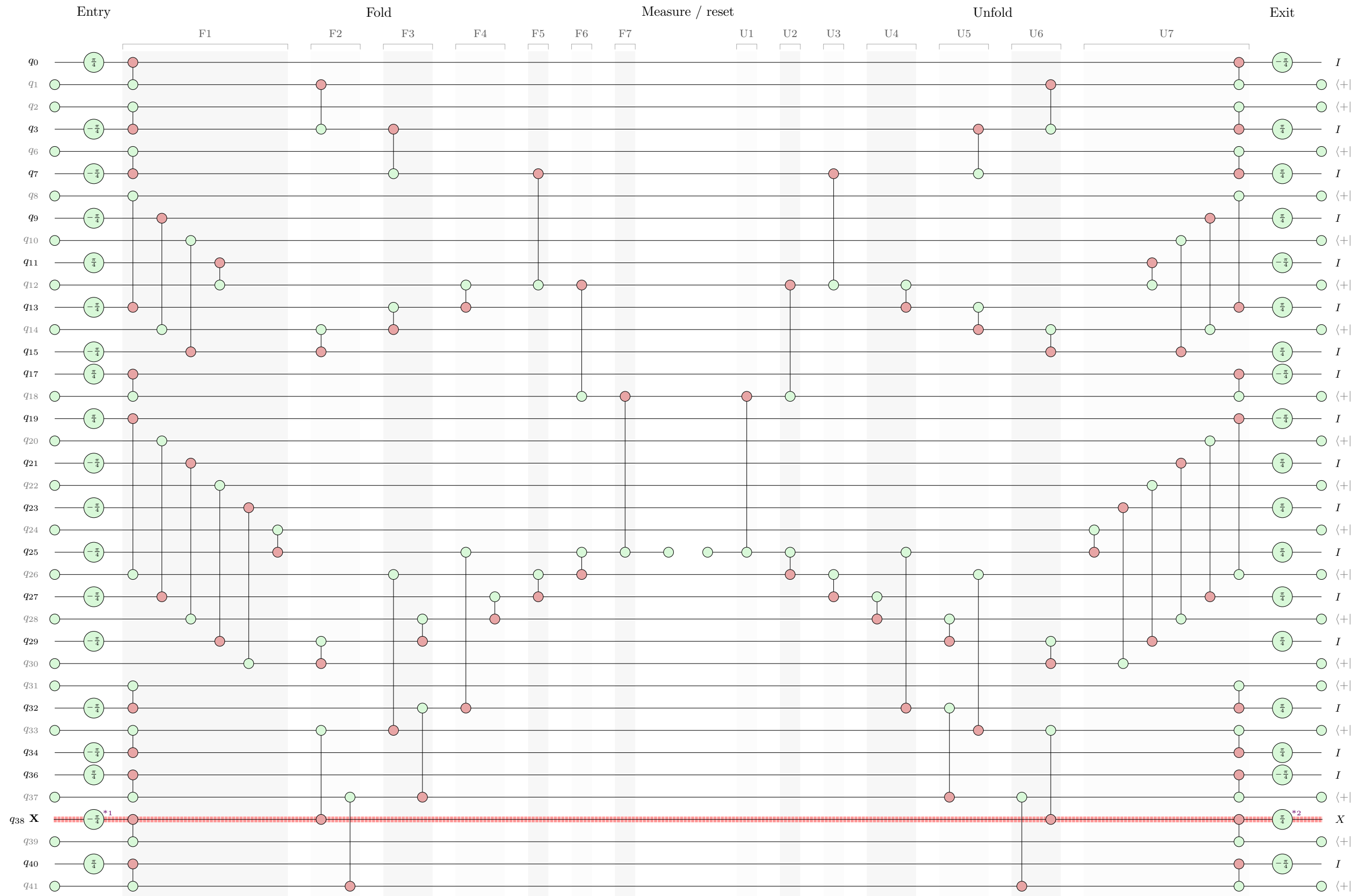
$$P_{\text{in}} = Z_{36}, \quad P_{\text{out}} = Z_{25}Z_{36}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_034.tikz

$d = 5$: incoming X_{38}

Incoming Paulis. 2 phase defects.



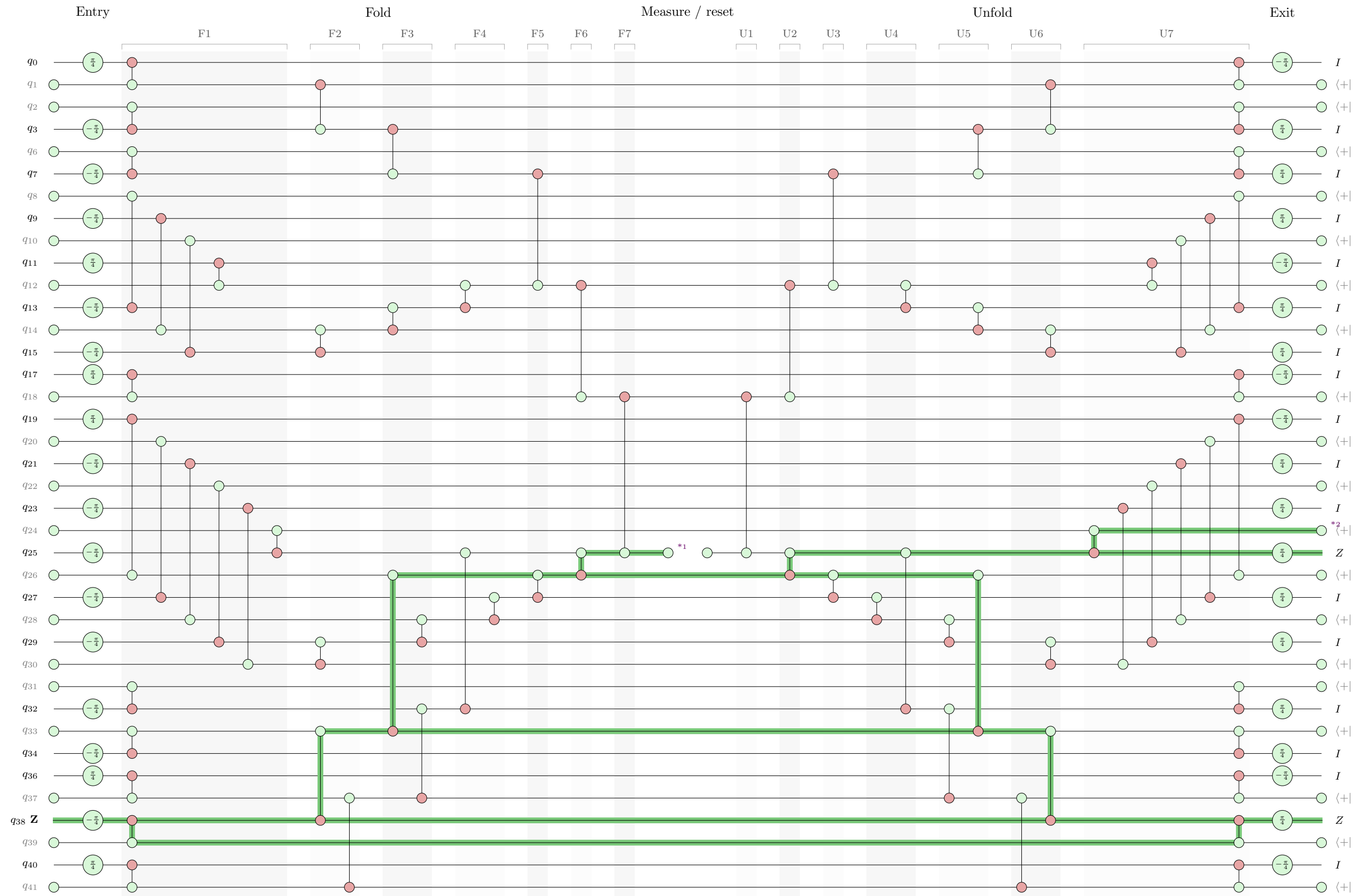
$$P_{\text{in}} = X_{38}, \quad P_{\text{out}} = X_{38}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \frac{\pi}{2}$; $\delta_2 = -\frac{\pi}{2}$.

d5_035.tikz

$d = 5$: incoming Z_{38}

Incoming Paulis. 2 phase defects.

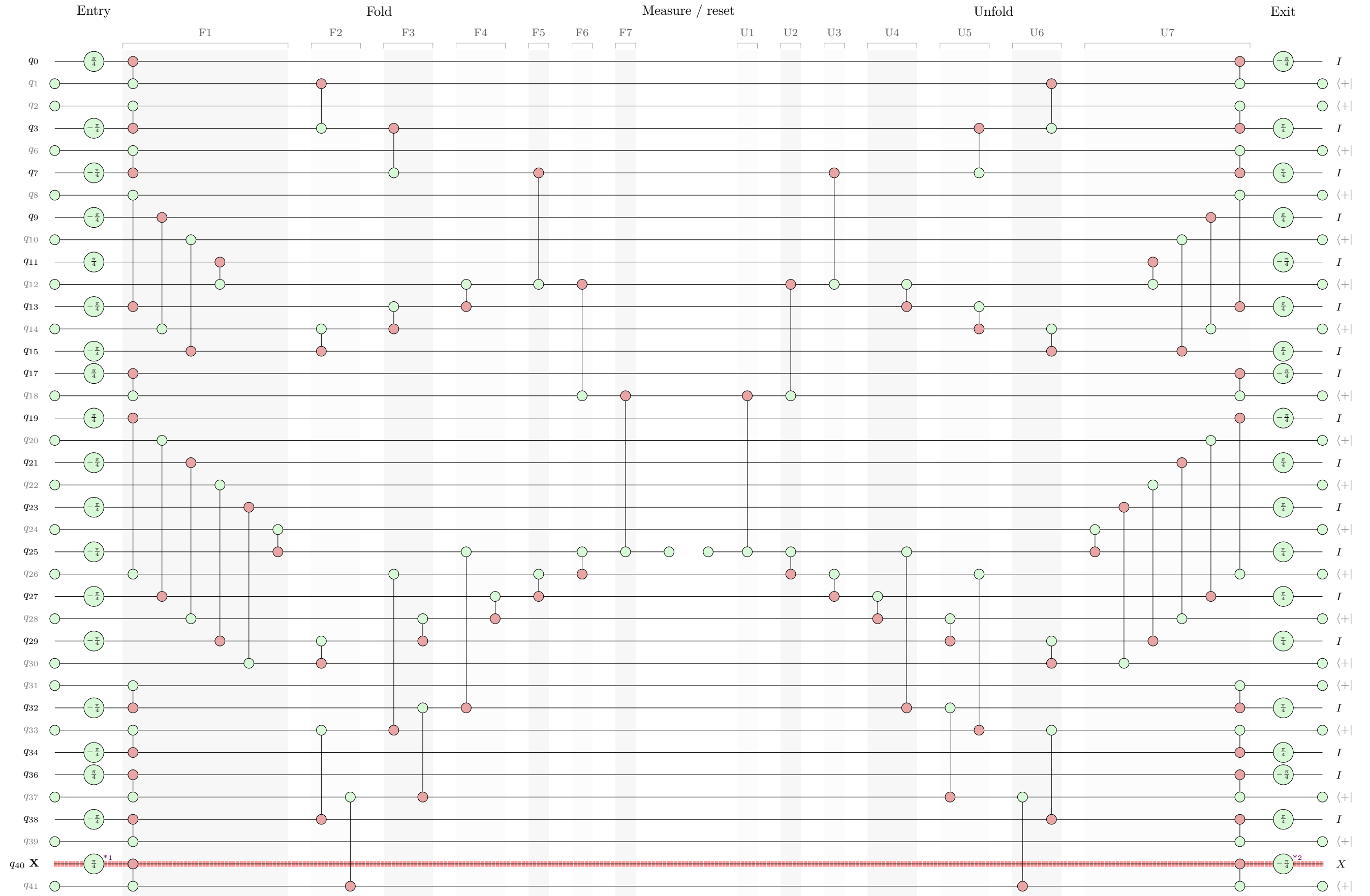


$$P_{\text{in}} = Z_{38}, \quad P_{\text{out}} = Z_{25}Z_{38}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_036.tikz

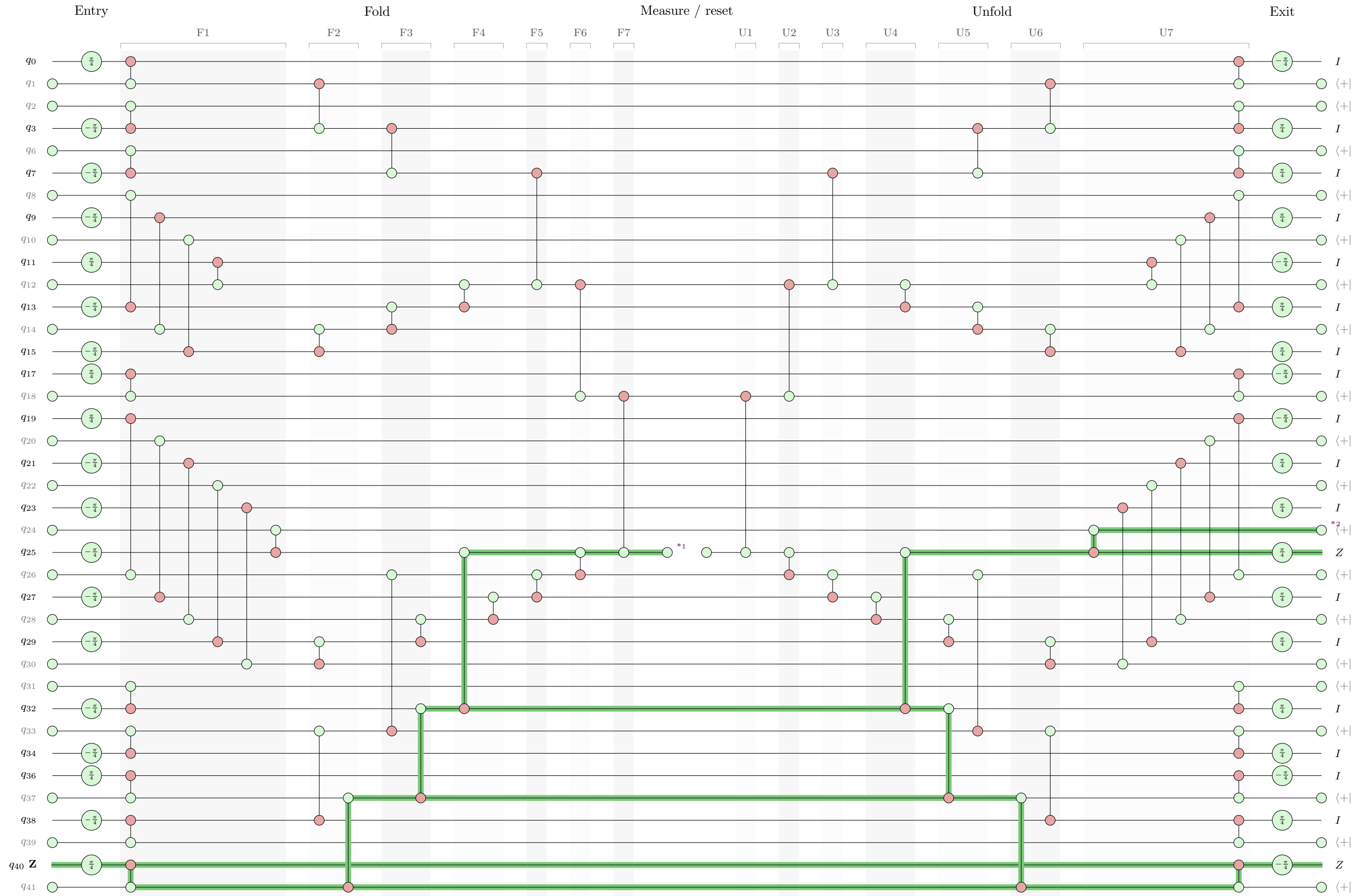
$d = 5$: incoming X_{40}
Incoming Paulis. 2 phase defects.



$$P_{\text{in}} = X_{40}, \quad P_{\text{out}} = X_{40}, \quad \lambda = 1, \quad K P_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = -\frac{\pi}{2}$; $\delta_2 = \frac{\pi}{2}$.

$d = 5$: incoming Z_{40}
Incoming Paulis. 2 phase defects.

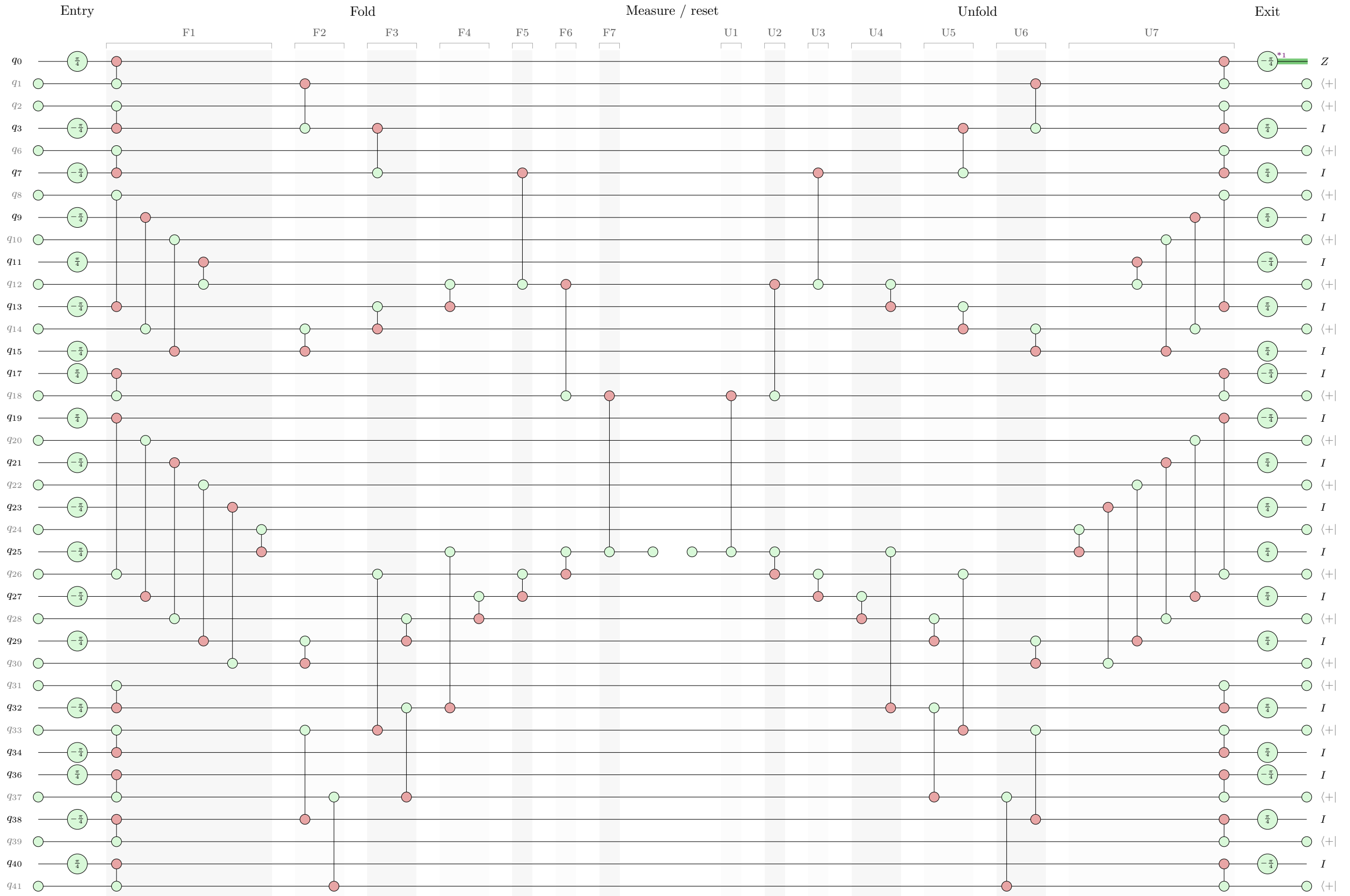


$$P_{\text{in}} = Z_{40}, \quad P_{\text{out}} = Z_{25}Z_{40}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}}K\delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: basis element 39

Additional output support. 1 phase defect.



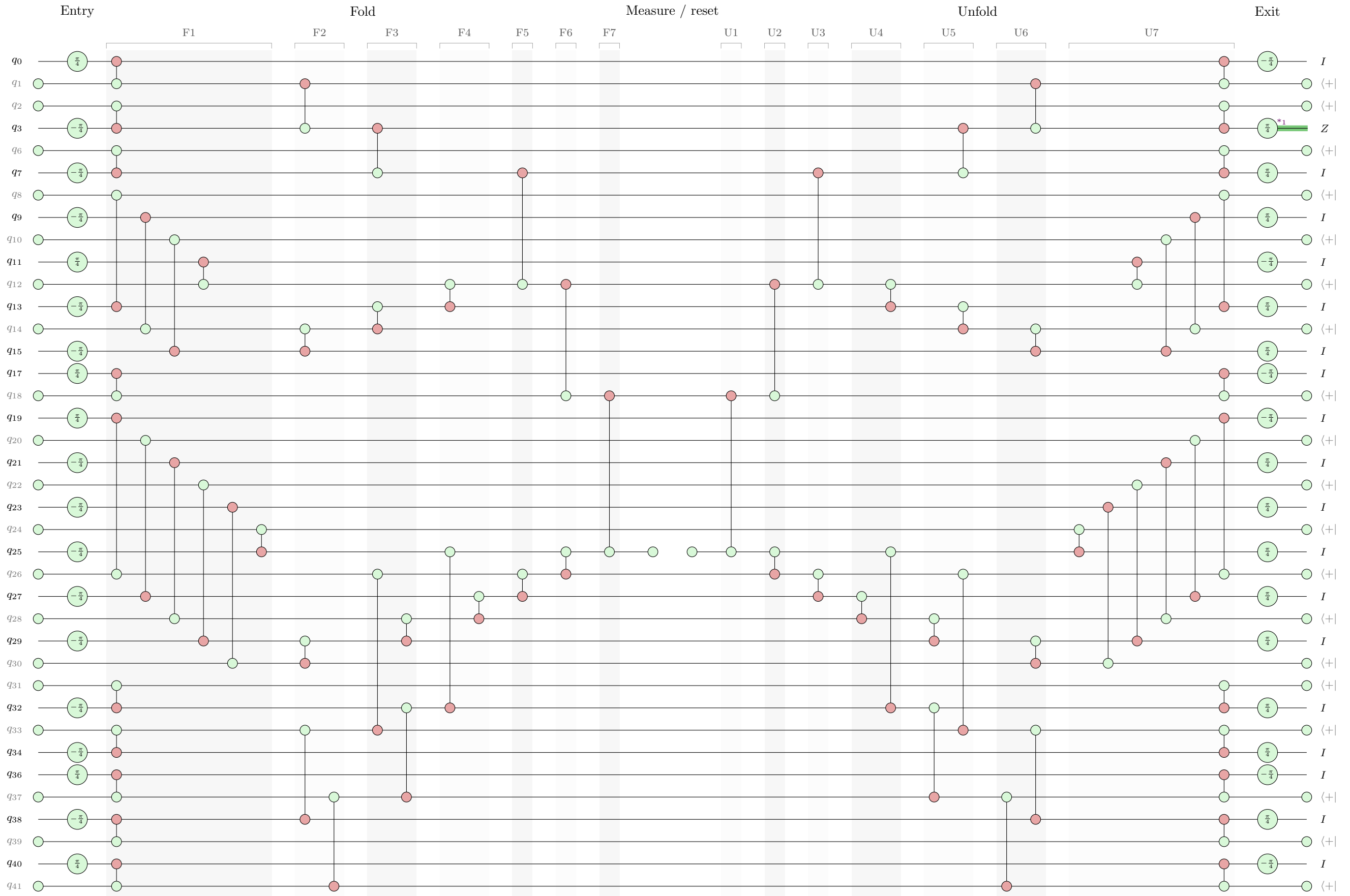
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_0, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$.

d5_039.tikz

$d = 5$: basis element 40

Additional output support. 1 phase defect.



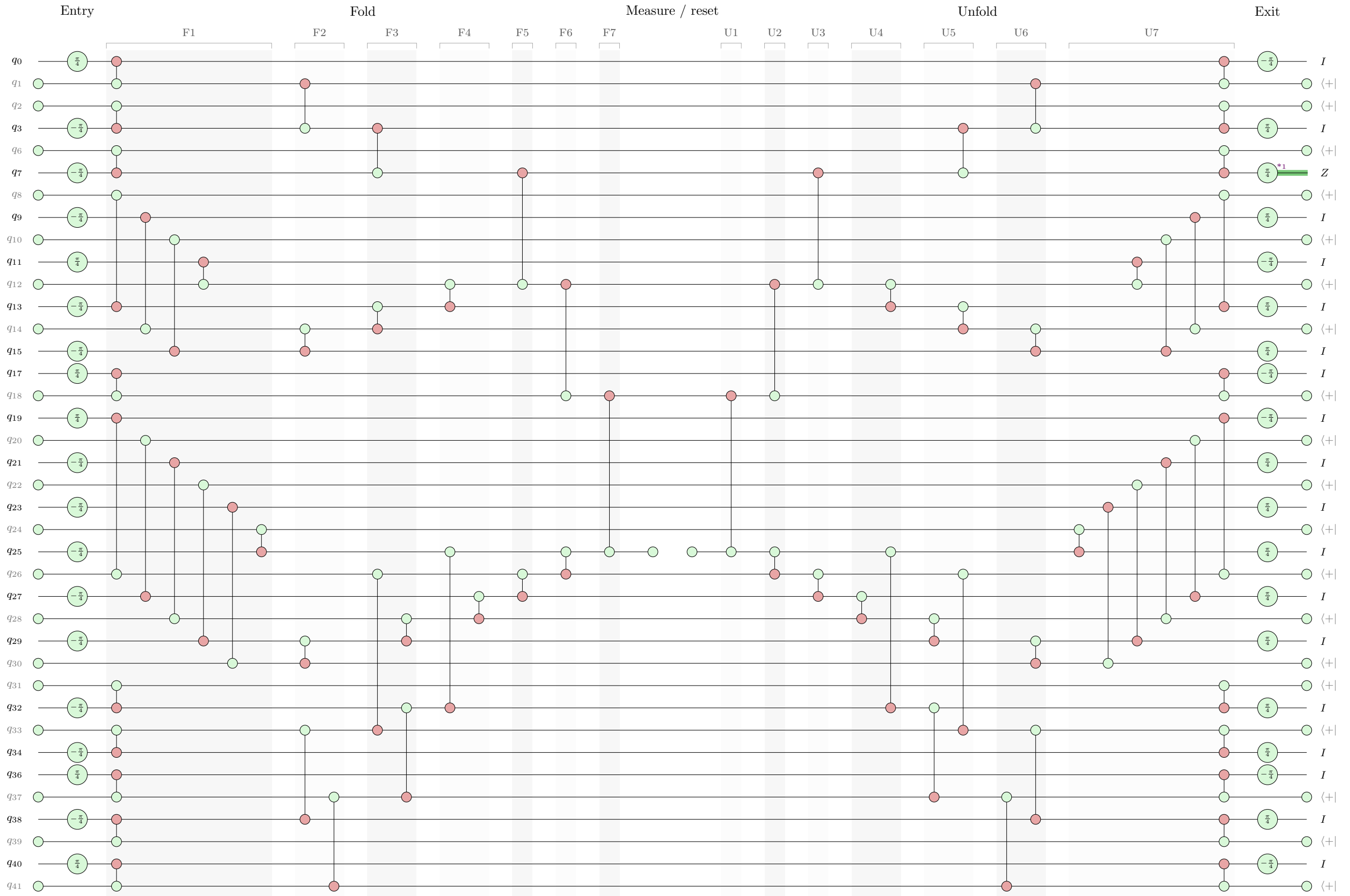
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_3, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$.

d5_040.tikz

$d = 5$: basis element 41

Additional output support. 1 phase defect.



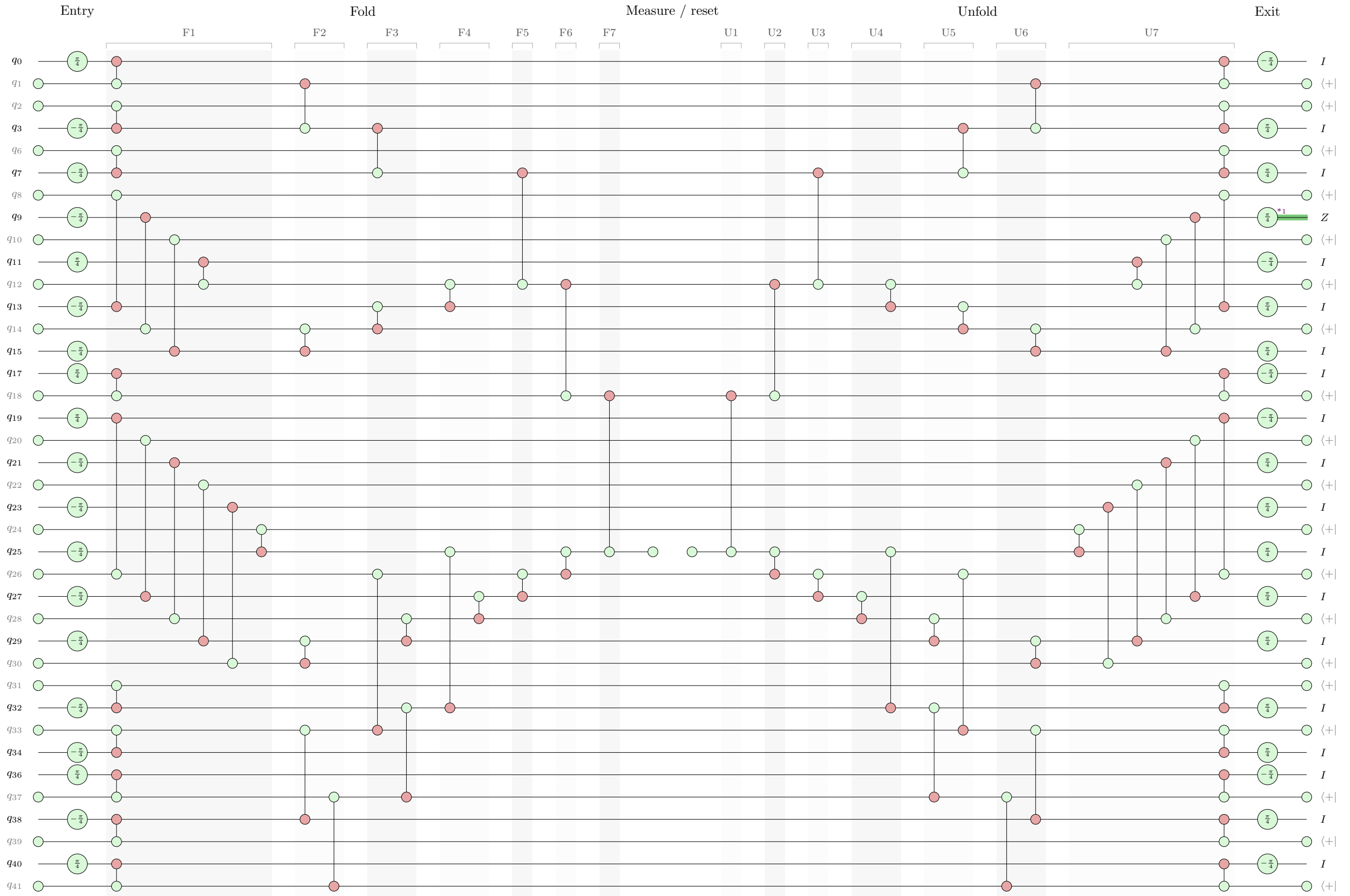
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_7, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$.

d5_041.tikz

$d = 5$: basis element 42

Additional output support. 1 phase defect.



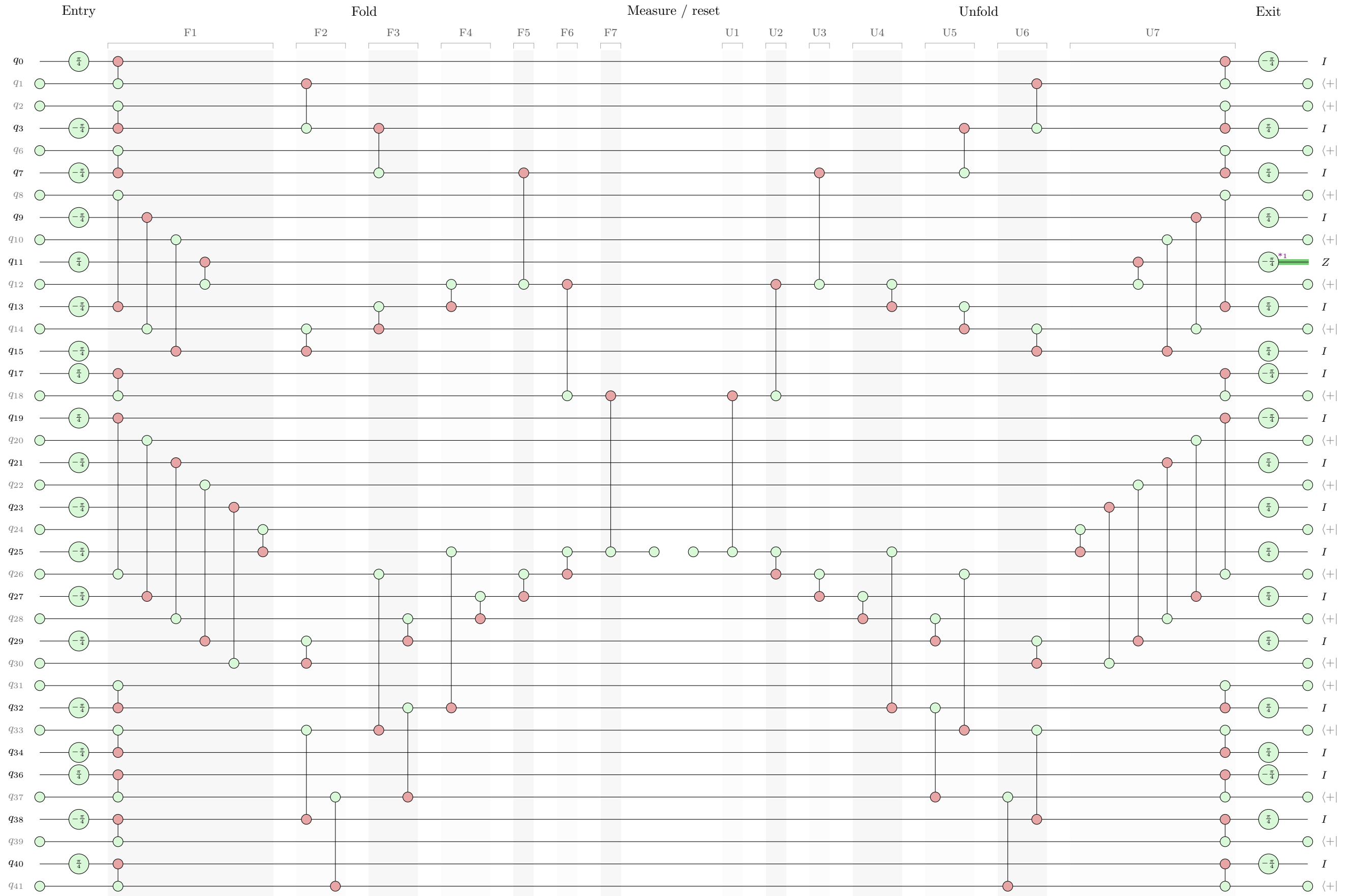
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_9, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$.

d5_042.tikz

$d = 5$: basis element 43

Additional output support. 1 phase defect.



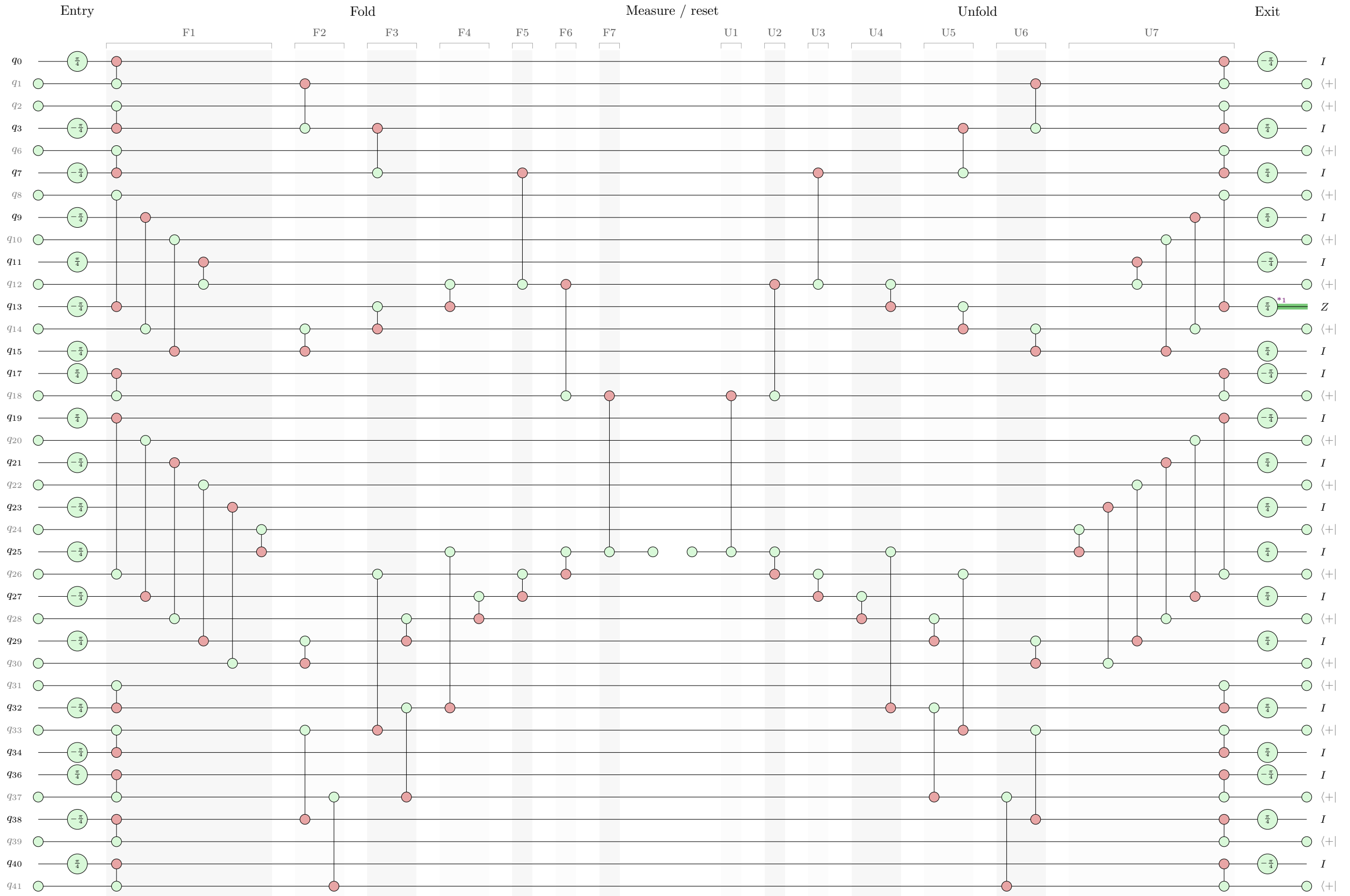
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{11}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_043.tikz

$d = 5$: basis element 44

Additional output support. 1 phase defect.



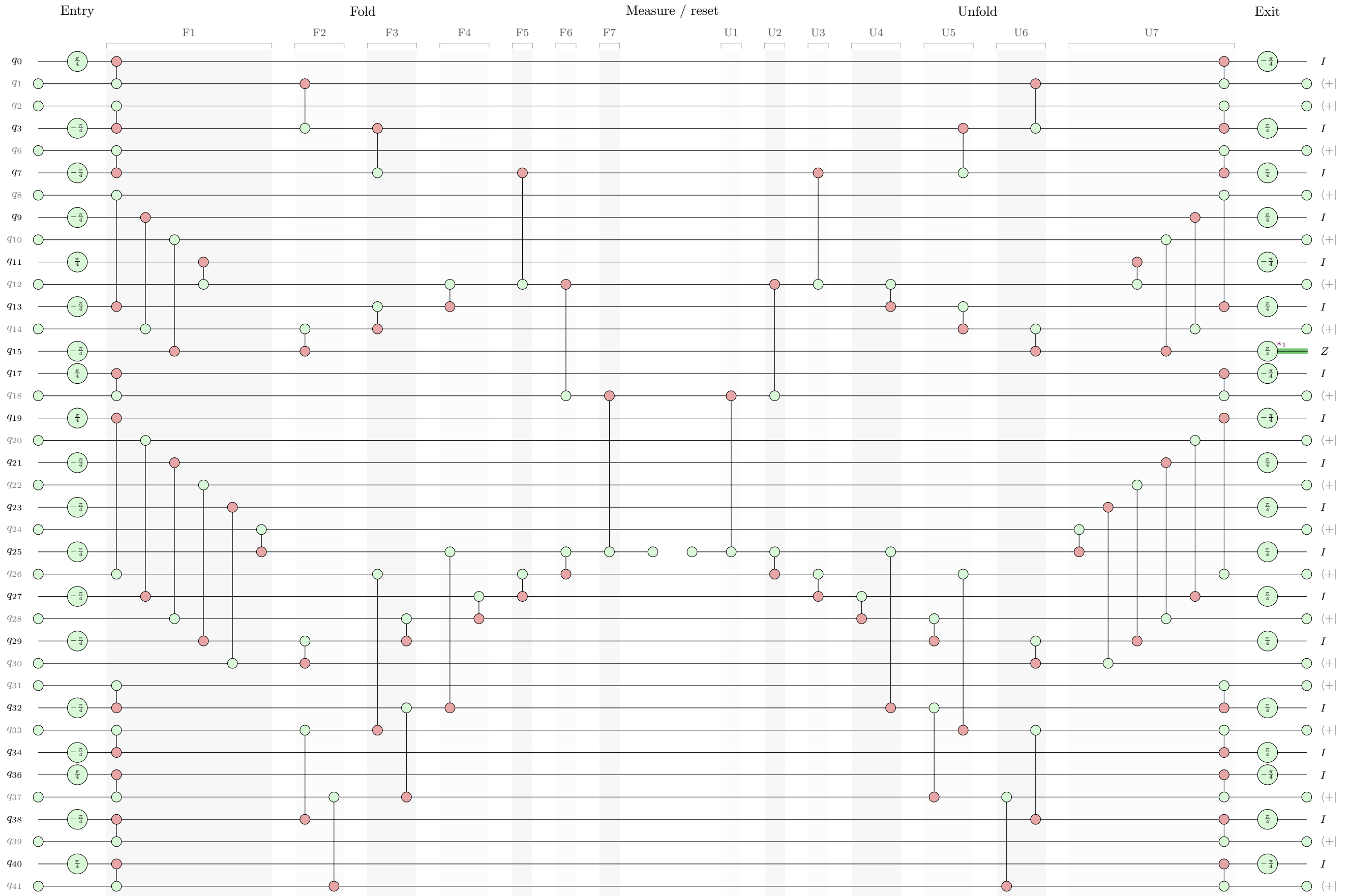
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{13}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_044.tikz

$d = 5$: basis element 45

Additional output support. 1 phase defect.



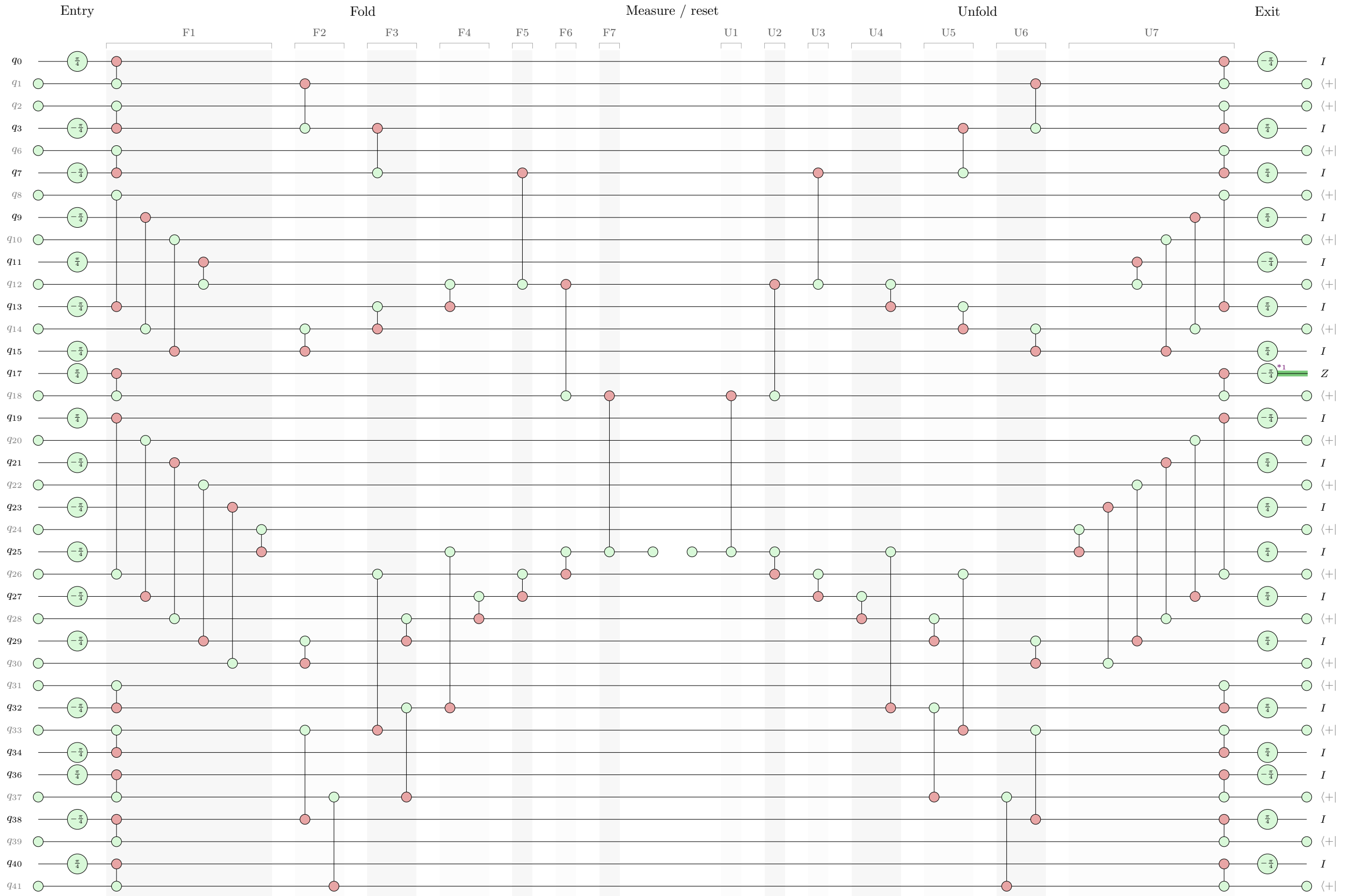
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{15}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_045.tikz

$d = 5$: basis element 46

Additional output support. 1 phase defect.



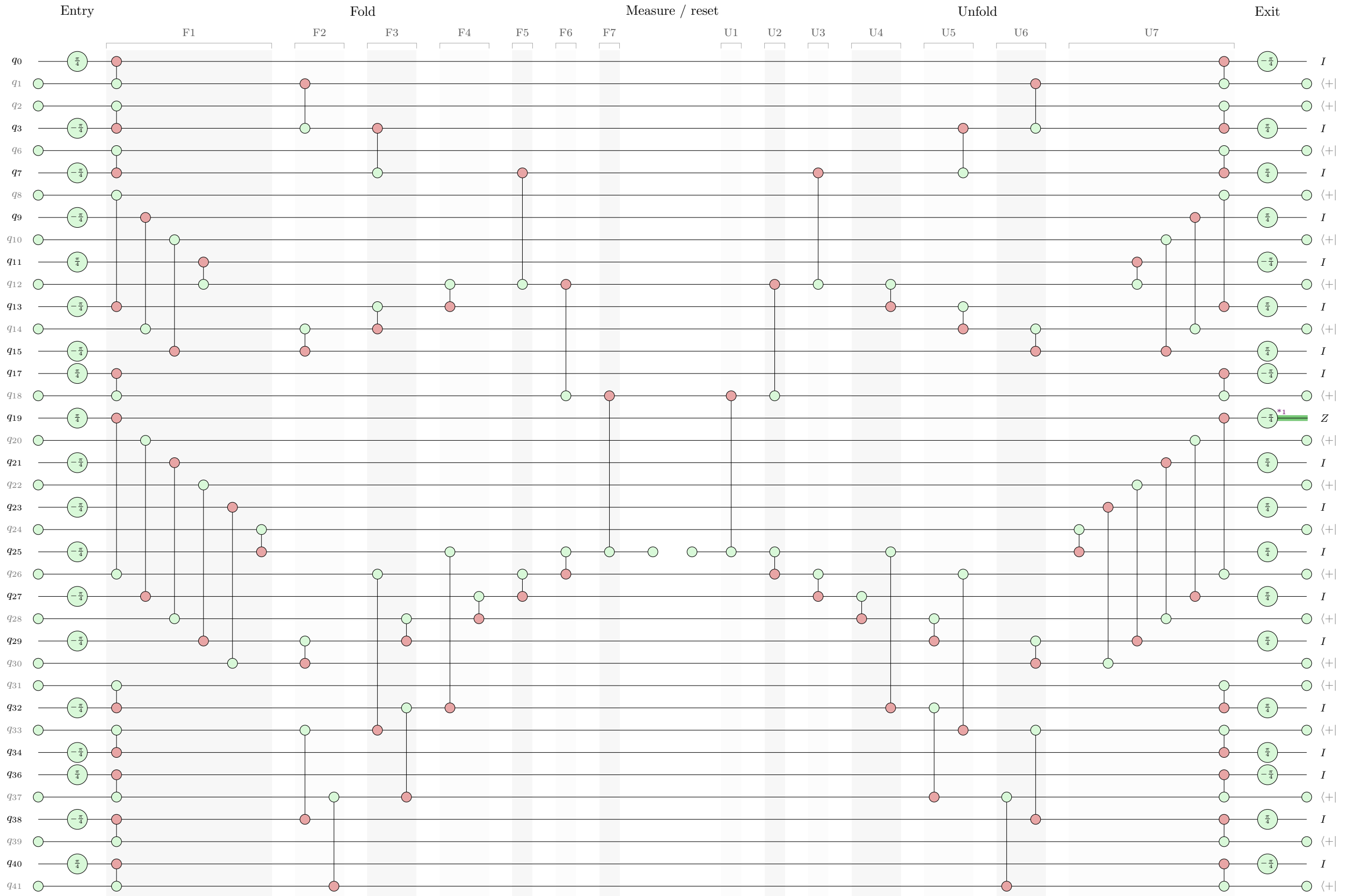
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{17}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_046.tikz

$d = 5$: basis element 47

Additional output support. 1 phase defect.



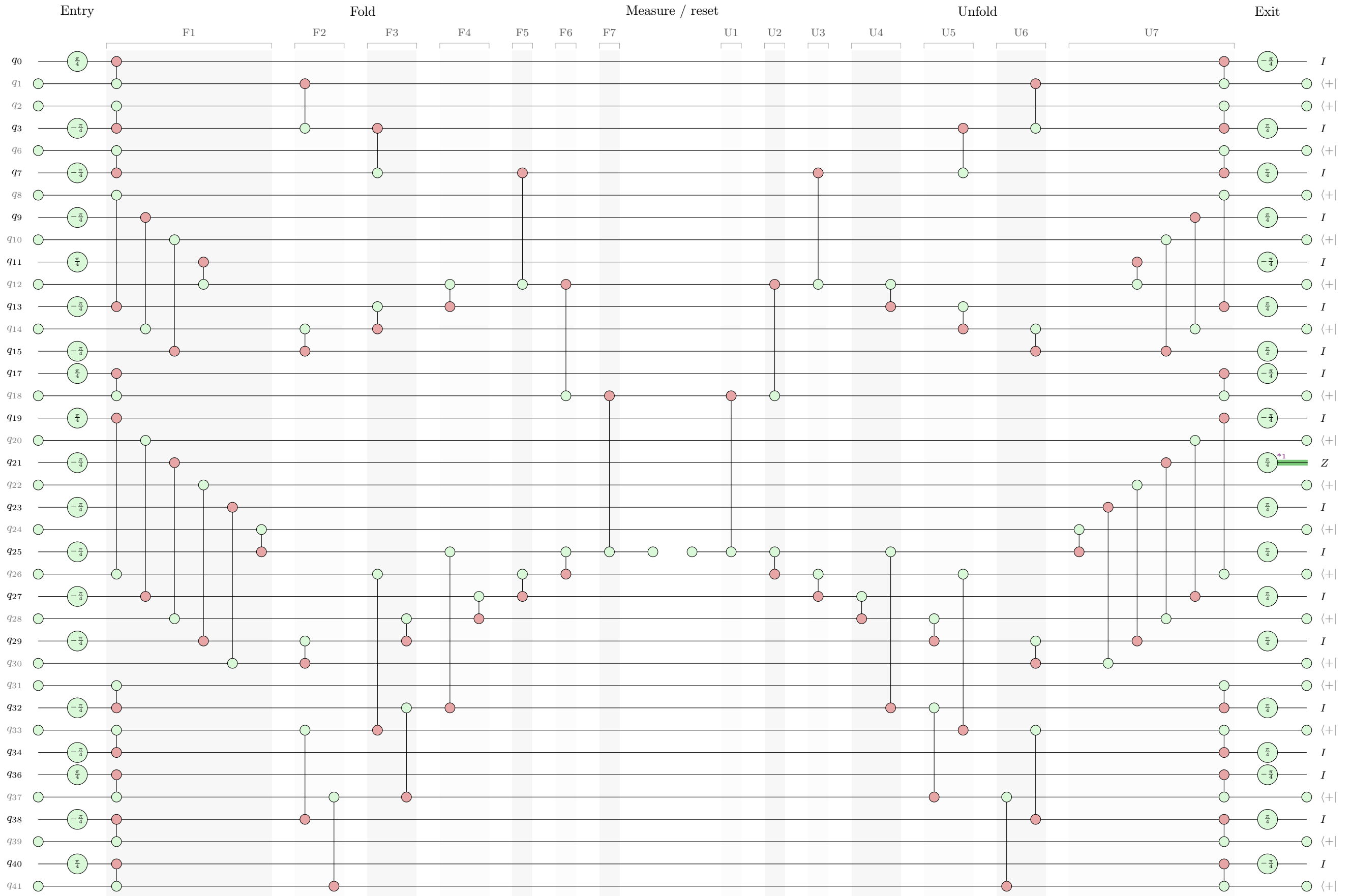
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{19}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_047.tikz

$d = 5$: basis element 48

Additional output support. 1 phase defect.



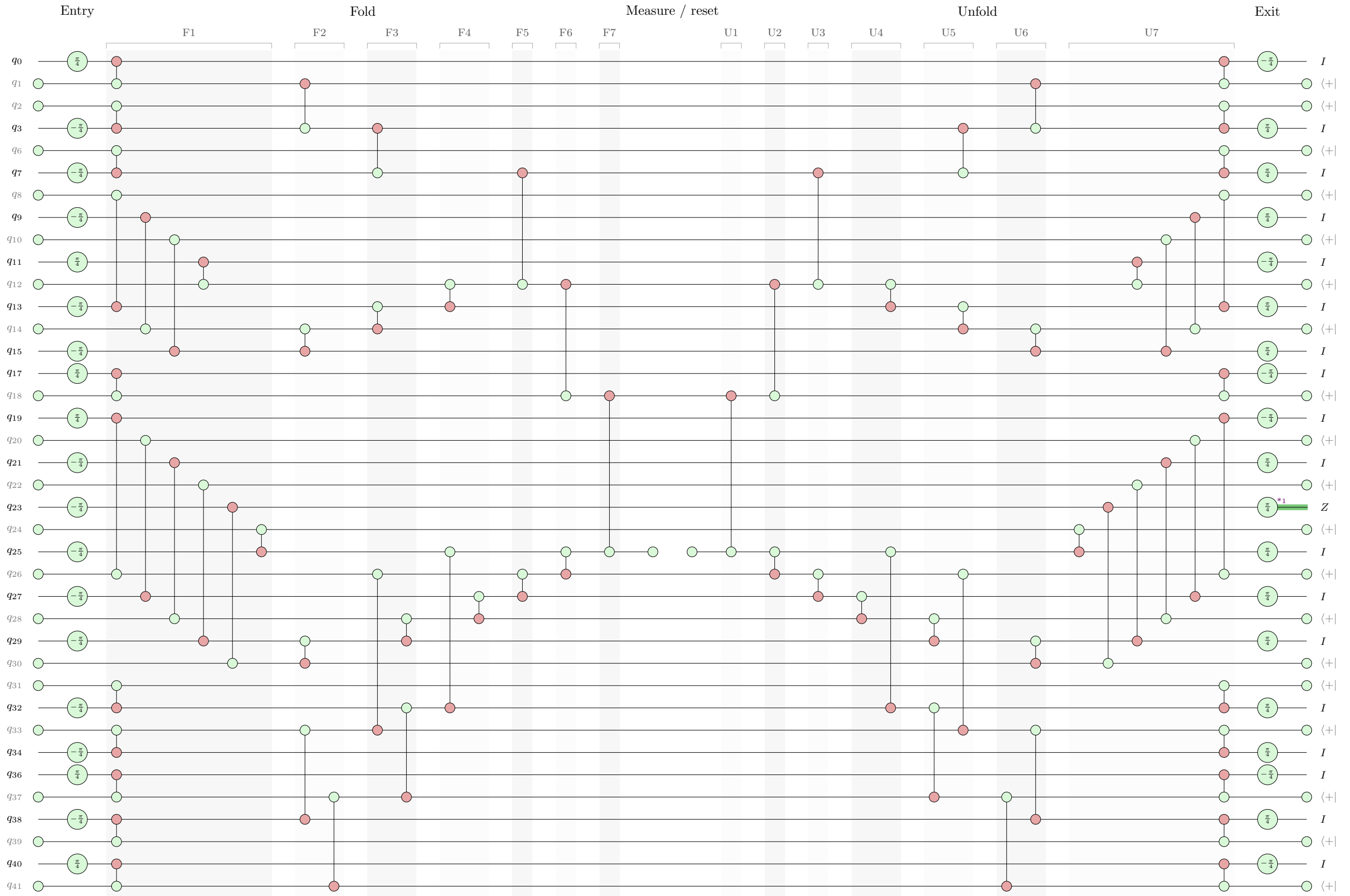
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{21}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_048.tikz

$d = 5$: basis element 49

Additional output support. 1 phase defect.



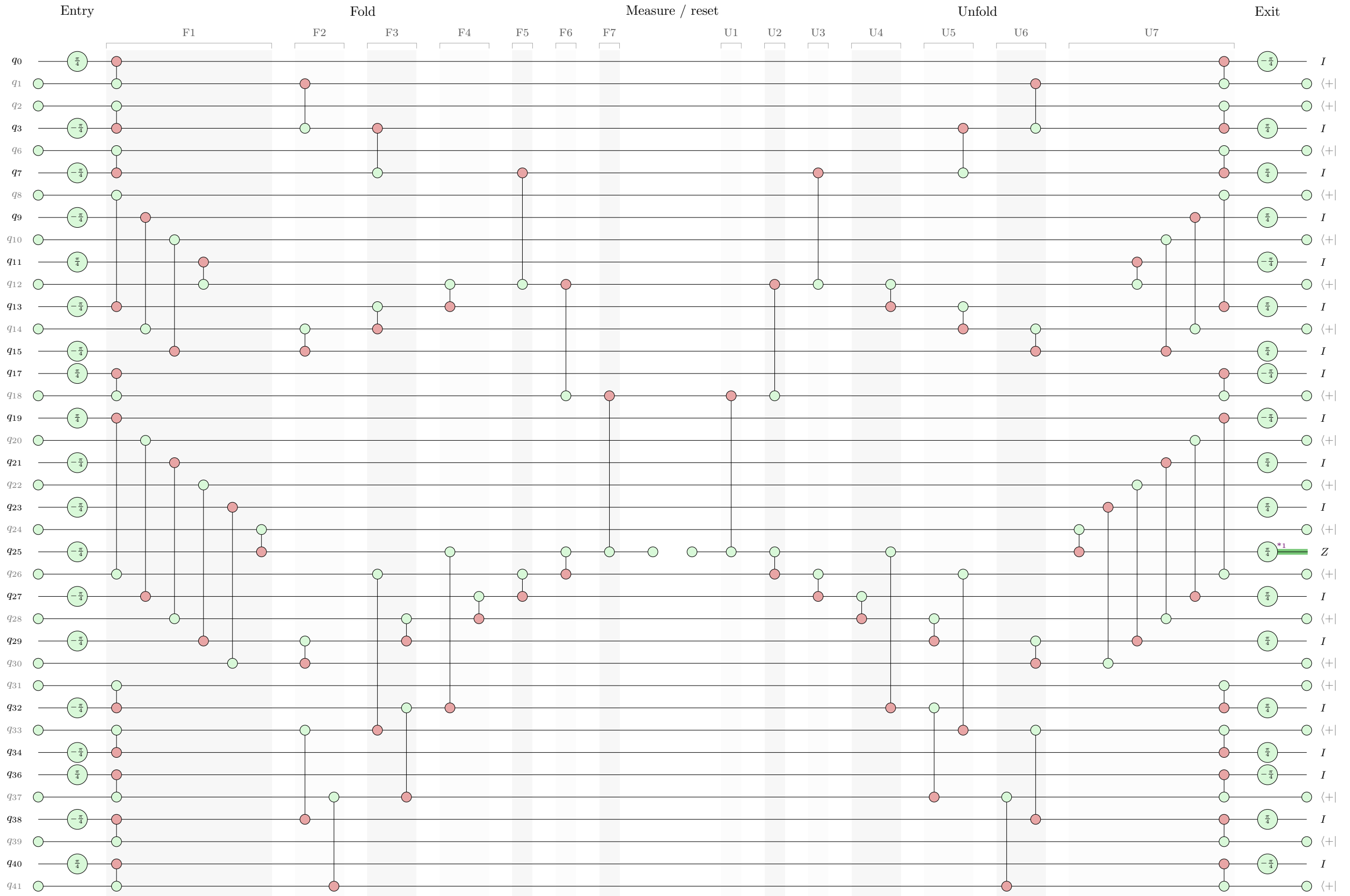
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{23}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_049.tikz

$d = 5$: basis element 50

Additional output support. 1 phase defect.



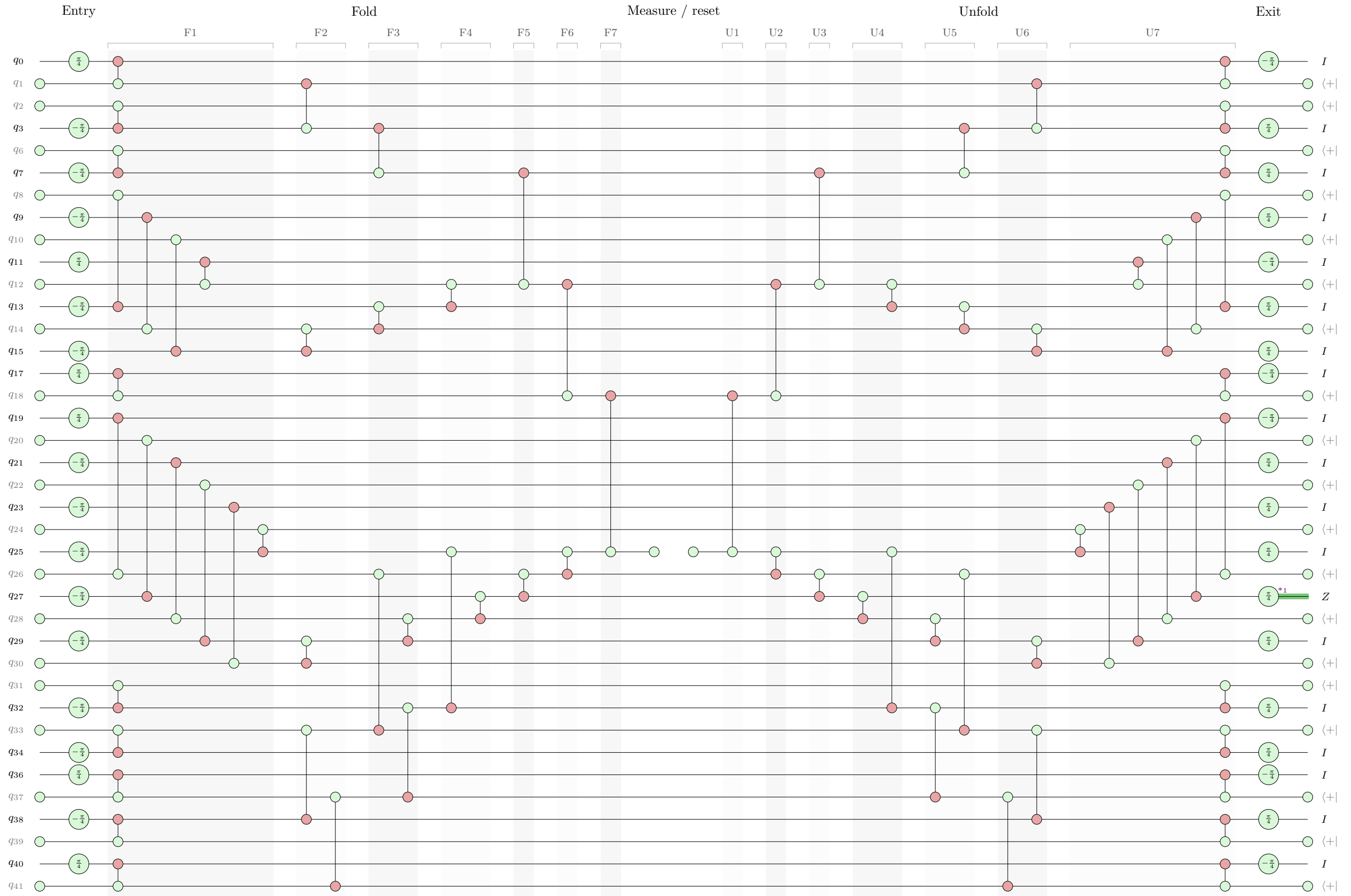
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{25}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_050.tikz

$d = 5$: basis element 51

Additional output support. 1 phase defect.



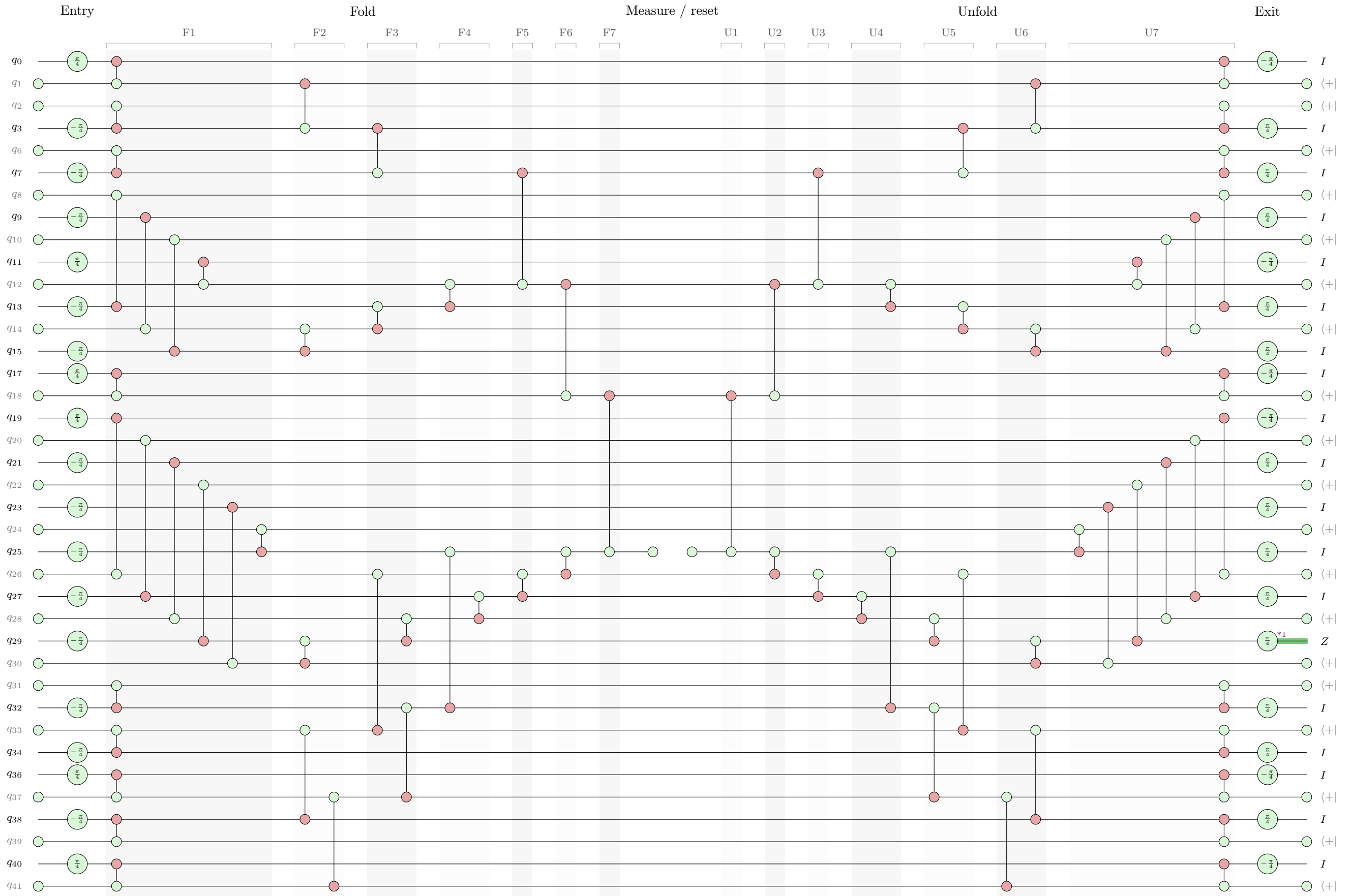
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{27}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_051.tikz

$d = 5$: basis element 52

Additional output support. 1 phase defect.



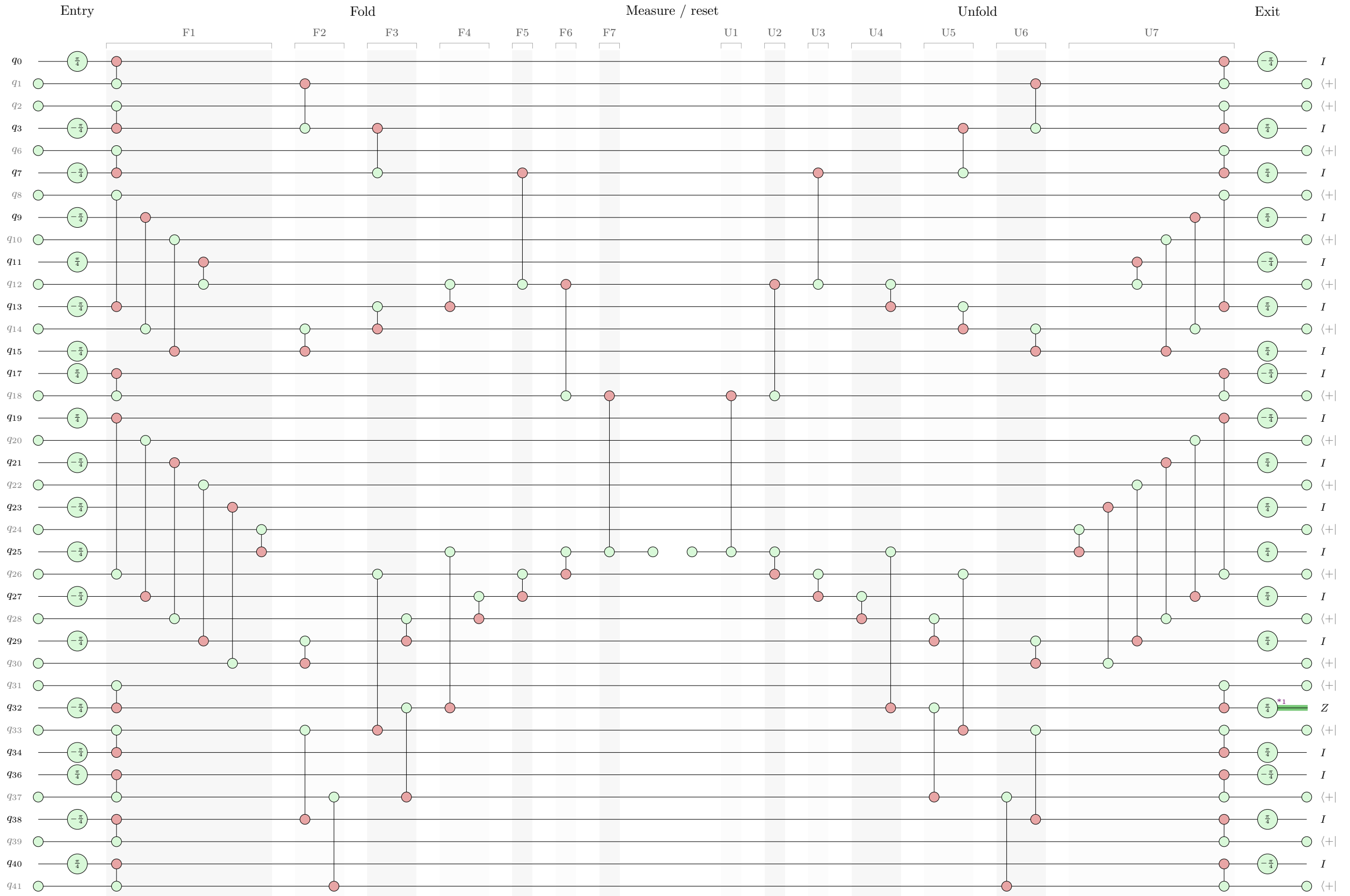
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{29}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_052.tikz

$d = 5$: basis element 53

Additional output support. 1 phase defect.

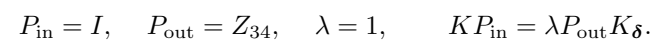


$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{32}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_053.tikz

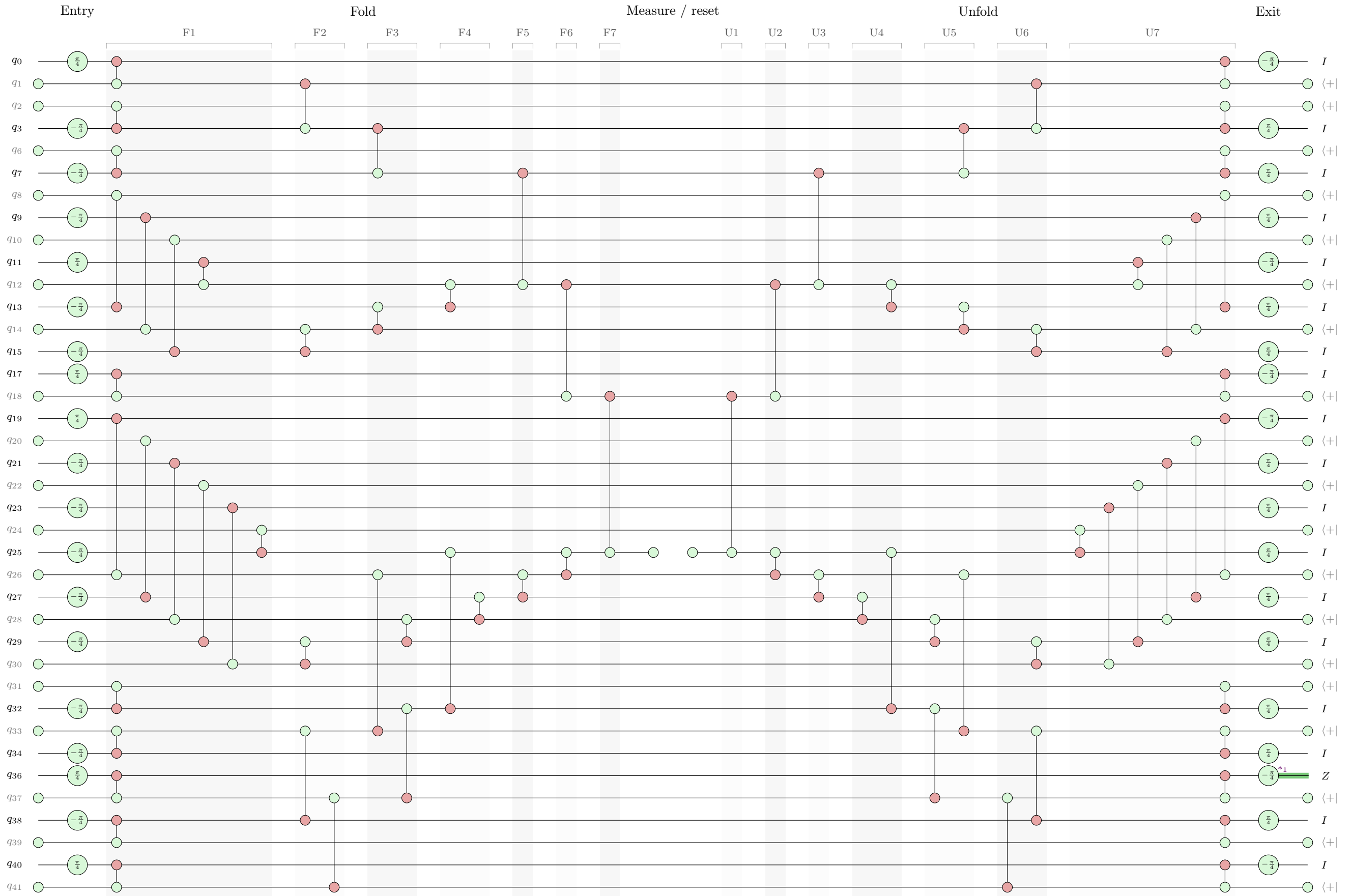
Additional output support. 1 phase defect.



d5_054.tikz

$d = 5$: basis element 55

Additional output support. 1 phase defect.



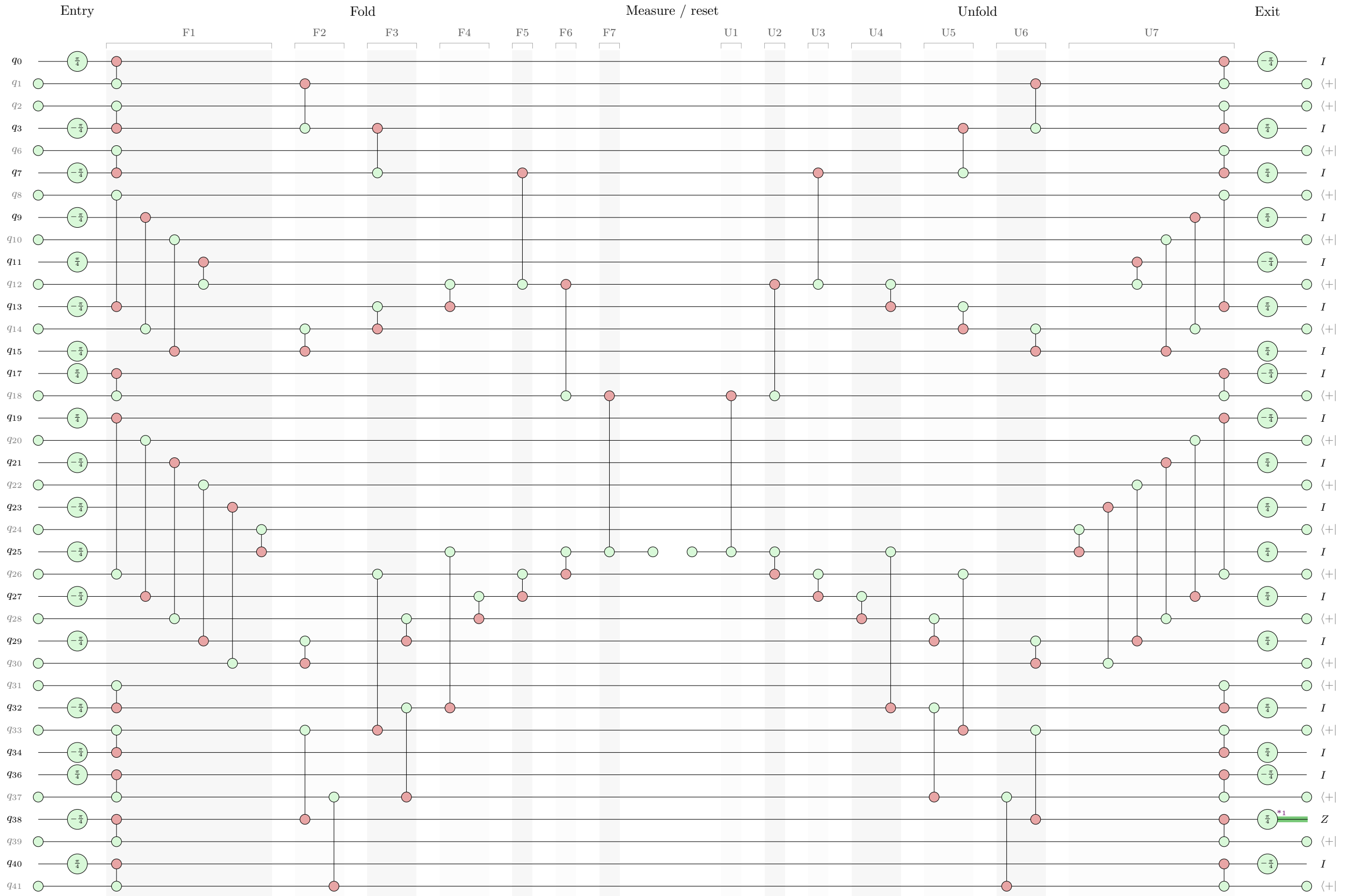
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{36}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_055.tikz

$d = 5$: basis element 56

Additional output support. 1 phase defect.



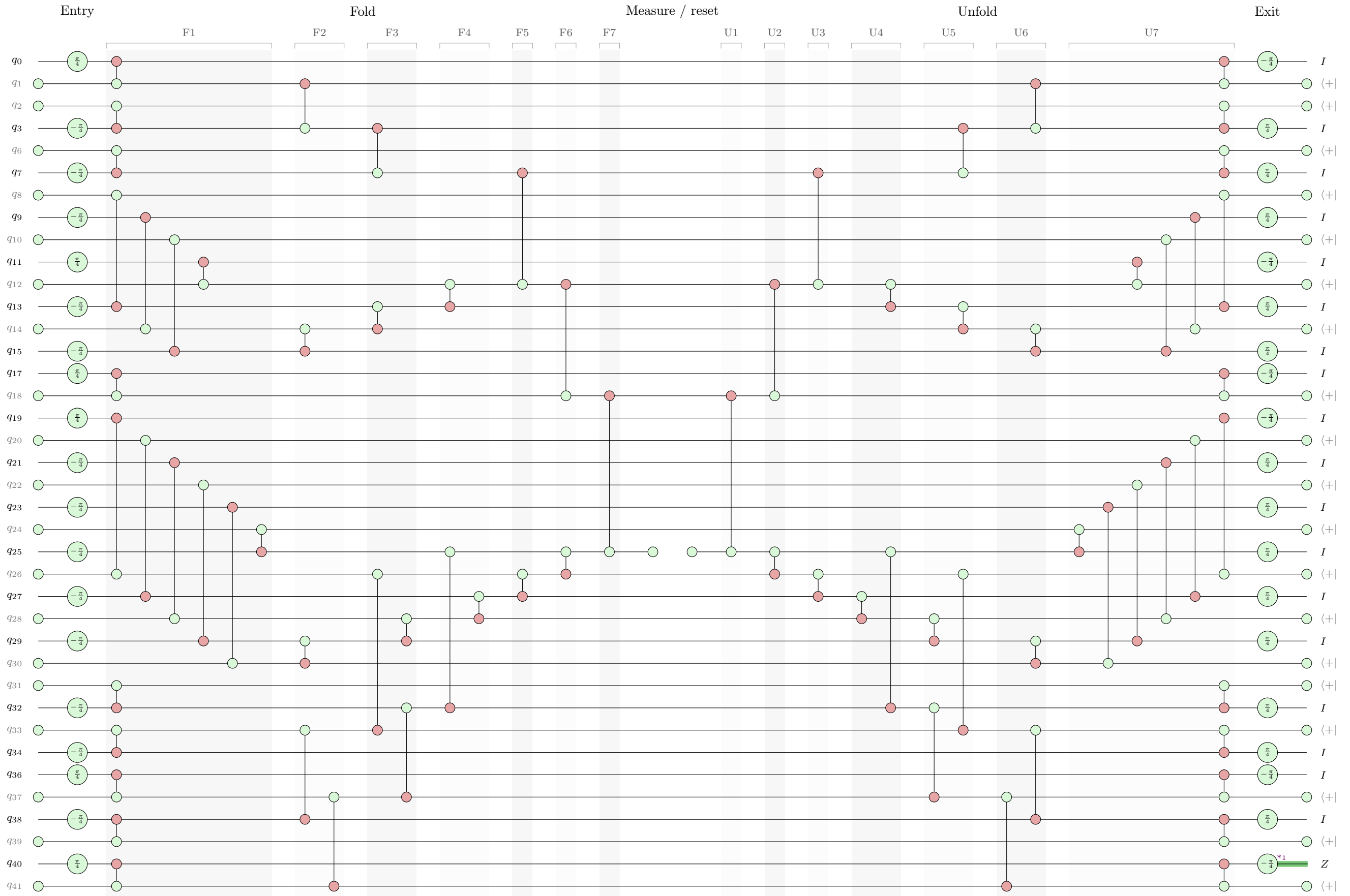
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{38}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_056.tikz

$d = 5$: basis element 57

Additional output support. 1 phase defect.



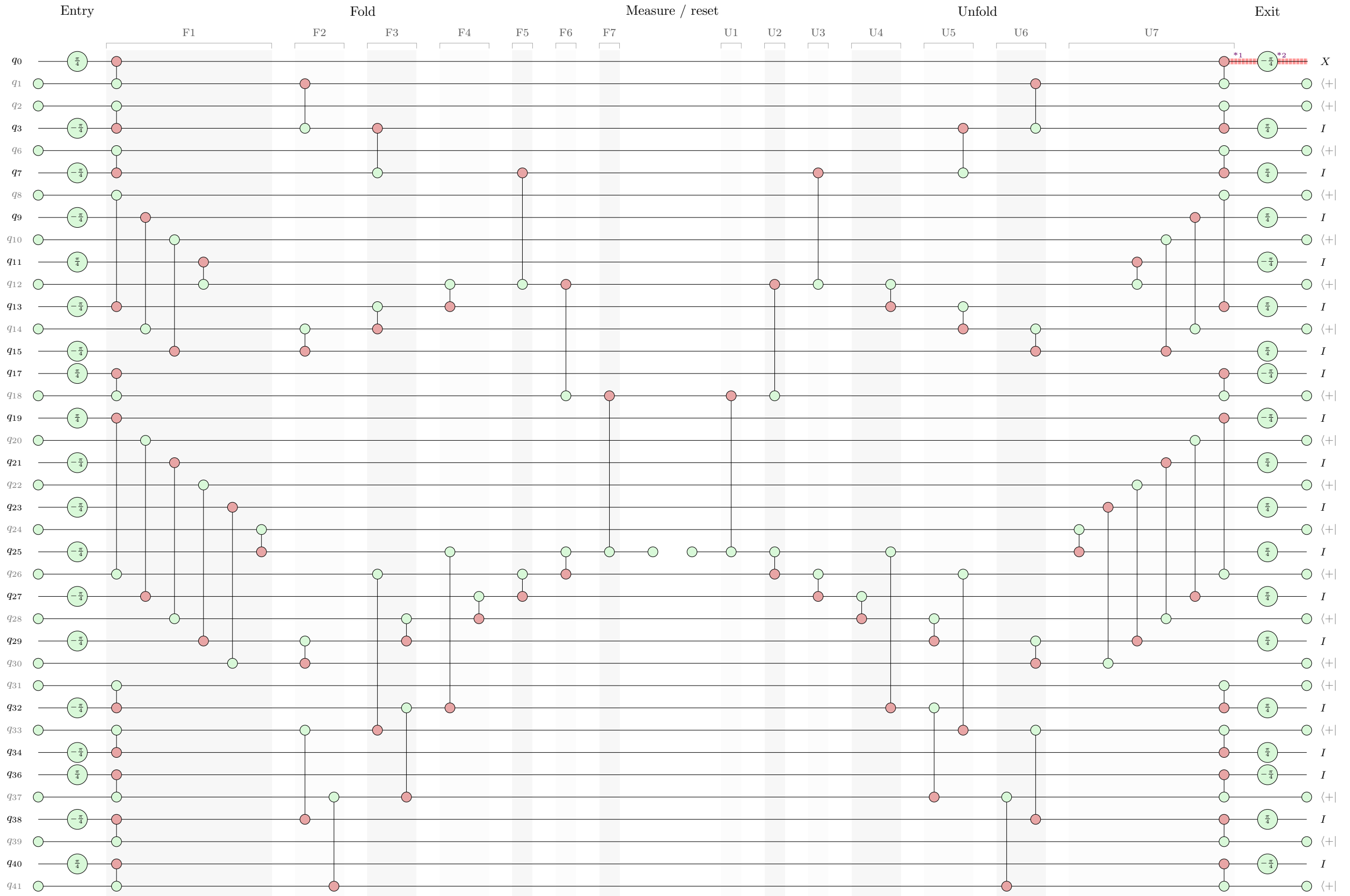
$$P_{\text{in}} = I, \quad P_{\text{out}} = Z_{40}, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$.

d5_057.tikz

$d = 5$: basis element 58

Additional output support. 2 phase defects.



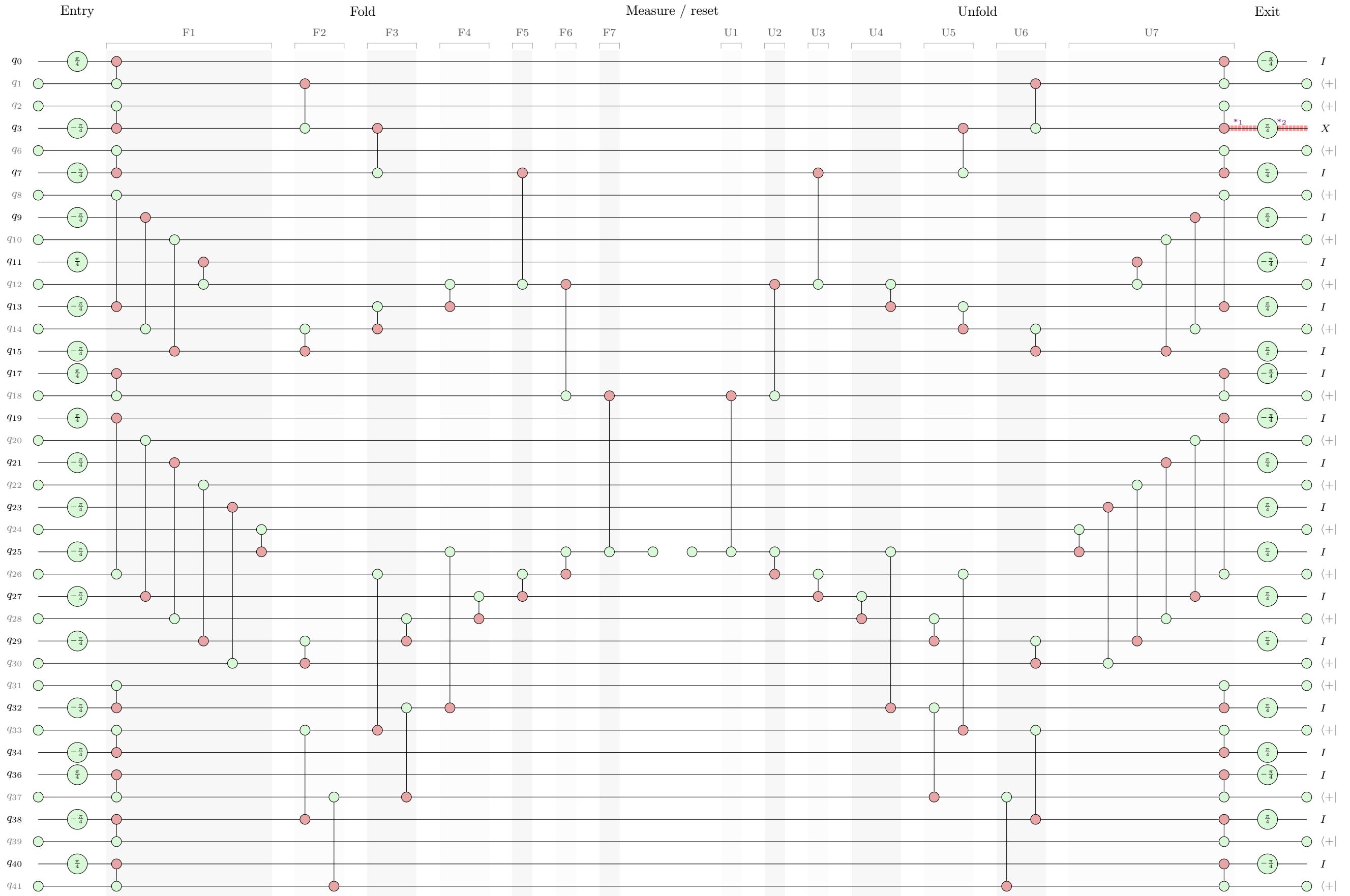
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_0, \quad \lambda = e^{i\frac{7\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \frac{\pi}{2}$.

d5_058.tikz

$d = 5$: basis element 59

Additional output support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = X_3, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_059.tikz

$d = 5$: basis element 60

Additional output support. 2 phase defects.

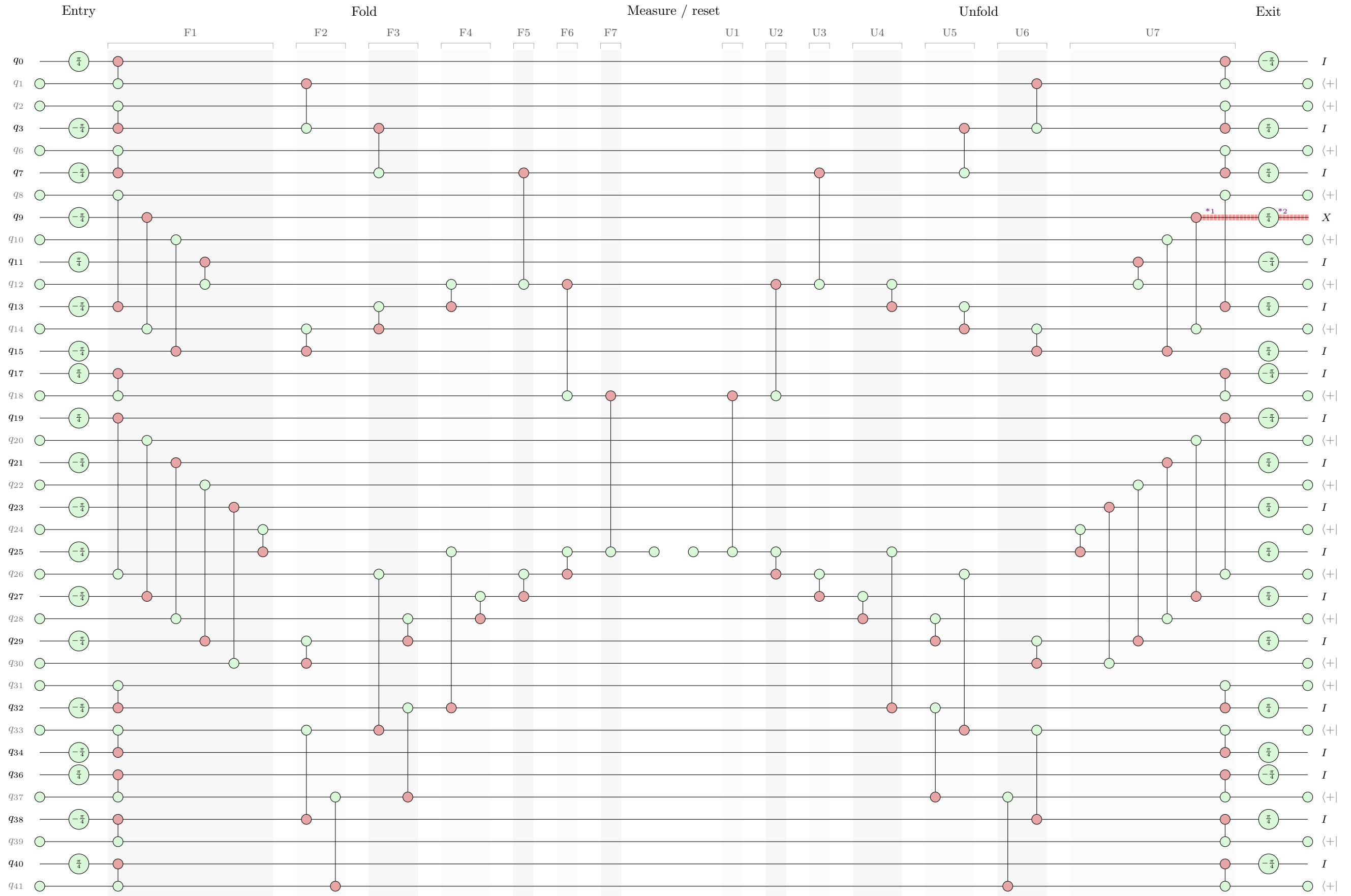


$$P_{\text{in}} = I, \quad P_{\text{out}} = X_7, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

$d = 5$: basis element 61

Additional output support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = X_9, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_061.tikz

$d = 5$: basis element 62

Additional output support. 2 phase defects.



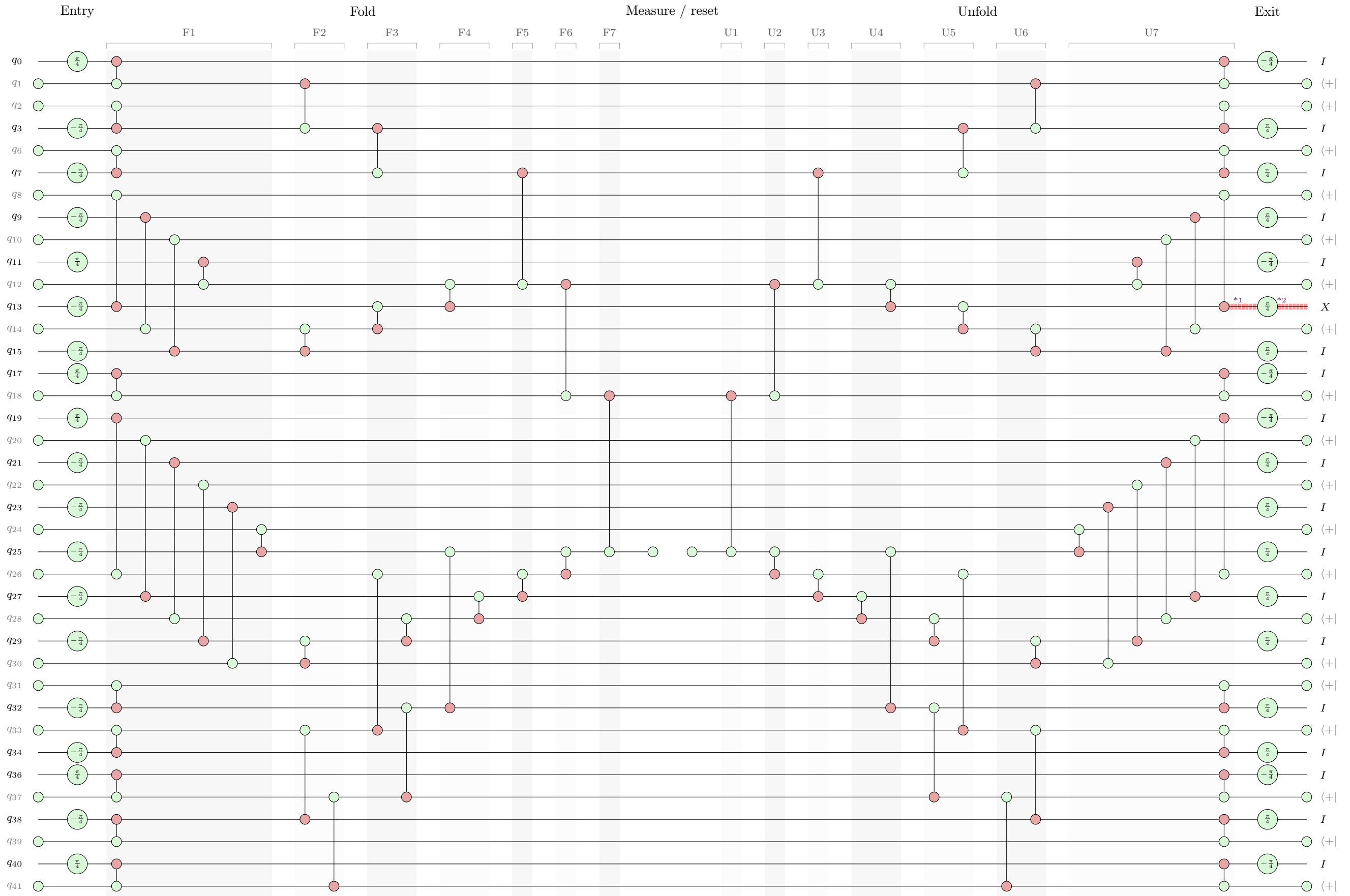
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{11}, \quad \lambda = e^{i\frac{7\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \frac{\pi}{2}$.

d5_062.tikz

$d = 5$: basis element 63

Additional output support. 2 phase defects.



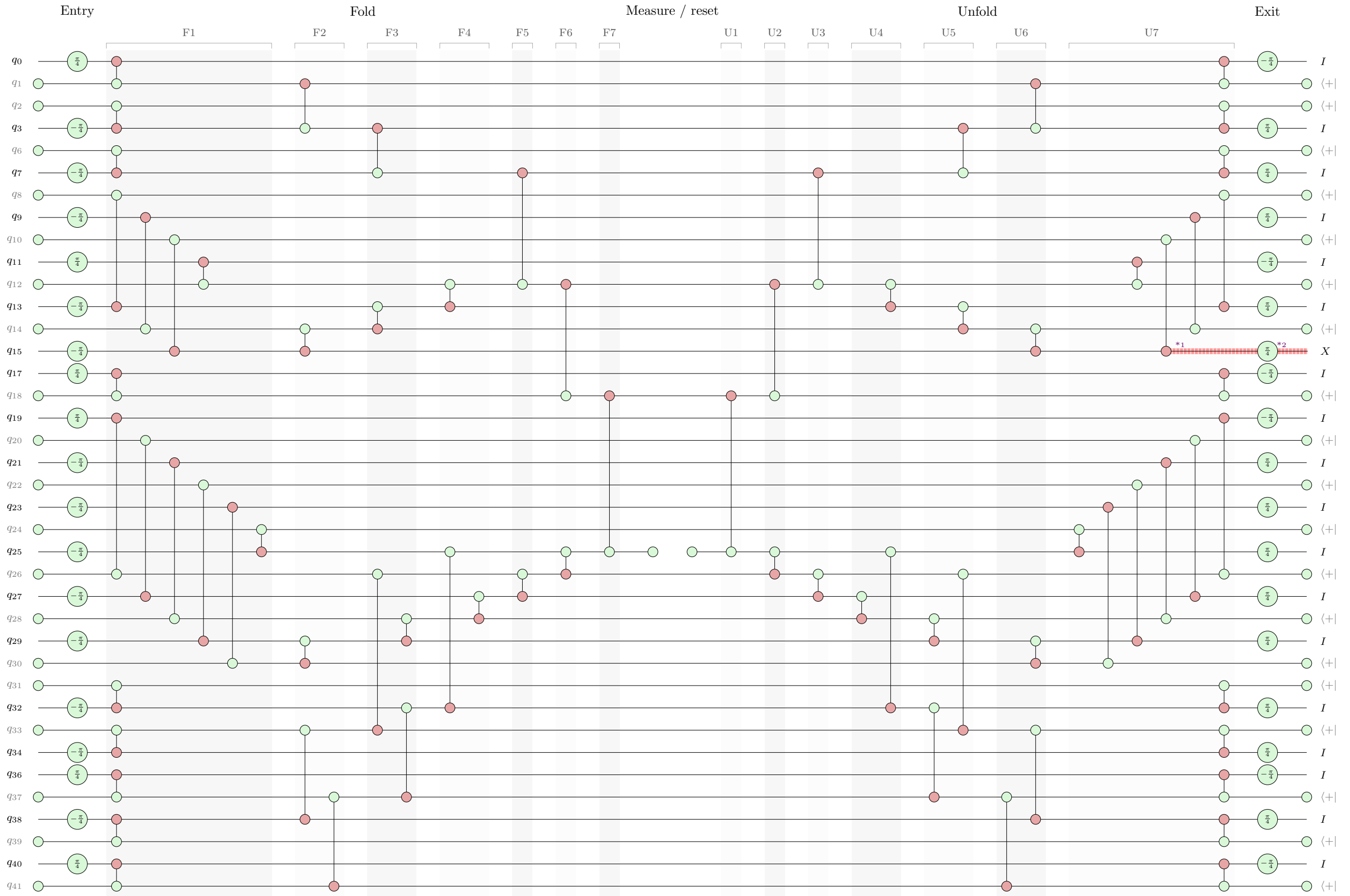
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{13}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_063.tikz

$d = 5$: basis element 64

Additional output support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{15}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_064.tikz

$d = 5$: basis element 65

Additional output support. 2 phase defects.



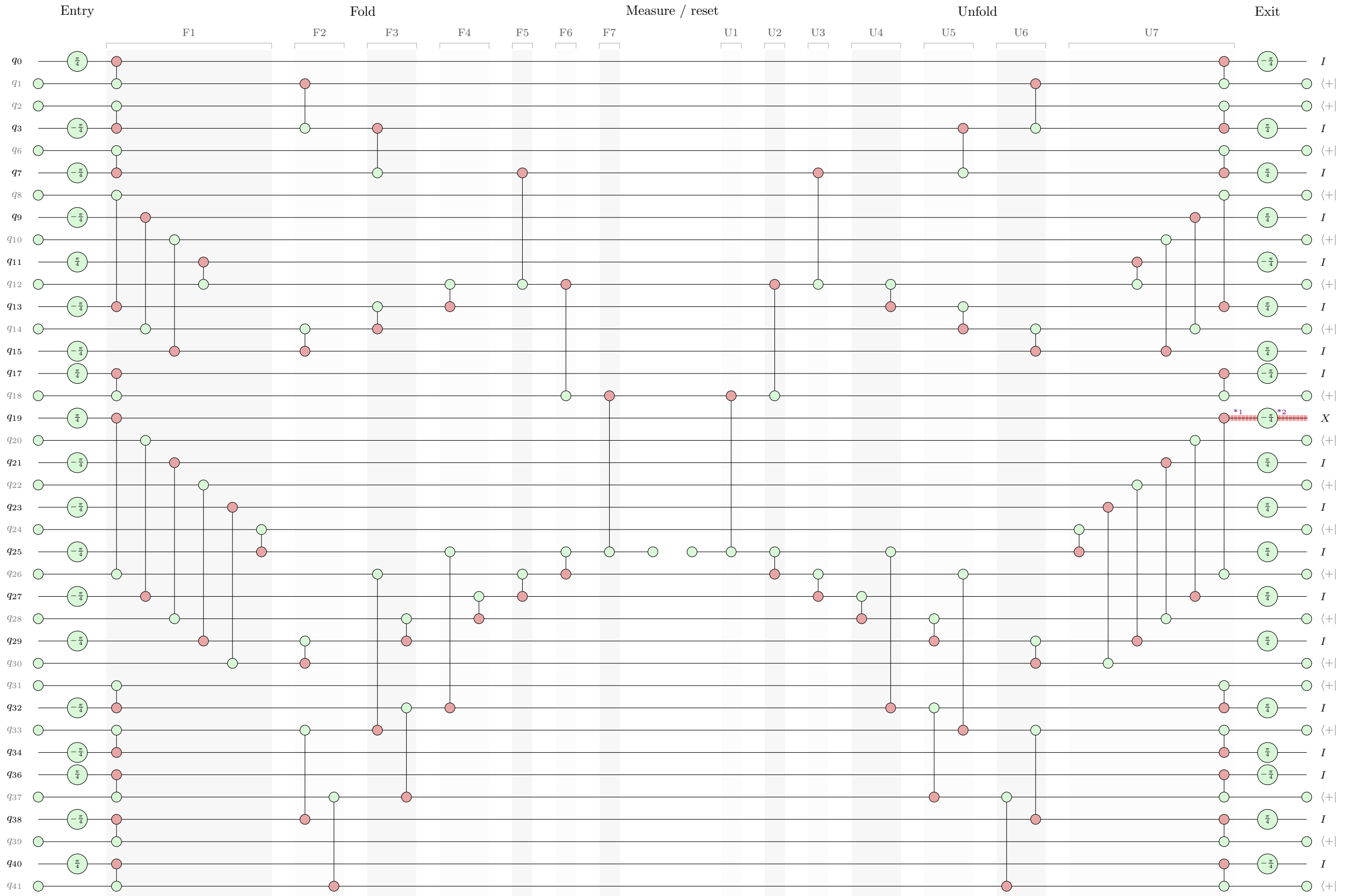
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{17}, \quad \lambda = e^{i\frac{7\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \frac{\pi}{2}$.

d5_065.tikz

$d = 5$: basis element 66

Additional output support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{19}, \quad \lambda = e^{i\frac{7\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \frac{\pi}{2}$.

d5_066.tikz

$d = 5$: basis element 67

Additional output support. 2 phase defects.



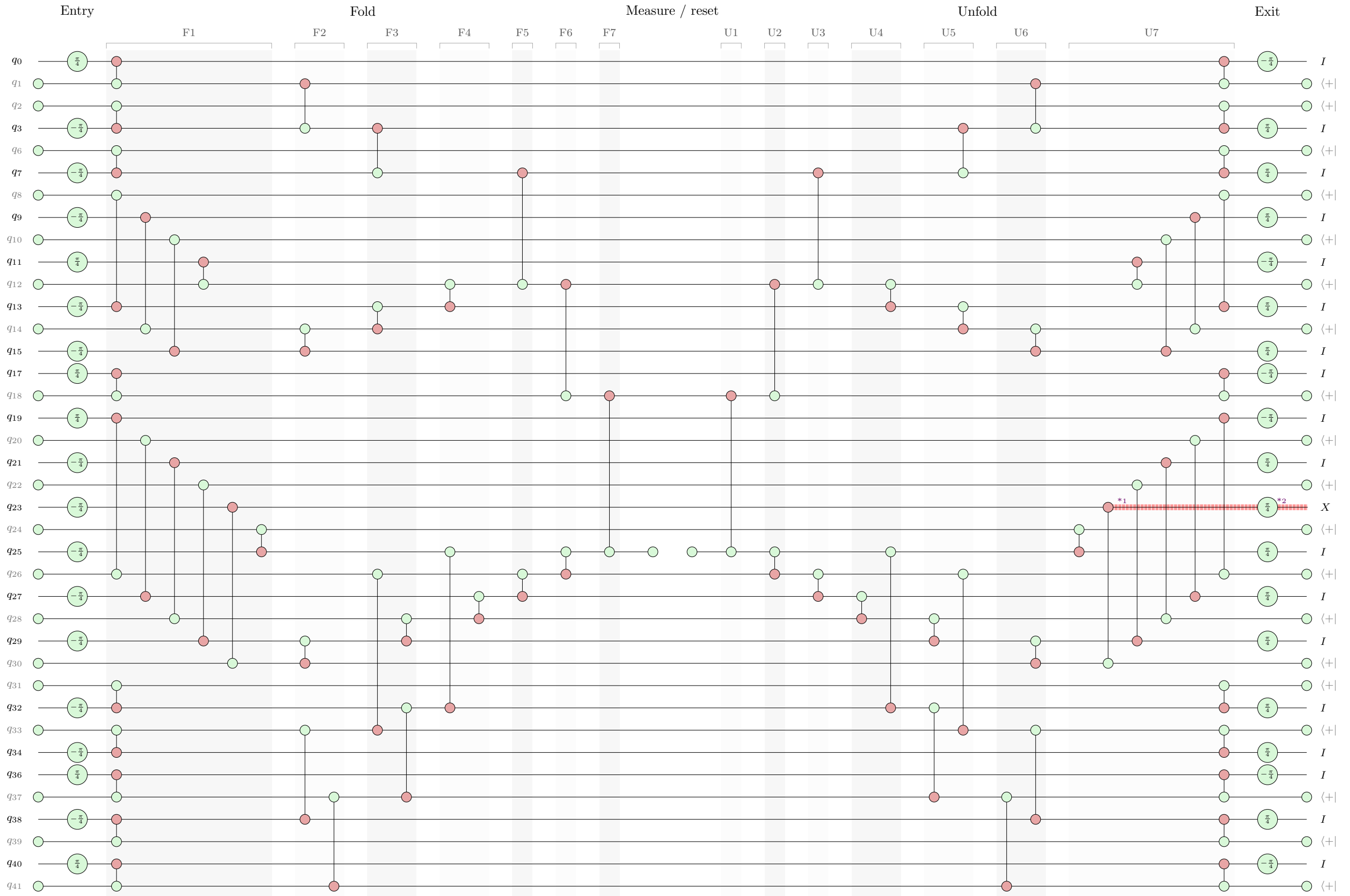
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{21}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_067.tikz

$d = 5$: basis element 68

Additional output support. 2 phase defects.



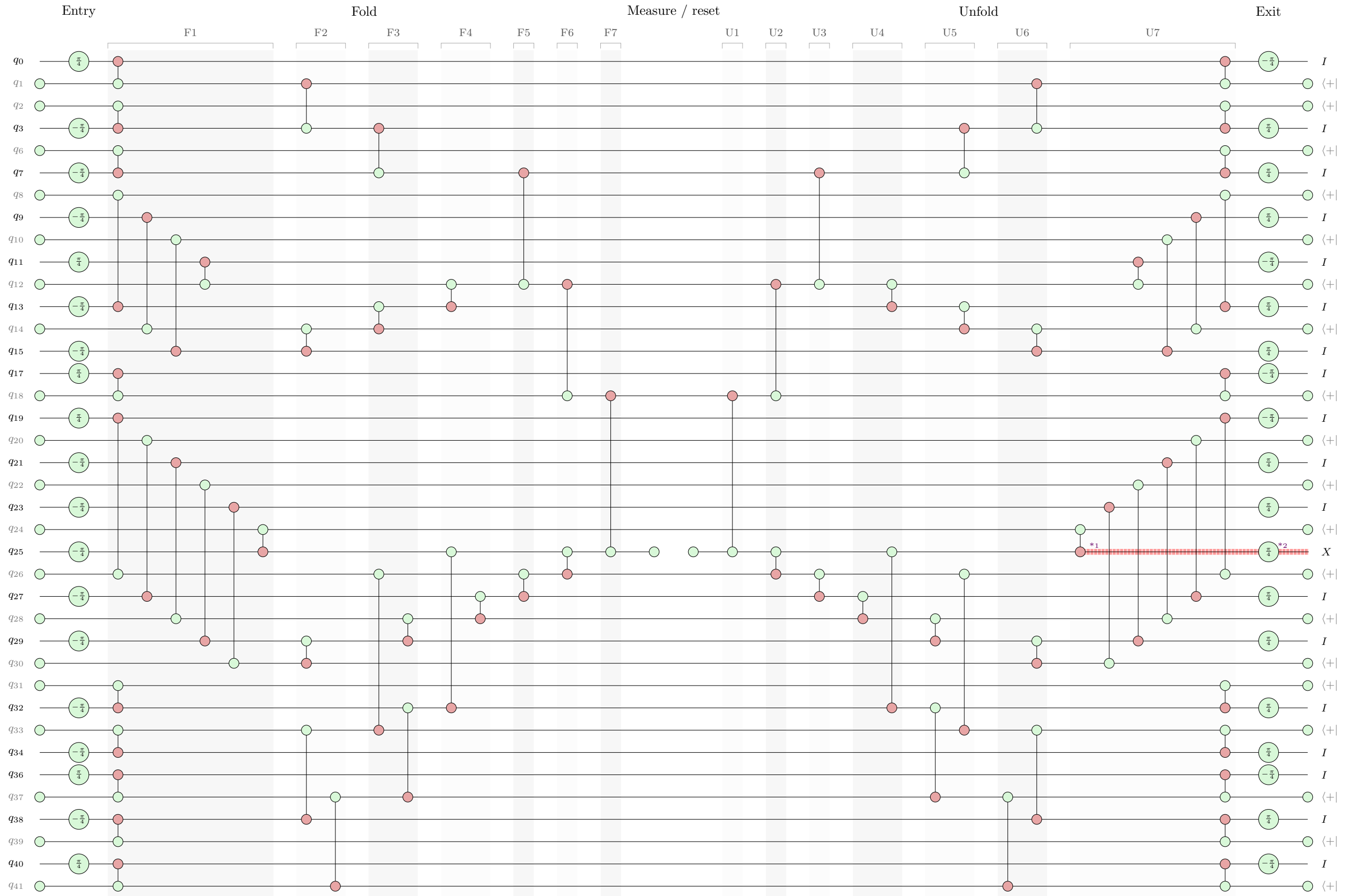
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{23}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_068.tikz

$d = 5$: basis element 69

Additional output support. 2 phase defects.



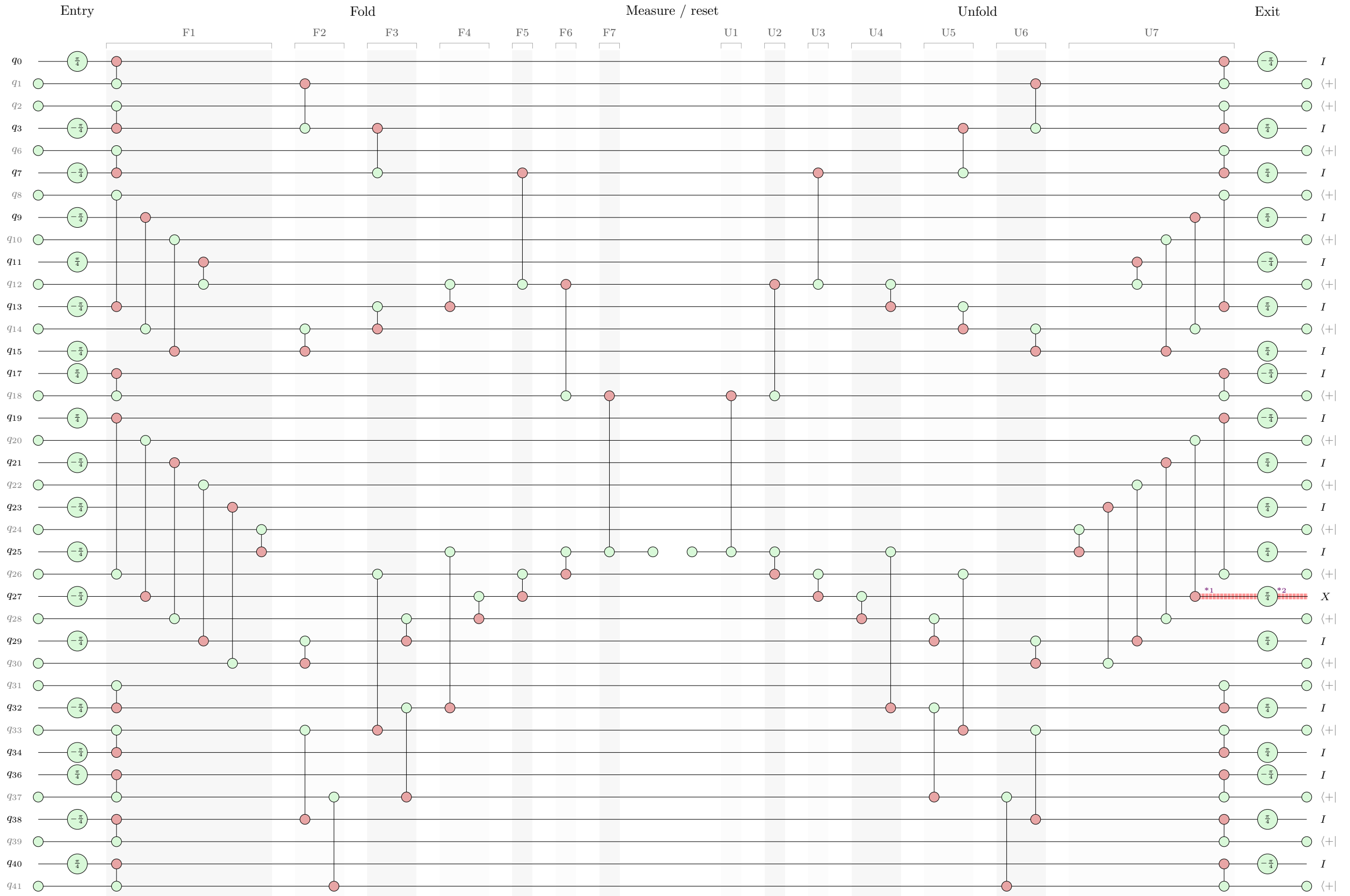
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{25}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_069.tikz

$d = 5$: basis element 70

Additional output support. 2 phase defects.



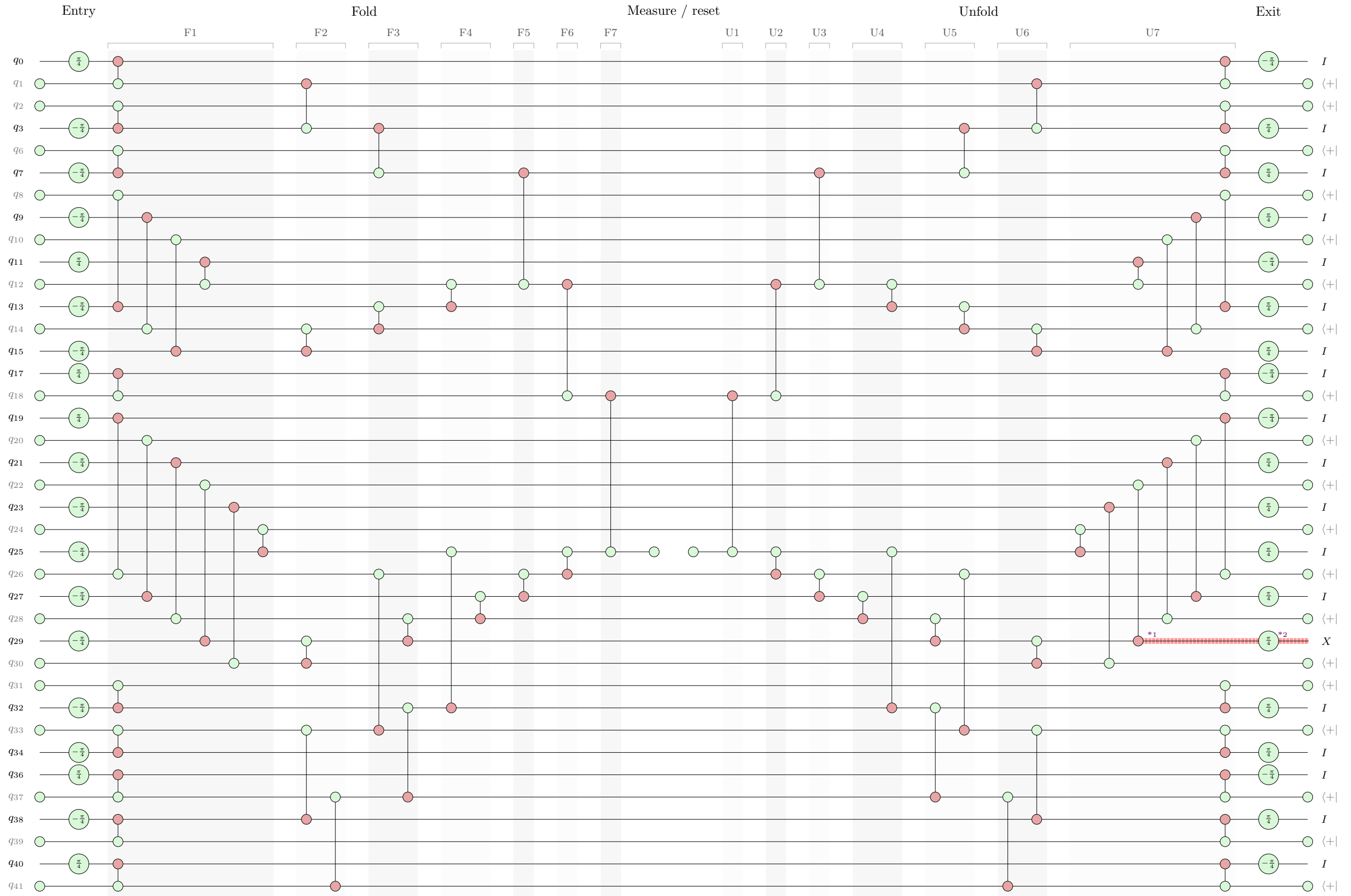
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{27}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_070.tikz

$d = 5$: basis element 71

Additional output support. 2 phase defects.



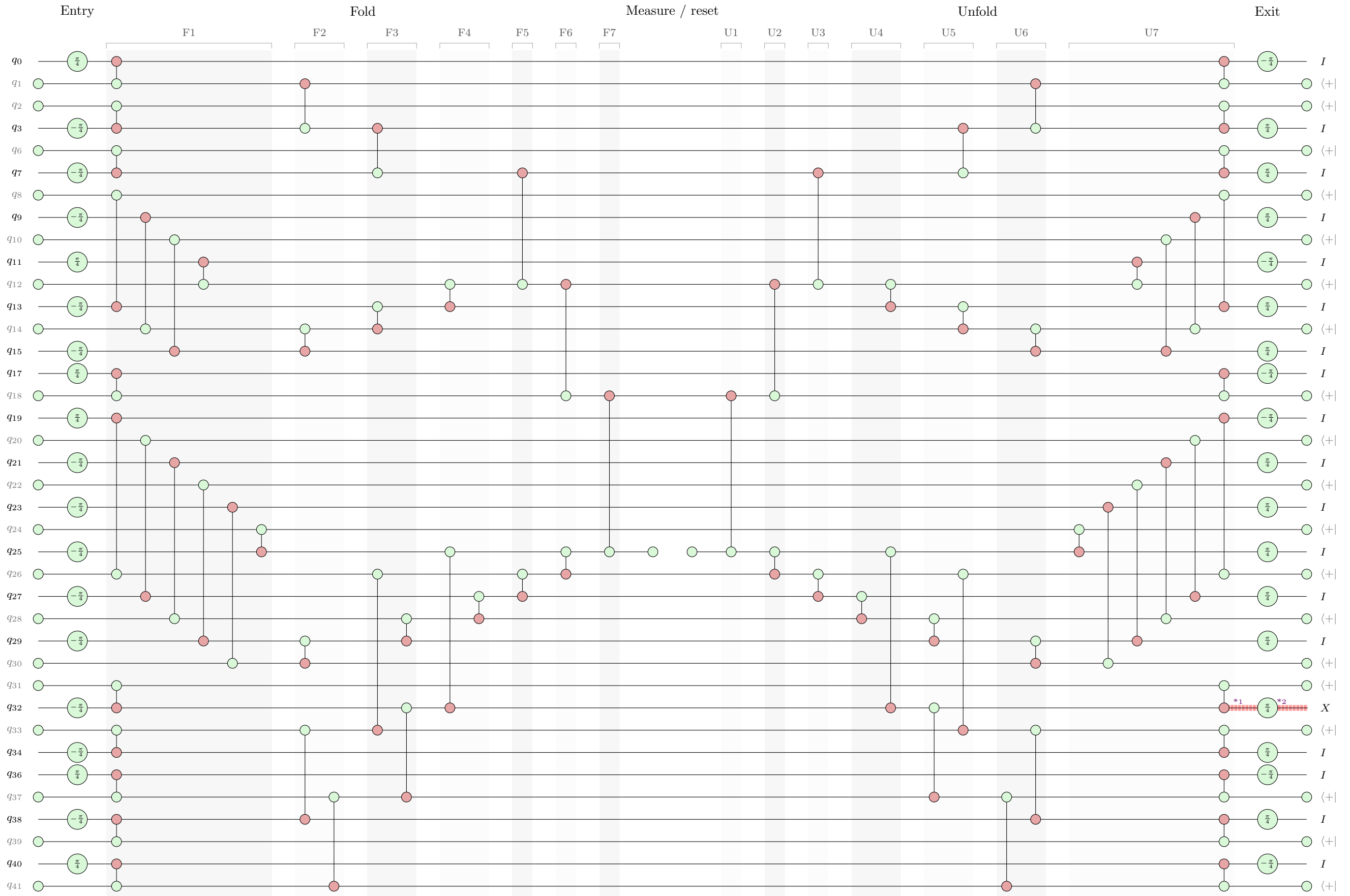
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{29}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_071.tikz

$d = 5$: basis element 72

Additional output support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{32}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_072.tikz

$d = 5$: basis element 73

Additional output support. 2 phase defects.



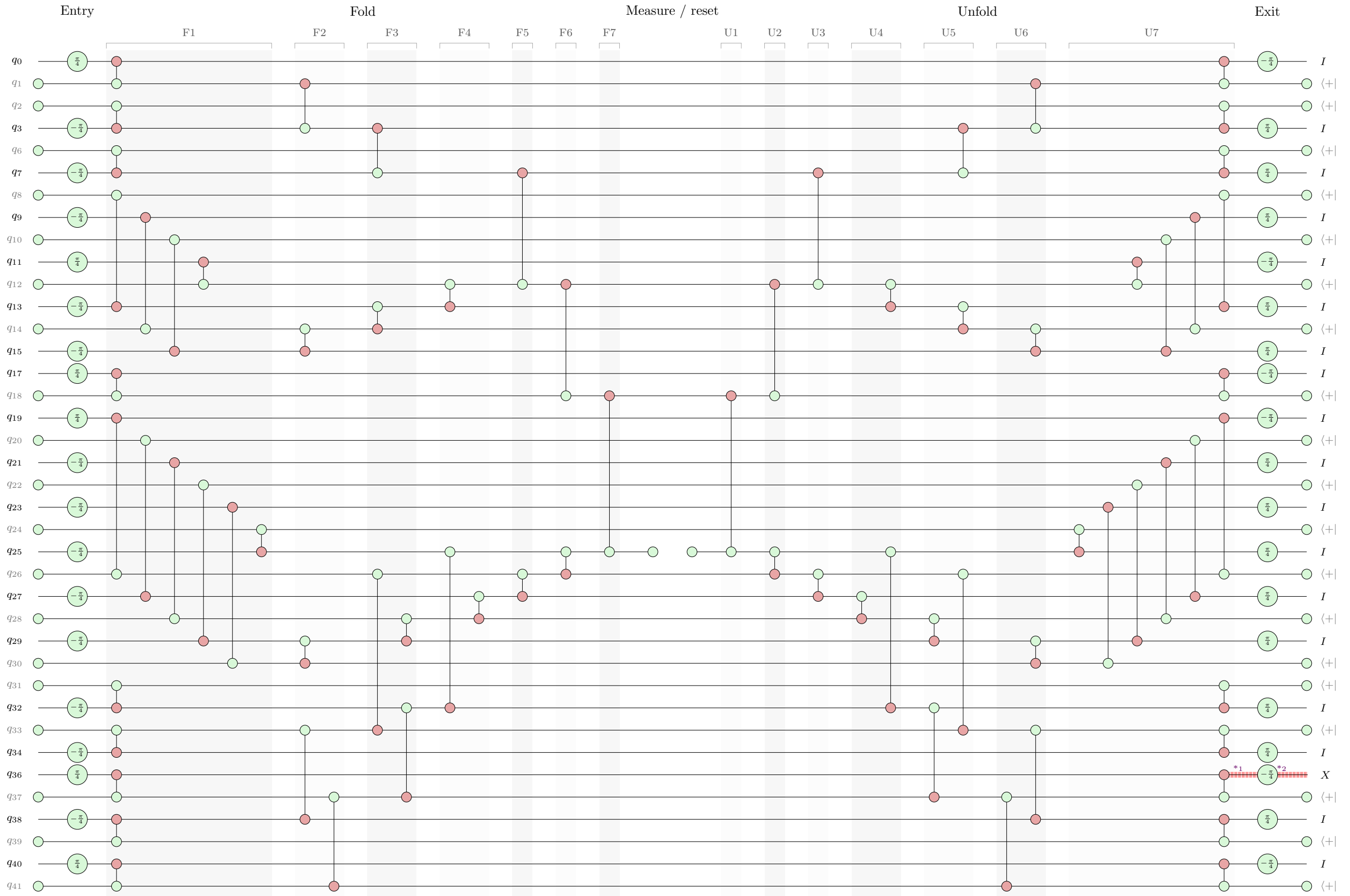
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{34}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_073.tikz

$d = 5$: basis element 74

Additional output support. 2 phase defects.



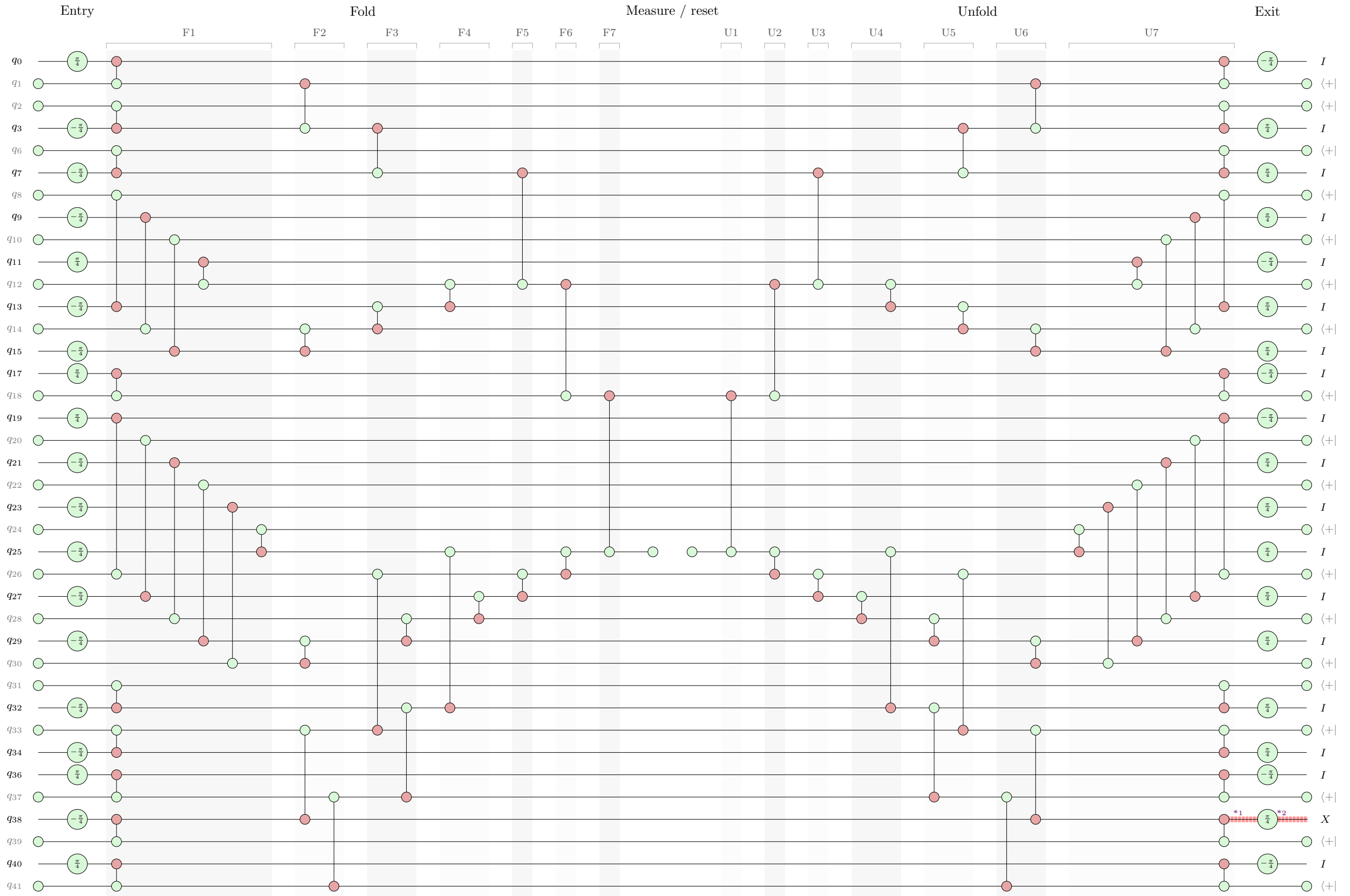
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{36}, \quad \lambda = e^{i\frac{7\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \frac{\pi}{2}$.

d5_074.tikz

$d = 5$: basis element 75

Additional output support. 2 phase defects.



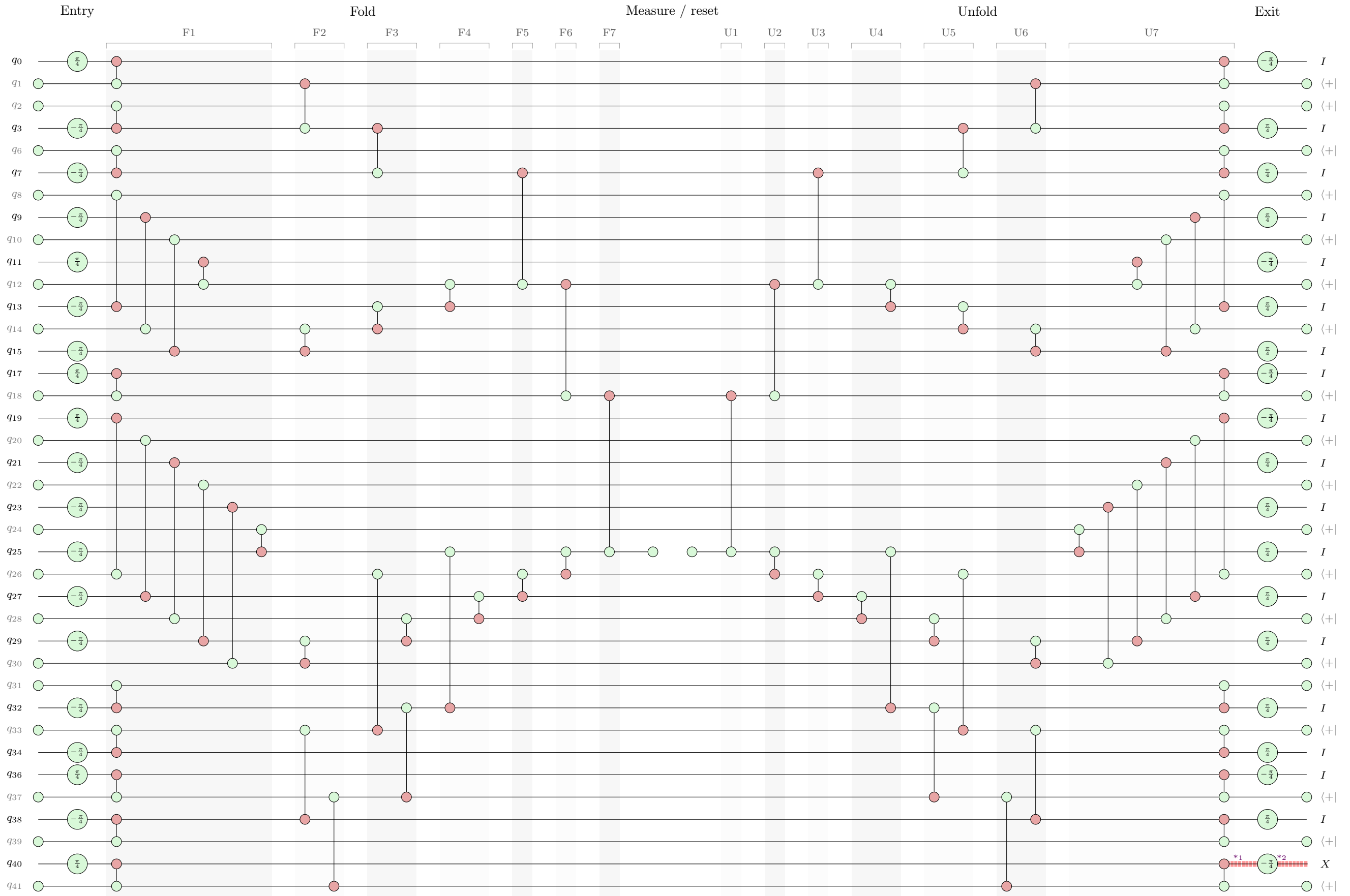
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{38}, \quad \lambda = e^{i\frac{\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = -\frac{\pi}{2}$.

d5_075.tikz

$d = 5$: basis element 76

Additional output support. 2 phase defects.



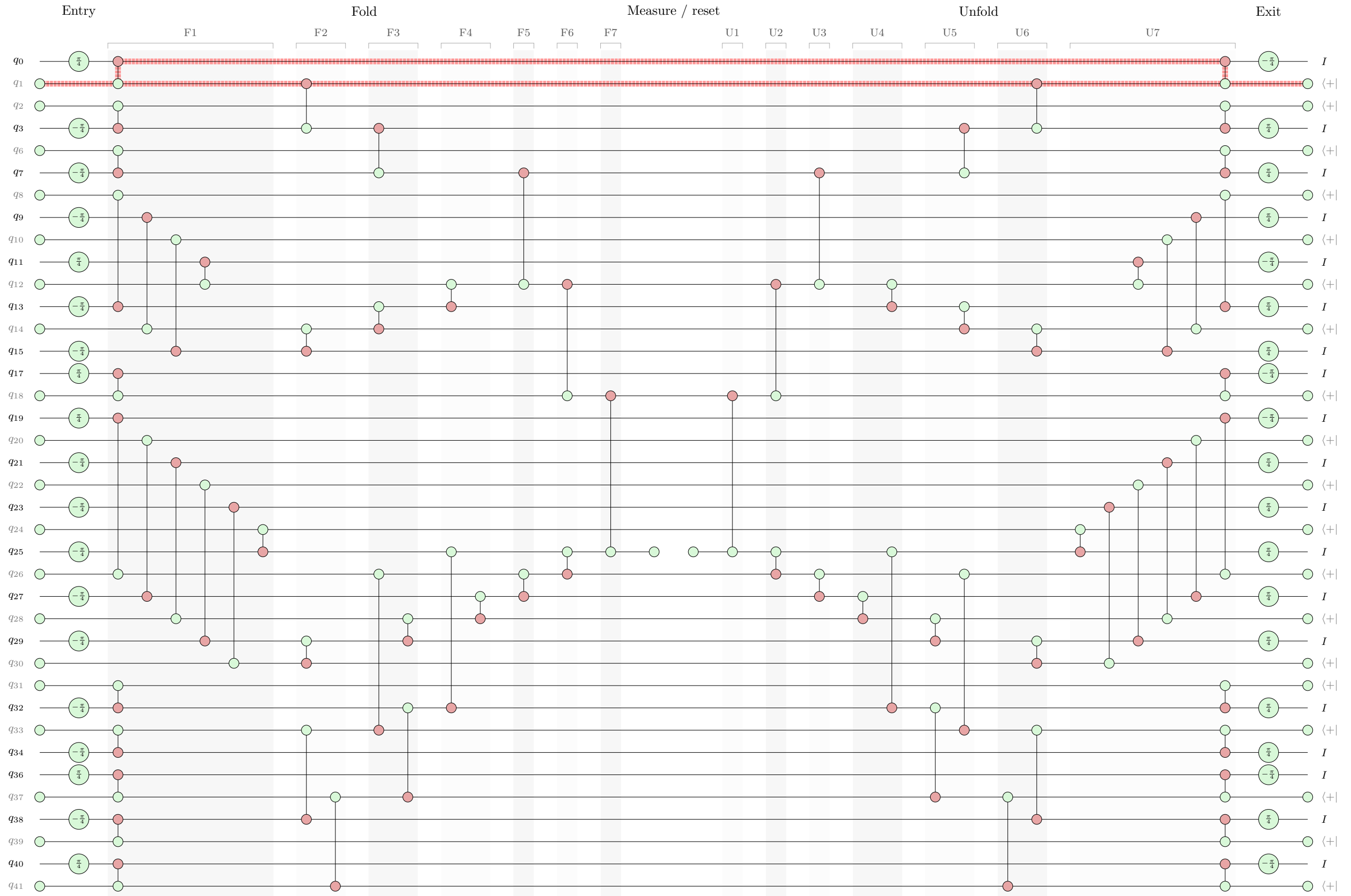
$$P_{\text{in}} = I, \quad P_{\text{out}} = X_{40}, \quad \lambda = e^{i\frac{7\pi}{4}}, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \frac{\pi}{2}$.

d5_076.tikz

$d = 5$: basis element 77

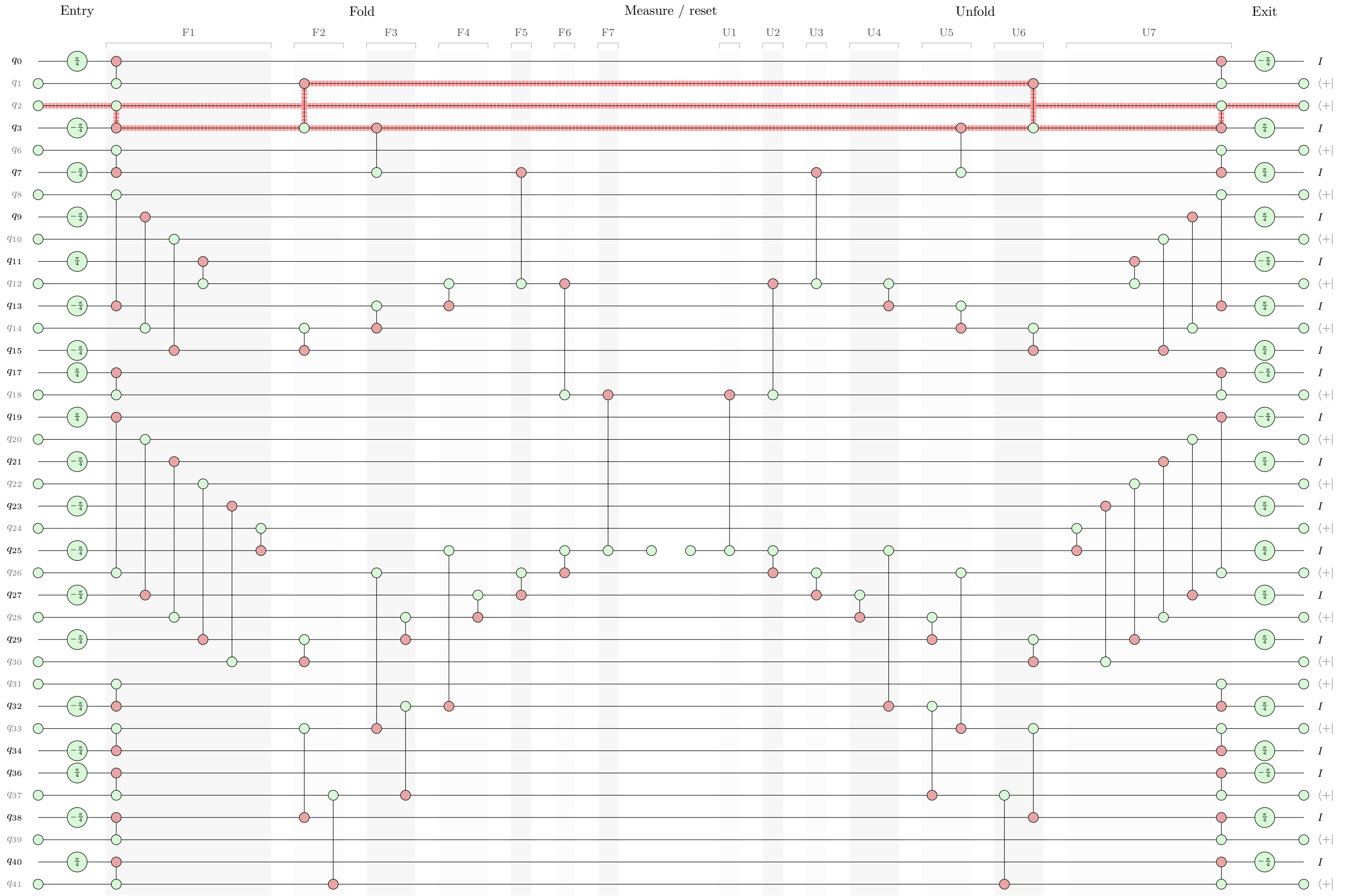
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 78

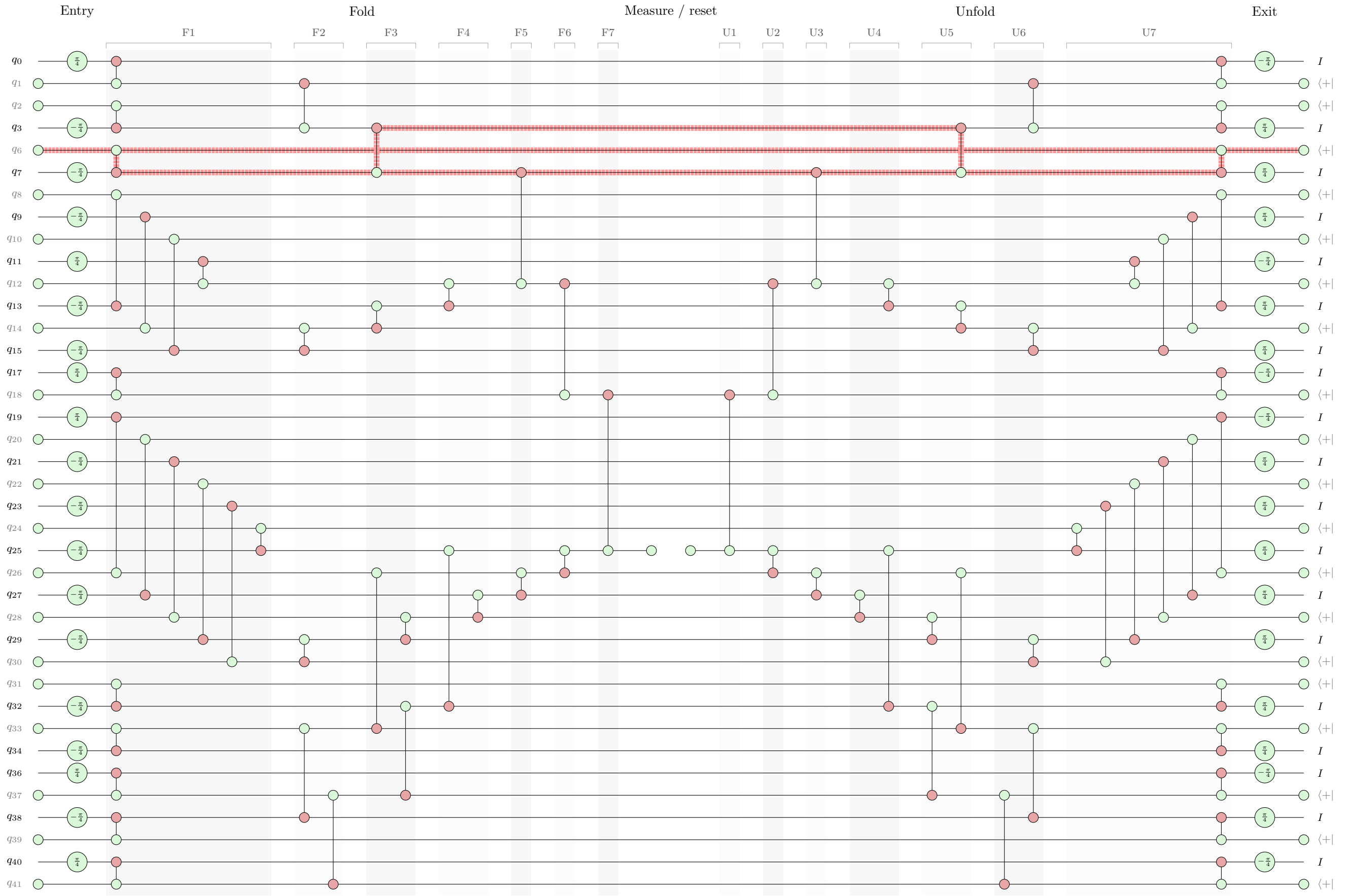
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 79

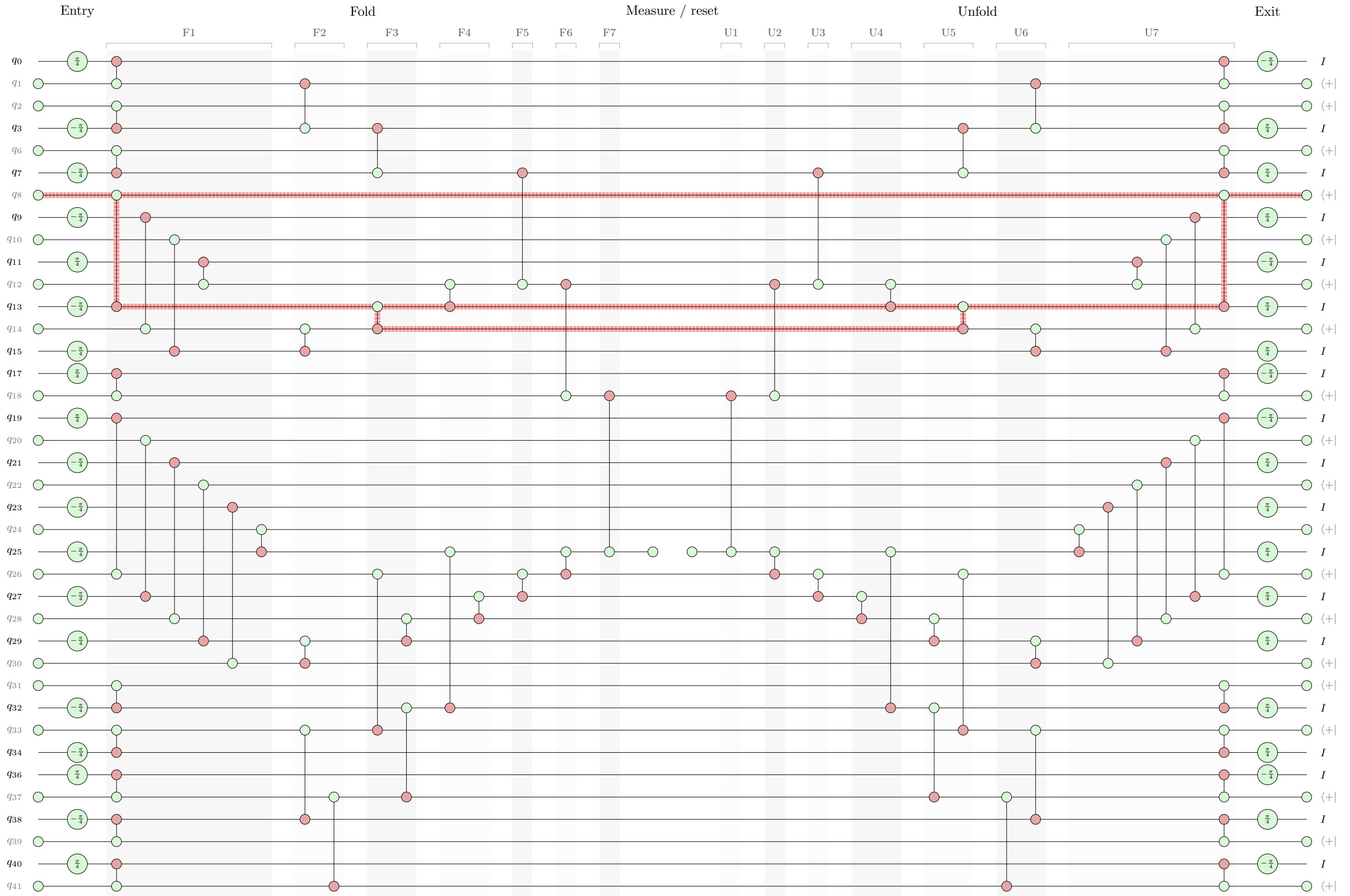
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 80

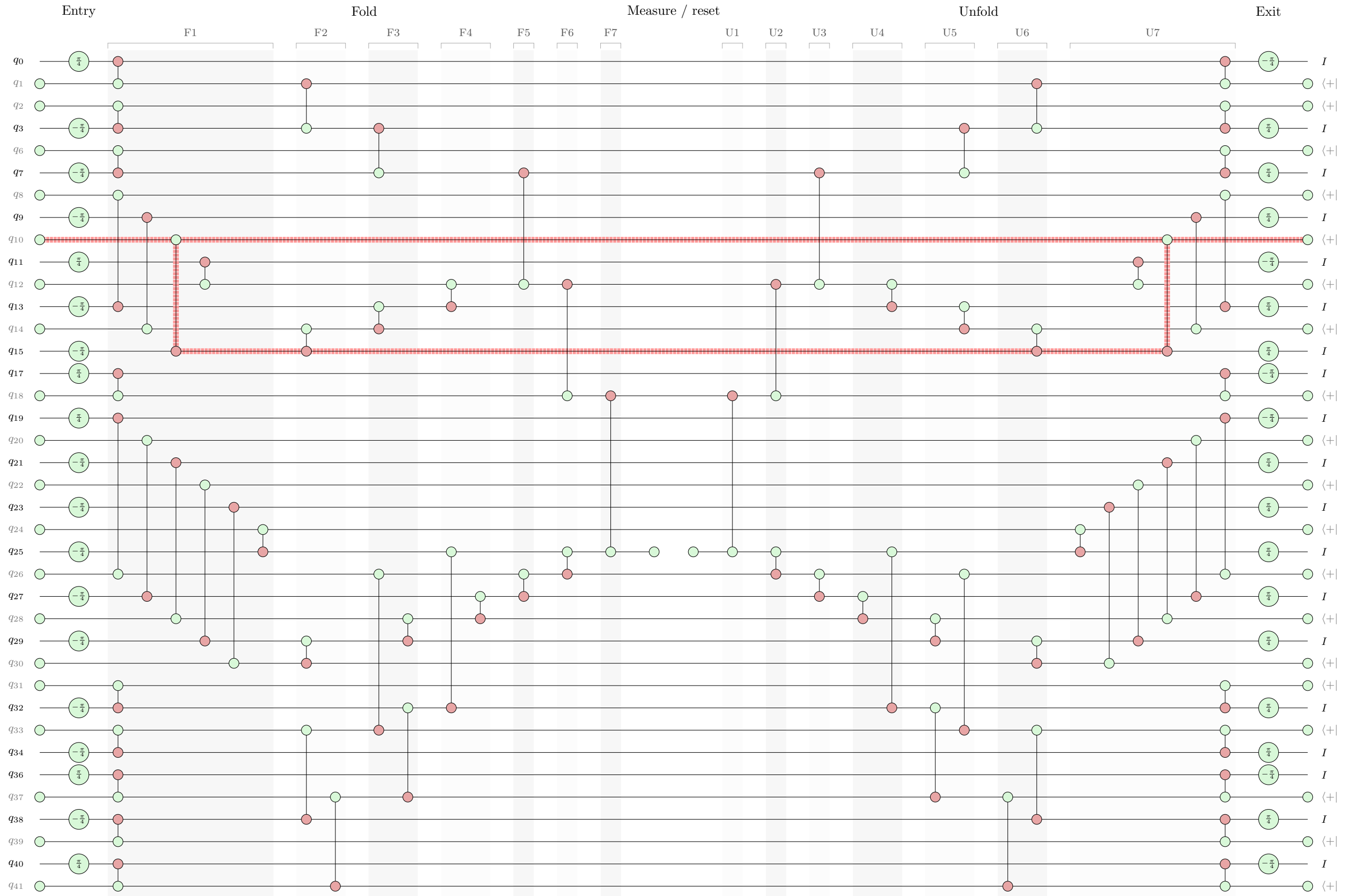
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 81

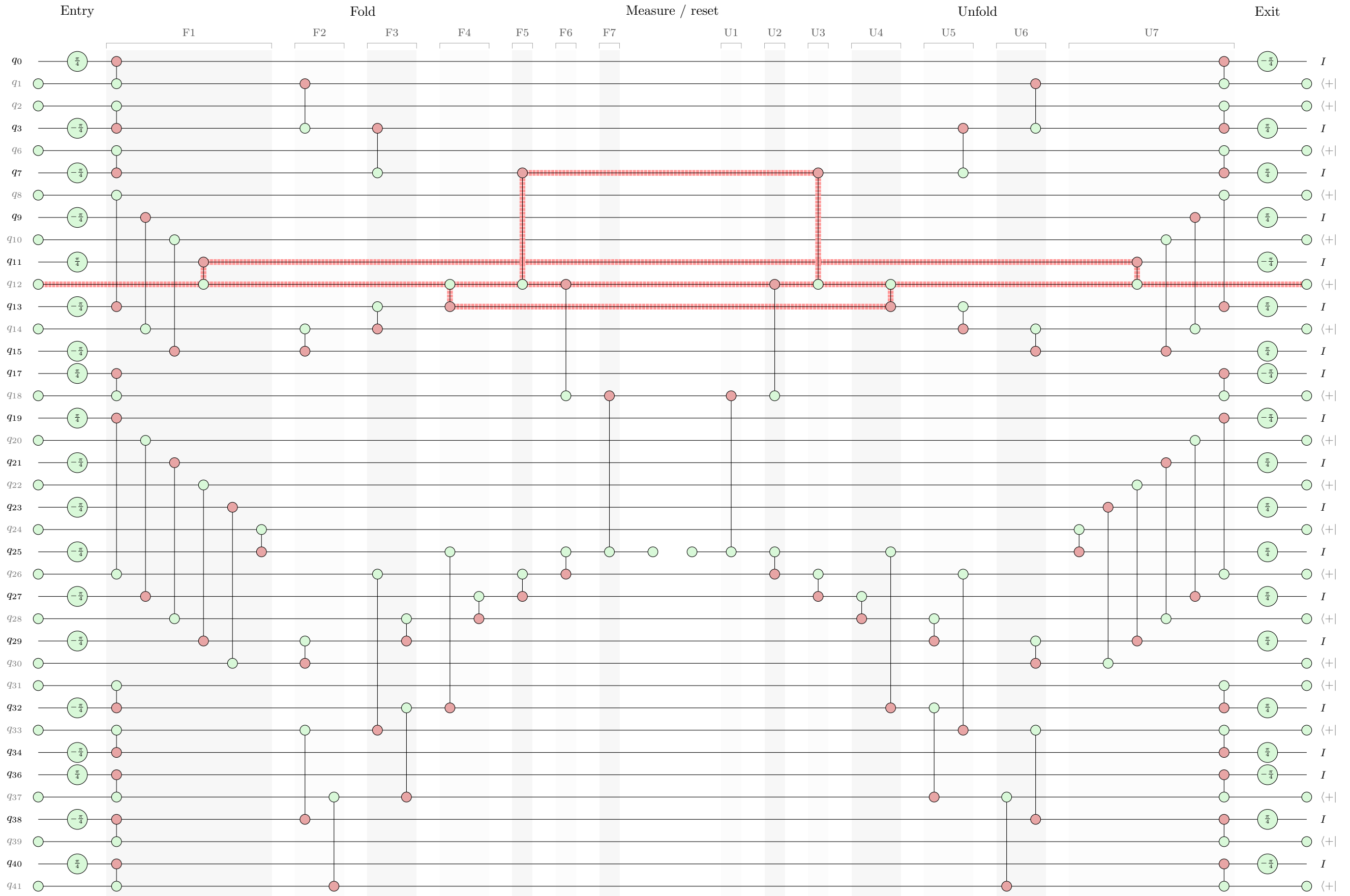
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 82

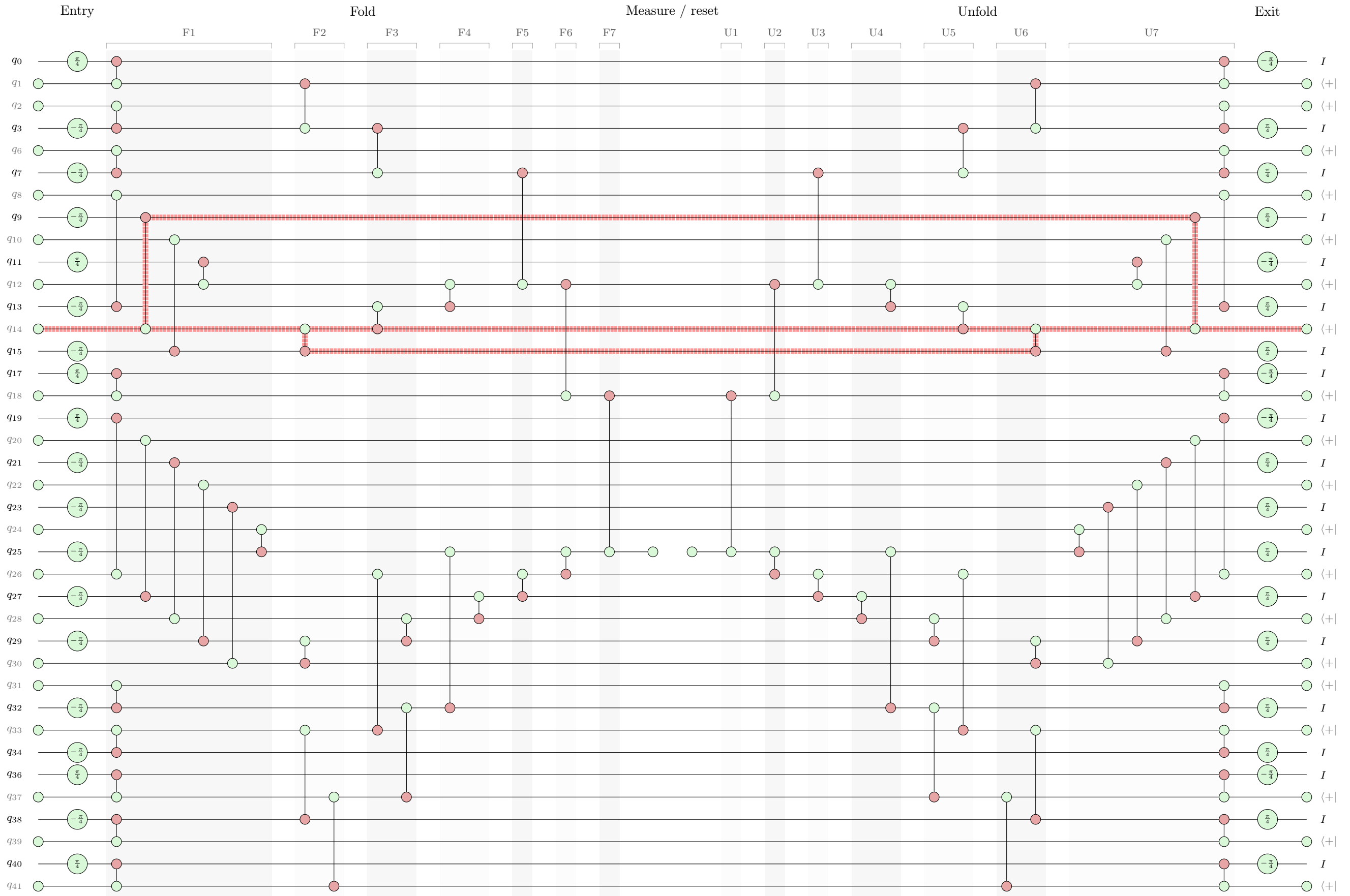
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 83

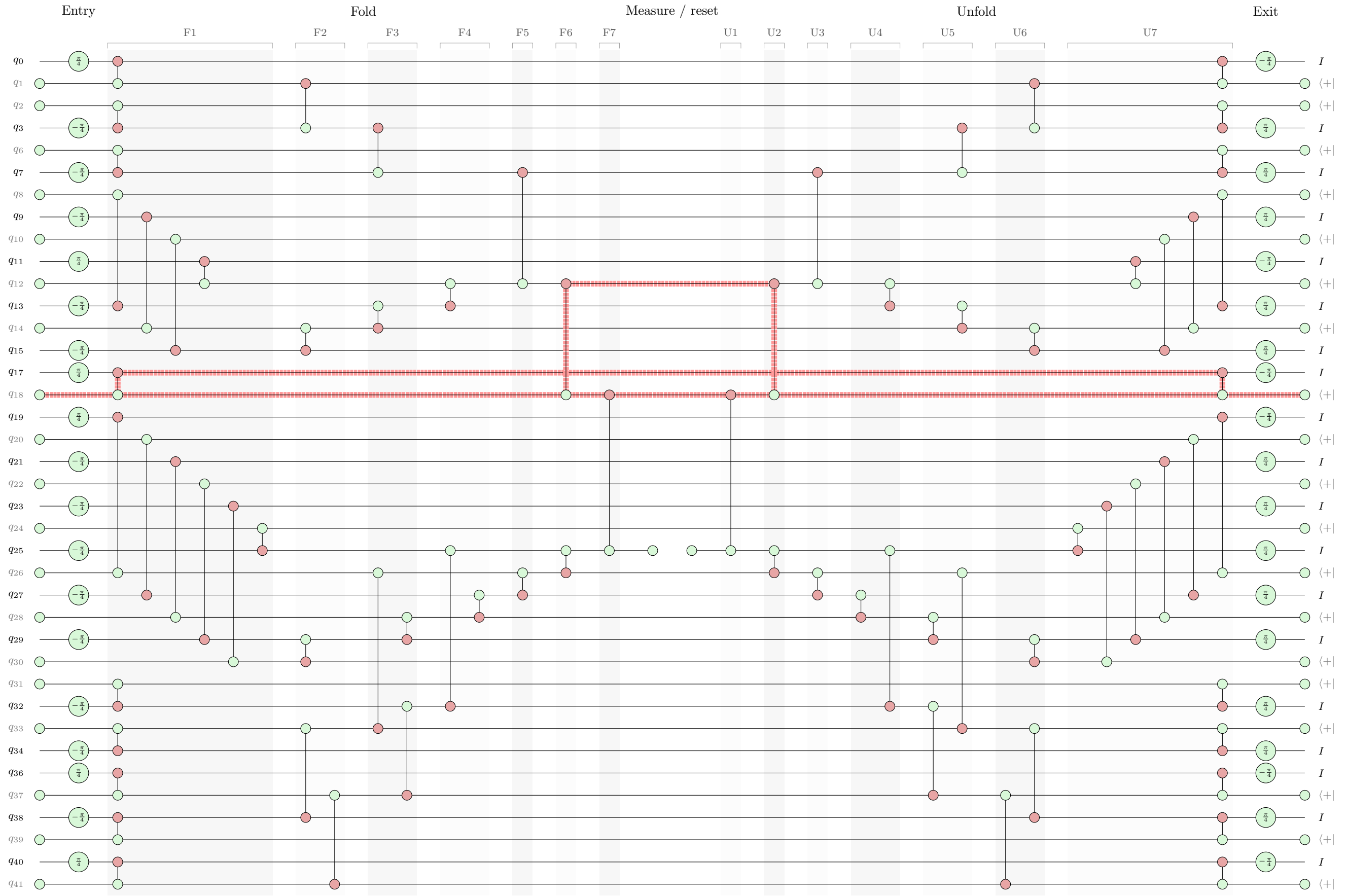
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 84

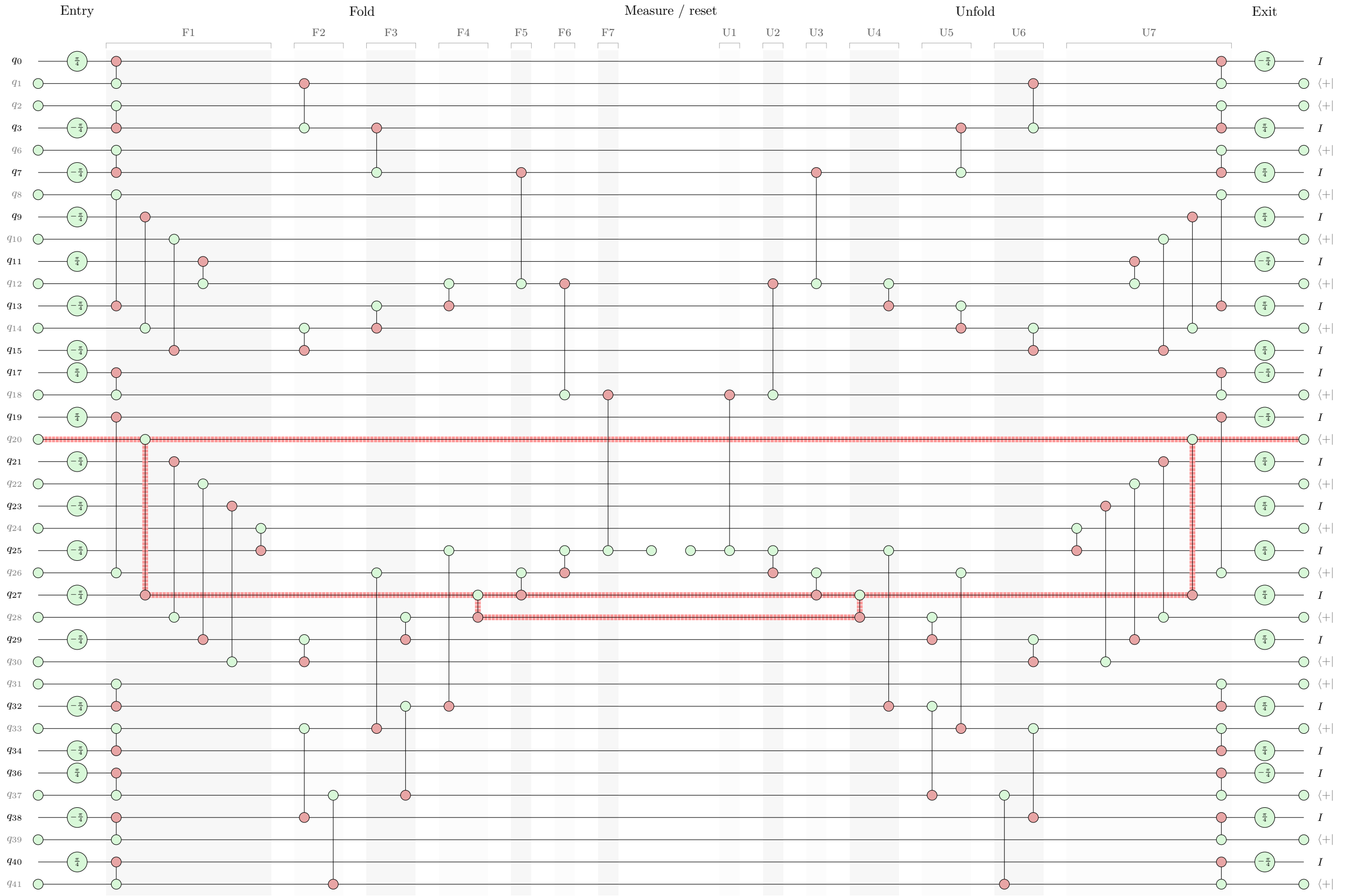
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 85

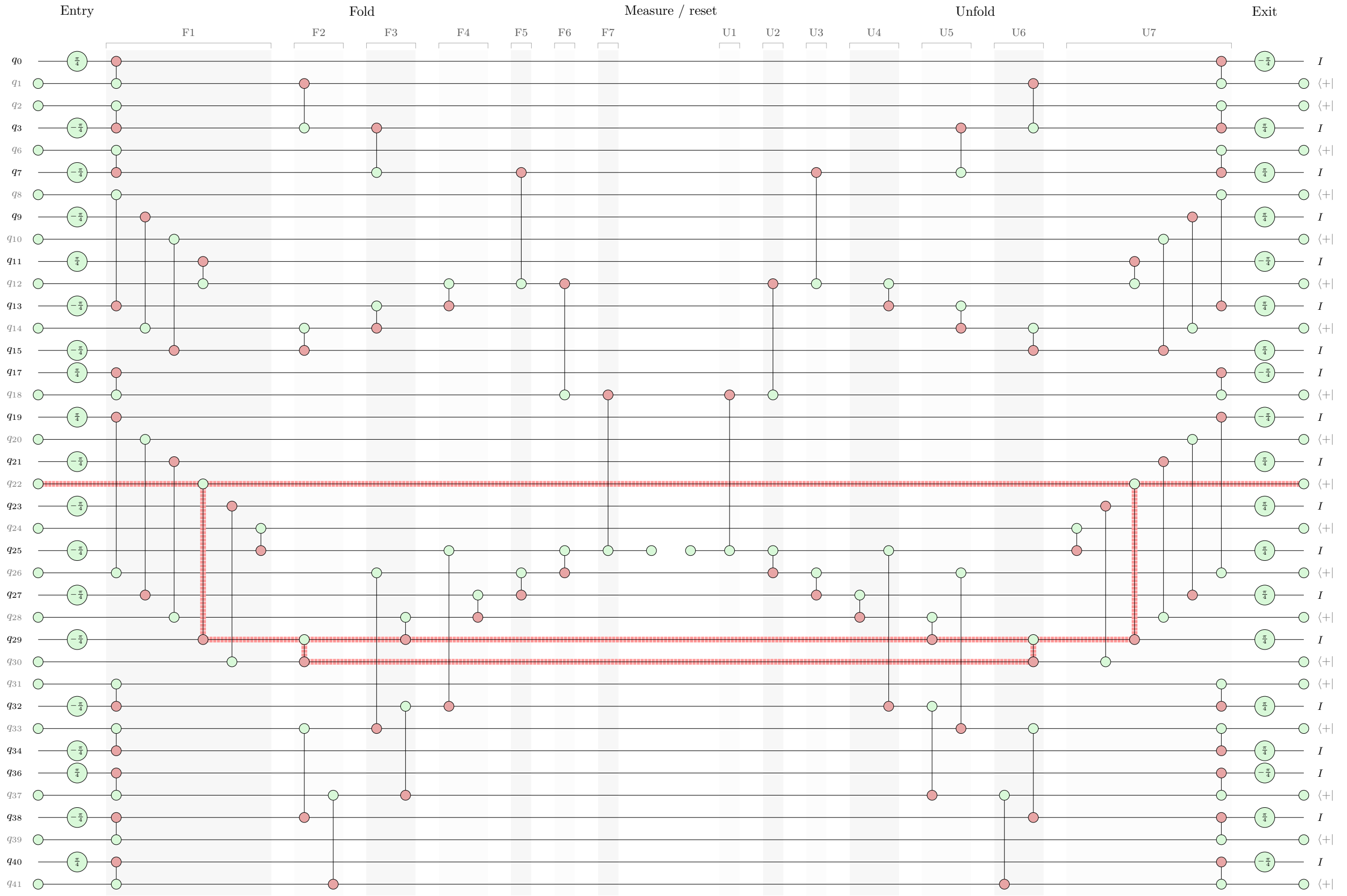
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 86

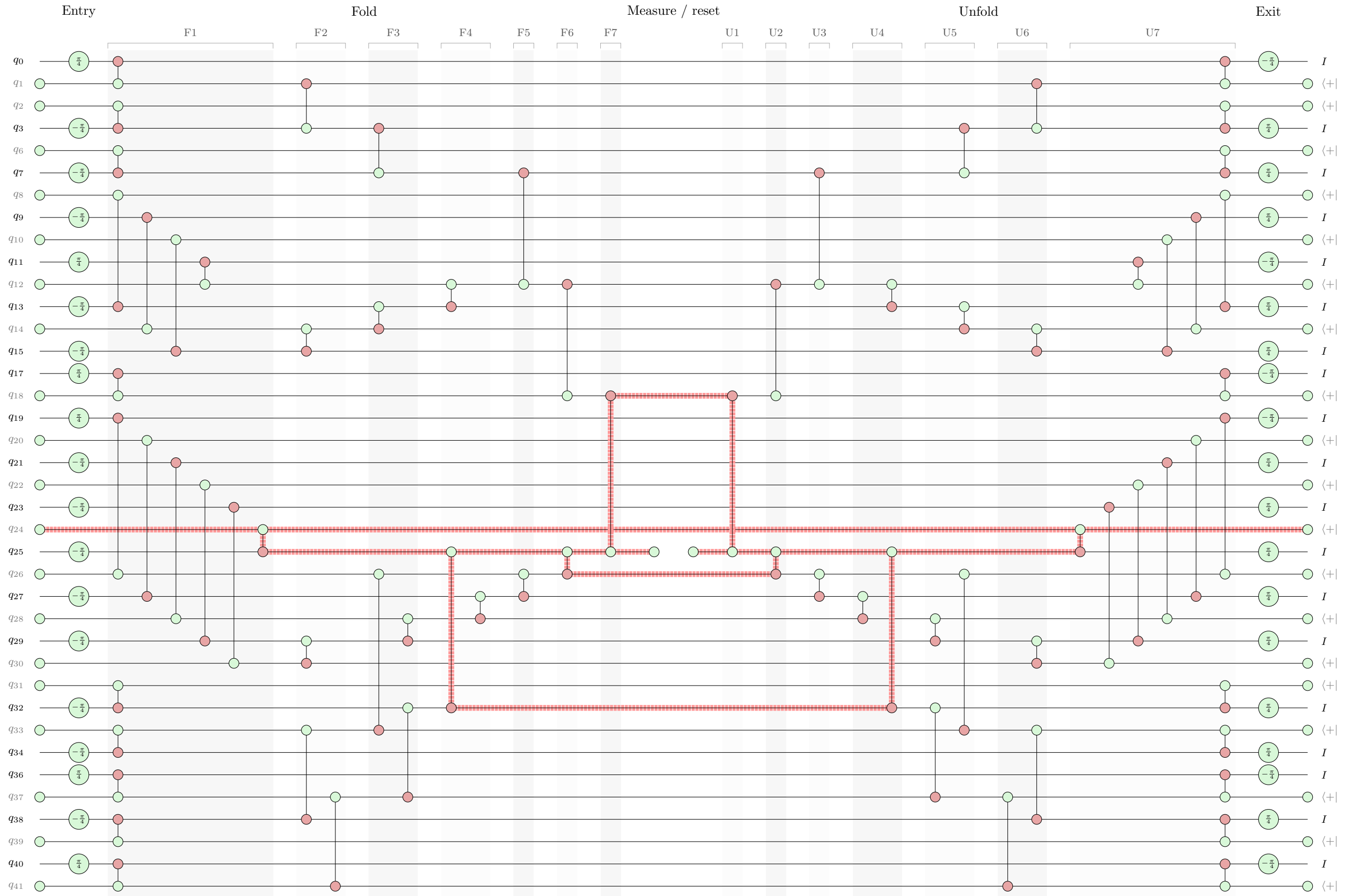
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 87

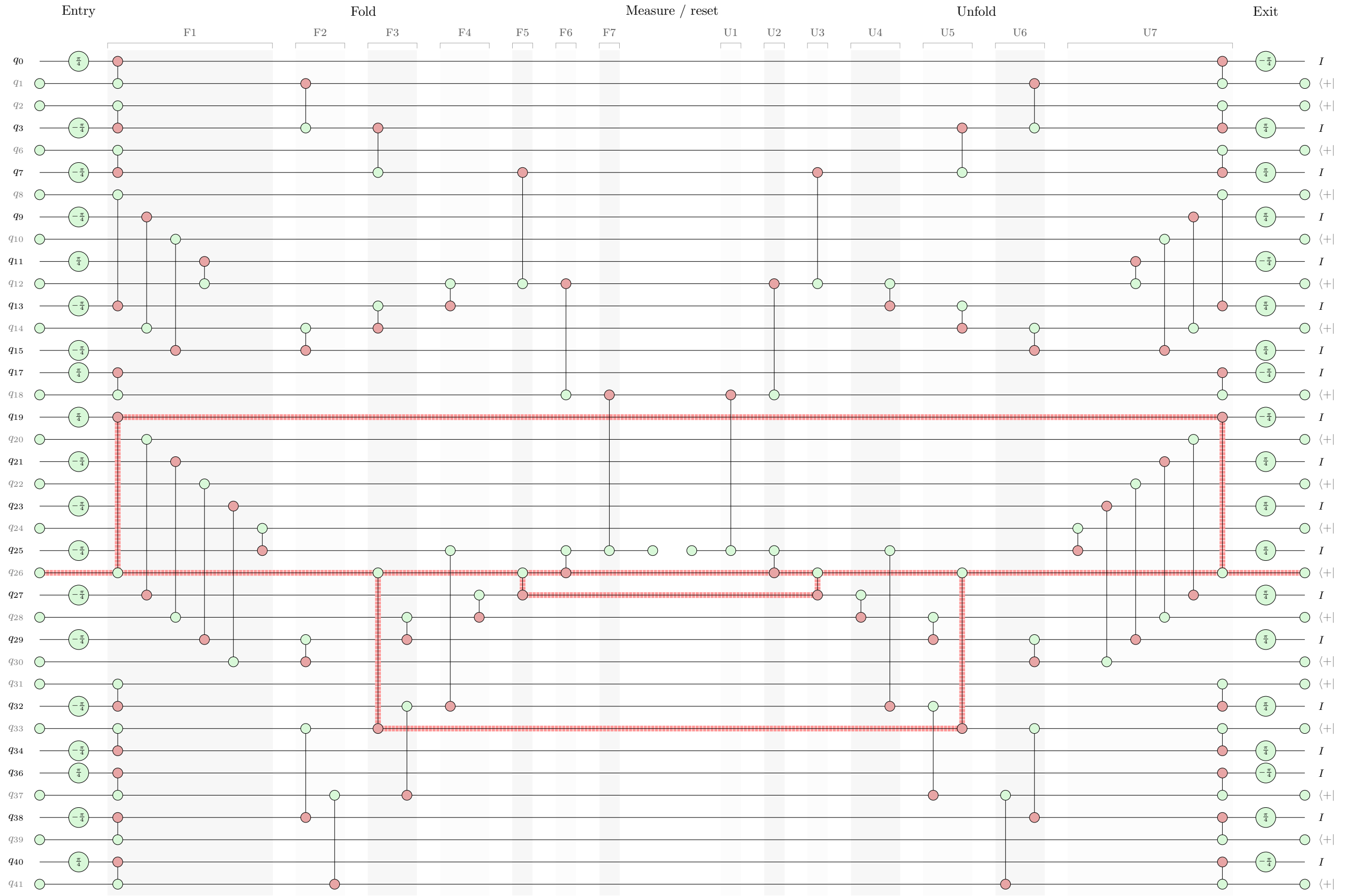
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 88

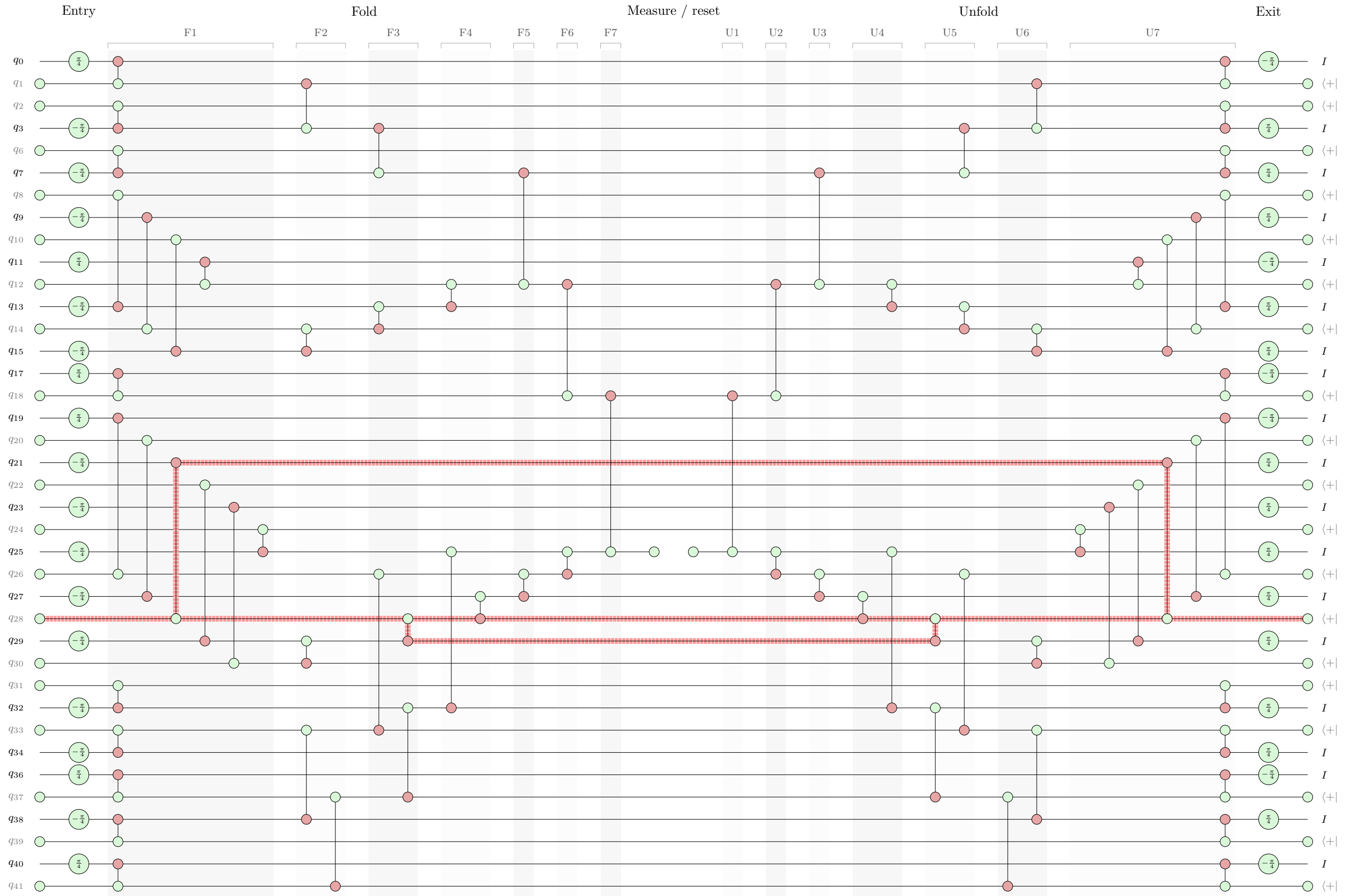
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 89

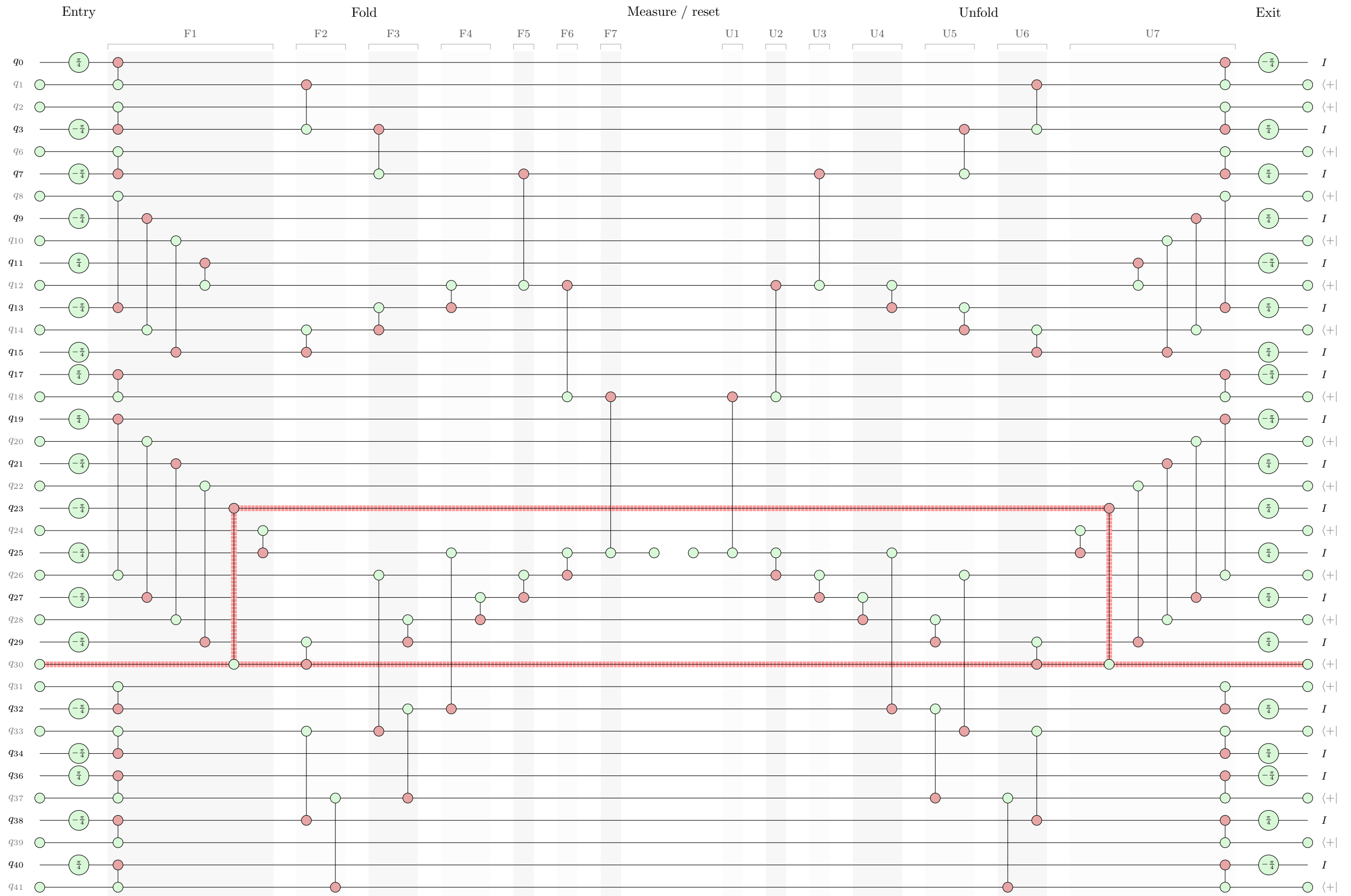
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 90

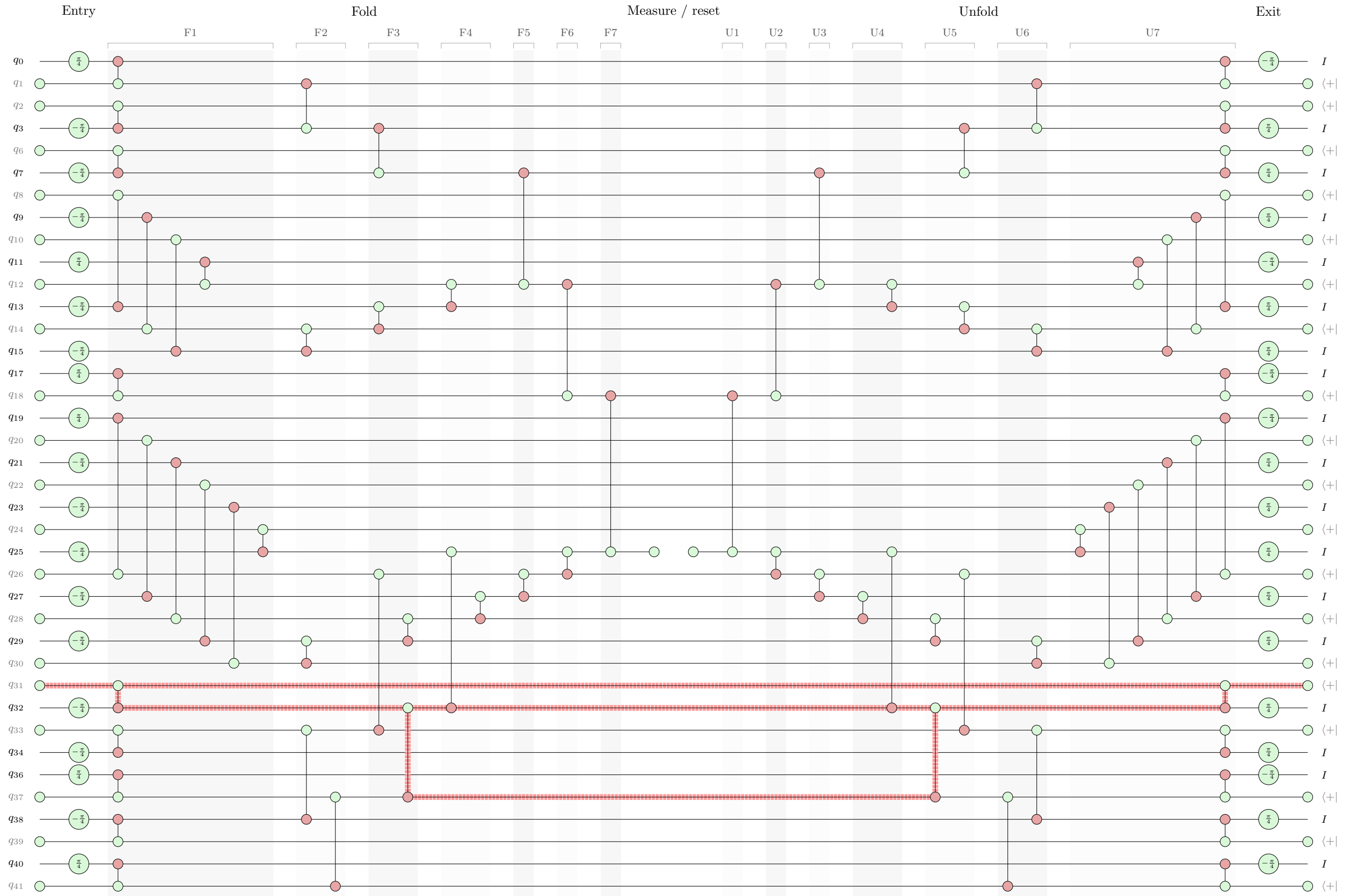
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 91

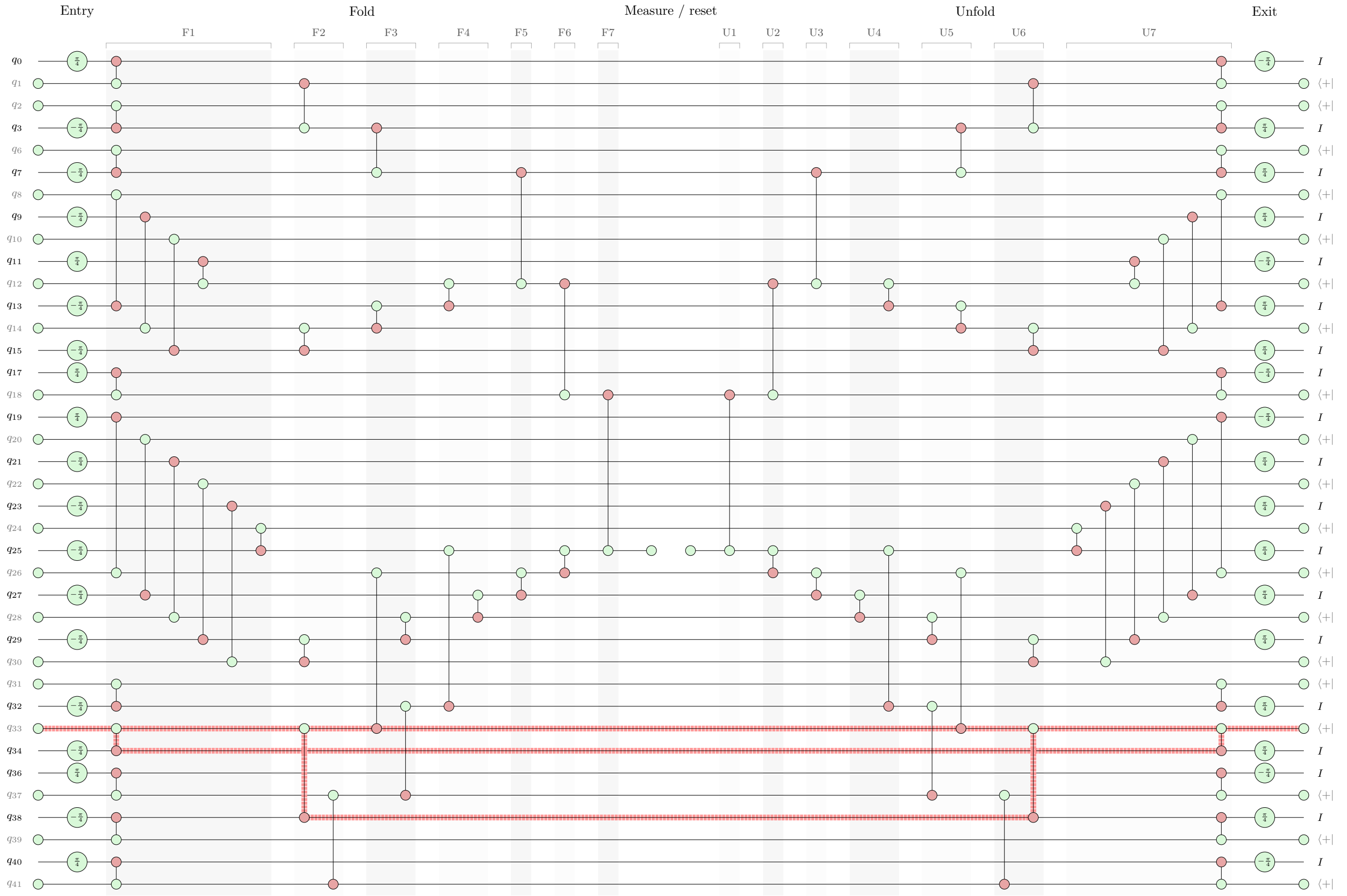
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 92

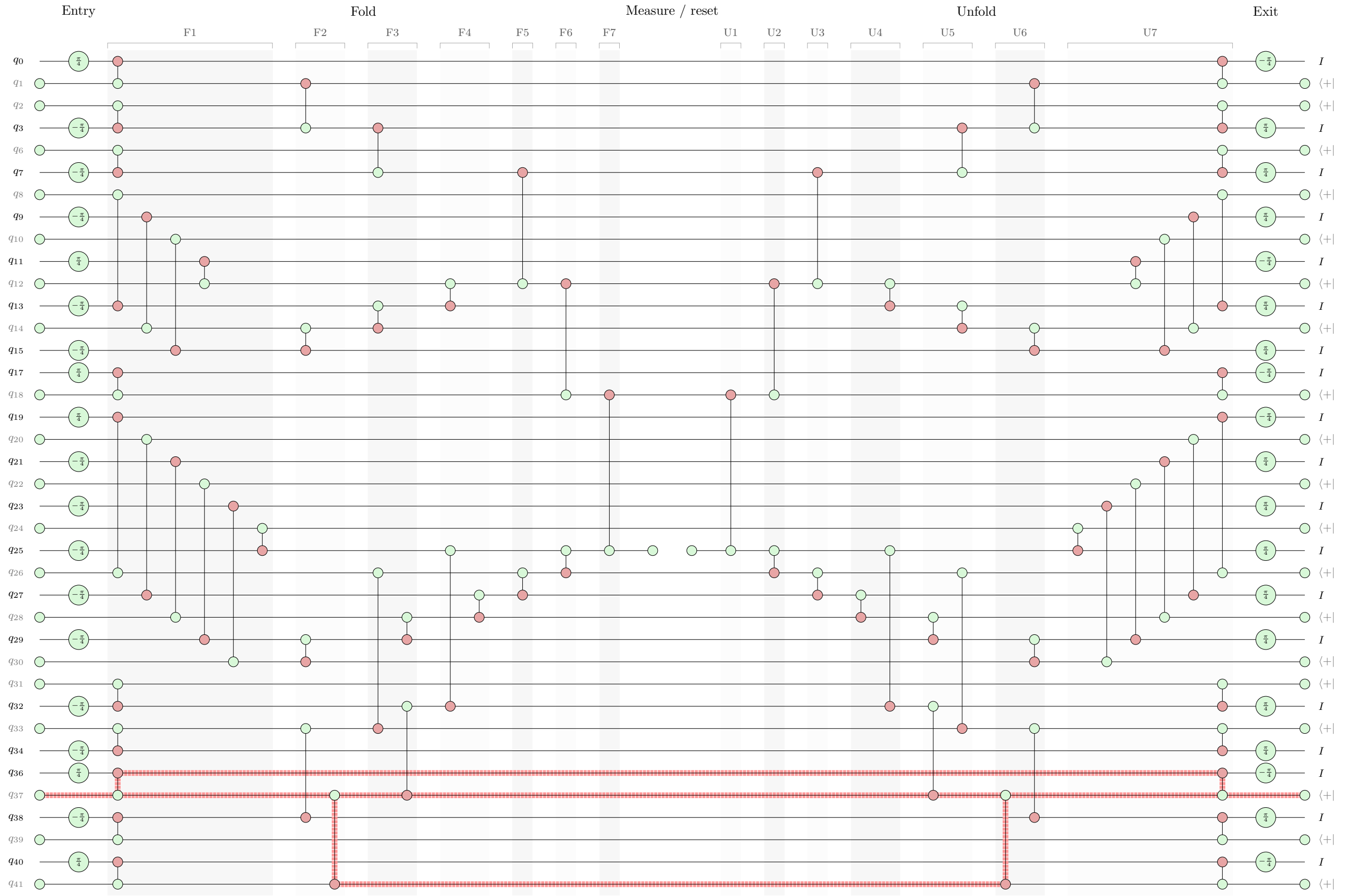
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 93

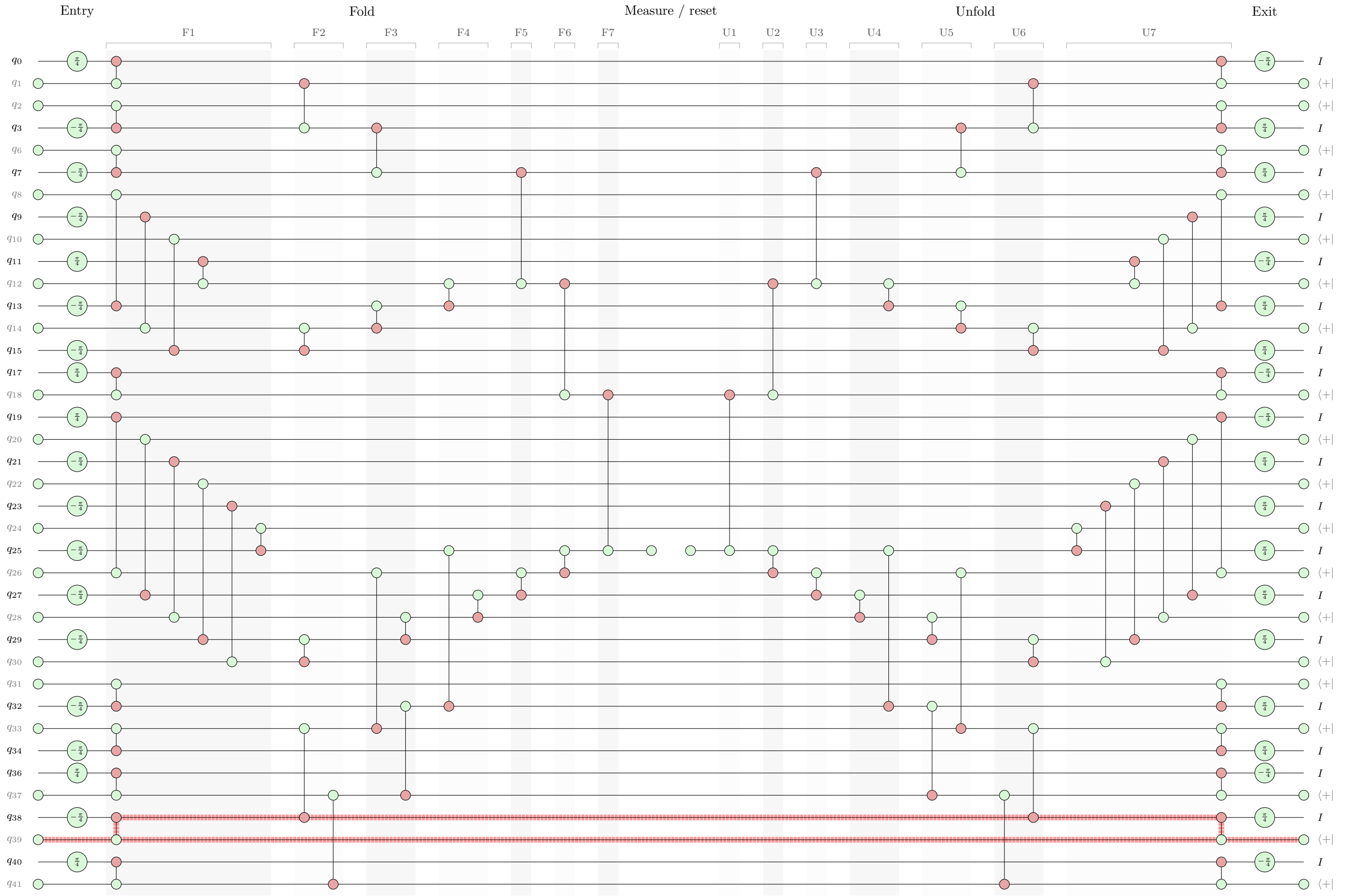
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 94

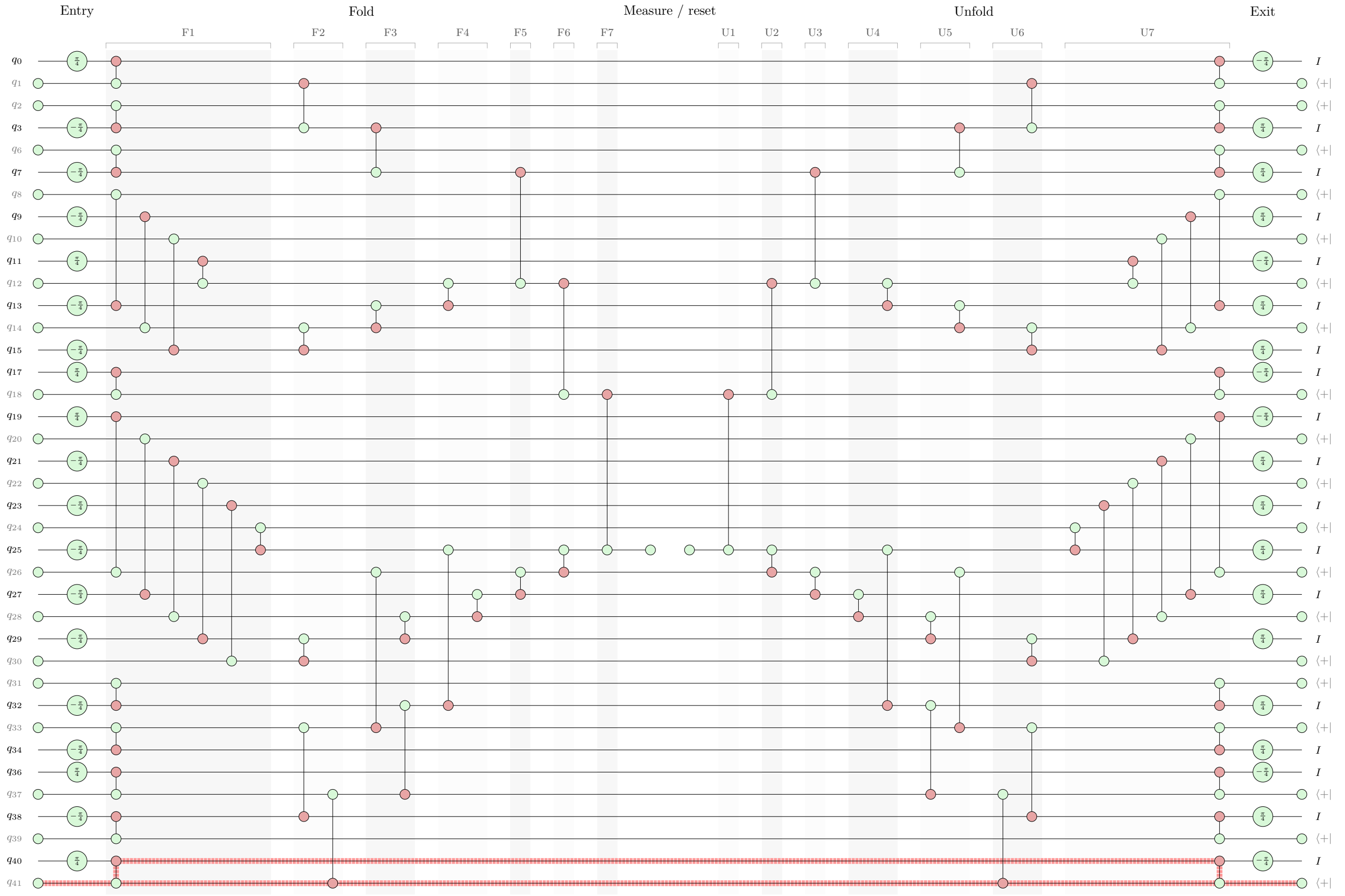
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 95

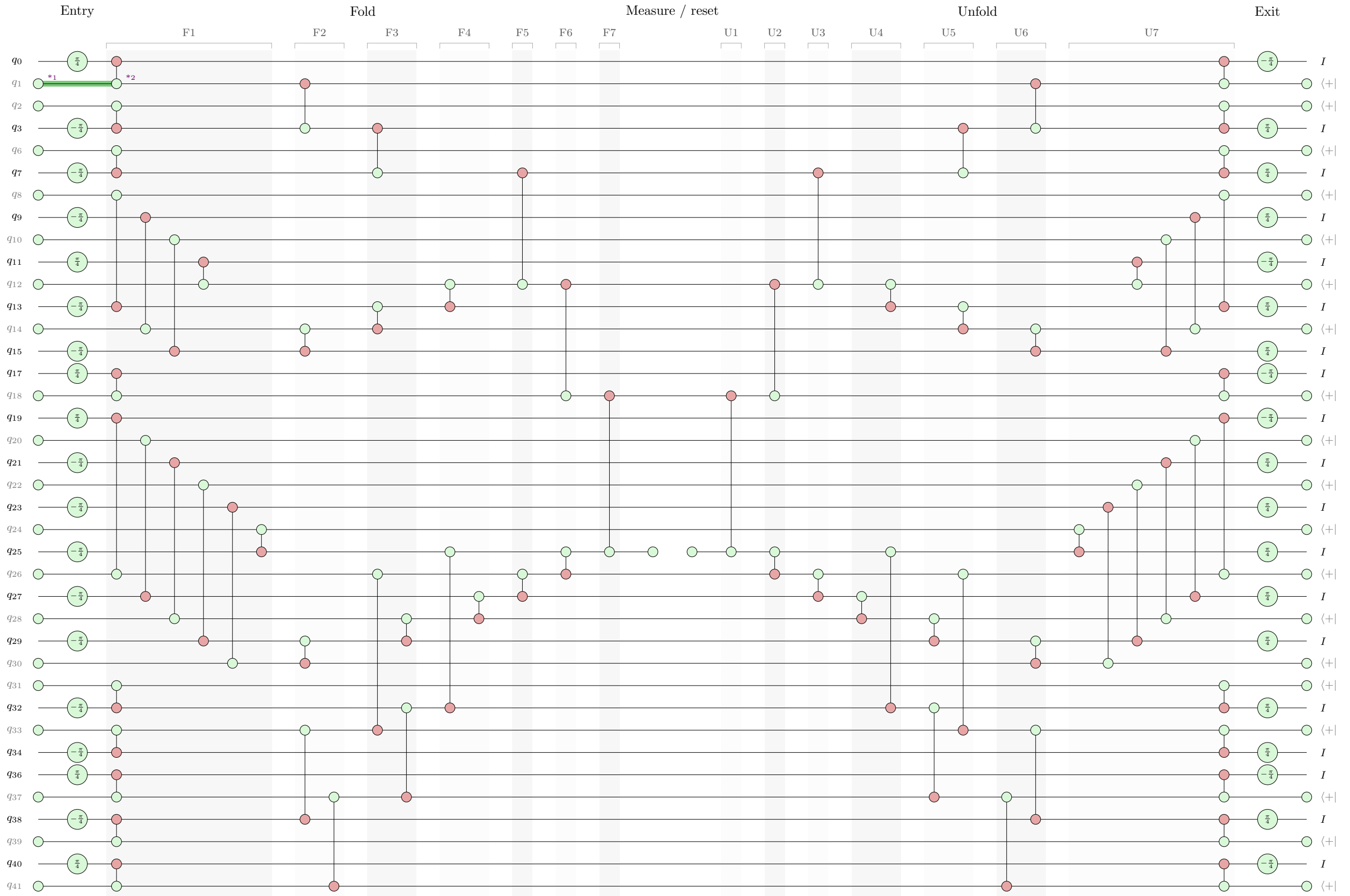
No boundary support. Pauli web: no defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

$d = 5$: basis element 96

No boundary support. 2 phase defects.



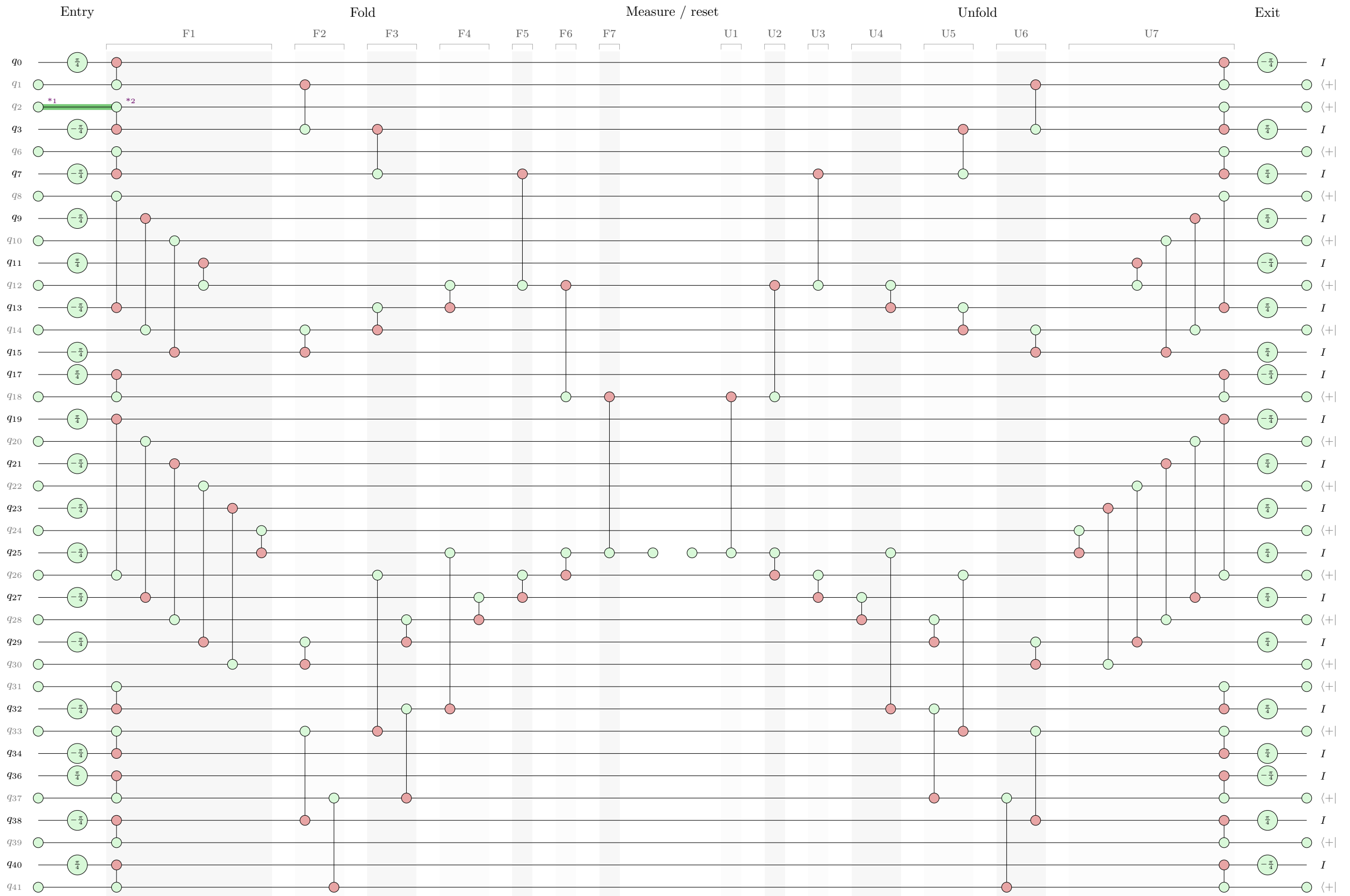
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_096.tikz

$d = 5$: basis element 97

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_097.tikz

$d = 5$: basis element 98

No boundary support. 2 phase defects.



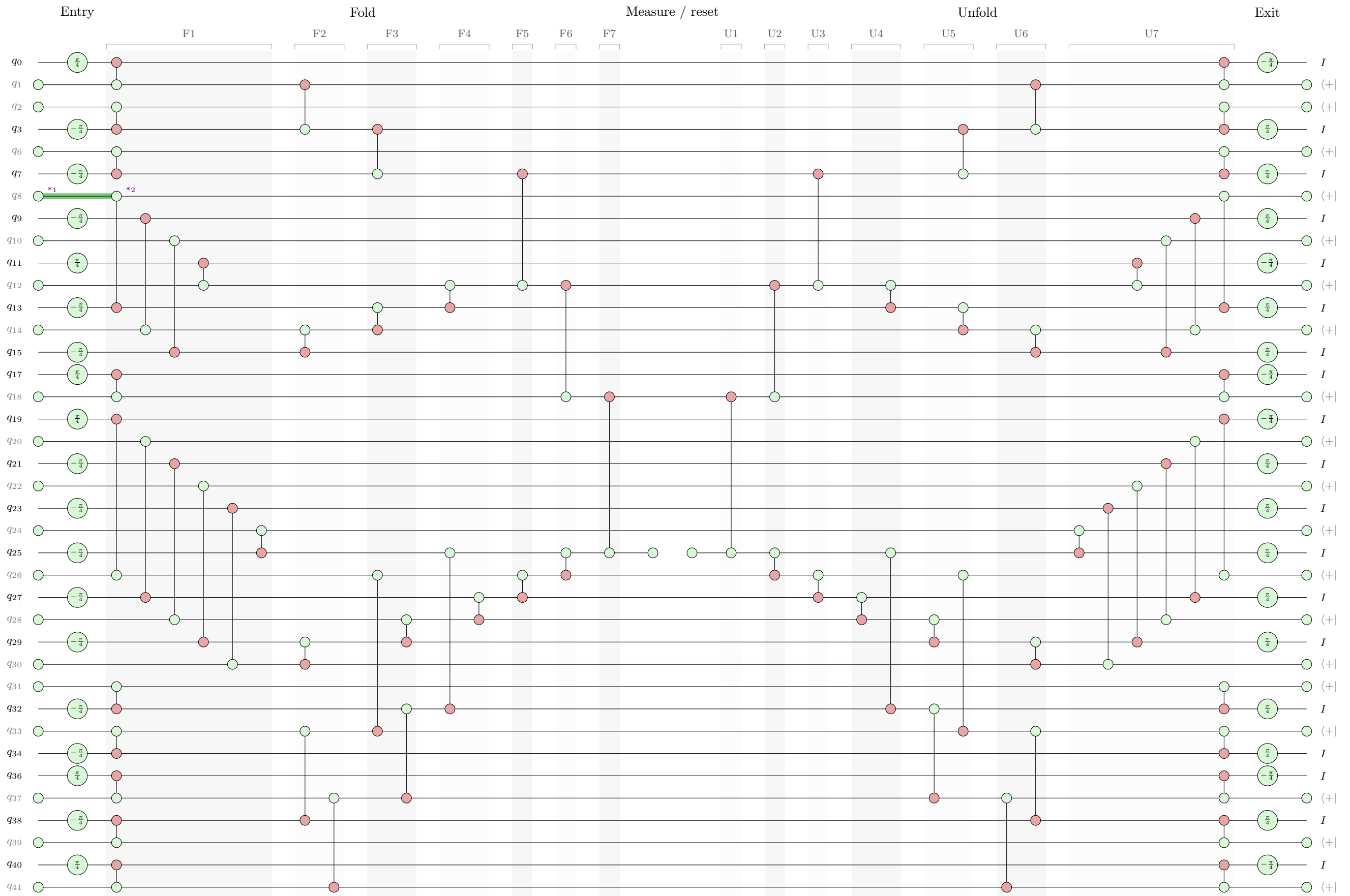
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_098.tikz

$d = 5$: basis element 99

No boundary support. 2 phase defects.



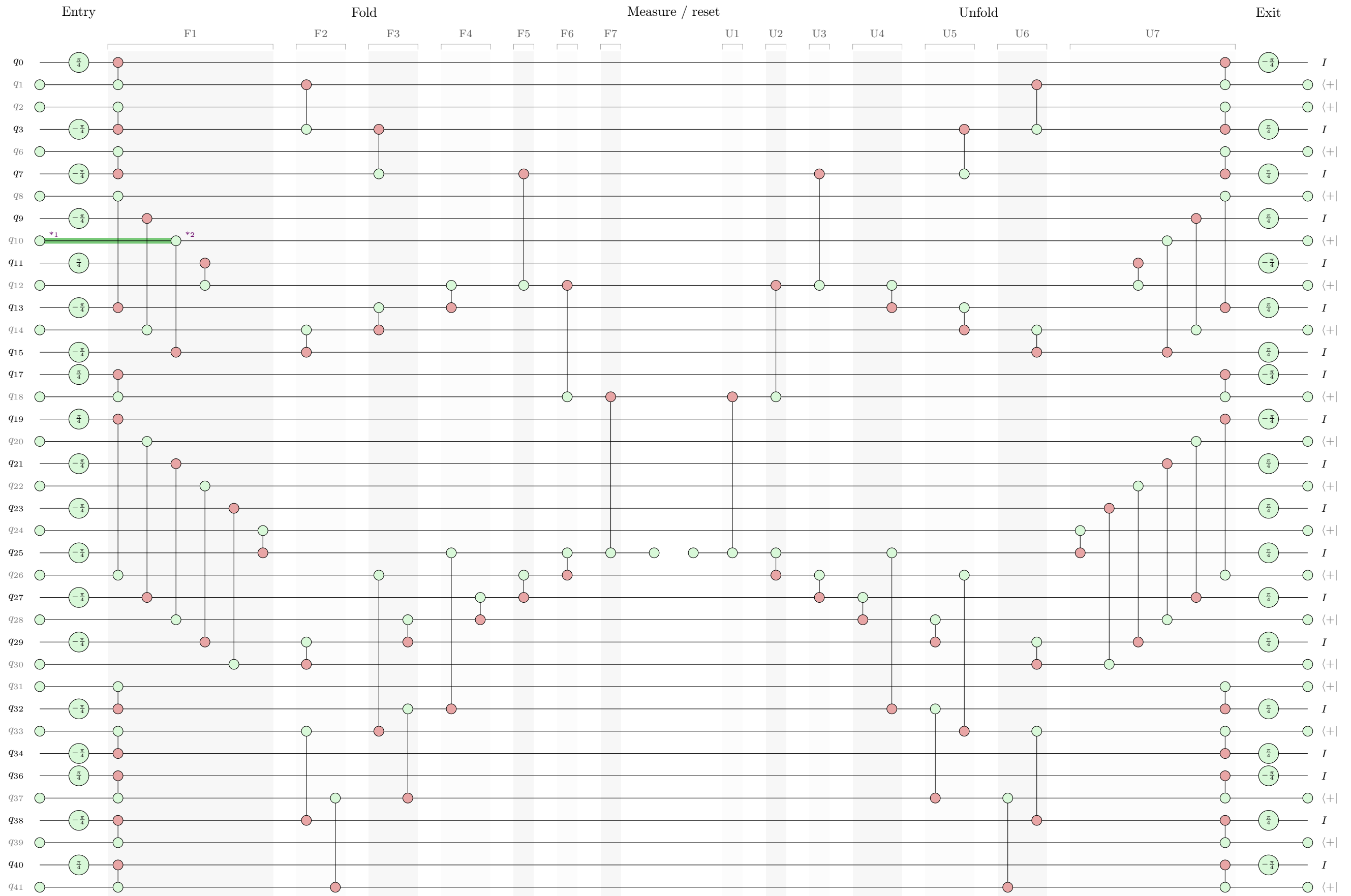
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_099.tikz

$d = 5$: basis element 100

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_100.tikz

$d = 5$: basis element 101

No boundary support. 2 phase defects.



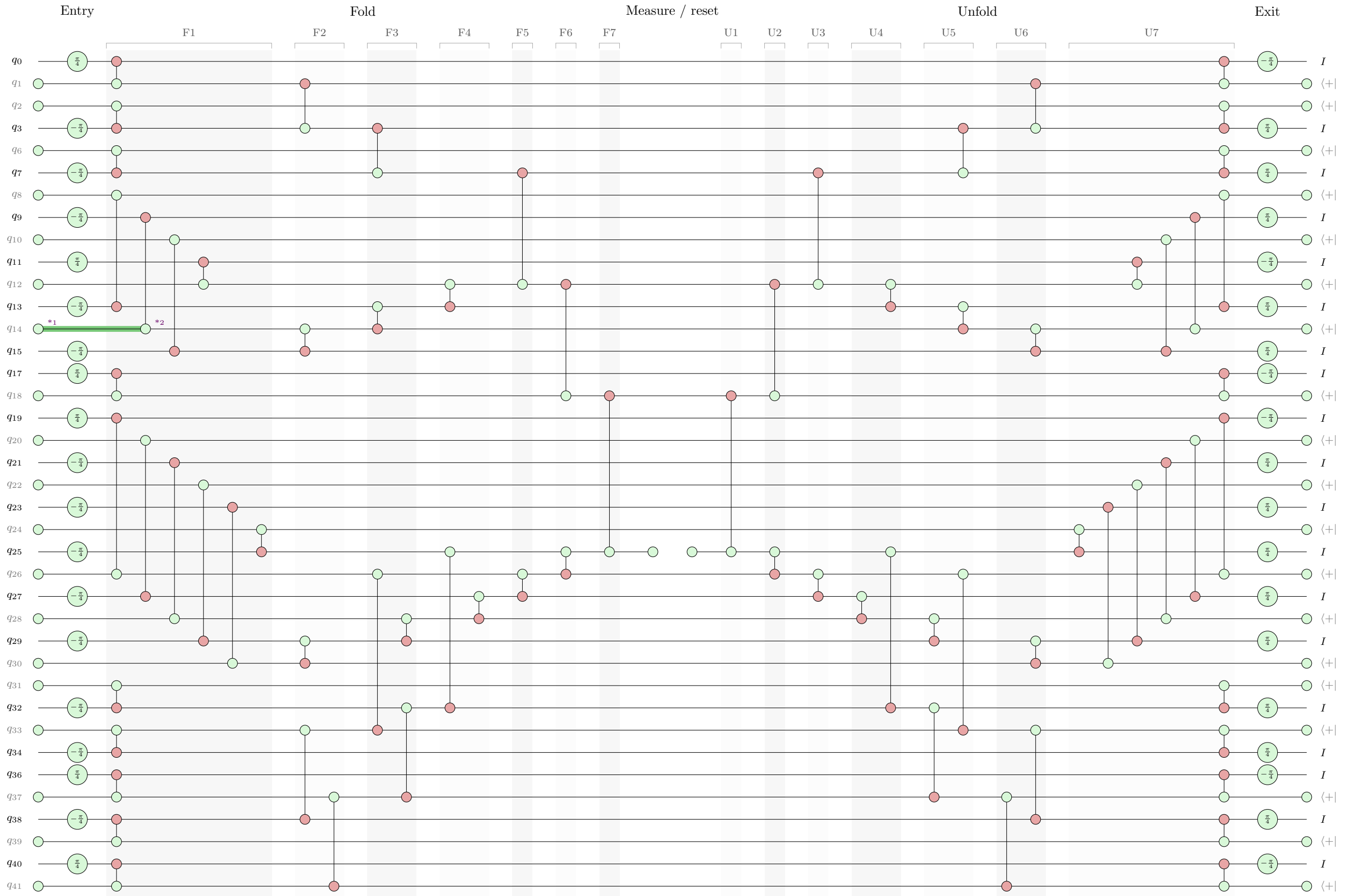
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_101.tikz

$d = 5$: basis element 102

No boundary support. 2 phase defects.

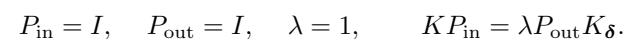


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_102.tikz

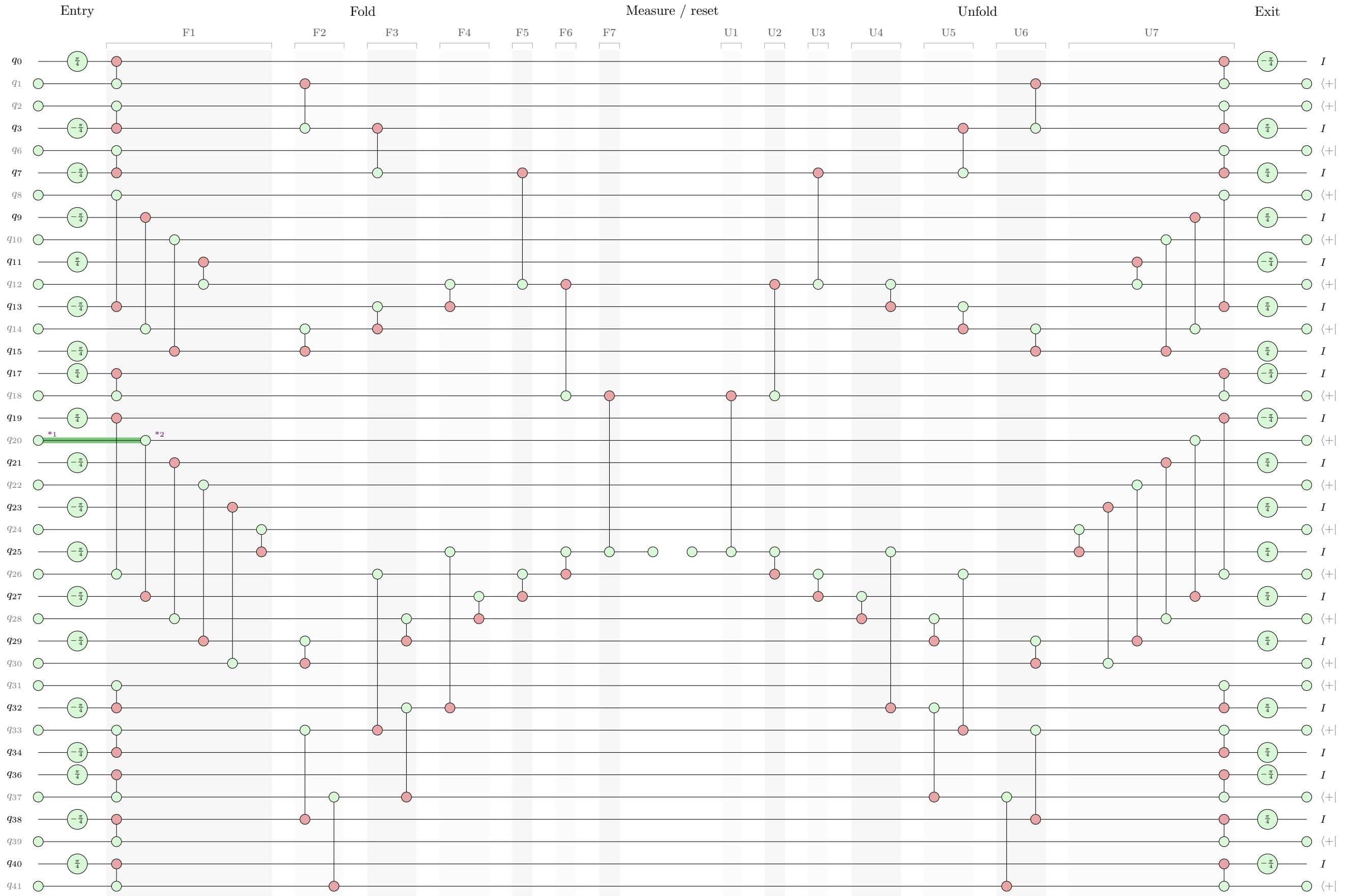
No boundary support. 2 phase defects.



d5_103.tikz

$d = 5$: basis element 104

No boundary support. 2 phase defects.



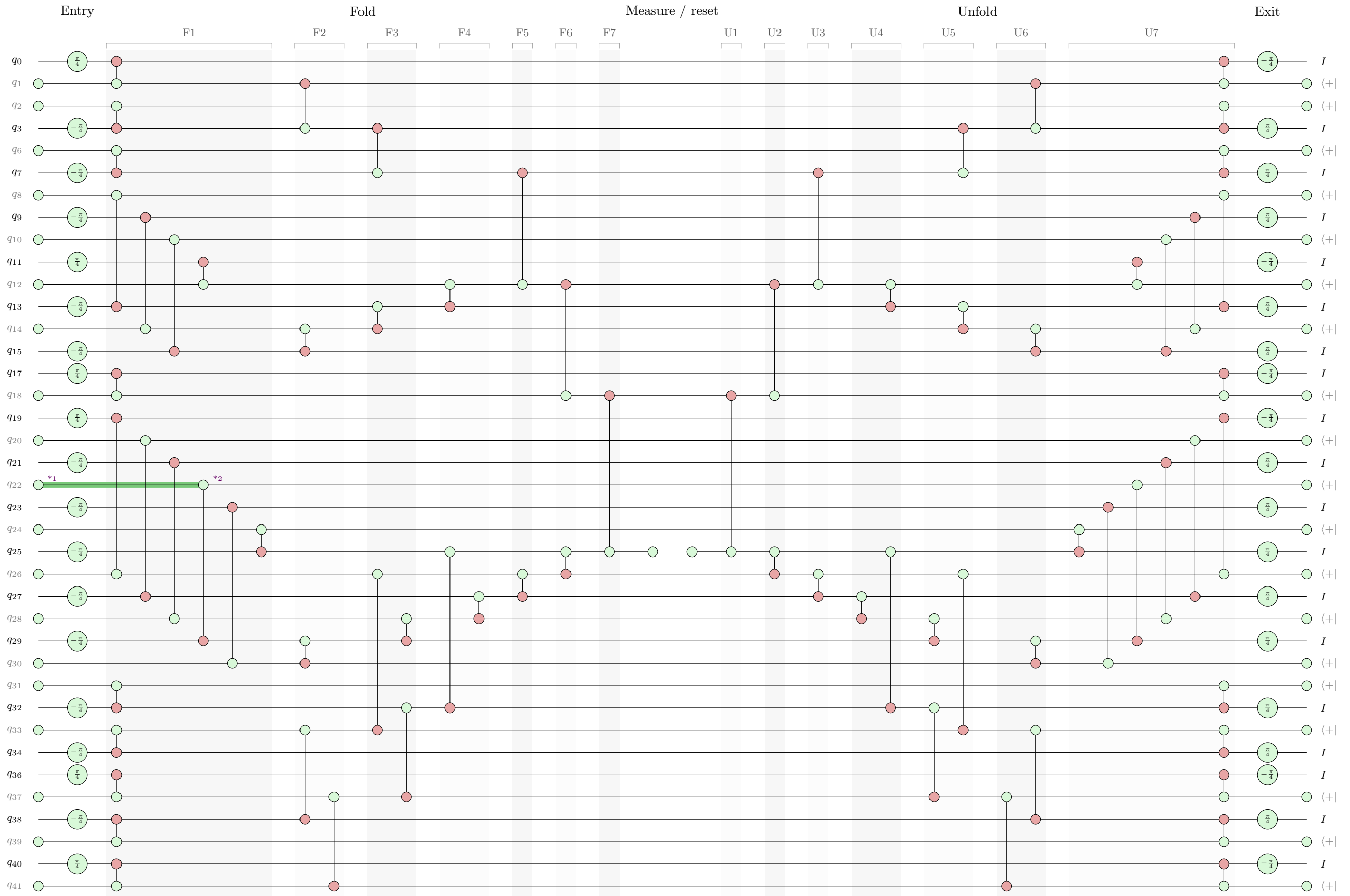
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_104.tikz

$d = 5$: basis element 105

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_105.tikz

$d = 5$: basis element 106

No boundary support. 2 phase defects.



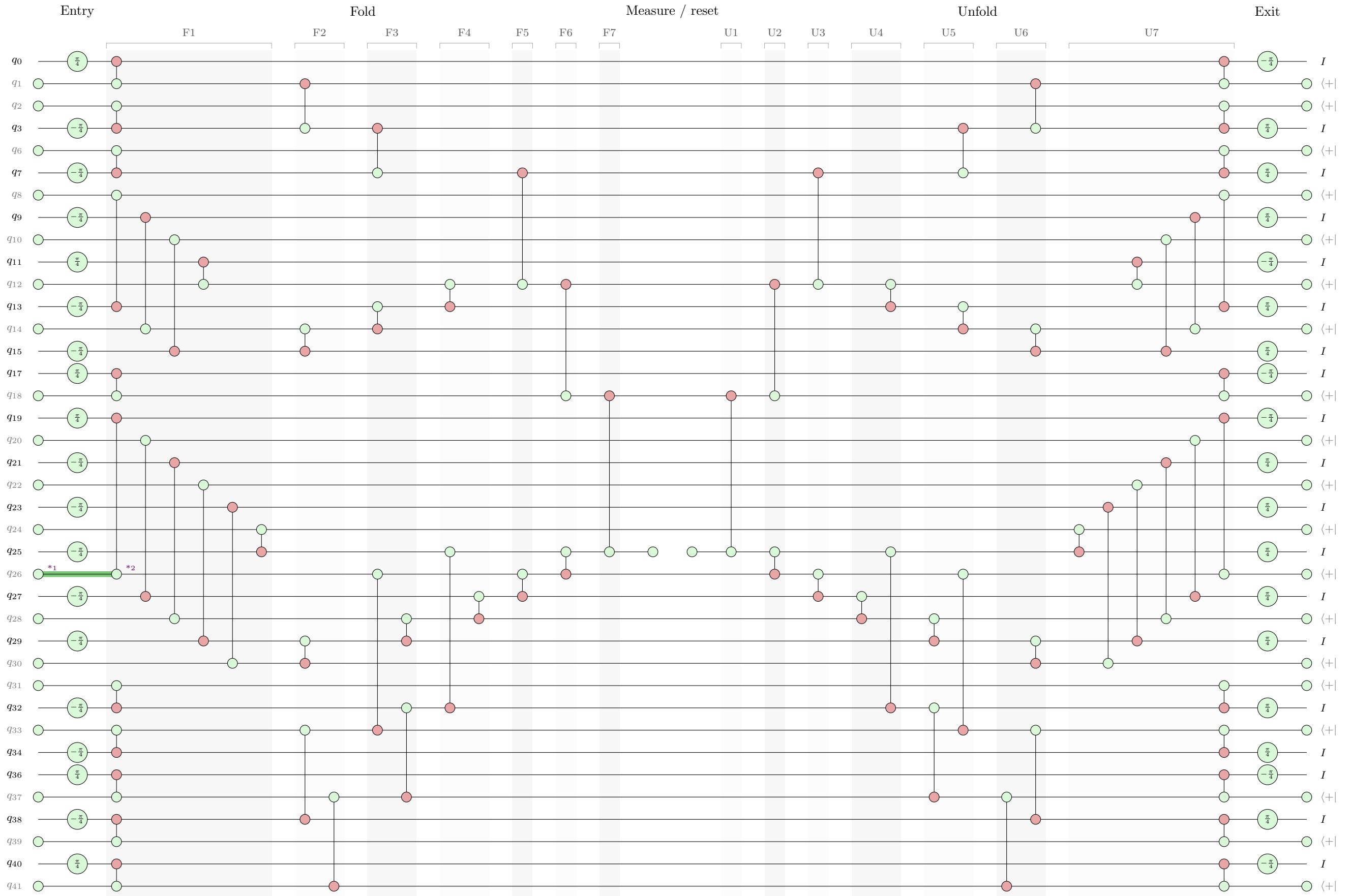
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_106.tikz

$d = 5$: basis element 107

No boundary support. 2 phase defects.



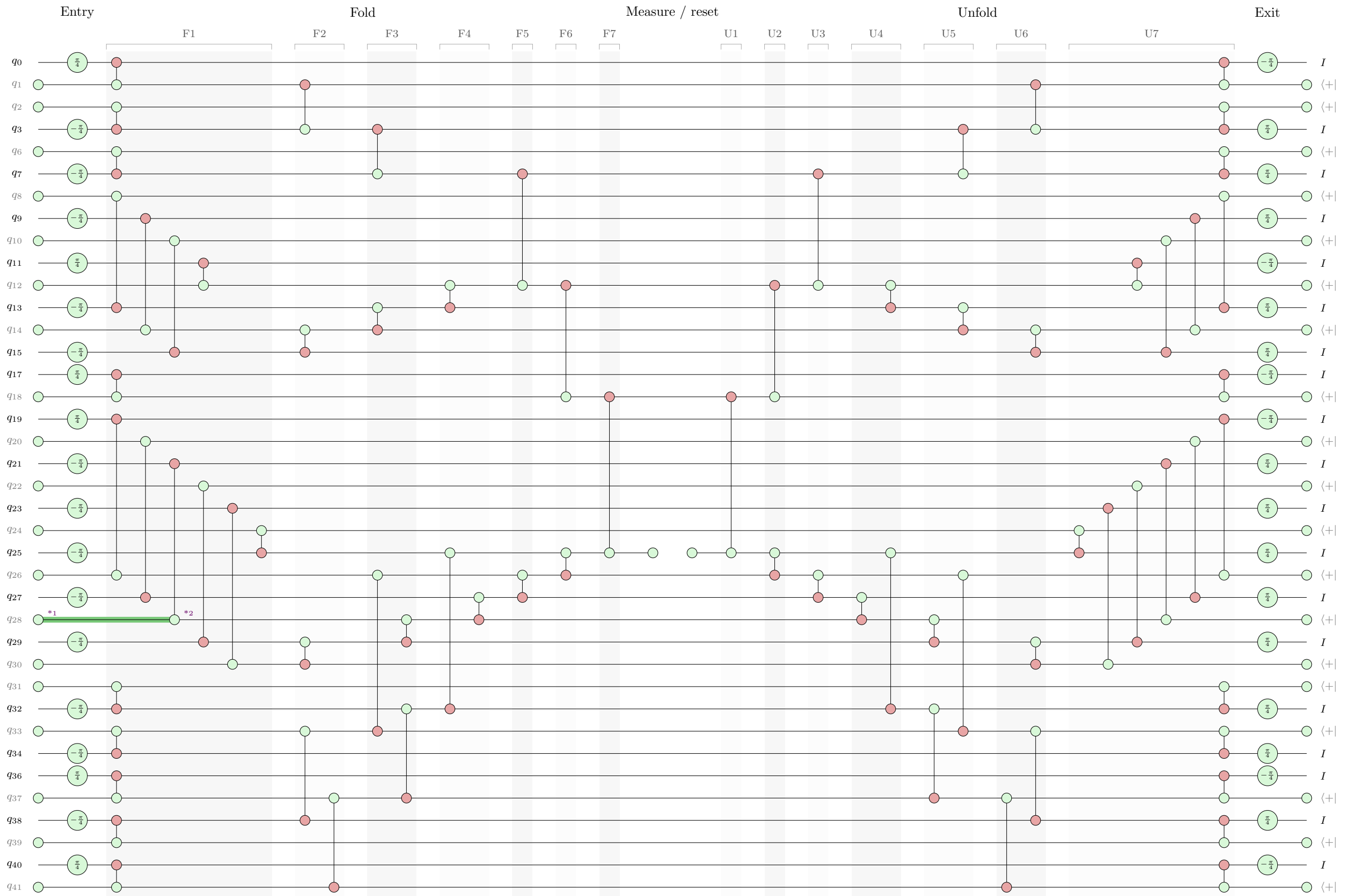
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_107.tikz

$d = 5$: basis element 108

No boundary support. 2 phase defects.



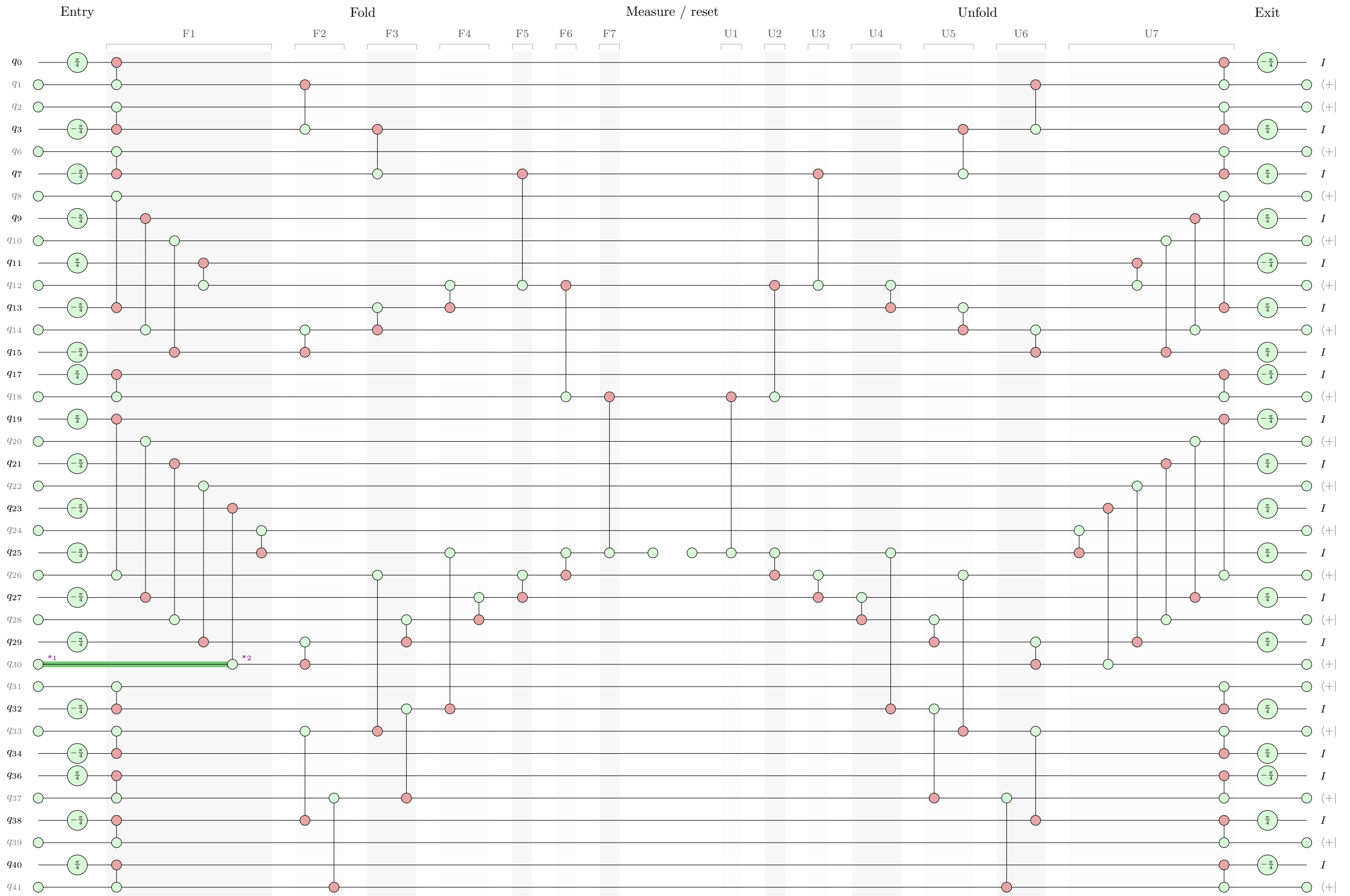
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_108.tikz

$d = 5$: basis element 109

No boundary support. 2 phase defects.



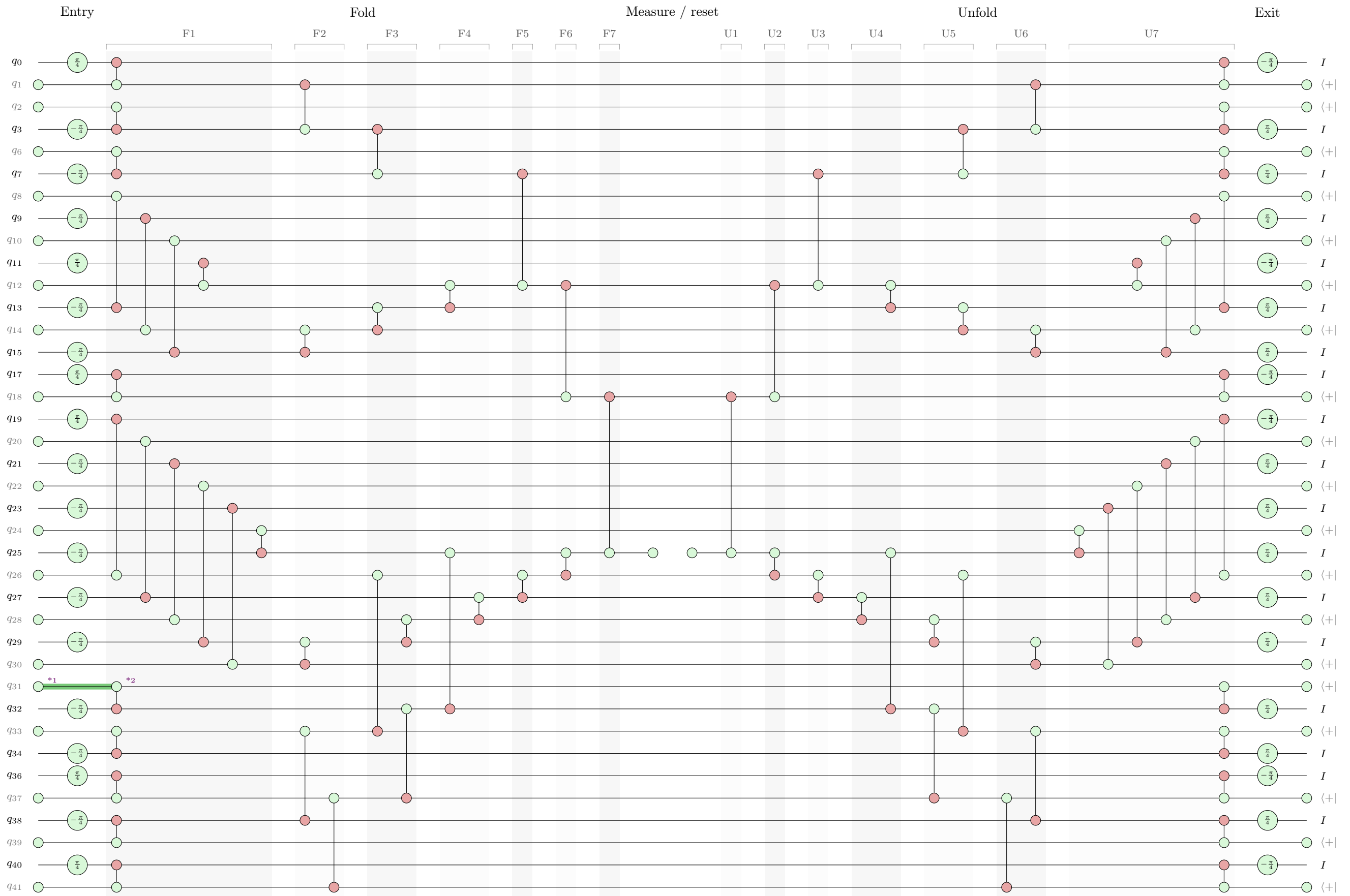
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_109.tikz

$d = 5$: basis element 110

No boundary support. 2 phase defects.



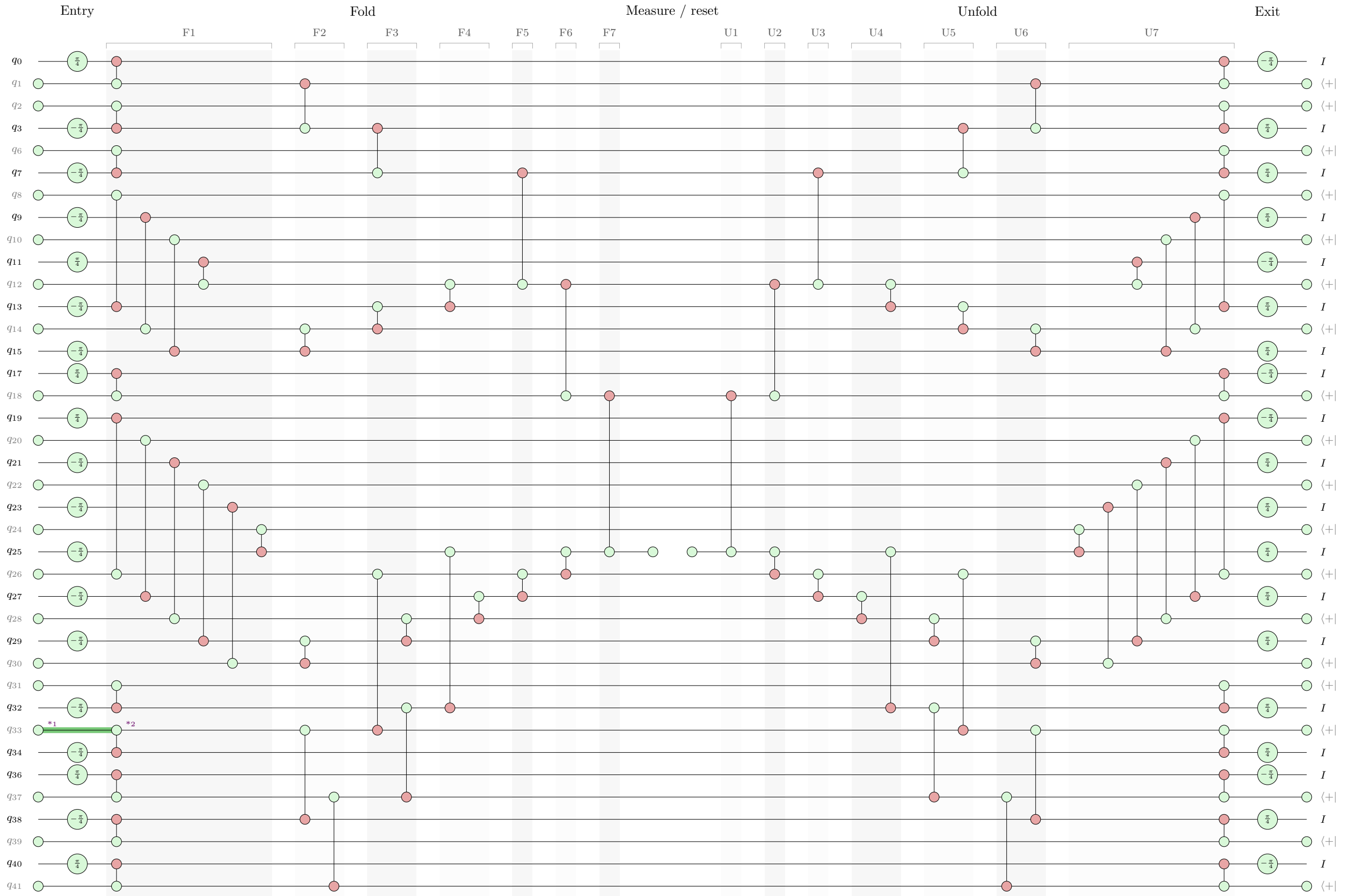
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_110.tikz

$d = 5$: basis element 111

No boundary support. 2 phase defects.



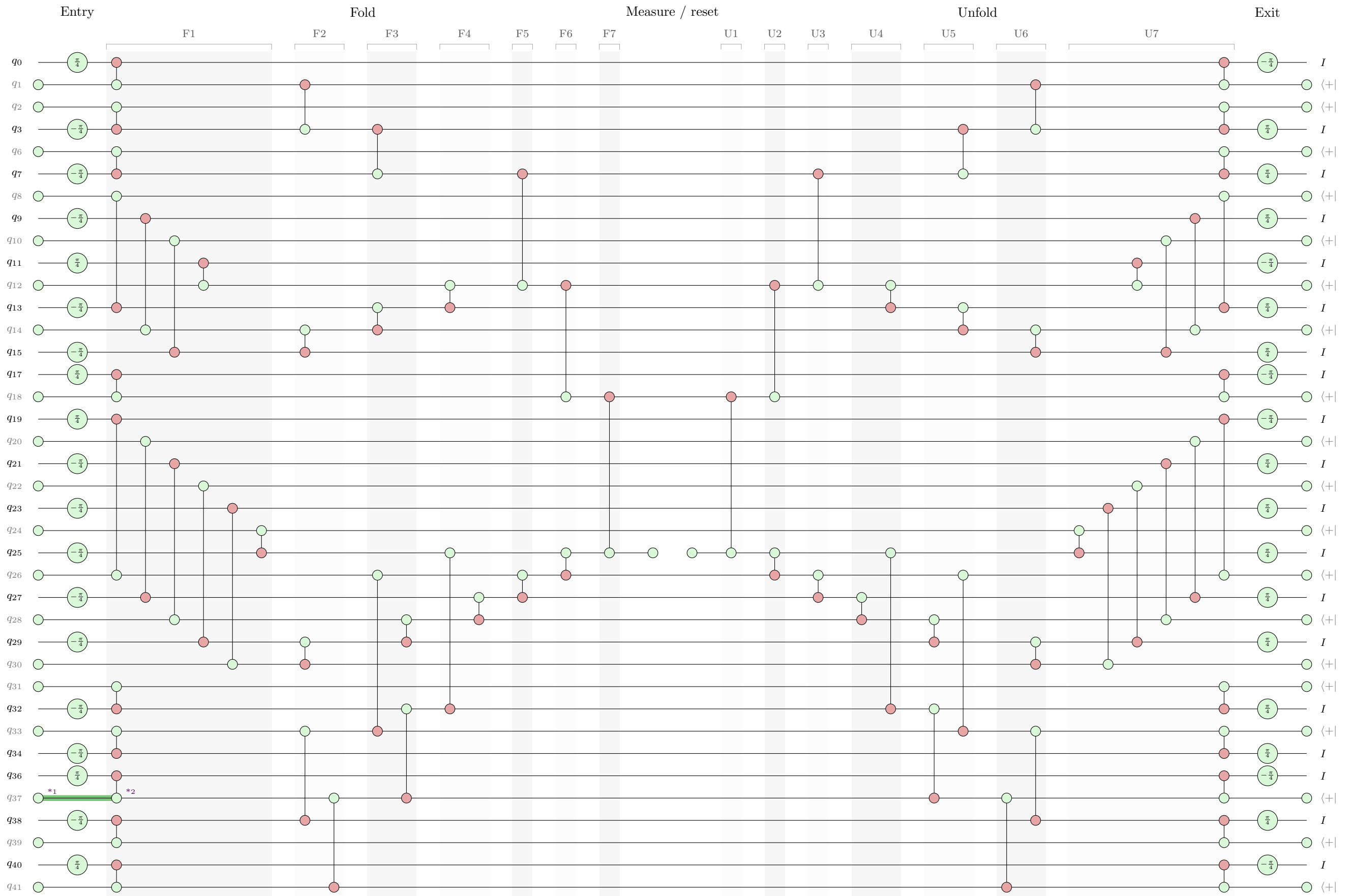
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_111.tikz

$d = 5$: basis element 112

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_112.tikz

$d = 5$: basis element 113

No boundary support. 2 phase defects.



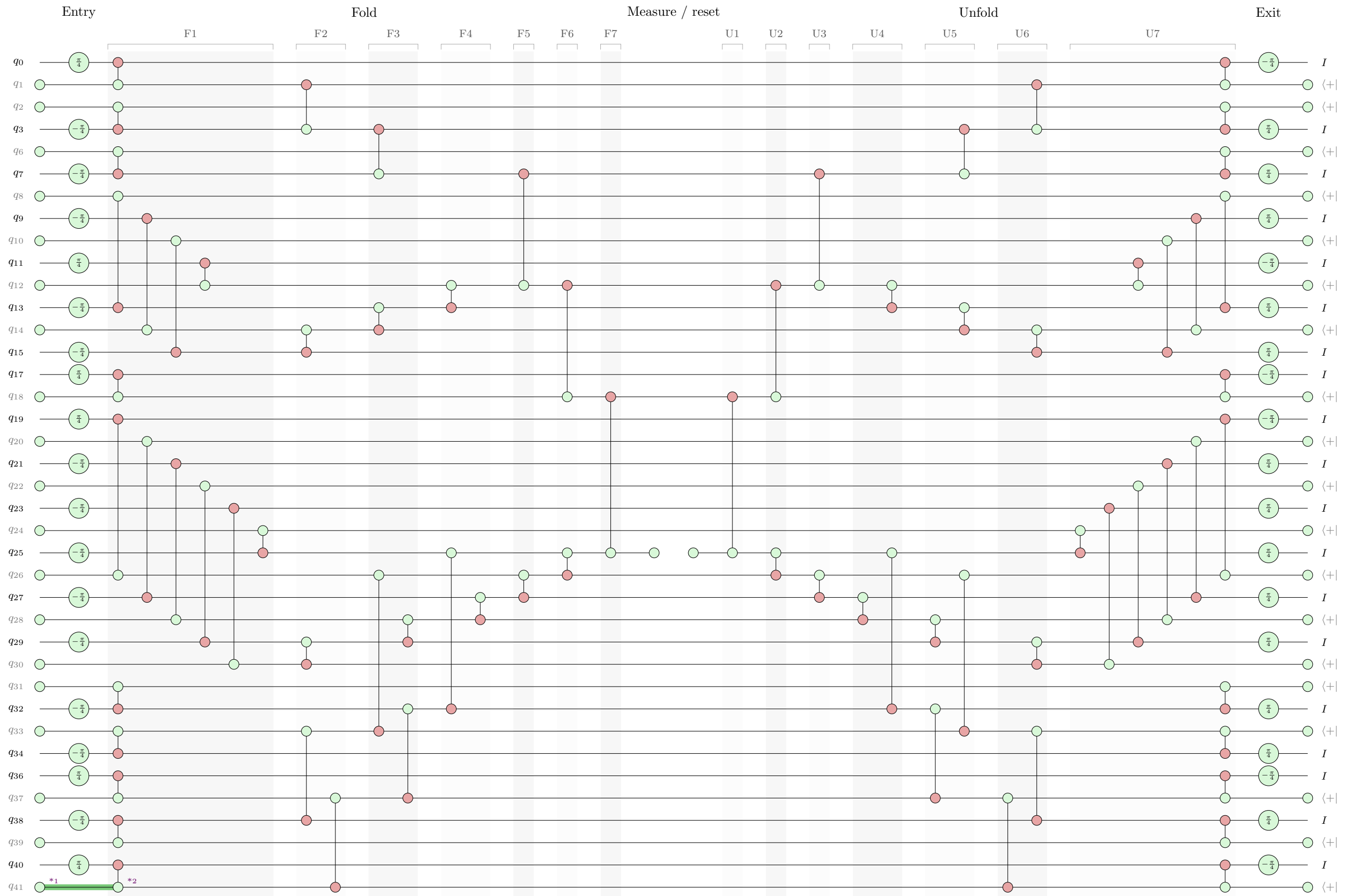
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_113.tikz

$d = 5$: basis element 114

No boundary support. 2 phase defects.



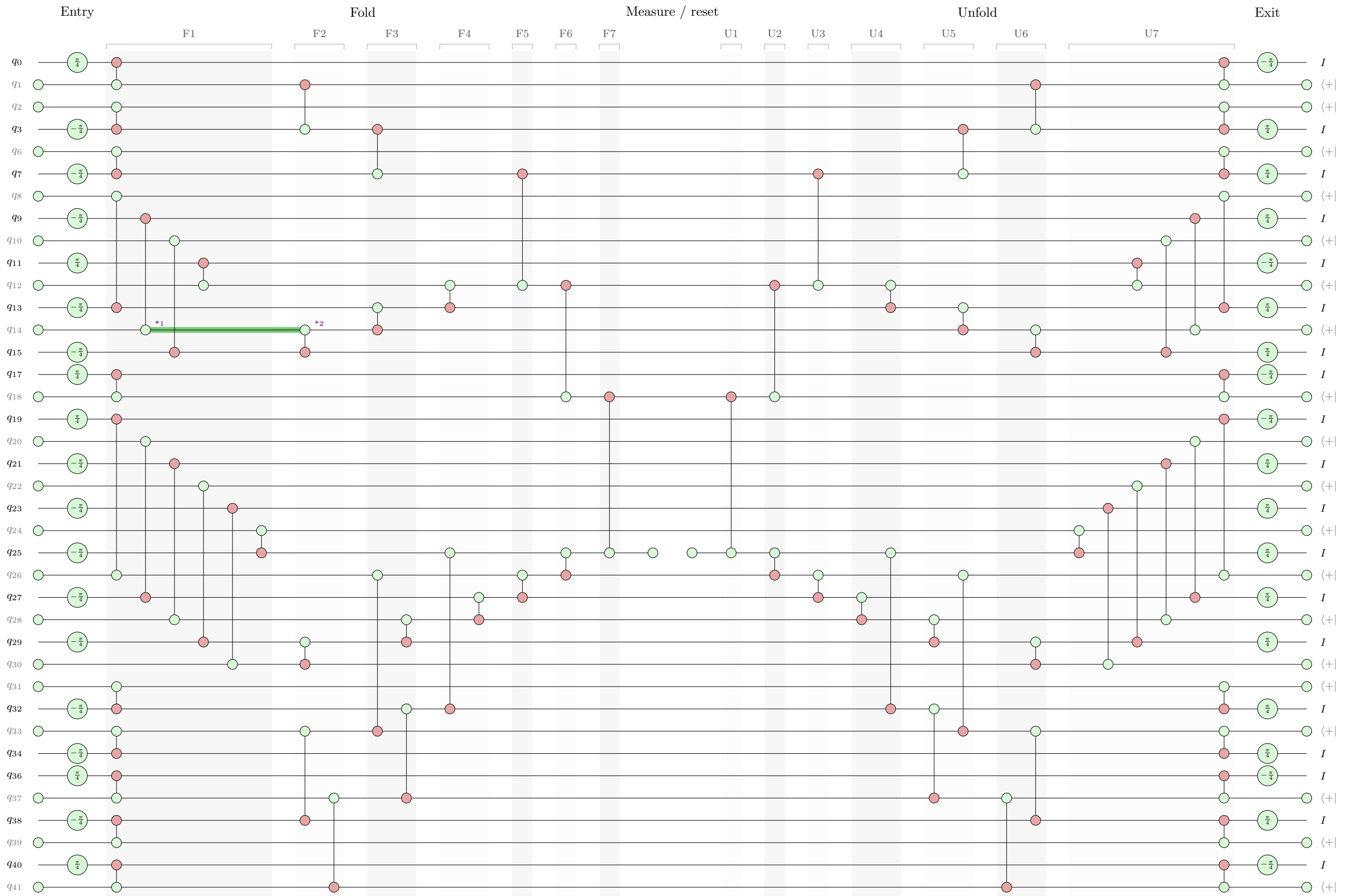
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_114.tikz

$d = 5$: basis element 115

No boundary support. 2 phase defects.



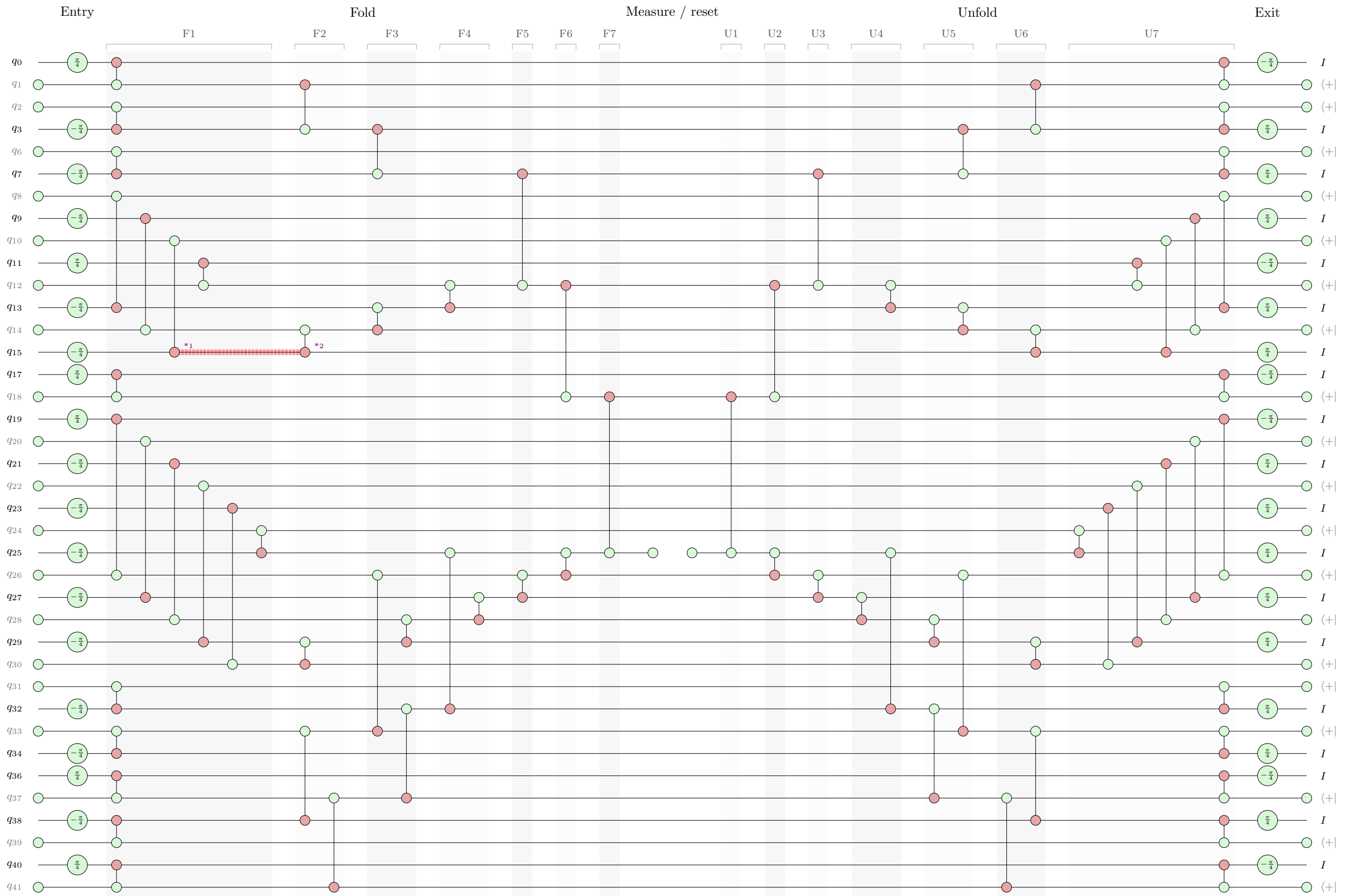
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_115.tikz

$d = 5$: basis element 116

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_116.tikz

$d = 5$: basis element 117

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_117.tikz

$d = 5$: basis element 118

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_118.tikz

$d = 5$: basis element 119

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_119.tikz

$d = 5$: basis element 120

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_120.tikz

$d = 5$: basis element 121

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_121.tikz

$d = 5$: basis element 122

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_122.tikz

$d = 5$: basis element 123

No boundary support. 2 phase defects.



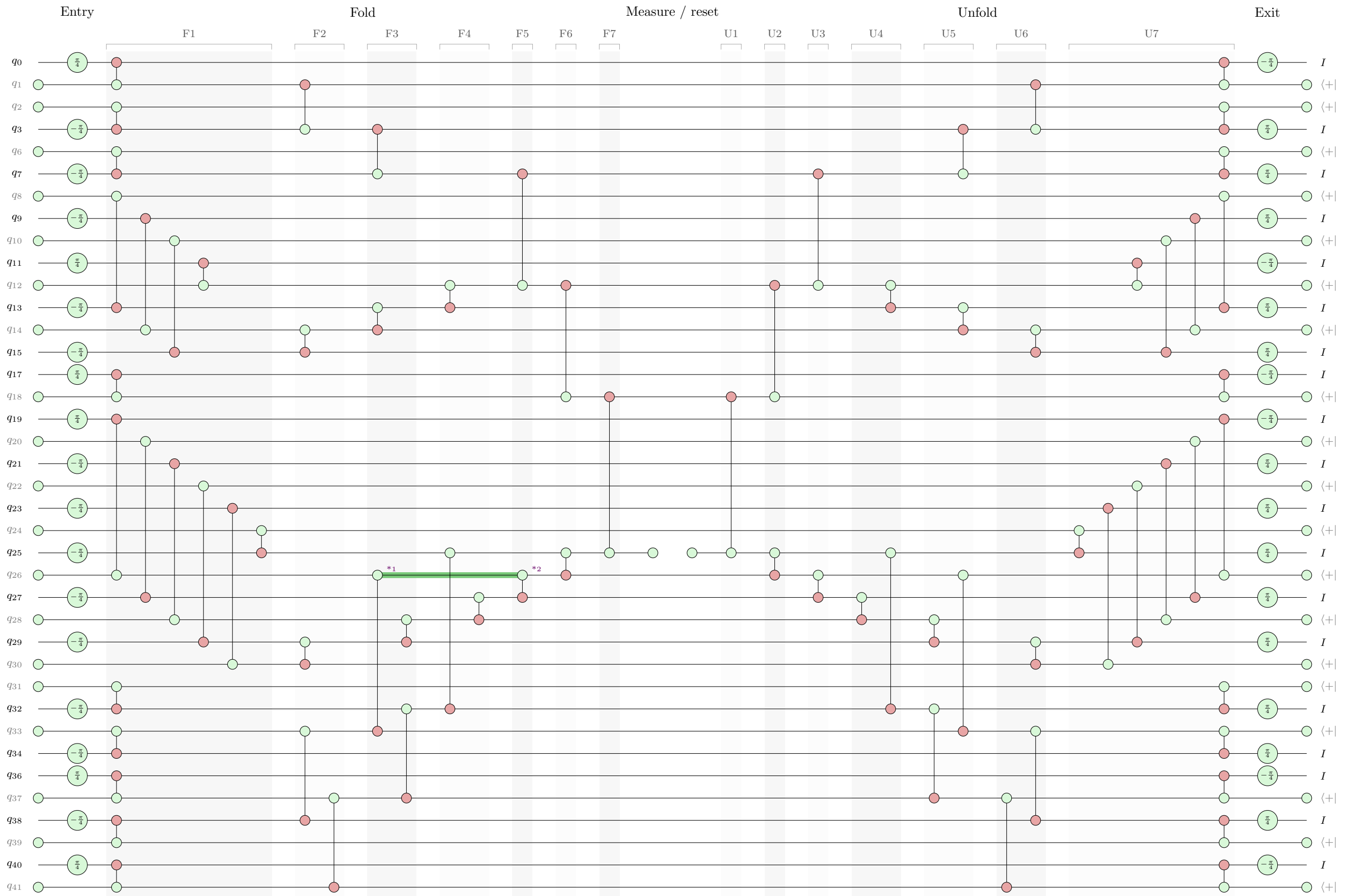
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_123.tikz

$d = 5$: basis element 124

No boundary support. 2 phase defects.



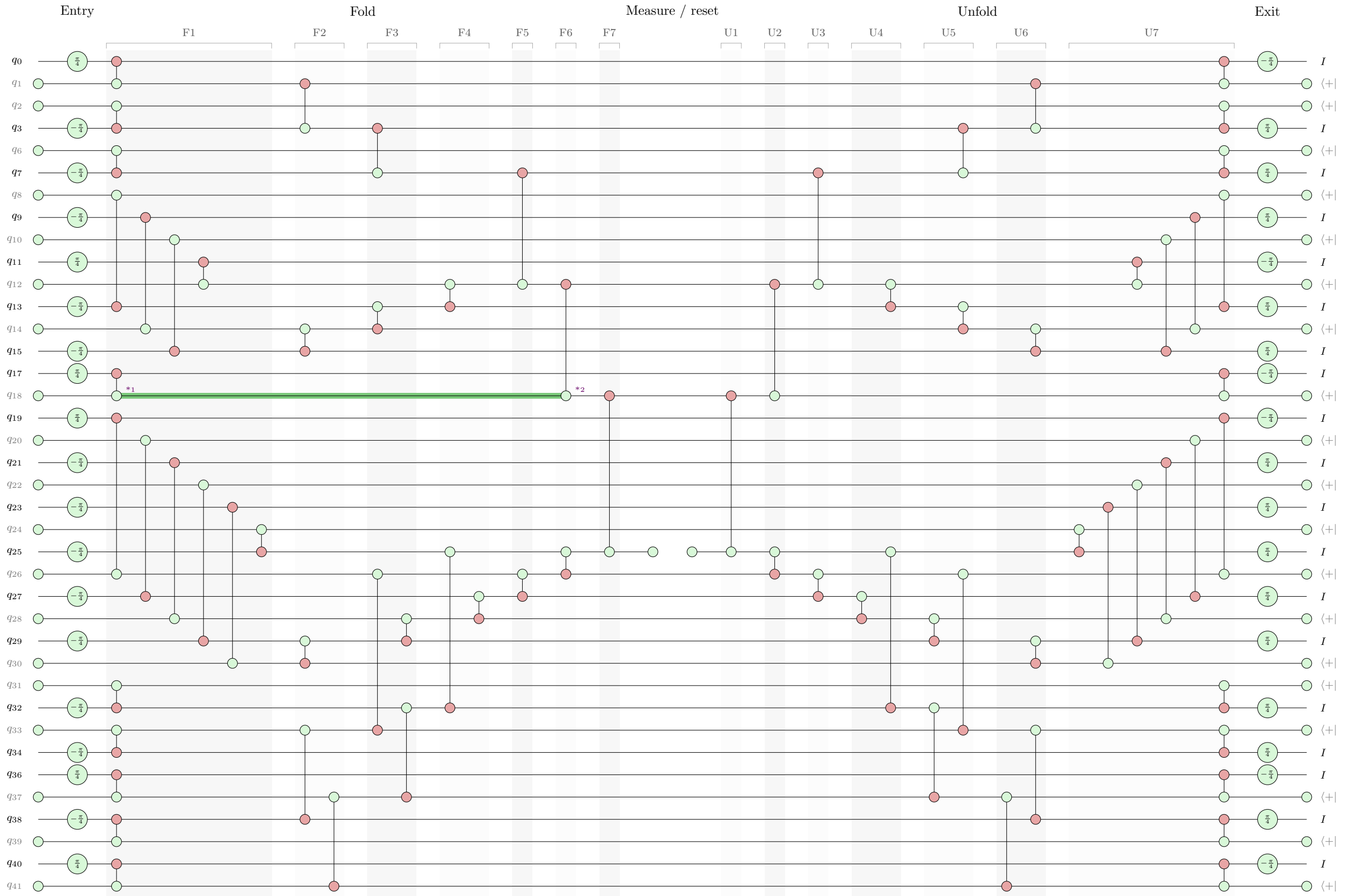
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_124.tikz

$d = 5$: basis element 125

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_125.tikz

$d = 5$: basis element 126

No boundary support. 2 phase defects.



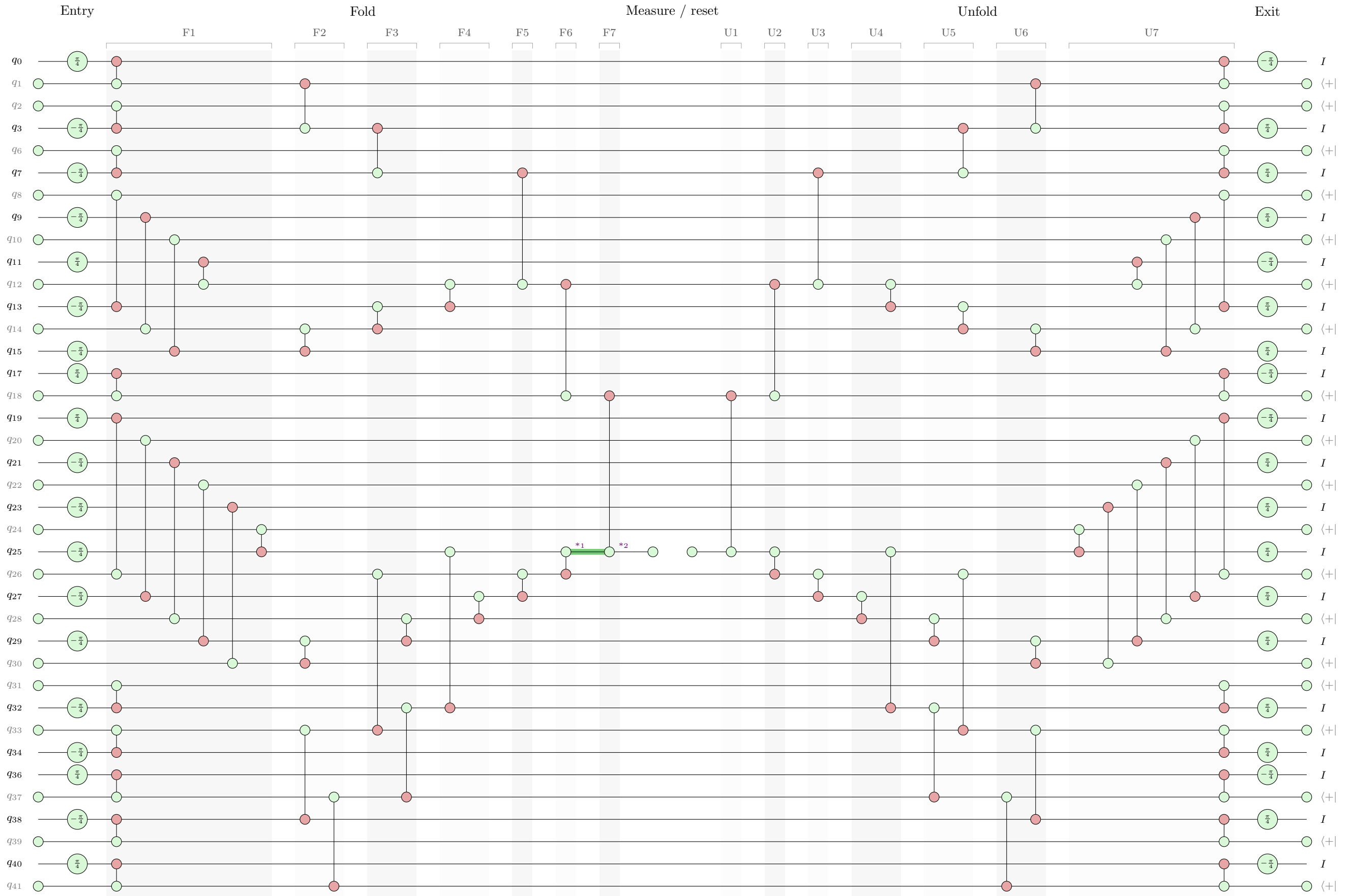
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_126.tikz

$d = 5$: basis element 127

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_127.tikz

$d = 5$: basis element 128

No boundary support. 2 phase defects.



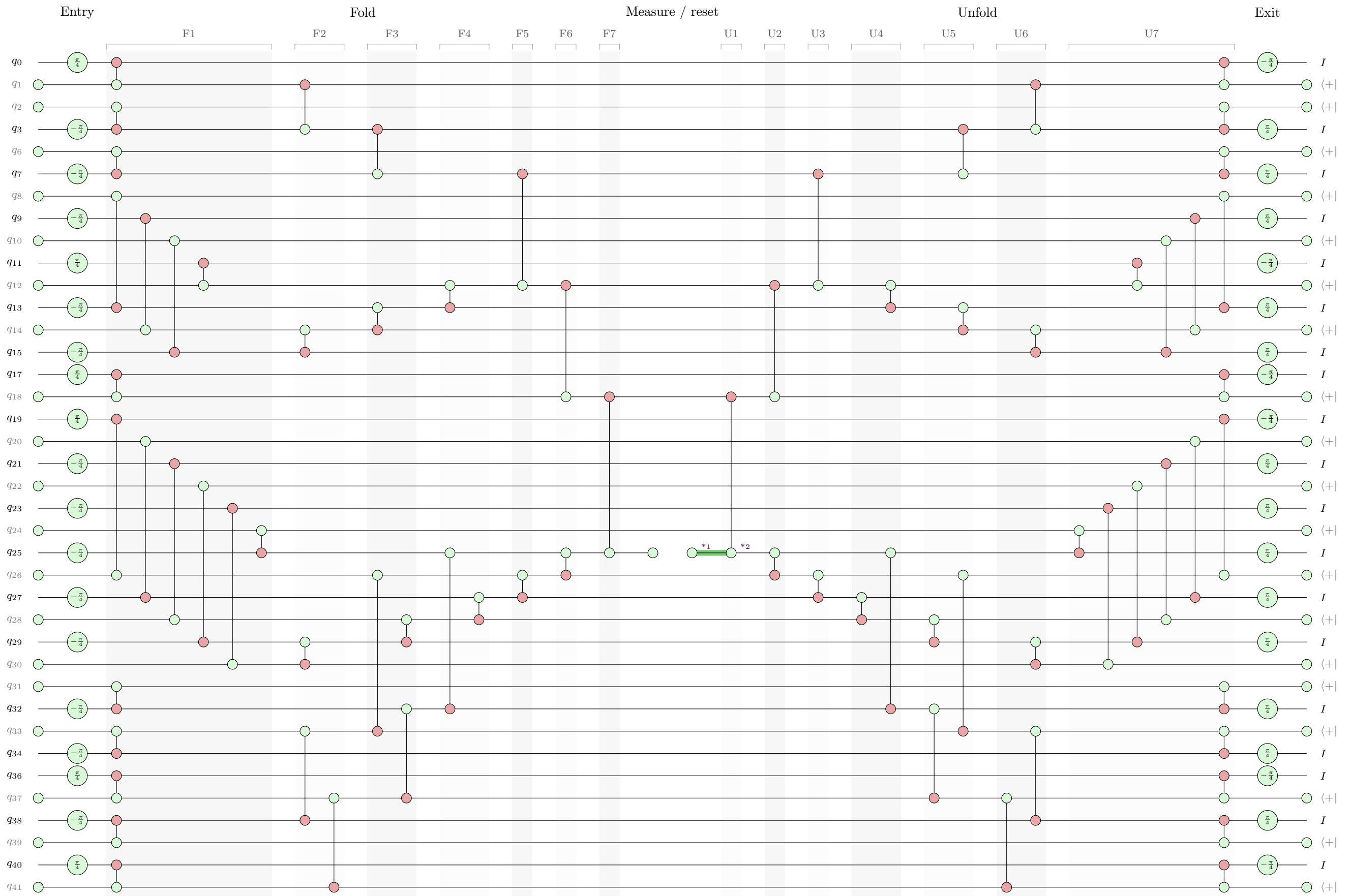
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_128.tikz

$d = 5$: basis element 129

No boundary support. 2 phase defects.



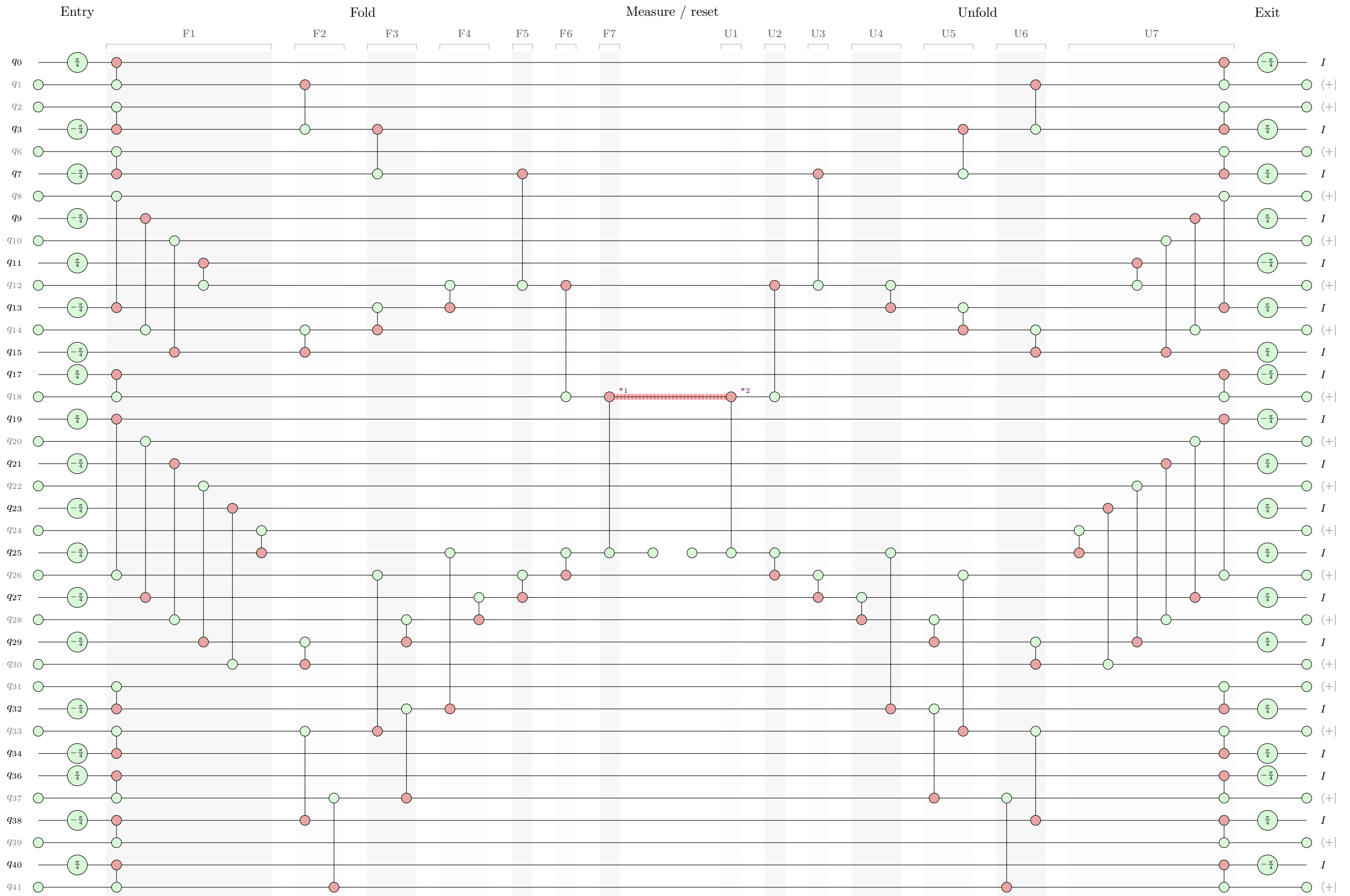
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_129.tikz

$d = 5$: basis element 130

No boundary support. 2 phase defects.



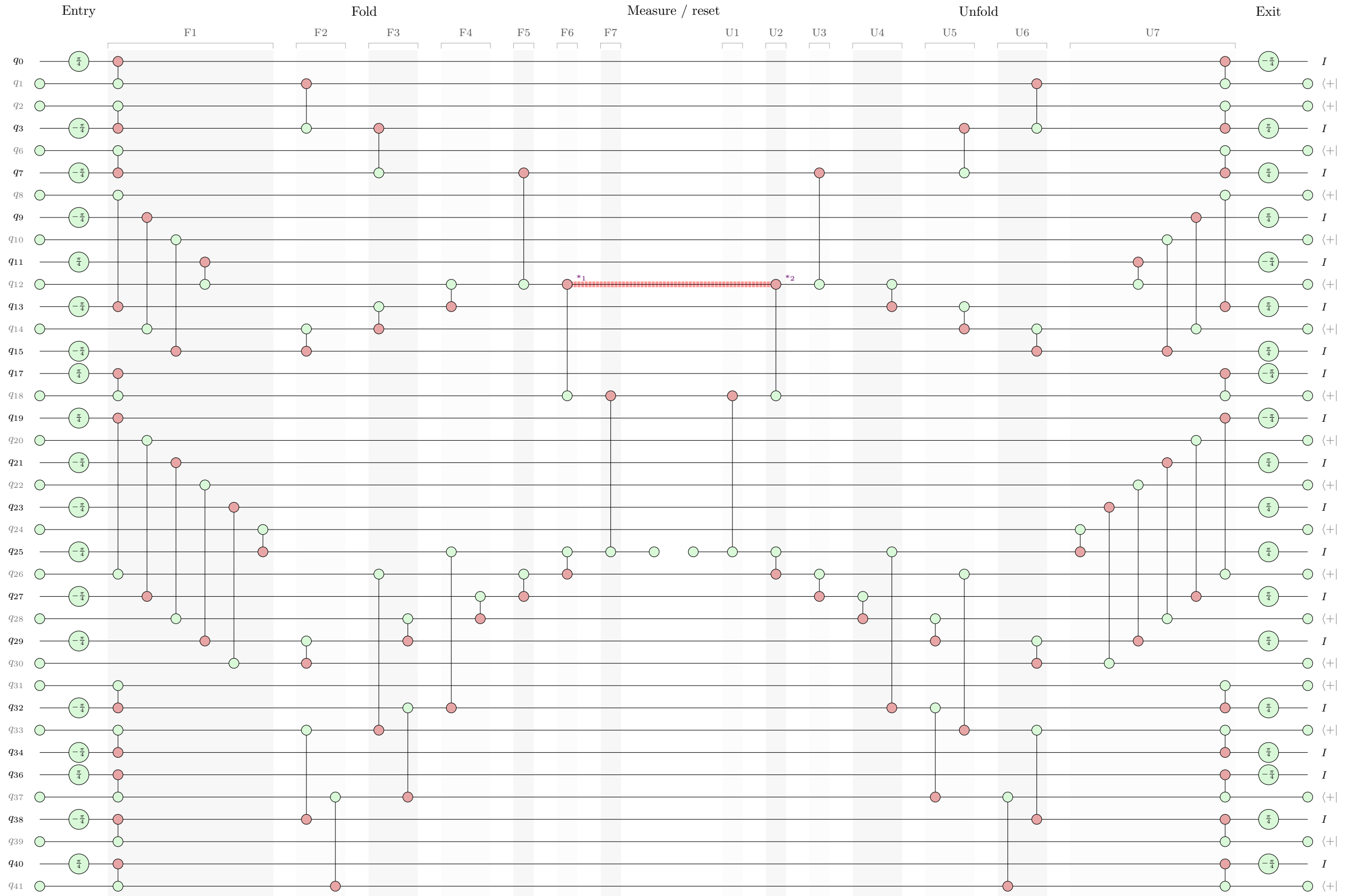
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_130.tikz

$d = 5$: basis element 131

No boundary support. 2 phase defects.



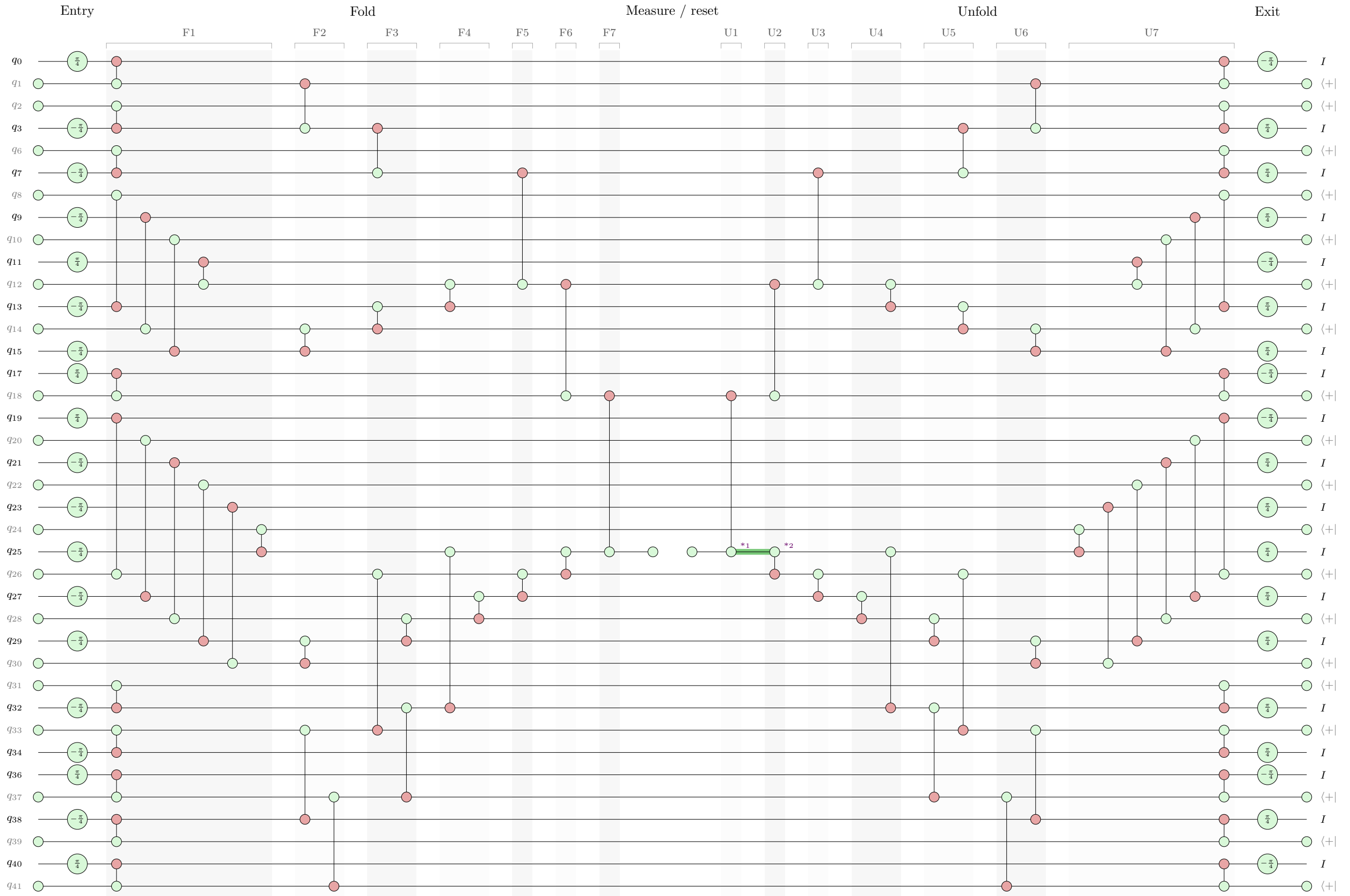
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_131.tikz

$d = 5$: basis element 132

No boundary support. 2 phase defects.



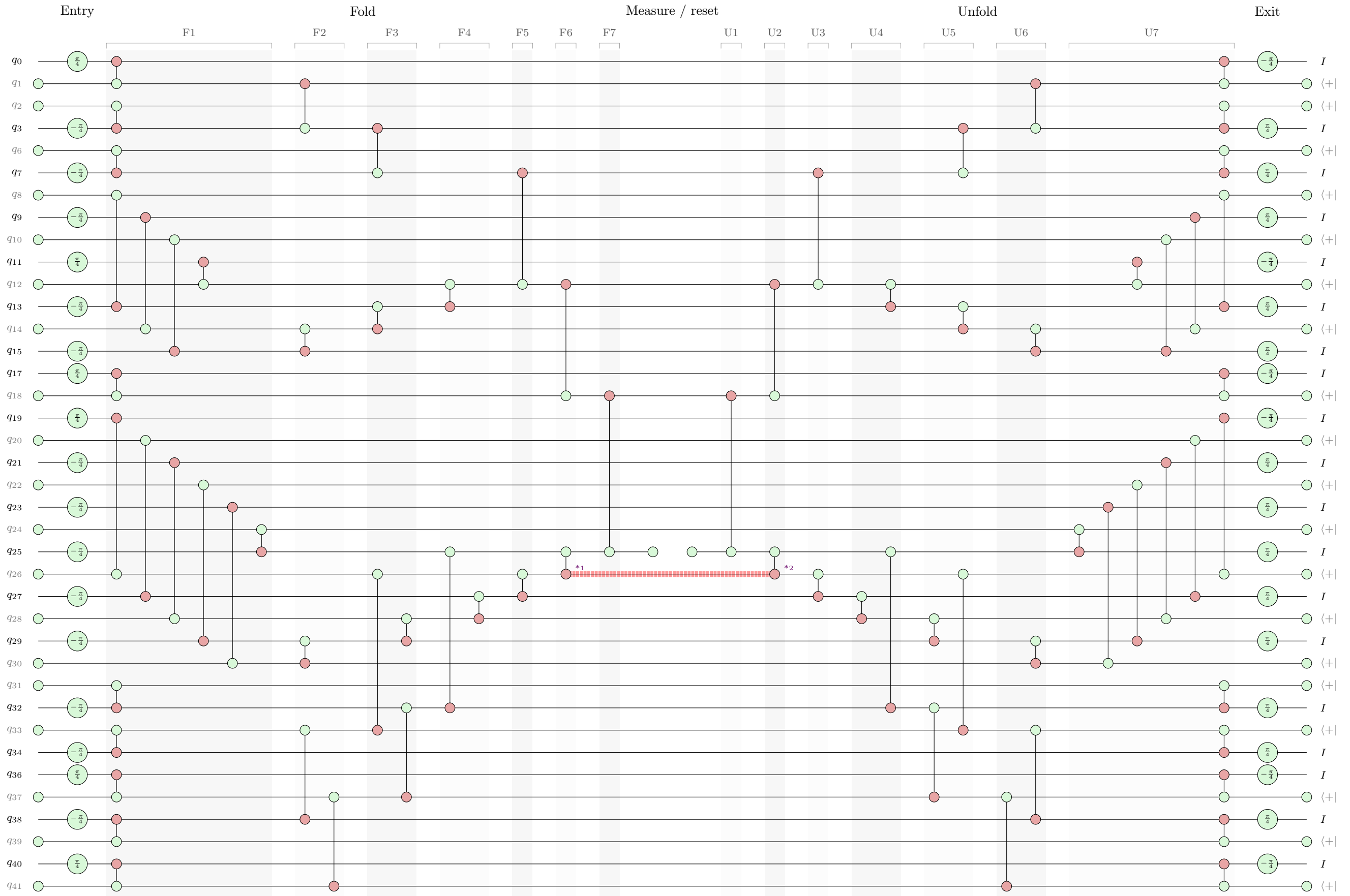
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_132.tikz

$d = 5$: basis element 133

No boundary support. 2 phase defects.



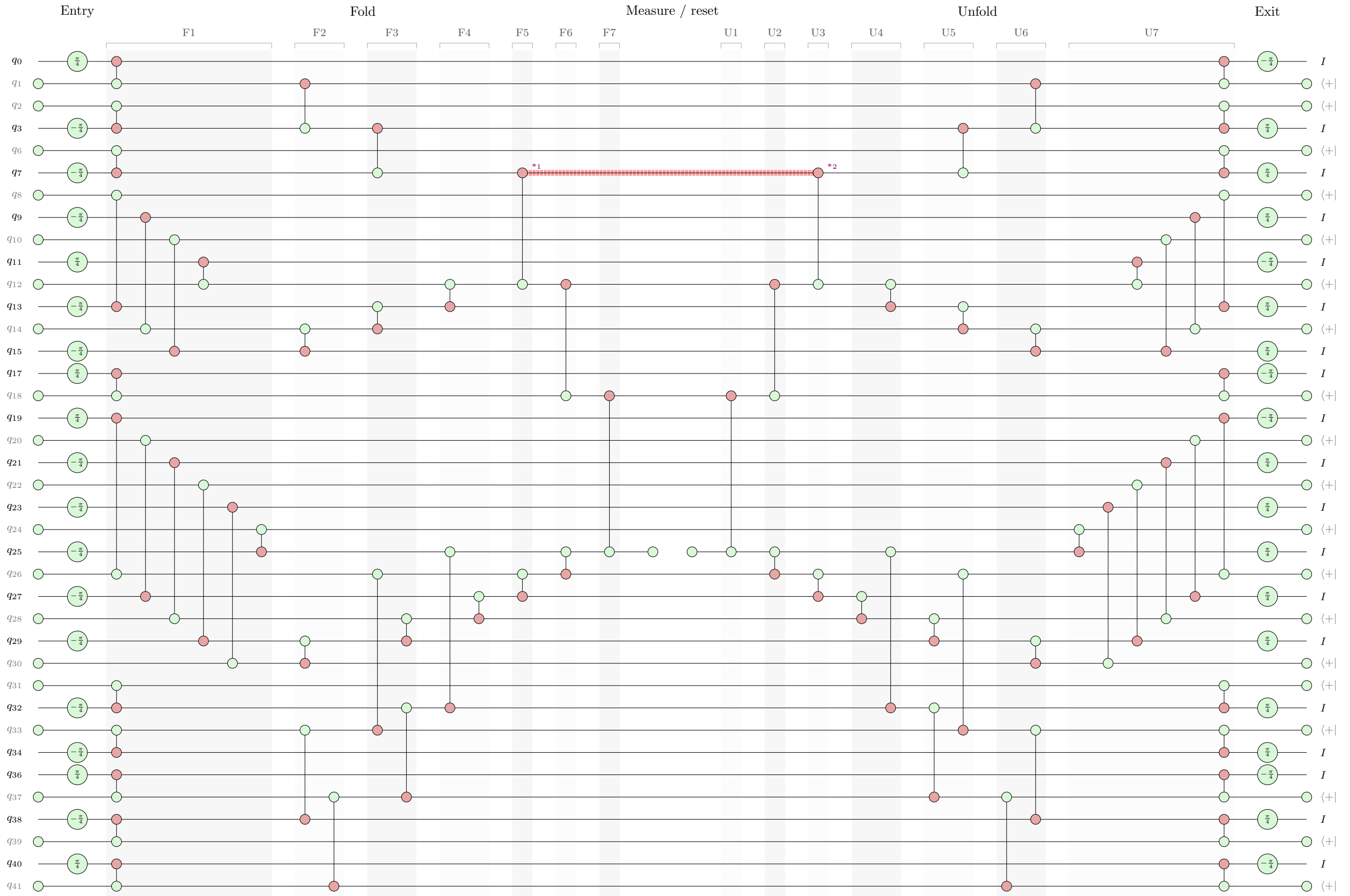
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_133.tikz

$d = 5$: basis element 134

No boundary support. 2 phase defects.



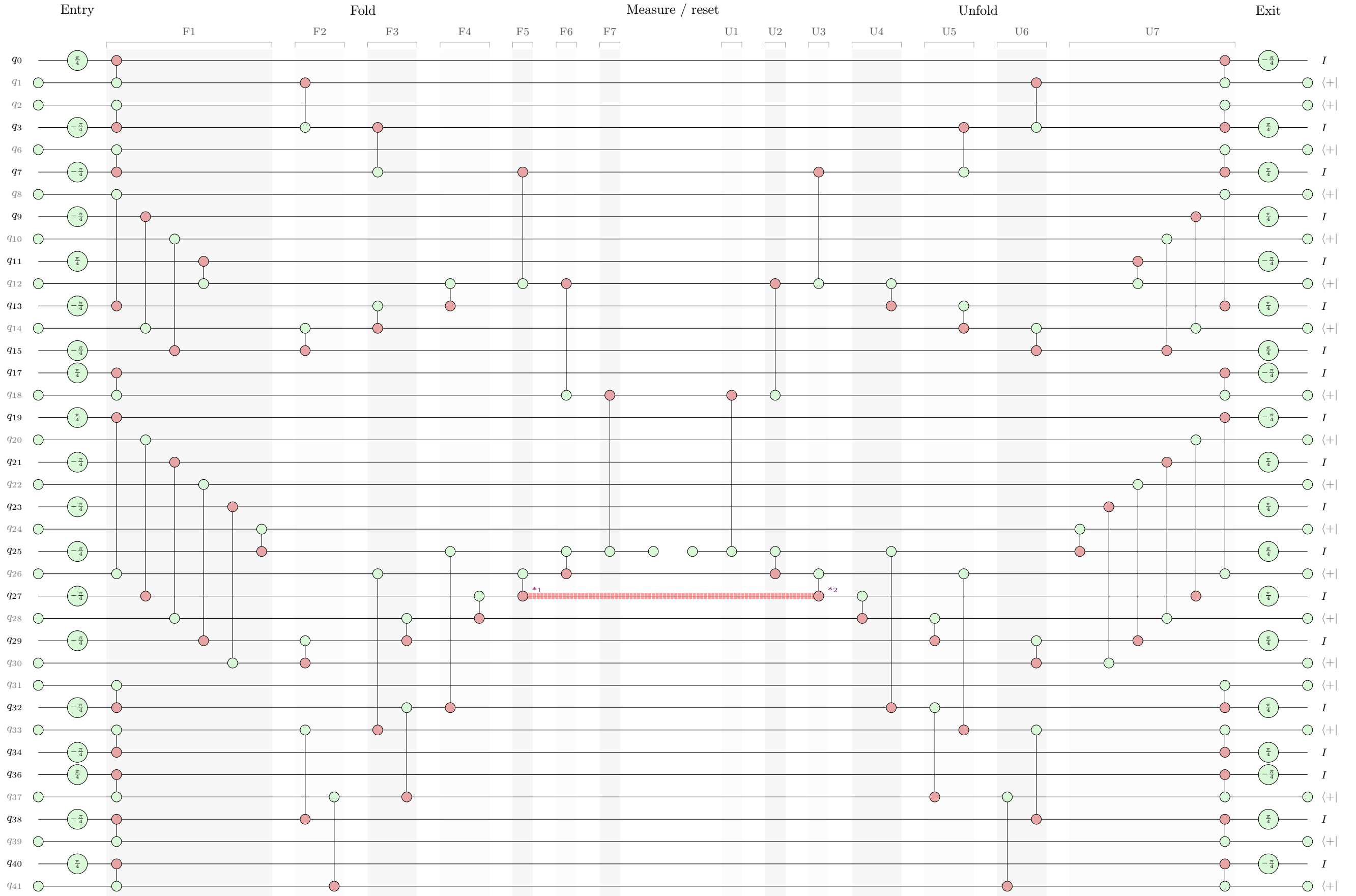
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_134.tikz

$d = 5$: basis element 135

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_135.tikz

$d = 5$: basis element 136

No boundary support. 2 phase defects.



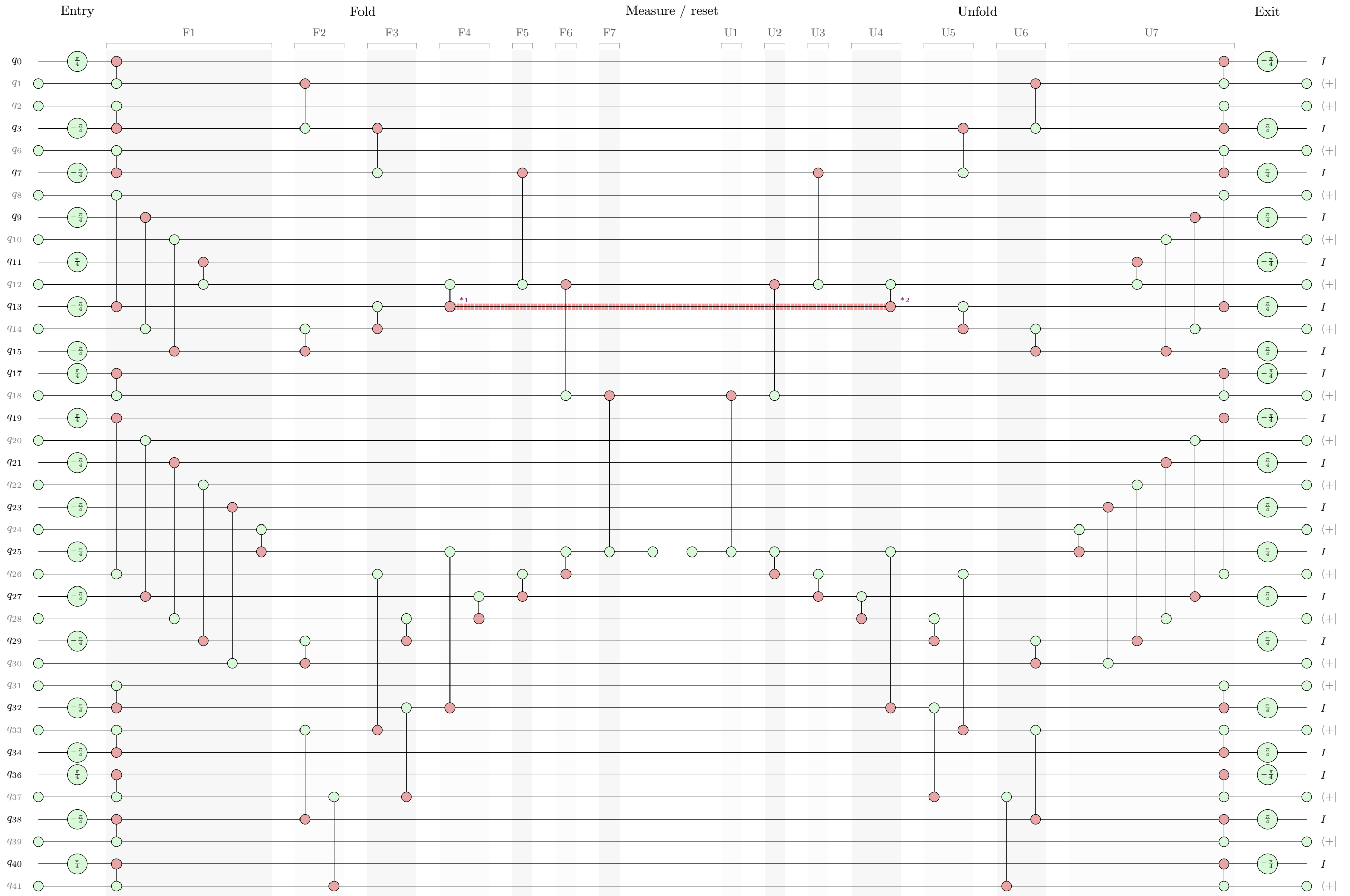
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_136.tikz

$d = 5$: basis element 137

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_137.tikz

$d = 5$: basis element 138

No boundary support. 2 phase defects.



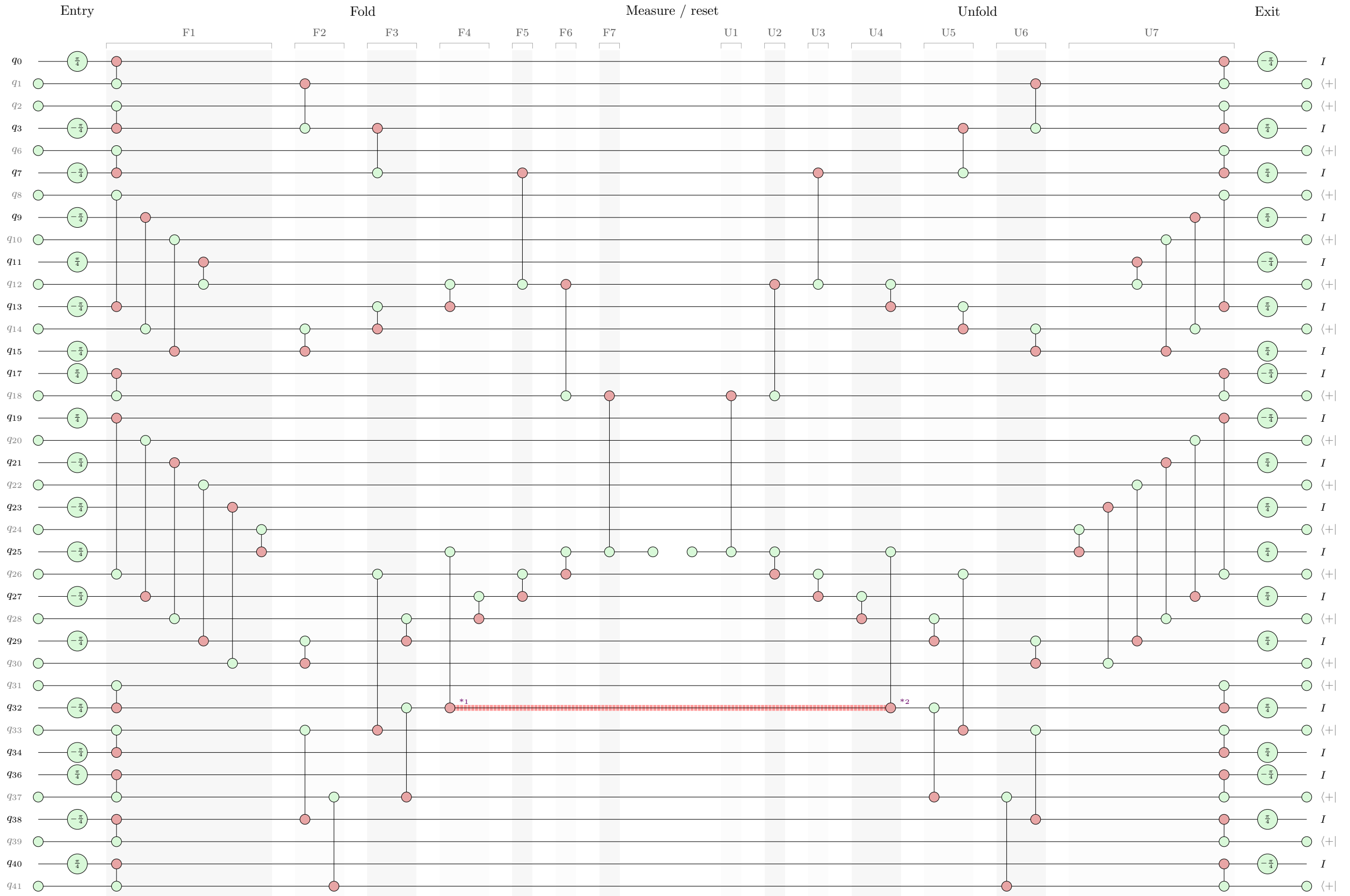
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_138.tikz

$d = 5$: basis element 139

No boundary support. 2 phase defects.



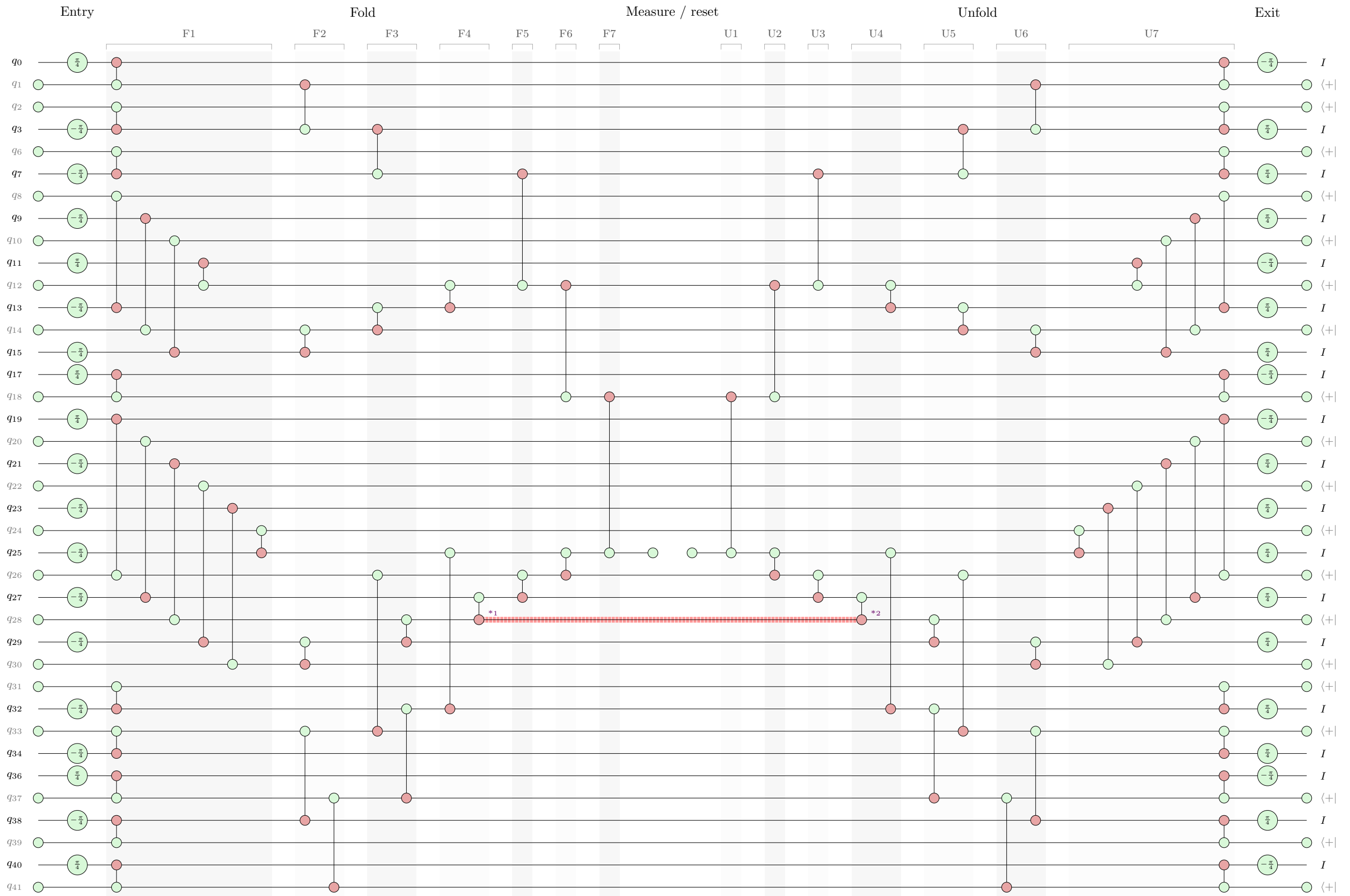
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_139.tikz

$d = 5$: basis element 140

No boundary support. 2 phase defects.



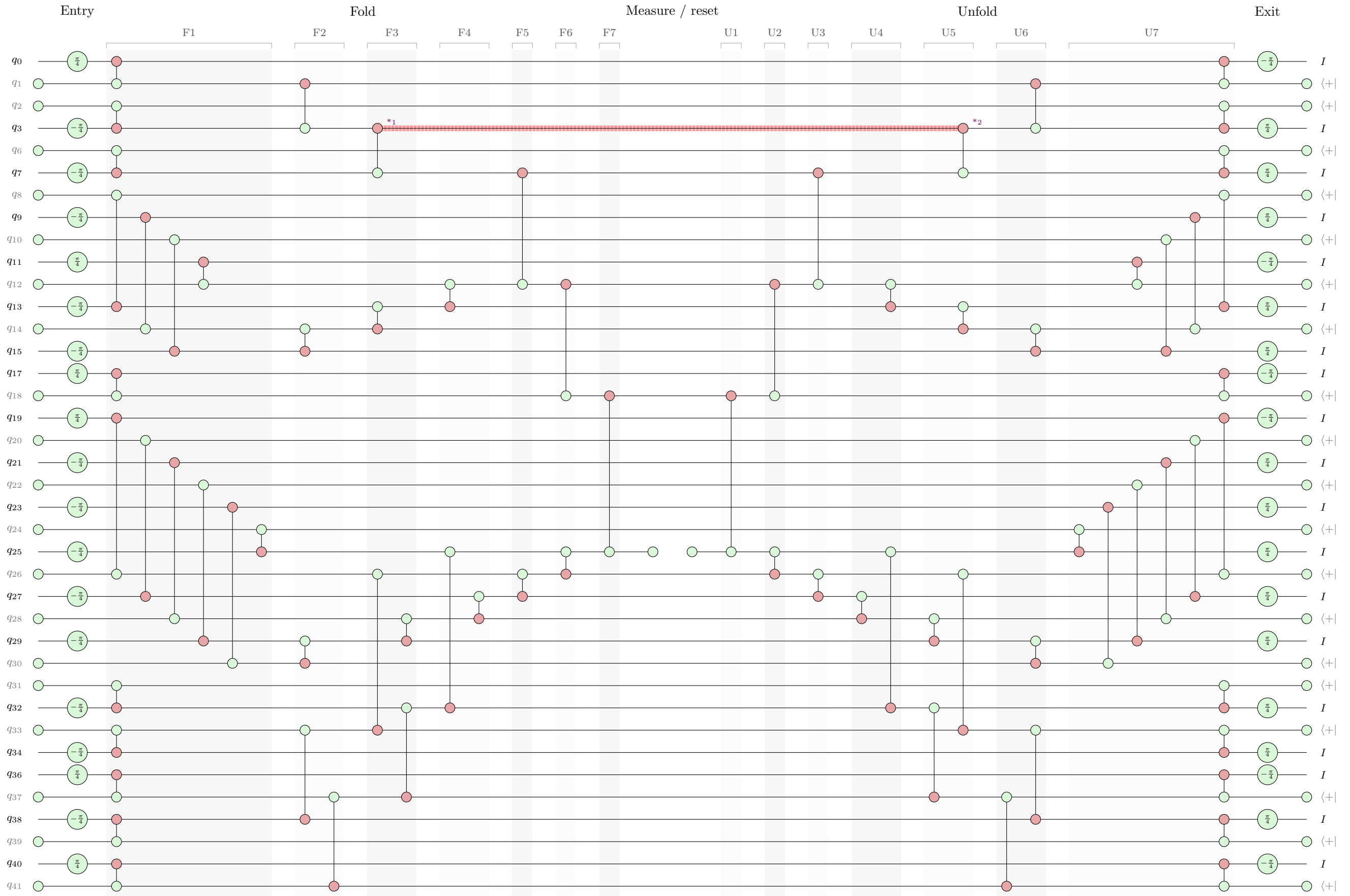
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_140.tikz

$d = 5$: basis element 141

No boundary support. 2 phase defects.



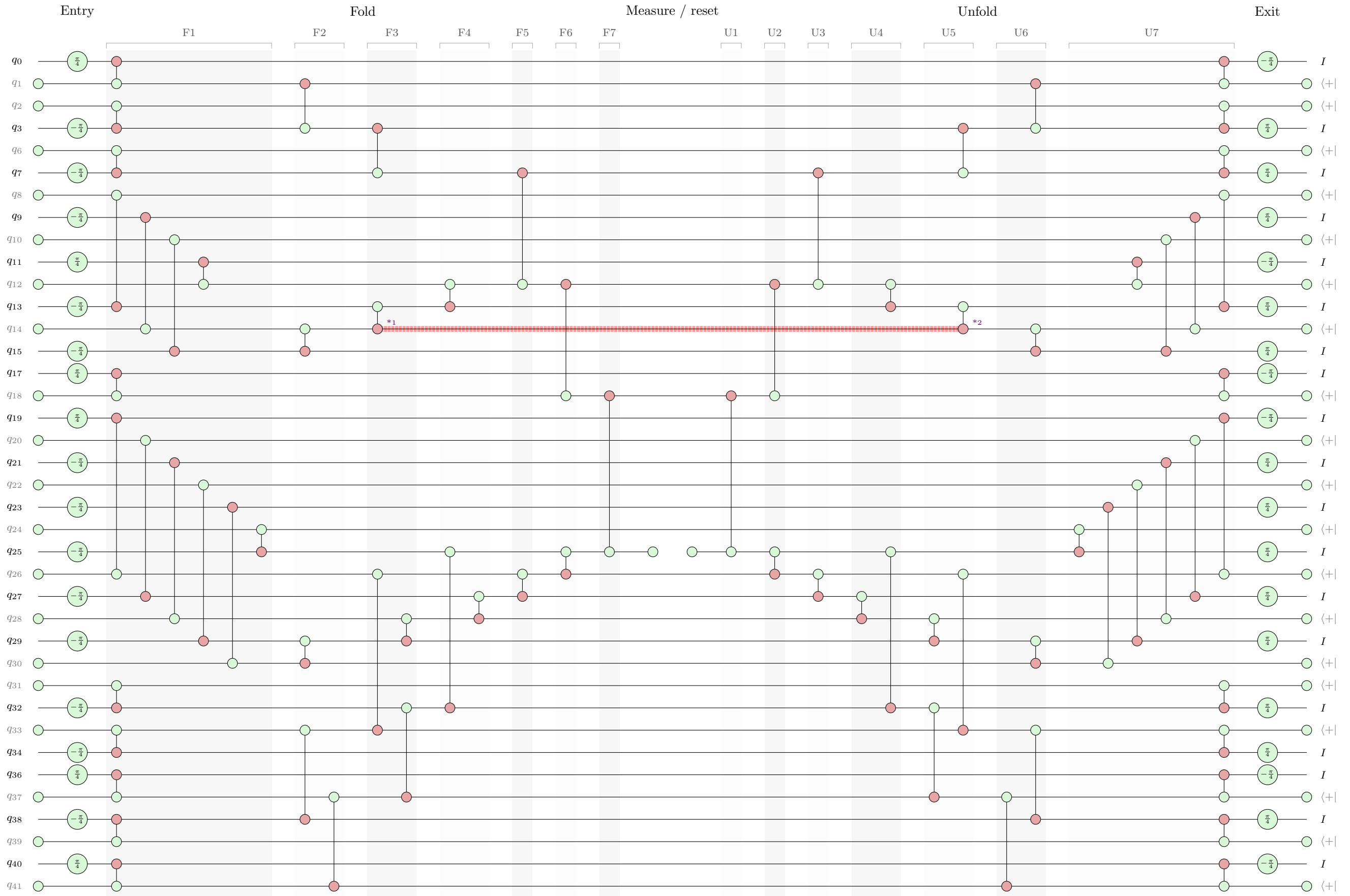
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_141.tikz

$d = 5$: basis element 142

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_142.tikz

$d = 5$: basis element 143

No boundary support. 2 phase defects.



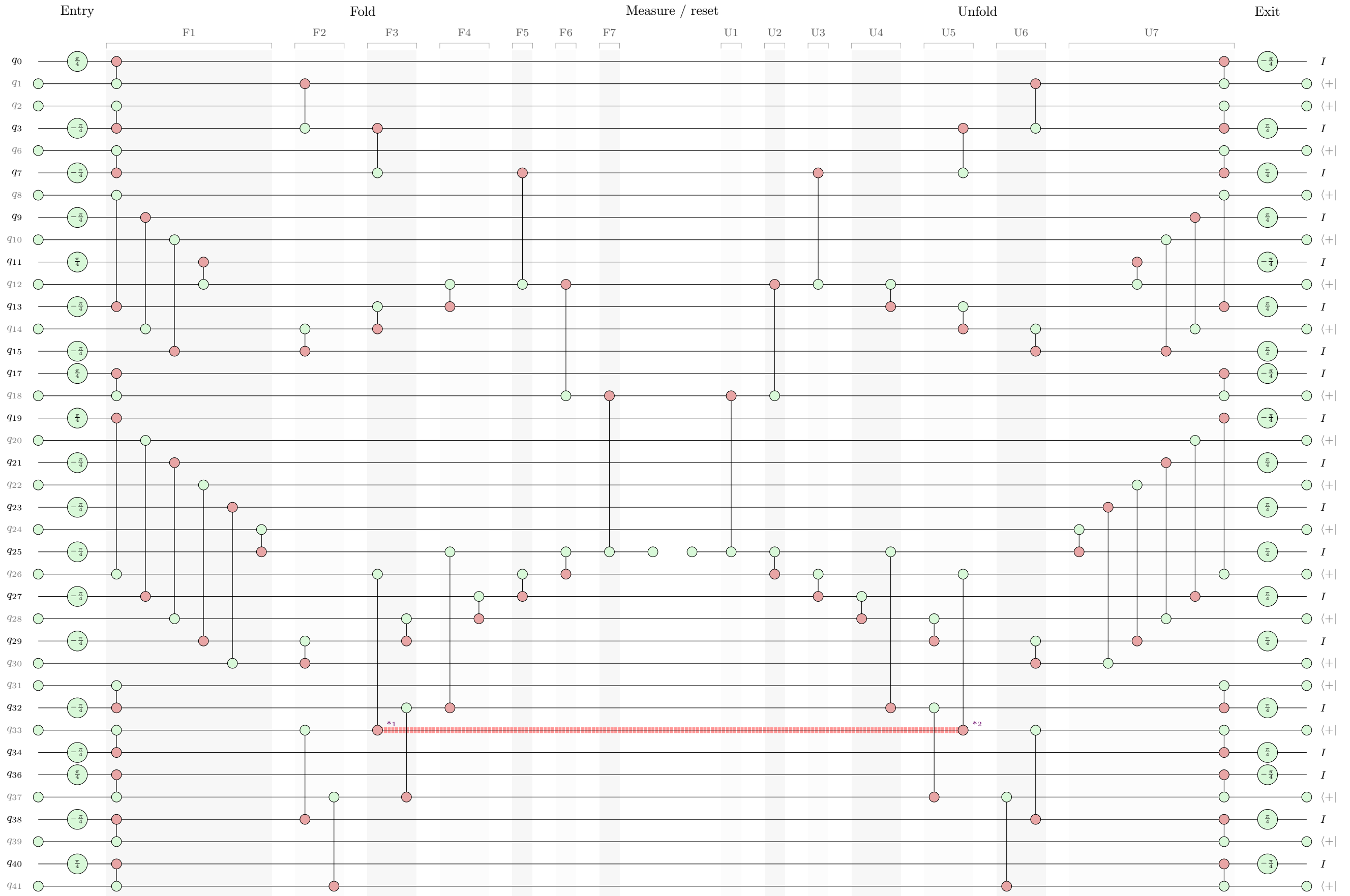
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_143.tikz

$d = 5$: basis element 144

No boundary support. 2 phase defects.



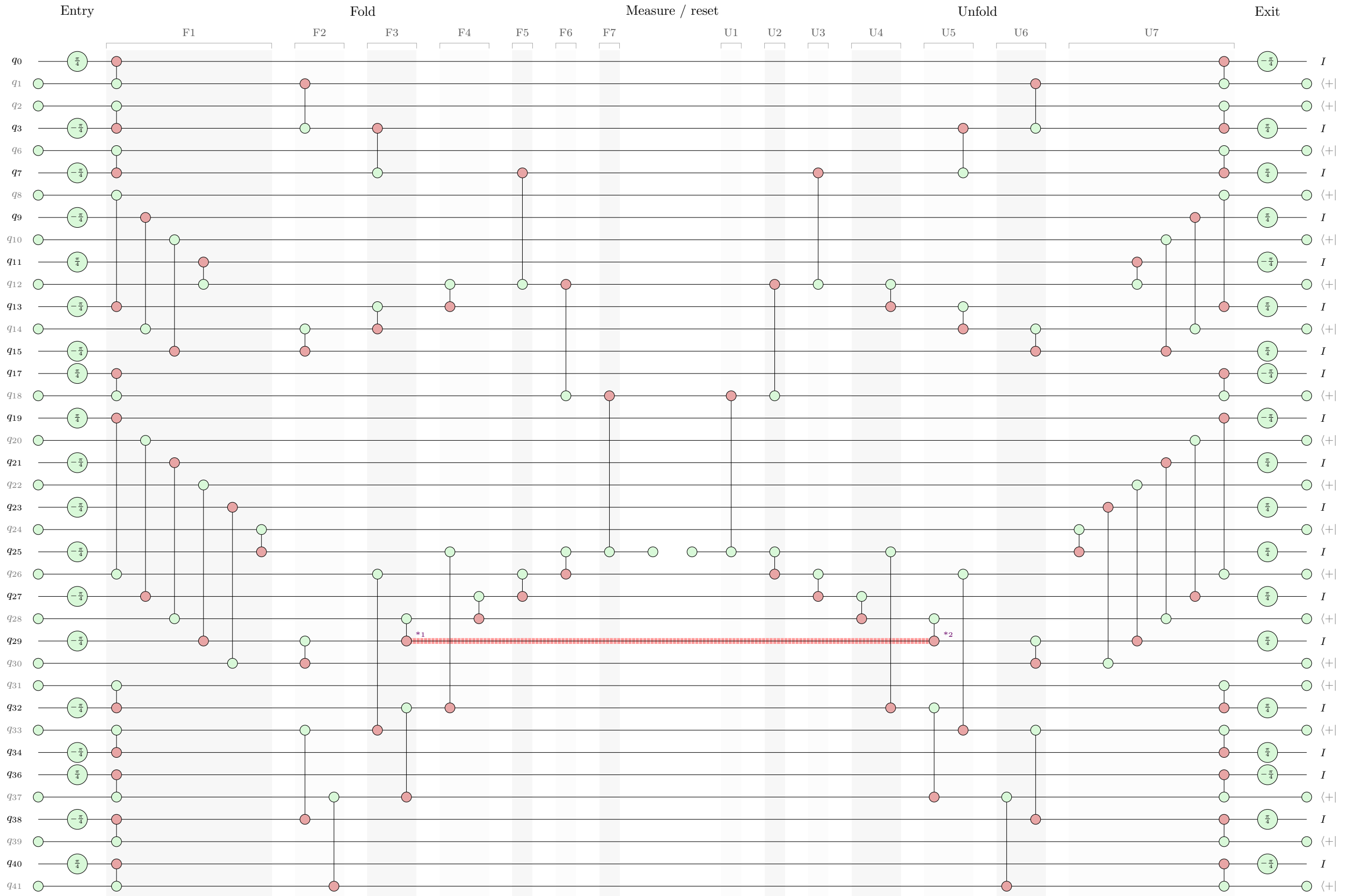
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_144.tikz

$d = 5$: basis element 145

No boundary support. 2 phase defects.



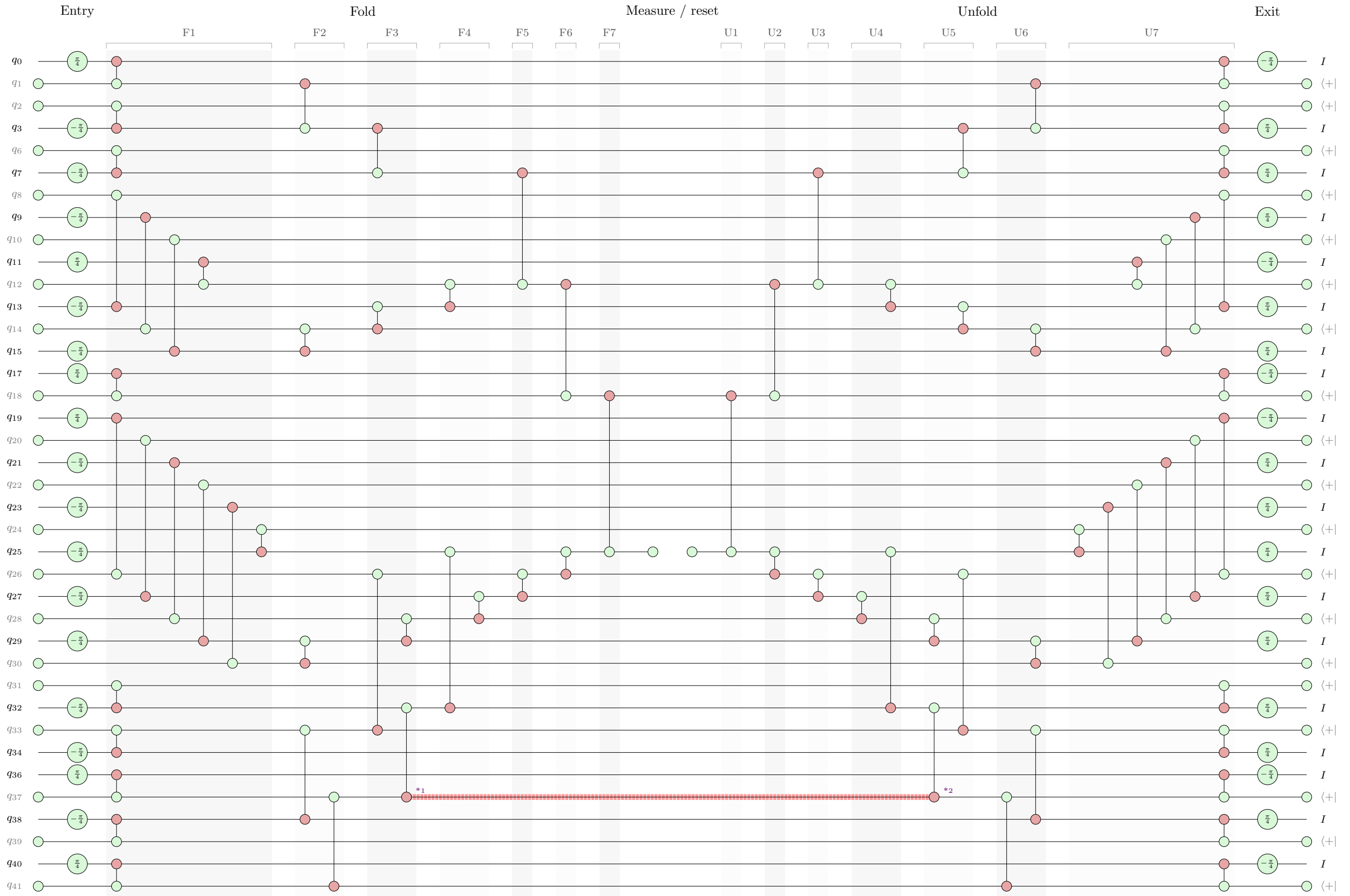
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_145.tikz

$d = 5$: basis element 146

No boundary support. 2 phase defects.



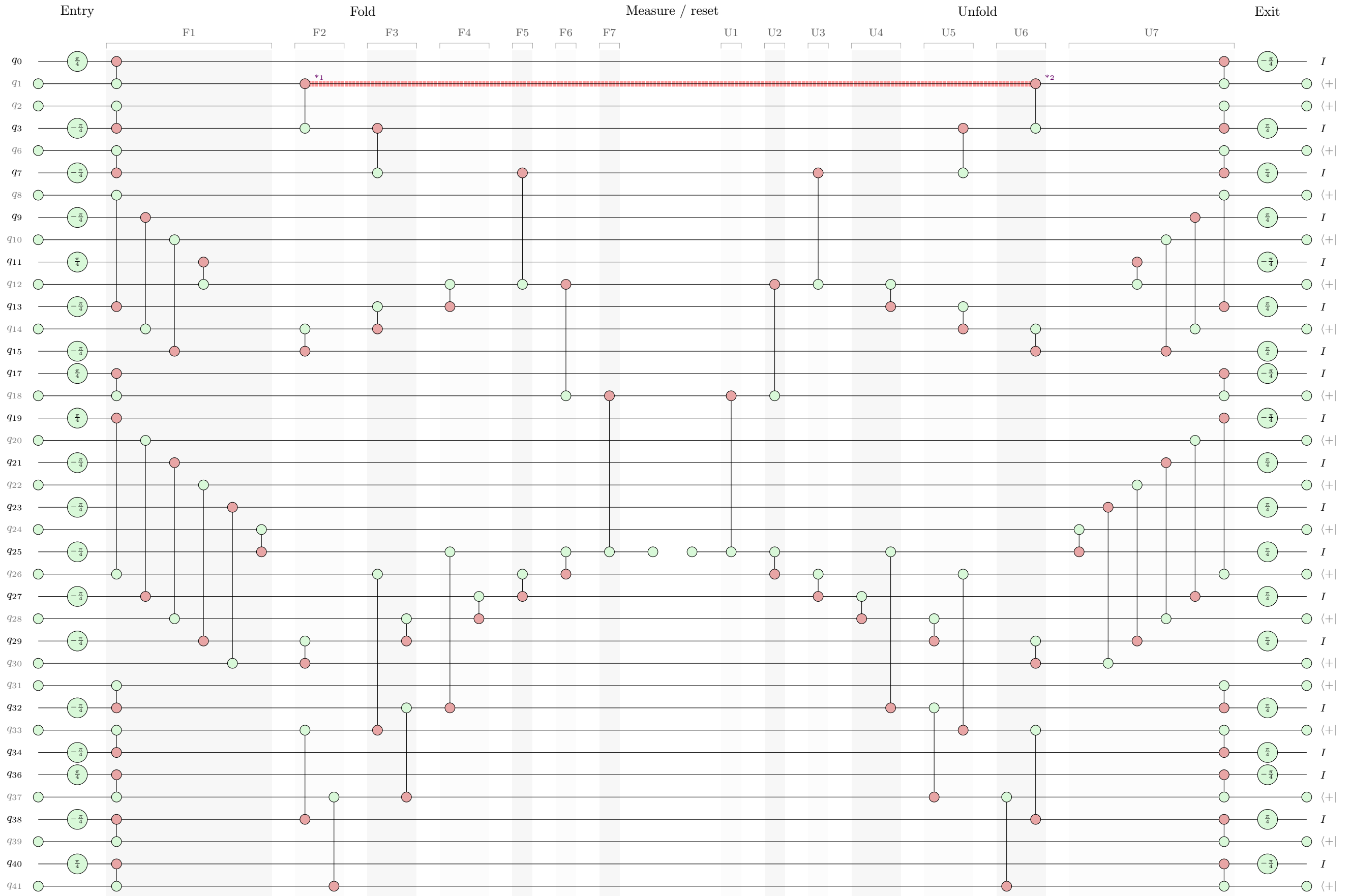
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_146.tikz

$d = 5$: basis element 147

No boundary support. 2 phase defects.



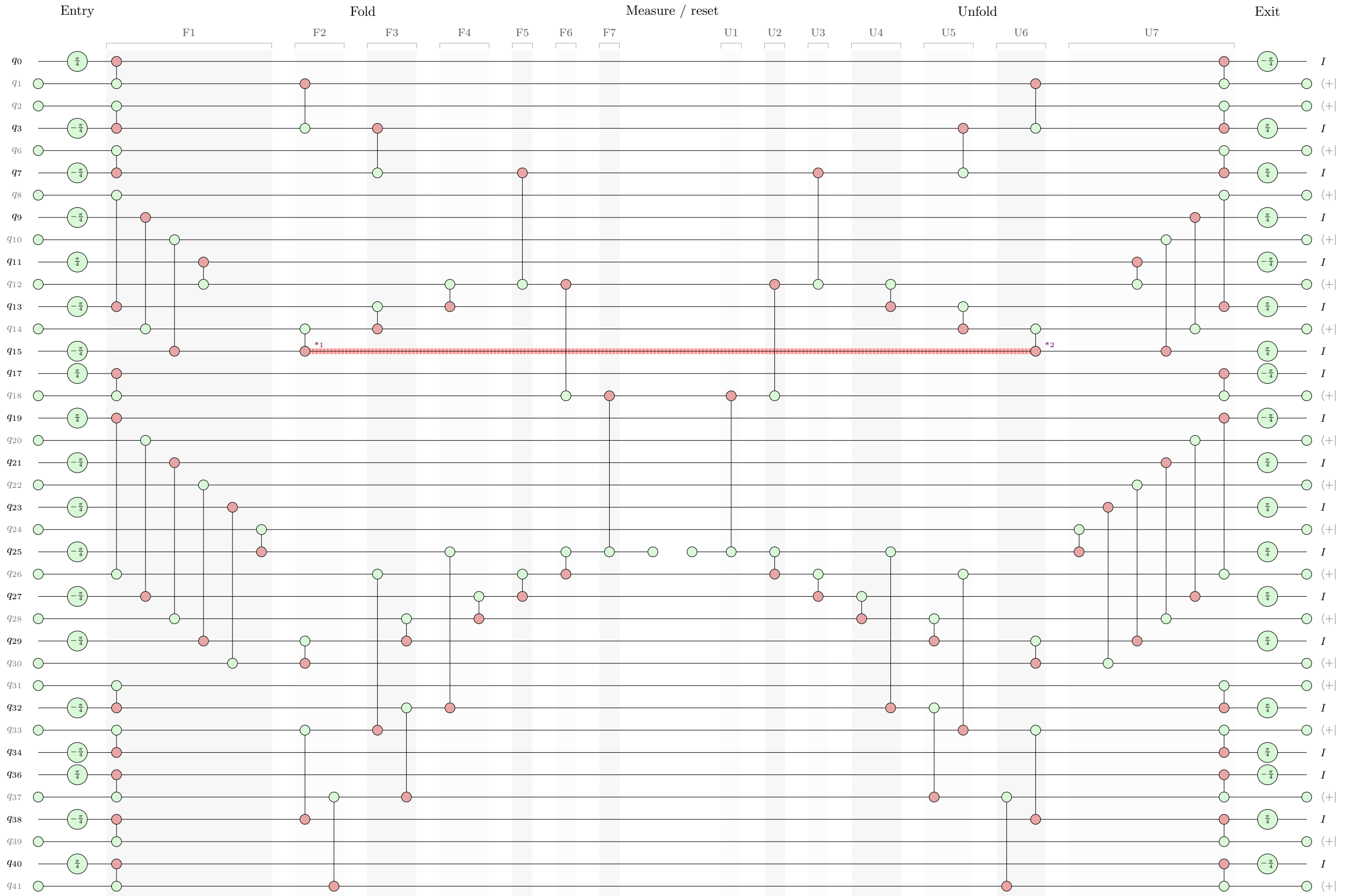
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_147.tikz

$d = 5$: basis element 148

No boundary support. 2 phase defects.



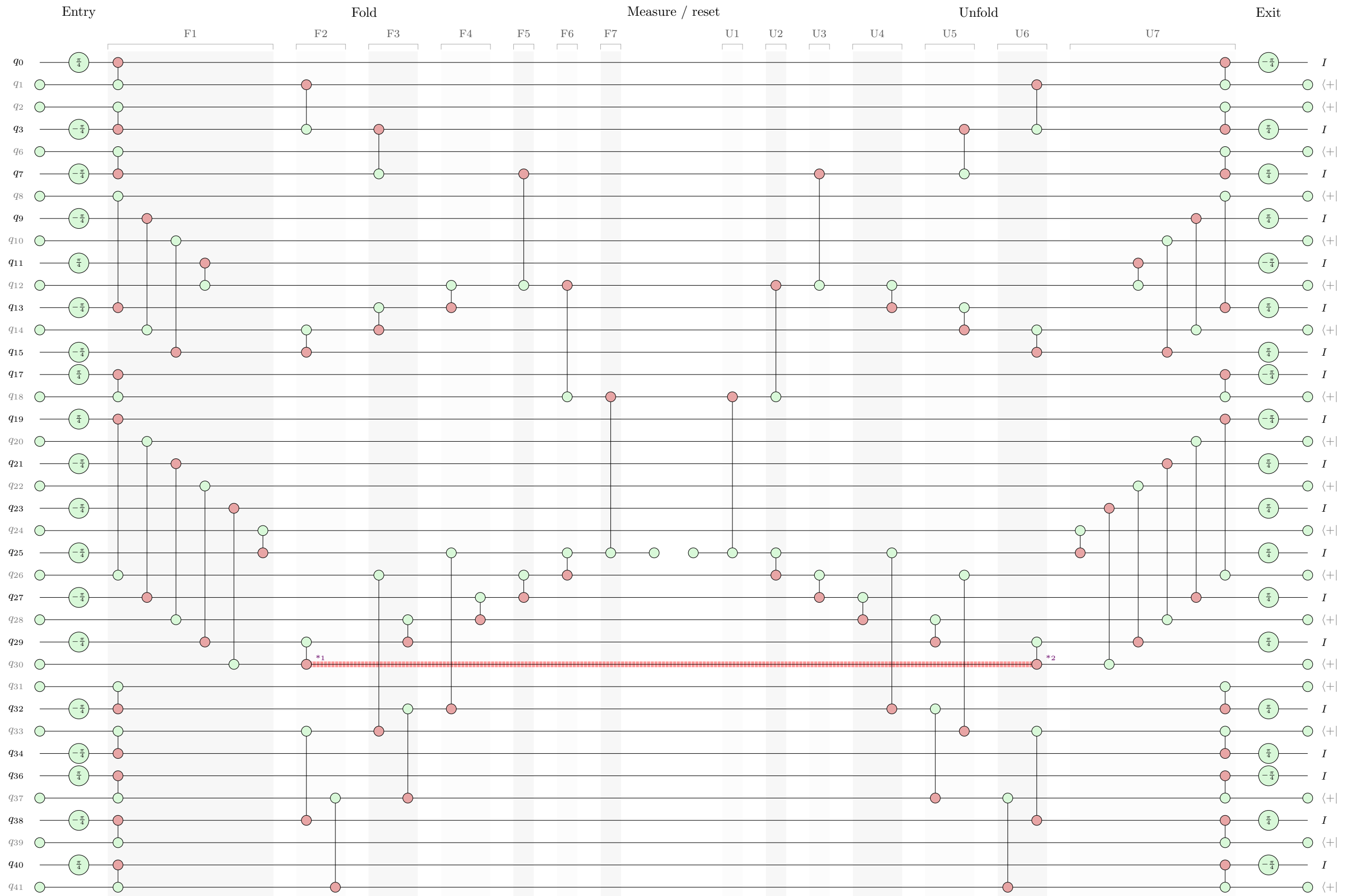
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_148.tikz

$d = 5$: basis element 149

No boundary support. 2 phase defects.



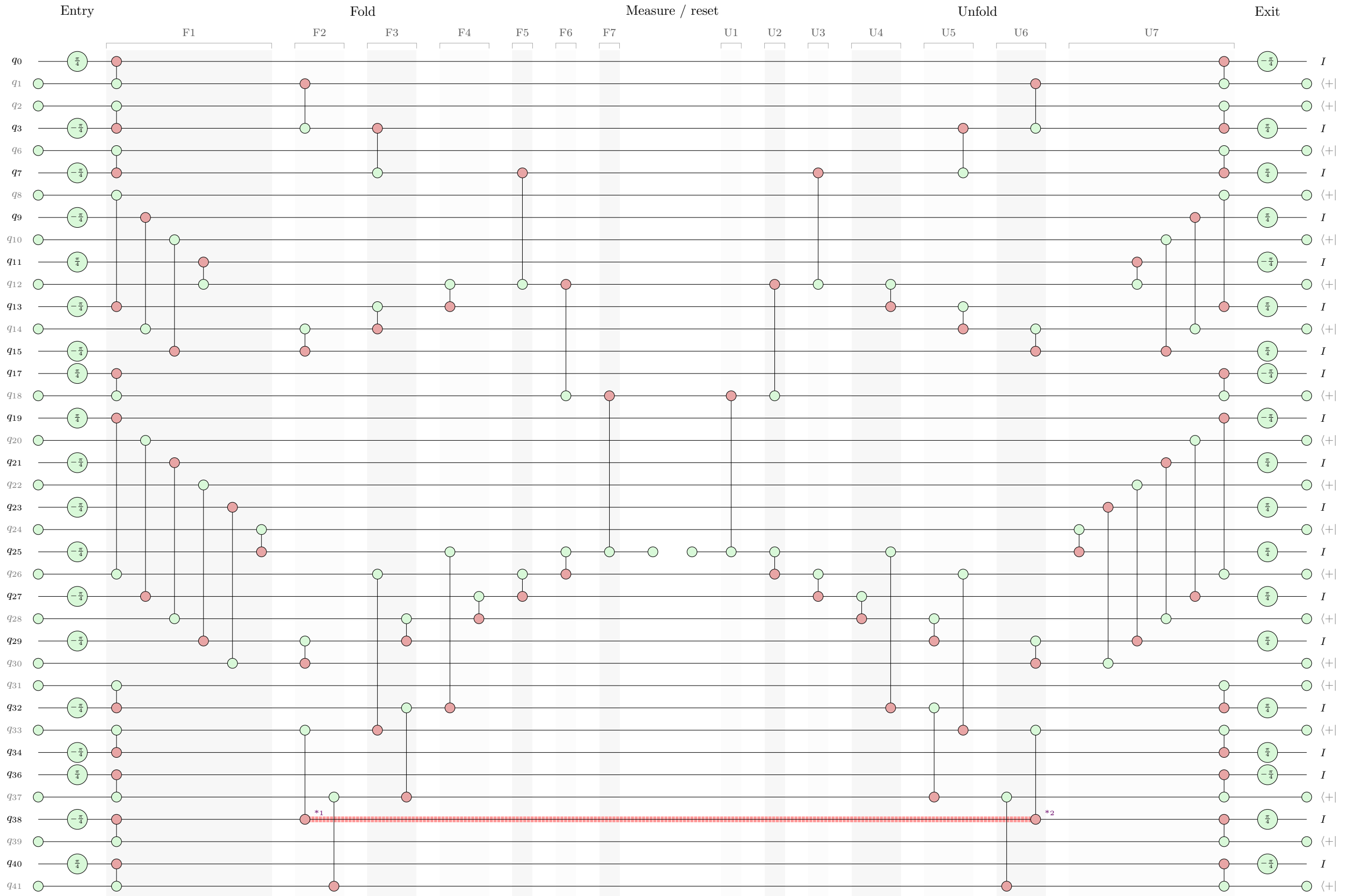
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_149.tikz

$d = 5$: basis element 150

No boundary support. 2 phase defects.



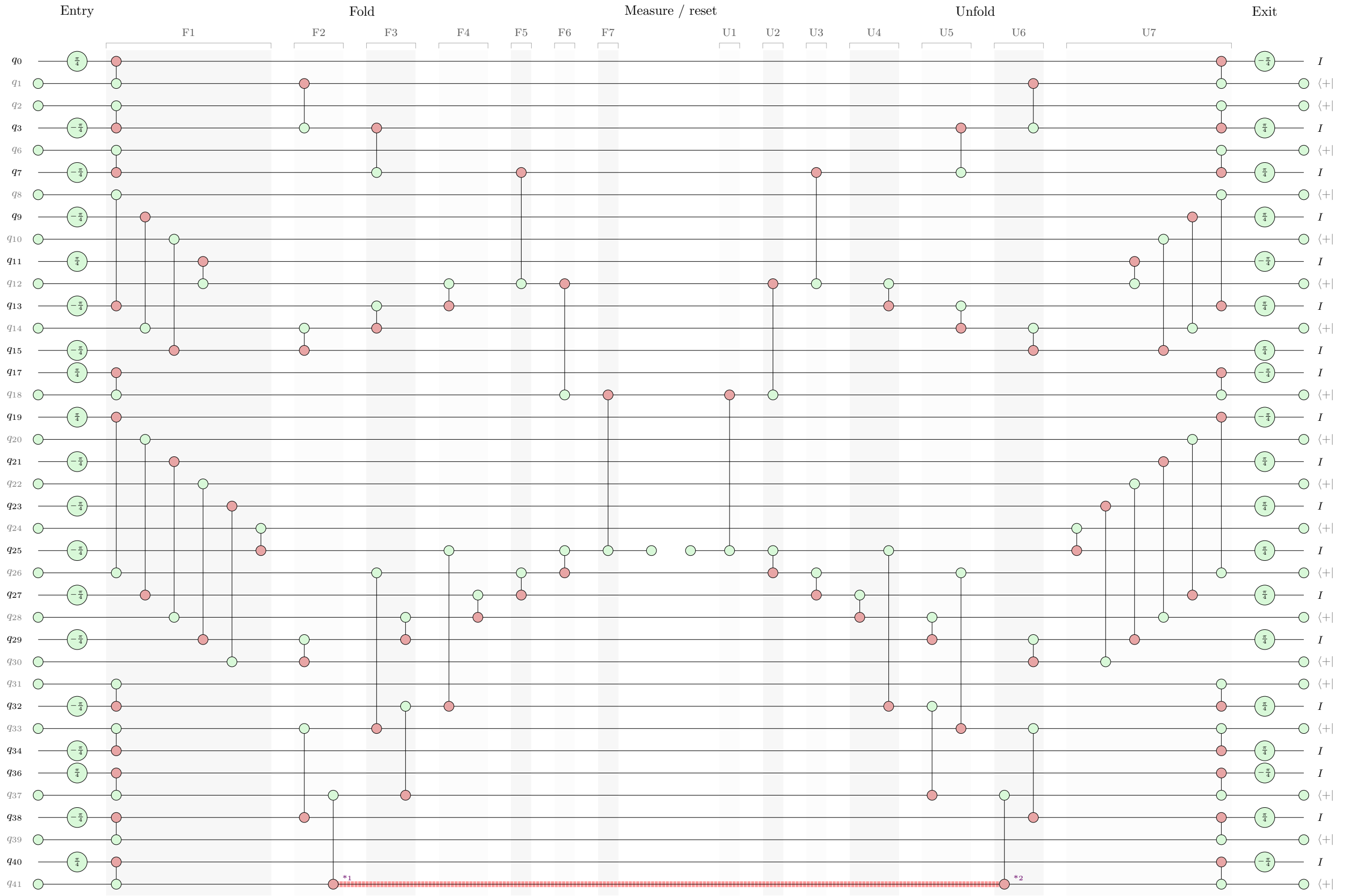
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_150.tikz

$d = 5$: basis element 151

No boundary support. 2 phase defects.



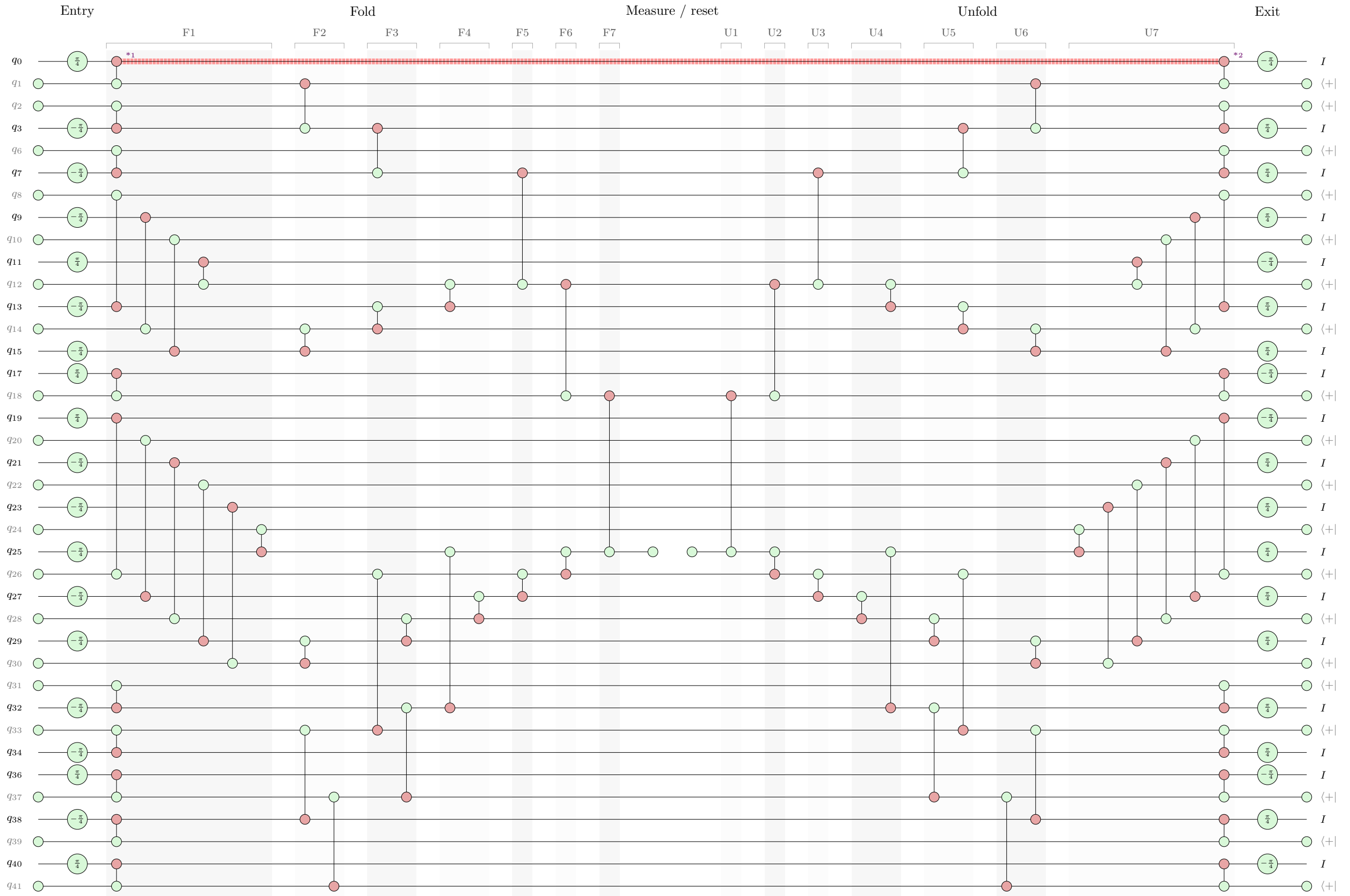
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_151.tikz

$d = 5$: basis element 152

No boundary support. 2 phase defects.



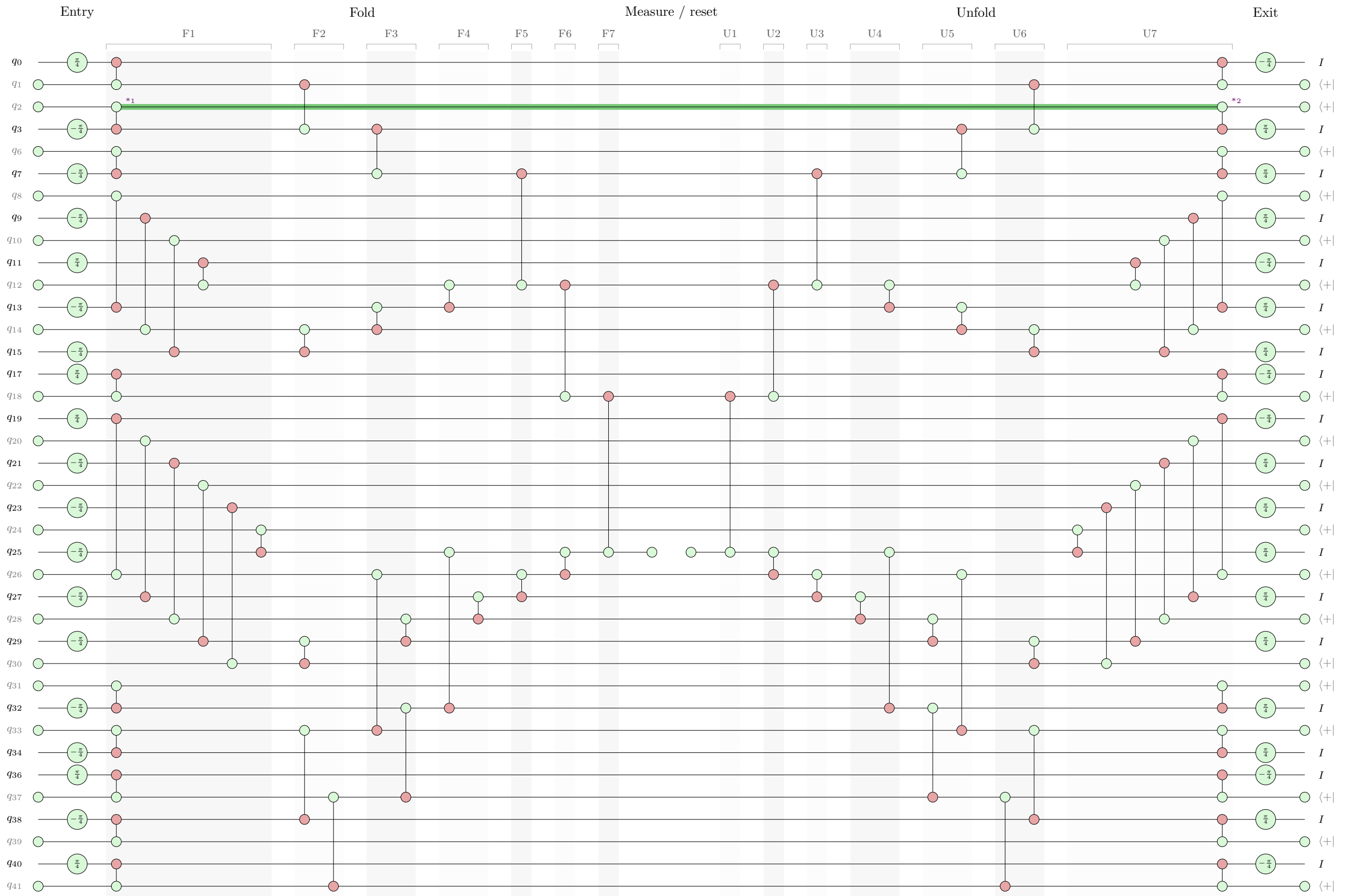
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_152.tikz

$d = 5$: basis element 153

No boundary support. 2 phase defects.



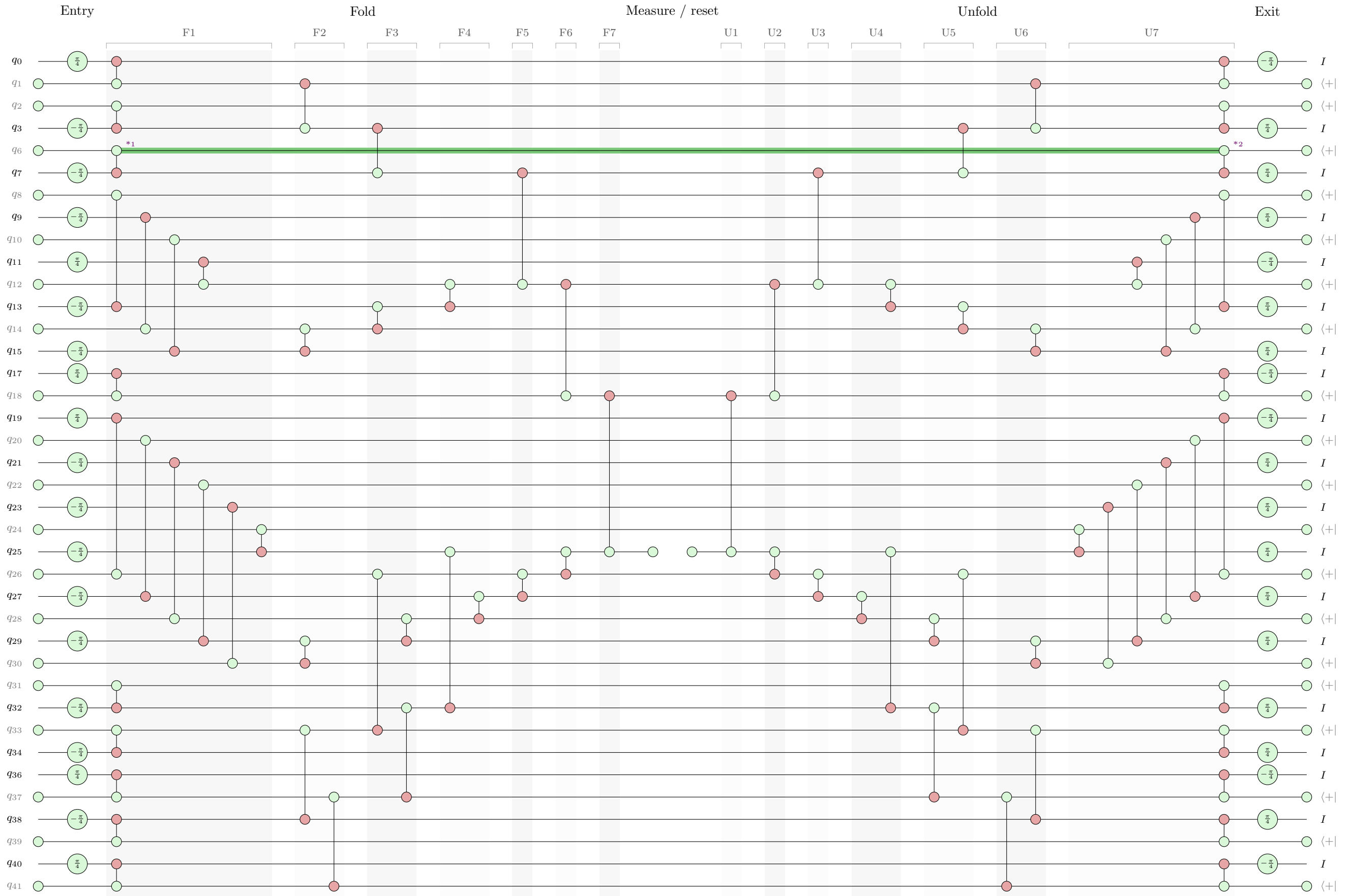
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_153.tikz

$d = 5$: basis element 154

No boundary support. 2 phase defects.



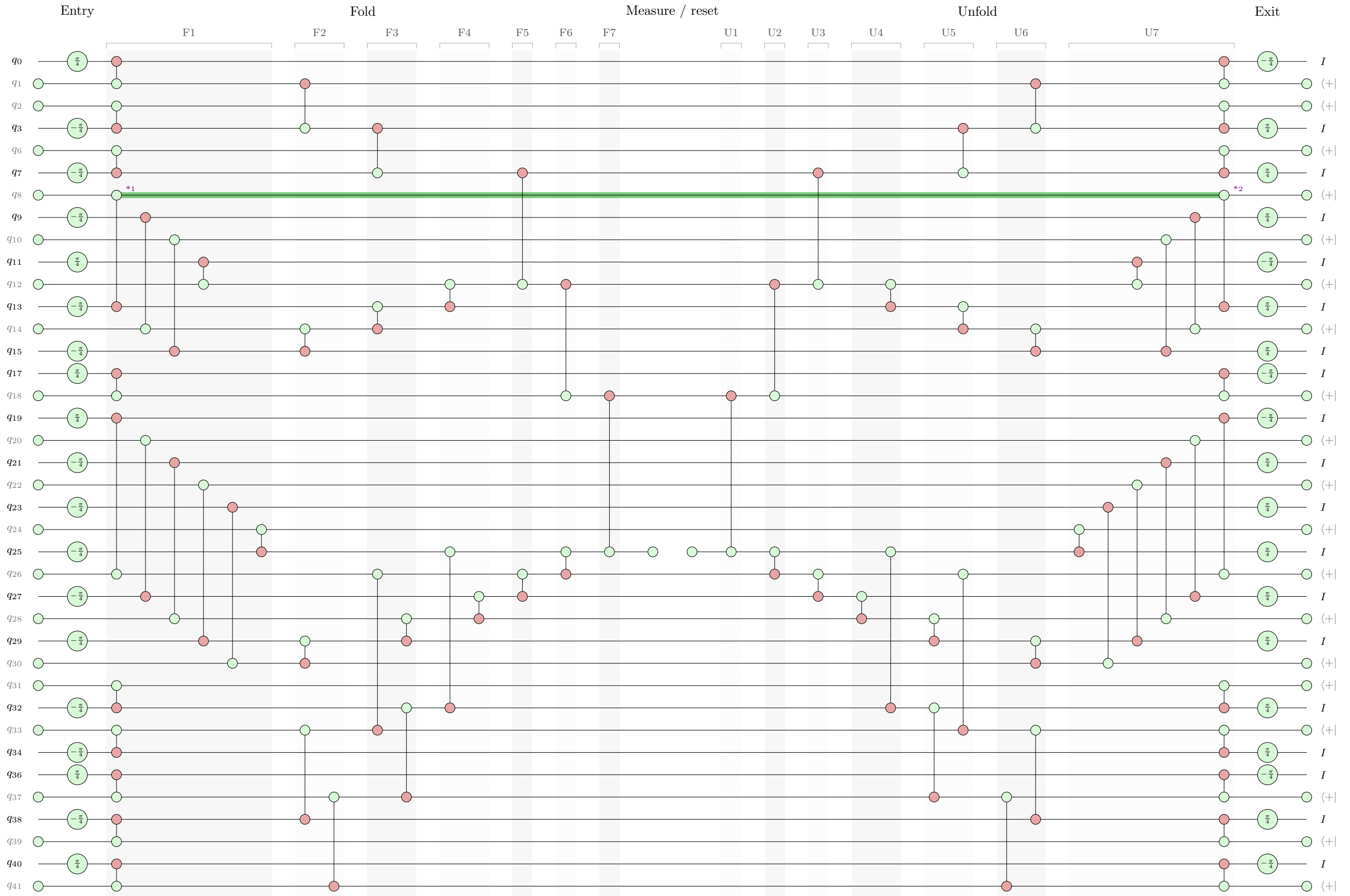
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_154.tikz

$d = 5$: basis element 155

No boundary support. 2 phase defects.



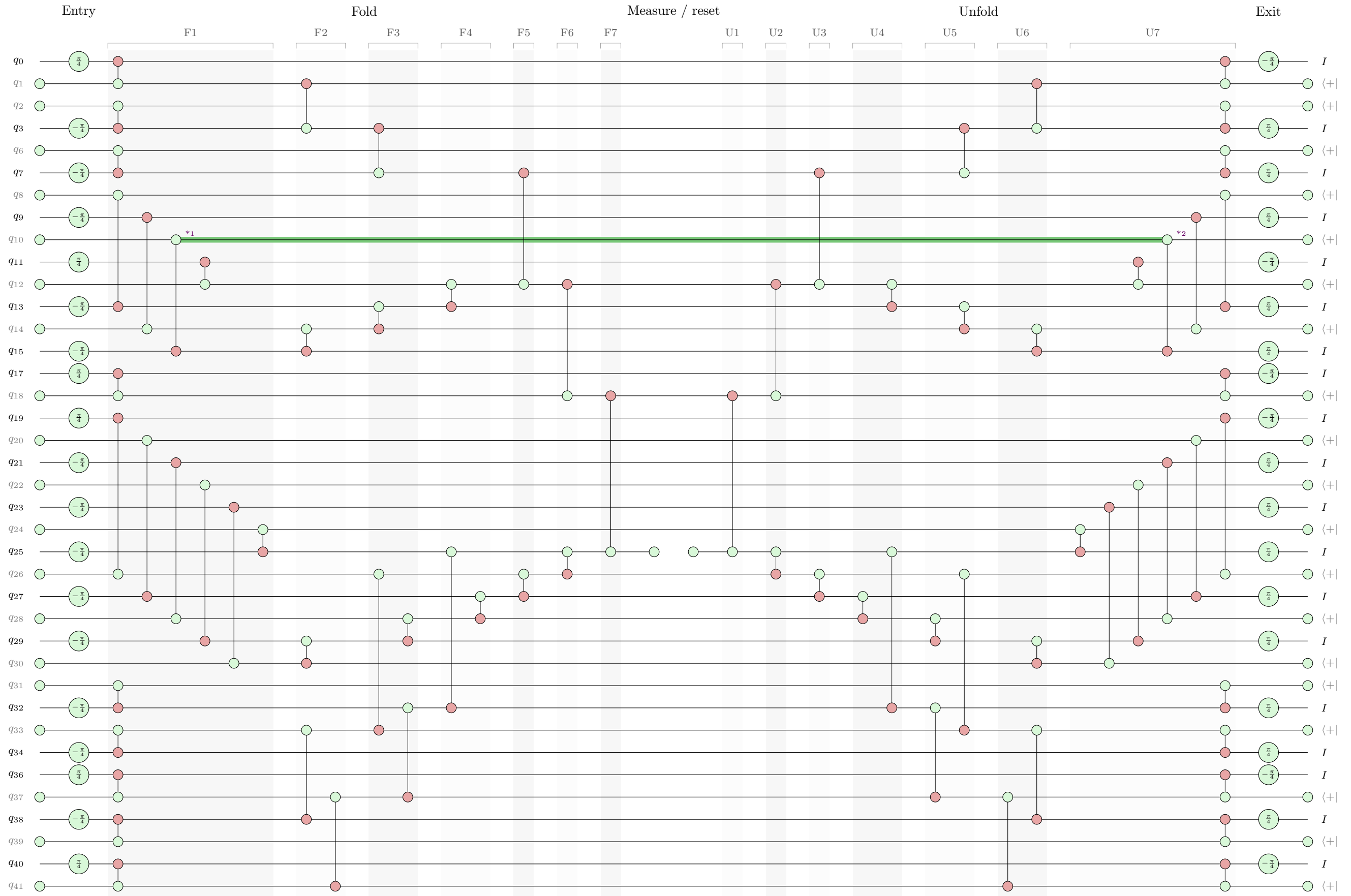
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_155.tikz

$d = 5$: basis element 156

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_156.tikz

$d = 5$: basis element 157

No boundary support. 2 phase defects.



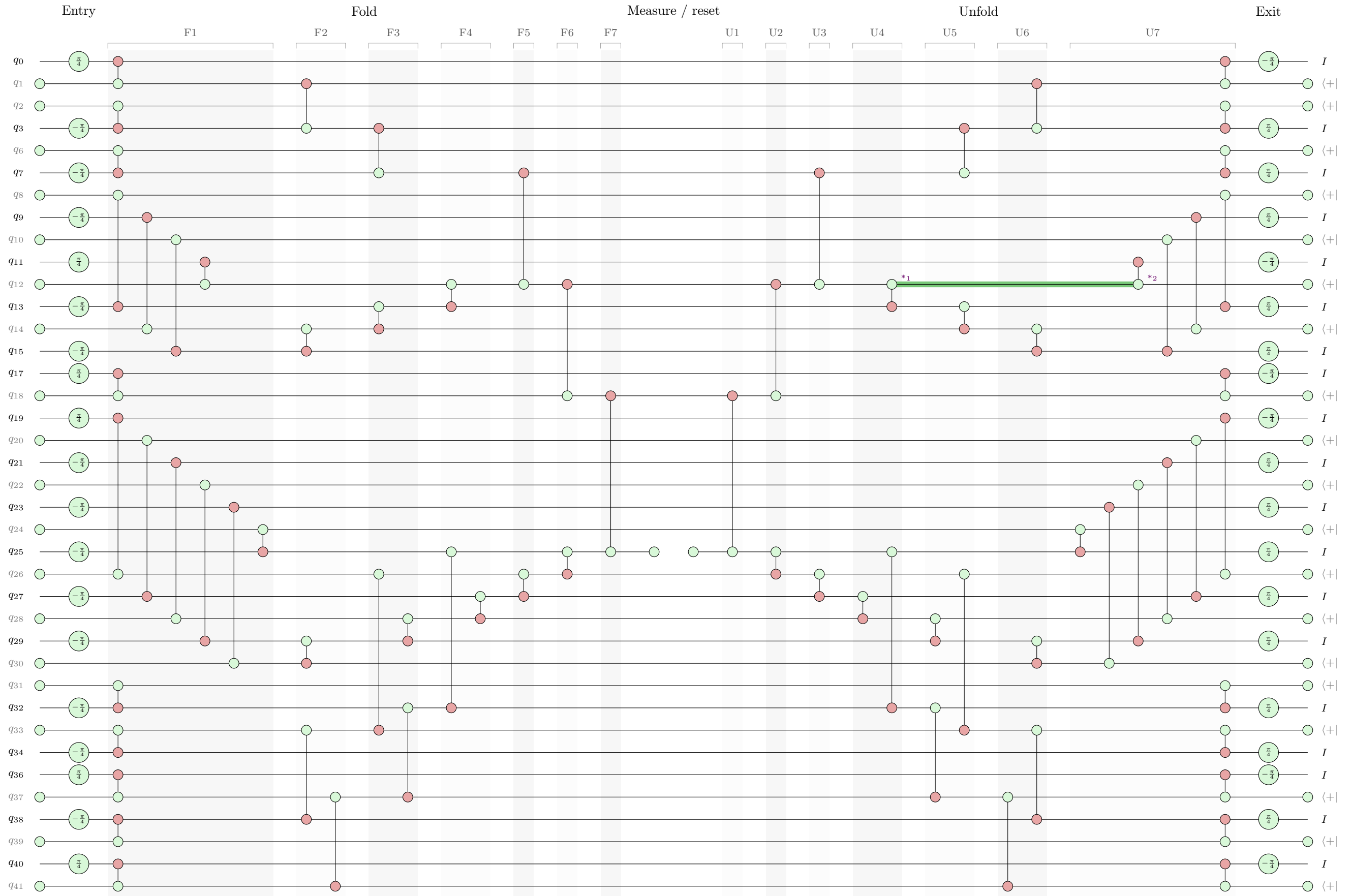
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_157.tikz

$d = 5$: basis element 158

No boundary support. 2 phase defects.



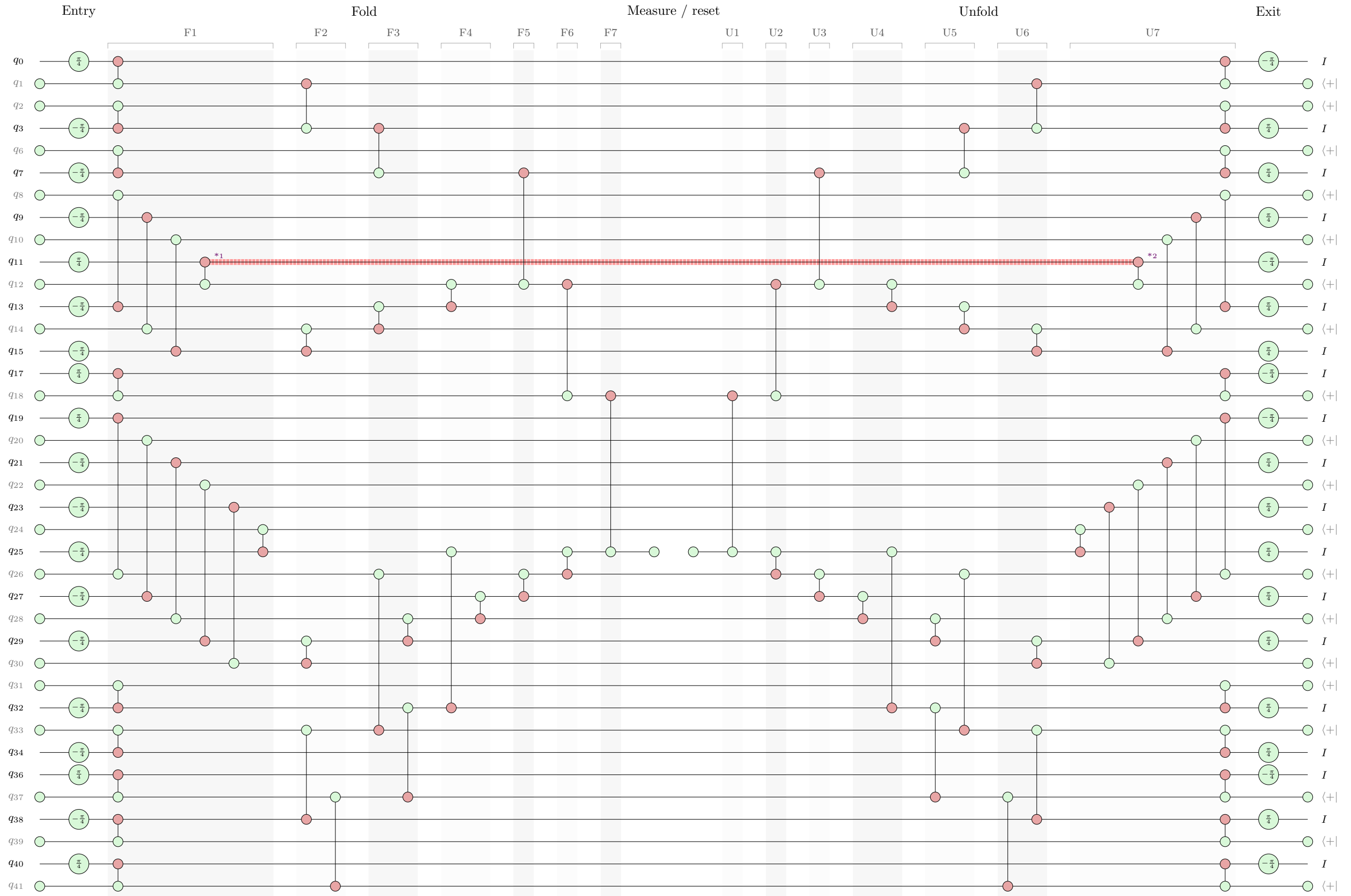
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_158.tikz

$d = 5$: basis element 159

No boundary support. 2 phase defects.



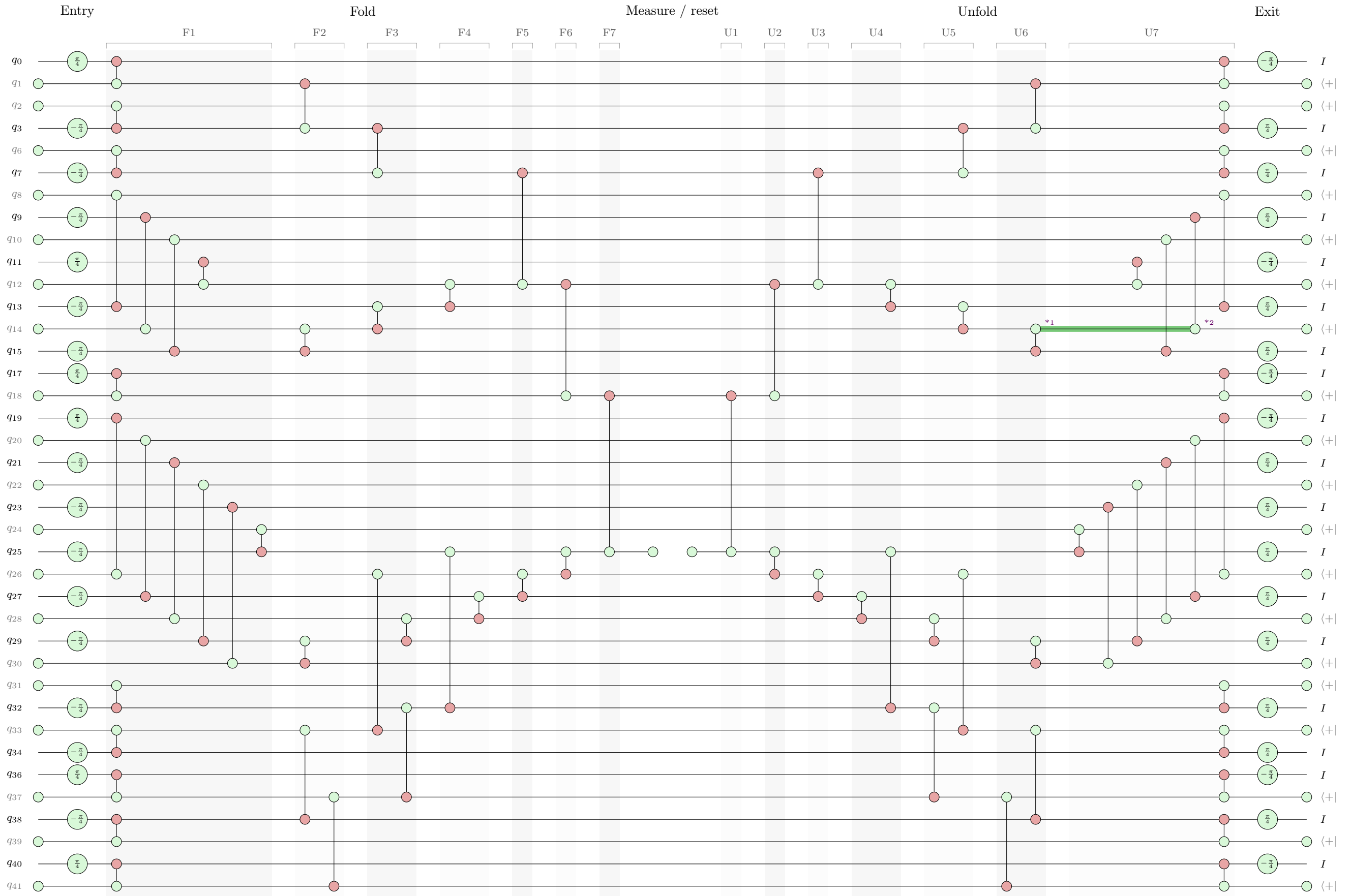
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_159.tikz

$d = 5$: basis element 160

No boundary support. 2 phase defects.



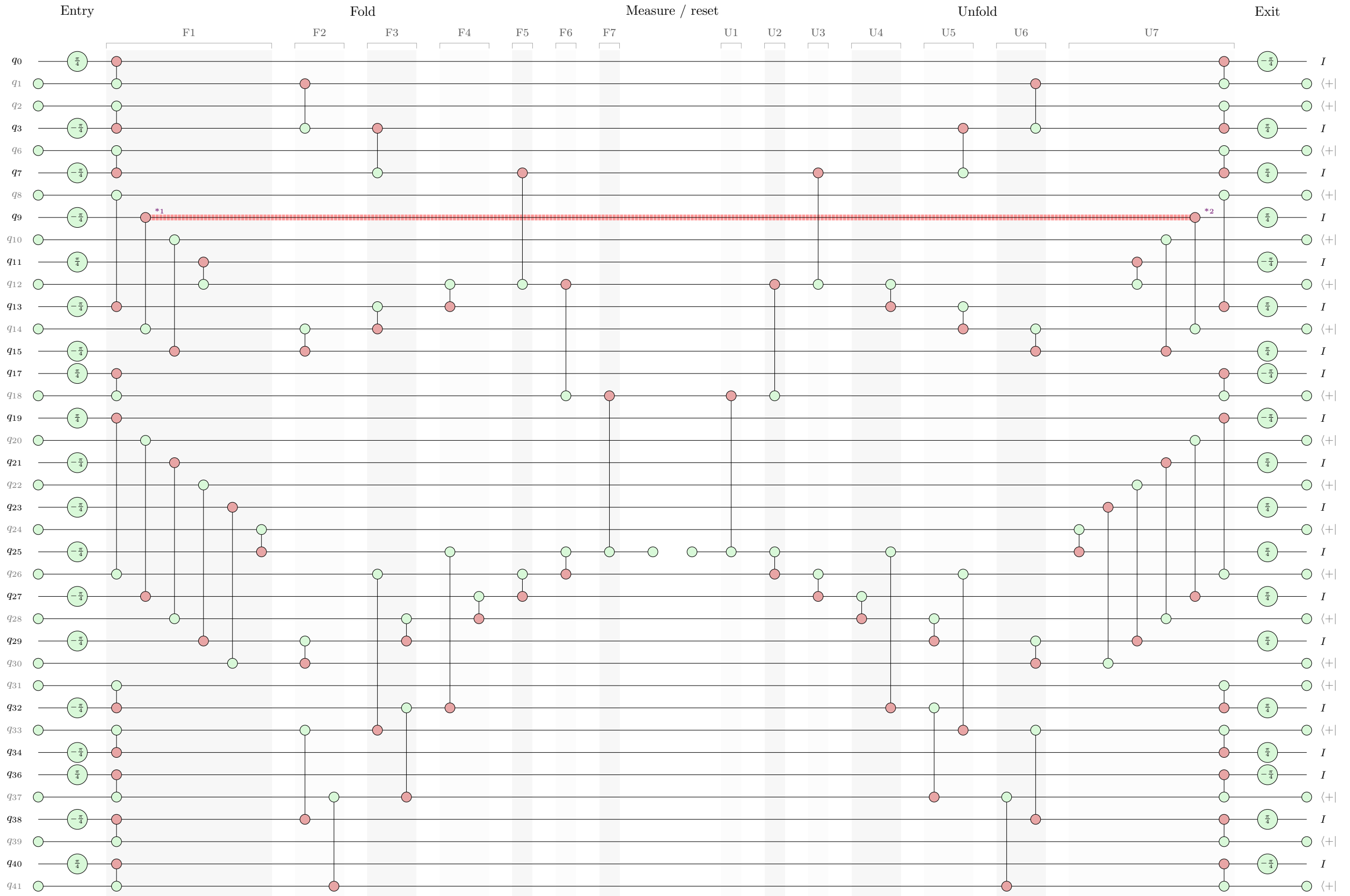
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_160.tikz

$d = 5$: basis element 161

No boundary support. 2 phase defects.



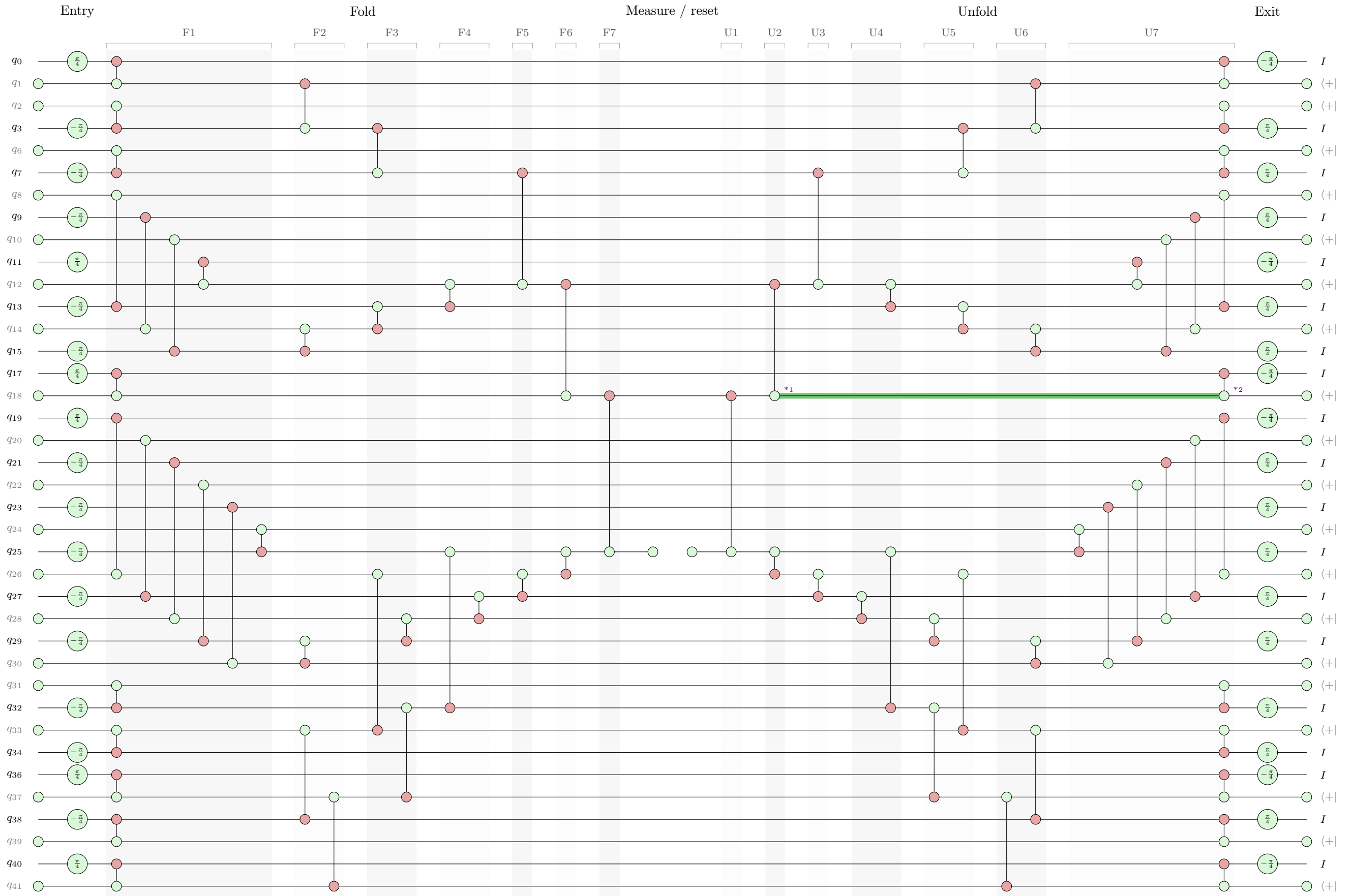
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_161.tikz

$d = 5$: basis element 162

No boundary support. 2 phase defects.



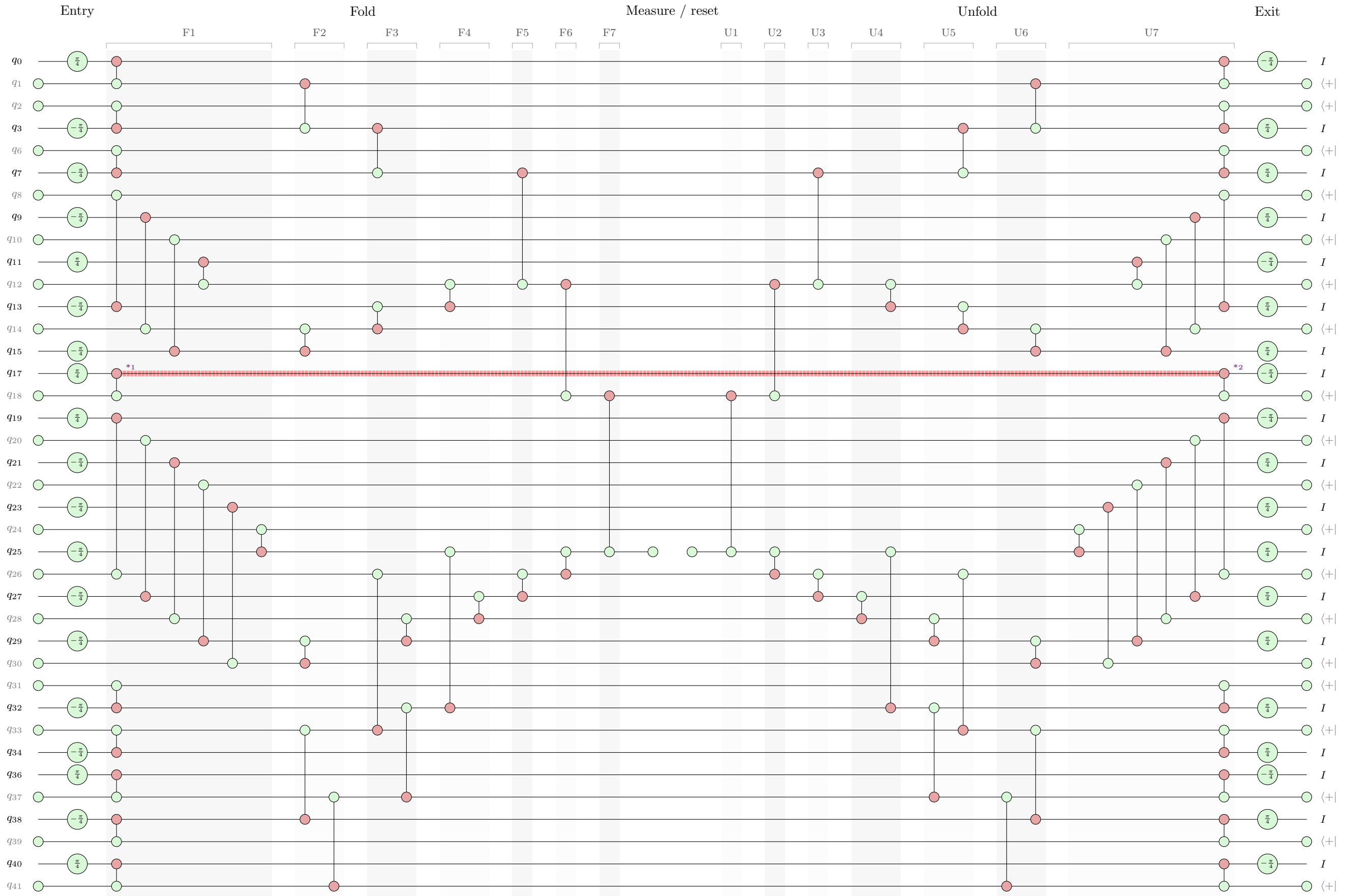
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_162.tikz

$d = 5$: basis element 163

No boundary support. 2 phase defects.



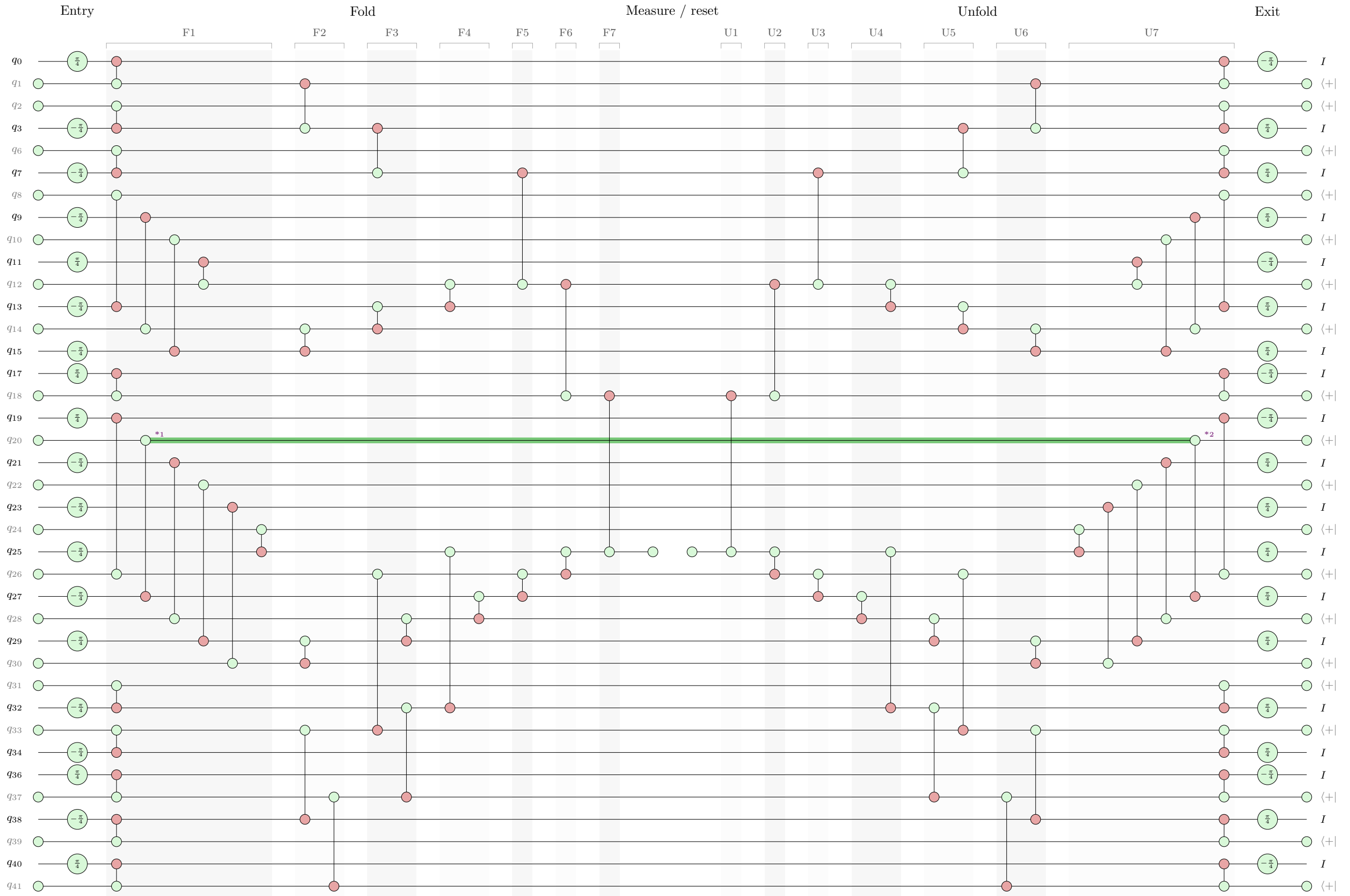
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_163.tikz

$d = 5$: basis element 164

No boundary support. 2 phase defects.



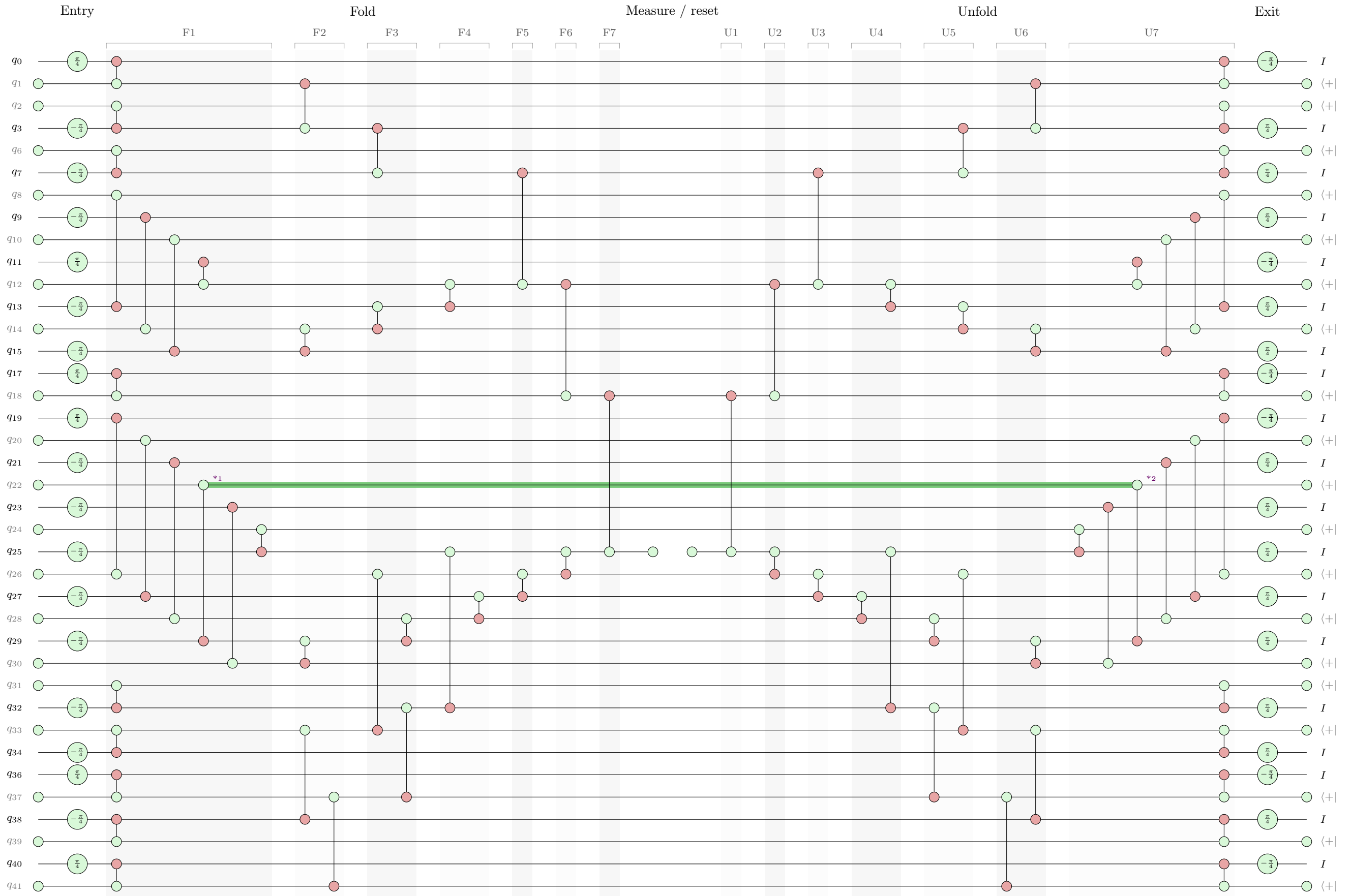
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_164.tikz

$d = 5$: basis element 165

No boundary support. 2 phase defects.



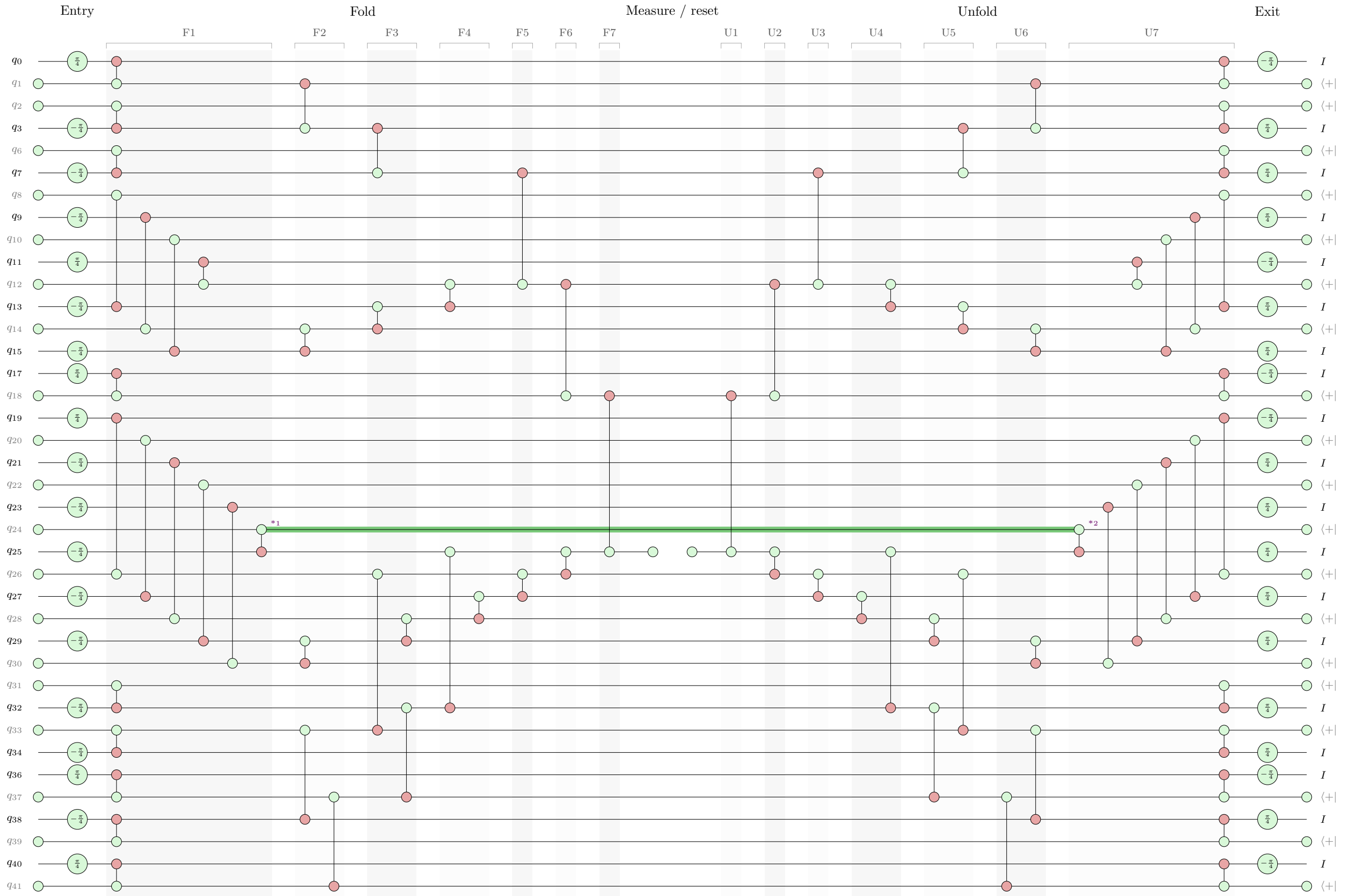
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_165.tikz

$d = 5$: basis element 166

No boundary support. 2 phase defects.



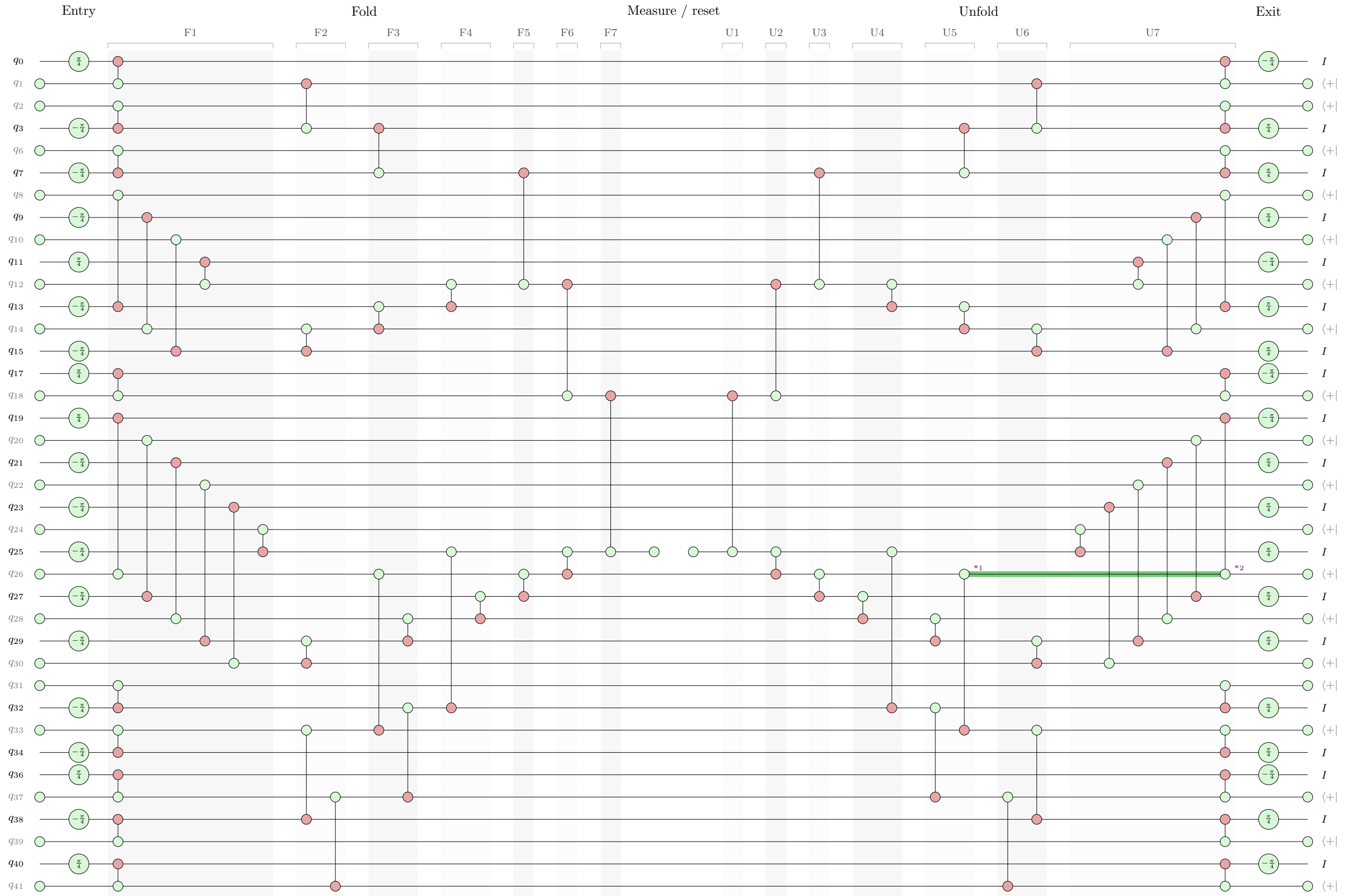
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_166.tikz

$d = 5$: basis element 167

No boundary support. 2 phase defects.



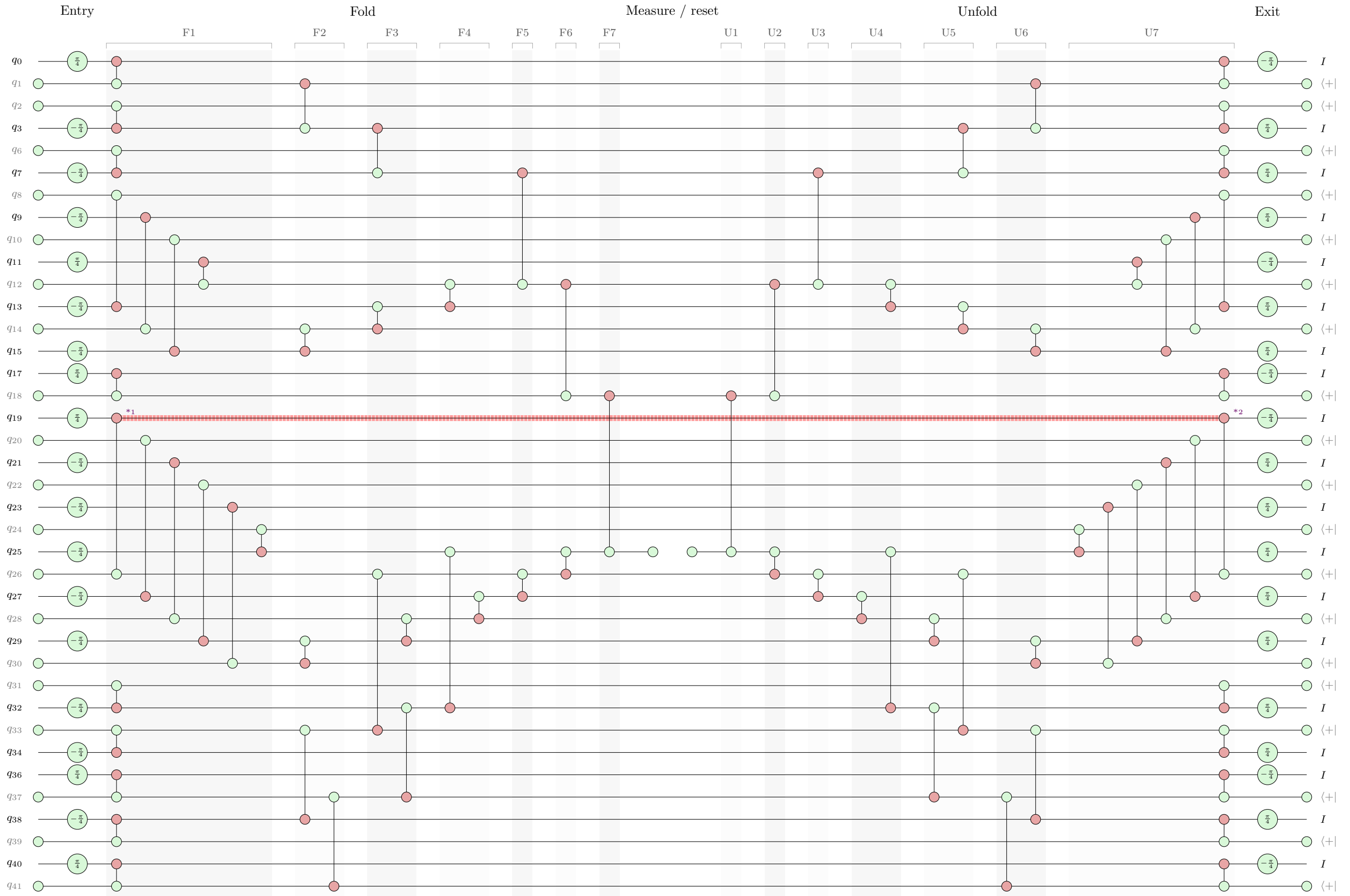
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_167.tikz

$d = 5$: basis element 168

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_168.tikz

$d = 5$: basis element 169

No boundary support. 2 phase defects.



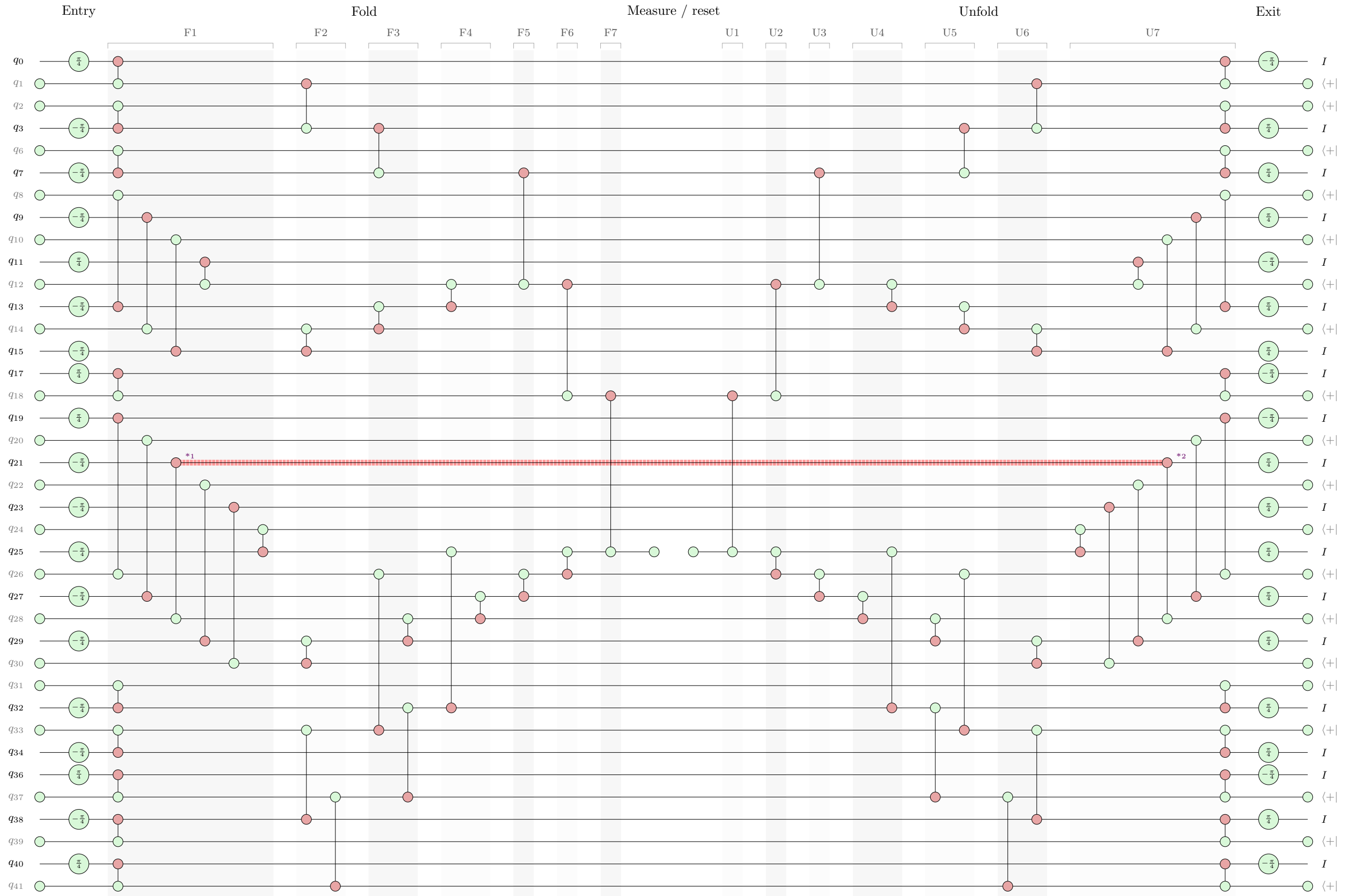
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_169.tikz

$d = 5$: basis element 170

No boundary support. 2 phase defects.



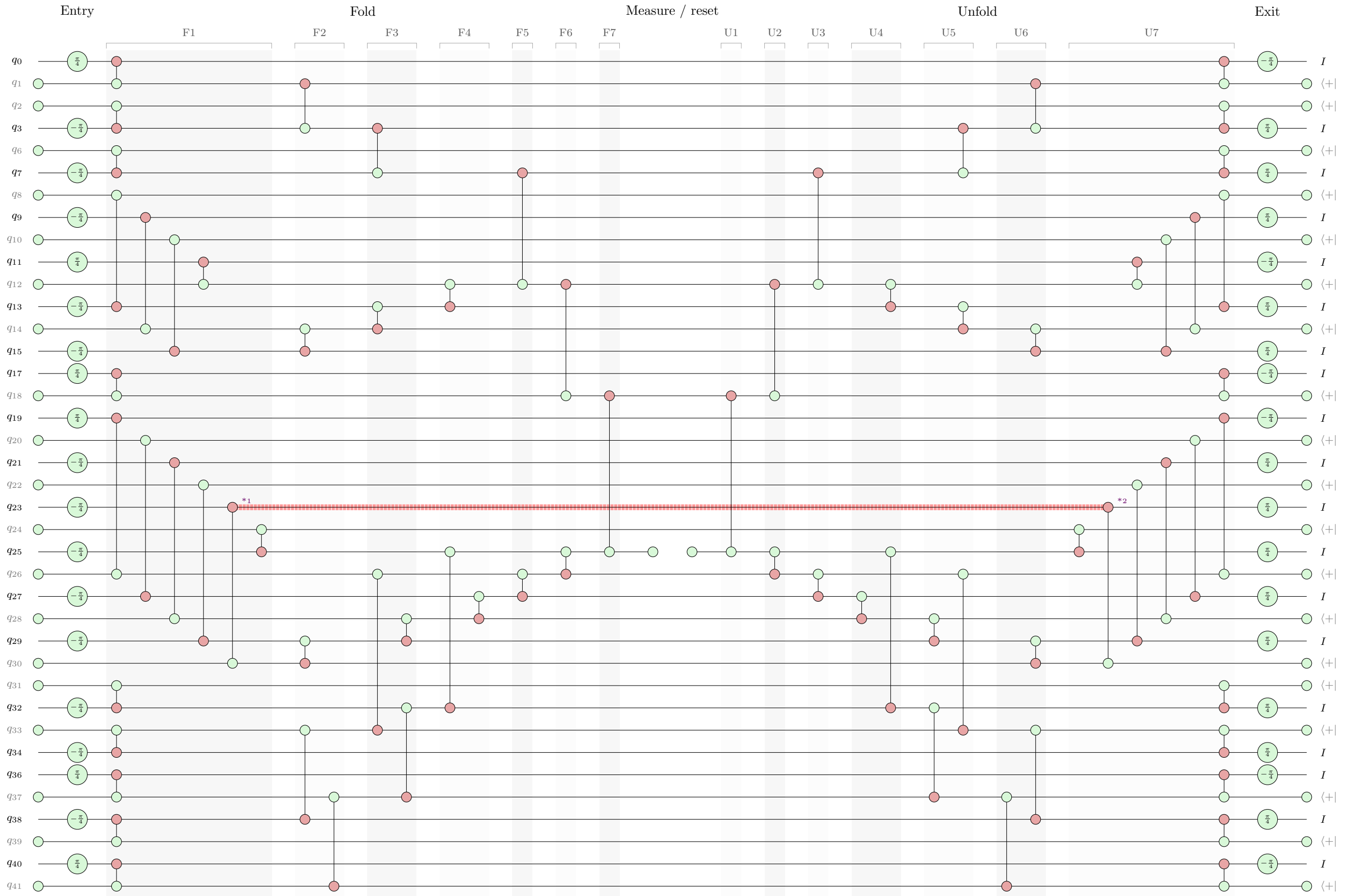
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_170.tikz

$d = 5$: basis element 171

No boundary support. 2 phase defects.



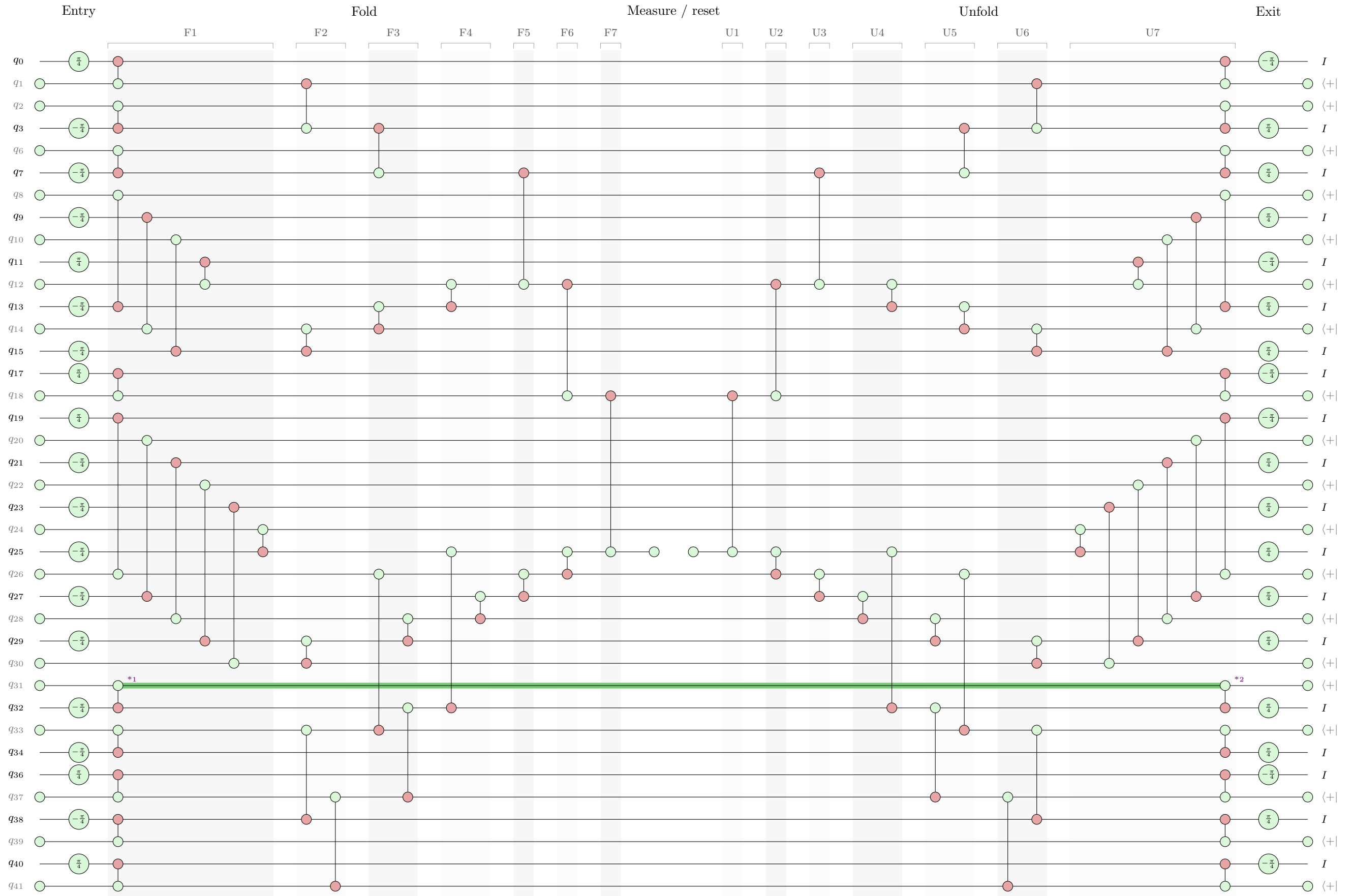
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_171.tikz

$d = 5$: basis element 172

No boundary support. 2 phase defects.



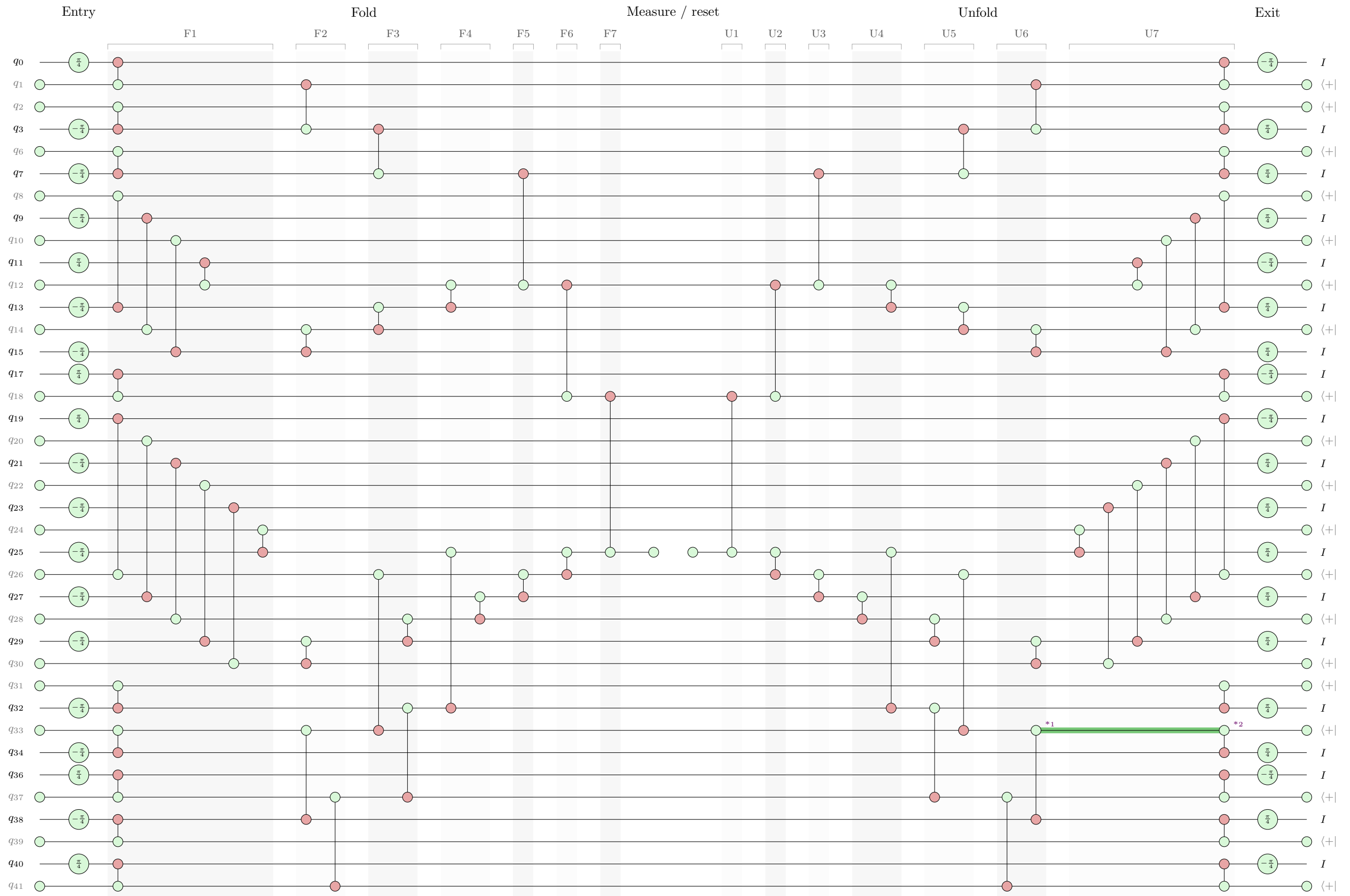
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_172.tikz

$d = 5$: basis element 173

No boundary support. 2 phase defects.



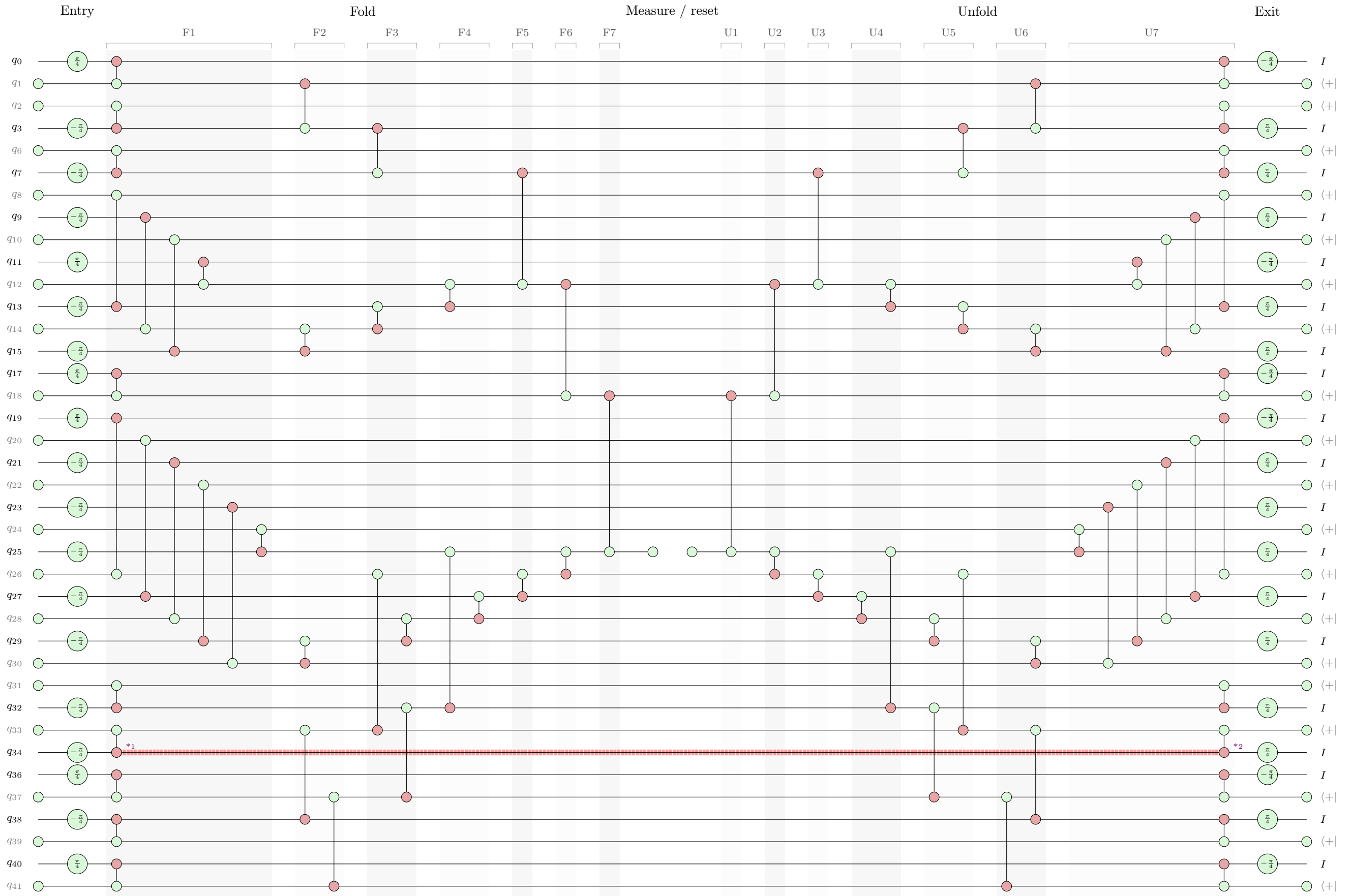
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_173.tikz

$d = 5$: basis element 174

No boundary support. 2 phase defects.



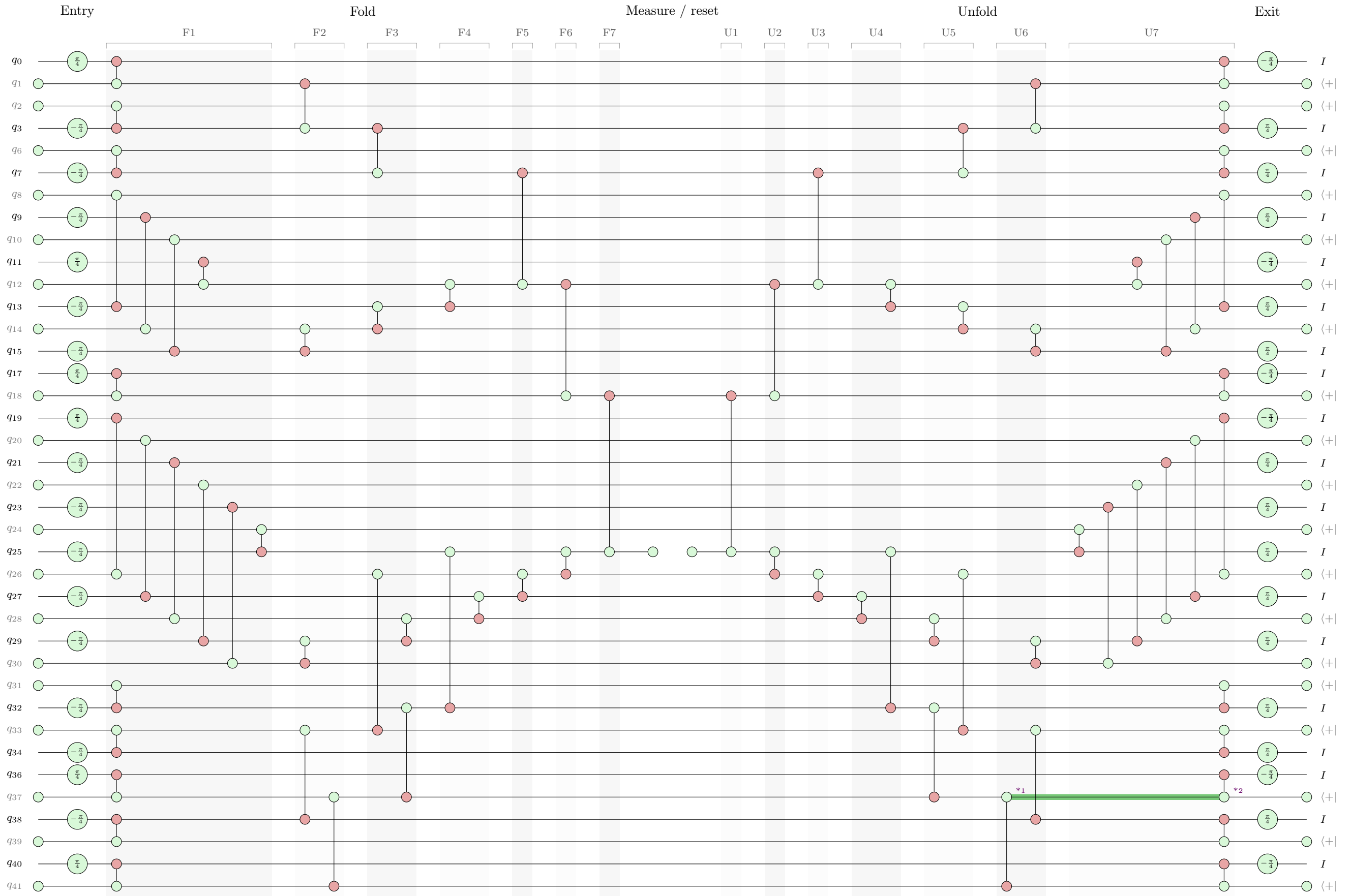
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_174.tikz

$d = 5$: basis element 175

No boundary support. 2 phase defects.



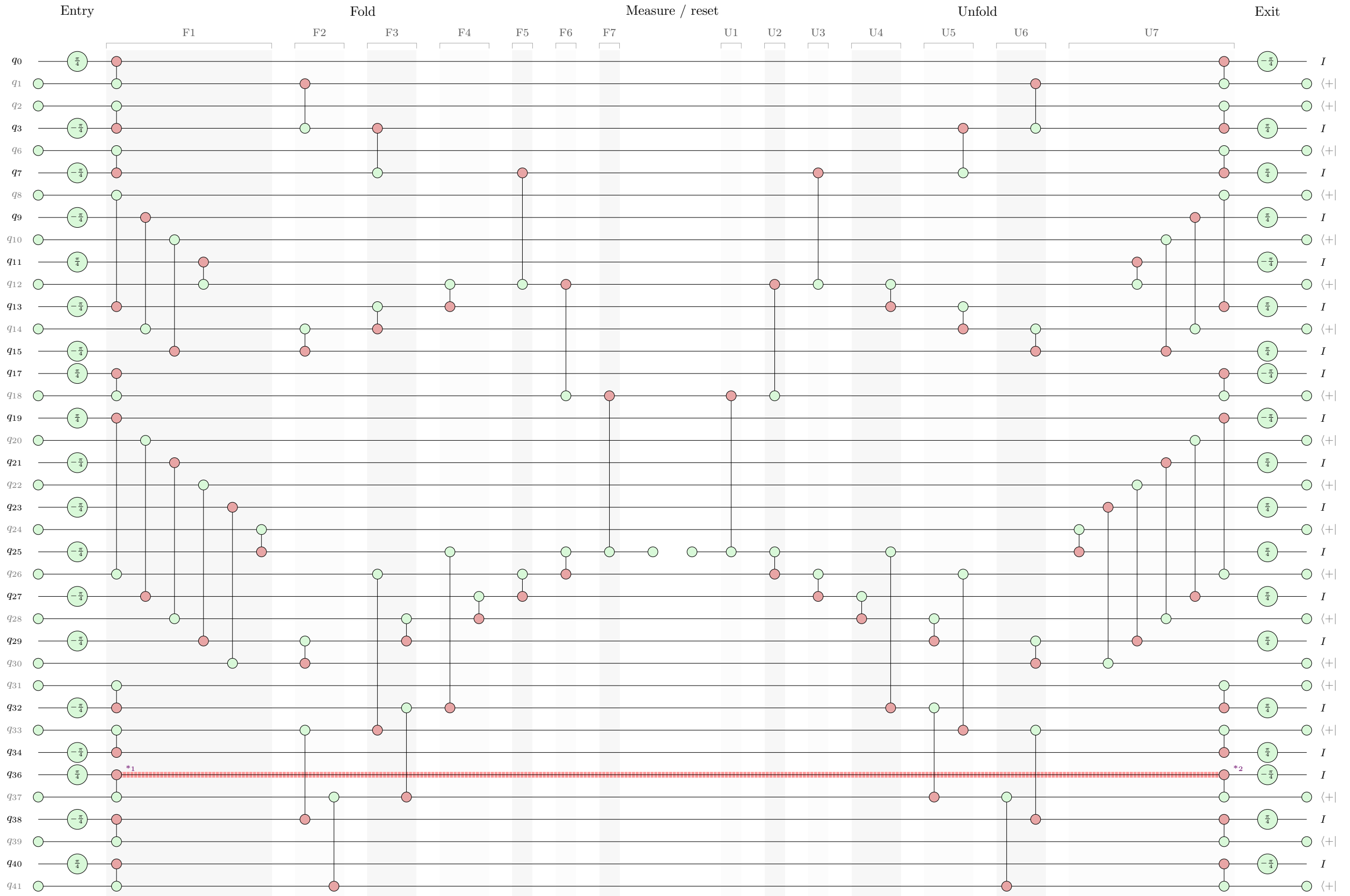
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_175.tikz

$d = 5$: basis element 176

No boundary support. 2 phase defects.



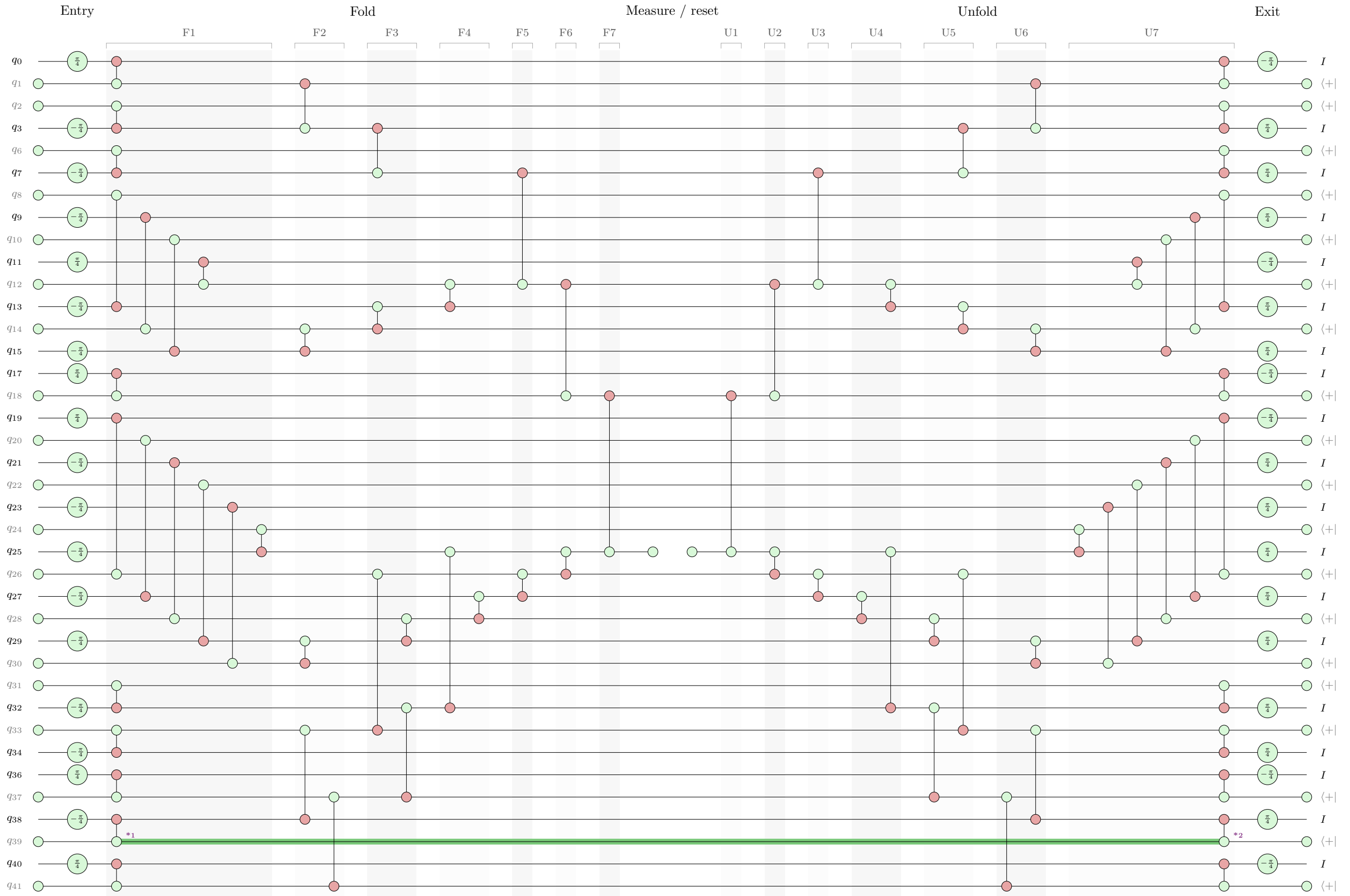
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_176.tikz

$d = 5$: basis element 177

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_177.tikz

$d = 5$: basis element 178

No boundary support. 2 phase defects.



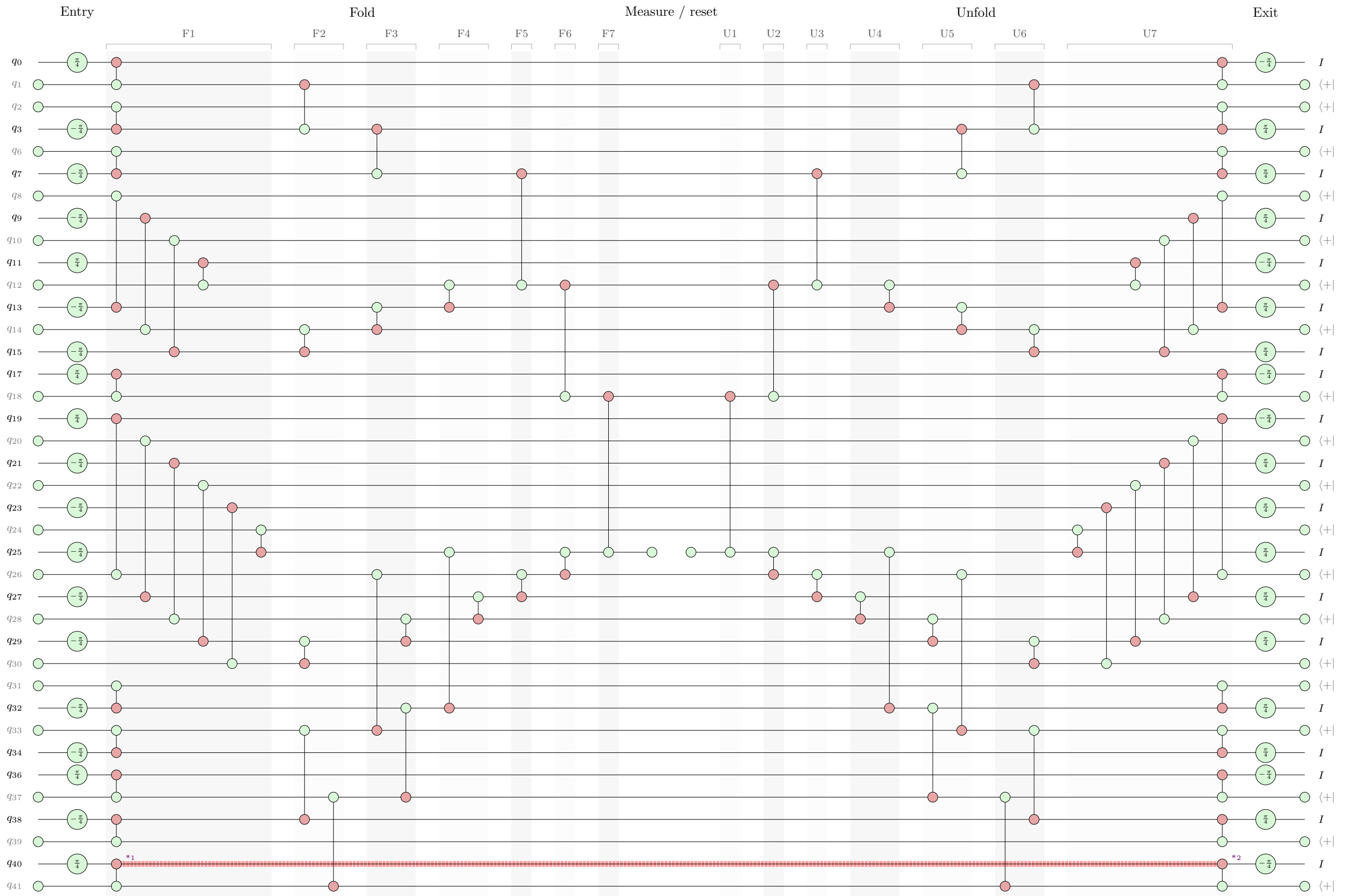
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_178.tikz

$d = 5$: basis element 179

No boundary support. 2 phase defects.



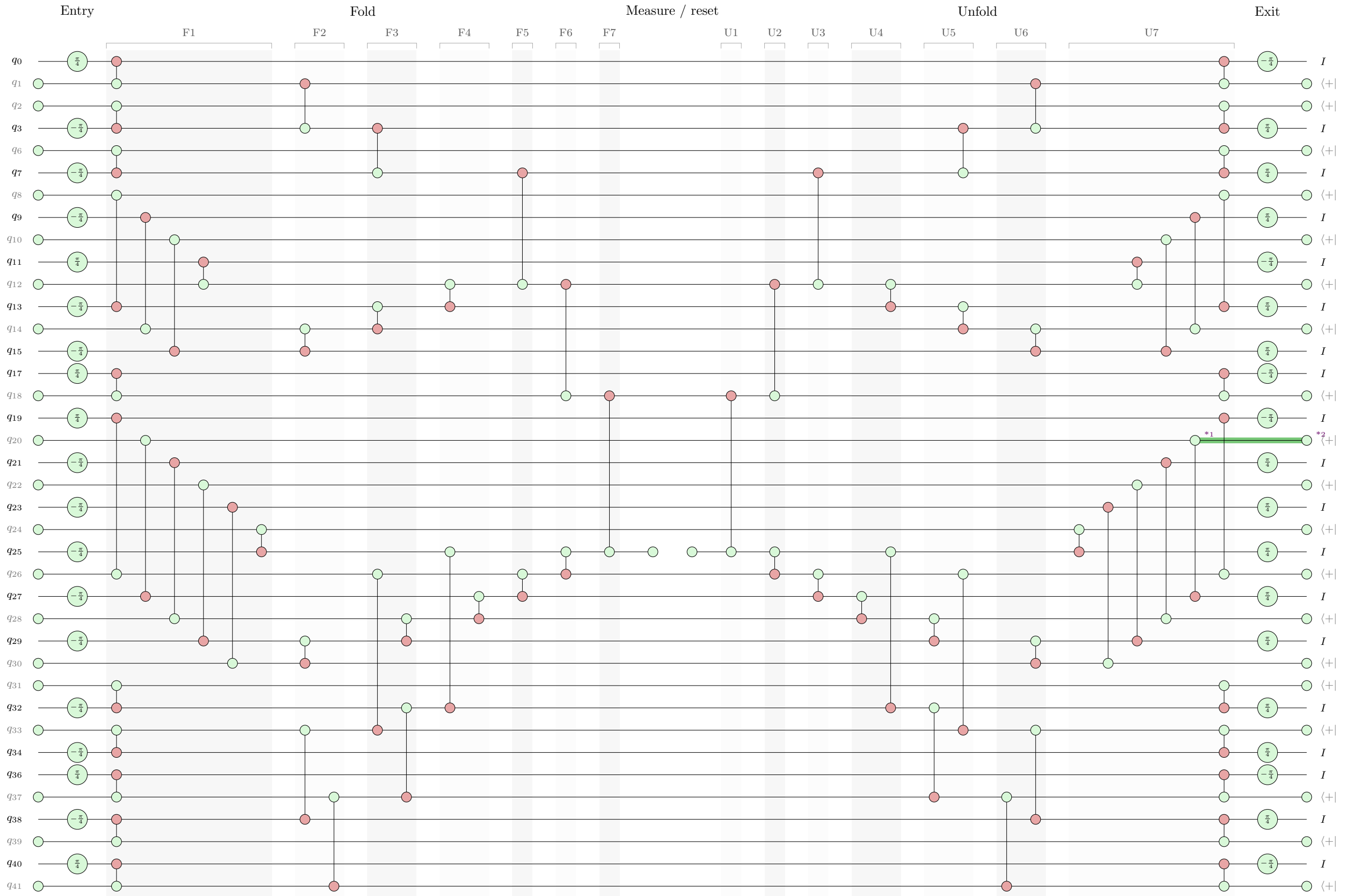
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_179.tikz

$d = 5$: basis element 180

No boundary support. 2 phase defects.



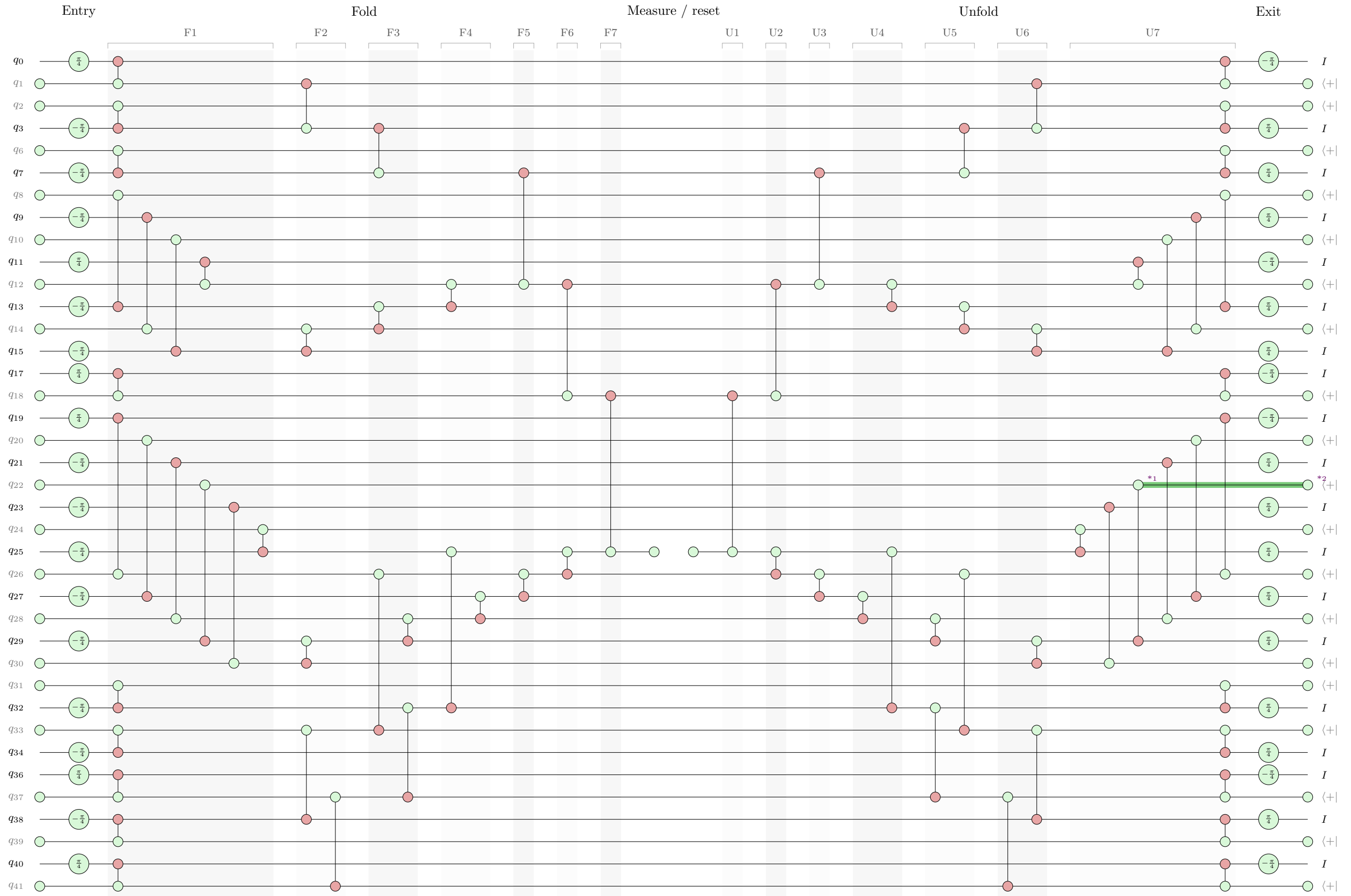
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_180.tikz

$d = 5$: basis element 181

No boundary support. 2 phase defects.



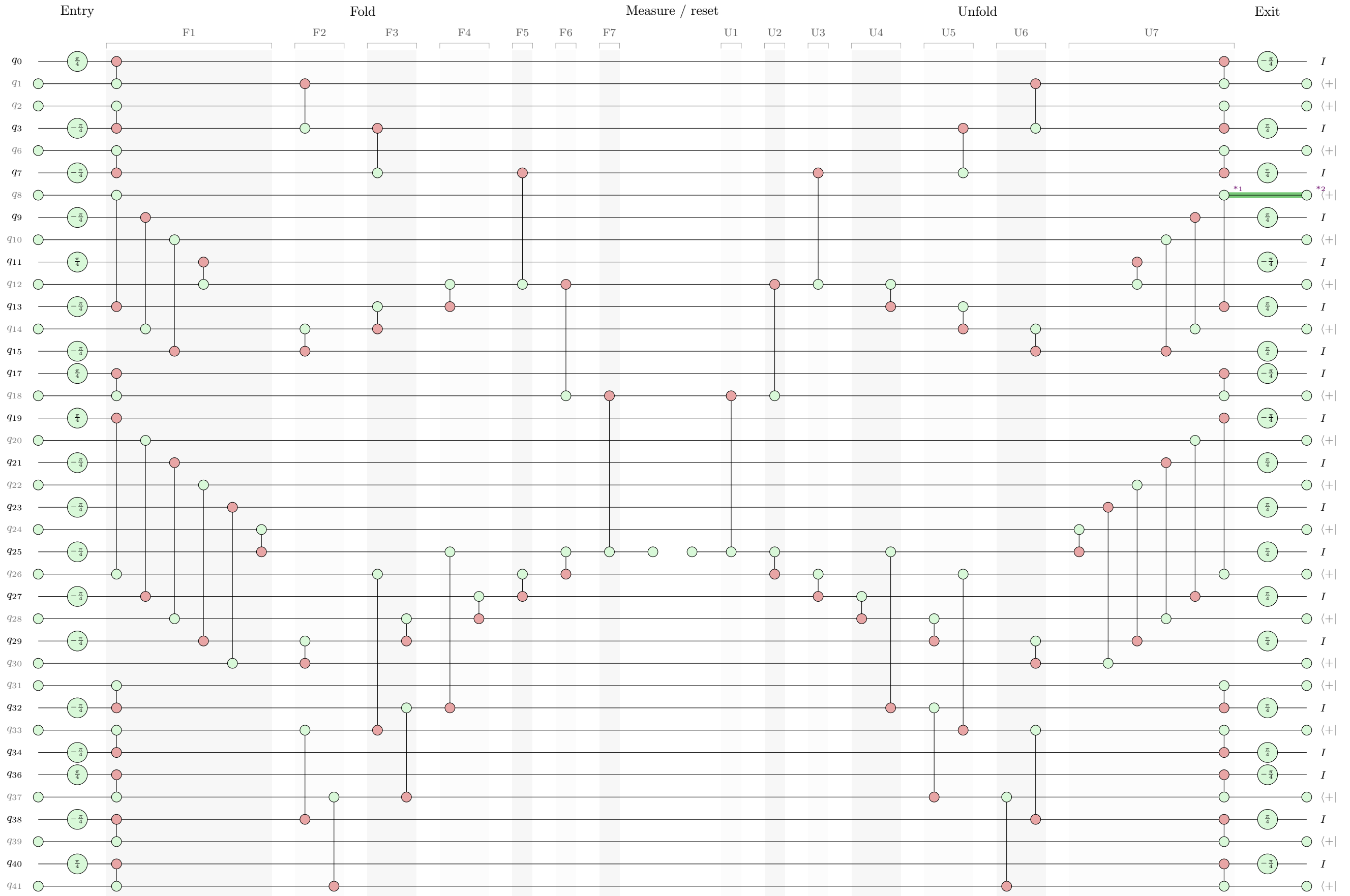
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_181.tikz

$d = 5$: basis element 182

No boundary support. 2 phase defects.



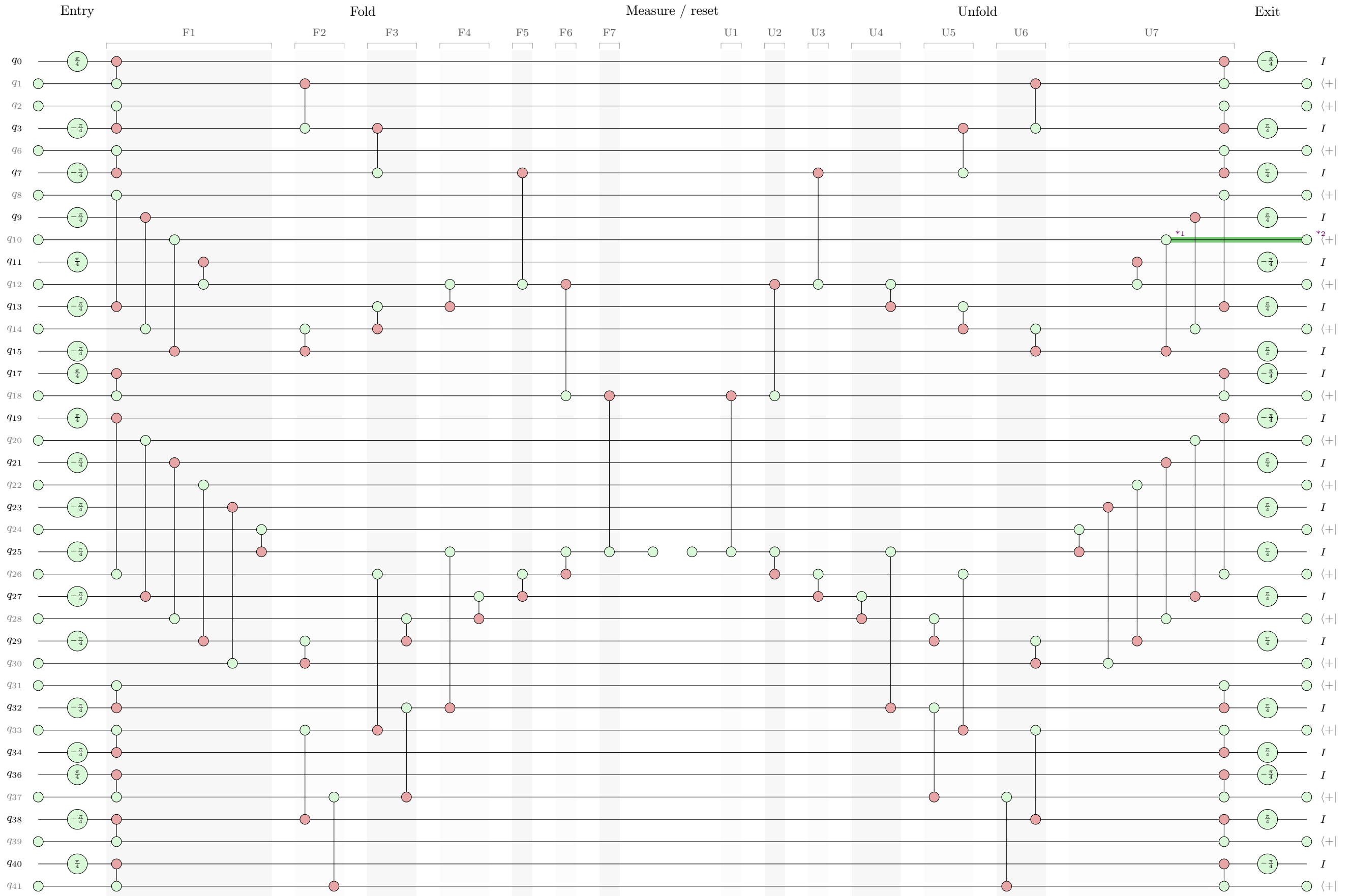
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_182.tikz

$d = 5$: basis element 183

No boundary support. 2 phase defects.



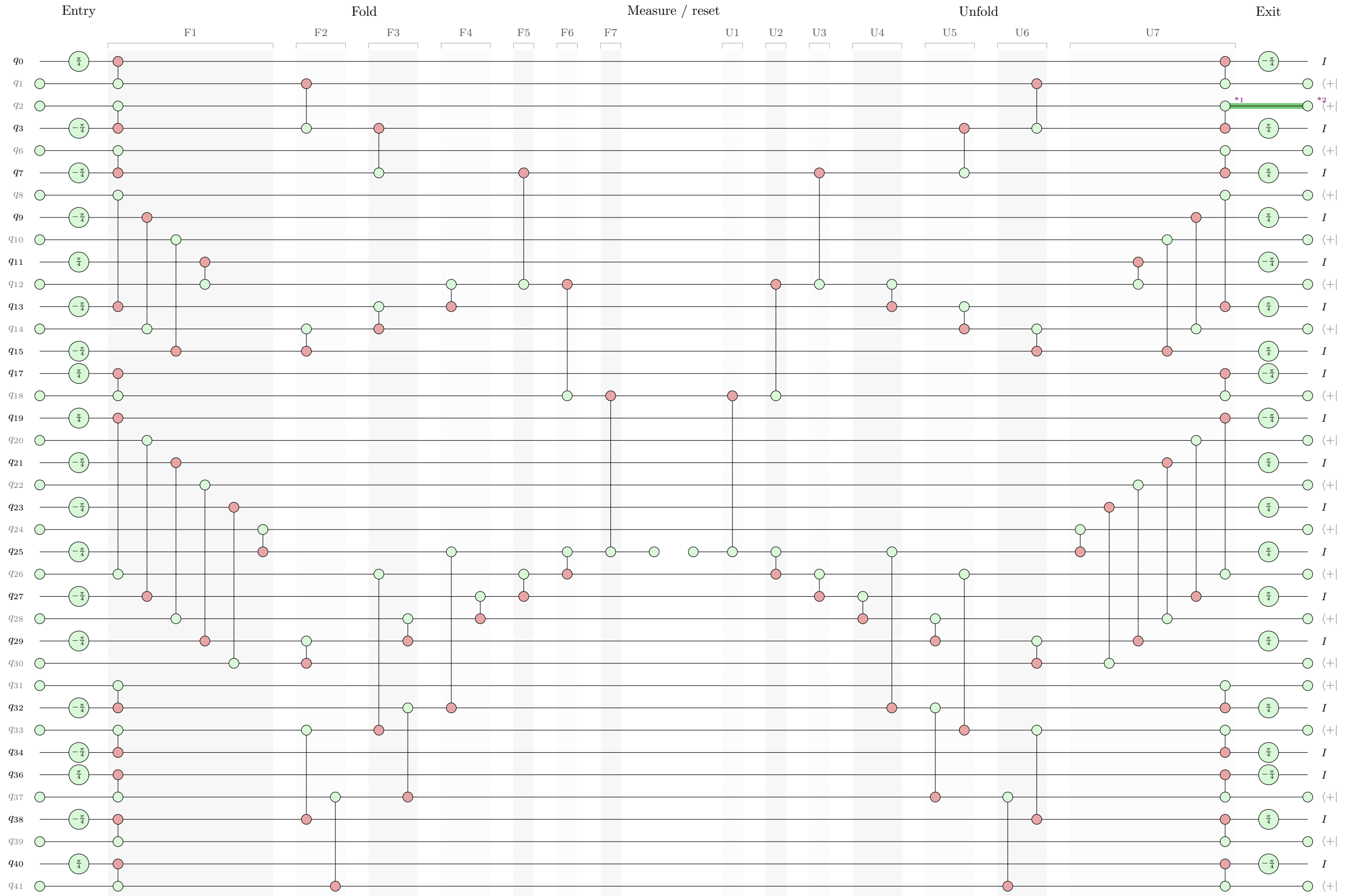
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_183.tikz

$d = 5$: basis element 184

No boundary support. 2 phase defects.



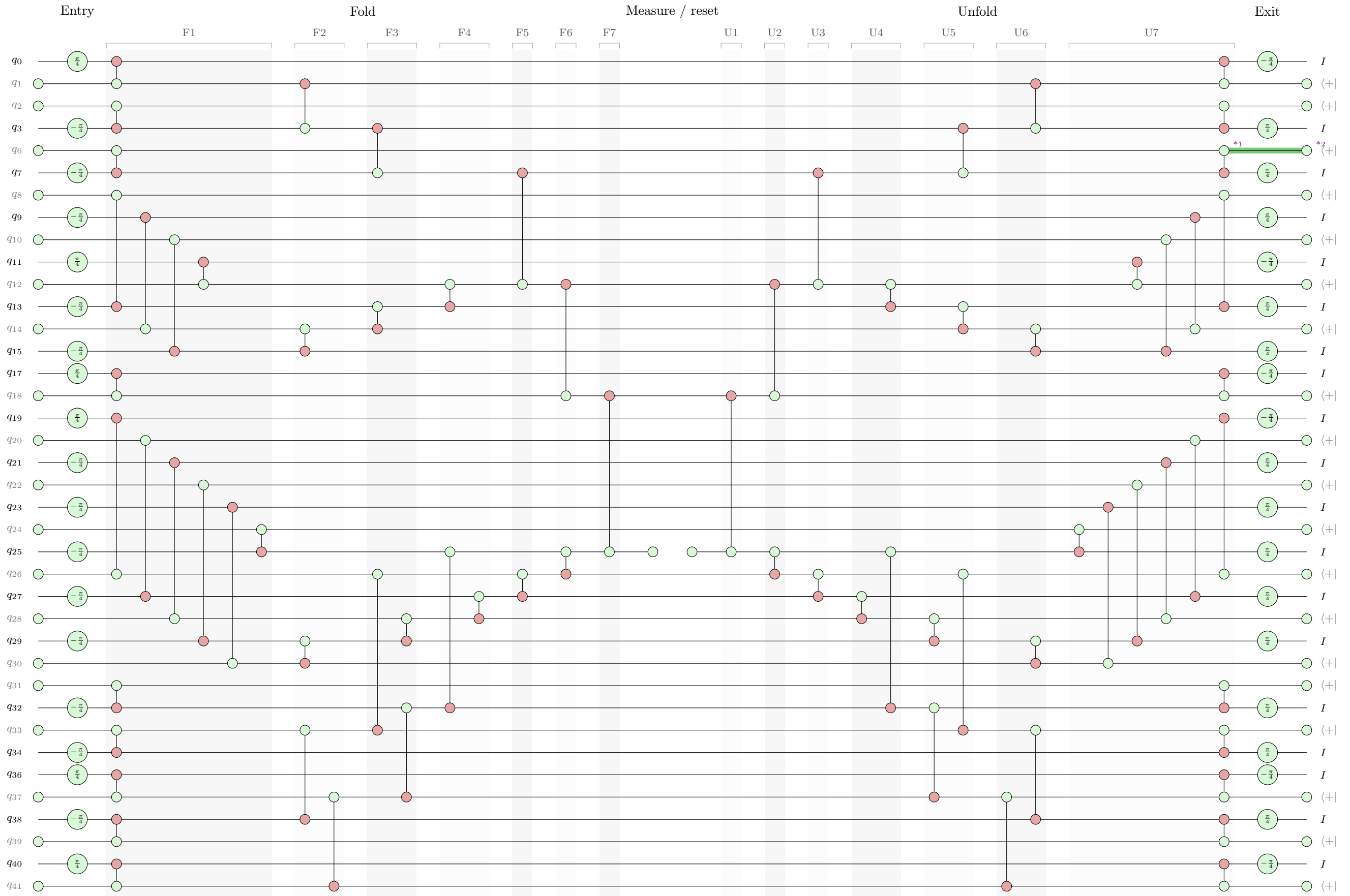
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_184.tikz

$d = 5$: basis element 185

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_185.tikz

$d = 5$: basis element 186

No boundary support. 2 phase defects.



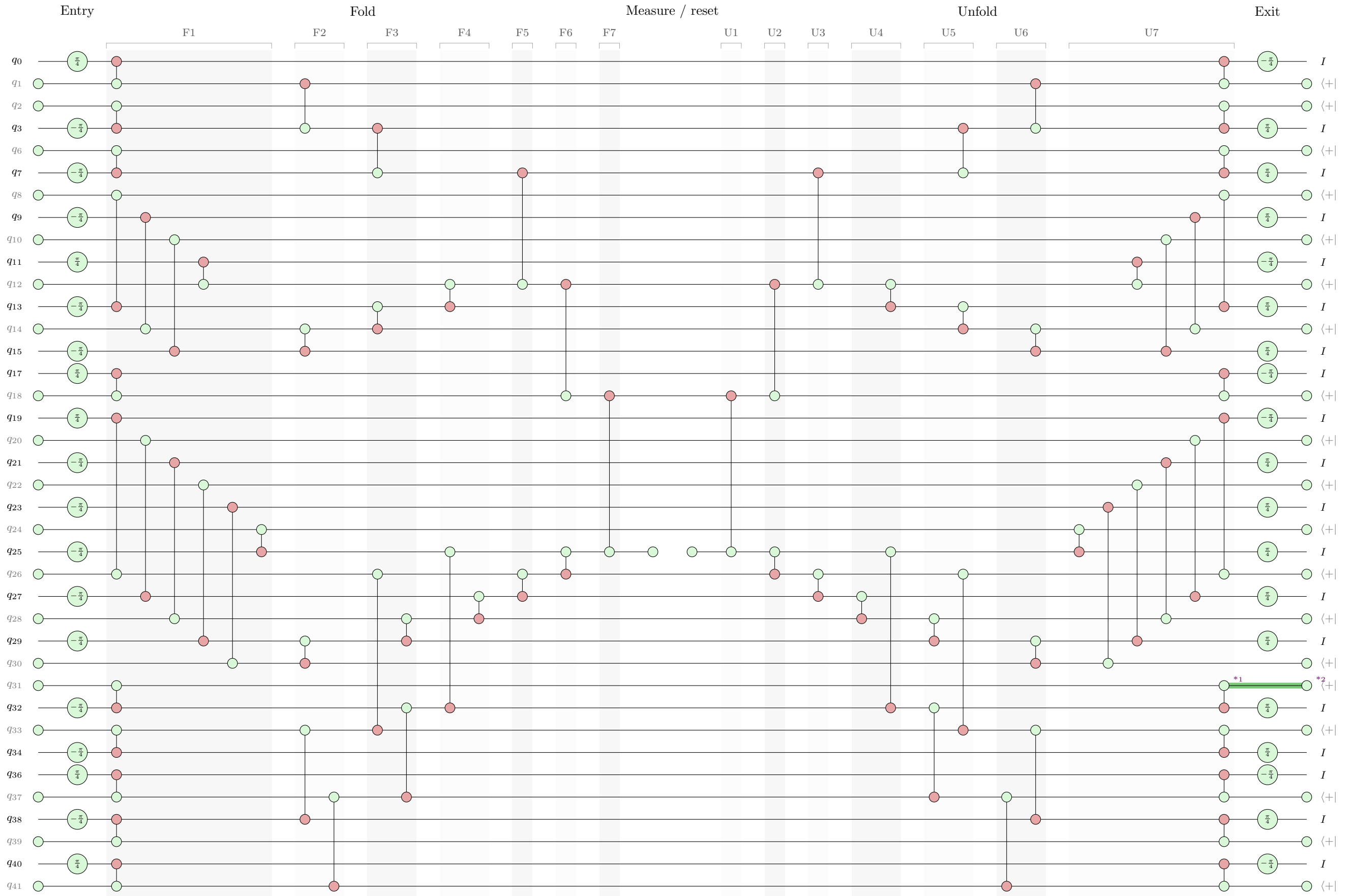
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_186.tikz

$d = 5$: basis element 187

No boundary support. 2 phase defects.



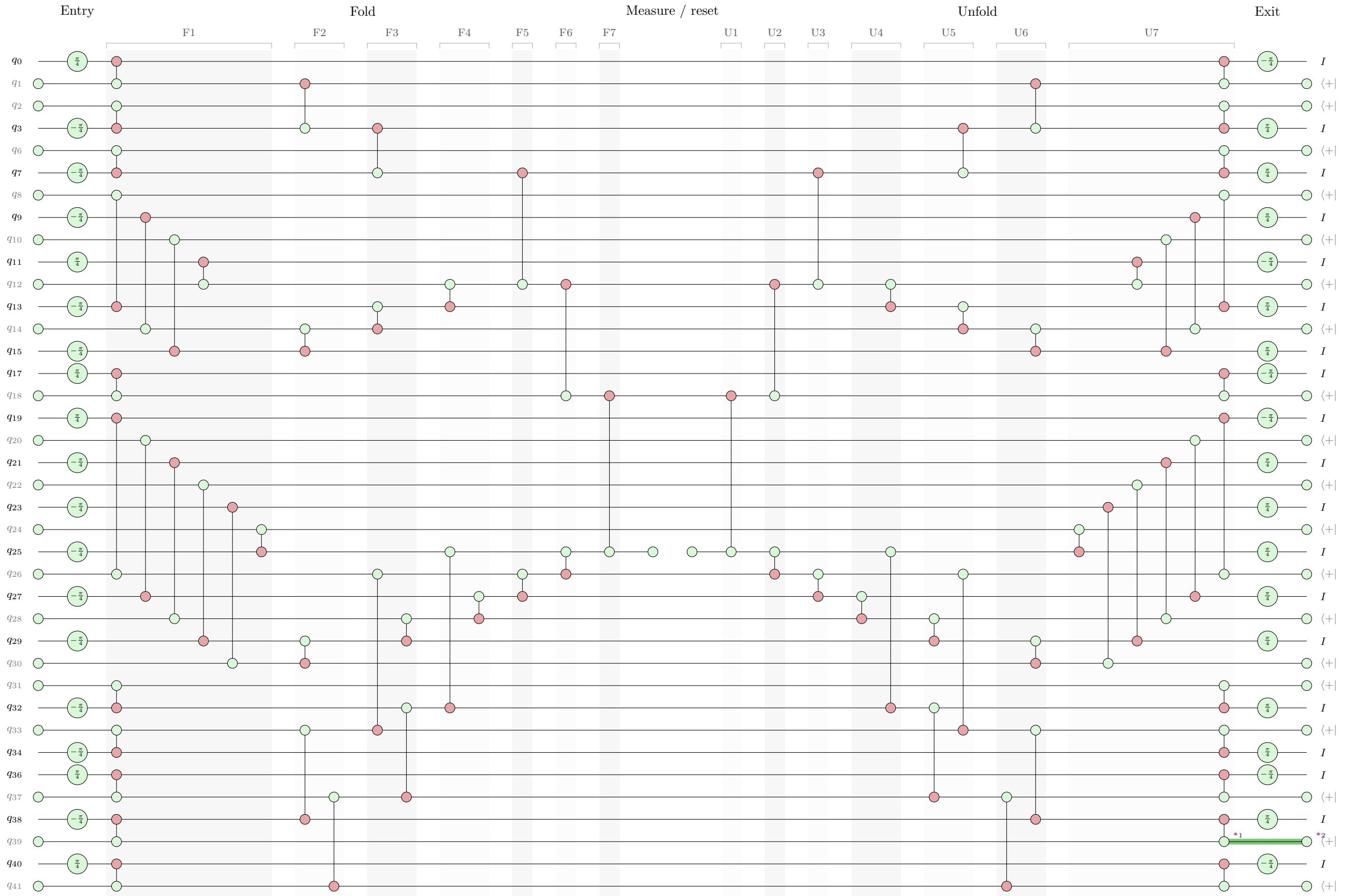
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_187.tikz

$d = 5$: basis element 188

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_188.tikz

$d = 5$: basis element 189

No boundary support. 2 phase defects.



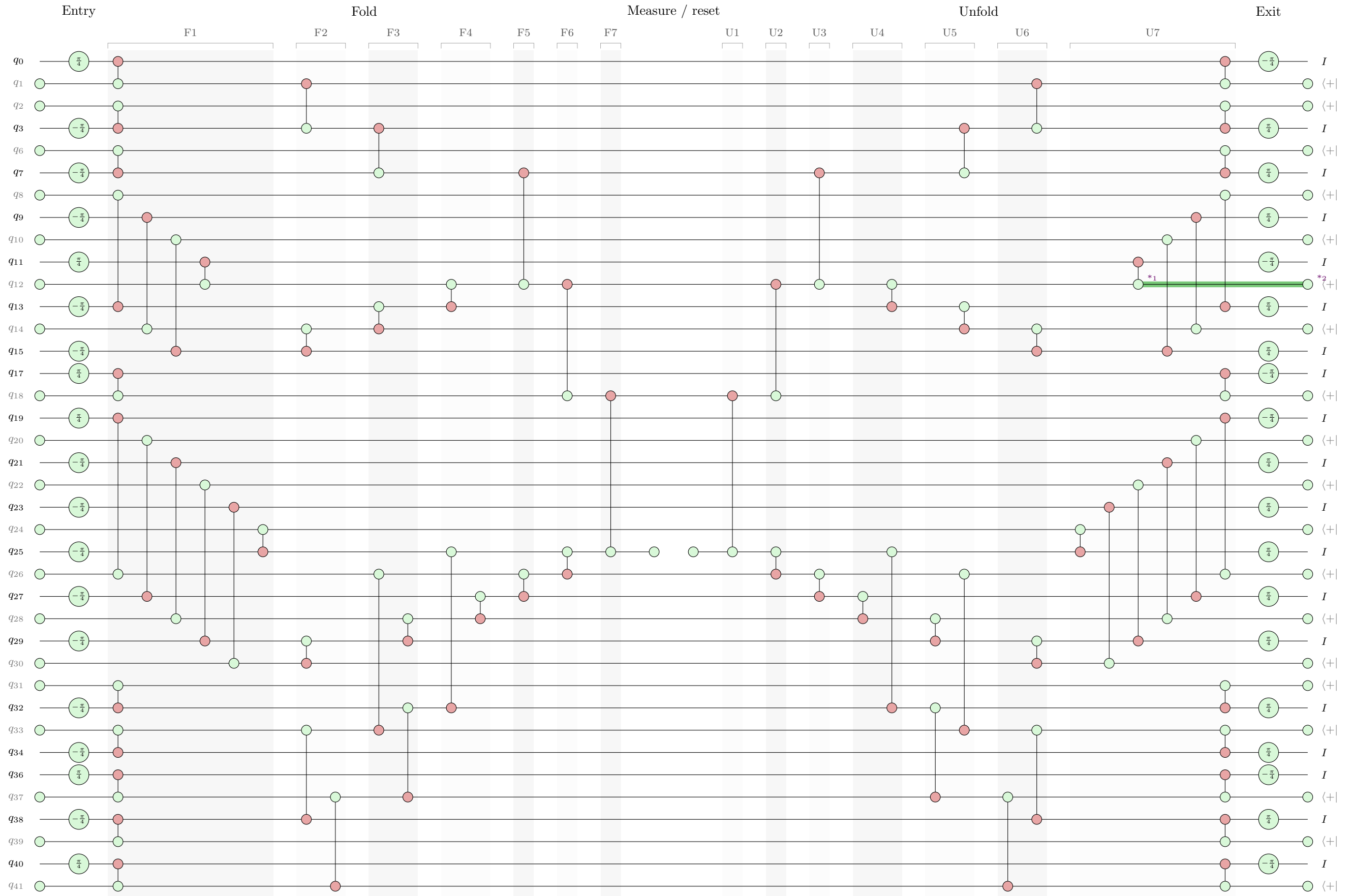
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_189.tikz

$d = 5$: basis element 190

No boundary support. 2 phase defects.



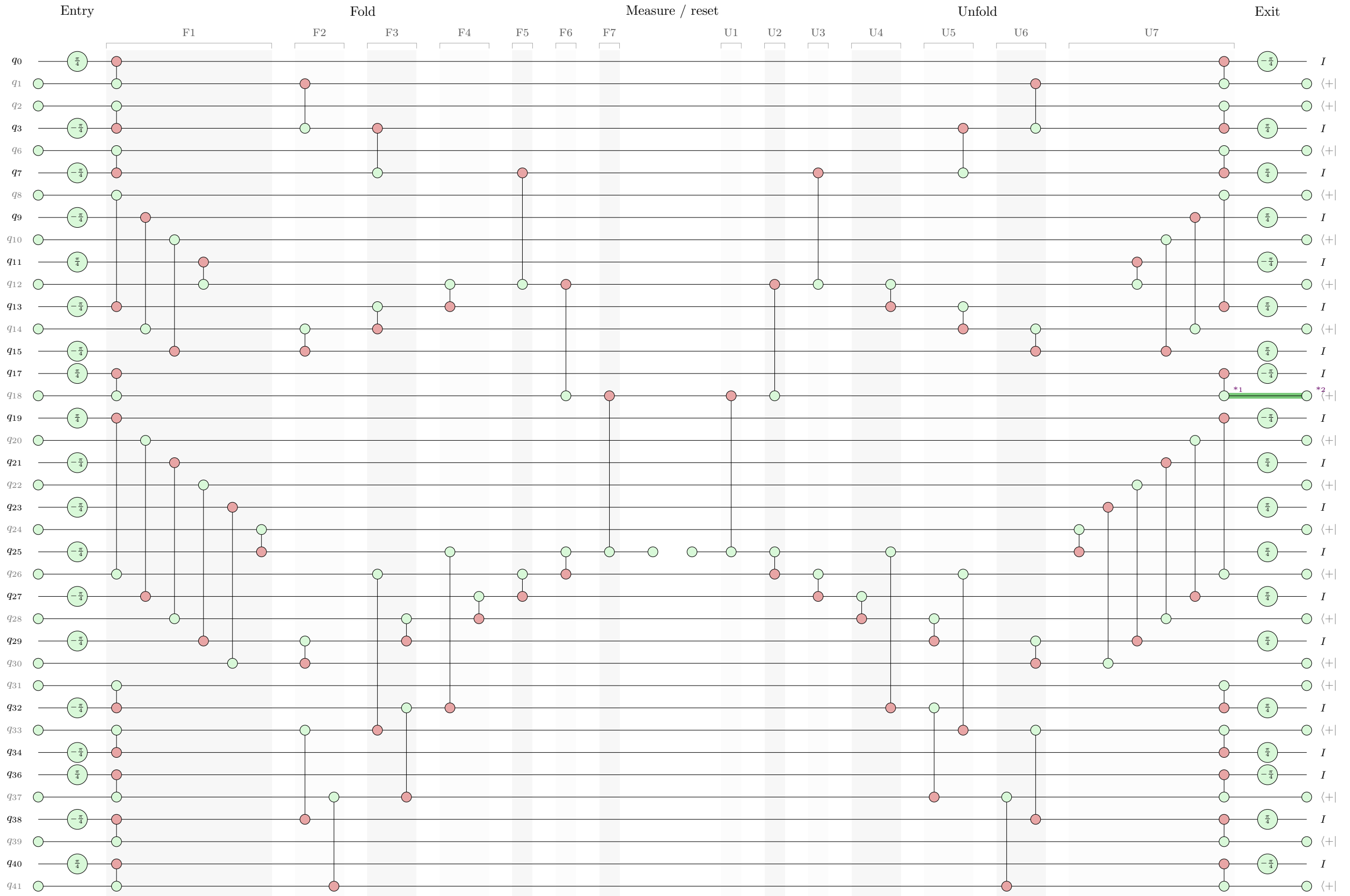
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_190.tikz

$d = 5$: basis element 191

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_191.tikz

$d = 5$: basis element 192
 No boundary support. 2 phase defects.

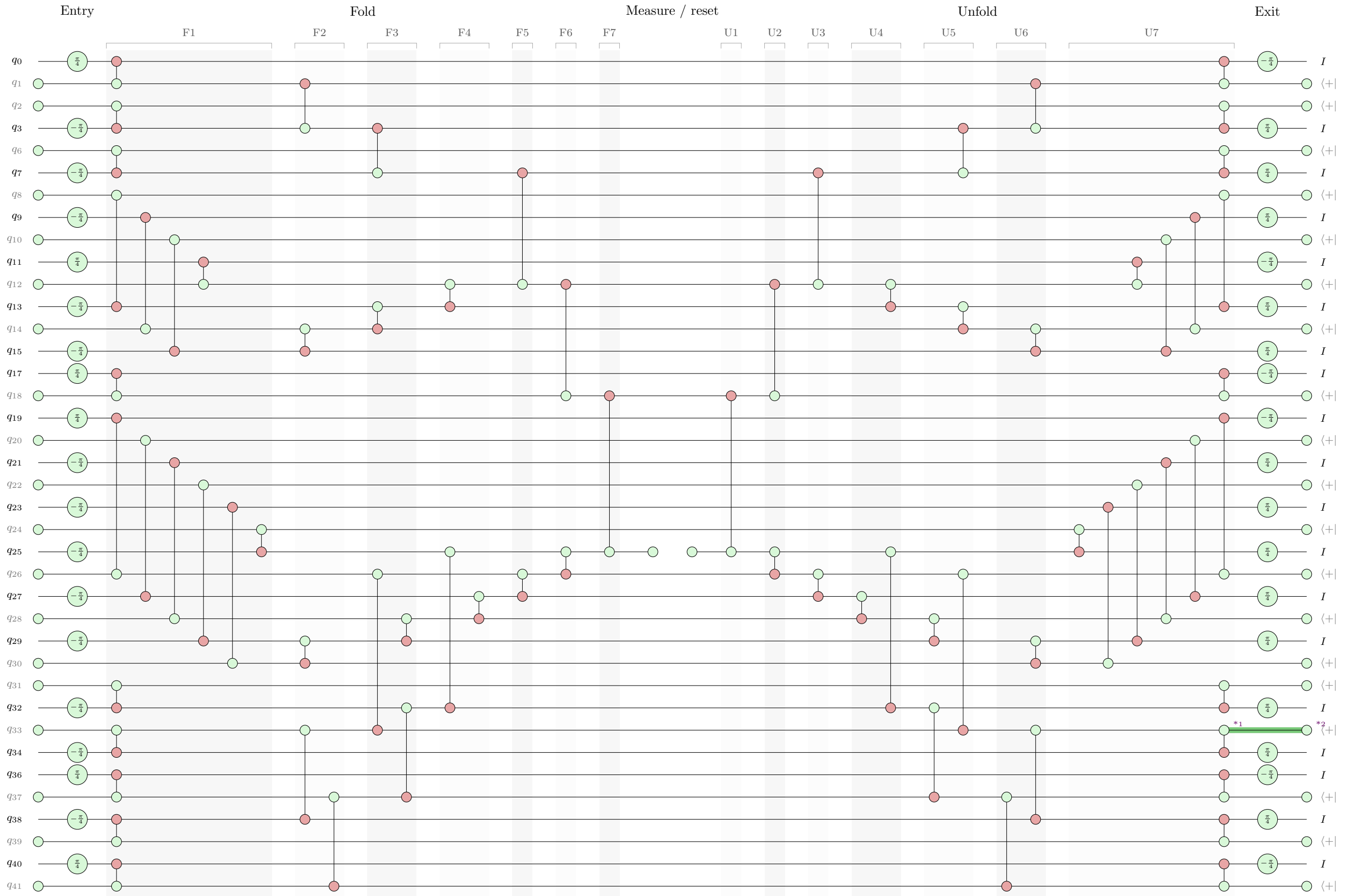


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: basis element 193

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_193.tikz

$d = 5$: basis element 194
 No boundary support. 2 phase defects.

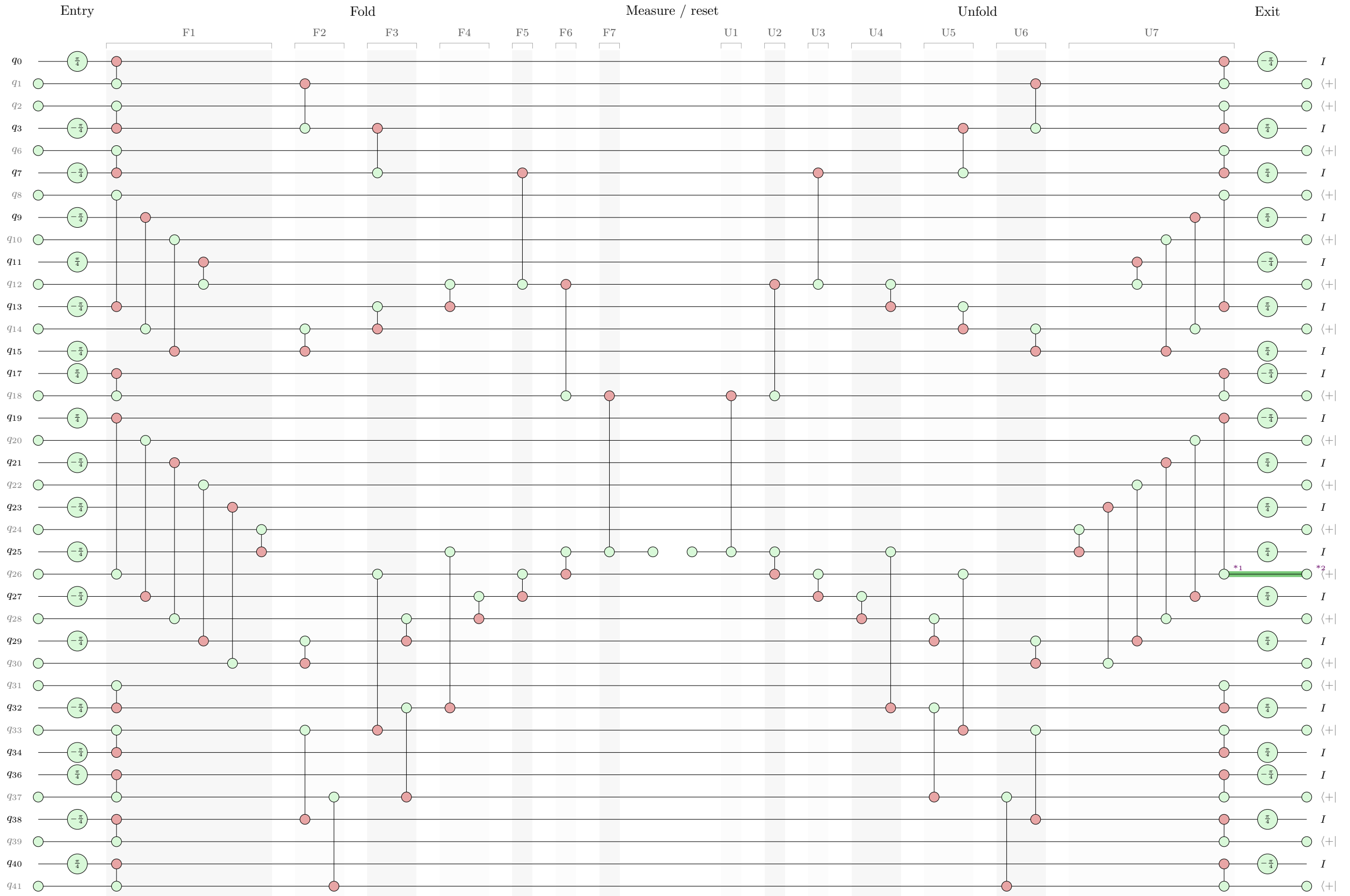


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: basis element 195

No boundary support. 2 phase defects.



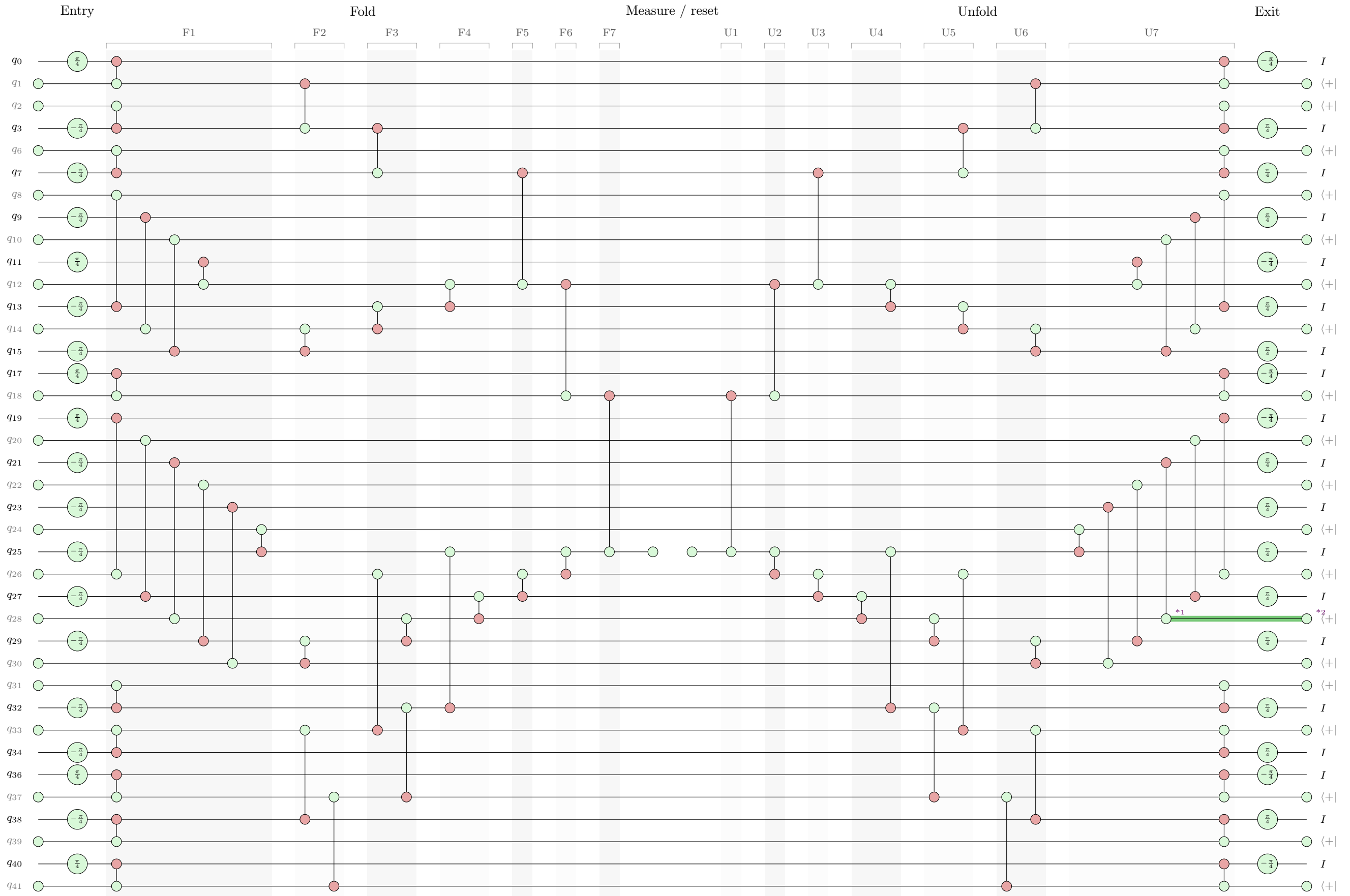
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_195.tikz

$d = 5$: basis element 196

No boundary support. 2 phase defects.



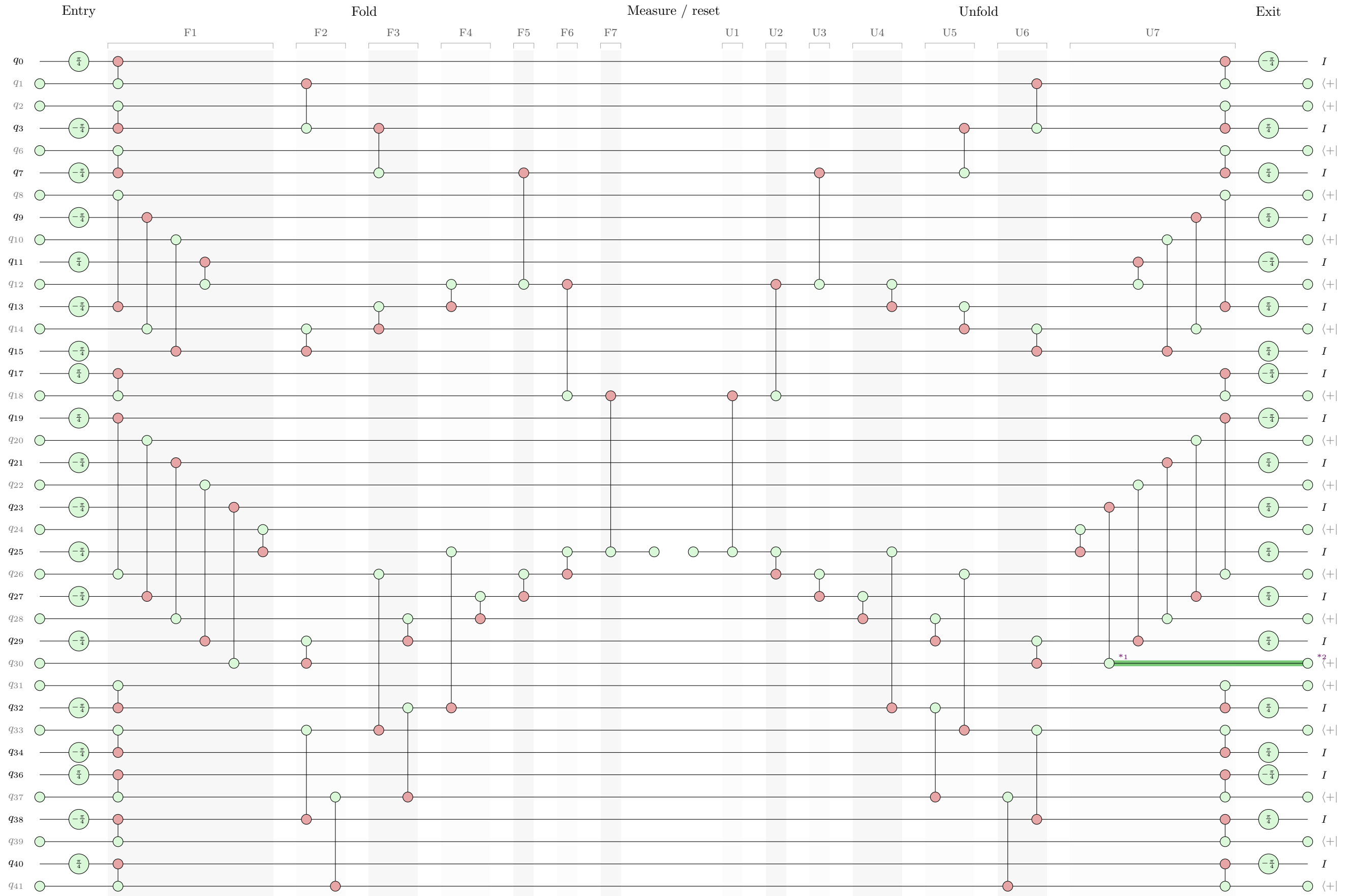
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_196.tikz

$d = 5$: basis element 197

No boundary support. 2 phase defects.



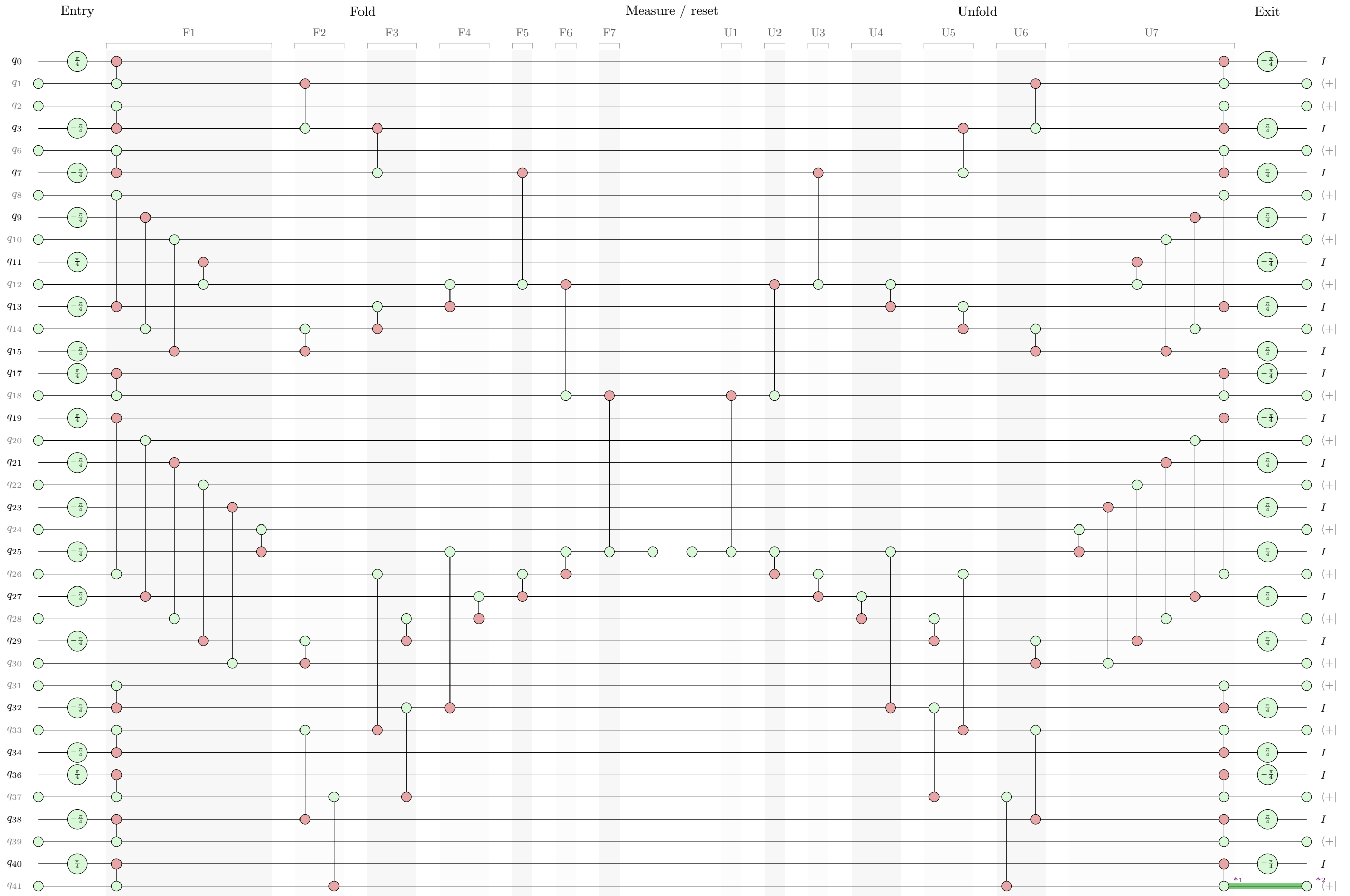
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_197.tikz

$d = 5$: basis element 198

No boundary support. 2 phase defects.



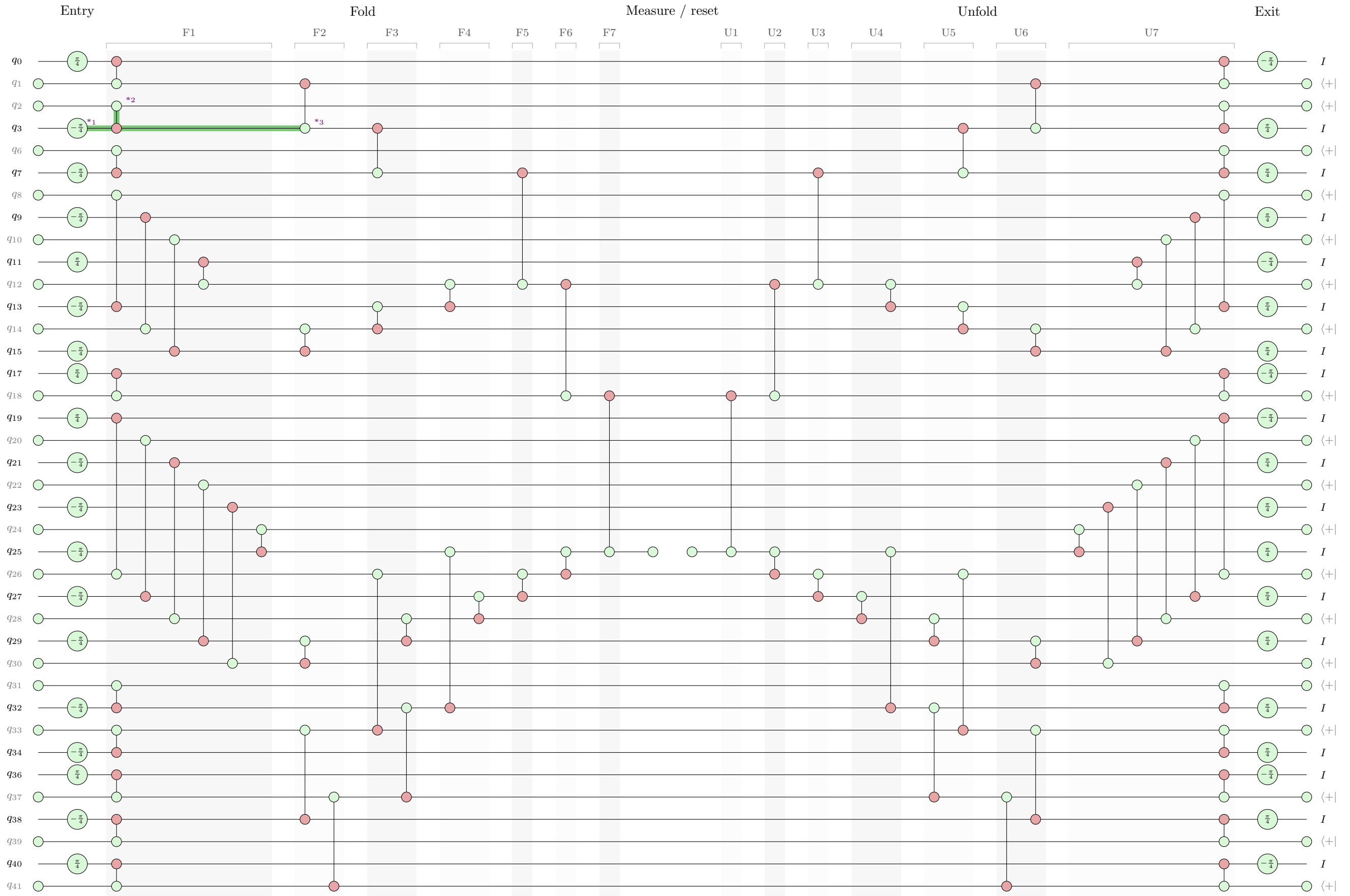
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_198.tikz

$d = 5$: basis element 199

No boundary support. 3 phase defects.



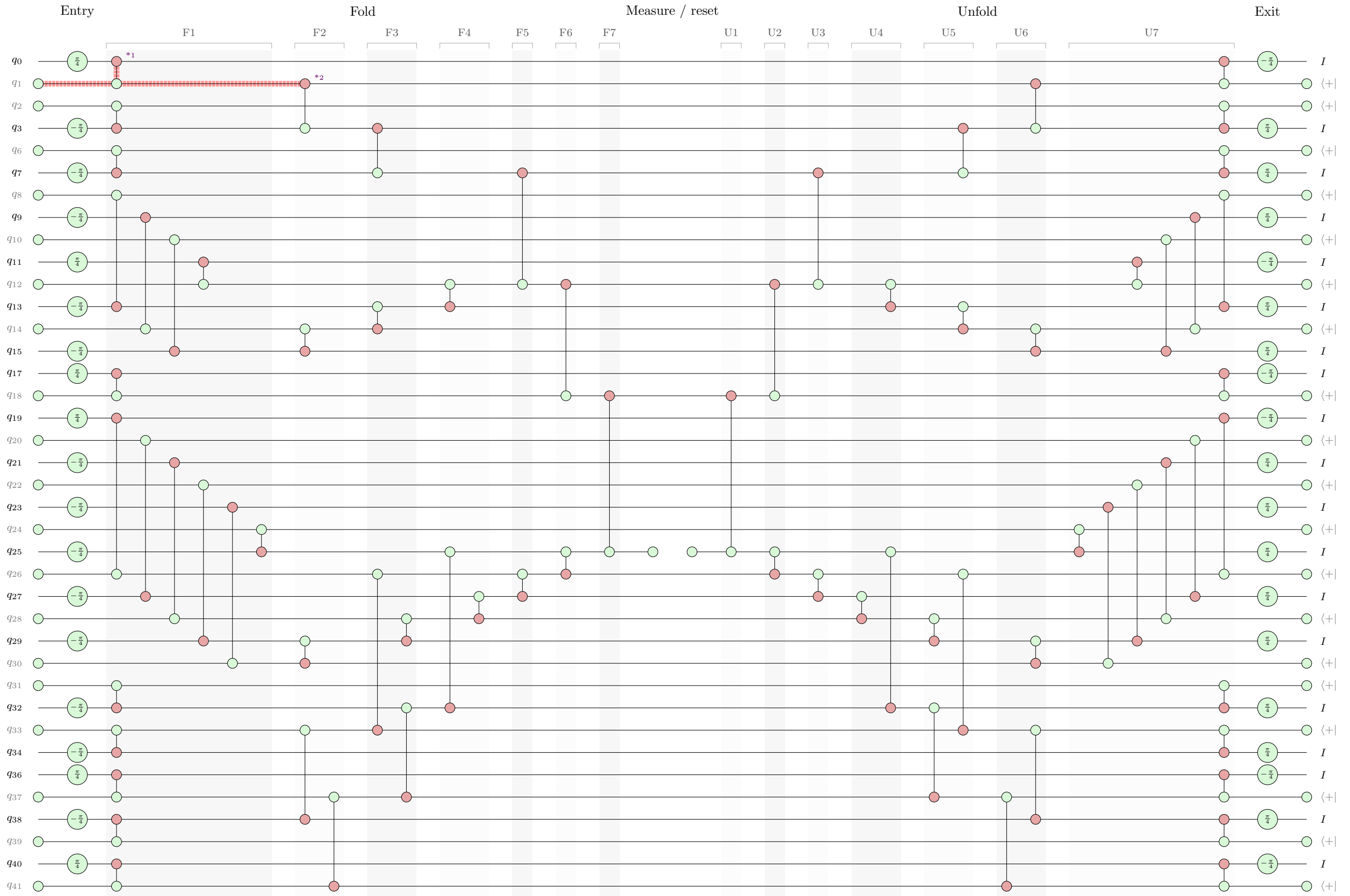
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_199.tikz

$d = 5$: basis element 200

No boundary support. 2 phase defects.



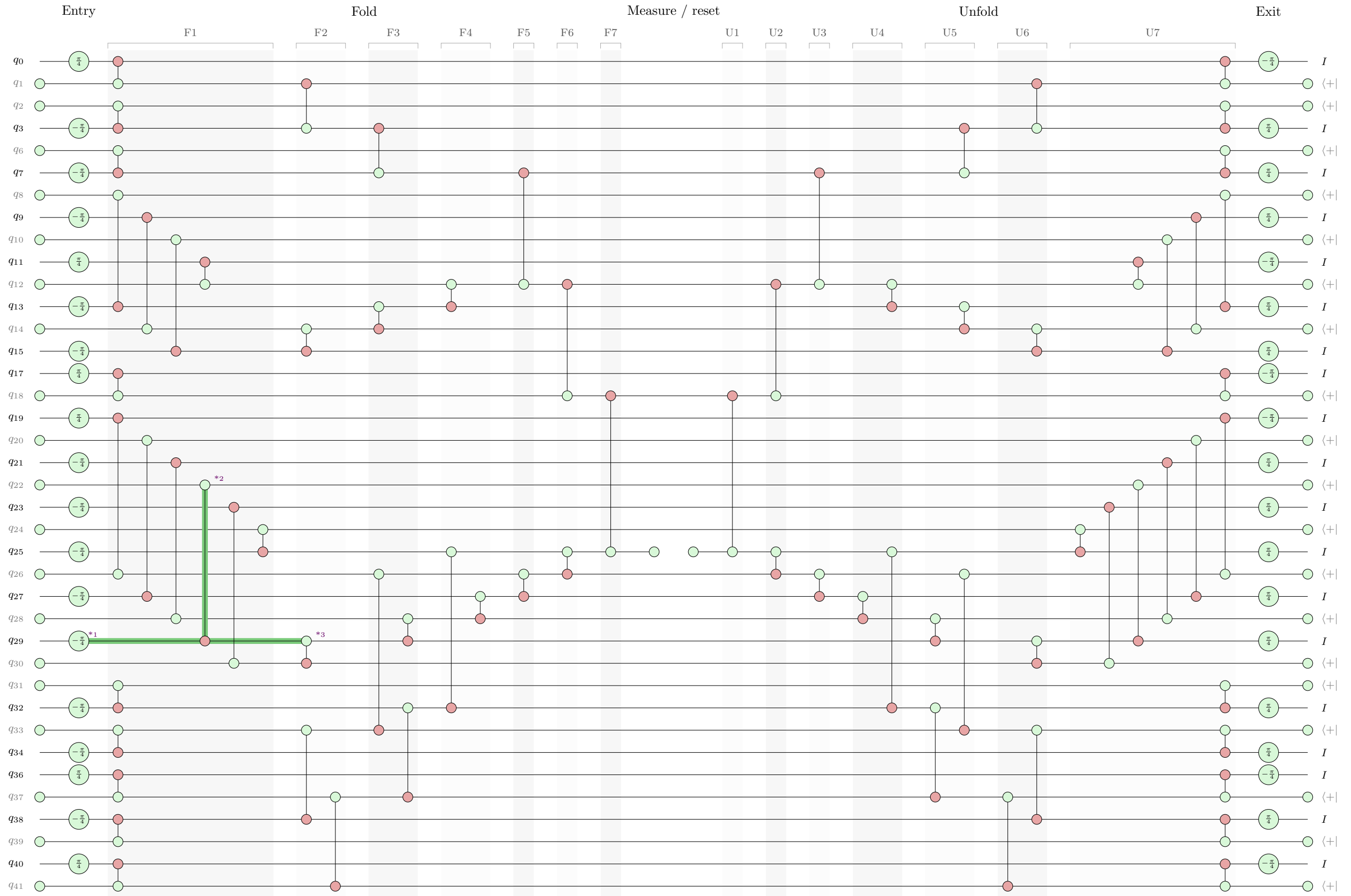
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_200.tikz

$d = 5$: basis element 201

No boundary support. 3 phase defects.



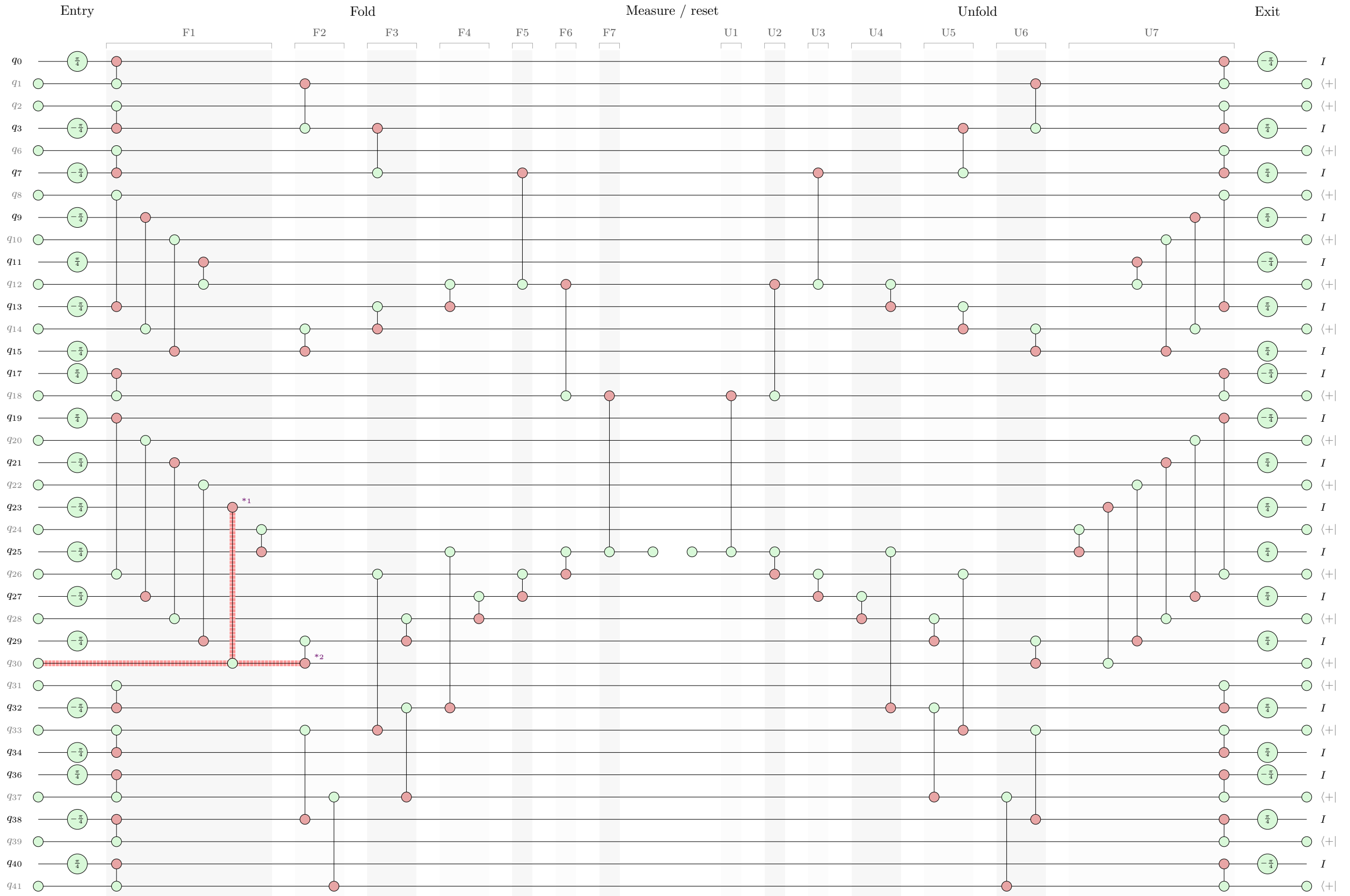
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_201.tikz

$d = 5$: basis element 202

No boundary support. 2 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_202.tikz

$d = 5$: basis element 203

No boundary support. 2 phase defects.

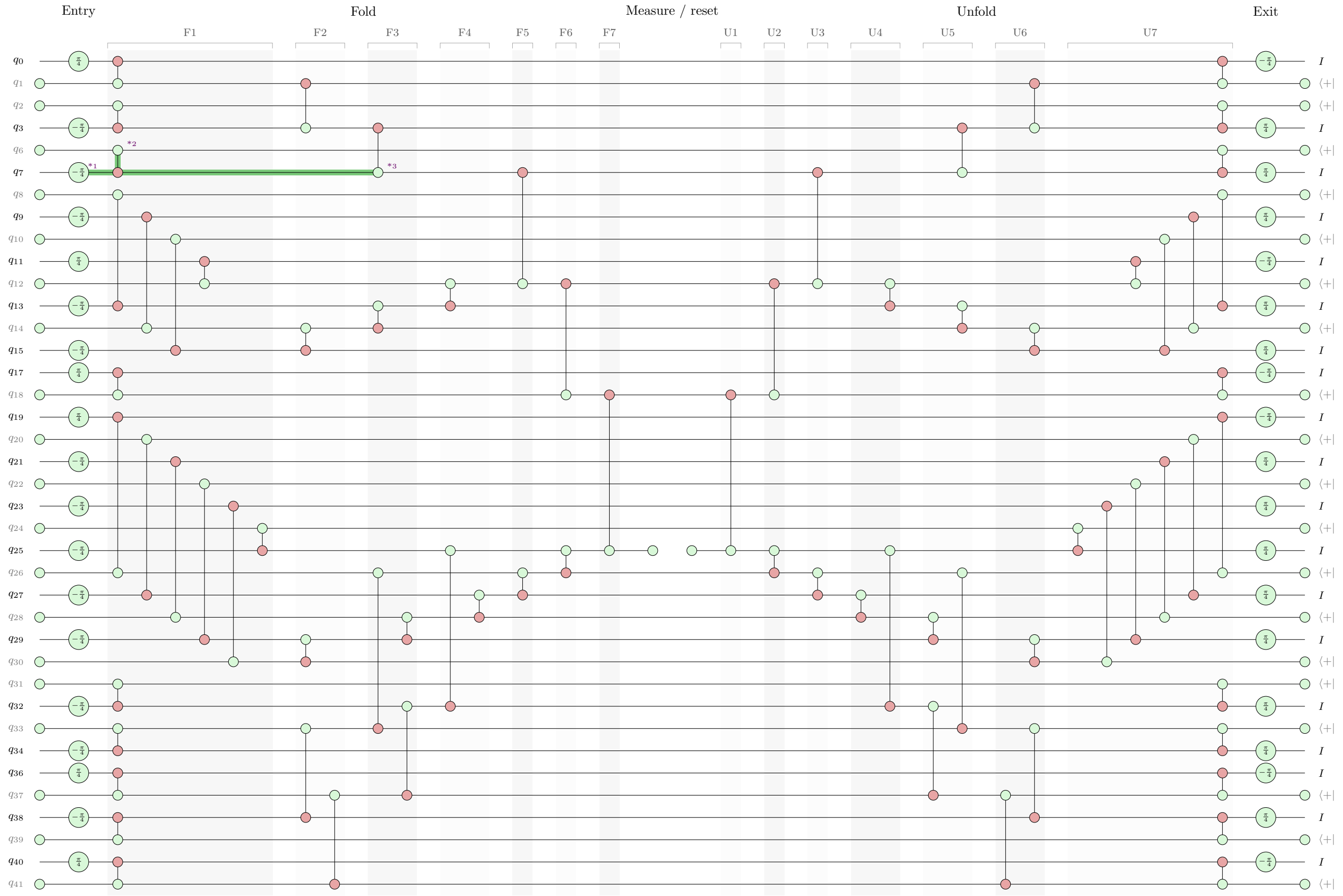


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

d5_203.tikz

$d = 5$: basis element 204
 No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

$d = 5$: basis element 205

No boundary support. 3 phase defects.



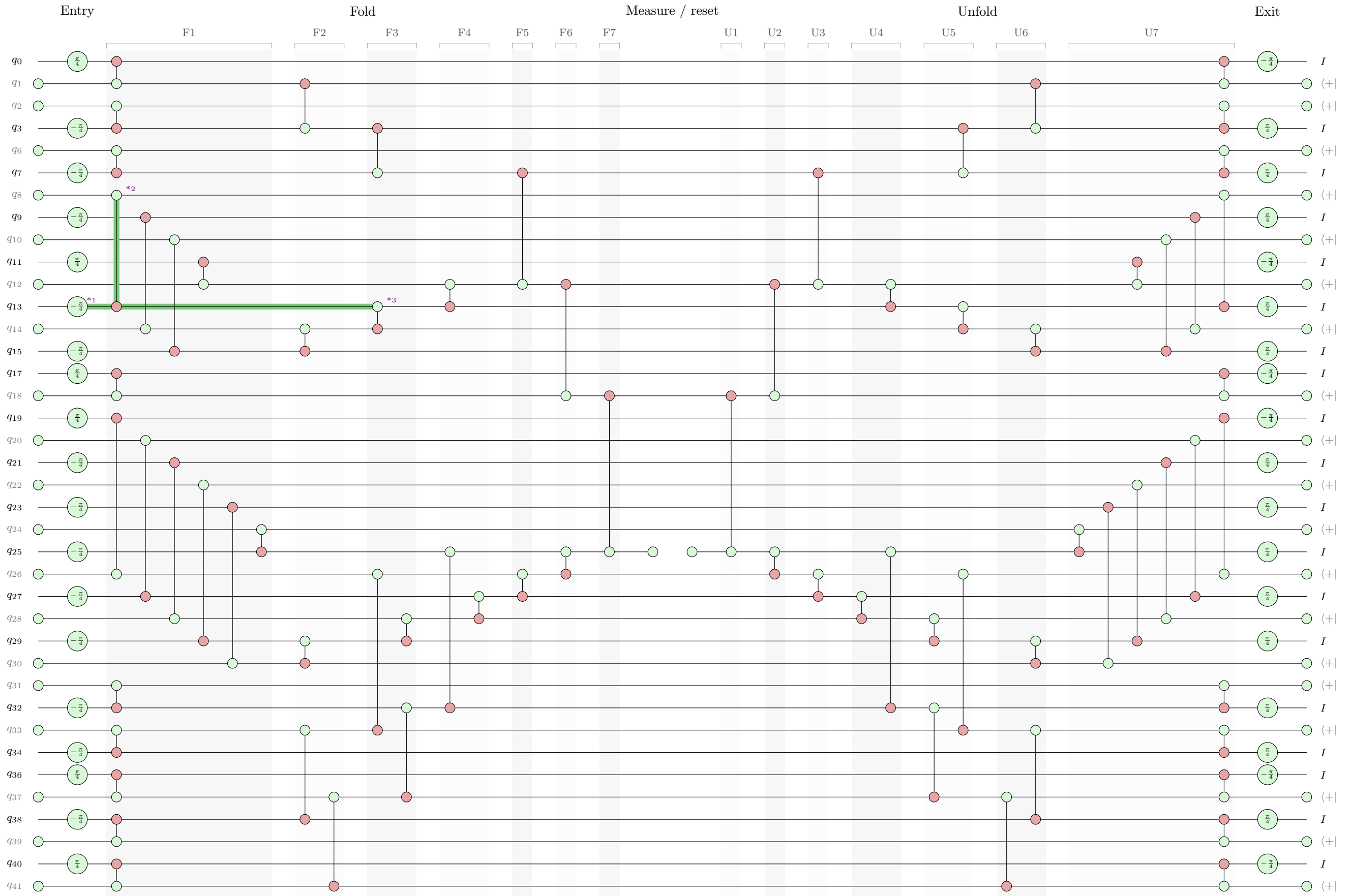
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_205.tikz

$d = 5$: basis element 206

No boundary support. 3 phase defects.



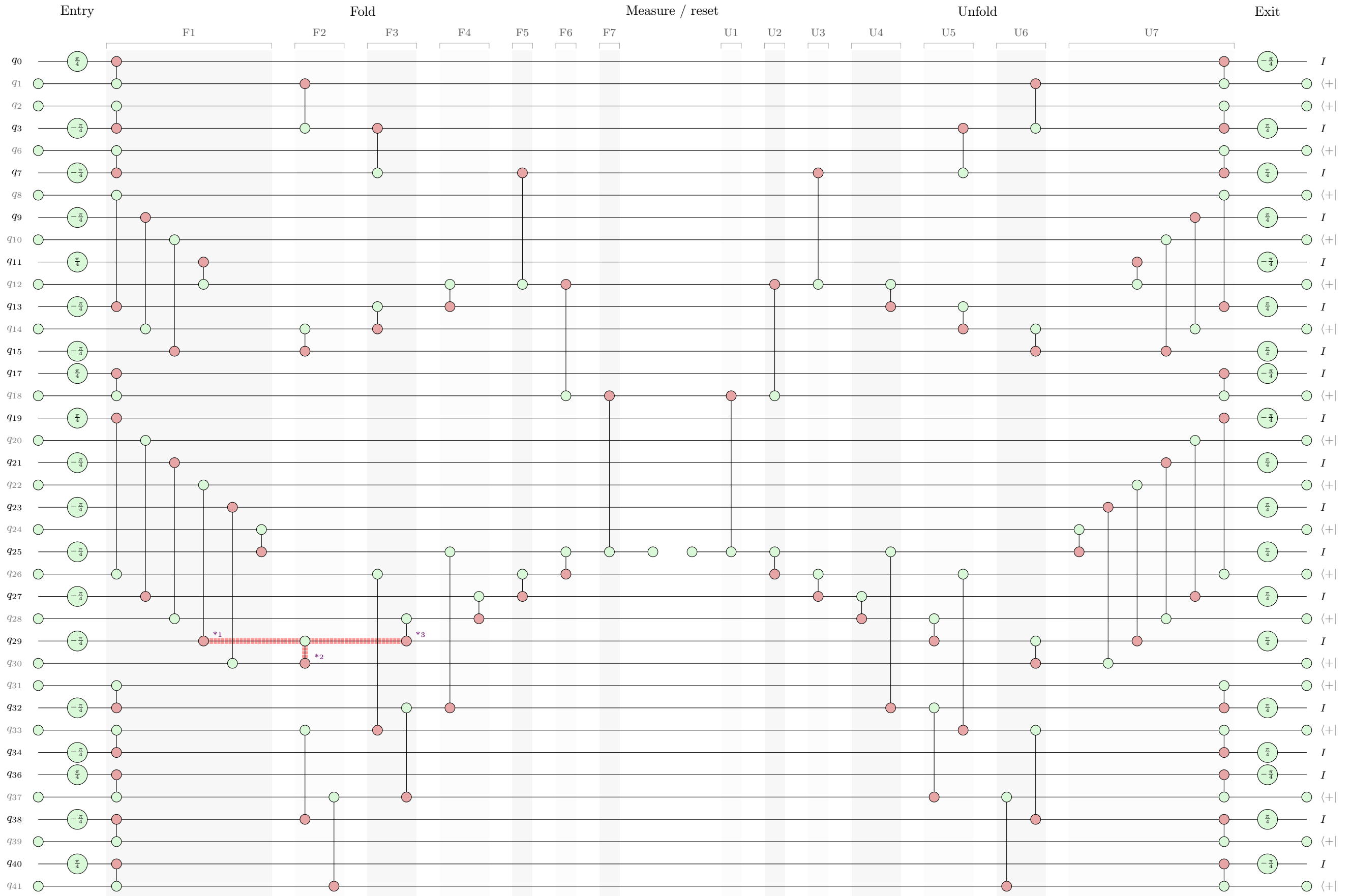
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_206.tikz

$d = 5$: basis element 207

No boundary support. 3 phase defects.



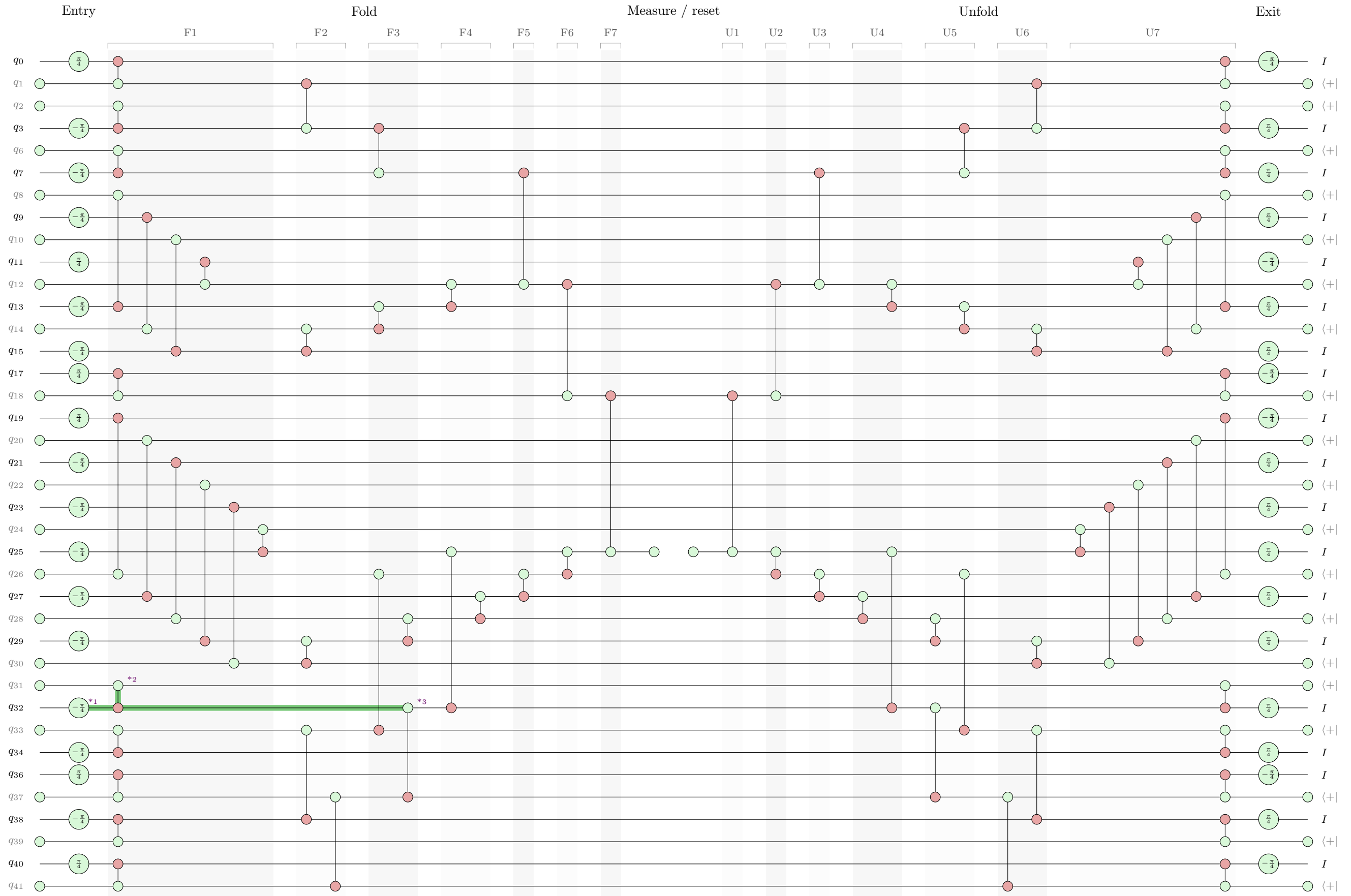
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_207.tikz

$d = 5$: basis element 208

No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_208.tikz

$d = 5$: basis element 209

No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_209.tikz

$d = 5$: basis element 210

No boundary support. 3 phase defects.



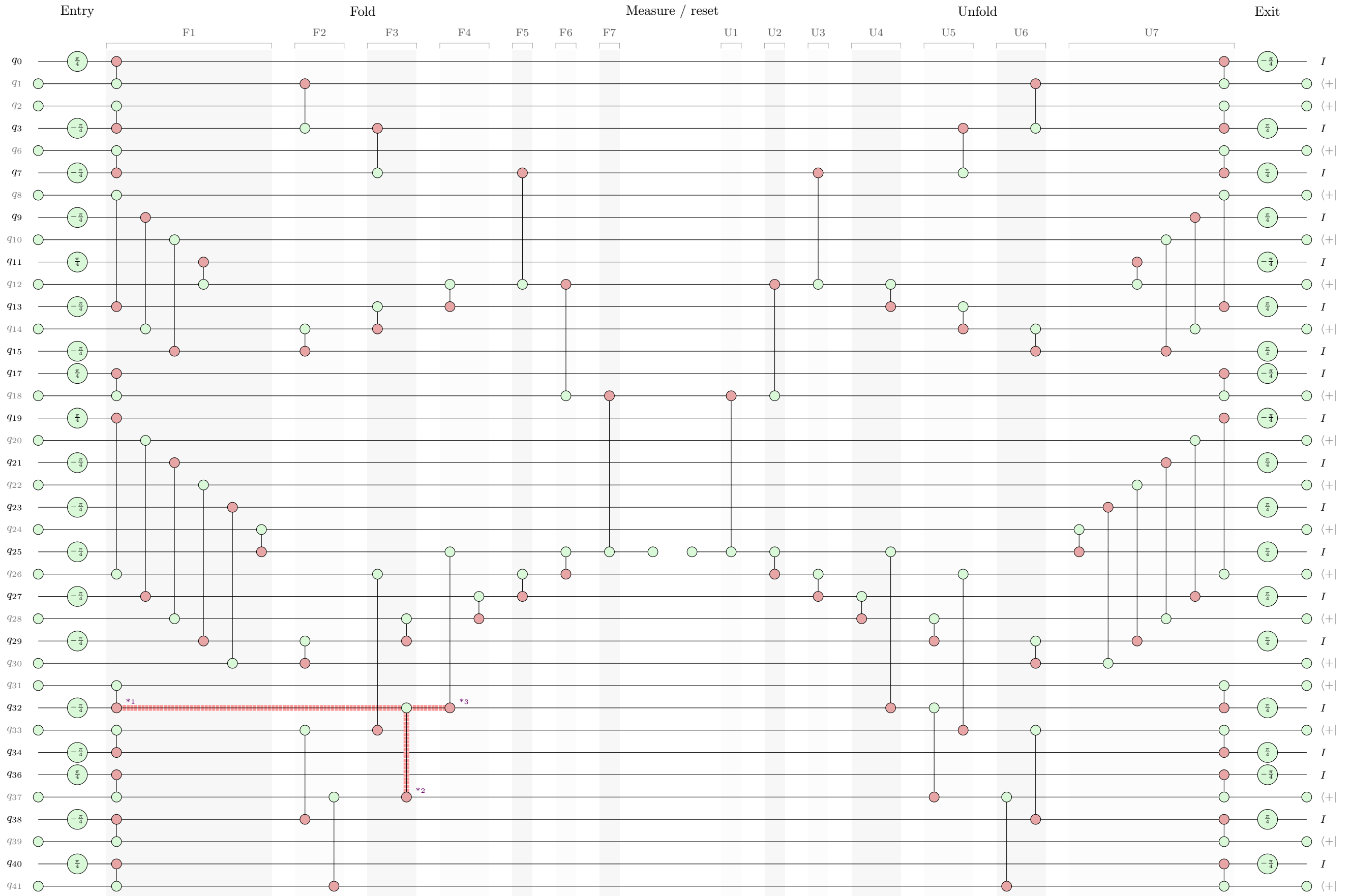
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_210.tikz

$d = 5$: basis element 211

No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_211.tikz

$d = 5$: basis element 212

No boundary support. 3 phase defects.



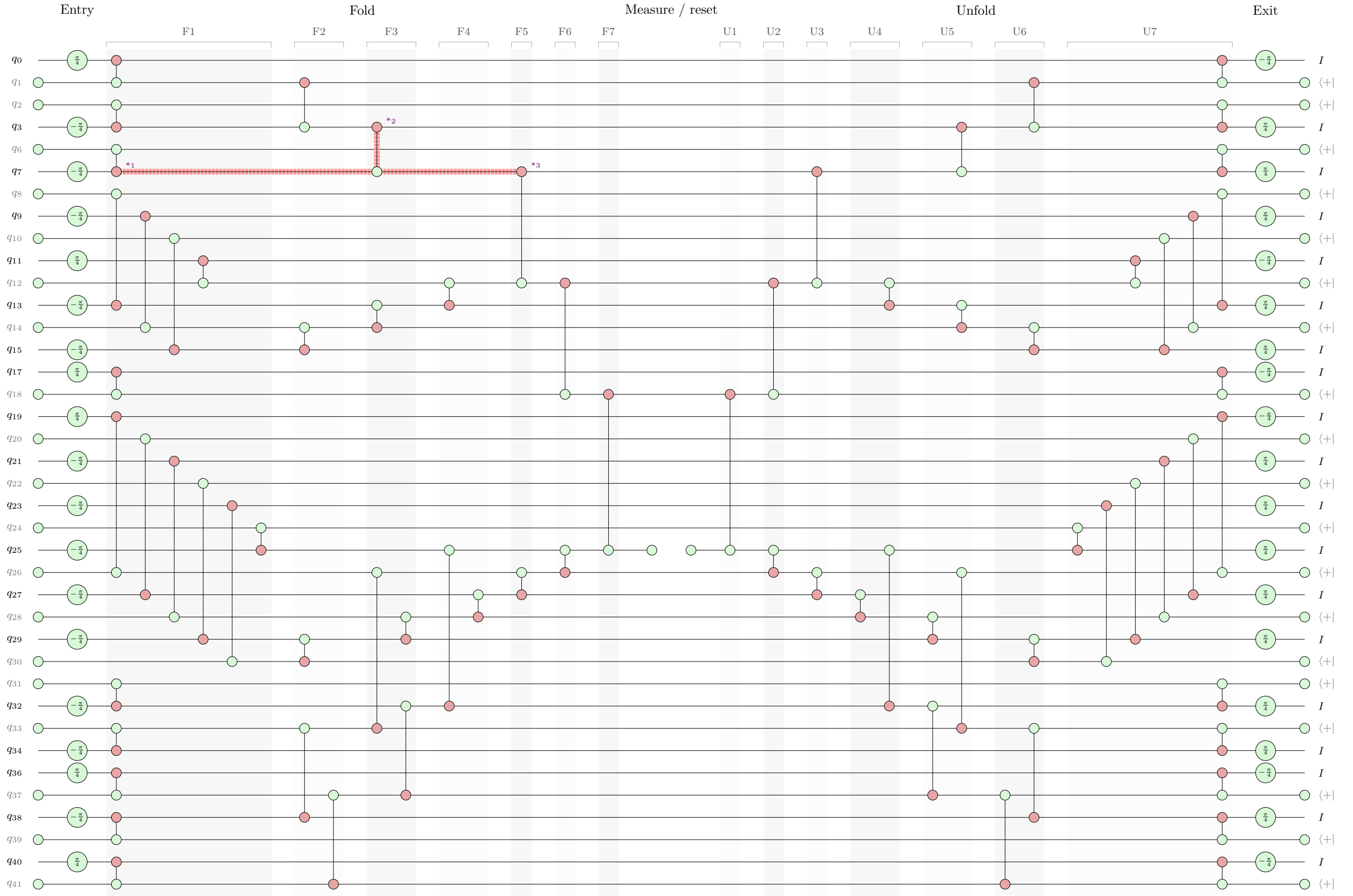
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_212.tikz

$d = 5$: basis element 213

No boundary support. 3 phase defects.



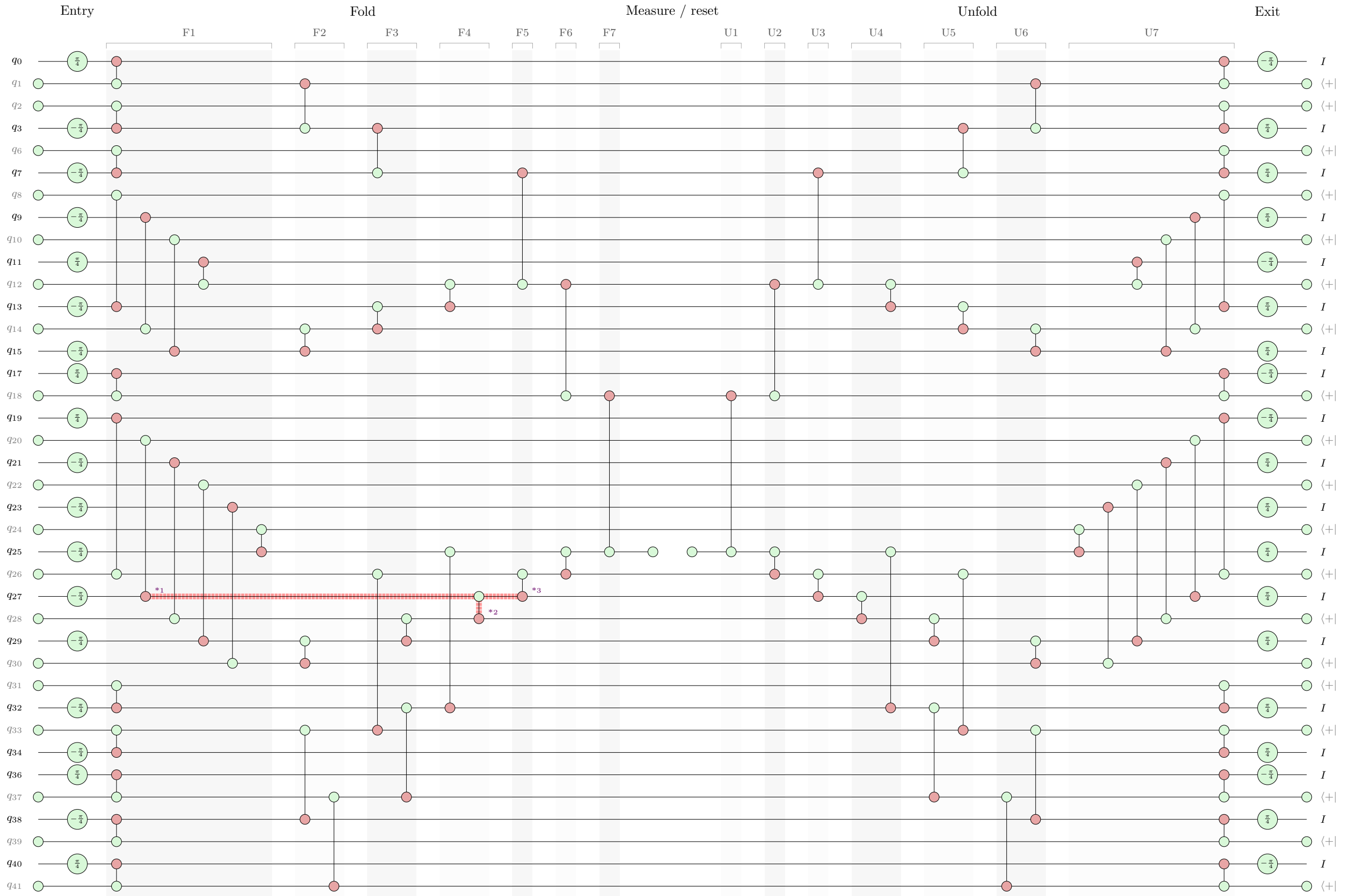
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_213.tikz

$d = 5$: basis element 214

No boundary support. 3 phase defects.



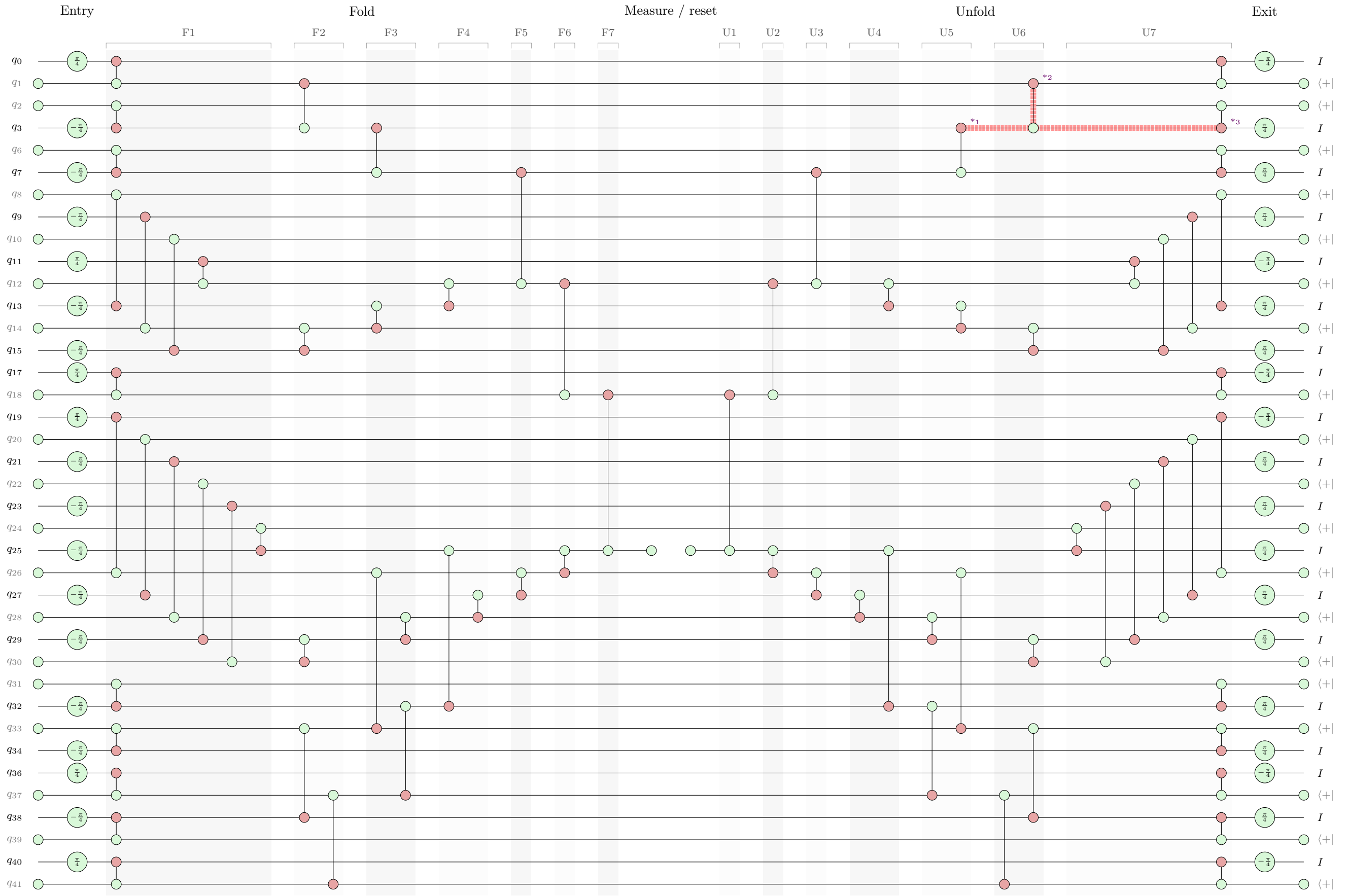
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_214.tikz

$d = 5$: basis element 215

No boundary support. 3 phase defects.



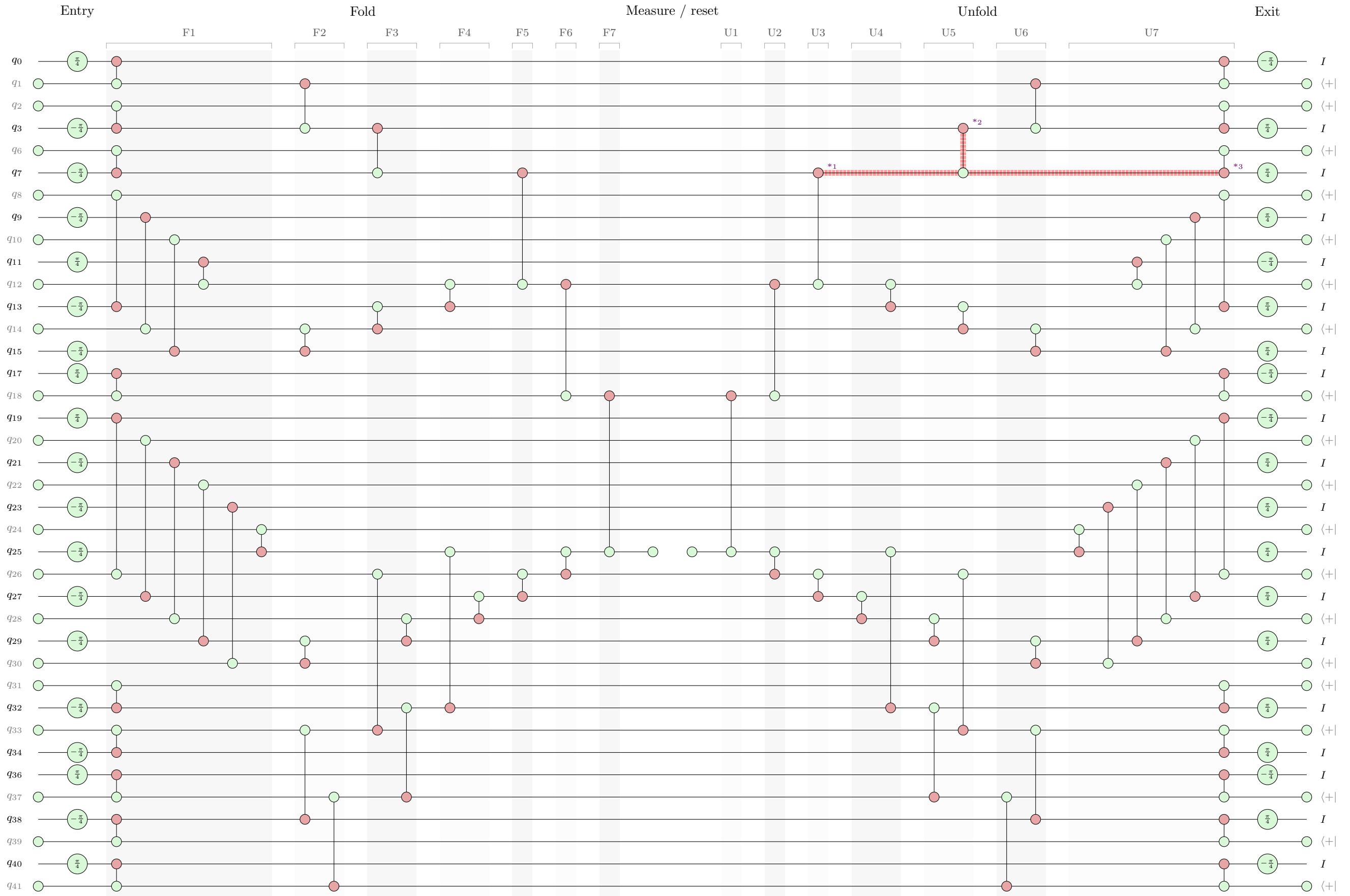
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_215.tikz

$d = 5$: basis element 216

No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_216.tikz

$d = 5$: basis element 217

No boundary support. 3 phase defects.



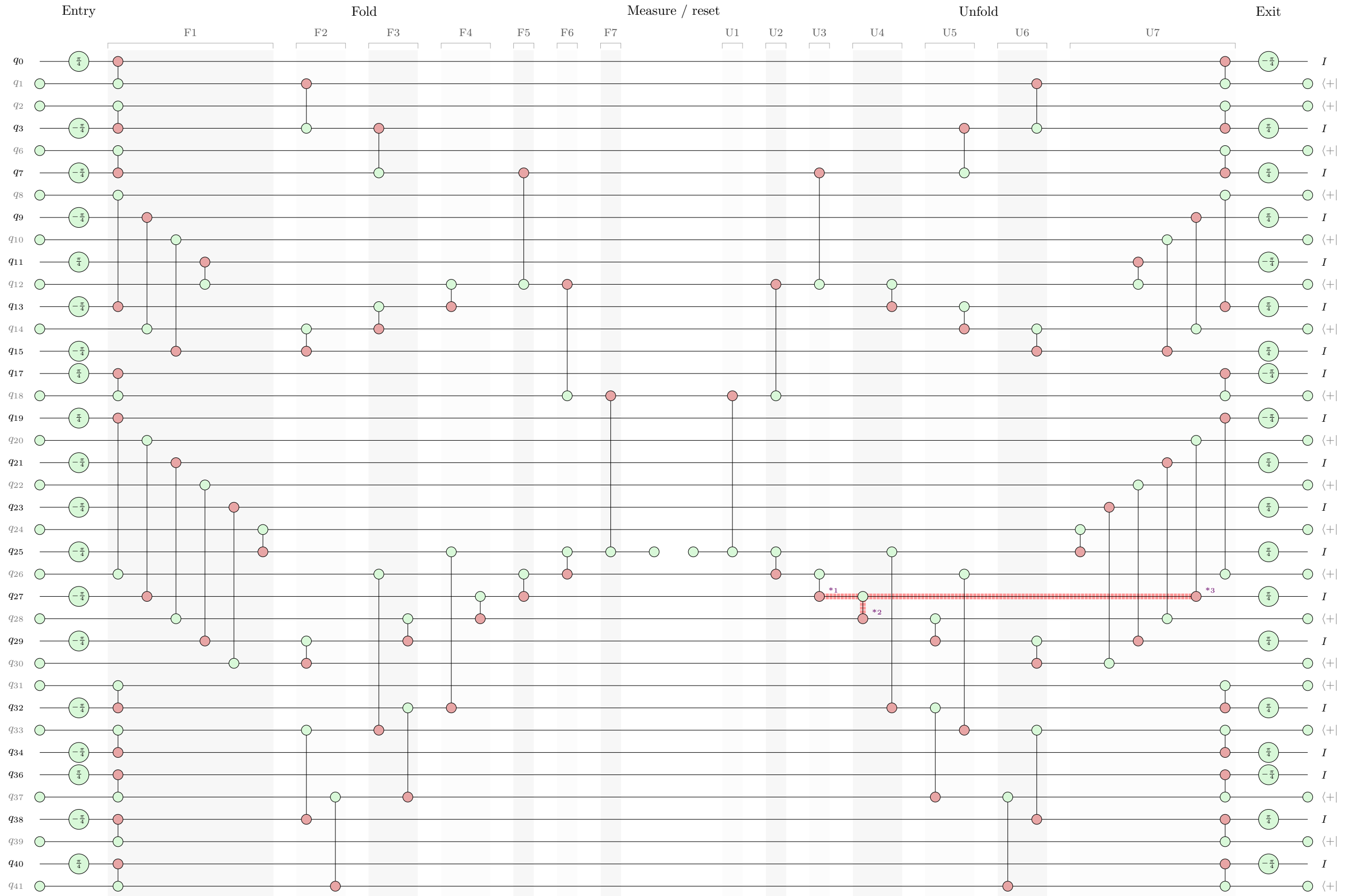
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_217.tikz

$d = 5$: basis element 218

No boundary support. 3 phase defects.



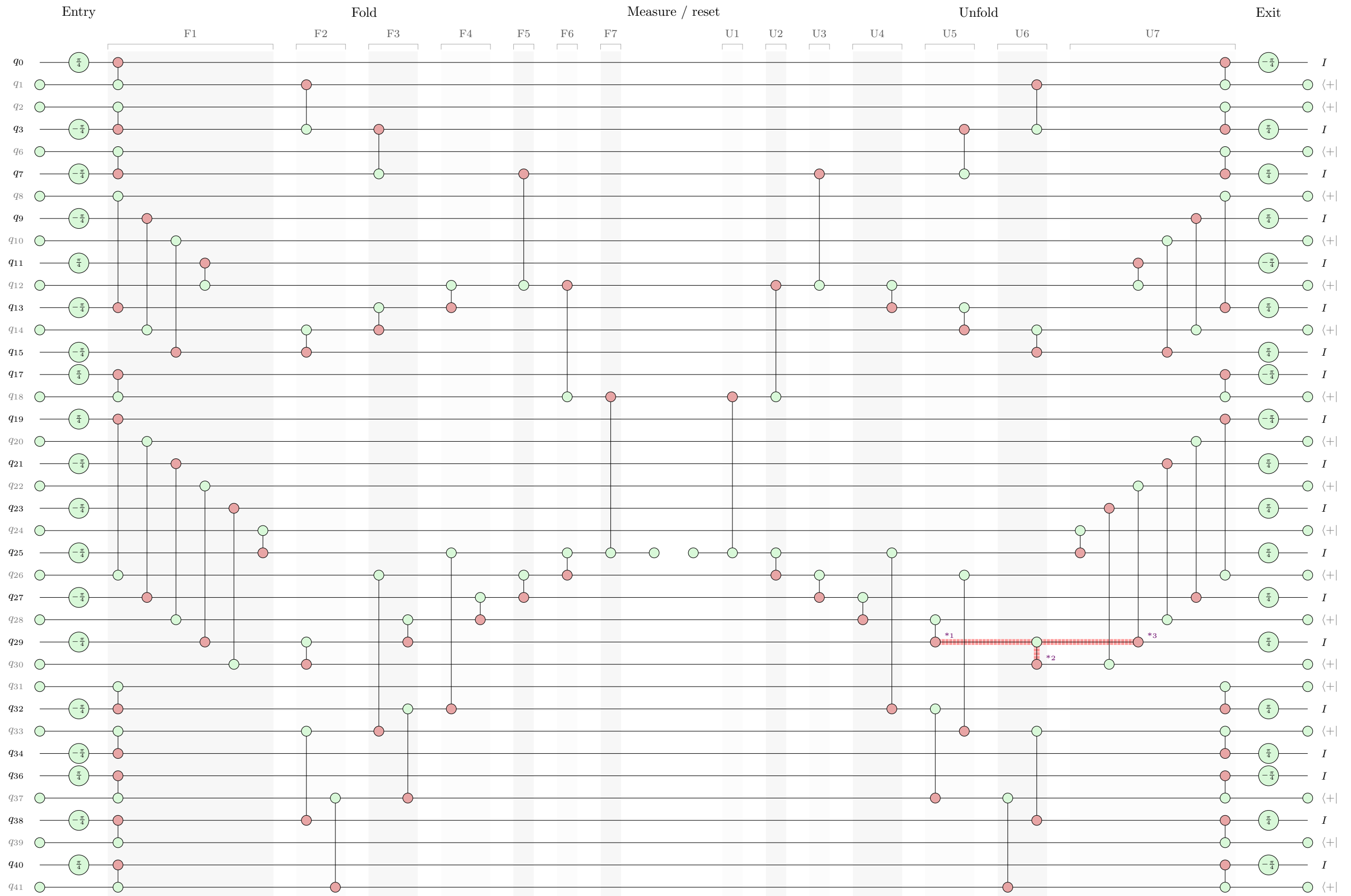
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_218.tikz

$d = 5$: basis element 219

No boundary support. 3 phase defects.



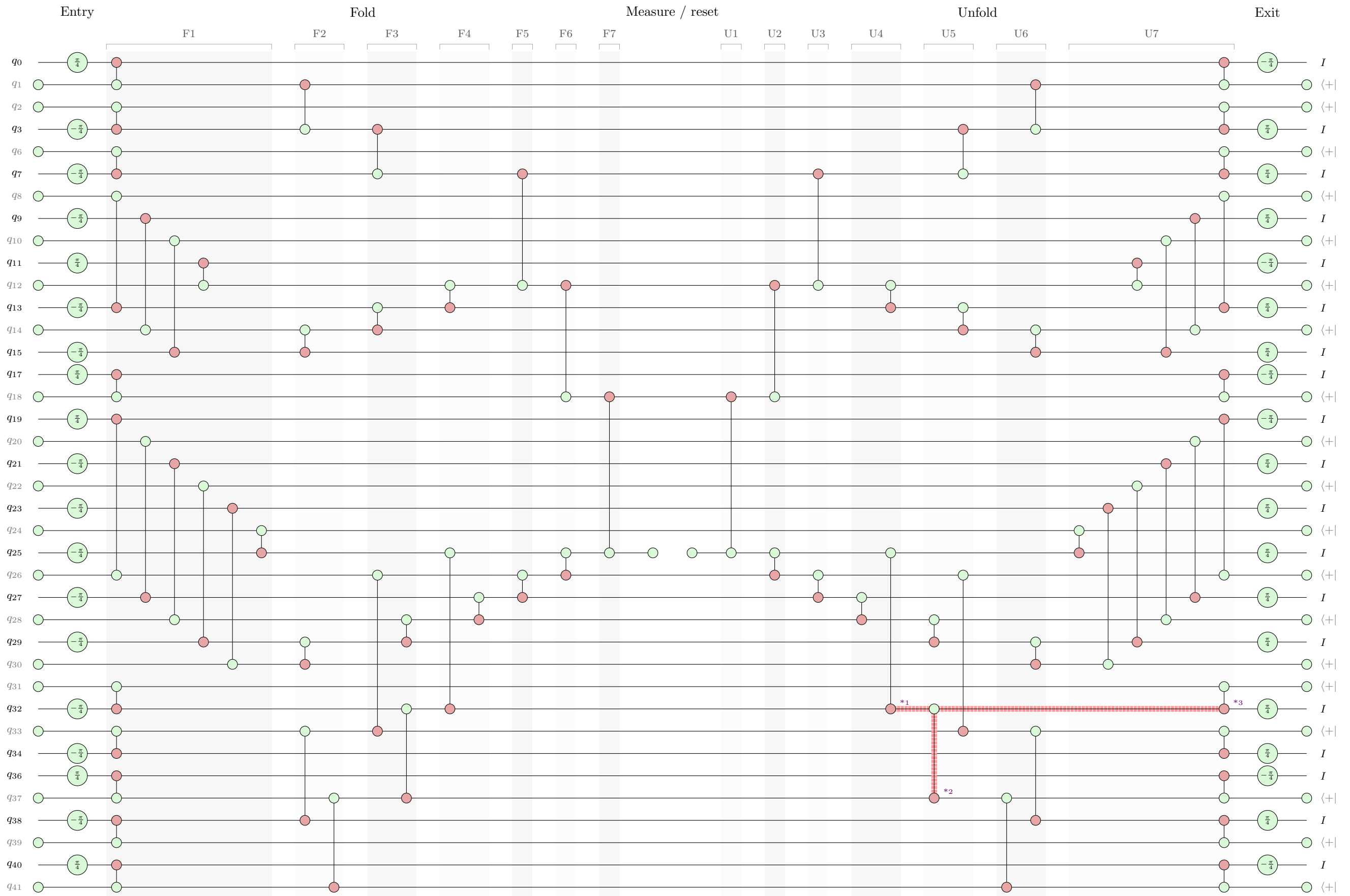
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_219.tikz

$d = 5$: basis element 220

No boundary support. 3 phase defects.



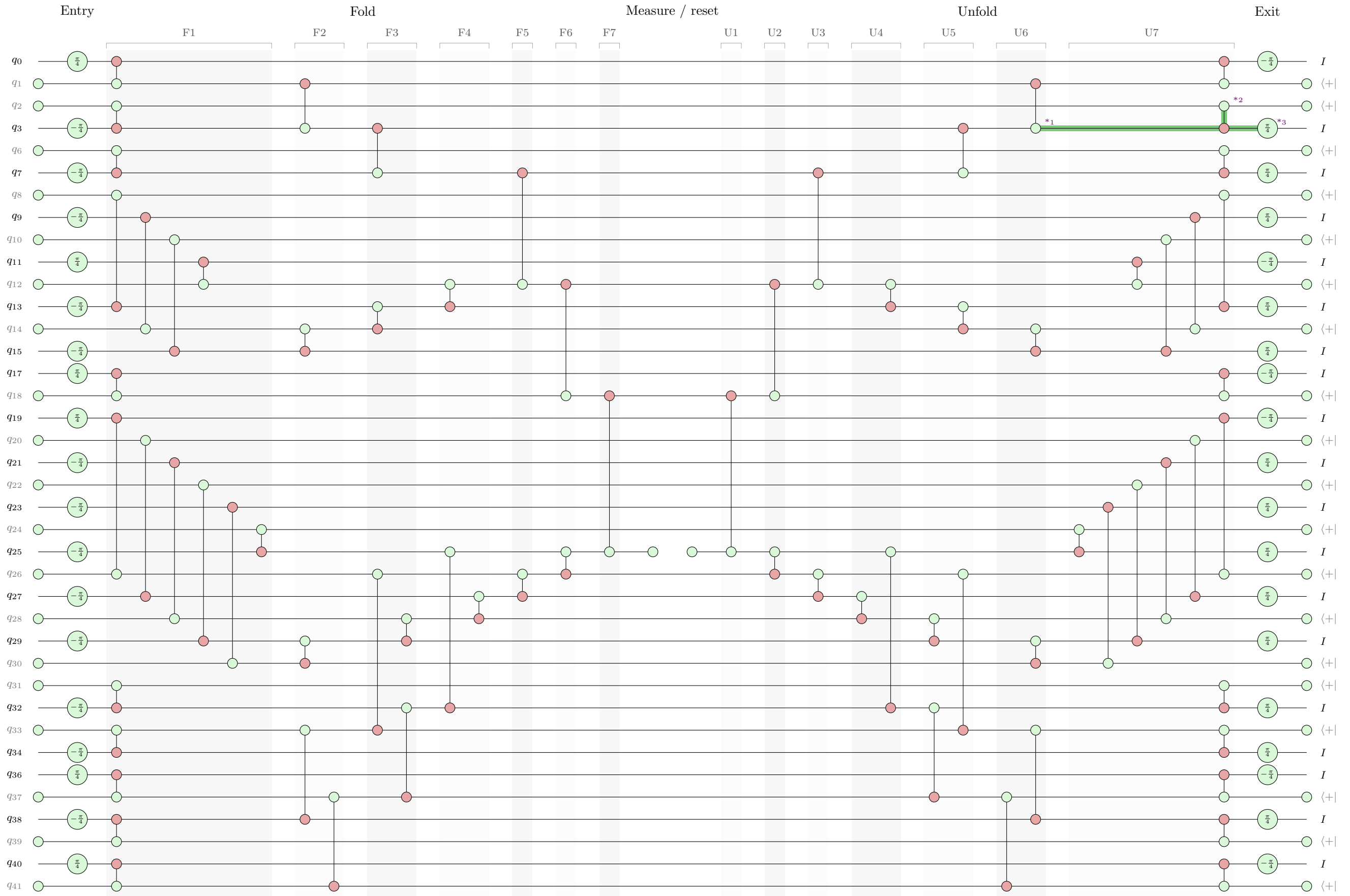
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_220.tikz

$d = 5$: basis element 221

No boundary support. 3 phase defects.



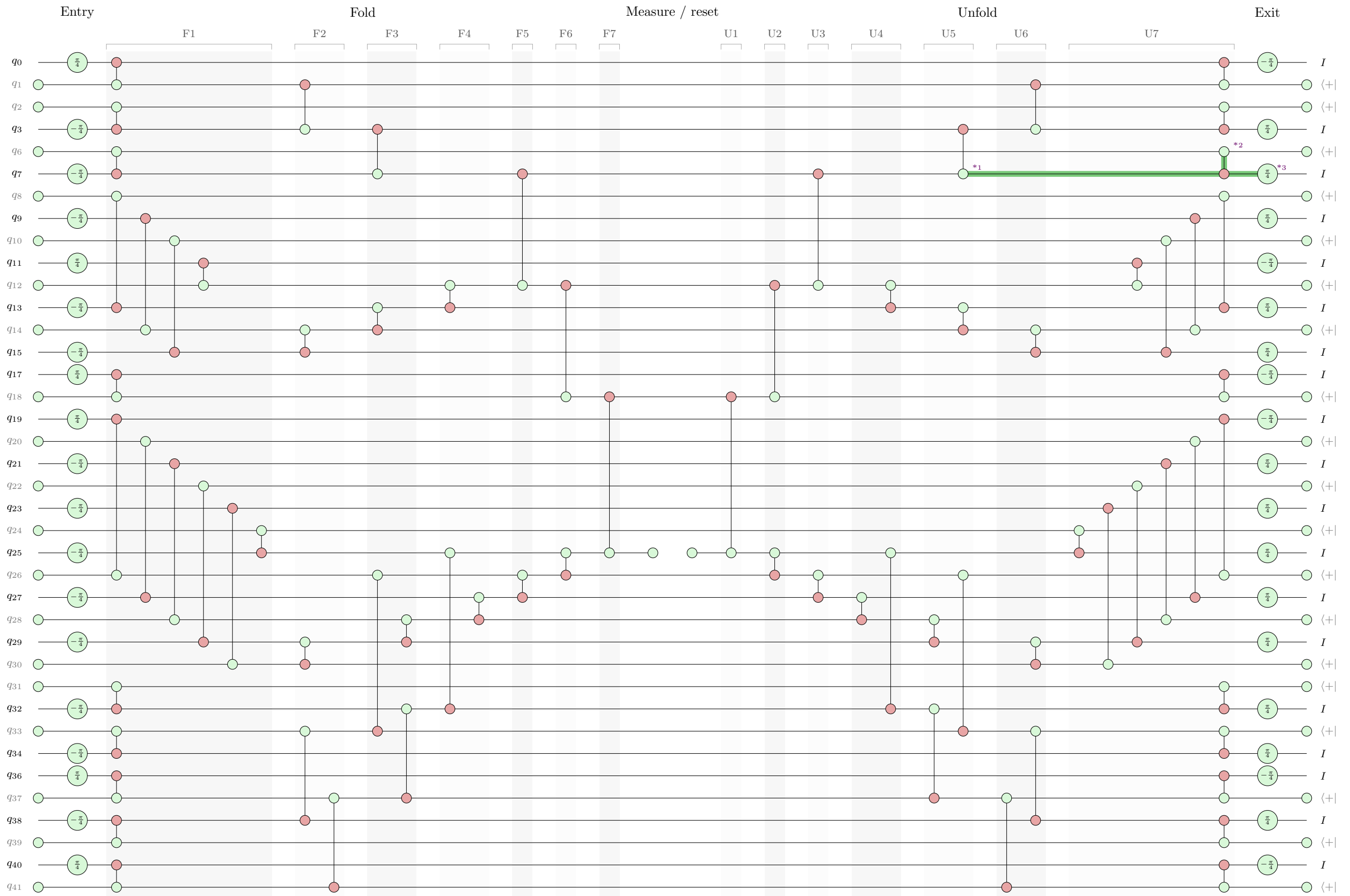
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_221.tikz

$d = 5$: basis element 222

No boundary support. 3 phase defects.



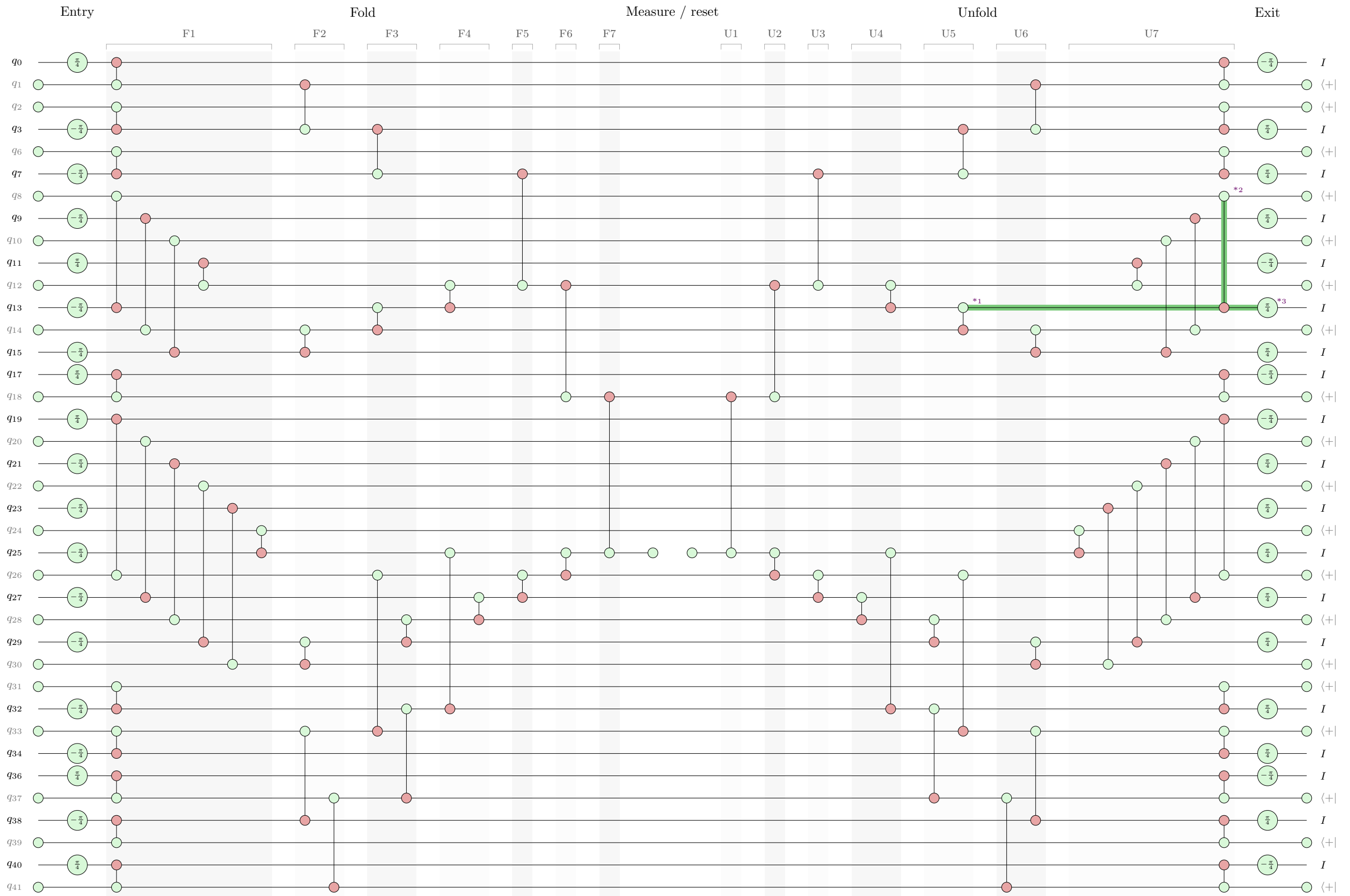
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_222.tikz

$d = 5$: basis element 223

No boundary support. 3 phase defects.



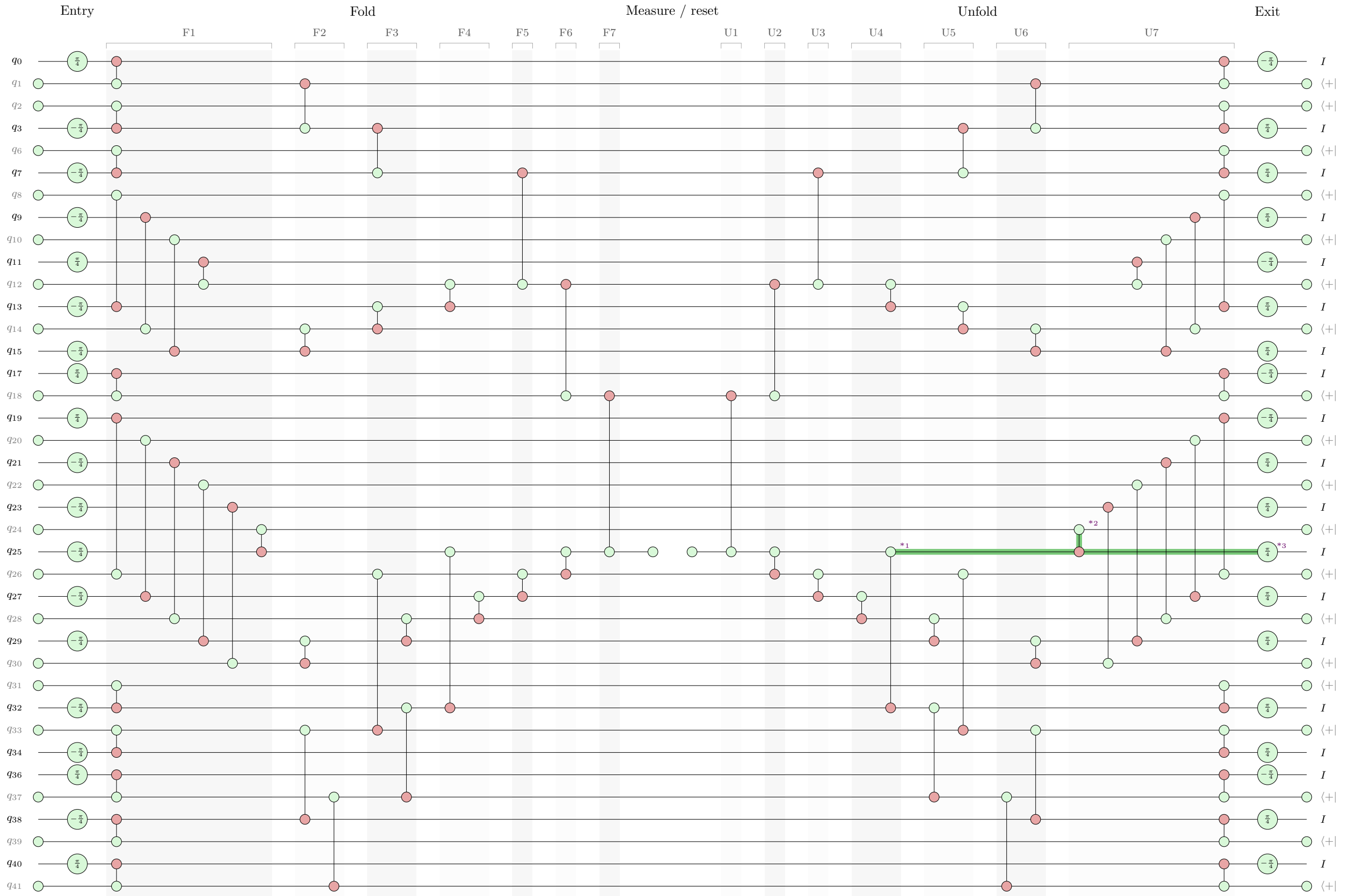
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_223.tikz

$d = 5$: basis element 224

No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_224.tikz

$d = 5$: basis element 225

No boundary support. 3 phase defects.



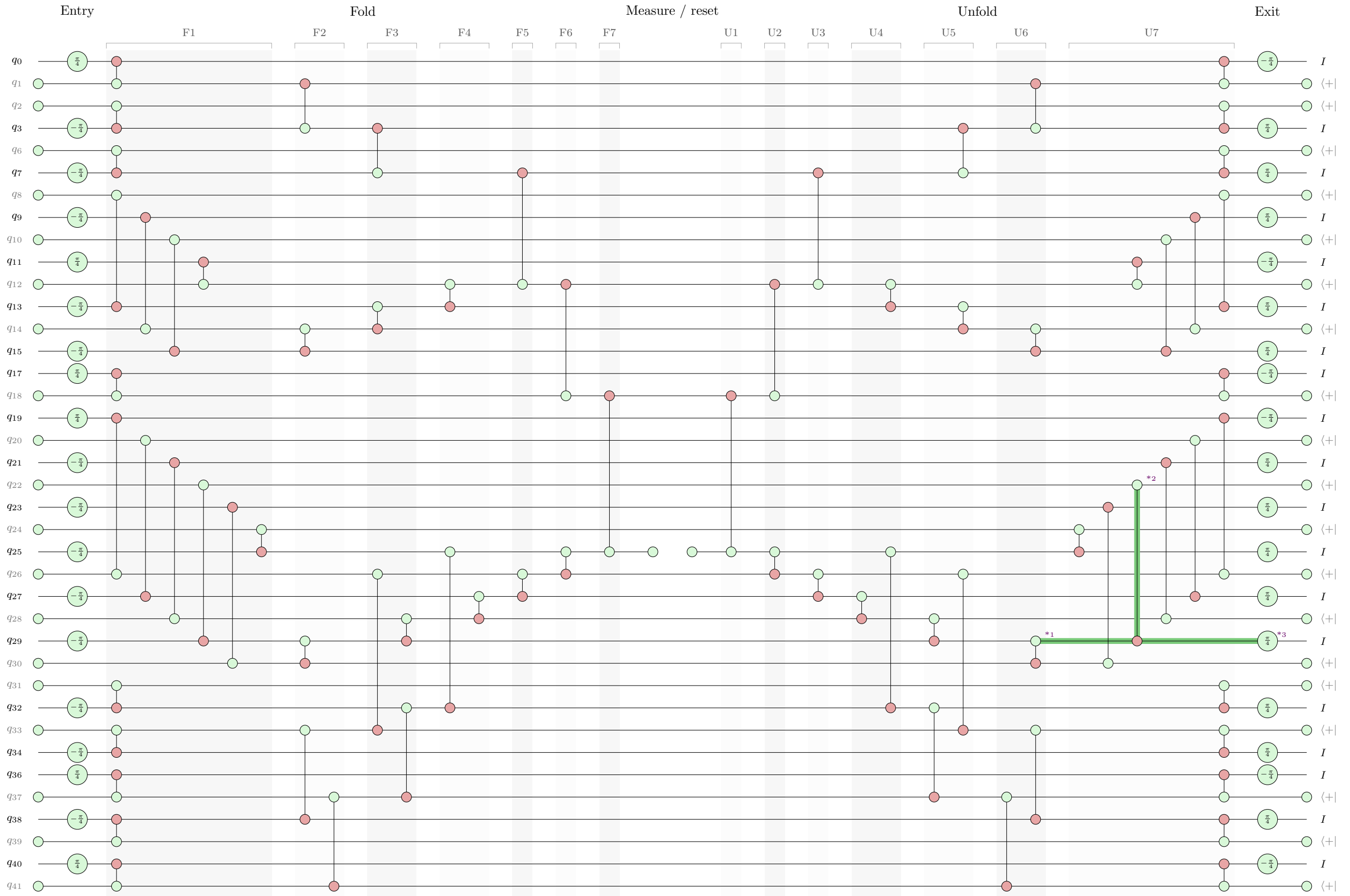
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_225.tikz

$d = 5$: basis element 226

No boundary support. 3 phase defects.



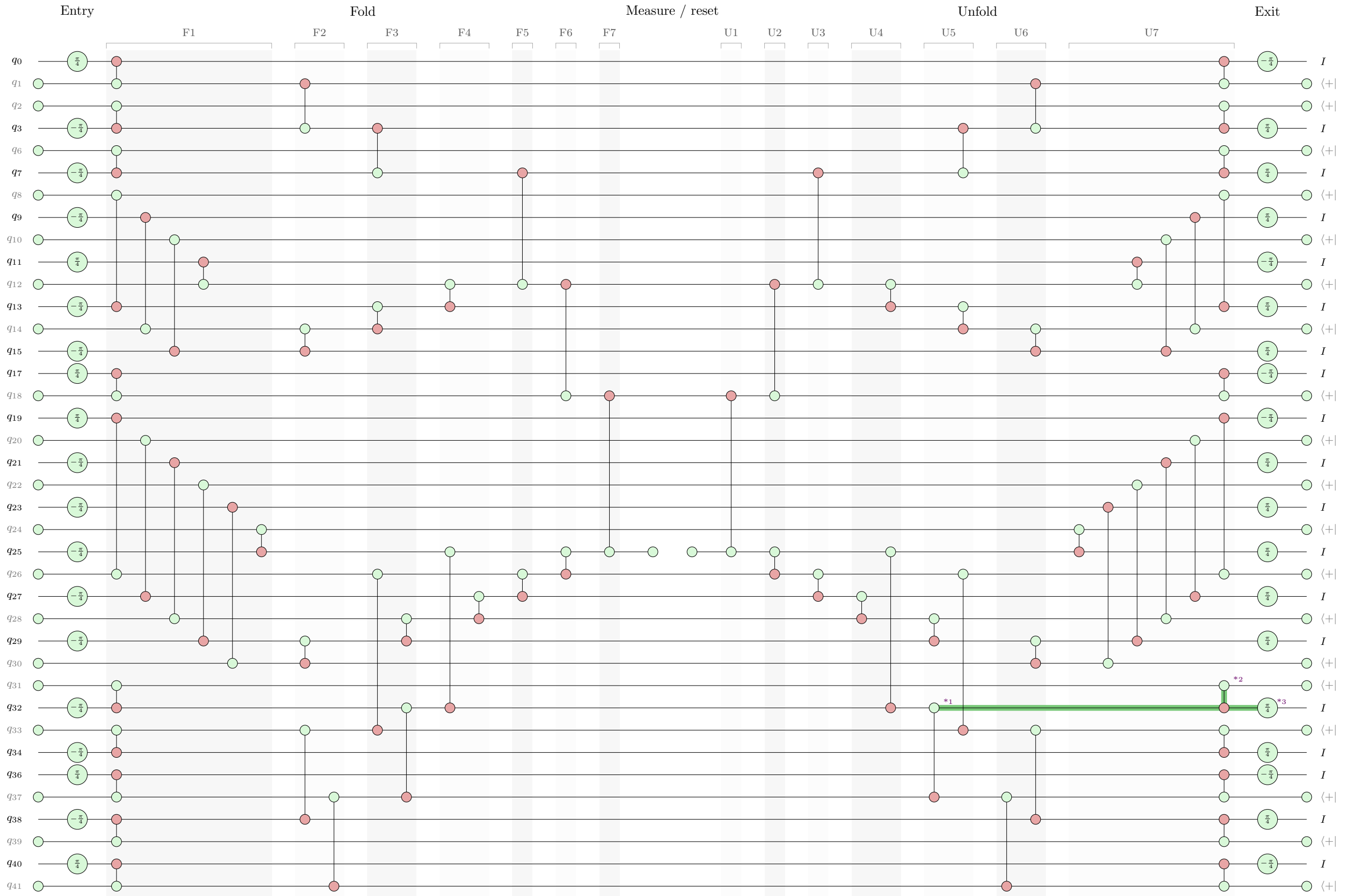
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_226.tikz

$d = 5$: basis element 227

No boundary support. 3 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_227.tikz

$d = 5$: basis element 228

No boundary support. 3 phase defects.



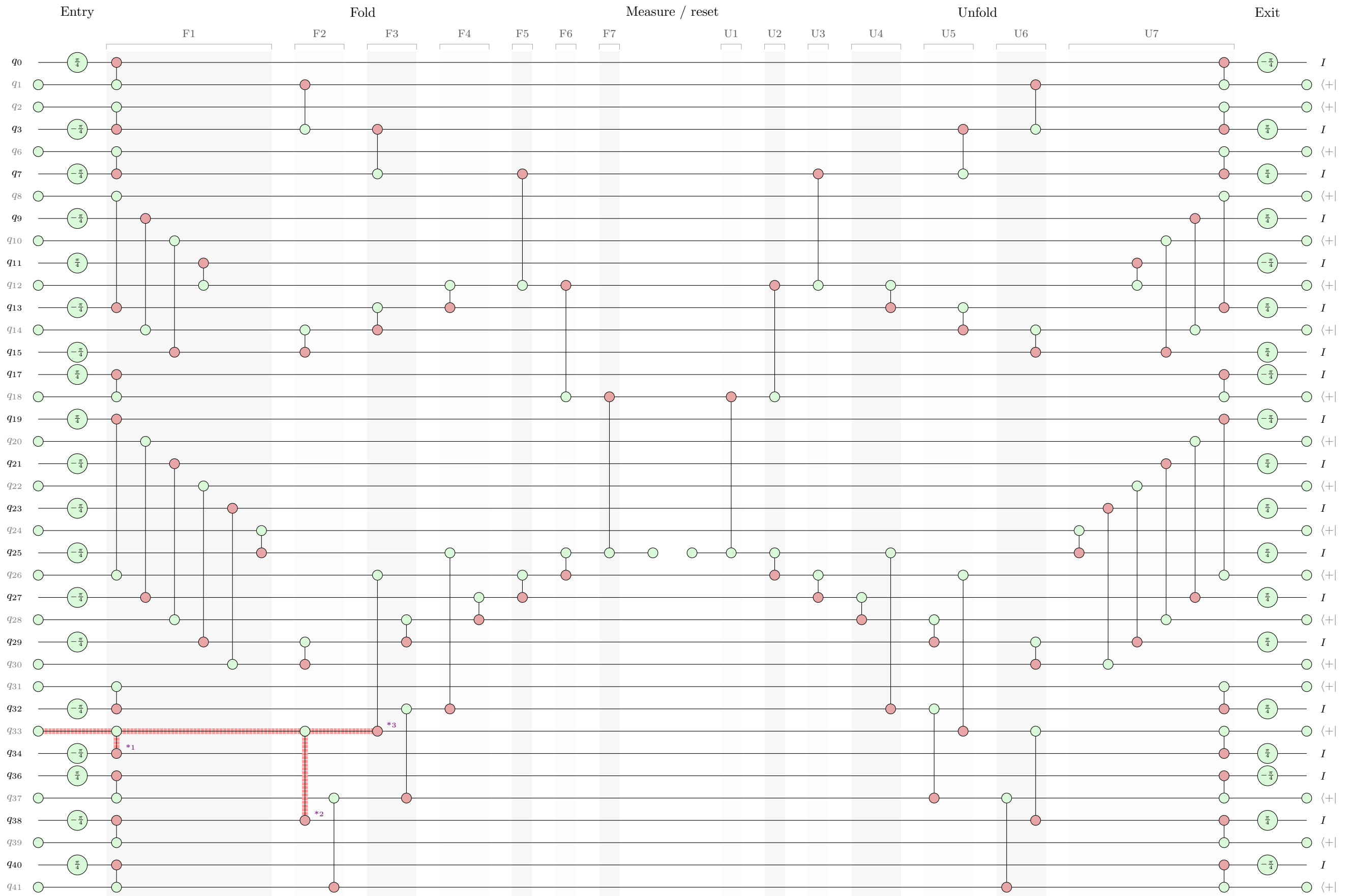
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_228.tikz

$d = 5$: basis element 229

No boundary support. 3 phase defects.

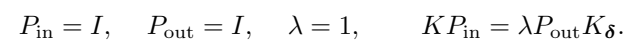


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_229.tikz

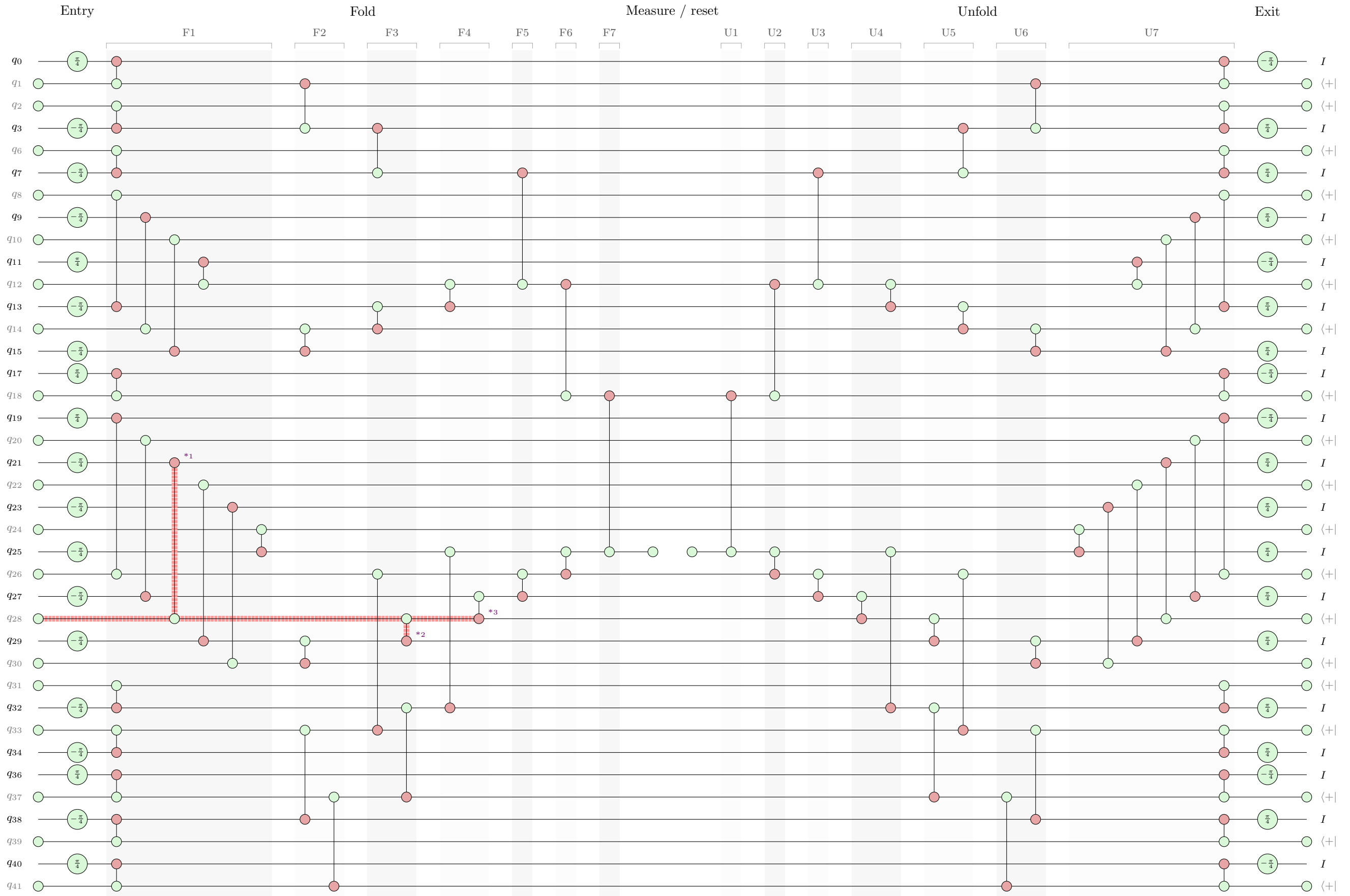
No boundary support. 3 phase defects.



d5_230.tikz

$d = 5$: basis element 231

No boundary support. 3 phase defects.



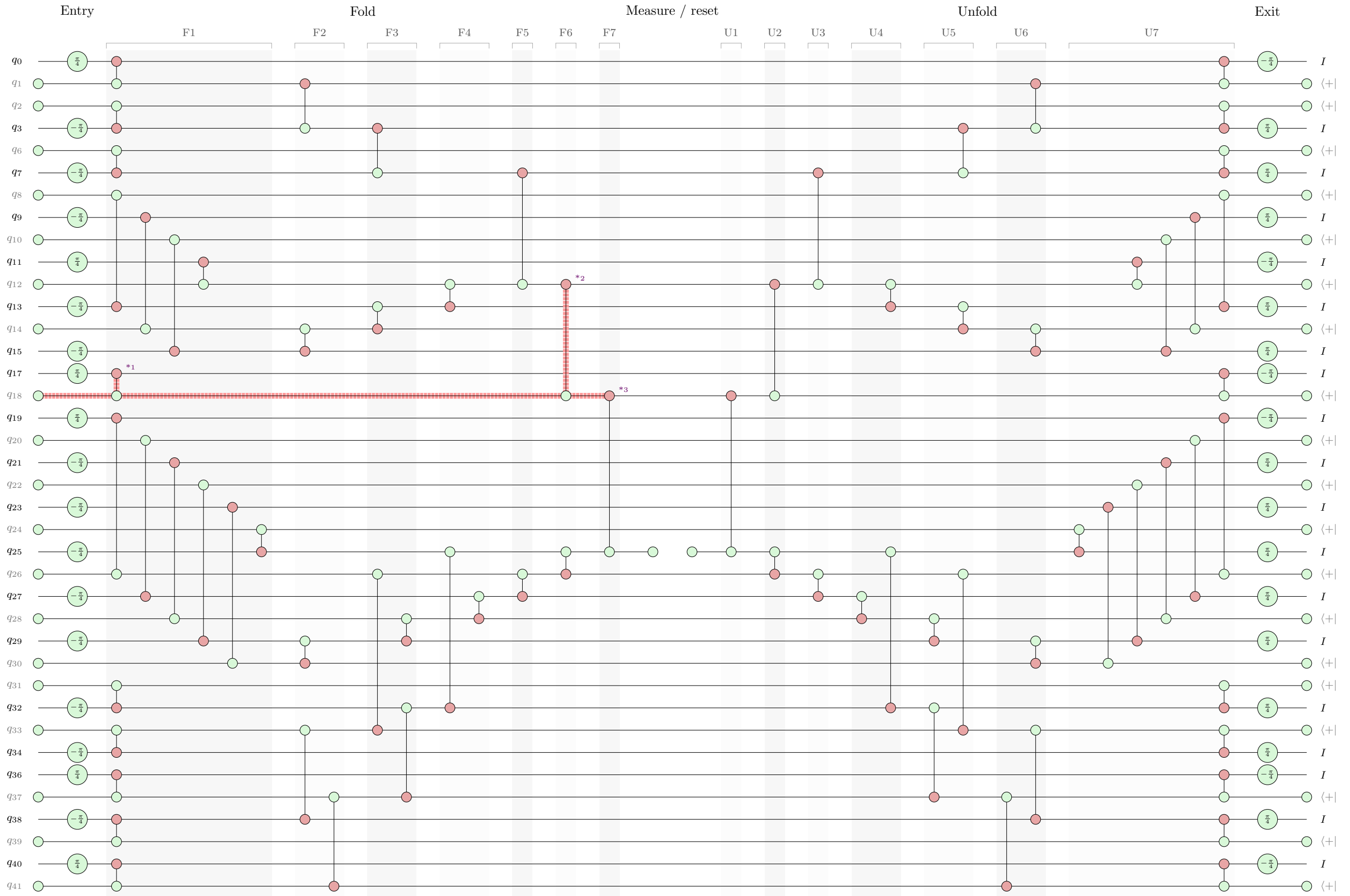
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_231.tikz

$d = 5$: basis element 232

No boundary support. 3 phase defects.



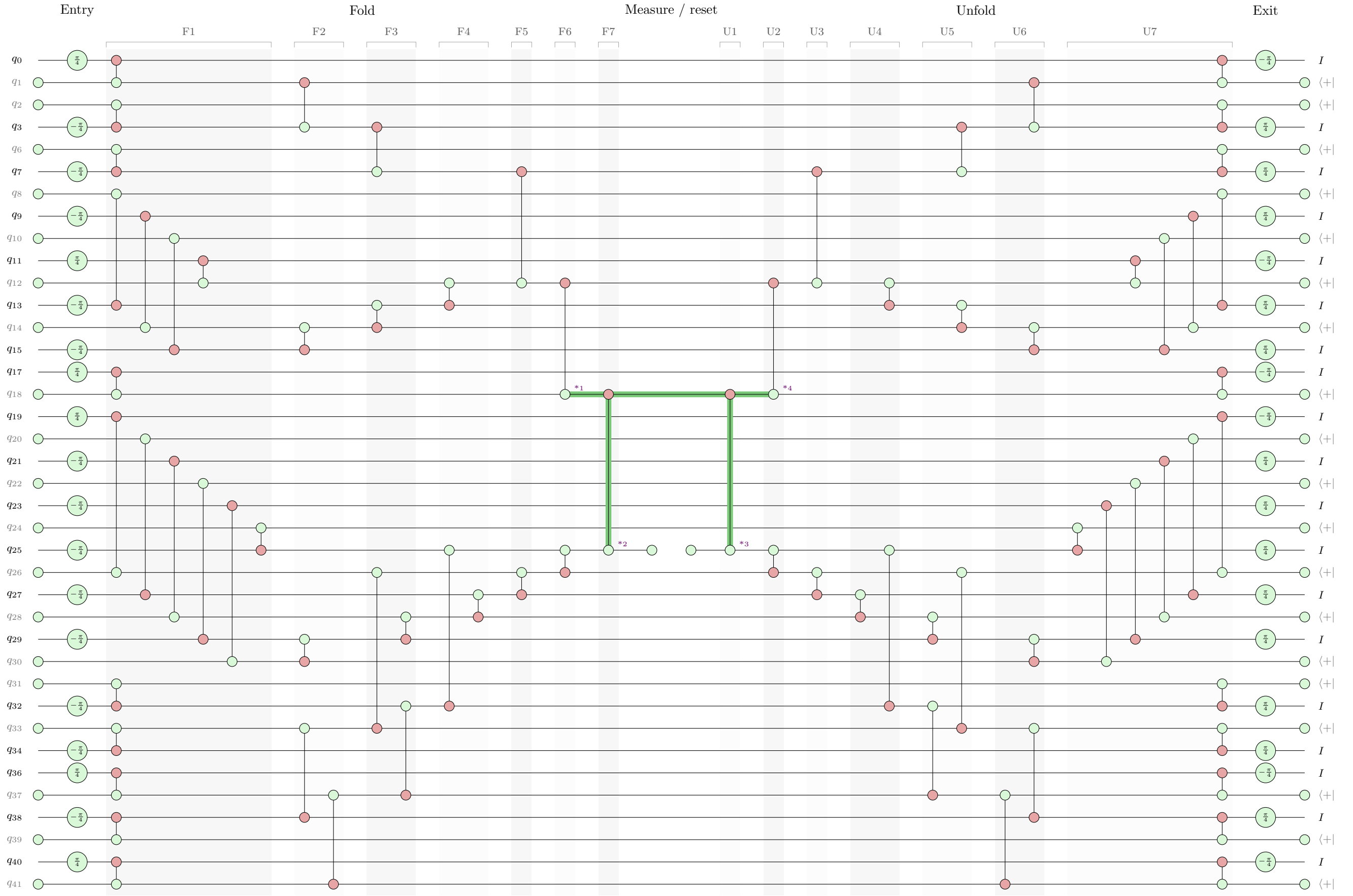
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$.

d5_232.tikz

$d = 5$: basis element 233

No boundary support. 4 phase defects.

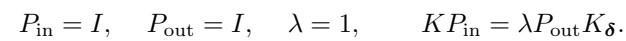


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_233.tikz

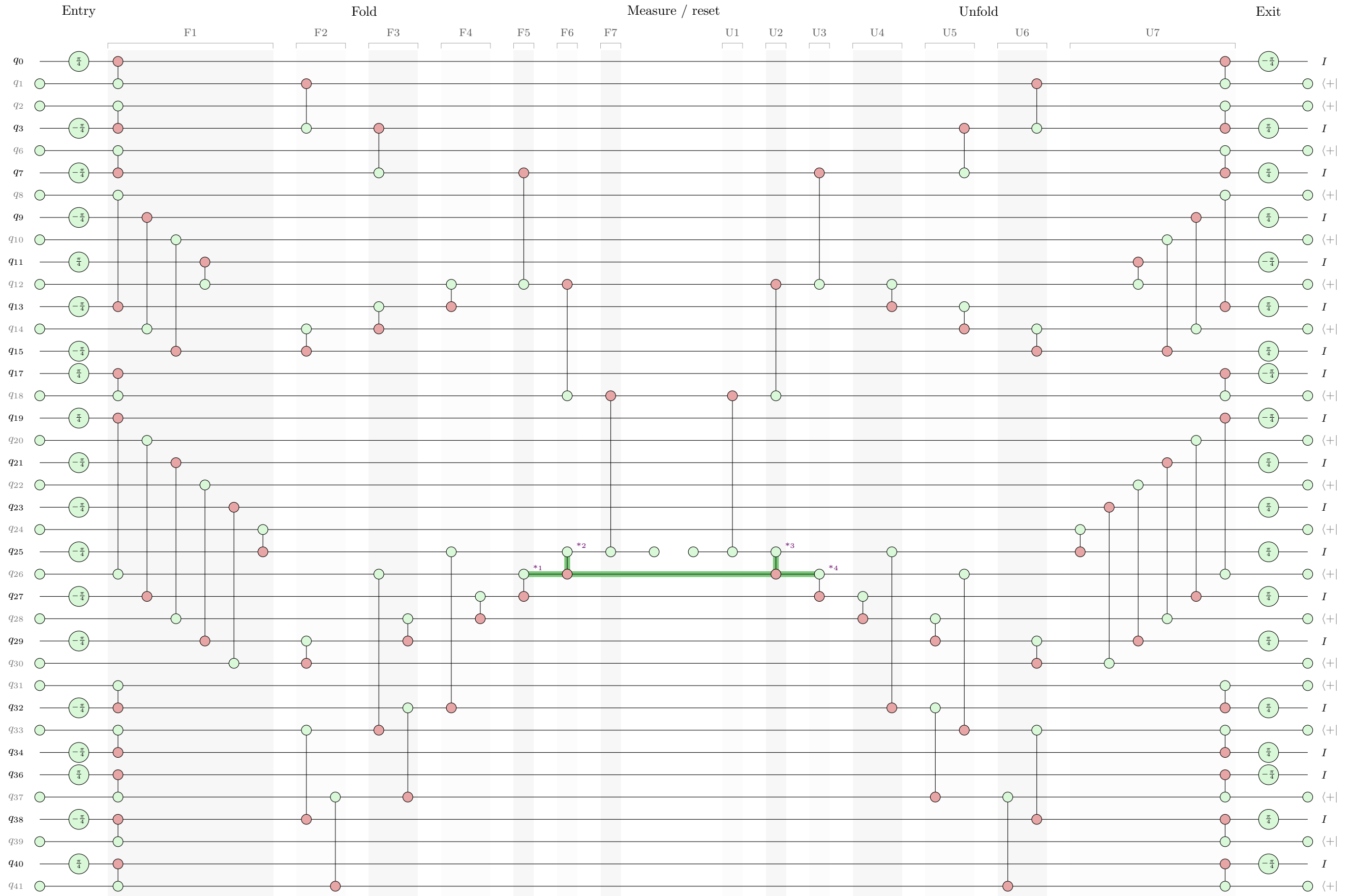
No boundary support. 4 phase defects.



d5_234.tikz

$d = 5$: basis element 235

No boundary support. 4 phase defects.



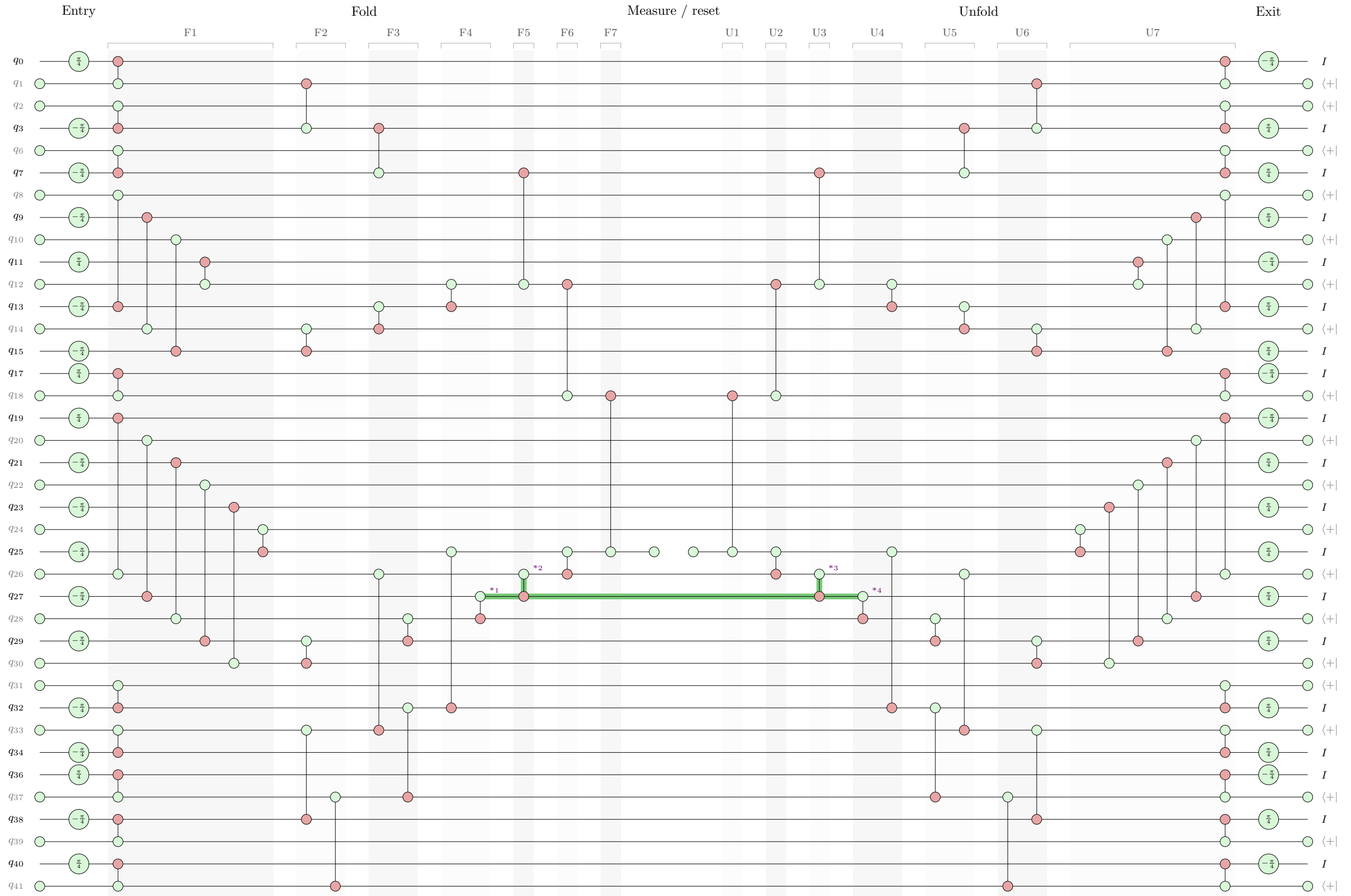
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_235.tikz

$d = 5$: basis element 236

No boundary support. 4 phase defects.



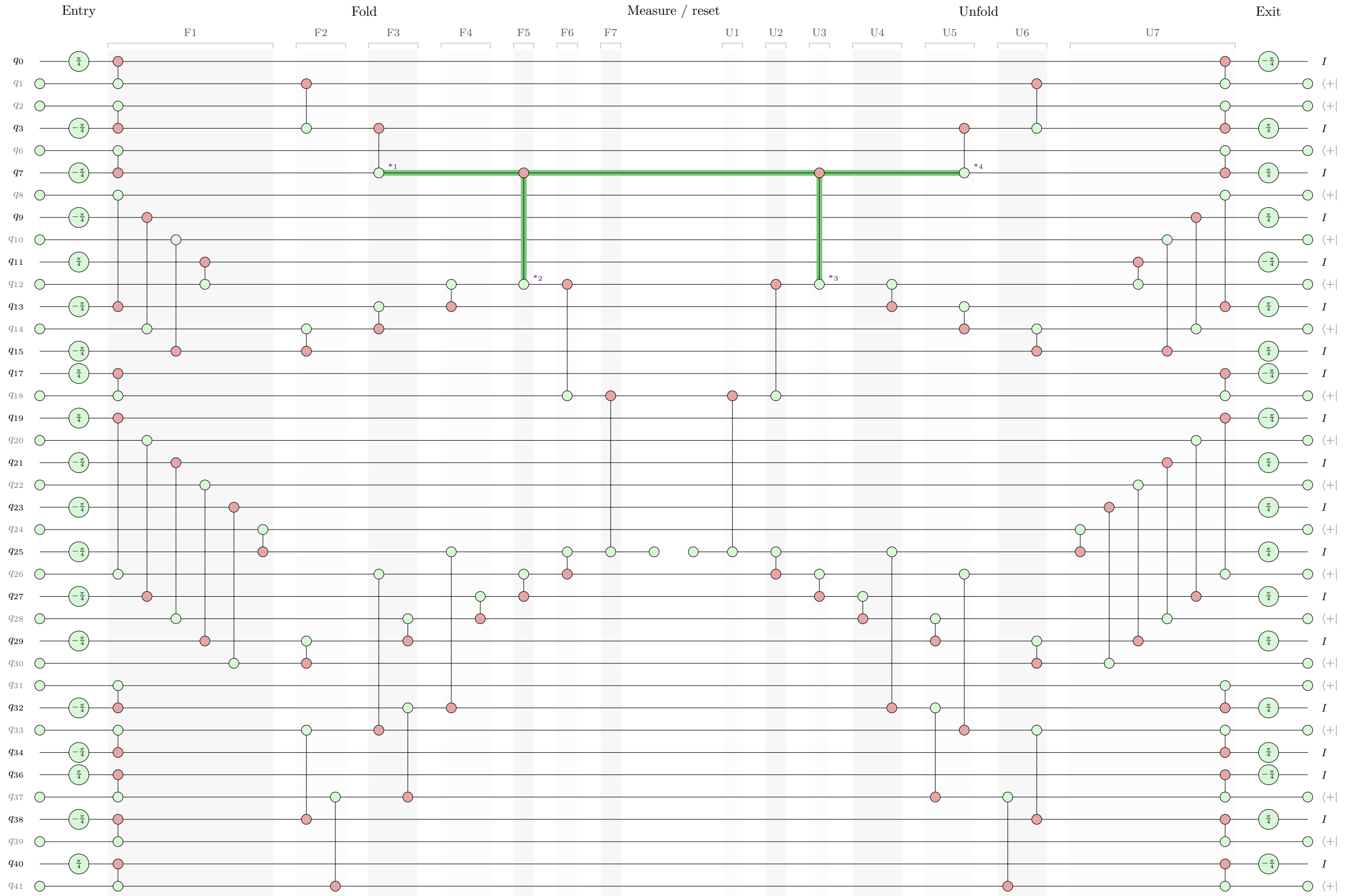
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_236.tikz

$d = 5$: basis element 237

No boundary support. 4 phase defects.



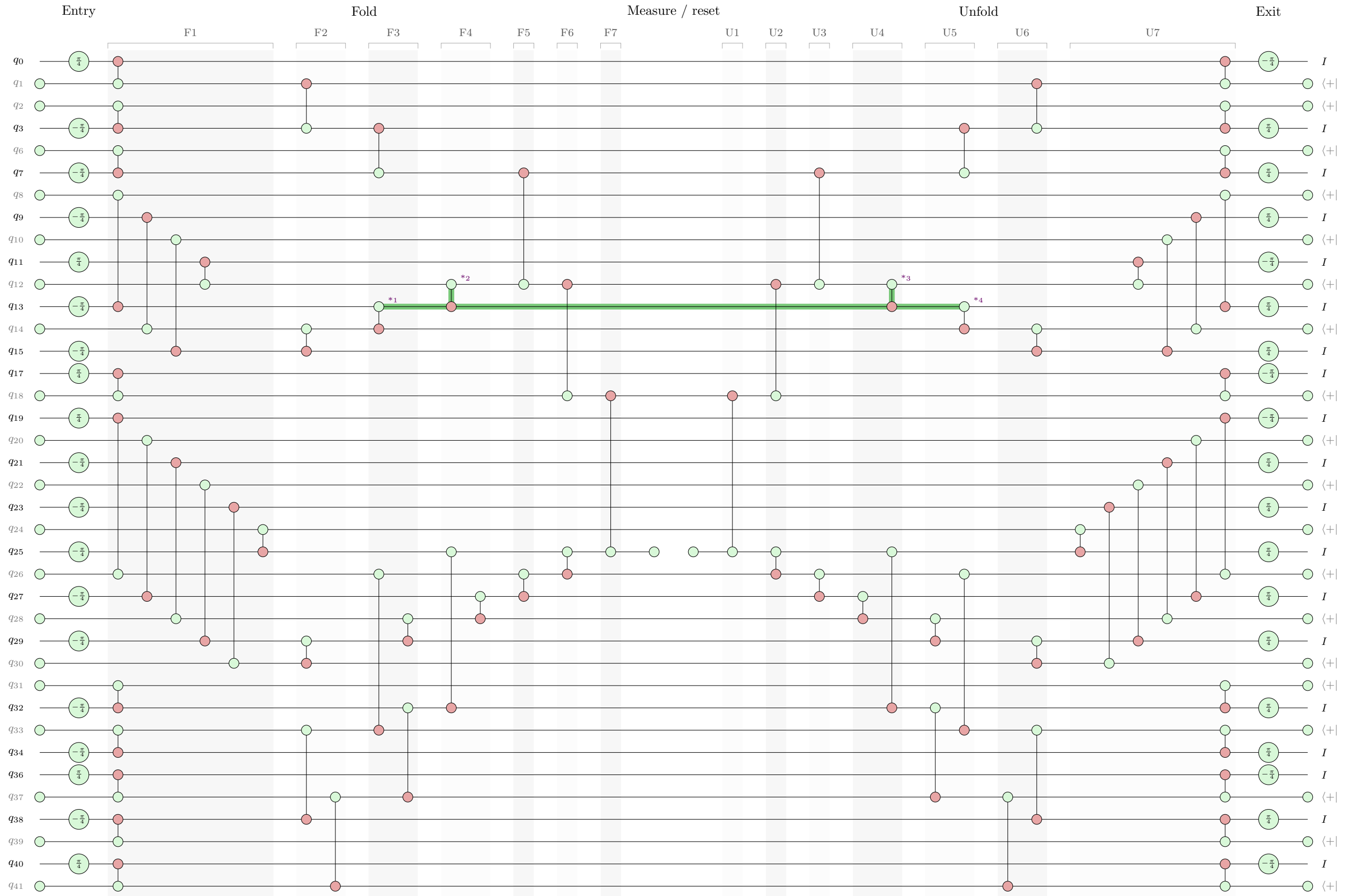
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_237.tikz

$d = 5$: basis element 238

No boundary support. 4 phase defects.



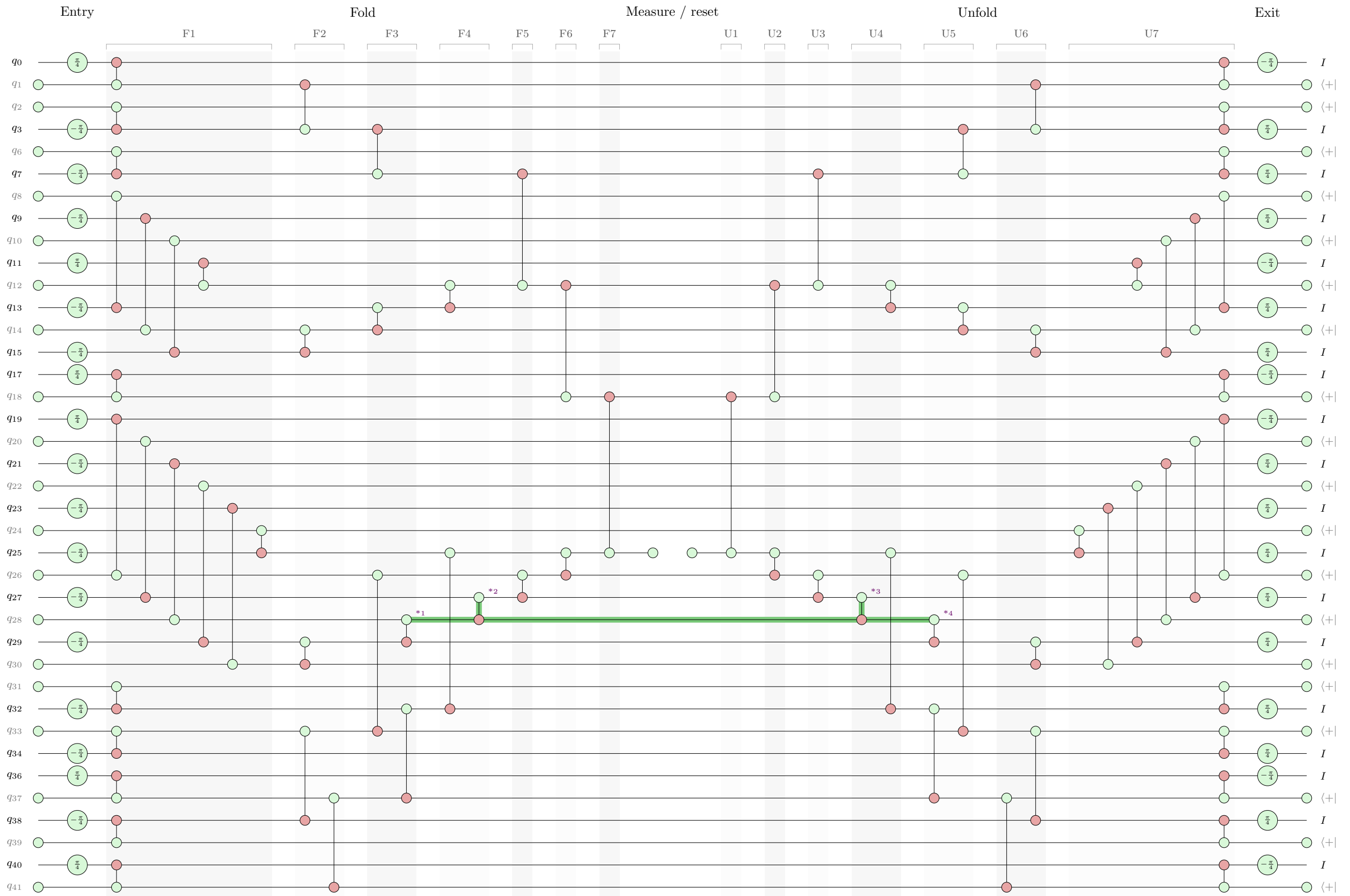
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_238.tikz

$d = 5$: basis element 239

No boundary support. 4 phase defects.



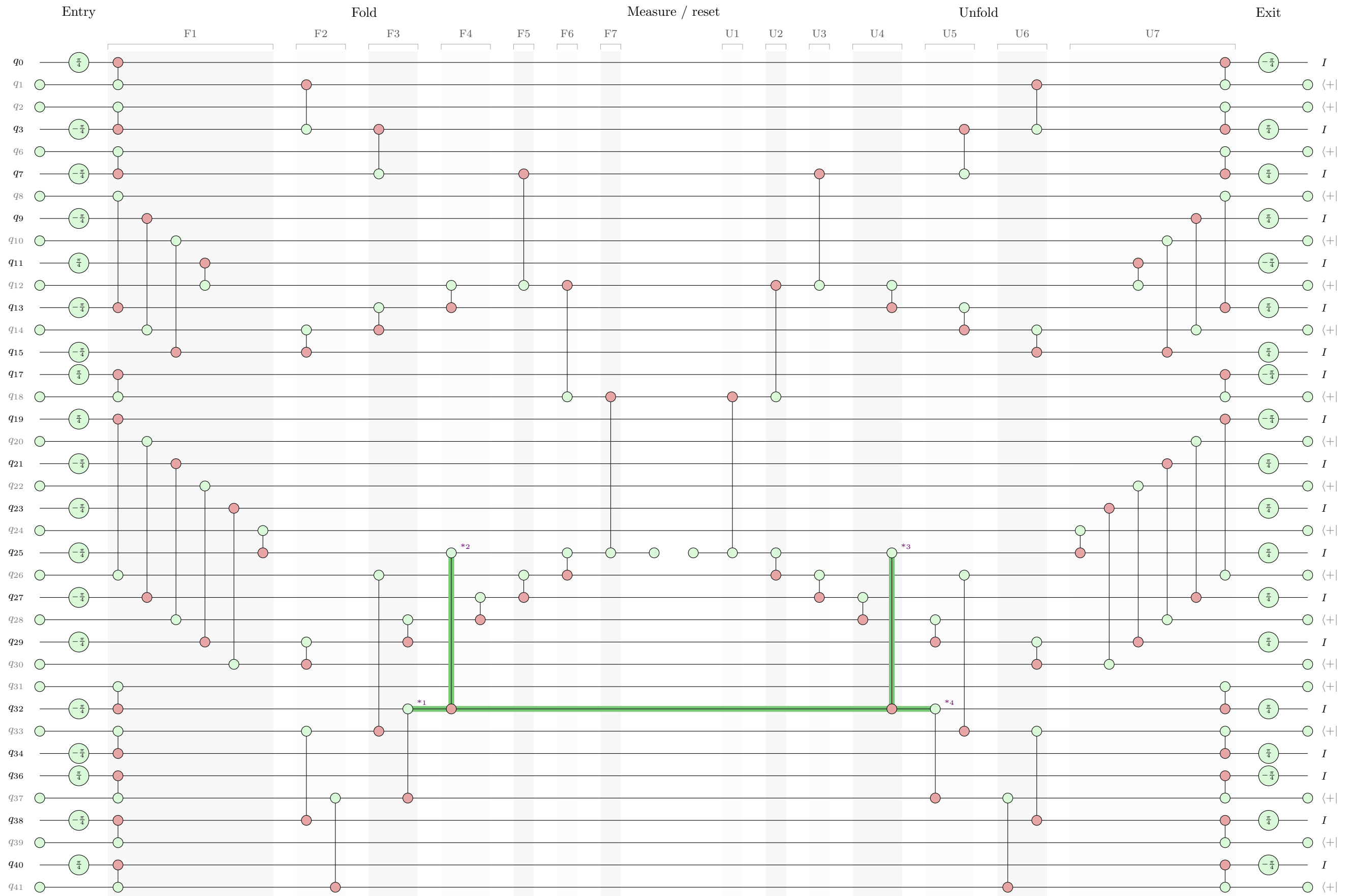
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_239.tikz

$d = 5$: basis element 240

No boundary support. 4 phase defects.



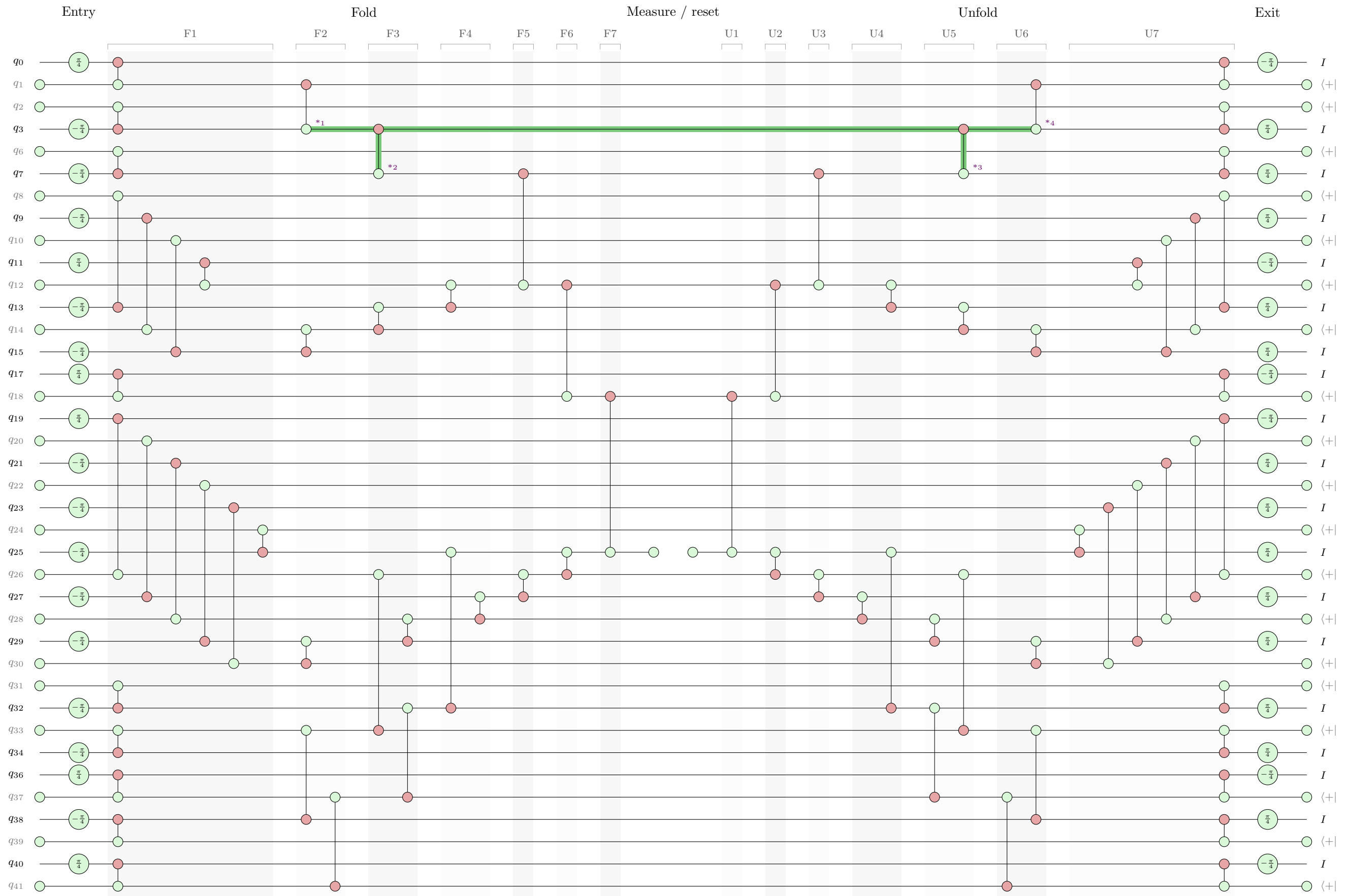
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_240.tikz

$d = 5$: basis element 241

No boundary support. 4 phase defects.



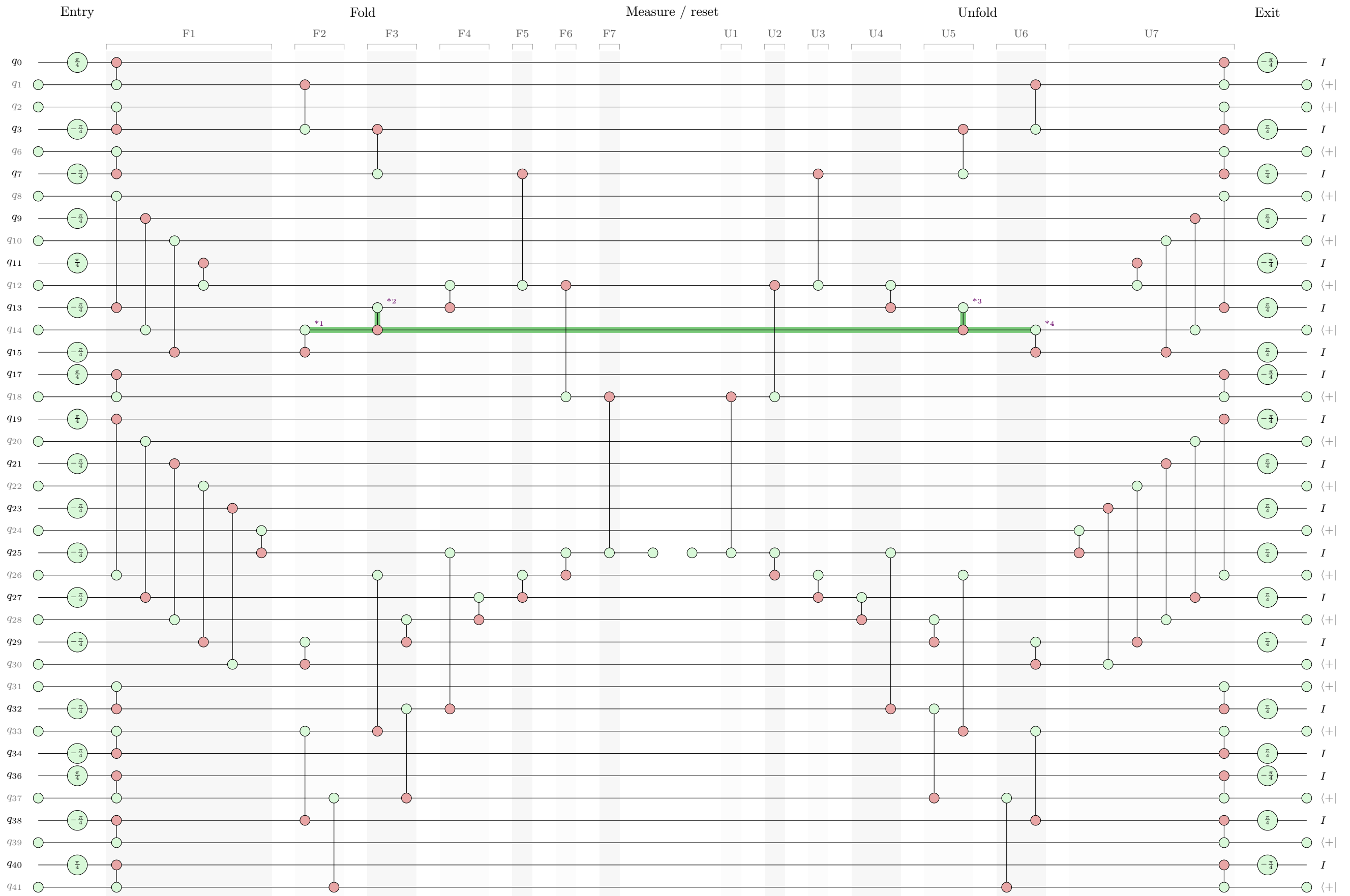
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_241.tikz

$d = 5$: basis element 242

No boundary support. 4 phase defects.



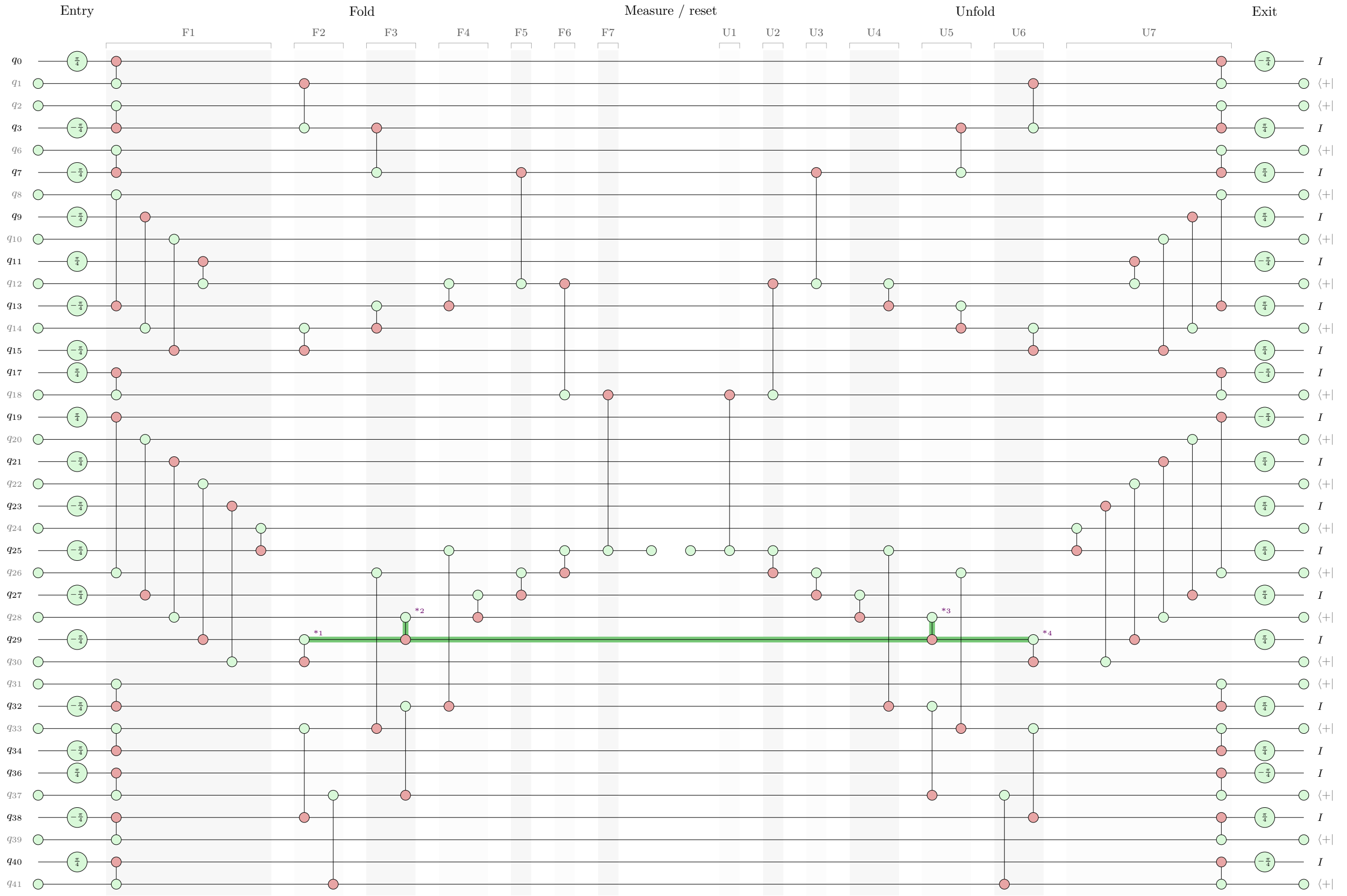
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_242.tikz

$d = 5$: basis element 243

No boundary support. 4 phase defects.



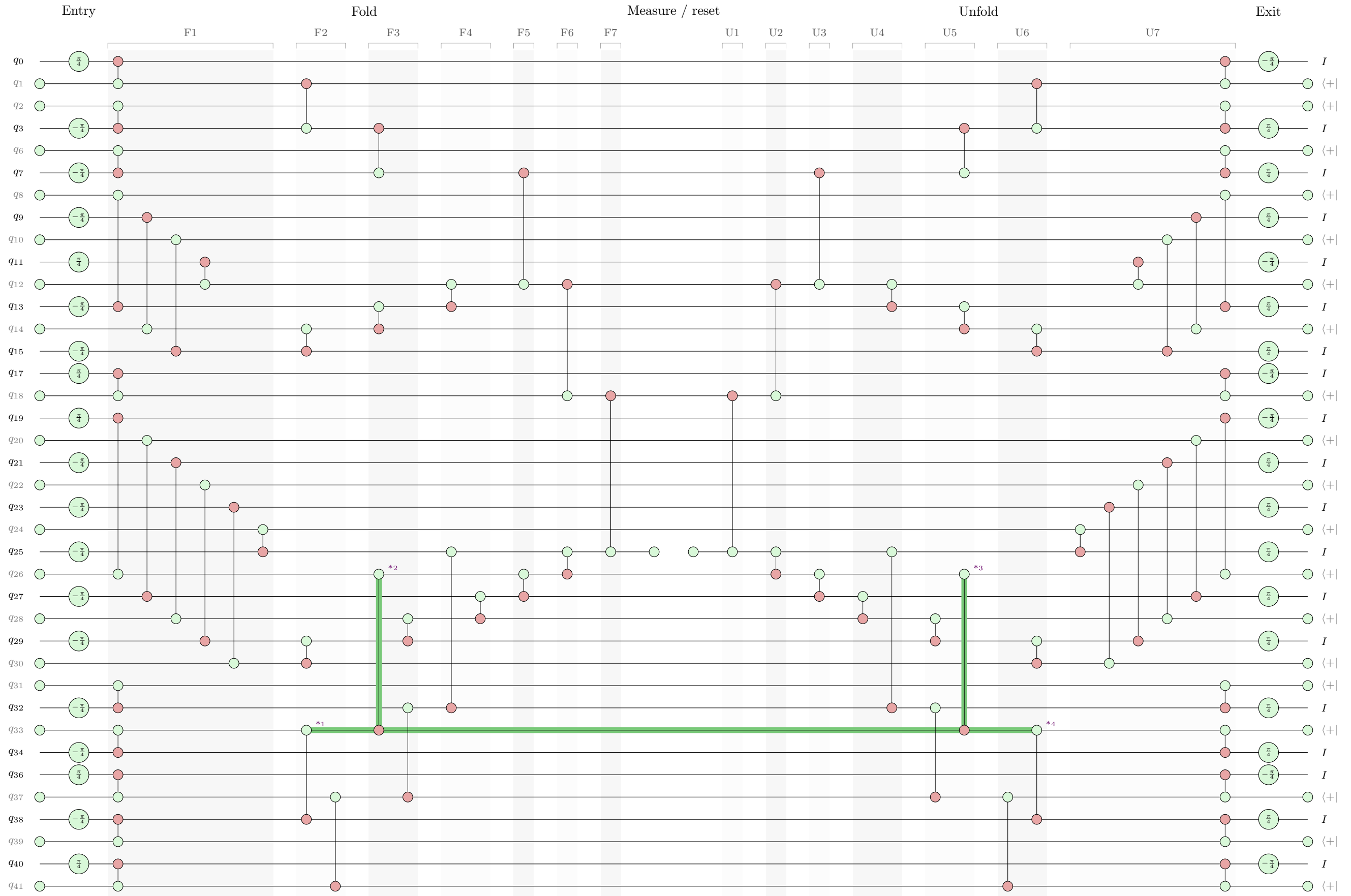
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_243.tikz

$d = 5$: basis element 244

No boundary support. 4 phase defects.



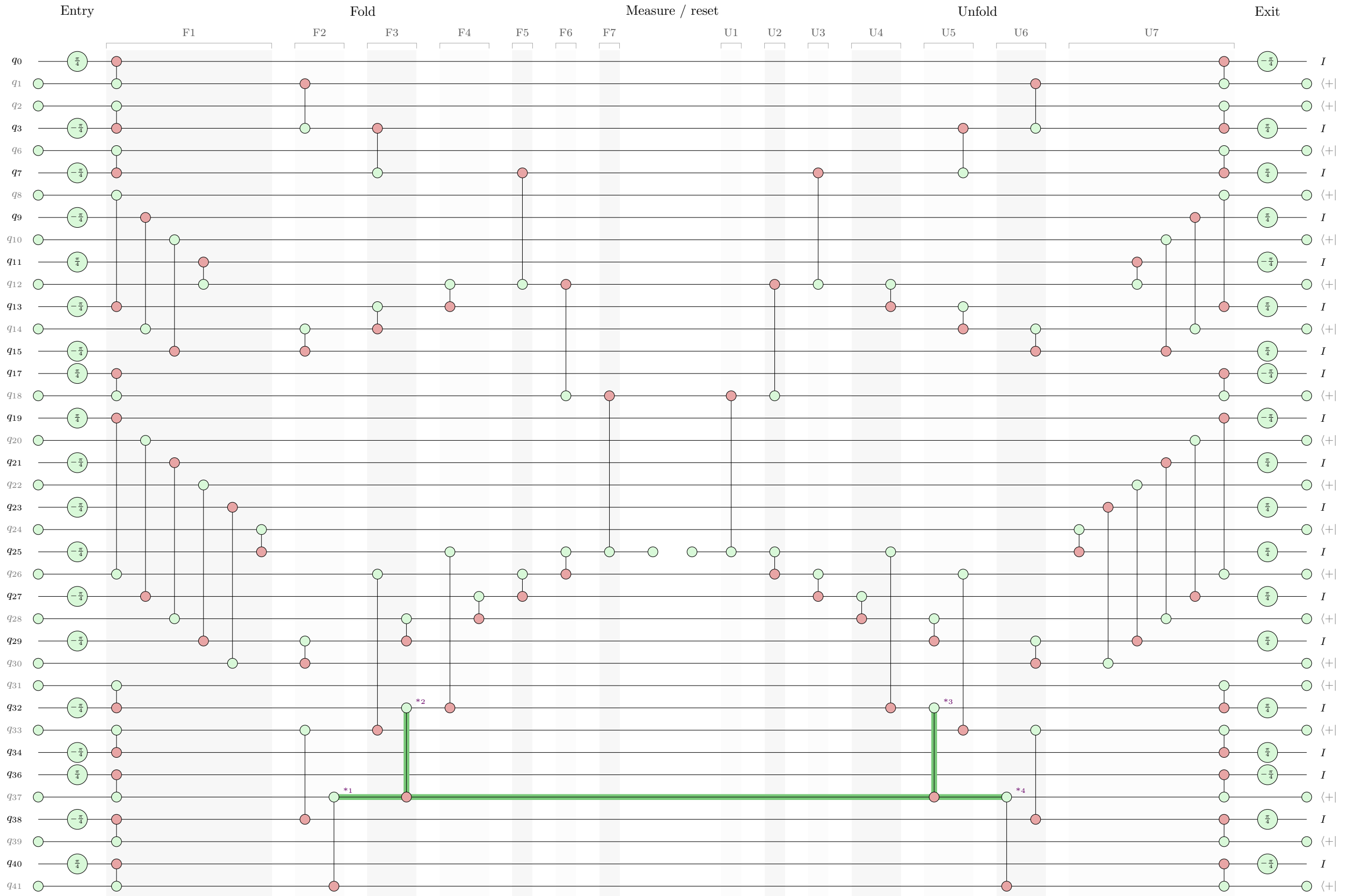
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_244.tikz

$d = 5$: basis element 245

No boundary support. 4 phase defects.



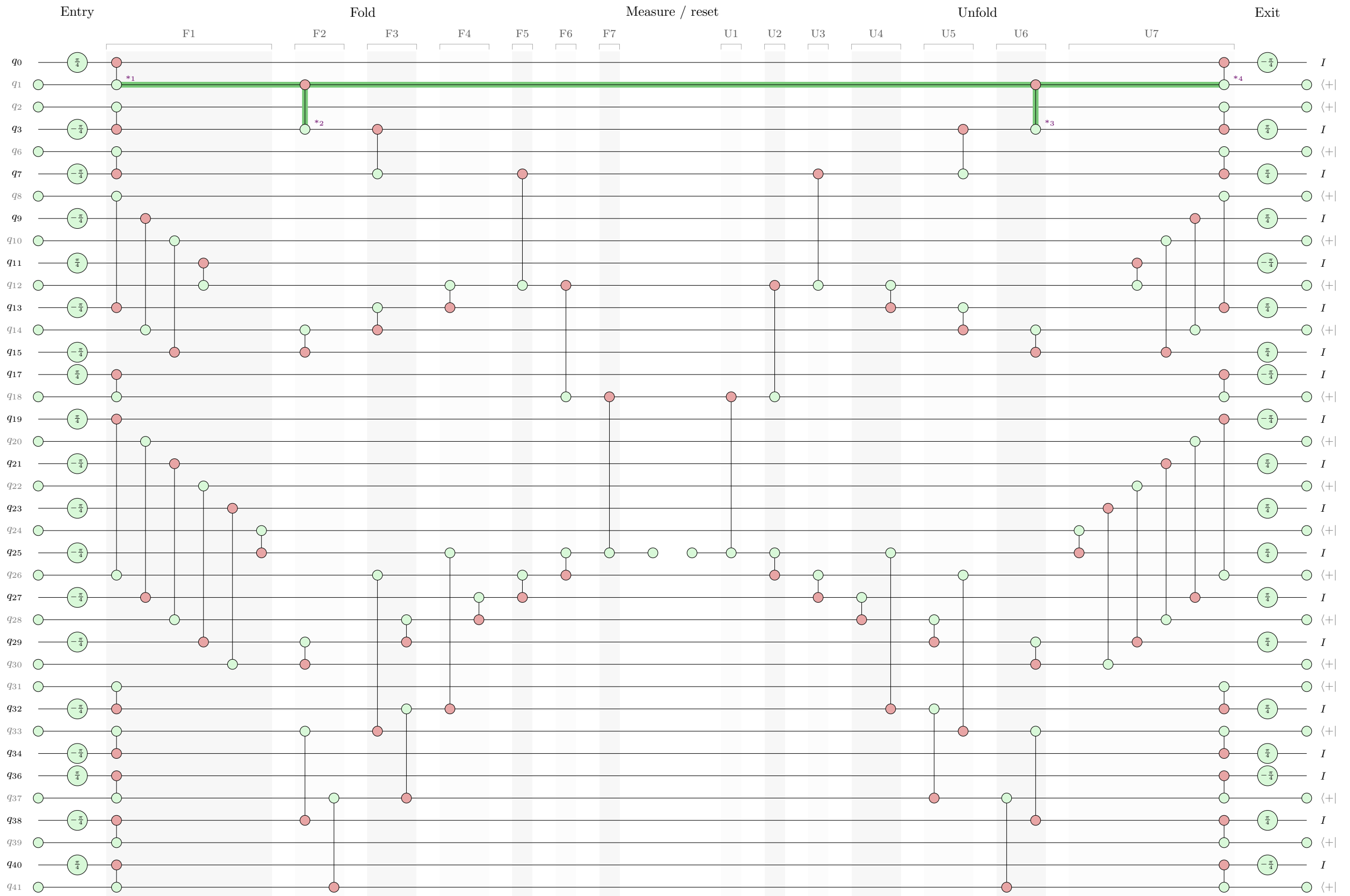
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_245.tikz

$d = 5$: basis element 246

No boundary support. 4 phase defects.



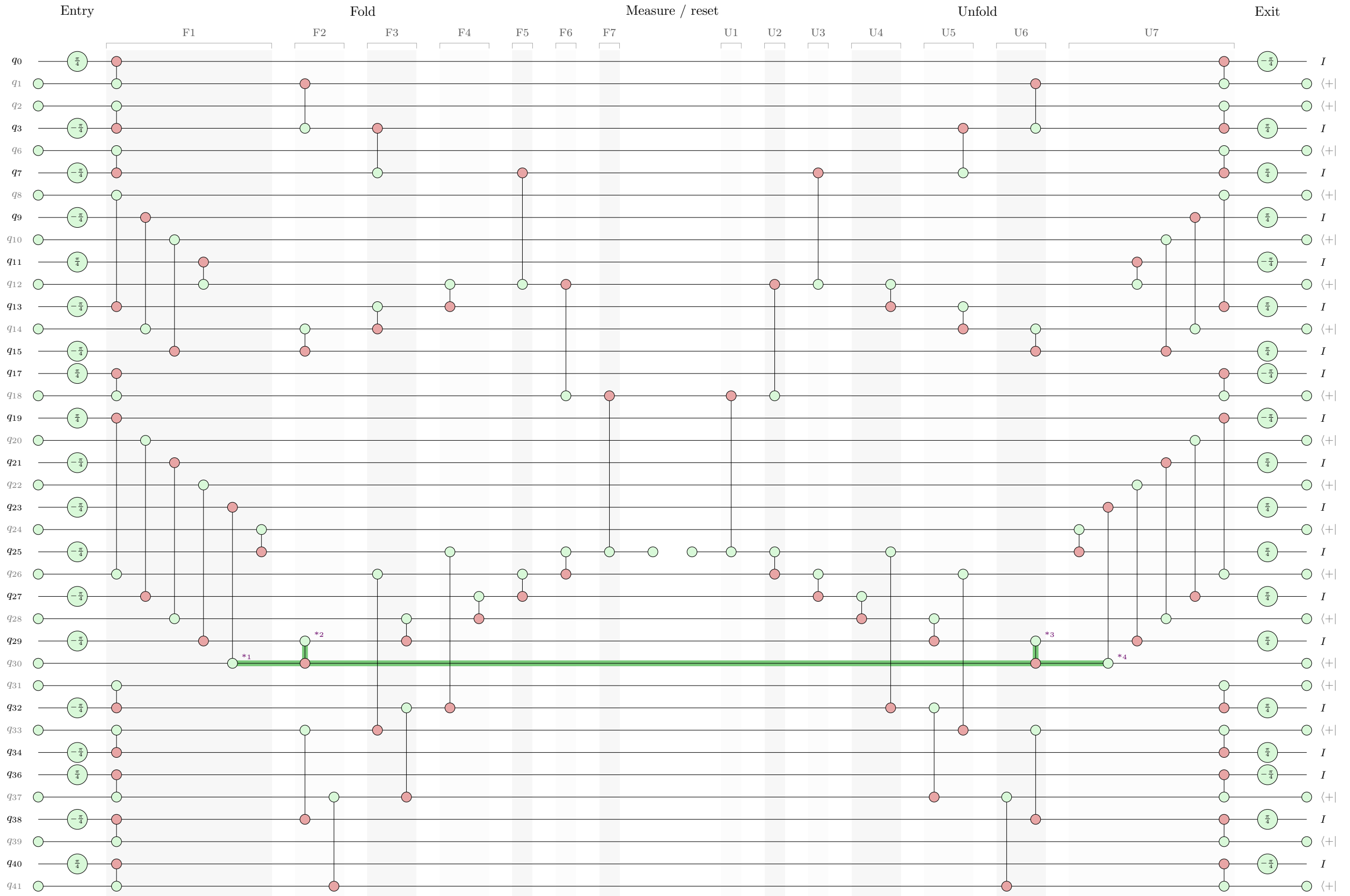
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_246.tikz

$d = 5$: basis element 247

No boundary support. 4 phase defects.



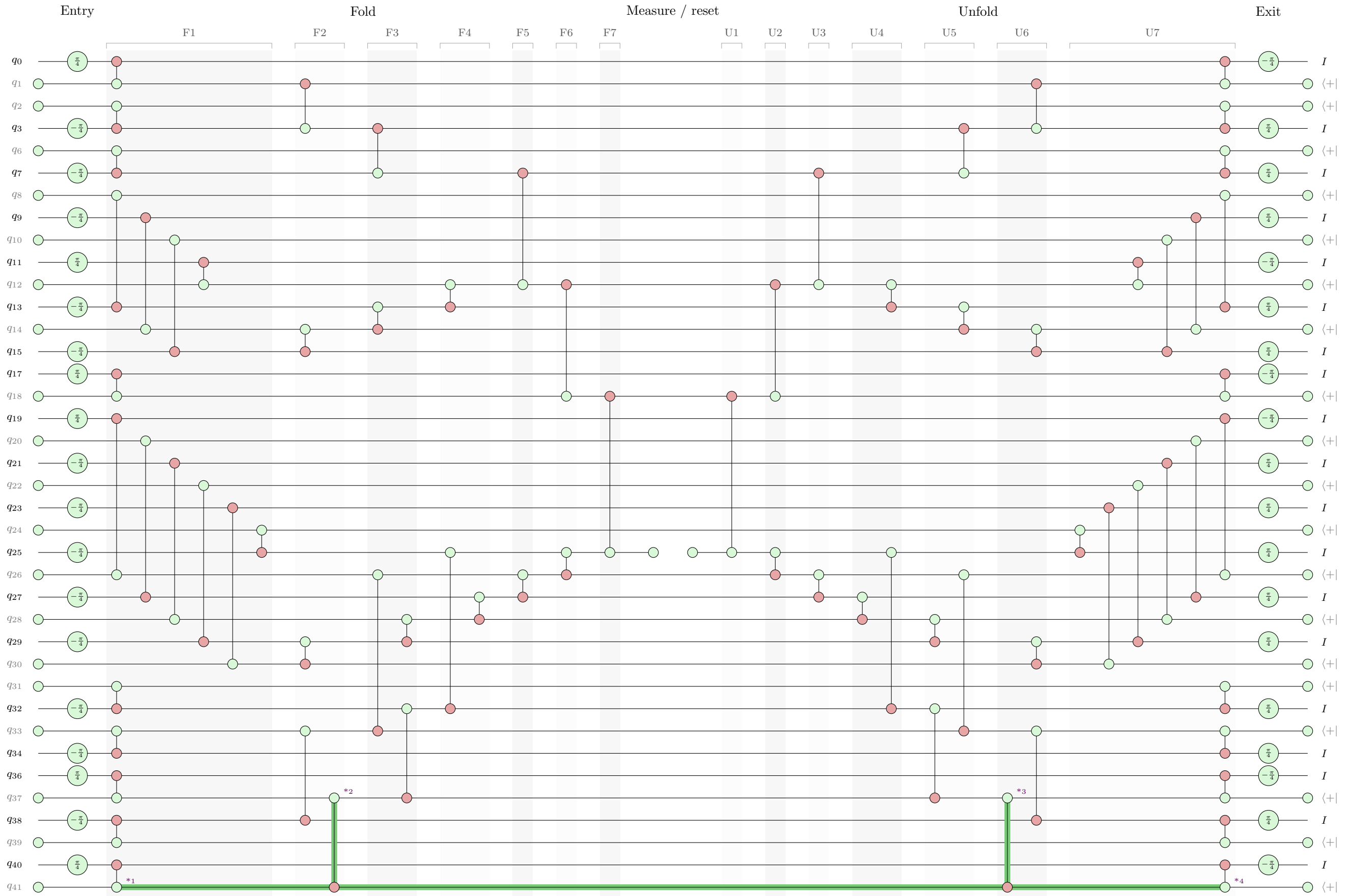
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_247.tikz

$d = 5$: basis element 248

No boundary support. 4 phase defects.



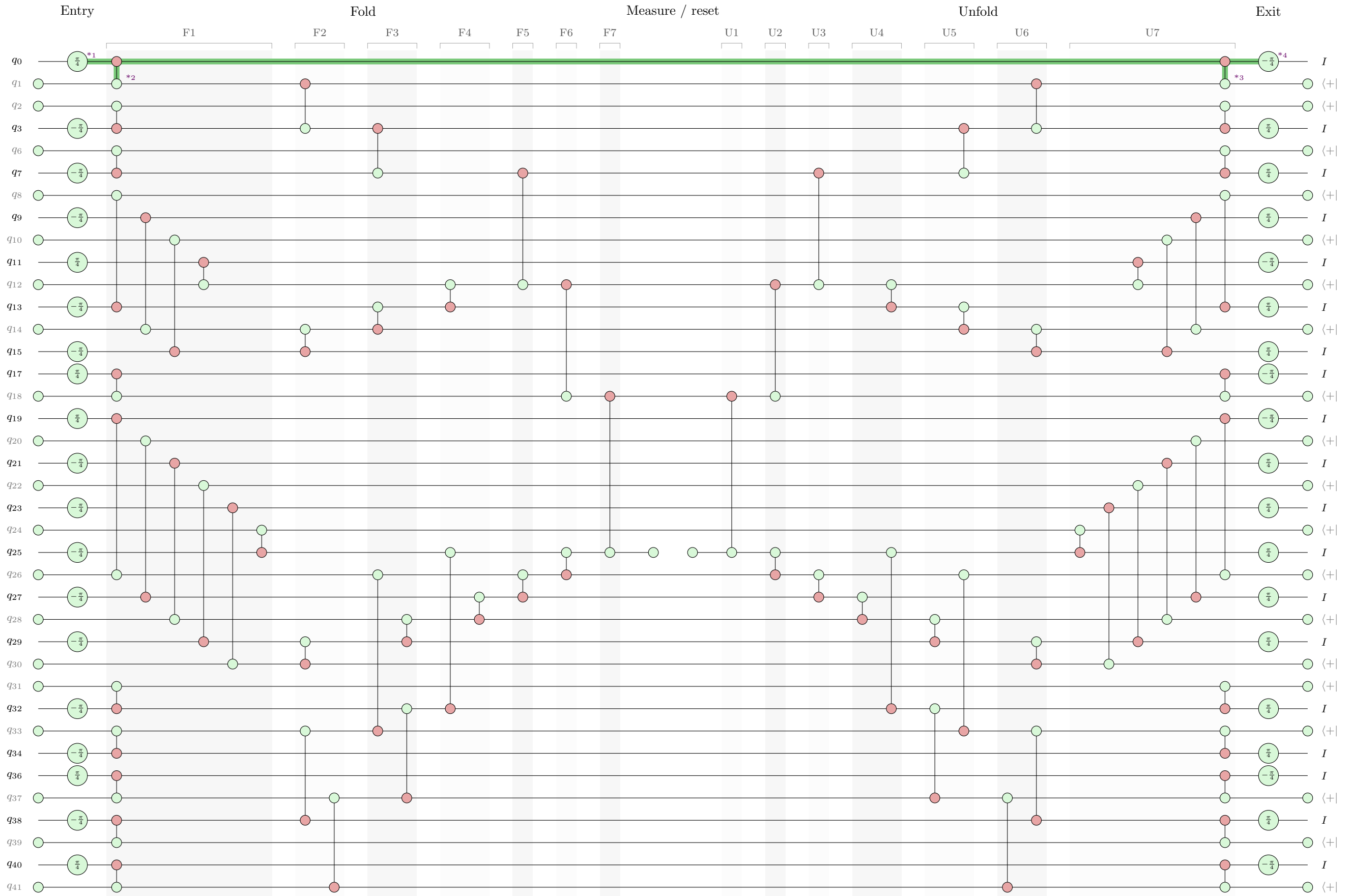
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_248.tikz

$d = 5$: basis element 249

No boundary support. 4 phase defects.



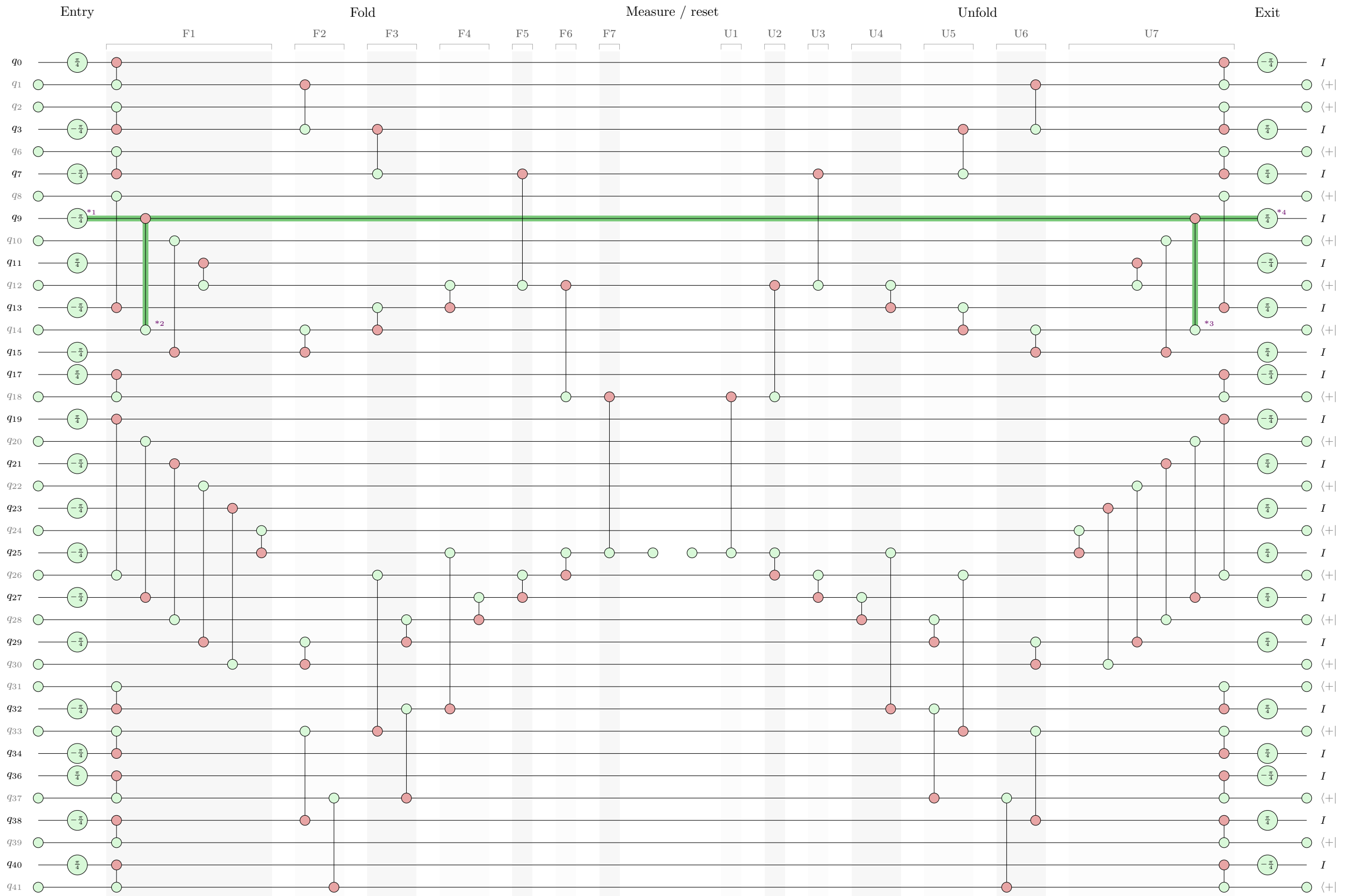
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_249.tikz

$d = 5$: basis element 250

No boundary support. 4 phase defects.

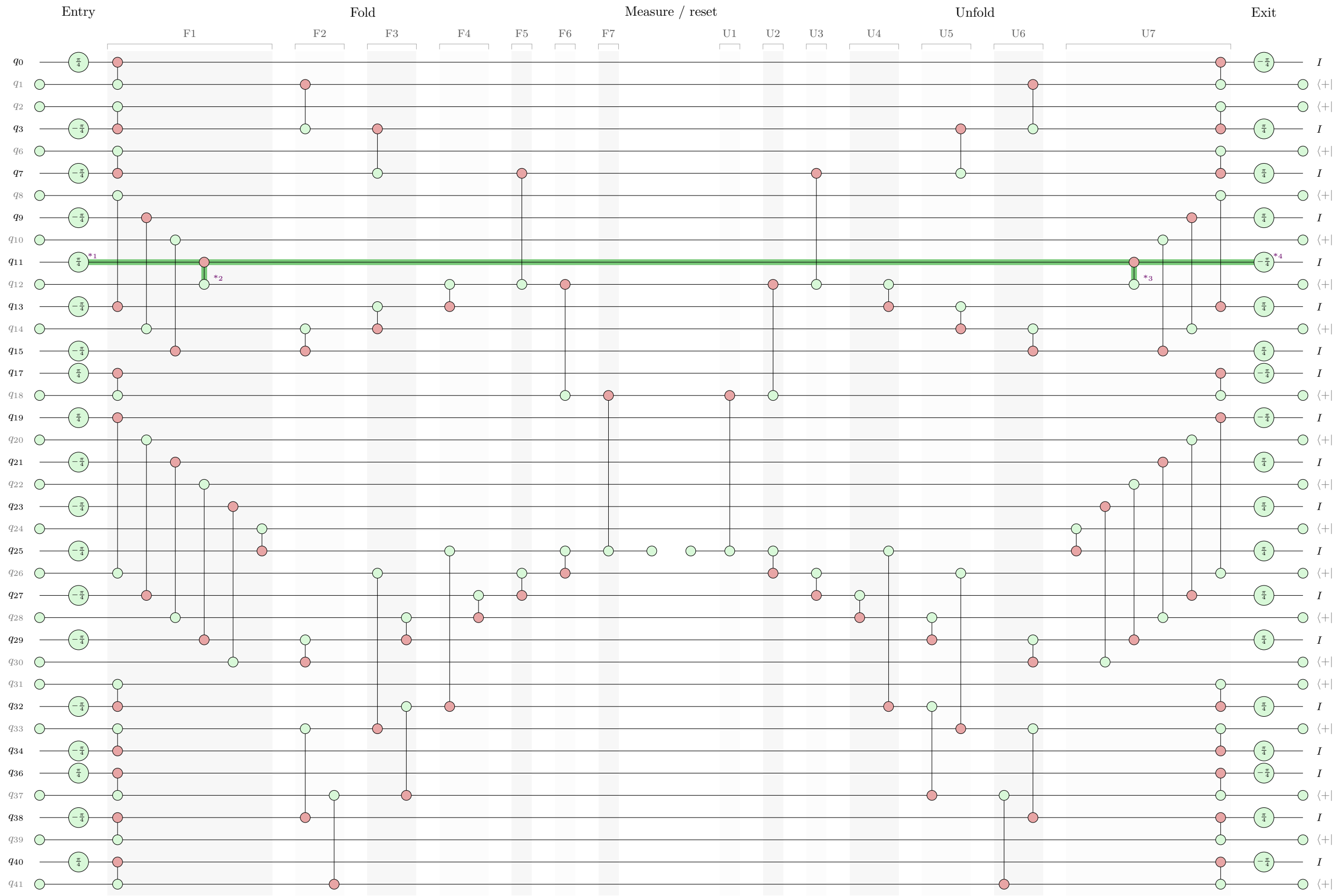


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_250.tikz

$d = 5$: basis element 251
 No boundary support. 4 phase defects.

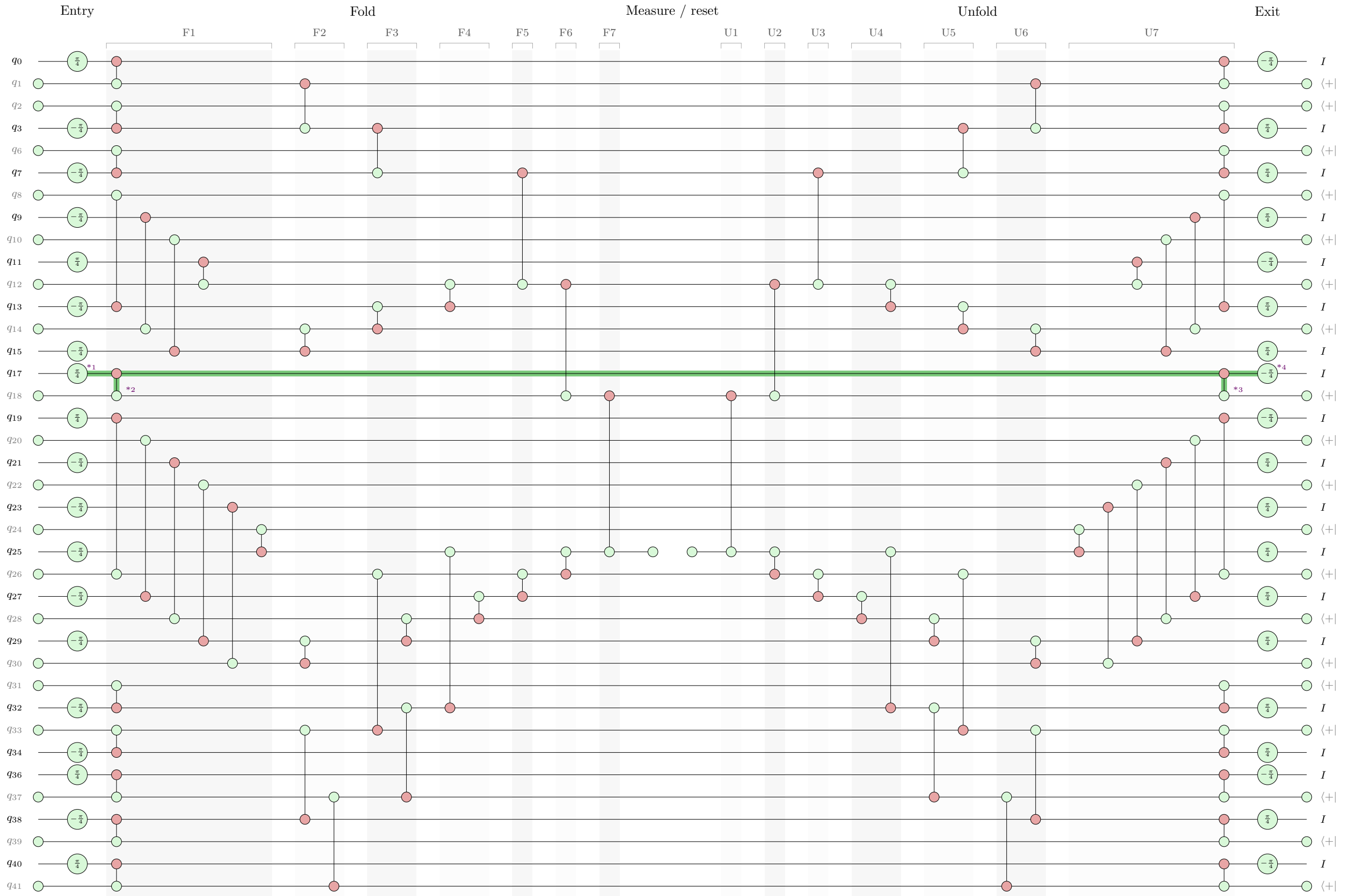


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

$d = 5$: basis element 252

No boundary support. 4 phase defects.



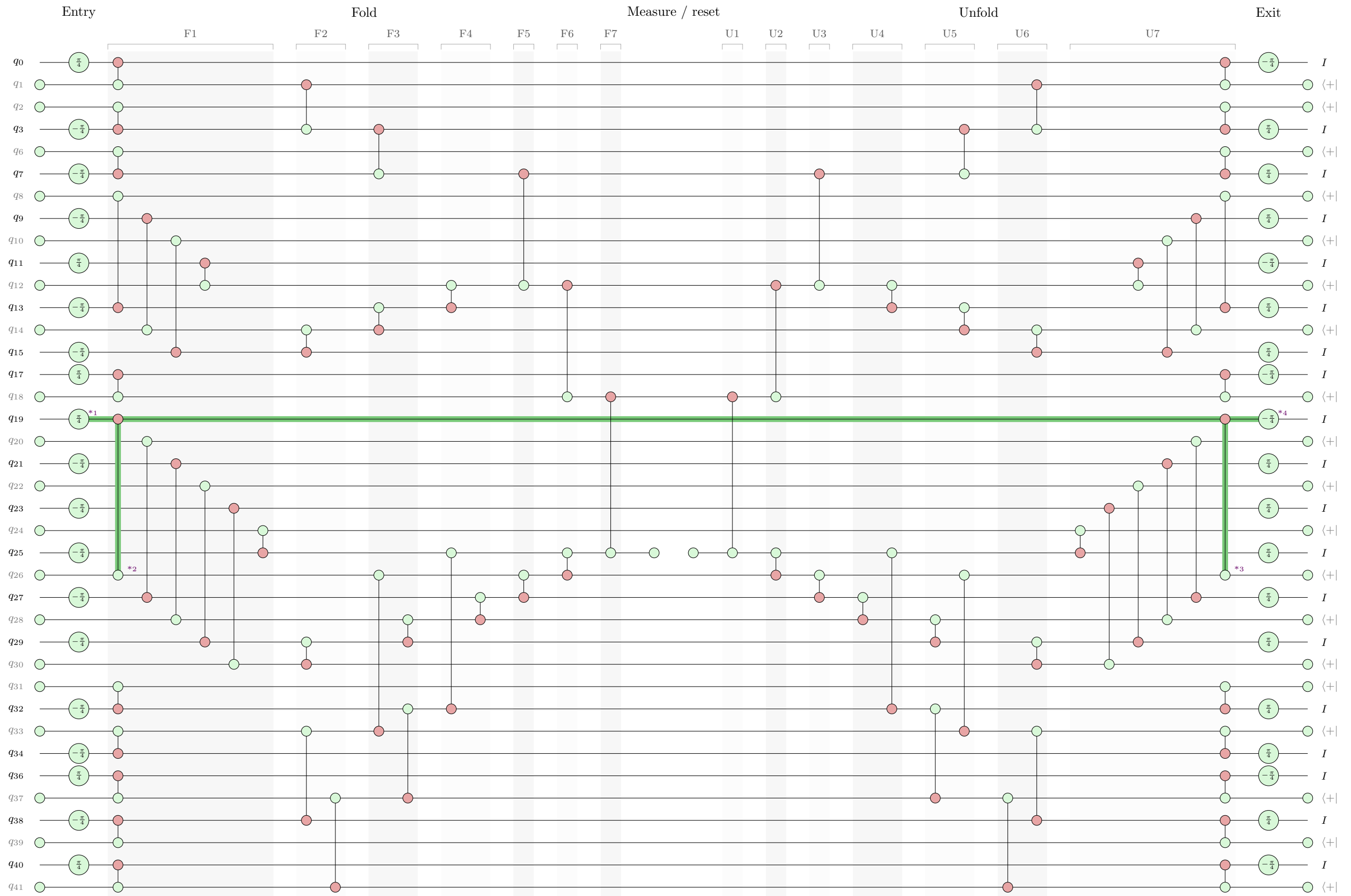
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_252.tikz

$d = 5$: basis element 253

No boundary support. 4 phase defects.



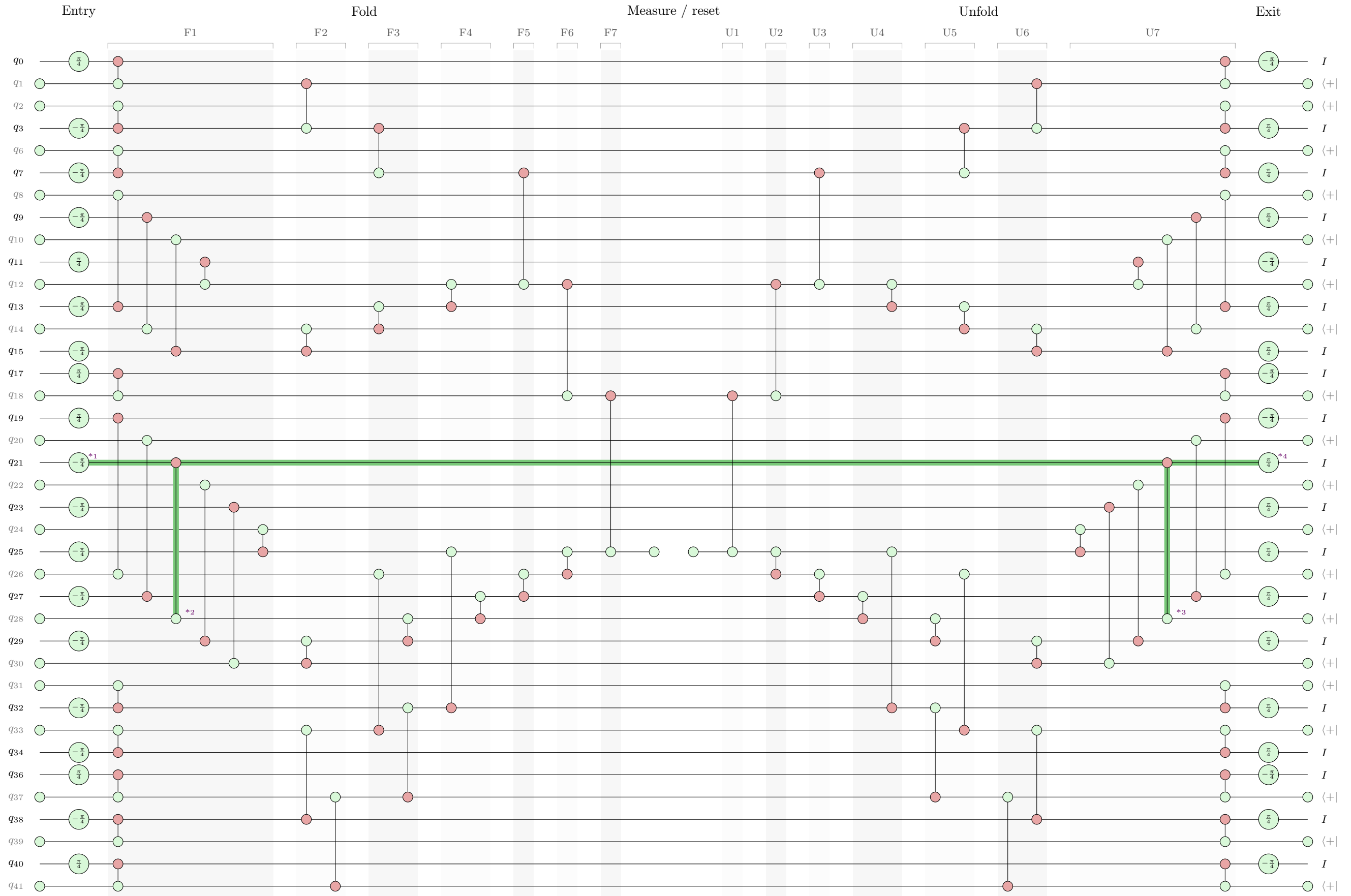
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_253.tikz

$d = 5$: basis element 254

No boundary support. 4 phase defects.



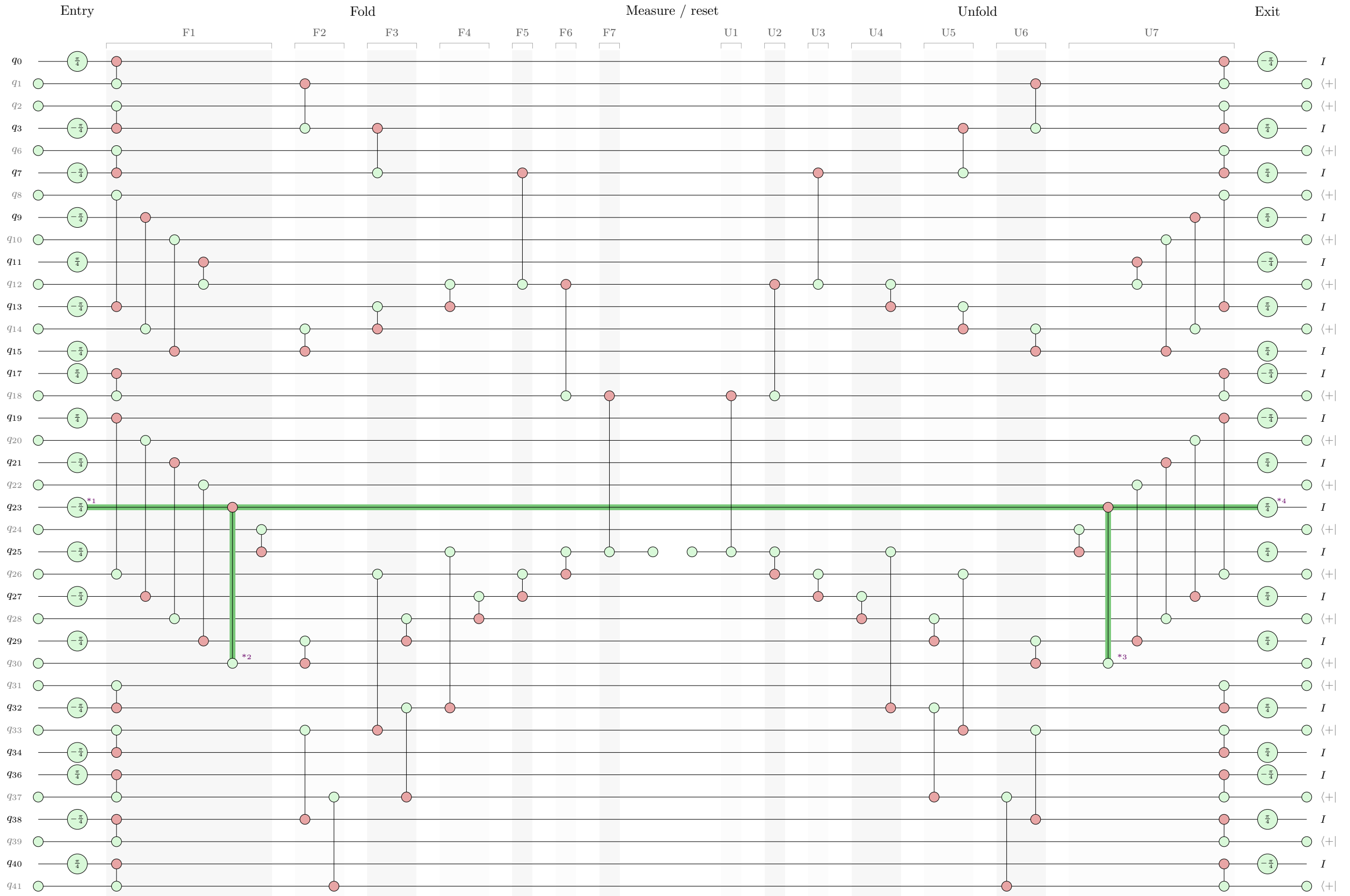
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_254.tikz

$d = 5$: basis element 255

No boundary support. 4 phase defects.



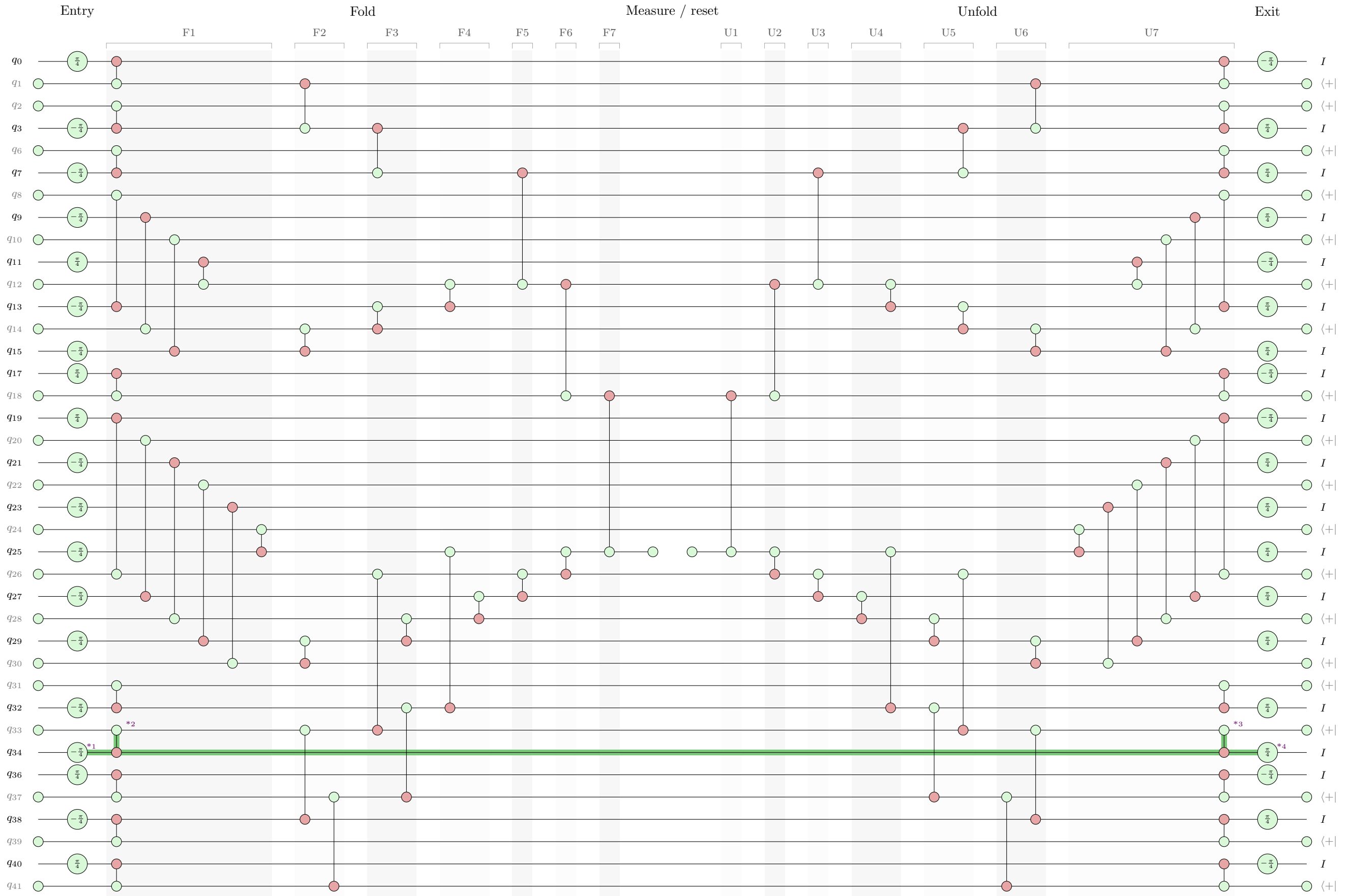
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_255.tikz

$d = 5$: basis element 256

No boundary support. 4 phase defects.



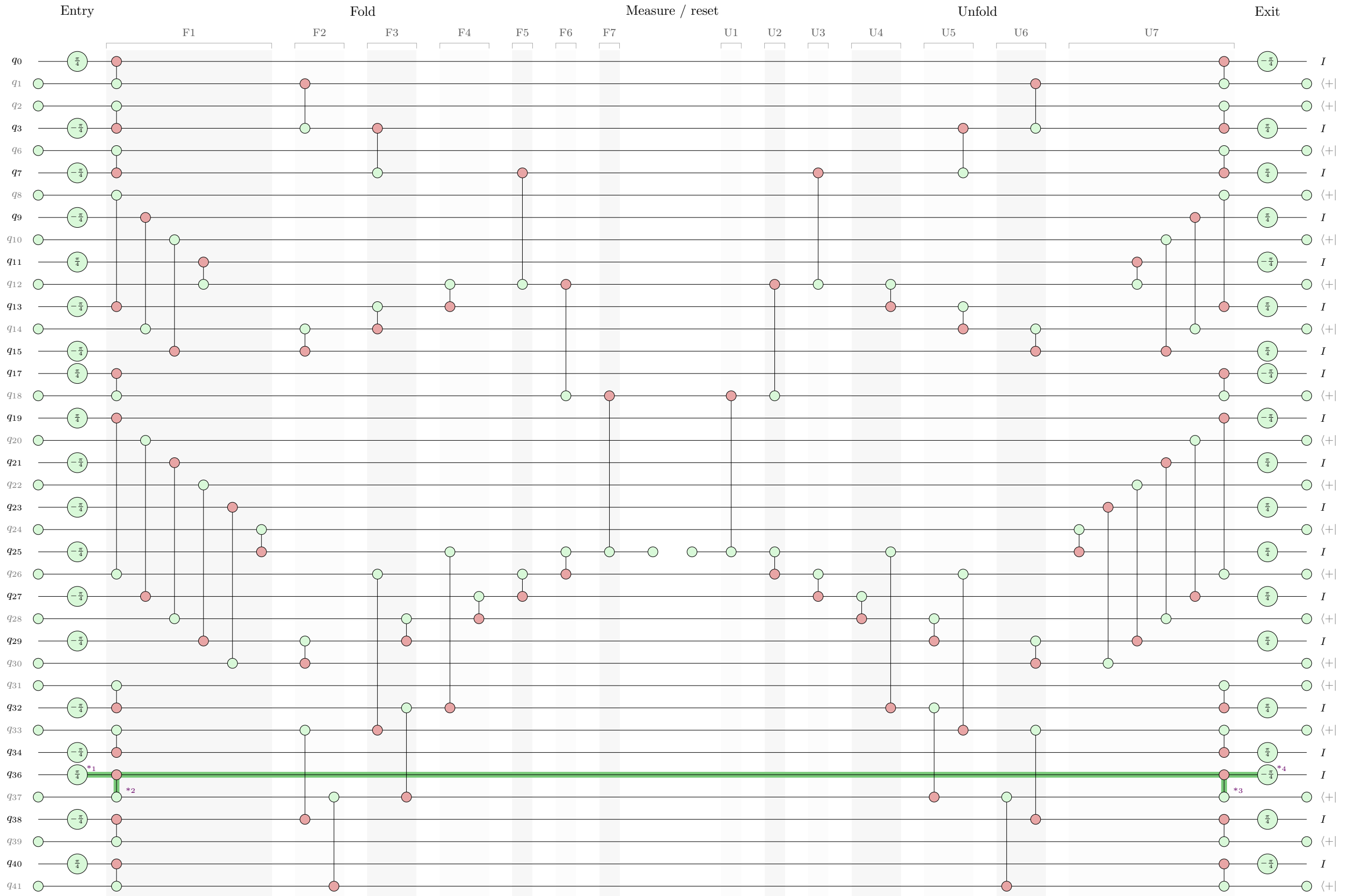
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_256.tikz

$d = 5$: basis element 257

No boundary support. 4 phase defects.



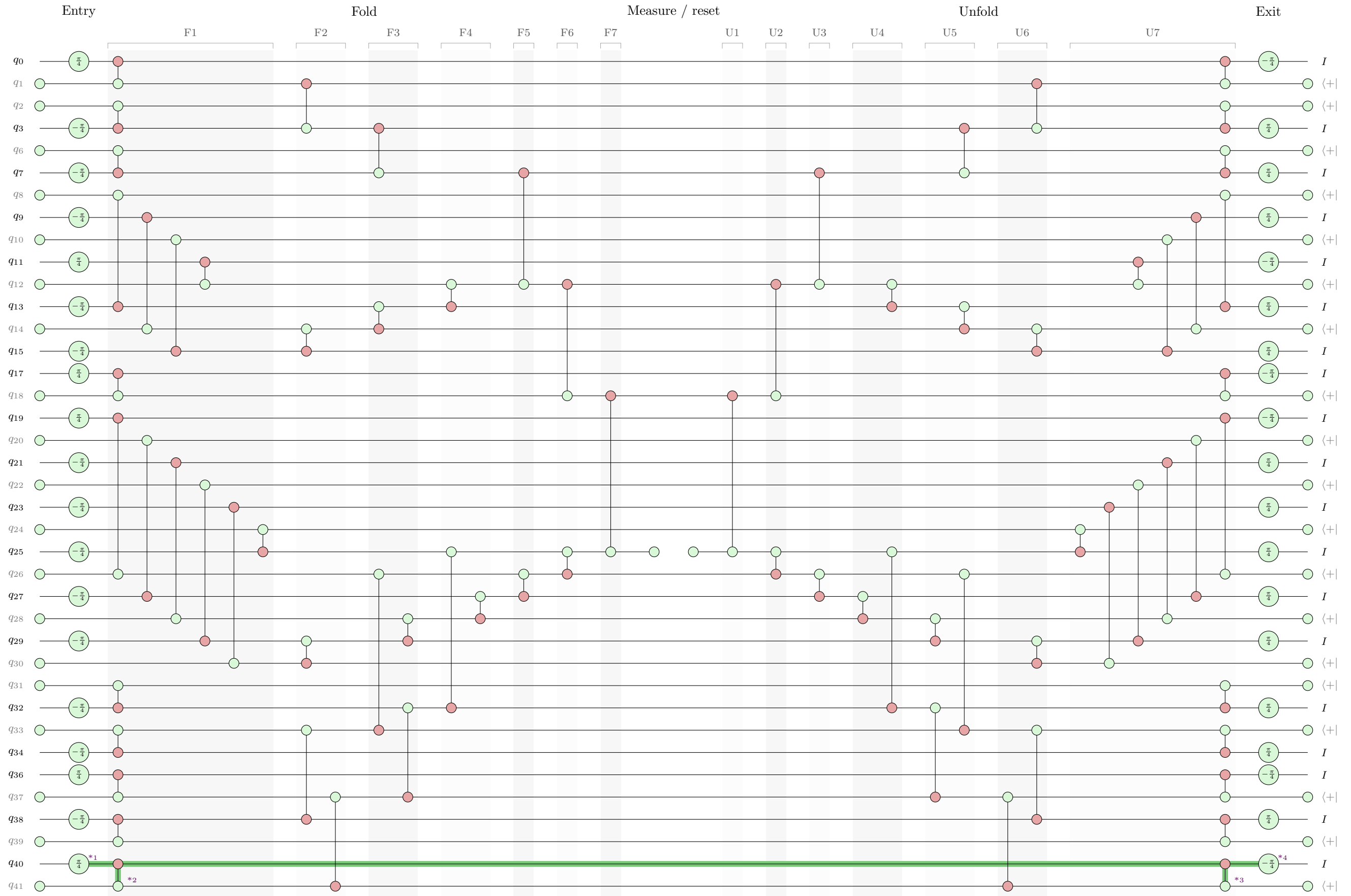
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_257.tikz

$d = 5$: basis element 258

No boundary support. 4 phase defects.

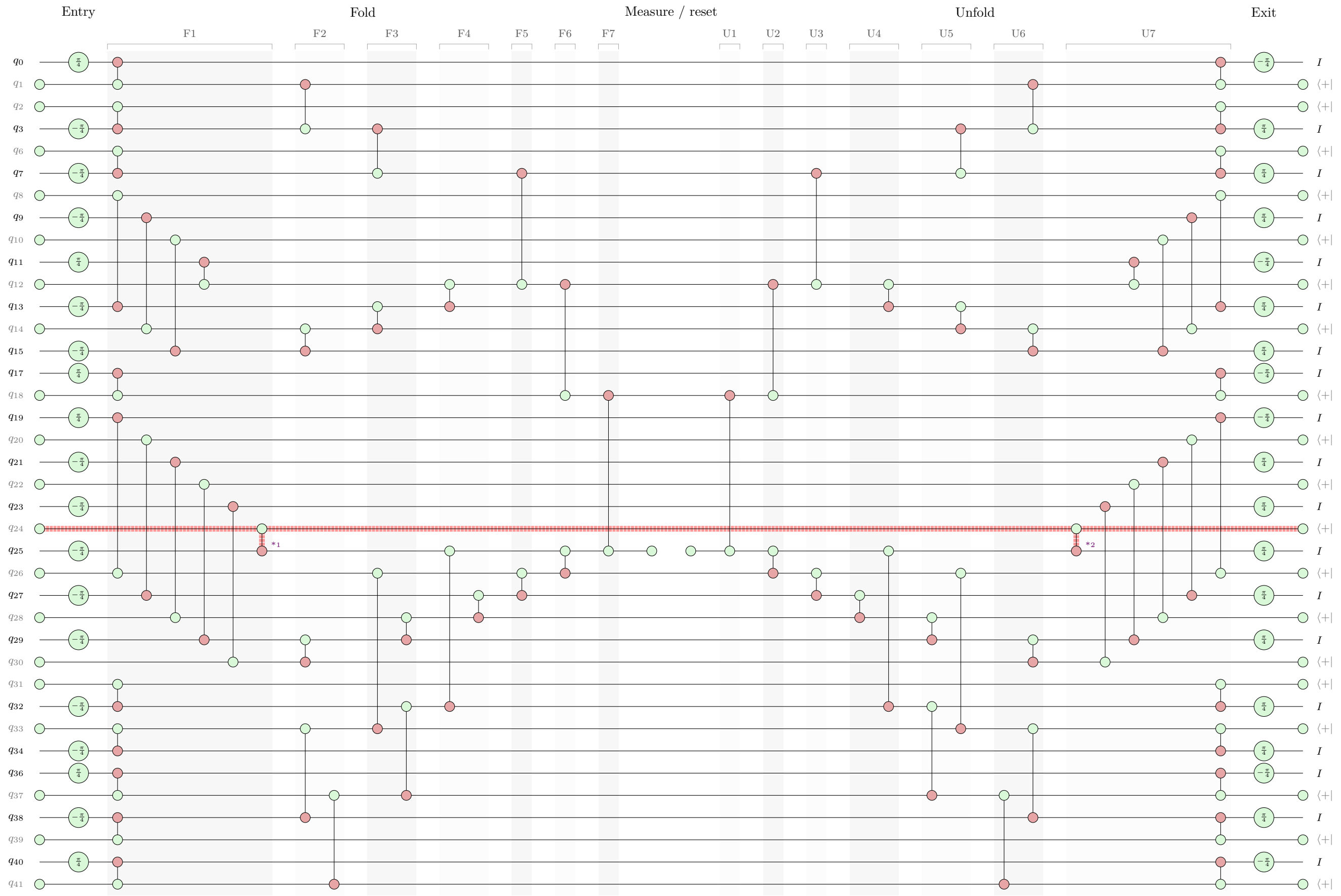


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_258.tikz

$d = 5$: basis element 259
 No boundary support. 2 phase defects.

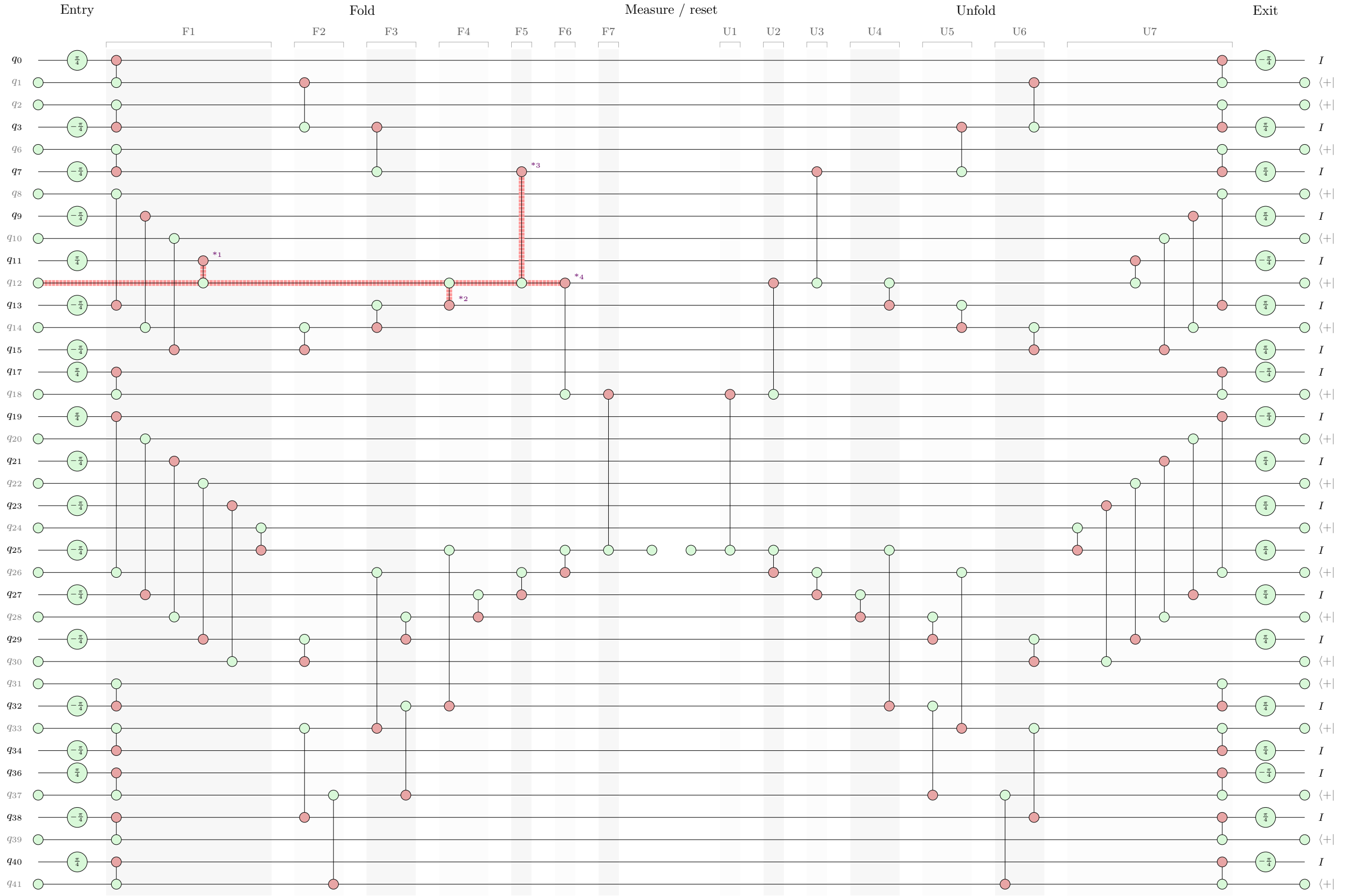


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$.

$d = 5$: basis element 260

No boundary support. 4 phase defects.



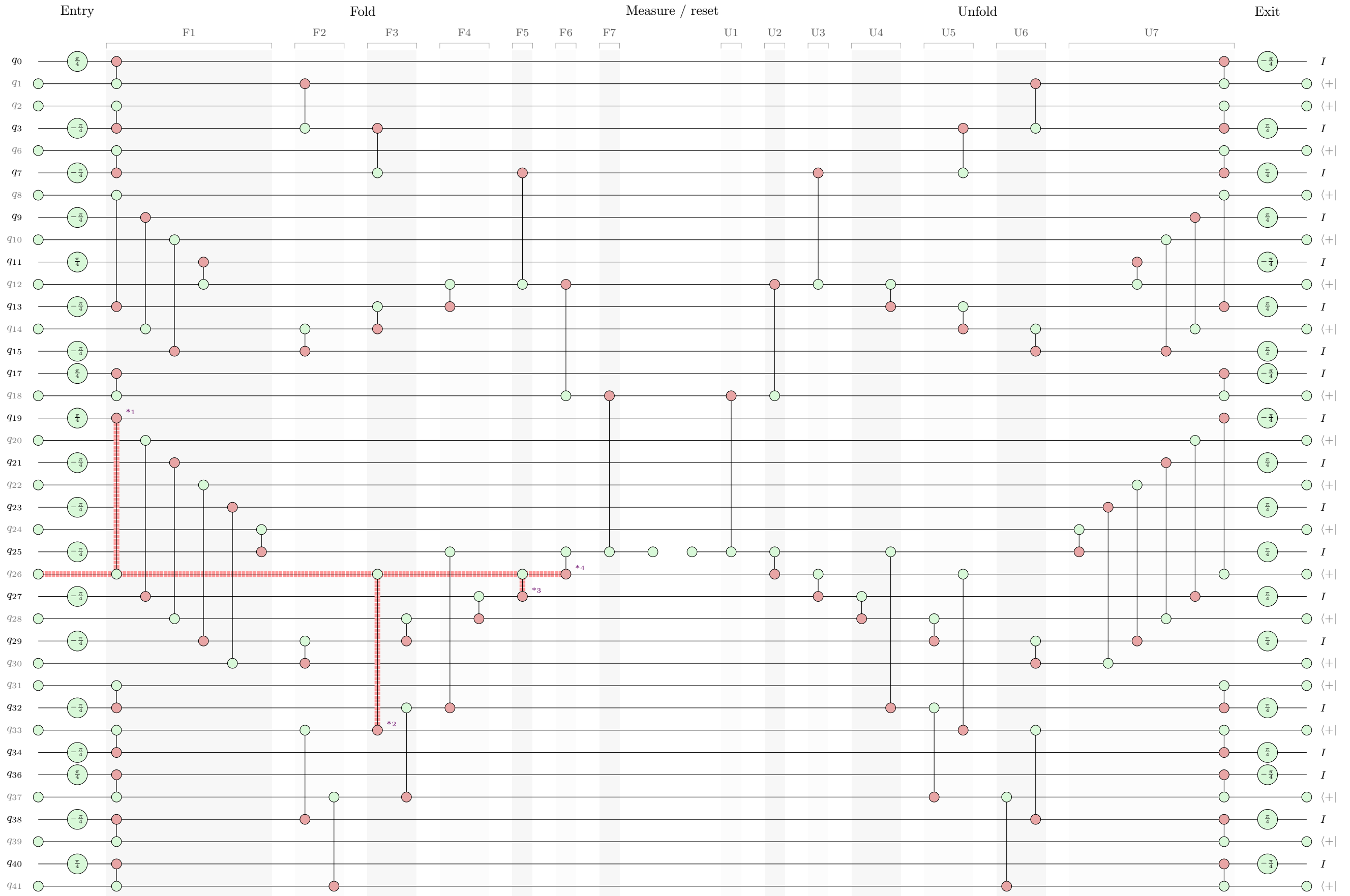
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_260.tikz

$d = 5$: basis element 261

No boundary support. 4 phase defects.



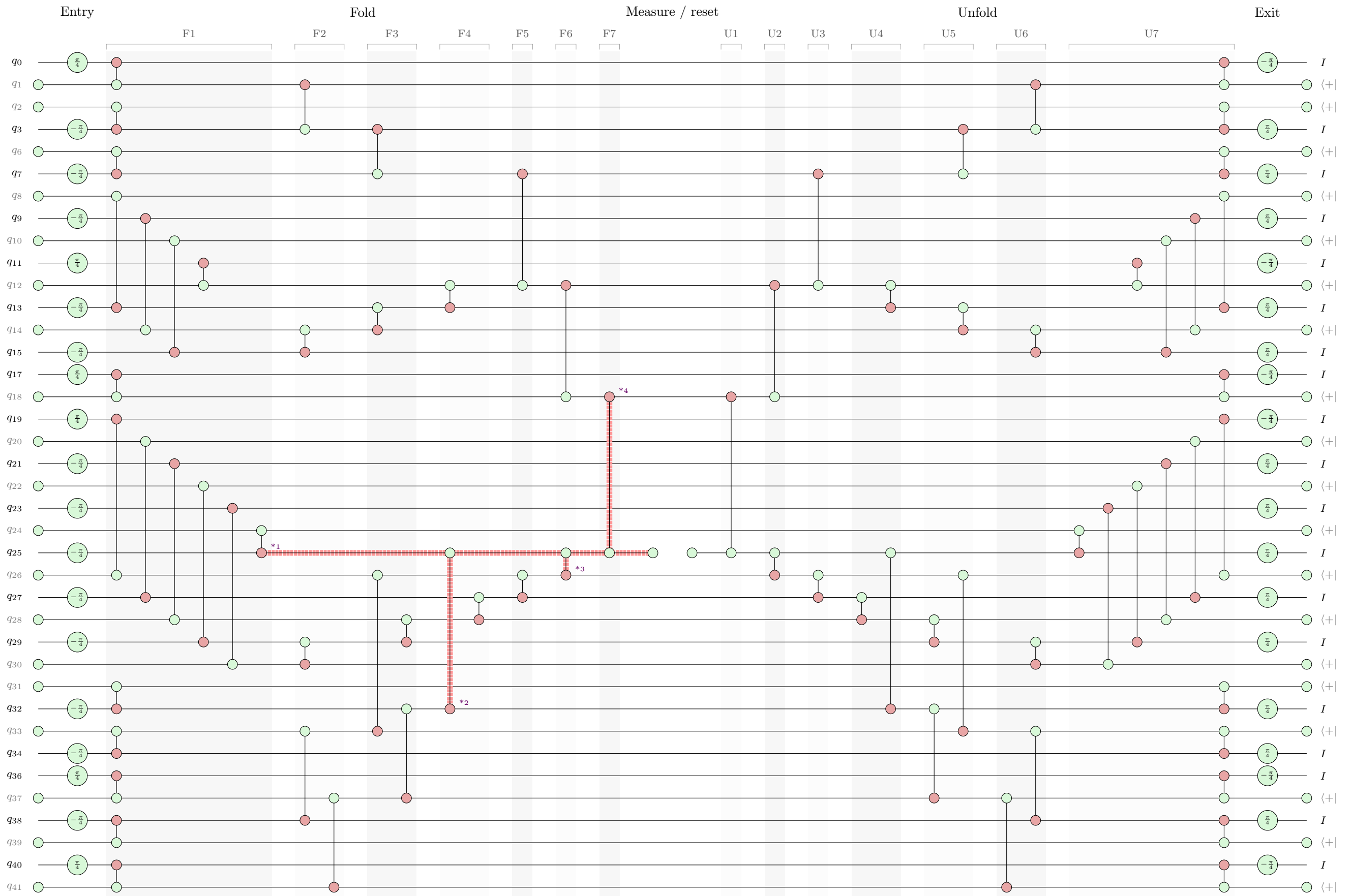
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K \delta.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_261.tikz

$d = 5$: basis element 262

No boundary support. 4 phase defects.



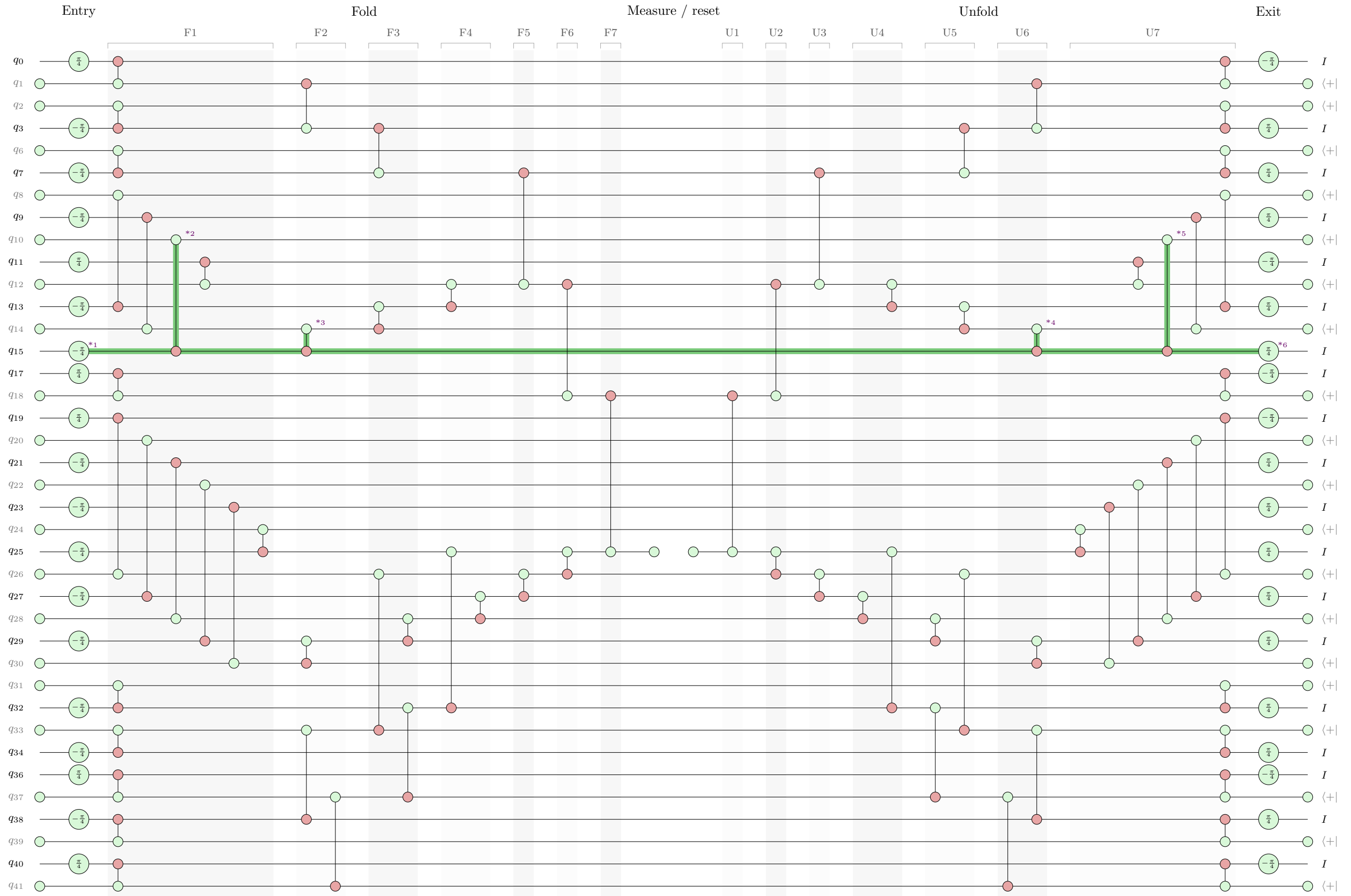
$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$.

d5_262.tikz

$d = 5$: basis element 263

No boundary support. 6 phase defects.

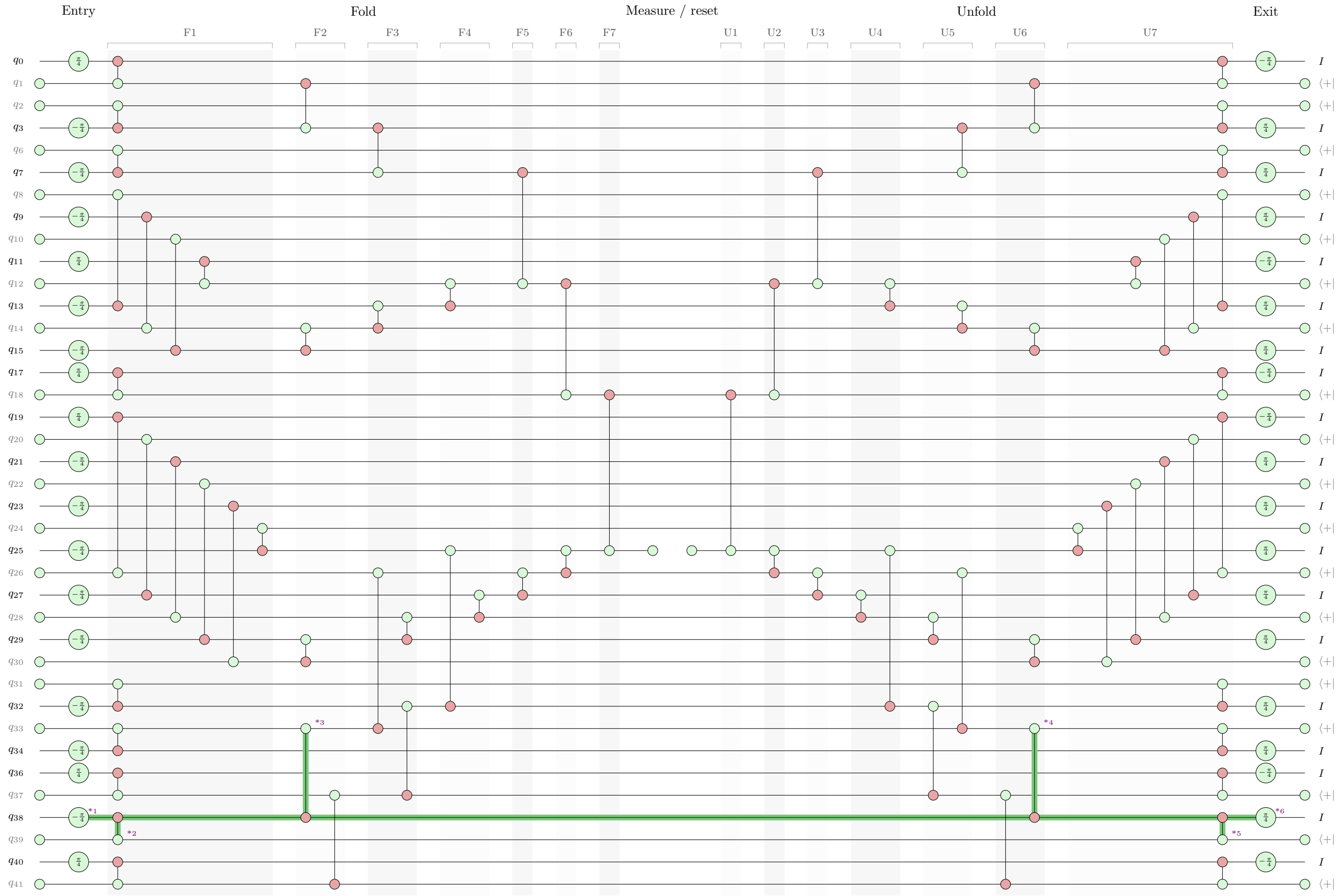


$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi$; $\delta_2 = \pi$; $\delta_3 = \pi$; $\delta_4 = \pi$; $\delta_5 = \pi$; $\delta_6 = \pi$.

d5_263.tikz

$d = 5$: basis element 264
 No boundary support. 6 phase defects.



$$P_{\text{in}} = I, \quad P_{\text{out}} = I, \quad \lambda = 1, \quad KP_{\text{in}} = \lambda P_{\text{out}} K_{\delta}.$$

Phase changes: $\delta_1 = \pi; \quad \delta_2 = \pi; \quad \delta_3 = \pi; \quad \delta_4 = \pi; \quad \delta_5 = \pi; \quad \delta_6 = \pi.$